FRM PART II • 2026 • EXAM 25 NOVEMBER 2026

Current Issues in Financial Markets

Notes + Question Workbook

Every learning objective, in syllabus order

All 31 learning objectives from the 8 Current Issues readings, CI-1 to CI-8, with notes and 8 to 10 practice questions each.

Built from the consolidated Current Issues class notes, SchweserNotes 2026 Book 5 (Readings 100 to 107) and the Tier A and B–D Recall Workbooks. GARP publishes no book for this area; SchweserNotes is the final arbiter.

How to use this book

- The book follows the **syllabus order**, CI-1 a to CI-8 c. Current Issues is covered in full: the readings change every year and the questions are qualitative, so every LO matters.
- The badge in each LO header (P1 P2 P3) is the LO's priority on the Master LO Map v3. It tells you where the reported exam shapes sit; it does not mean a lower-priority LO can be skipped.
- Each LO is one block: **focus box** (why the LO has its priority, Vardeez's pointer and the reported exam shapes), **notes** covering the full LO wording, a **results card** or comparison table, **exam traps**, then **8 to 10 practice questions**. **Solutions** sit at the end of each block, so attempt the questions first.
- LO wording is verbatim from the official GARP 2026 learning objectives, as listed in the Master LO Map. The class notes and SchweserNotes number the readings 100 to 107: Reading n is CI-$(n-99)$.
- Question sources, shown on every question: *Workbook* questions (Tier A to D, August 2026 recall and mapped shapes) appear unchanged; *Adapted* questions take a CI question-bank item (CI QB VI or Imp) and rebuild it to GARP level; *New* questions are written in GARP style from the class notes and SchweserNotes. Every question was checked by an independent reviewer.
- Boxes headed *Correction to the source notes* mark errors found in the class notes, SchweserNotes or a workbook. Figures are drawn for black-and-white printing.

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CI-1 a

P1 CONCEPT

Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A14 Artificial Intelligence and GenAI: AI came up in the August 2026 sitting (two questions recalled), and Map #20 counts it in 6 of 16 sources. Two of the four A14 shapes sit on this LO: S1 (AI versus GenAI) and S3 (AI in market risk). This LO also carries the shared background for the whole reading; CI-1 b to d recap it in a line.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A14·S1** AI vs GenAI: the IMF's two model groups, what GenAI adds, and its signature failure mode (Questions CI-1 a.7 and CI-1 a.9; a.9 is workbook A14.1).
- **A14·S3** Application of AI in market risk: volatility and liquidity models, risk scenarios, dynamic risk sensing (Question CI-1 a.8; workbook A14.3 tests the same shape and is worked under CI-1 c).

What the questions look like: no calculations. Statement audits ("which statement is consistent with the IMF reading"), placing a tool in the right model group or a use case in the right function, and today against tomorrow: what AI does now, what the IMF expects next, and where the human stays.

1. The IMF outreach and the two model groups

The reading is the IMF's study of how AI and ML are used in capital markets. Its evidence comes mainly from an IMF staff market outreach dedicated to AI, alongside several other surveys. Participants included banks and dealers, asset managers, rating agencies, AI vendors, academic institutions and market infrastructure firms. **27 firms** formally responded, from the *buy side* (mutual funds, hedge funds, asset managers, private equity firms, pension funds, institutional investors) and the *sell side* (brokerage firms, research analysts, market makers, investment banks). The aim was to capture how institutions apply AI and ML now and how they expect that use to grow.

Throughout the reading, models fall into two groups:

1. **AI/ML models**: established predictive analytics tools, such as natural language processing (NLP), neural networks, decision trees and clustering algorithms.
2. **Sophisticated AI models**: the most advanced innovations, including deep learning, reinforcement learning and large language models (LLMs).

GenAI runs on LLMs, so it sits on the sophisticated side. Across the financial sector it is already applied through use cases that *build on* existing investment strategies and financial models: investment idea generation, code development for analysts, and customer-facing activities. On the predictive side, LLMs are increasingly integrated into analytical frameworks to support more accurate forecasting.

The failure mode that goes with GenAI. IMF survey respondents list model hallucinations and lack

of explainability among the main areas of concern they raised (CI-1 c). The definition comes from the FSB reading (CI-2 c): a hallucination is a type of model inaccuracy associated with GenAI, in which a model produces a highly confident but incorrect response. Because LLM outputs are typically unstructured text, effective outcome analysis and error rates are often difficult to establish. Workbook A14.1 (Question CI-1 a.9) turns on exactly this split.

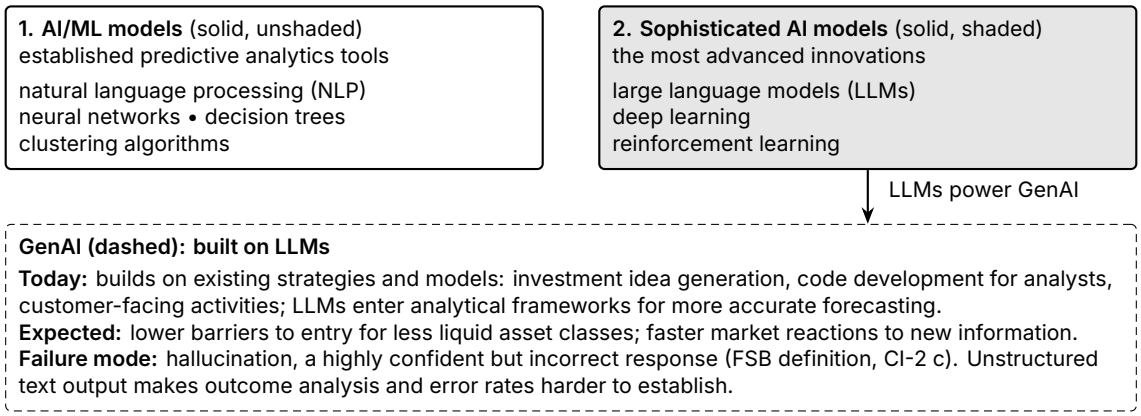


Figure 1. The IMF's two model groups and where GenAI sits. The split is by generation, not by data type: NLP works on text and stays in the established group, and neural networks are established tools even though deep learning is sophisticated.

Correction to the source notes. (1) The class notes list the established group as "NLP, decision trees, clustering" and leave out *neural networks*. SchweserNotes Book 5, LO 100.a, puts neural networks in the established AI/ML group; deep learning is in the sophisticated group. A question that lists neural networks as sophisticated is a sibling swap, not a fact. (2) The notes say AI adoption "is strongest in liquid markets such as equities, FX, and bonds". The direction is right (Schweser reports a strong correlation between market liquidity and AI adoption among investment managers), but the list and the ranking need fixing: the most promising areas for *value creation* are liquid public markets such as equities and listed derivatives, with *government* bonds and foreign exchange also offering significant opportunities. Listed derivatives belong in the answer, equities and listed derivatives lead, and "bonds" means government bonds.

2. Benefits and the four broad risk areas

In the IMF's framing, AI and ML are expected to change capital market structure, driven by faster execution speeds and stronger network effects. On the positive side they can enhance financial stability, efficiency, productivity and forecasting accuracy, refine investment frameworks and improve the quantification of market crash risk. On the negative side they may introduce new financial stability risks and amplify existing concerns about liquidity, leverage and interconnectedness. The reading groups the risks into **four broad areas**: (1) increased operational risks, (2) greater market volatility and trading speed under stress, (3) heightened cyber and market manipulation risks, and (4) challenges in monitoring and transparency (opacity). The same four themes return, in a different order, as the four financial stability challenges of CI-1 c.

3. Where AI sits: the four functions

The IMF maps AI's uses onto four capital markets processes (Figure 2).

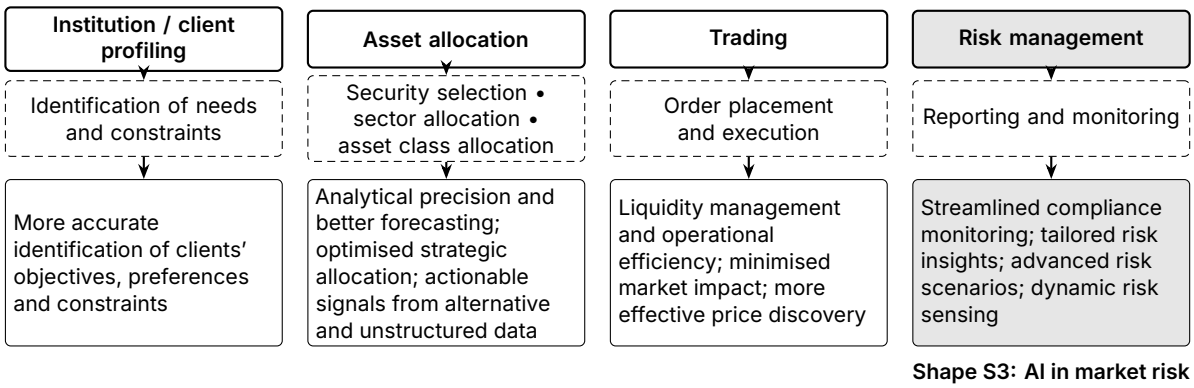


Figure 2. The IMF's four processes (solid top row), the steps inside each (dashed) and what AI adds (bottom row); the shaded column is shape S3. Each benefit belongs to one column: signals from alternative data are asset allocation, not trading.

In that map, **risk management** is shape S3: AI streamlines compliance monitoring, generates tailored risk insights, develops advanced risk scenarios and enables dynamic risk sensing. The FSB reading adds volatility and liquidity risk models that better predict how heightened volatility may affect positions (CI-2 b); workbook A14.3 combines the two.

4. What AI does today

Survey benefits. Respondents named four key benefits of AI and ML: (1) enhanced productivity and efficiency, (2) improved processing and analysis of *unstructured* data, (3) *narrower* bid-ask spreads, and (4) cost savings in the design and execution of trading algorithms. High fixed costs are not on the list: they are the main cost of adoption (Section 5).

Use cases. The survey's list runs from forward-looking indicators, alternative datasets, new asset classes and trading signals to portfolio allocation and optimisation, back-office operations, customer service, pricing and forecasting, risk assessment and trade automation; respondents also cite code generation, prototyping, information extraction and summarisation. In asset management, one use analyses text and alternative data to uncover causal relationships in previously unexplored markets; another uses traditional AI and ML to make existing models more accurate and robust.

Who decides. AI in trading and investment decision execution is still at an early stage. Sophisticated AI has mainly been used to generate *signals* that feed existing systems; human traders still make the final decisions, rather than fully autonomous AI trading agents. Respondents agreed that humans must remain "in the loop", understand AI-generated strategies, and actively manage the related regulatory and ethical risks.

Not new. Robo-advisors and AI-powered ETFs (in which AI systems build and adjust portfolios) have been in use for quite some time. Robo-advisory assets under management have grown significantly; AI-powered ETFs are still a relatively small share of the overall market.

Two prefixes. Supervisors use AI through supervisory technology (*SupTech*) to monitor institutions and markets, detect anomalies in large datasets and support regulatory compliance; banks increasingly adopt AI-powered regulatory technology (*RegTech*) to improve efficiency and compliance management. The supervisory side is developed in CI-1 d.

5. Where the value is, and what it costs

Adoption has accelerated markedly in recent years. The most promising areas for value creation are **liquid asset classes traded in public markets**: equities and listed derivatives first, with government bonds and foreign exchange also offering significant opportunities. These markets are highly dynamic, so AI systems can more readily learn from and adapt to changing conditions, and the survey found a strong correlation between market liquidity and AI adoption among investment managers. Private credit and emerging markets may offer new opportunities *in future*.

Costs and concentration. Better hardware and algorithmic efficiency have cut the *unit* cost of training models, but growing model complexity has pushed *overall* costs up. The high fixed costs of infrastructure and specialised talent may concentrate the market among a few private-sector developers, and firms with extensive trading and client data are more likely to keep their advantage. The pattern repeats across countries: North America clearly leads in developing foundational models.

6. Future uses

Area	Today	Expected next
Trading and investment decisions	Sophisticated AI mainly generates signals for existing systems; humans make the final decisions	Greater autonomy in AI-driven decisions, particularly in markets such as equities; a human-in-the-loop approach likely to remain
GenAI	Idea generation, analysts' code, customer-facing work; LLMs in forecasting frameworks	Lower barriers to entry for less liquid asset classes; faster market reactions to new information
Asset classes	Value greatest in liquid public markets (equities, listed derivatives; then government bonds and FX)	Private credit and emerging markets may offer new opportunities
Patents	Significant rise in AI and ML patents for algorithmic and high-frequency trading; asset management filings up too	Trading: high-throughput, low-latency signals and more efficient platforms. Asset management: automated rebalancing, tailored capital requirements, liquidity and cash flow management, forecasting and valuation

Patent filings also target systems that process alternative data sources and interpret unstructured data, and wider access to (and management of) alternative asset classes, including digital assets, emissions trading and signed transaction validation. **Talent:** the pool of AI-skilled professionals in financial services is growing, demand keeps rising, and job postings requiring AI skills have increased, particularly for roles that directly influence market transactions and investment decisions.

Emerging and developing economies. AI could create opportunities by lowering barriers to entry, enhancing liquidity, increasing capital flows and stimulating investment, through better access to financial services, loan origination, portfolio construction, credit scoring and robo-advice. GenAI can interpret and structure unorganised or fragmented data, and AI-generated *synthetic data* could help train investment models where data are scarce. Synthetic data carry two considerations: biases can be *amplified*, and some values can be *over- or underrepresented* relative to real-world distributions. Differences among emerging markets matter, particularly between larger and smaller economies. Respondents acknowledged the risk of automating lower-skilled jobs but were optimistic about financial inclusion, access to credit and lower barriers to entry; they cautioned that the high fixed costs of AI adoption may be a significant challenge for these markets.

Results card: CI-1 a

Evidence	IMF staff market outreach on AI: 27 firms responded, buy side and sell side
Model groups	AI/ML: NLP, neural networks, decision trees, clustering. Sophisticated: LLMs, deep learning, reinforcement learning. GenAI runs on LLMs
Four functions	Client profiling; asset allocation (signals from alternative and unstructured data); trading (market impact, price discovery); risk management (compliance monitoring, tailored insights, advanced scenarios, dynamic risk sensing)
Survey benefits	Productivity and efficiency; unstructured data; narrower bid-ask spreads; cheaper design and execution of trading algorithms
Value creation	Equities and listed derivatives, then government bonds and FX; strong liquidity and adoption link. Future: private credit, emerging markets
Costs	Unit costs down, overall costs up; high fixed costs concentrate the market; data-rich firms keep their edge; North America leads foundational models
Today versus next	Signals in, humans decide; more autonomy expected (equities), human in the loop stays
Emerging markets	Lower barriers, credit access, robo-advice; synthetic data (bias amplification, over- or underrepresentation); high fixed costs a hurdle

Exam traps

- **Sibling swap on the model groups.** Neural networks, NLP, decision trees and clustering are established AI/ML; LLMs, deep learning and reinforcement learning are sophisticated. Handling text does not promote NLP.
- **Function swap.** Signals from alternative data are asset allocation; market impact and price discovery are trading; advanced risk scenarios and dynamic risk sensing are risk management.
- **Polarity flips on the survey benefits.** *Narrower* spreads, *unstructured* data; high fixed costs are a burden, never a benefit.
- **Unit cost for overall cost.** Unit training costs fell; overall costs rose with complexity.
- **Forecast stated as fact.** Fully autonomous trading is not today's practice; more autonomy is *expected*, particularly in equities, with a human kept in the loop.
- **Future stated as current.** Private credit and emerging markets are *future* opportunities; value today is greatest in liquid public markets.
- **Robo against AI ETF.** Robo-advisory AUM has grown significantly; AI-powered ETFs remain a small share.
- **Synthetic data cut both ways.** They help where data are scarce, but can amplify bias and mis-weight values in either direction.

Practice questions: CI-1 a

Question CI-1 a.1 Adapted from CI QB VI Q2 • SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

An AI vendor is pitching a trade-execution product to the head of trading at a Johannesburg brokerage and cites the IMF's outreach to market participants. Which statement in the pitch is consistent with the benefits that the survey respondents identified?

A. Respondents cited the low fixed cost of AI as a key benefit, which is why smaller firms tend to build their AI capabilities in-house.

B. Respondents cited better processing of structured data as a key benefit, since unstructured text remains beyond current models.

C. Respondents cited narrower bid-ask spreads and cost savings in the design and execution of trading algorithms as key benefits.

D. Respondents cited the rapid growth of AI-powered ETFs, which now hold a dominant share of the overall ETF market.

Question CI-1 a.2 Adapted from CI QB VI Q4 • SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

The CIO of a Hong Kong multi-asset manager asks the firm's head of data science where AI is most likely to create value, based on the IMF survey. Which response is most consistent with the survey findings?

A. Government bonds and foreign exchange ahead of equities, because sovereign and currency markets are the deepest and most data-rich.

B. Equities and listed derivatives, because AI systems learn and adapt most readily in liquid, dynamic public markets.

C. Private credit and emerging markets, which the survey identified as the segments where AI already creates the most value.

D. Over-the-counter corporate bonds, because AI adds most where less than 10% of trading is algorithmic and prices are slow to adjust.

Question CI-1 a.3 Adapted from CI QB VI Q7 • SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

A fintech founder in Bengaluru tells investors that falling training costs will soon let small firms everywhere build their own foundational AI models. A risk analyst reviewing the pitch draws on the IMF reading. Which response is most accurate?

A. Falling unit costs have cut overall costs as well, so the only barrier to entry left is access to specialised AI talent.

B. High fixed costs weigh most on firms with extensive trading and client data, so data-rich incumbents are losing their edge.

C. Foundational model development is spread evenly across regions, so the reading raises no concern about concentration by country.

D. Unit training costs have fallen, but model complexity has raised overall costs, and high fixed costs favour a few developers.

Question CI-1 a.4 Adapted from CI QB VI Q5 • SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

The investment committee of a Toronto pension fund asks its head of quantitative strategies how far AI has gone in trading and investment decisions, and where the IMF expects it to go. Which answer is most consistent with the reading?

A. Fully autonomous AI agents already execute most equity trades without human input, and human oversight is expected to be phased out as the models improve.

B. Sophisticated AI mainly generates signals for existing systems while humans decide; more autonomy is expected, notably in equities, with a human kept in the loop.

C. AI is confined to back-office tasks today and generates no trading signals, and the IMF expects it to stay out of investment and trading decisions altogether.

D. Greater autonomy is expected first in fixed income, where AI use is most mature, while human traders keep the final decision in equities for the foreseeable future.

Question CI-1 a.5 Adapted from CI QB VI Q10 • SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

A strategist at a Chicago market infrastructure firm treats AI-related patent filings and job postings as indicators of where AI innovation is heading. Which observation is consistent with the IMF reading?

A. Recent trading filings target high-throughput, low-latency signal computation, while asset management filings target automated rebalancing and forecasting.

B. Recent trading filings target automated asset class rebalancing and tailored capital requirements, while asset management filings target low-latency signals.

C. Asset management filings have fallen in volume as the field has matured, while trading filings have moved away from referencing AI and ML techniques.

D. Job postings that require AI skills have risen mainly in back-office roles, with little change in roles that directly influence investment decisions.

Question CI-1 a.6 New • built from SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

An adviser at a development finance institution in Nairobi is assessing what AI could do for local capital markets, where data are thin and many firms lack access to credit. Which assessment best reflects the IMF reading?

A. AI could lower entry barriers and widen credit access through credit scoring and robo-advice, though high fixed adoption costs may be a significant challenge.

B. AI-generated synthetic data would remove the biases present in thin local samples, so models could be trained on them without adding new sources of model risk.

C. Respondents saw the automation of lower-skilled jobs as the main benefit of AI for these markets, because it cuts the cost of delivering financial services.

D. The benefits should be much the same across all emerging markets, since GenAI's ability to structure fragmented data does not depend on the size of the economy.

Question CI-1 a.7 New • built from SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

The head of research at a Seoul brokerage is briefing the board on what GenAI adds to the firm's existing AI/ML toolkit. Which statement is consistent with the IMF reading?

A. GenAI replaces the firm's existing models and strategies with new architectures, so its use cases do not build on current investment processes.

B. GenAI is confined to customer chatbots for now; LLMs are not yet being integrated into analytical or forecasting frameworks.

C. GenAI is expected to raise barriers to entry in less liquid asset classes and to slow market reactions while LLMs parse full documents.

D. GenAI builds on existing strategies and models, for example in idea generation and analysts' code, and LLMs are entering forecasting work.

Question CI-1 a.8 New • built from SchweserNotes Book 5, LO 100.a; CI notes CI-1 a

The CRO of a Zurich private bank proposes an AI project that would generate tailored risk insights for desk heads, build advanced stress scenarios and update risk measures continuously as market conditions change. Where does the IMF's map of capital markets processes place this project?

A. Risk management (reporting and monitoring), whose AI benefits also include streamlined compliance monitoring and dynamic risk sensing.

B. Trading (order placement and execution), whose AI benefits include minimising market impact and supporting more effective price discovery.

C. Asset allocation (security, sector and asset class selection), whose AI benefits include actionable signals from alternative and unstructured data.

D. Institution and client profiling (needs and constraints), whose AI benefits include identifying clients' objectives, preferences and constraints.

Question CI-1 a.9 Workbook A14.1 • S1 • Aug-2026 • Map #20

A junior analyst at a Singapore asset manager is presenting to newly hired analysts on the firm's model inventory, which contains a decision-tree credit model, a clustering tool for client segmentation, an NLP news classifier, and a recently licensed LLM-based assistant that drafts research notes and code. She wants one slide that correctly distinguishes established AI/ML from GenAI. Which of the following statements should she include?

A. Decision trees, clustering and NLP belong to the sophisticated AI group, while LLMs and reinforcement learning belong to the group of established predictive analytics tools.

B. GenAI is now used mainly as fully autonomous trading agents that execute without human intervention, taking over the final decisions that human traders made with earlier AI/ML signals.

C. Decision trees, clustering and NLP are established predictive analytics tools, while the LLM is a GenAI tool whose characteristic inaccuracy is a hallucination: a highly confident but incorrect response.

D. Because LLM outputs are structured numerical forecasts, outcome analysis and error rates are easier to establish for the GenAI assistant than for the established AI/ML models.

Solutions • CI-1 a

Solution CI-1 a.1 — Answer: C

Adapted from CI QB VI Q2 • SchweserNotes Book 5, LO 100.a

Step 1 – Recall the survey's four benefits

(1) Enhanced productivity and efficiency; (2) improved processing and analysis of *unstructured* data; (3) *narrower* bid-ask spreads; (4) cost savings in the design and execution of trading algorithms.

Step 2 – Match the pitch

Narrower spreads and cheaper algorithm design and execution are two of the four, and both fit a trade-execution product.

$\Rightarrow$ C is the only statement that the survey supports.

Interpretation

The benefits compress frictions (spreads, algorithm costs) and unlock messy data. A pitch that turns any of them around, or presents a cost of AI as a benefit, has left the reading.

Why the other options are wrong

A is incorrect. Polarity flip: AI involves *high* fixed costs, and those costs mean that larger firms benefit most while smaller firms often outsource their AI capabilities (CI-1 b).

B is incorrect. Wrong-input twin: the cited benefit is processing *unstructured* data, which is what AI unlocks.

D is incorrect. Polarity flip: AI-powered ETFs still represent a relatively small share of the overall market; it is robo-advisory AUM that has grown significantly.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024), Advances in Artificial Intelligence: Implications for Capital Markets Activities; SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.2 — Answer: B

Adapted from CI QB VI Q4 • SchweserNotes Book 5, LO 100.a

Step 1 – Where the survey puts the value

The most promising areas for value creation are liquid asset classes traded in public markets, such as equities and listed derivatives, with government bonds and foreign exchange also offering significant opportunities.

Step 2 – Why

These markets are highly dynamic, so AI systems can more readily learn from and adapt to changing conditions. The survey found a strong correlation between market liquidity and AI adoption among investment managers.

$\Rightarrow$ Liquidity is the driver: the more liquid and dynamic the market, the more AI can learn.

Interpretation

Government bonds and FX are on the list, but in second place. Private credit and emerging markets are the reading's *future* opportunities. The correction box in Section 1 explains why "bonds" in the class notes should be read as government bonds.

Why the other options are wrong

A is incorrect. Ranking swap: government bonds and FX "also" offer significant opportunities; equities and listed derivatives lead.

C is incorrect. Future stated as current: the survey says these segments may present new oppor-

tunities in future, not that they lead today.
D is incorrect. Wrong-input twin with a polarity flip: under 10% is the US fixed-income algorithmic trading share (CI-1 b), and AI value is greatest in liquid public markets, not in less liquid OTC bonds.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a and 100.b

Solution CI-1 a.3 — Answer: D

Adapted from CI QB VI Q7 • SchweserNotes Book 5, LO 100.a

Step 1 – Separate unit cost from overall cost

Advances in computing hardware and algorithmic efficiency have reduced the *unit* cost of training AI models, but the growing complexity of the models has driven *overall* costs up.

Step 2 – The consequence

The high fixed costs of infrastructure and specialised talent may lead to greater market concentration, with a few private-sector developers dominating. Firms with extensive trading and client data are more likely to keep their advantage, and North America clearly leads in foundational models.
⇒ The founder has taken a fall in unit cost for a fall in total cost.

Interpretation

Cheaper bricks, much bigger buildings. The reading expects concentration among developers and data-rich firms, not a democratised field of model builders.

Why the other options are wrong

A is incorrect. Scope error: the unit-cost decline does not carry over to overall costs, and specialised talent is part of the high fixed cost, not the only remaining barrier.
B is incorrect. Polarity flip: data-rich firms are *more* likely to maintain their competitive advantage and reinforce concentration.
C is incorrect. Absolute claim: the pattern repeats at country level, with North America clearly leading in foundational models.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.4 — Answer: B

Adapted from CI QB VI Q5 • SchweserNotes Book 5, LO 100.a

Step 1 – Today

Use of AI in trading and investment decision execution is at an early stage. Sophisticated AI has mainly generated signals that serve as inputs to existing systems, with human traders making the final decisions, not fully autonomous AI trading agents.

Step 2 – Next

AI is expected to play a larger role in investment and trading, with greater autonomy in AI-driven decisions, particularly in markets such as equities. A human-in-the-loop approach is likely to remain, so that models operate within established rules and under human oversight.
⇒ More autonomy, but still a human in the loop.

Interpretation

The question separates a forecast from current practice. Autonomy is the direction of travel; the human stays in the decision seat both now and in the IMF's expected future.

Why the other options are wrong

A is incorrect. Forecast stated as fact, plus a polarity flip: autonomous execution is not today's practice, and respondents expect human oversight to remain.
C is incorrect. Polarity flip: signal generation is precisely where sophisticated AI is used today.
D is incorrect. Market swap: the reading names equities as the market where greater autonomy is expected.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.5 Answer: A

Adapted from CI QB VI Q10 • SchweserNotes Book 5, LO 100.a

Step 1 – Trading patents

There has been a significant rise in patents referencing AI and ML for algorithmic and high-frequency trading. Recent filings focus on computing trading signals with high throughput and low latency, and on the operational efficiency of trading and brokerage platforms.

Step 2 – Asset management patents

Filings have also increased in volume. Many apply ML to automate asset class rebalancing, set capital requirements tailored to individual needs, improve liquidity management and cash flow efficiency, and improve forecasting and valuation.
⇒ Trading patents chase speed; asset management patents chase automation.

Interpretation

The talent data point the same way: job postings requiring AI skills have increased, particularly for roles that directly influence market transactions and investment decisions.

Why the other options are wrong

B is incorrect. Sibling swap: it exchanges the two domains' themes. Low-latency signals are trading; rebalancing and tailored capital requirements are asset management.
C is incorrect. Polarity flip on both halves: asset management filings have increased, and AI and ML references in trading filings have risen significantly.
D is incorrect. Scope error: the rise in AI-skilled postings is concentrated in roles that directly influence transactions and investment decisions, not in back-office roles.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.6 Answer: A

New • built from SchweserNotes Book 5, LO 100.a

Step 1 – The opportunity

AI could create opportunities for emerging and developing markets by lowering barriers to entry, enhancing liquidity, increasing capital flows and stimulating investment, through better access to financial services, loan origination, portfolio construction, credit scoring and robo-advice.

Step 2 – The hurdle

Respondents cautioned that the high fixed costs of AI adoption may pose a significant challenge

for these markets.
⇒ Benefit and cost, both stated in the reading.

Interpretation

For a data-poor market the reading also offers synthetic data, with two caveats: it can amplify biases and over- or underrepresent values relative to real-world distributions. It is a tool with model risk of its own, not a cure.

Why the other options are wrong

B is incorrect. Polarity flip: synthetic data can *amplify* biases, and mis-weighting of values is a second risk.
C is incorrect. Polarity flip: respondents acknowledged the automation of lower-skilled jobs as a risk; their optimism was about financial inclusion, access to credit and lower barriers to entry.
D is incorrect. Absolute claim: the reading stresses potential differences among emerging markets, particularly between larger and smaller economies.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.7 — Answer: D

New • built from SchweserNotes Book 5, LO 100.a

Step 1 – What GenAI does today

GenAI is already applied through use cases that build on existing investment strategies and financial models: investment idea generation, code development for analysts and customer-facing activities. LLMs are increasingly integrated into analytical frameworks to support more accurate forecasting.

Step 2 – What it may do next

Potential benefits include lowering barriers to entry for less liquid asset classes and accelerating market reactions to new information.
⇒ GenAI extends the existing toolkit; it does not replace it.

Interpretation

This is the "what GenAI adds" half of shape S1. The other half, its signature failure mode, is the hallucination tested in Question CI-1 a.9.

Why the other options are wrong

A is incorrect. Polarity flip: GenAI use cases *build on* existing strategies and models.
B is incorrect. Scope error: customer-facing work is only one use; idea generation and analysts' code are others, and LLMs are already entering analytical frameworks for forecasting.
C is incorrect. Polarity flip on both expected benefits: GenAI may *lower* barriers to entry for less liquid asset classes and *accelerate* market reactions to new information.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.8 — Answer: A

New • built from SchweserNotes Book 5, LO 100.a

Step 1 – Read the project

Tailored risk insights, advanced stress scenarios and continuously updated risk measures are

monitoring and reporting tasks for the risk function.

Step 2 – Find the column

In the IMF's map, risk management (reporting and monitoring) is where AI streamlines compliance monitoring, generates tailored risk insights, develops advanced risk scenarios and enables dynamic risk sensing.
⇒ The project is the risk management column: shape S3, AI in market risk.

Interpretation

All four options pair a process with a benefit that the reading really assigns to it, so the question is won on the project, not on the pairing. Workbook A14.3 (worked under CI-1 c) adds the FSB's view of the same use: volatility and liquidity risk models that predict how heightened volatility may affect positions.

Why the other options are wrong

B is incorrect. Sibling swap: minimising market impact and price discovery are trading benefits, and the project does not execute orders.
C is incorrect. Sibling swap: signals from alternative and unstructured data drive asset allocation, not risk monitoring.
D is incorrect. Sibling swap: client profiling identifies objectives, preferences and constraints; it does not build stress scenarios.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a

Solution CI-1 a.9 — Answer: C

Workbook A14.1 • S1 • Aug-2026 • Map #20

Step 1 – Place each tool in the IMF grouping

AI/ML models are established predictive analytics tools: NLP, neural networks, decision trees, clustering. Sophisticated AI models are the most advanced innovations: LLMs, deep learning, reinforcement learning.
⇒ Three of the four inventory items are established AI/ML; the LLM assistant is the sophisticated, generative one.

Step 2 – What makes the LLM generative

LLMs, built on the Transformer architecture (attention mechanism), generate content: research drafts, code, chatbot answers. GenAI's characteristic inaccuracy is the hallucination, a highly confident but incorrect response.
⇒ The discriminator between the groups is not "better accuracy" but a different output (generated text) and a different failure mode.

Step 3 – How such models are used today

Sophisticated AI has mainly generated signals that feed existing systems; human traders still make the final decisions, and a human-in-the-loop approach is expected to remain.

Interpretation

C is correct. For model governance the slide matters: the LLM's text outputs need different validation (outcome analysis is harder) and different controls (hallucination checks) from the three predictive models.

Why the other options are wrong

A is incorrect. Sibling swap: the two groups are reversed. NLP, decision trees and clustering are the established AI/ML tools; LLMs and reinforcement learning are sophisticated AI.

B is incorrect. Role misassignment: the final decision still rests with human traders; sophisticated AI has been applied mainly to generate signals for existing systems, not as autonomous agents.

D is incorrect. Polarity flip: LLM outputs are typically unstructured text, which makes outcome analysis and error rates harder to determine, not easier.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. • CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: third-party dependencies and service provider concentration; market correlations; cyber risks; model risk, data, and governance; other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** IMF (October 2024), Advances in Artificial Intelligence; FSB (November 2024), The Financial Stability Implications of AI; SchweserNotes Book 5, Readings 100.1 and 101.1, 101.3

CI-1 • Advances in Artificial Intelligence: Implications for Capital Markets Activities (IMF, October 2024) • Vardeez: VERY IMPORTANT

CI-1 b

P2 CONCEPT

Explain the implications of further adoption of AI on market dynamics.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall (reading section; no shape on this LO)

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter. The A14 workbook section covers the reading, but none of its four reported shapes sits on this LO, so the questions follow the class notes and Schweser.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes: none on this LO (Vardeez focus only).

What the questions look like: no calculations. The examinable pattern is *direction under a condition*: almost every effect here is a benefit in normal conditions and the reverse under stress. Expect the ranking of US algorithmic trading shares, the "flighty liquidity" combination (cancellations down, hidden orders up), and the IMF's negative and positive scenarios.

Recap. The IMF outreach (27 firms), the two model groups and the four broad risk areas are set out in CI-1 a. This LO asks what further adoption does to the way markets behave.

1. Who gains: NBFIs, large firms and algorithmic traders

AI is expected to influence both market dynamics and market structure, and it may help **nonbank financial institutions (NBFIs)** grow. NBFIs hold *more than half* of global financial market assets, and their agility and relatively less restrictive operating environments position them to benefit significantly from AI integration. Larger banks may face greater challenges because of legacy infrastructure and stricter regulatory constraints.

Algorithmic trading is a major area of growth in advanced economies. In the United States alone it accounts for about 70% of equities trading, over 50% of futures trading, more than 40% of options trading, and less than 10% of fixed-income trading (Figure 1). In the markets where it is most prevalent (the United States, Europe and Asia), activity tends to be concentrated among a relatively small number of dominant participants. The high fixed costs of developing and deploying AI mean that *larger* firms benefit most, while smaller firms often need to outsource their AI capabilities.

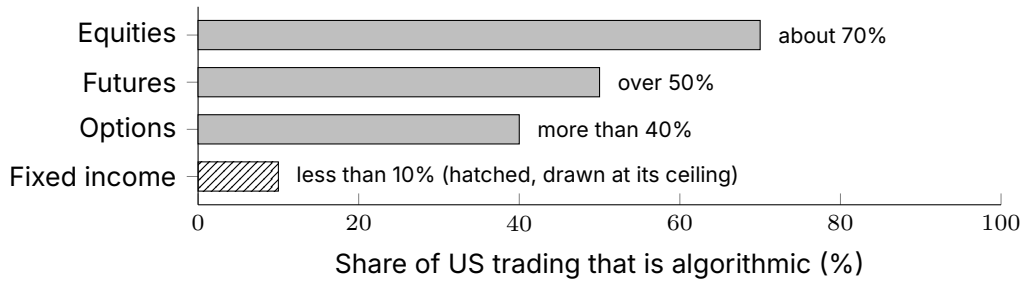


Figure 1. US algorithmic trading shares. Grey bars sit at the stated level or floor; the hatched fixed-income bar sits at its ceiling, since only "less than 10%" is given. The order is what gets examined: equities, futures, options, then a long drop.

2. Algorithmic trading: normal times against stress

Algorithmic trading delivers its benefits through AI's ability to extract value from, and rapidly process, massive quantities of unstructured and high-frequency data. The reading lists four outcomes, and the first three share one shape: a benefit in normal conditions, the reverse under stress (Figure 2).

1. **Market efficiency and liquidity.** AI can keep improving efficiency and liquidity, but it carries the risk of higher volatility under stress. Volatility can follow negative economic news, when informed traders withdraw and efficiency falls. Liquidity can also deteriorate if high-frequency traders step back, which has been observed in US Treasuries and other bond markets.
2. **Price behaviour.** Over the past two decades, algorithmic trading has helped reduce price swings *unrelated to new information* during normal liquidity conditions; it has also been associated with occasional large price swings in illiquid or stressed environments.
3. **Market stability and risk limits.** The risk controls that come with algorithmic trading can sometimes *reduce* stability under stress. Protective mechanisms such as risk reduction and automatic shutdowns in high-volatility periods may create feedback loops, trigger trading limits and drain liquidity. Research finds that in such periods high-frequency traders *reduce* their order cancellations as they step back from the market, while institutional investors *increase* their use of hidden orders (orders partially or fully concealed from the order book, used to limit price impact). The reading calls this pattern **"flighty liquidity under stress"**.
4. **Impact of GenAI.** GenAI in algorithmic trading could lower barriers to entry through improved testing, coding and trade automation. It may also help overcome the difficulty of automating trading across diverse asset classes, by processing complex text-based data and enabling more standardised risk analyses.

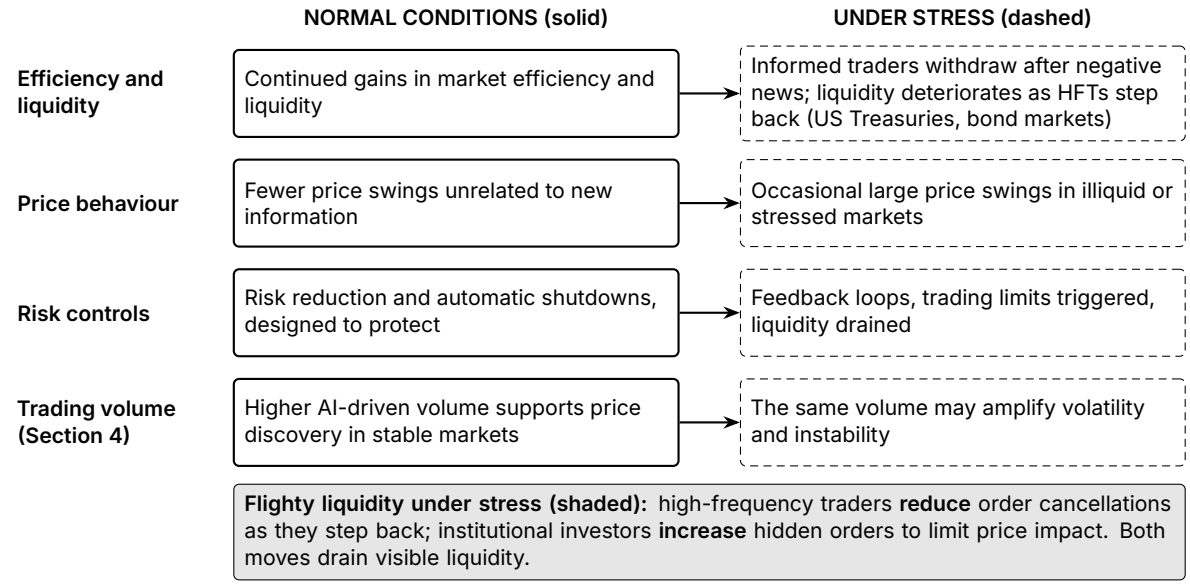


Figure 2. Each row reads left to right: the effect in normal conditions (solid box) turns into the stress effect (dashed box). The condition, not the topic, is the discriminating word in most questions. The shaded panel is the two-actor combination the reading names.

3. The IMF's two scenarios

The IMF staff assessment sets out a negative and a positive scenario for AI adoption in algorithmic trading.

	Negative scenario	Positive scenario
Market liquidity	AI magnifies existing algorithmic trading risks by facilitating its growth: it could "democratise" algorithmic trading and expand it to a broader set of assets and geographic areas, exacerbating sudden liquidity withdrawal under stress	AI makes algorithmic trading more stable under stress: AI-driven algorithms could operate in a wider set of market conditions than traditional algorithms, with lower flash-crash risk and less liquidity withdrawal under stress
Leverage	AI-driven strategies boost short-term leverage: as more advanced algorithms exploit arbitrage opportunities more efficiently, the remaining opportunities might require higher leverage to deliver similar returns	AI improves the management of leverage and related risks: more frequent and automated management of leveraged positions, based on more inputs, mitigating operational lags
Interconnectedness	*In both scenarios* AI increases interconnectedness: it could spread algorithmic trading to other asset classes, regions and trading venues, and operate between market segments (a multi-asset, multi-venue approach). Here: higher correlations between capital market segments, spillovers and transmission of stress	*The same rise in interconnectedness.* Here: market access, efficiency and liquidity improve for some market segments, including emerging markets

The third row is the one to remember: interconnectedness rises in *both* scenarios, and only its consequence differs. An answer that turns on whether interconnectedness rises cannot be settled by the scenario alone.

4. New market dynamics

(a) **Faster market reactions to news.** Reviews of market responses to the release of FOMC minutes find that initial reactions now reflect the eventual market impact more accurately than before LLMs were introduced.

(b) **Higher and procyclical trading volumes.** AI's ability to process and act on new information quickly can lead to more frequent and larger portfolio adjustments, and so to higher volumes. ETFs that use AI-driven strategies show significantly higher turnover than ETFs following other active or passive strategies. Higher volume can support price discovery in stable markets, but may amplify volatility and instability in stressed conditions.

(c) **Potential for AI-driven manipulation or collusion.** Theoretical models of interacting AI trading algorithms suggest possible inefficiencies, such as persistent gaps between fundamental values and market prices, particularly when one algorithm holds a strategic advantage over the others. That could create opportunities for manipulation, or even *unintended* collusion among AI systems.

Results card: CI-1 b

Structure	NBFIs (more than half of global financial market assets) gain; banks held back by legacy systems and stricter regulation; large firms benefit most, small firms outsource
US algorithmic shares	Equities about 70% > futures over 50% > options more than 40% > fixed income less than 10%; activity concentrated among few participants
Normal against stress	Efficiency and liquidity up, then withdrawal; fewer non-news price swings, then occasional large swings; risk controls protect, then feedback loops
Flighty liquidity	HFT order cancellations *down*; institutional hidden orders *up*
GenAI	Lower entry barriers (testing, coding, automation); automation across asset classes; more standardised risk analyses
Scenarios	Liquidity: wider algo trading and sudden withdrawal, or more stable algorithms. Leverage: more needed for similar returns, or better managed. Interconnectedness: up in both
New dynamics	FOMC reactions closer to final impact; AI ETFs higher turnover (procyclical); persistent price gaps, manipulation or unintended collusion

Exam traps

- **Role misassignment on who gains.** NBFIs and large firms gain; banks face legacy and regulatory constraints; smaller firms outsource.
- **Sibling swap in the ranking.** Futures (over 50%) sit above options (more than 40%); equities lead at about 70%; fixed income trails at under 10%.
- **Polarity flip on flighty liquidity.** "Panic means more cancellations" is the intuitive answer and the wrong one: HFT cancellations fall because the HFTs have stepped back.
- **Dropped condition.** Every benefit is stated for normal conditions and every harm for stress. An option that states either for "all conditions" is wrong.
- **Safety mechanism read as always safe.** Automatic shutdowns can destabilise under stress through feedback loops.
- **Scenario row swap.** Wider algorithmic trading and sudden liquidity withdrawal is the *liquidity* row; needing more leverage for similar returns is the *leverage* row.
- **Interconnectedness falls in the positive scenario.** It does not: it rises in both.
- **Collusion needs intent.** The reading's word is *unintended*.

Practice questions: CI-1 b

Question CI-1 b.1 Adapted from CI QB VI Q11 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

A supervisor at a European securities regulator is assessing which institutions are best placed to gain from further AI adoption in capital markets. Which assessment is most consistent with the IMF reading?

A. Large banks, because stricter regulation and established infrastructure have already given them the data systems that AI requires.

B. NBFIs, which hold more than half of global financial market assets and gain from their agility and less restrictive operating environments.

C. Smaller firms, because the high fixed costs of AI fall mainly on large firms, which are therefore pushed to outsource their AI capabilities.

D. A wide field of participants, because in the US, Europe and Asia algorithmic trading activity is spread across many firms rather than concentrated.

Question CI-1 b.2 New • built from SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

A market structure consultant advising an Asian exchange group cites the IMF's figures on algorithmic trading in the United States to show where automated trading is most and least established. Which statement is correct?

A. Futures have the highest algorithmic share, at about 70%, ahead of equities at more than 50%.

B. Options have a higher algorithmic share than futures, at more than 50% against more than 40%.

C. Equities have the highest algorithmic share, at about 70%, and fixed income the lowest, at less than 10%.

D. Fixed income has an algorithmic share of more than 40%, close to the shares reported for options and futures.

Question CI-1 b.3 Adapted from CI QB VI Q16 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

An academic adviser to a securities regulator summarises two decades of evidence on algorithmic trading and price behaviour, as described in the IMF reading. Which summary is most accurate?

A. It has cut price swings unrelated to new information in normal liquidity, but is linked to occasional large swings in illiquid or stressed markets.

B. It has amplified price swings unrelated to new information in normal liquidity conditions, but has damped price swings in illiquid or stressed markets.

C. It has reduced price swings in all market conditions, because embedded risk controls such as automatic shutdowns stabilise markets under stress.

D. It has slowed the pace at which prices reflect new information, and in doing so has damped the volatility that follows negative economic news.

Question CI-1 b.4 Adapted from CI QB VI Q13 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

During a stress episode, a market surveillance officer at an Australian exchange reviews order-book data for signs of the "flighty liquidity under stress" pattern described in the IMF reading. Which combination is she most likely to find?

A. High-frequency traders cancel more orders as they reprice risk, and institutional investors post more visible orders to attract liquidity.

B. High-frequency traders cancel fewer orders as they step back, and institutional investors cut their use of hidden orders to signal demand.

C. High-frequency traders cancel more orders as they reprice risk, and institutional investors make greater use of hidden orders to limit price impact.

D. High-frequency traders cancel fewer orders as they step back, and institutional investors make greater use of hidden orders to limit price impact.

Question CI-1 b.5 Adapted from CI QB VI Q20 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

The head of electronic trading at a Tokyo dealer is assessing what GenAI could add to the firm's algorithmic trading business. Which statement best reflects the IMF reading?

A. GenAI raises barriers to entry, since only firms with proprietary exchange data can train it, which further entrenches existing algorithmic traders.

B. GenAI adds little under stress, because it cannot process the complex text-based data that moves prices when markets are volatile.

C. GenAI could lower entry barriers through better testing, coding and trade automation, and help automate trading across diverse asset classes.

D. GenAI could make risk analysis less standardised across asset classes, because each model reads and weights text-based data in its own way.

Question CI-1 b.6 New • CI notes Table CI-1.3 (the notes' Table 3.1); SchweserNotes Book 5, LO 100.b

The CRO of a London hedge fund is reviewing the IMF's scenario analysis of AI adoption in algorithmic trading. Which statement correctly gives the IMF's negative and positive scenarios for leverage?

A. Negative: AI-driven strategies cut short-term leverage as arbitrage gaps close. Positive: lower flash-crash risk and less liquidity withdrawal under stress.

B. Negative: efficient arbitrage means similar returns may need more leverage. Positive: AI manages leveraged positions more often, cutting operational lags.

C. Negative: efficient arbitrage means similar returns may need more leverage. Positive: better market access and liquidity, including in emerging markets.

D. Negative: AI manages leveraged positions more often, which speeds deleveraging. Positive: efficient arbitrage means less leverage is needed for the same return.

Question CI-1 b.7 New • CI notes Table CI-1.3 (the notes' Table 3.1); SchweserNotes Book 5, LO 100.b

A financial stability analyst at the Bank of Thailand asks how wider AI adoption in algorithmic trading would affect interconnectedness across capital markets, and whether the effect would help or harm the country's markets. Based on the IMF's scenario analysis, which response is correct?

A. It rises only in the negative scenario; in the positive scenario AI spreads trading across more segments and so lowers the correlations between them.

B. It rises in both scenarios, but the gains in the positive scenario go to advanced economies, while emerging markets are left to bear the stress spillovers.

C. It falls in both scenarios, because AI keeps algorithmic trading within the asset classes, regions and venues where it is already well established.

D. It rises in both scenarios: in one it raises correlations and spreads stress; in the other it improves access and liquidity, emerging markets included.

Question CI-1 b.8 Adapted from CI QB VI Q17 and Q18 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

A Federal Reserve economist is writing a note on the new market dynamics that the IMF links to wider AI adoption. Which pair of observations is consistent with the reading?

A. Initial reactions to FOMC minutes now track the eventual impact more closely than before LLMs, and AI-driven ETFs show significantly higher turnover than other ETFs.

B. Initial reactions to FOMC minutes now diverge further from the eventual impact than before LLMs, and AI-driven ETFs show significantly higher turnover than other ETFs.

C. Initial reactions to FOMC minutes now track the eventual impact more closely than before LLMs, and AI-driven ETFs show lower turnover because models minimise trading costs.

D. Initial reactions to FOMC minutes have not changed since LLMs arrived, and AI-driven ETFs show higher turnover that impairs price discovery even in stable markets.

Question CI-1 b.9 Adapted from CI QB VI Q19 • SchweserNotes Book 5, LO 100.b; CI notes CI-1 b

A market integrity team at a Canadian securities regulator reviews theoretical models that simulate interactions among AI trading algorithms. Which conclusion do these models support, according to the IMF reading?

A. Collusion requires explicit coordination by the firms that deploy the algorithms, so the models find manipulation risk only where their staff coordinate.

B. Persistent gaps between fundamentals and prices can arise, notably when one algorithm has a strategic edge, inviting manipulation or unintended collusion.

C. Price gaps between fundamental values and market prices close faster when one algorithm holds a strategic advantage, because it then arbitrages the others.

D. Competition among AI algorithms removes persistent mispricing in all market conditions, so their interactions can only raise overall market efficiency.

Solutions • CI-1 b

Solution CI-1 b.1 — Answer: B

Adapted from CI QB VI Q11 • SchweserNotes Book 5, LO 100.b

Step 1 – Who the reading says gains

AI adoption may facilitate the growth of NBFIs. They hold more than half of global financial market assets, and their agility and relatively less restrictive operating environments position them to benefit significantly.

Step 2 – Who is held back

Larger banks may face greater challenges because of legacy infrastructure and stricter regulatory constraints. Among firms generally, the high fixed costs of AI mean that larger firms benefit most while smaller firms often outsource.

$\Rightarrow$ The structural winners are NBFIs and, within any sector, large firms.

Interpretation

The size of the NBFI sector is what makes this a supervisory question: activity migrating to lightly overseen NBFIs becomes the opacity challenge of CI-1 c.

Why the other options are wrong

A is incorrect. Polarity flip: stricter regulation and legacy infrastructure are the reasons banks may face *greater* challenges.

C is incorrect. Role misassignment: high fixed costs favour larger firms; it is the smaller firms that often need to outsource.

D is incorrect. Polarity flip: in the US, Europe and Asia, algorithmic trading activity tends to be concentrated among a relatively small number of dominant participants.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024), Advances in Artificial Intelligence: Implications for Capital Markets Activities; SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.2 — Answer: C

New • built from SchweserNotes Book 5, LO 100.b

Step 1 – Anchor the four figures

US algorithmic trading accounts for about 70% of equities trading, over 50% of futures, more than 40% of options and less than 10% of fixed income.

Step 2 – Read off the ends of the ranking

Equities lead and fixed income trails, with a long drop after options.

$\Rightarrow$ Only C states both ends correctly.

Interpretation

Equities lead and fixed income trails by a long way: the class notes call this uneven adoption across markets. The middle pair (futures above options) is where candidates slip.

Why the other options are wrong

A is incorrect. Sibling swap: it exchanges the equities and futures figures; equities are about 70%, futures over 50%.

B is incorrect. Sibling swap on the middle pair: futures (over 50%) are above options (more than 40%).

D is incorrect. Wrong-input twin: more than 40% is the options figure; fixed income is under 10%.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.3 — Answer: A

Adapted from CI QB VI Q16 • SchweserNotes Book 5, LO 100.b

Step 1 – Normal conditions

Over the past two decades, algorithmic trading has helped reduce price swings unrelated to new information during normal liquidity conditions.

Step 2 – Stressed or illiquid conditions

It has also been associated with occasional large price swings in illiquid or stressed environments.
⇒ Damped noise in calm markets, occasional large swings in stressed ones.

Interpretation

The same two-regime shape runs through the whole LO. Risk controls follow it too: designed to protect, they can create feedback loops and drain liquidity under stress.

Why the other options are wrong

B is incorrect. Polarity flip: it swaps the two regimes' effects.
C is incorrect. Dropped condition and a polarity flip: the reduction in price swings is a normal-conditions finding, and automatic shutdowns can *reduce* stability under stress through feedback loops.
D is incorrect. Polarity flip: AI speeds the processing of information (initial reactions to FOMC minutes now track the eventual impact more closely), and volatility after negative news arises when informed traders withdraw.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.4 — Answer: D

Adapted from CI QB VI Q13 • SchweserNotes Book 5, LO 100.b

Step 1 – High-frequency traders

In high-volatility periods, high-frequency traders *reduce* their order cancellations as they step back from the market: with less standing activity, there is less to cancel.

Step 2 – Institutional investors

Institutional investors *increase* their use of hidden orders, which are partially or fully concealed from the order book, to limit price impact.
⇒ Cancellations down, hidden orders up: visible liquidity drains on both sides.

Interpretation

The reading calls the combination "flighty liquidity under stress". It sits under the risk-limits outcome: protective mechanisms such as automatic shutdowns can create feedback loops, trigger trading limits and drain liquidity.

Why the other options are wrong

A is incorrect. Polarity flip on both actors: cancellations fall, and institutions conceal rather than display orders.
B is incorrect. Polarity flip on the institutional half: hidden order use rises, to limit price impact.
C is incorrect. Polarity flip on the HFT half: "panic means more cancellations" is intuitive but wrong; cancellations fall as HFTs withdraw.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024);

SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.5 — Answer: C

Adapted from CI QB VI Q20 • SchweserNotes Book 5, LO 100.b

Step 1 – Barriers to entry

Integrating GenAI into algorithmic trading could bring substantial benefits by lowering barriers to entry through improved testing, coding and trade automation.

Step 2 – Reach across asset classes

GenAI may help overcome the challenges of automating trading across diverse asset classes, by processing complex text-based data and enabling more standardised risk analyses.
$\Rightarrow$ GenAI widens who can trade algorithmically and what can be traded that way.

Interpretation

Lower barriers cut both ways. The negative scenario in Section 3 is the same widening seen from the risk side: AI could "democratise" algorithmic trading across more assets and regions and exacerbate sudden liquidity withdrawal under stress.

Why the other options are wrong

A is incorrect. Polarity flip: GenAI is expected to *lower* barriers to entry.
B is incorrect. Polarity flip: processing complex text-based data is one of GenAI's stated contributions.
D is incorrect. Polarity flip: GenAI enables *more* standardised risk analyses.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.6 — Answer: B

New • built from CI notes CI-1 b, Table CI-1.3 (the notes' Table 3.1); SchweserNotes Book 5, LO 100.b

Step 1 – Negative scenario for leverage

AI-driven strategies boost short-term leverage. As more advanced algorithms exploit arbitrage opportunities more efficiently, the remaining opportunities might require higher leverage to deliver similar returns.

Step 2 – Positive scenario for leverage

AI improves the management of leverage and related risks: it could support more frequent and automated management of leveraged positions, based on more inputs, and mitigate operational lags.
$\Rightarrow$ More leverage for the same return, or better-managed leverage.

Interpretation

Both scenarios accept that AI changes leverage; they differ on whether it is a source of fragility or a tool for control. The CI-1 c stability challenges take the negative side: higher leverage could amplify volatility and deleveraging under stress.

Why the other options are wrong

A is incorrect. Polarity flip and row swap: the negative scenario has short-term leverage *rising*, and lower flash-crash risk with less liquidity withdrawal is the positive scenario for *market liquidity*.
C is incorrect. Row swap on the positive half: better market access, efficiency and liquidity, emerging markets included, is the positive consequence of rising *interconnectedness*, not the leverage scenario.

D is incorrect. Scenario swap: frequent automated management of leveraged positions is the positive scenario, and efficient arbitrage is what pushes leverage up in the negative one.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024), staff assessment (CI notes Table CI-1.3, the notes' Table 3.1); SchweserNotes 2026 Book 5, Reading 100, LO 100.b, Figure 100.1 (not reproduced in text)

Solution CI-1 b.7 — Answer: D

New • built from CI notes CI-1 b, Table CI-1.3 (the notes' Table 3.1); SchweserNotes Book 5, LO 100.b

Step 1 – What happens to interconnectedness

In *both* scenarios AI increases interconnectedness: it could spread algorithmic trading to other asset classes, geographic regions and trading venues, and operate between market segments in a multi-asset, multi-venue approach.

Step 2 – What differs

Negative: higher correlations between capital market segments, facilitating spillovers and the transmission of stress. Positive: market access, efficiency and liquidity improve for some segments, including emerging markets.

$\Rightarrow$ The analyst's answer is "both", depending on which consequence dominates.

Interpretation

This row is a structural oddity of the source table, kept on purpose: the driver is the same in both columns and only the consequence differs. Emerging markets appear on the *positive* side.

Why the other options are wrong

A is incorrect. Polarity flip: interconnectedness does not fall in the positive scenario; its benefit there is access, efficiency and liquidity.

B is incorrect. Role misassignment: the positive scenario names emerging markets among the segments whose access, efficiency and liquidity improve.

C is incorrect. Polarity flip: AI is expected to spread algorithmic trading to new asset classes, regions and venues.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024), staff assessment (CI notes Table CI-1.3, the notes' Table 3.1); SchweserNotes 2026 Book 5, Reading 100, LO 100.b, Figure 100.1 (not reproduced in text)

Solution CI-1 b.8 — Answer: A

Adapted from CI QB VI Q17 and Q18 • SchweserNotes Book 5, LO 100.b

Step 1 – Faster reactions to news

Reviews of market responses to FOMC minutes show that initial reactions now reflect the eventual market impact more accurately than before LLMs were introduced.

Step 2 – Higher and procyclical volumes

AI's ability to process and act on new information quickly leads to more frequent and larger portfolio adjustments. ETFs using AI-driven strategies show significantly higher turnover than those following other active or passive strategies.

$\Rightarrow$ Both halves of A are stated in the reading.

Interpretation

The volume finding is procyclical: elevated trading volume can support price discovery in stable markets but may amplify volatility and instability in stressed conditions.

Why the other options are wrong

B is incorrect. Polarity flip on the first half: initial reactions have moved *closer* to the eventual impact.
C is incorrect. Polarity flip on the second half: AI-driven ETFs show significantly *higher* turnover.
D is incorrect. Polarity flip on both halves: reactions have changed, and in stable markets higher volume supports price discovery rather than impairing it.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b

Solution CI-1 b.9 — Answer: B

Adapted from CI QB VI Q19 • SchweserNotes Book 5, LO 100.b

Step 1 – What the models show

Theoretical models of interacting AI trading algorithms suggest possible market inefficiencies, such as persistent gaps between fundamental values and market prices.

Step 2 – The condition and the consequence

The gaps are most likely when one algorithm holds a strategic advantage over the others, and they could create opportunities for manipulation or even unintended collusion among AI systems.
$\Rightarrow$ Collusion can emerge from the algorithms' interaction without anyone agreeing to it.

Interpretation

For a market integrity team the point is that surveillance built to detect human coordination may miss collusive outcomes that no person intended.

Why the other options are wrong

A is incorrect. Absolute claim: the reading's concern is collusion that is *unintended*, arising among AI systems themselves.
C is incorrect. Polarity flip: a strategic advantage is the condition under which gaps *persist*.
D is incorrect. Absolute claim and polarity flip: the models suggest persistent mispricing, not its removal in all conditions.

LO: CI-1 b: Explain the implications of further adoption of AI on market dynamics. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b

CI-1 • Advances in Artificial Intelligence: Implications for Capital Markets Activities (IMF, October 2024) • Vardeez: VERY IMPORTANT

CI-1 c

P1 CONCEPT

Explain the implications of further adoption of AI on financial stability.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A14 Artificial Intelligence and GenAI (August 2026 recall, Map #20, 6 of 16 sources). Map #20 names the financial-stability implications (herding, model monoculture, third-party concentration, explainability) as one of the two examinable axes, and shapes S2 and S4 sit on this LO.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A14·S2** Disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors (Questions CI-1 c.4 and c.5, the IMF's side). The FSB's side, including the short-term edge of bad actors and prompt injection against poisoning, is workbook A14.2, worked under CI-2 c.
- **A14·S4** Financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk (Questions CI-1 c.3, c.7, c.8 and c.9; c.9 is workbook A14.3, tagged CI-2 b, c and CI-1 a, c, and placed here).

What the questions look like: no calculations. Tier questions (main against lesser concerns), label matching (which concern or challenge a scenario illustrates), and transmission channels: from a deepfake to a deposit run, from similar models to a flight to safety.

Recap. The four broad risk areas of CI-1 a (operational; volatility and speed under stress; cyber and manipulation; monitoring and opacity) are the skeleton of this LO: the survey's concerns are the firm-level view, and the four financial stability challenges are the same themes at system level.

1. What the survey respondents worried about

Participants in the IMF outreach highlighted five areas of concern about AI's effect on financial stability:

(a) **Concentration risk**, identified as a *principal* concern, particularly if investment and trading strategies are built mainly on open-source AI models trained on data from a limited group of vendors. Vendor concentration is a vulnerability in itself: a small number of dominant providers could create systemic risk if any of them failed. (The class notes label this concern "herding", because similar models trained on similar data tend to act alike.)

(b) **Market fragility**: excess volatility, flash crashes, illiquidity, cyberattacks and large-scale data poisoning, arising from ineffective guardrails and rapid, automated decision-making. Respondents also worried about the availability of professionals with the expertise to operate in AI-driven environments.

(c) **Model hallucinations and lack of explainability**, which could undermine market confidence in AI-generated outputs and make it harder for participants to rely on automated systems.

(d) **Market manipulation**: misuse of AI for misinformation campaigns or deepfakes could disrupt

market integrity and investor trust.

(e) **High fixed costs**: the infrastructure and resources AI requires may give large firms a competitive advantage, further entrenching concentration and market imbalances.

Cyberattacks in particular. Within market fragility, cyberattacks were identified as a significant vulnerability, with their frequency and sophistication rising in recent years. The threats include audio and video impersonations, system disruptions, deepfakes, and misinformation spread through social media and other channels. They could erode confidence in financial institutions and trigger market selloffs and/or deposit runs. Because advanced economies have been the earliest and heaviest adopters of AI, most reported cyber incidents have occurred there.

Lesser concerns. Respondents ranked three items lower: data access and unauthorised use; the potential for customer fraud; and the transmission of economic and market shocks *from* advanced economies *to* emerging and developing markets, which could heighten capital flow volatility.

> **Correction to the source notes.** (1) The class notes fold the last three items into "additional concerns include data misuse, fraud, and the transmission of shocks across economies". SchweserNotes Book 5, LO 100.c, ranks them as *lesser* concerns, below the five main areas of concern (concentration risk is the one Schweser calls principal), and gives the direction: shocks travel from advanced economies to emerging and developing markets and heighten capital flow volatility. Question CI-1 c.1 turns on the tier. (2) The notes' Figure CI-1.3 caption says that concentration and high fixed costs each "show up as volatility and as third-party operational risk". That holds for concentration, but the figure itself draws high fixed costs to the opacity and third-party channels, and neither the notes' text nor Schweser links fixed costs to volatility or to opacity. Figure 1 below draws a solid arrow only where the IMF text names the same mechanism on both sides, keeps the notes' other pairings dashed, drops the notes' fixed-costs-to-opacity arrow, and adds the fragility-to-cyber link (cyberattacks and data poisoning sit inside market fragility).

2. What respondents want from regulators

Survey participants expect regulators to provide greater guidance and clarity on AI-related risk management, including requirements for *transparent disclosures* and *extreme-scenario stress testing*. They stressed accountability frameworks, guidance on AI in customer-facing applications and on regulatory structures across industries, and the need to balance innovation, customer protection and responsible AI governance. Overall they saw **best practices and flexible guidelines** as more effective for an evolving technology than strict, rules-based regulation. What regulators actually recommend is CI-1 d.

3. The four financial stability challenges

1. **Higher volatility and increased market speed under stress.** Algorithmic trading can bring efficiency gains, faster price adjustments, narrower margins and greater liquidity, but the potential for higher leverage could amplify volatility and deleveraging under stress. Herding and similar outputs across AI models in stressed conditions may together trigger a flight to safe assets and large-scale selloffs in other asset classes. AI models may also shut down, or generate complex and inconsistent outputs, during significant adverse events such as the COVID-19 pandemic.
2. **Greater monitoring and opacity challenges.** As AI adoption expands, capital market, trading and investment activity shifts from the banking sector to NBFIs. Banks face stringent transparency and model validation requirements; NBFIs operate under relatively lighter oversight, so transparency may fall. Correlations and interconnectedness may also rise as AI-generated portfolios link regions, trading venues and asset classes in ways not seen before.
3. **Elevated cyber risks and potential market manipulation.** Markets and asset prices could become

targets for manipulation through disinformation, deepfakes or fraud. AI models themselves may be exposed to compromised data integrity or confidentiality, leading to flawed investment and trading decisions.

4. **Operational risks from dependence on third-party AI service providers.** Overreliance on a small number of AI model and IT service providers could heighten systemic vulnerability; the failure or disruption of any major provider could have severe and widespread consequences for capital markets.

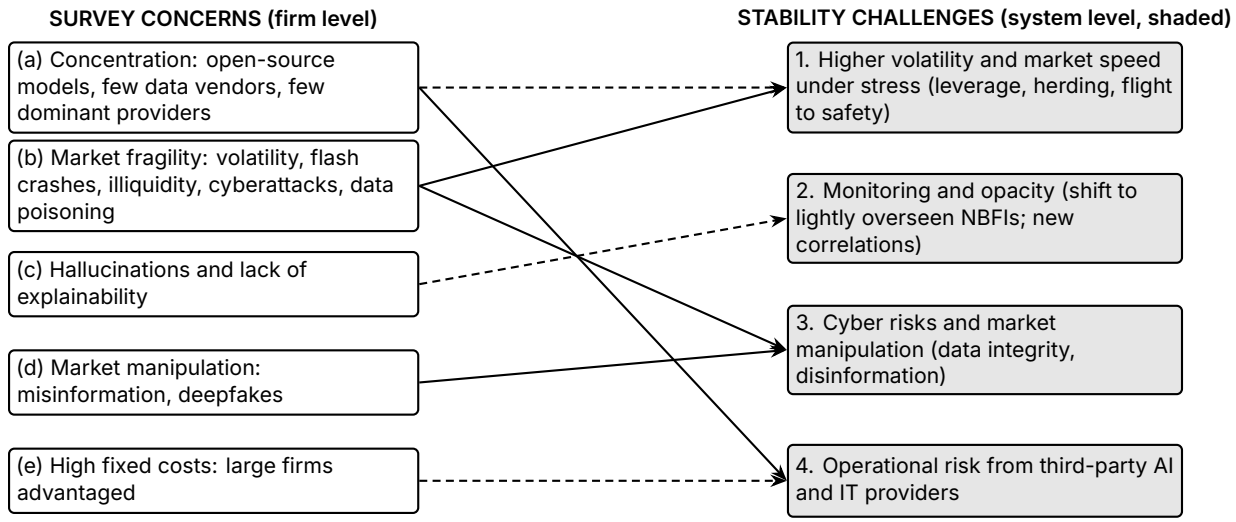


Figure 1. The five survey concerns (left, unshaded) and the four stability challenges (right, shaded) are the same material at firm and at system level. Solid arrows join items whose texts name the same mechanism; dashed arrows are pairings found only in the class notes' figure (concentration with herding, explainability with opacity, fixed costs with reliance on a few providers, since smaller firms outsource, CI-1 b). The notes' arrow from high fixed costs to opacity is dropped. Concentration and market fragility each feed two challenges.

4. Shapes S2 and S4: the IMF's version and the FSB's

Both readings cover AI and financial stability, and the workbook shapes draw on both. Keep the sources apart: a question set on the IMF paper is answered from the IMF paper.

Shape	IMF (CI-1 c, this LO)	FSB (CI-2 c; workbook A14.2, A14.3)
S2: GenAI's disadvantages	Manipulation through misinformation and deepfakes (main concern); cyberattacks by audio and video impersonation, system disruption, deepfakes and social-media misinformation, eroding confidence into selloffs or deposit runs; data poisoning (fragility); compromised data integrity or confidentiality; customer fraud a *lesser* concern	Video and voice generation enable deepfakes, fake profiles and IDs and bypassed security checks; generating fraudulent content may currently be easier than detecting it; bad actors may benefit more in the short term; prompt injection extracts information, poisoning manipulates weights or training data
S4: stability axis	Concentration on open-source models and few vendors; herding and similar outputs under stress; models that shut down or give inconsistent outputs (COVID-19); hallucinations and lack of explainability; reliance on a few AI and IT providers	Pre-trained core models share architectures and data, so outputs correlate; correlation risk higher when LLMs inform risk management or investment strategy, far lower for back-office uses; model risk handled in the institution's existing framework

Results card: CI-1 c

Main concerns (5)	Concentration (the principal one); market fragility (incl. cyberattacks, data poisoning, skills shortage); hallucinations and lack of explainability; manipulation; high fixed costs
Lesser concerns (3)	Data access and unauthorised use; customer fraud; shock transmission from advanced to emerging and developing economies (capital flow volatility)
Cyber channel	Impersonation, deepfakes, misinformation → loss of confidence → selloffs and/or deposit runs; most incidents in advanced economies (earliest, heaviest adopters)
What respondents want	More guidance: transparent disclosures, extreme-scenario stress tests, accountability; balance innovation and customer protection; flexible guidelines over strict rules
Four challenges	(1) Volatility and speed under stress; (2) monitoring and opacity (NBFIs); (3) cyber and manipulation; (4) third-party operational risk
One worry, two labels	Few providers = *concentration risk* on the survey list = *operational risk from third-party dependence* on the challenge list

Exam traps

- **Tier swap.** Customer fraud, data access and shock transmission are *lesser* concerns; high fixed costs and market fragility are among the five main concerns.
- **Label swap.** Excess volatility, flash crashes, illiquidity, cyberattacks and data poisoning from weak guardrails is *market fragility*, not concentration or manipulation.

- **Which list?** A provider failure is concentration risk in the survey and operational risk in the challenges. Read which list the stem invokes.
- **Geography flip.** Most reported cyber incidents are in *advanced* economies, because they adopted AI earliest and most heavily.
- **Channel swap.** Deepfakes work through confidence (selloffs, deposit runs), not through a direct capital charge.
- **Role flip on oversight.** Banks carry the stringent transparency and validation rules; NBFIs are lightly overseen.
- **Polarity on stress behaviour.** Under stress, AI models may shut down or give inconsistent outputs; similar outputs mean herding, not offsetting positions.
- **Rules for guidelines.** Respondents prefer best practices and flexible guidelines to strict rules, and want innovation *and* customer protection.
- **Paper swap.** "Bad actors gain more in the short term" and prompt injection are FSB material (CI-2 c), not IMF findings.

Practice questions: CI-1 c

Question CI-1 c.1 New • built from SchweserNotes Book 5, LO 100.c; CI notes CI-1 c (correction box)

The CRO of a Toronto asset manager is building a risk register from the IMF survey, tagging each concern as one of the five main concerns or as one of the lesser concerns. Which tagging is correct?

A. The potential for customer fraud, alongside data access and the unauthorised use of data: a main concern.

B. Transmission of shocks from advanced economies to emerging and developing markets: a main concern.

C. High fixed costs that give large firms an advantage and entrench concentration: a lesser concern.

D. Market fragility, including a shortage of experts able to work in AI-driven markets: a main concern.

Question CI-1 c.2 Adapted from CI QB VI Q26 • SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

The head of regulatory affairs at a Singapore exchange is drafting a consultation response that reflects what the IMF survey respondents expect from regulators on AI. Which position is most consistent with those expectations?

A. Authorities should replace guidance with strict, rules-based regulation, since flexible guidelines cannot keep pace with the speed of AI development in markets.

B. Authorities should give priority to innovation over customer protection until the technology has matured and its risks are better understood.

C. Authorities should give more guidance on AI risk management, including transparent disclosures and extreme-scenario stress tests, via flexible guidelines.

D. Authorities should leave AI risk management to each firm's discretion, with accountability frameworks designed by firms rather than set out in guidance.

Question CI-1 c.3 New • built from SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

The CRO of a Dublin fund manager reports that portfolio managers have stopped acting on an AI system's trade recommendations after it produced outputs that proved wrong and that neither the vendor nor the fund's own validation team could explain. Which concern from the IMF survey does this episode best illustrate?

A. Market fragility, because rapid automated decisions with ineffective guardrails produce excess volatility and flash crashes.

B. Model hallucinations and lack of explainability, which undermine confidence in AI outputs and make automated systems harder to rely on.

C. Concentration risk, because the funds' strategies are built on open-source models trained on data from a limited group of vendors.

D. High fixed costs, because smaller funds cannot afford the specialised talent that is needed to validate and monitor an AI system's outputs.

Question CI-1 c.4 Adapted from CI QB VI Q24 and Q25 • SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

A deepfake video showing the chief executive of a large Hong Kong bank announcing heavy trading losses spreads on social media before markets open. A financial stability analyst at the central bank is asked how the IMF reading would assess such an event. Which assessment is most consistent with the reading?

A. It could erode confidence and trigger market selloffs or deposit runs; most reported incidents are in advanced economies, the earliest and heaviest AI adopters.

B. Its losses stay with the targeted bank, because impersonation attacks do not spread to other institutions or move market prices beyond the bank's own share price.

C. It could hit the bank's regulatory capital directly through operational losses, which the IMF treats as the main link from cyber events to stability.

D. It is mainly an emerging-market problem, since most reported AI-related cyber incidents have occurred in the economies where cyber defences are weakest.

Question CI-1 c.5 New • built from SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

An internal auditor at a Paris asset manager is reviewing controls over the data that feed the firm's AI trading models. Which statement about the IMF's cyber and market manipulation challenge should the audit report include?

A. AI models can be exposed to compromised data integrity or confidentiality, which can lead to flawed investment and trading decisions.

B. The threat to AI models is limited to breaches of confidentiality, since the integrity of their training data cannot be tampered with.

C. Disinformation and deepfakes can threaten firms' reputations, but in deep markets asset prices themselves are not a realistic target.

D. Cyber risk sits outside the four stability challenges, because the survey respondents ranked it among their lesser concerns.

Question CI-1 c.6 Adapted from CI QB VI Q28 • SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

A macroprudential analyst at the Bank of England notes that a growing share of AI-driven trading and investment activity is moving from banks to hedge funds and other NBFIs. Which concern does the IMF reading attach to this shift?

A. Concentration risk in the survey's sense, because NBFIs build their strategies on open-source models trained on data from only a few vendors.

B. Lower system-wide correlation, because AI-generated NBFI portfolios spread their exposures across more regions, trading venues and asset classes.

C. Monitoring and opacity: NBFIs face lighter oversight than banks' transparency and validation rules, and AI portfolios link markets in new ways.

D. Little change in transparency, because NBFIs face the same transparency and model validation requirements that already apply to banks.

Question CI-1 c.7 Adapted from CI QB VI Q27 • SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

A stress-testing team at the Reserve Bank of Australia is designing a severe-shock scenario for markets in which AI is widely used. Which assumption is most consistent with the financial stability challenges described in the IMF reading?

A. AI models keep giving consistent outputs through the shock, so amplification comes mainly from human traders overriding the models' signals.

B. Similar outputs across AI models produce herding and a flight to safe assets, while some models shut down or give inconsistent outputs.

C. AI's faster price adjustments add liquidity under stress, so volatility rises mainly in normal conditions rather than during the shock.

D. The shift of trading from banks to NBFIs removes leverage from the system, so the deleveraging in the shock is smaller than before AI.

Question CI-1 c.8 New • built from SchweserNotes Book 5, LO 100.c; CI notes CI-1 c

At a risk committee of a Frankfurt bank, one member argues that the IMF treats dependence on a few AI service providers as a concentration problem; another argues that it treats it as an operational problem. Which response best reflects the IMF reading?

A. Only the first is right: the reading treats dependence on a few providers as concentration risk and never as a source of operational risk.

B. Only the second is right: the reading treats it as operational risk, and its concentration concern is limited to large banks' market share.

C. Neither is right: dependence on a few providers belongs to market fragility, alongside flash crashes, illiquidity and large-scale data poisoning.

D. Both are right: the survey lists vendor concentration under concentration risk, and the challenges frame it as third-party operational risk.

Question CI-1 c.9 Workbook A14.3 • S3, S4 • Map #20

The head of market risk at an Australian bank plans to use a pre-trained LLM, the same core model licensed by most of the bank's peers, in two ways: (1) to build volatility and liquidity risk models that predict how heightened volatility would affect the trading book, to generate advanced stress scenarios, and to suggest hedges and risk limits; (2) to draft internal policy documents and retrieve information for staff training. An internal auditor reviewing the plan must state the financial-stability implication. Which of the following assessments is correct?

A. Use (2) carries higher market correlation risk than use (1), because back-office drafting and retrieval are the most widespread uses of LLMs across the industry.

B. Circuit breakers and transparency requirements fully eliminate the correlation vulnerability, so the bank need not monitor how the LLM's risk outputs move with those of its peers.

C. Validation and outcome analysis of the LLM's risk models should be governed by the vendor's own model risk standards rather than by the bank's existing model risk framework.

D. Use (1) raises market correlation risk relative to use (2), because models sharing architectures and training data produce correlated outputs that can amplify volatility and liquidity crunches in stress.

Solutions • CI-1 c

Solution CI-1 c.1 — Answer: D

New • built from SchweserNotes Book 5, LO 100.c

Step 1 – The five main concerns

Concentration risk (the principal one), market fragility, model hallucinations and lack of explainability, market manipulation, and high fixed costs.

Step 2 – The three lesser concerns

Data access and unauthorised use, the potential for customer fraud, and the transmission of shocks from advanced economies to emerging and developing markets.

Step 3 – Test each tag

Market fragility is a main concern, and it explicitly includes respondents' worry about the availability of professionals with the expertise to operate in AI-driven environments.

$\Rightarrow$ Only D carries the right tier.

Interpretation

Every item in the options is a real concern from the survey; the question is won on the tier. The class notes blur it by calling the three lesser items "additional concerns" (see the correction box).

Why the other options are wrong

A is incorrect. Tier swap: customer fraud and data access are lesser concerns.

B is incorrect. Tier swap: shock transmission to emerging and developing markets is the third lesser concern.

C is incorrect. Tier swap: high fixed costs are a main concern, because they entrench concentration and market imbalances.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024), Advances in Artificial Intelligence: Implications for Capital Markets Activities; SchweserNotes 2026 Book 5, Reading 100, LO 100.c

Solution CI-1 c.2 — Answer: C

Adapted from CI QB VI Q26 • SchweserNotes Book 5, LO 100.c

Step 1 – What respondents asked for

Greater guidance and clarity from authorities on AI-related risk management, including transparent disclosures and extreme-scenario stress testing, plus accountability frameworks.

Step 2 – In what form

Best practices and flexible guidelines, which respondents saw as more effective for an evolving technology than strict, rules-based regulation, balancing innovation, customer protection and responsible AI governance.

$\Rightarrow$ More guidance, delivered flexibly.

Interpretation

This is the respondents' view, not the regulators' recommendations; those (circuit breakers, margin reviews, provider mapping, NBFI disclosure, OTC resilience) are CI-1 d.

Why the other options are wrong

A is incorrect. Polarity flip: respondents preferred best practices and flexible guidelines to strict, rules-based regulation.

B is incorrect. Polarity flip: respondents wanted innovation and customer protection balanced, not one ranked above the other.

D is incorrect. Polarity flip: respondents expected authorities to provide *more* guidance and clarity, including on accountability frameworks.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c

Solution CI-1 c.3 — Answer: B

New • built from SchweserNotes Book 5, LO 100.c

Step 1 – Read the episode

The outputs are wrong and nobody can explain them, and the consequence is that participants stop relying on the system.

Step 2 – Match the concern

Model hallucinations and lack of explainability could undermine market confidence in AI-generated outputs, making it harder for participants to rely on automated systems.
⇒ The channel is confidence: unreliable and opaque outputs stop being used.

Interpretation

This is the explainability strand of shape S4. The FSB adds the definition (a hallucination is a highly confident but incorrect response) and the model-risk consequence: unstructured text outputs make outcome analysis harder (CI-2 c, workbook A14.1).

Why the other options are wrong

A is incorrect. Label swap: market fragility is about excess volatility, flash crashes and illiquidity from weak guardrails and speed; the episode has no market move.
C is incorrect. Label swap: nothing in the stem says the fund's strategies rest on open-source models or a few data vendors; the defect is in the outputs.
D is incorrect. Label swap: high fixed costs are about large firms' advantage and concentration, not about unexplainable outputs.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c

Solution CI-1 c.4 — Answer: A

Adapted from CI QB VI Q24 and Q25 • SchweserNotes Book 5, LO 100.c

Step 1 – The threat

Within market fragility, cyberattacks were identified as a significant and growing vulnerability. The threats include audio and video impersonations, system disruptions, deepfakes, and misinformation spread through social media.

Step 2 – The channel

Such threats could erode confidence in financial institutions, potentially triggering market selloffs and/or deposit runs.

Step 3 – The geography

Advanced economies have been the earliest and heaviest adopters of AI, so most reported cyber incidents have occurred there.
⇒ A deepfake does not need to cause a loss; it needs to be believed.

Interpretation

This is the IMF's side of shape S2. The FSB adds that generating fraudulent AI content may currently be easier than detecting it, and that bad actors may benefit more than legitimate users in the short term (CI-2 c, workbook A14.2).

Why the other options are wrong

B is incorrect. Absolute claim: the reading's concern is precisely that such threats spread through confidence to selloffs and runs.

C is incorrect. Channel swap: the stated systemic channel runs through confidence and funding behaviour, not a direct capital charge.

D is incorrect. Geography flip: most reported incidents are in advanced economies, which adopted AI earliest and most heavily.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c

Solution CI-1 c.5 — Answer: A

New • built from SchweserNotes Book 5, LO 100.c

Step 1 – The challenge in full

Financial markets and asset prices could become targets for manipulation through disinformation, deepfakes or fraud. AI models themselves may be exposed to compromised data integrity or confidentiality, leading to flawed investment and trading decisions.

Step 2 – Apply it to the audit

Controls over model input data are controls over integrity and confidentiality, and the consequence the reading names is flawed investment and trading decisions.

$\Rightarrow$ A is the reading's own statement of the risk.

Interpretation

The survey already flagged large-scale data poisoning under market fragility, and the regulators' priorities in CI-1 d note that AI systems are particularly vulnerable to attacks on training data or on the models themselves, so safeguards are needed at every stage of the AI lifecycle.

Why the other options are wrong

B is incorrect. Scope error: the reading names integrity *and* confidentiality; dropping integrity halves the threat.

C is incorrect. Scope error: the reading says markets and asset prices could themselves become targets.

D is incorrect. Tier error: cyberattacks sit inside market fragility, a main concern, and cyber risk is one of the four challenges; the lesser concerns are data access, customer fraud and shock transmission.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c and 100.d

Solution CI-1 c.6 — Answer: C

Adapted from CI QB VI Q28 • SchweserNotes Book 5, LO 100.c

Step 1 – Name the challenge

As AI adoption expands, capital market, trading and investment activity shifts from banks to NBFIs. This is the monitoring and opacity challenge.

Step 2 – Why transparency falls

Banks face stringent transparency and model validation requirements; NBFIs operate under relatively lighter oversight.

Step 3 – The second strand

Correlations and interconnectedness may rise as AI-generated portfolios link regions, trading

venues and asset classes in ways not seen before.
⇒ Less visibility over a more connected system.

Interpretation

CI-1 b explains why the shift happens: NBFIs hold more than half of global financial market assets and their agility suits AI. CI-1 d gives the response: NBFIs should be required to identify themselves and disclose AI-related information.

Why the other options are wrong

A is incorrect. Sibling swap: concentration risk is the survey's concern about open-source models and few vendors; the reading attaches opacity, not concentration, to the move to NBFIs.
B is incorrect. Polarity flip: linking regions, venues and asset classes *raises* correlations and interconnectedness.
D is incorrect. Role misassignment: the stringent transparency and validation requirements are the banks'; NBFIs are lightly overseen.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.b and 100.c

Solution CI-1 c.7 — Answer: B

Adapted from CI QB VI Q27 • SchweserNotes Book 5, LO 100.c

Step 1 – Herding

Herding behaviour and similar outputs across AI models under stress may together trigger a flight to safe assets and large-scale selloffs in other asset classes.

Step 2 – The models themselves

AI models may shut down, or generate complex and inconsistent outputs, during significant adverse events such as the COVID-19 pandemic.
⇒ The scenario should assume correlated selling and unreliable models at the same moment.

Interpretation

Leverage is the amplifier in the same challenge: the potential for higher leverage from algorithmic strategies could amplify volatility and deleveraging under stress. Workbook A14.3 (Question CI-1 c.9) gives the FSB's version of the correlation channel.

Why the other options are wrong

A is incorrect. Polarity flip: the reading expects some models to shut down or give inconsistent outputs in severe events.
C is incorrect. Regime flip: efficiency, faster price adjustments and liquidity are the normal-time gains; the challenge is higher volatility and speed *under stress*.
D is incorrect. Sibling swap with a polarity flip: the move to NBFIs is the opacity challenge, and the reading warns that higher leverage could amplify deleveraging.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c

Solution CI-1 c.8 — Answer: D

New • built from SchweserNotes Book 5, LO 100.c

Step 1 – The survey list

Concentration risk was the principal survey concern, and vendor concentration is named within it: a small number of dominant providers could create systemic risk if any failed.

Step 2 – The challenge list

The fourth stability challenge is operational risk from dependence on third-party AI service providers: overreliance on a small number of AI model and IT providers, whose failure or disruption could have severe and widespread consequences.
⇒ Same worry, two labels, depending on which list the question uses.

Interpretation

Schweser's module quiz (Module Quiz 100.2, Question 4) answers "concentration risk" for a provider failure, from options that do not include third-party operational risk. When the options come from the challenge list, the label is operational risk. Read the options for the list.

Why the other options are wrong

A is incorrect. Absolute claim: the fourth challenge frames the same dependence as an operational risk.

B is incorrect. Scope error: the survey's concentration concern is about open-source models, data vendors and dominant providers, not banks' market share.

C is incorrect. Label swap: market fragility covers volatility, flash crashes, illiquidity, cyberattacks and data poisoning from weak guardrails, not provider dependence.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c, Module Quiz 100.2 Q4

Solution CI-1 c.9 — Answer: D

Workbook A14.3 • S3, S4 • Map #20

Step 1 – Classify the two uses

Use (1) is AI in market risk: the FSB lists volatility and liquidity risk management (models predicting how heightened volatility affects positions) among operations use cases, and the IMF lists advanced risk scenarios and dynamic risk sensing under risk management. Use (2) is back-office: information retrieval, employee training, document drafting.

Step 2 – Apply the correlation channel

Pre-trained core models share architectures and training data, so their outputs are naturally correlated. If LLMs inform risk management or investment strategy, correlation risk is higher; if they address internal back-office inefficiencies, it is far lower.
⇒ When most peers hedge and set limits from the same model, they react alike in stress: herding, flash crashes, amplified volatility and liquidity crunches.

Step 3 – Governance and mitigants

Model risk for AI is addressed through the institution's existing framework (data quality, validation, performance and outcome analysis, ongoing monitoring). Circuit breakers, transparency, monitoring and surveillance mitigate correlation; they do not remove it.

Interpretation

D is correct. The auditor's recommendation: treat use (1) as a model under full validation, monitor how its signals co-move with peers, and keep a human decision on hedges and limits; use (2) is low correlation risk.

Why the other options are wrong

A is incorrect. Polarity flip: correlation risk is higher when LLMs inform risk management and far lower for back-office uses; how widespread a use is does not reverse that.

B is incorrect. Absolute claim: circuit breakers and transparency are mitigants, complemented by consistent monitoring and surveillance; nothing in the reading says they eliminate the vulnerability.

C is incorrect. Role misassignment: AI-related data quality challenges and model risk must

be addressed by the regulated institution through its existing risk management framework, not delegated to the vendor.

LO: CI-1 c: Explain the implications of further adoption of AI on financial stability. • CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: third-party dependencies and service provider concentration; market correlations; cyber risks; model risk, data, and governance; other vulnerabilities, such as fraud, disinformation, and misalignment. • CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024), market correlations and model risk sections; IMF (October 2024), uses of AI in risk management; SchweserNotes Book 5, Readings 101.2, 101.3 and 100.1

CI-1 • Advances in Artificial Intelligence: Implications for Capital Markets Activities (IMF, October 2024) • Vardeez: VERY IMPORTANT

CI-1 d

P2 CONCEPT

Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall (reading section; no shape on this LO)

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter. The A14 workbook section covers the reading, but none of its four reported shapes sits on this LO.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes: none on this LO (Vardeez focus only). Workbook A14.2, worked under CI-2 c, also tags this LO: its distractor C tests the rule that a bank stays fully responsible for a vendor's model.

What the questions look like: no calculations. The LO splits into two actors and two verbs: what *supervisors* use AI for (a tool) and what policy asks of *regulated entities* (a rule). Expect role misassignment (authority against institution, NIST against IOSCO), timing (initial margin in normal conditions, the rest under stress) and polarity flips on the policy menu.

Recap. The four financial stability challenges of CI-1 c (volatility and speed under stress, monitoring and opacity, cyber and manipulation, third-party operational risk) are what the regulatory priorities below respond to.

1. Scope: who is responsible

The regulatory and supervisory frameworks that govern capital markets also extend to the use of AI systems. Institutions remain **fully responsible** for the AI systems they deploy, whether those systems are developed internally or obtained from external providers. Buying a model never transfers accountability for it.

2. Best practices for regulators

Standard-setting bodies expect financial sector authorities to stay attentive to the growing use of AI in capital markets and to be prepared for adoption to keep accelerating. Supervisors should make sure that their skills and tools can handle large volumes of detailed data and monitor increasingly complex investment strategies, and existing frameworks must be adaptable to ongoing AI developments.

Six focus areas. In responding to AI applications, standard-setting bodies focus on: (1) governance of AI use, (2) transparency and explainability, (3) outsourcing practices, (4) data quality and bias, (5) reporting requirements, and (6) maintaining meaningful human oversight.

Who is doing what. The FSB, the Basel Committee on Banking Supervision (BCBS), the BIS, the US National Institute of Standards and Technology (NIST), IOSCO and national regulators have taken steps to strengthen investor protection, maintain market integrity and safeguard financial stability. These efforts build on existing frameworks and in some cases introduce new ones, such as **NIST's 2024 AI framework**. **IOSCO** has focused on market volatility associated with algorithmic trading and on AI-related risks in market intermediation and asset management. Many governments have

also begun broader AI strategies emphasising cybercrime prevention, data protection and sound AI governance.

Recommendations for safe adoption. (1) Evaluate the adequacy of existing regulatory frameworks; (2) develop new risk management frameworks; (3) advance principles for AI governance through *public-private collaboration*. The review of current frameworks should test whether they address robustness, explainability, cybersecurity, and data bias and privacy.

3. Supervisors' own use of AI: the tool

For supervisors, AI brings efficiency gains through **automated data quality checks**, **integration of multiple data sources** and **detection of anomalies in trading patterns**, and further applications such as **real-time market monitoring** and **identification of misleading information**. GenAI adds more: **legacy code optimisation**, **content creation**, **improved information retrieval**, and **code generation, debugging and explanation**. (In CI-1 a terms this is SupTech; RegTech is the regulated firms' own compliance tooling.)

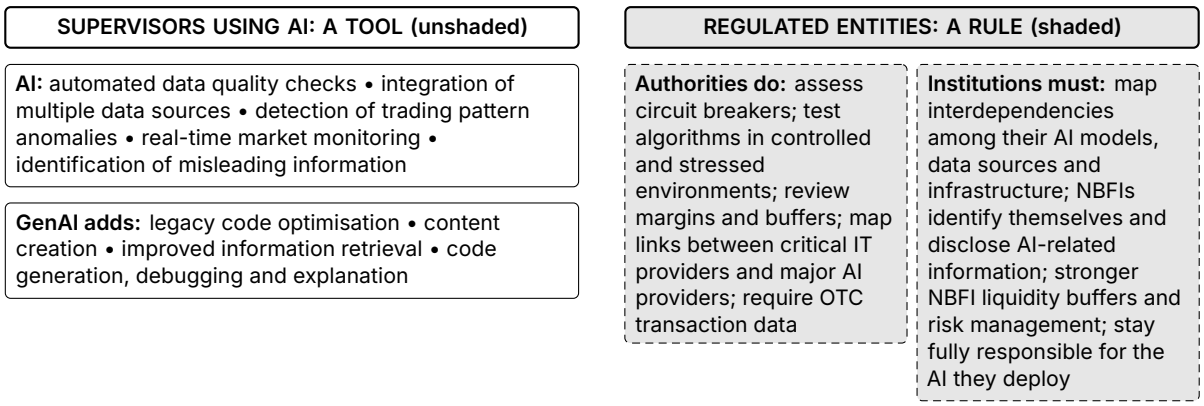


Figure 1. Two actors, two verbs. The left half (unshaded) is what supervisors gain by using AI themselves; the right half (shaded, two dashed boxes) is what policy asks of the AI used by banks and NBFIs, split between what authorities do and what institutions must do. Mapping appears on both sides of the right half: authorities map the links between critical IT and major AI providers, institutions map their own models, data and infrastructure.

4. Policy on AI use by regulated entities: the rule

Given the financial stability risks faced by both the banking and NBFI sectors, regulatory and supervisory frameworks for AI should be strengthened to balance realising the benefits with enhanced risk management. The key policy recommendations and risk-mapping priorities are to improve *cyber resilience*, to address dependencies on *models, data and third-party service providers*, to mitigate market volatility (particularly in *OTC* markets), and to improve the quality and timeliness of regulatory reporting. Four areas are identified as priorities; the second column names the CI-1 c challenge each one answers.

Correction to the source notes. The class notes list "initial margins, margin call volatility, variation margins, and the responsiveness of central counterparties' models during crises" as one block. SchweserNotes Book 5, LO 100.d, separates the timing: initial margin levels are reviewed *during normal conditions*, which set the baseline for stress; margin call volatility, variation margin collections and distributions, and CCP margin model responsiveness are the stress-period items (Schweser Module Quiz 100.3, Question 2, keys on it). The notes also test algorithms "in both normal and stressed conditions"; Schweser says "controlled and stressed environments". A controlled environment is a test setting, not normal market conditions; in

the reading, "normal conditions" attaches only to the initial margin review.

Priority	Challenge	Measures
1. Market speed and volatility under stress	Challenge 1	Modify existing volatility response mechanisms or develop new ones; assess the quality of circuit breakers; test algorithms in controlled *and* stressed environments; evaluate margin requirements and other buffers in large price swings. On IOSCO, BCBS and BIS proposals: review *initial margin levels during normal conditions*, and margin call volatility, variation margin collections and distributions, and the responsiveness of CCP margin models *under stress*
2. Operational risk from a few third-party AI providers	Challenges 4 and 3	*Authorities* map the interconnections between critical IT infrastructure providers and major AI service providers. Markets must be resilient: prevent attacks where possible, and be able to protect against, respond to and recover from them. AI systems are vulnerable to attacks on training data or on the models themselves, so cybersecurity safeguards are evaluated at every stage of the AI lifecycle
3. Opacity and monitoring	Challenge 2	*Institutions* are required to map interdependencies among AI models, data sources and technological infrastructure. Risks arise when data fail to capture a full financial cycle, or depend on a few external providers. NBFIs are required to identify themselves and disclose AI-related information. Large traders matter even more. Stronger liquidity buffers and risk management improve NBFI resilience against liquidity runs and asset mispricing
4. OTC market resilience	Challenge 1 (volatility in OTC markets)	Authorities could require the collection and dissemination of detailed OTC transaction information. Options: adjust liquidity frameworks; expand incentives for central clearing; enhance liquidity provision by market makers; introduce margin requirements for non-centrally cleared derivatives; develop liquidity and funding backstops for extreme stress

Results card: CI-1 d

Accountability	Frameworks extend to AI; institutions fully responsible, built or bought
Focus areas (6)	Governance; transparency and explainability; outsourcing; data quality and bias; reporting; meaningful human oversight
Bodies	FSB, BCBS, BIS, NIST, IOSCO, national regulators. NIST: 2024 AI framework. IOSCO: algorithmic trading volatility; AI in intermediation and asset management. Governments: cybercrime, data protection, AI governance
Safe adoption	Evaluate existing frameworks; develop new risk frameworks; governance principles through public-private collaboration; test robustness, explainability, cybersecurity, data bias and privacy
Supervisors' AI	Data quality checks, data integration, anomaly detection, real-time monitoring, misleading information. GenAI: legacy code, content, retrieval, code generation and debugging
Four priorities	Speed and volatility (circuit breakers, algorithm tests, margins); third-party providers (authorities map, cyber resilience); opacity (institutions map, NBFI disclosure, buffers); OTC (data, clearing, margin, backstops)
Margin timing	Initial margin: normal conditions. Margin call volatility, variation margin, CCP models: stress

Exam traps

- **Role misassignment on accountability.** The institution stays fully responsible for a bought model; neither the vendor nor the supervisor takes it over.
- **Body swap.** NIST issued the 2024 AI framework; IOSCO works on algorithmic trading volatility and AI in intermediation and asset management.
- **Tier swap on supervisors' AI.** Real-time monitoring and anomaly detection are general AI gains; legacy code and retrieval are the GenAI additions.
- **Who maps what.** Authorities map IT-to-AI provider links; institutions map their own models, data and infrastructure.
- **Timing swap on margins.** Initial margin in normal conditions; the flows and CCP responsiveness under stress. Maintenance margin is not on the list.
- **Dropped condition.** Algorithms are tested in controlled *and* stressed environments; resilience means protect, respond *and* recover, not prevention alone.
- **Polarity flips on the OTC menu.** More transaction data, more central clearing, more market-maker liquidity, margin for non-centrally cleared derivatives, and backstops developed, not removed.
- **Full cycle.** Data that miss a full financial cycle are a risk, not a feature.
- **Replace for review.** Frameworks are evaluated and built on, with new ones where needed; governance principles come from public-private collaboration.

Practice questions: CI-1 d

Question CI-1 d.1 Adapted from CI QB VI Q31 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

The COO of a Luxembourg asset manager plans to license an AI trading model from an external vendor rather than build one, and tells the board that the vendor contract will shift regulatory responsibility for the model. Which response best reflects the IMF reading?

A. Responsibility for model errors passes to the vendor, while the firm keeps responsibility only for how it uses the model's outputs.

B. Existing capital markets regulation does not reach AI systems, so responsibility rests on the contract until AI-specific rules exist.

C. The firm remains fully responsible for the AI systems it deploys, whether developed internally or obtained from external providers.

D. Responsibility passes to the supervisor once it has reviewed the model, since supervisors must approve the AI used in trading.

Question CI-1 d.2 New • built from SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

A policy adviser at a Brazilian securities regulator is drafting a gap analysis against the key focus areas that standard-setting bodies have set out in responding to AI applications. Each of the following names focus areas on that list EXCEPT:

A. Governance of AI use, and transparency and explainability.

B. Outsourcing practices, and data quality and bias.

C. Reporting requirements, and maintaining meaningful human oversight.

D. Presuming outsourced AI riskier, and ending human oversight.

Question CI-1 d.3 Adapted from CI QB VI Q33 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

A junior analyst at a Mumbai brokerage is drafting a slide that attributes AI-related regulatory work to the bodies named in the IMF reading. Which attribution is correct?

A. The Basel Committee introduced a new AI framework in 2024, building on the FSB's work on the volatility linked to algorithmic trading.

B. IOSCO has focused on volatility linked to algorithmic trading and on AI-related risks in market intermediation and asset management.

C. The FSB has led the work on AI risks in market intermediation and asset management, while IOSCO concentrates on bank capital standards.

D. National governments' AI strategies concentrate on capital markets conduct, leaving cybercrime prevention and data protection to IOSCO.

Question CI-1 d.4 Adapted from CI QB VI Q40 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

A financial stability directorate at the central bank of Chile is drafting a roadmap to support the safe adoption of AI in capital markets. Which step is most consistent with the IMF reading's recommendations?

A. Assess whether existing frameworks adequately address robustness, explainability, cybersecurity, and data bias and privacy.

B. Replace existing regulatory frameworks with AI-specific rulebooks, since current frameworks were not designed with AI in mind.

C. Develop AI governance principles within the authority alone, to keep them independent of the industry they will govern.

D. Leave explainability out of the framework review, because it is a design choice that each firm makes model by model.

Question CI-1 d.5 New • built from SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

The head of SupTech at a supervisory authority in Kenya wants AI to help modernise the authority's legacy surveillance code and to retrieve information faster from its archives. Which statement is consistent with the IMF reading?

A. These are RegTech uses, which the IMF assigns to regulated banks for their own compliance management rather than to the supervisors.

B. These are GenAI capabilities in the IMF reading, over and above automated data quality checks, data integration and anomaly detection.

C. These are the core supervisory gains from AI in general, while GenAI's added value for supervisors lies in real-time market monitoring.

D. These uses fall outside supervision, since the IMF sees the value of AI for supervisors only in detecting anomalies in trading patterns.

Question CI-1 d.6 Adapted from CI QB VI Q35 and Q36 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d (correction box)

A markets authority in Hong Kong is updating its response plan for dislocations caused by AI-driven trading, drawing on proposals from IOSCO, the Basel Committee and the BIS. Which review plan is most consistent with the IMF reading?

A. Review initial margin levels in normal conditions, and margin call volatility, variation margin flows and CCP model responsiveness under stress.

B. Review margin call volatility and variation margin flows in normal conditions, and initial margin levels and CCP margin models during stress.

C. Review maintenance margin minimums in normal conditions, and test algorithms only in controlled environments to avoid destabilising live markets.

D. Fix initial and variation margin at permanent levels, so that margin calls no longer change during the large price swings of a dislocation.

Question CI-1 d.7 New • built from SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

A supervisor at the Monetary Authority of Singapore is assigning the mapping tasks that the IMF recommends for AI-related risks. Which allocation is consistent with the reading?

A. Institutions map the links between critical IT infrastructure providers and major AI providers; authorities map each firm's AI models and data.

B. Authorities map both sets of links, because institutions cannot see their own dependencies on third-party AI service providers.

C. Institutions map both sets of links, because the authorities' role in the reading is limited to circuit breakers and margin rules.

D. Authorities map links between critical IT providers and major AI providers; institutions map their own AI models, data and infrastructure.

Question CI-1 d.8 Adapted from CI QB VI Q38 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

A regulator in Ireland is designing measures to address the opacity and monitoring challenges that come with AI adoption by NBFIs. Which measure is most consistent with the IMF reading?

A. Exempt large traders from additional monitoring, since AI spreads trading activity across many smaller and more agile participants.

B. Accept training data drawn from the latest expansion only, since data from earlier cycles reflect an outdated structure of the market.

C. Require NBFIs to identify themselves and disclose AI-related information, and strengthen their liquidity buffers and risk management.

D. Limit the new disclosure requirements to banks, since NBFIs operate outside the perimeter of capital markets regulation and supervision.

Question CI-1 d.9 Adapted from CI QB VI Q39 • SchweserNotes Book 5, LO 100.d; CI notes CI-1 d

As AI use expands, a securities authority in South Korea is reviewing options to maintain the integrity, efficiency and resilience of over-the-counter (OTC) markets. Which package is most consistent with the IMF reading?

A. Scale back the collection of OTC transaction data to cut compliance costs, and restrict market makers' liquidity provision during stress.

B. Require detailed OTC transaction data to be collected and disseminated, and develop liquidity and funding backstops for extreme stress.

C. Remove backstop liquidity and funding measures to avoid moral hazard, and rely on central clearing alone to absorb any extreme stress.

D. Exempt non-centrally cleared derivatives from margin requirements, so that incentives for central clearing are no longer needed.

Solutions • CI-1 d

Solution CI-1 d.1 — Answer: C

Adapted from CI QB VI Q31 • SchweserNotes Book 5, LO 100.d

Step 1 – Scope of regulation

The regulatory and supervisory frameworks that govern capital markets also extend to the use of AI systems.

Step 2 – Accountability

Institutions remain fully responsible for the AI systems they deploy, regardless of whether those systems are developed internally or obtained from external providers.

⇒ The contract cannot move regulatory responsibility.

Interpretation

Workbook A14.2 (worked under CI-2 c) uses the same rule as a distractor for a German retail bank buying a vendor's GenAI model. Outsourcing practices are also one of the standard setters' six focus areas.

Why the other options are wrong

A is incorrect. Role misassignment: no part of the responsibility passes to the vendor; the split between "errors" and "use" is invented.

B is incorrect. Polarity flip: existing capital markets frameworks already extend to AI systems.

D is incorrect. Role misassignment: supervisory review does not transfer accountability to the supervisor.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024), Advances in Artificial Intelligence: Implications for Capital Markets Activities; SchweserNotes 2026 Book 5, Reading 100, LO 100.d

Solution CI-1 d.2 — Answer: D

New • built from SchweserNotes Book 5, LO 100.d

Step 1 – The six focus areas

Governance of AI use; transparency and explainability; outsourcing practices; data quality and bias; reporting requirements; maintaining meaningful human oversight.

Step 2 – Find the option that is not on the list

A, B and C between them name all six. D pairs two positions the reading rejects: there is no indication that outsourced models are riskier than internally developed ones, and maintaining meaningful human oversight is itself a focus area (Schweser Module Quiz 100.3, Question 1, wrong options C and A).

⇒ D is the exception.

Interpretation

Outsourcing is a focus because institutions stay fully responsible for what they buy, not because bought models are presumed worse, and human oversight is to be maintained, not phased out. The risk the reading does attach to outsourcing is concentration: reliance on a few third-party providers (CI-1 c).

Why the other options are wrong

A is incorrect. On the list: governance of AI use, and transparency and explainability, are both focus areas.

B is incorrect. On the list: outsourcing practices, and data quality and bias.

C is incorrect. On the list: reporting requirements, and meaningful human oversight.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d, Module Quiz 100.3 Q1

Solution CI-1 d.3 — Answer: B

Adapted from CI QB VI Q33 • SchweserNotes Book 5, LO 100.d

Step 1 – The bodies named

The FSB, BCBS, BIS, NIST, IOSCO and national regulators have acted to strengthen investor protection, market integrity and financial stability, building on existing frameworks and in some cases introducing new ones.

Step 2 – Attach each initiative to its body

NIST: the 2024 AI framework. IOSCO: market volatility associated with algorithmic trading, and AI-related risks in market intermediation and asset management. Governments: broader AI strategies on cybercrime prevention, data protection and sound AI governance.
$\Rightarrow$ Only B attributes correctly.

Interpretation

Governance questions are won on *who*. IOSCO, the BCBS and the BIS reappear together as the source of the margin proposals in the first policy priority.

Why the other options are wrong

A is incorrect. Role misassignment: the 2024 AI framework is NIST's, and the algorithmic trading work is IOSCO's.
C is incorrect. Role misassignment: the intermediation and asset management workstream is IOSCO's; nothing in the reading gives IOSCO a bank capital role.
D is incorrect. Role misassignment and scope error: cybercrime prevention and data protection are what government AI strategies emphasise.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d

Solution CI-1 d.4 — Answer: A

Adapted from CI QB VI Q40 • SchweserNotes Book 5, LO 100.d

Step 1 – The three recommendations

Evaluate the adequacy of existing regulatory frameworks; develop new risk management frameworks; advance principles for AI governance through public-private collaboration.

Step 2 – What the review must test

Whether the current frameworks sufficiently address robustness, explainability, cybersecurity, and data bias and privacy.
$\Rightarrow$ A is the first recommendation with its stated test.

Interpretation

The approach is to build on what exists, add where needed and write governance principles with the industry. That matches the survey respondents' preference for best practices and flexible guidelines (CI-1 c).

Why the other options are wrong

B is incorrect. Absolute claim: frameworks are evaluated for adequacy and built on, with new ones introduced where needed, not replaced wholesale.
C is incorrect. Polarity flip: governance principles are to be advanced through public-private collaboration.
D is incorrect. Scope error: explainability is one of the key considerations the review must cover, and a focus area of the standard setters.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d

Solution CI-1 d.5 — Answer: B

New • built from SchweserNotes Book 5, LO 100.d

Step 1 – General AI for supervisors

Efficiency gains through automated data quality checks, integration of multiple data sources and detection of trading pattern anomalies, plus real-time market monitoring and identification of misleading information.

Step 2 – What GenAI adds

Legacy code optimisation, content creation, improved information retrieval, and advanced code generation, debugging and explanation.
⇒ Modernising legacy code and faster retrieval are on the GenAI list.

Interpretation

This is the "tool" half of the LO. The "rule" half, policy on AI used by banks and NBFIs, is the four priorities.

Why the other options are wrong

A is incorrect. Role misassignment: RegTech is the banks' compliance tooling; legacy code optimisation and retrieval are named among supervisors' GenAI gains.
C is incorrect. Tier swap: real-time market monitoring is a general AI gain; legacy code and retrieval are the GenAI additions.
D is incorrect. Absolute claim: anomaly detection is one of several supervisory gains, alongside data checks, data integration, real-time monitoring and the GenAI uses.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.a and 100.d

Solution CI-1 d.6 — Answer: A

Adapted from CI QB VI Q35 and Q36 • SchweserNotes Book 5, LO 100.d

Step 1 – The volatility toolkit

Authorities should assess the quality of circuit breakers, test algorithms in controlled and stressed environments, and evaluate margin requirements and other buffers during large price swings.

Step 2 – The margin agenda and its timing

Based on IOSCO, BCBS and BIS proposals: review initial margin levels during normal conditions; review margin call volatility, variation margin collections and distributions, and the responsiveness of CCP margin models during market stress.
⇒ Set the baseline in calm conditions; watch the flows in the storm.

Interpretation

Initial margin is the buffer posted before stress arrives, which is why it is reviewed in normal times. The class notes run the four items together; the correction box restores the timing.

Why the other options are wrong

B is incorrect. Timing swap: initial margin is the normal-conditions item; margin call volatility and variation margin are stress-period items.

C is incorrect. Wrong input and dropped condition: maintenance margin is not on the agenda, and algorithms are tested in stressed as well as controlled environments.

D is incorrect. Absolute claim: the reading asks authorities to evaluate the adequacy and responsiveness of margins, not to freeze them.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d, Module Quiz 100.3 Q2

Solution CI-1 d.7 — Answer: D

New • built from SchweserNotes Book 5, LO 100.d

Step 1 – The authorities' map

Under the third-party priority, financial sector authorities should map the interconnections between critical IT infrastructure providers and major AI service providers, since disruption at these entities could have widespread consequences.

Step 2 – The institutions' map

Under the opacity priority, financial institutions should be required to map interdependencies among their AI models, data sources and underlying technological infrastructure.

$\Rightarrow$ The system-wide map belongs to the authority; the firm-level map belongs to the firm.

Interpretation

The split mirrors the LO: what authorities do themselves, and what policy requires of regulated entities. A firm's map also has to flag data that miss a full financial cycle or depend on a few external providers.

Why the other options are wrong

A is incorrect. Role misassignment: it swaps the two maps between authorities and institutions.

B is incorrect. Role misassignment: institutions are required to map their own interdependencies.

C is incorrect. Role misassignment and scope error: authorities map provider interconnections, and their priorities go well beyond circuit breakers and margins.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d

Solution CI-1 d.8 — Answer: C

Adapted from CI QB VI Q38 • SchweserNotes Book 5, LO 100.d

Step 1 – Disclosure

NBFIs should be required to identify themselves and disclose AI-related information.

Step 2 – Resilience

Strengthening liquidity buffers and enhancing risk management could improve NBFI resilience, mitigating the risk of liquidity runs and asset mispricing.

$\Rightarrow$ C states both halves of the NBFI response.

Interpretation

The measure answers the opacity challenge in CI-1 c: activity is moving to NBFIs that are lightly overseen compared with banks.

Why the other options are wrong

A is incorrect. Polarity flip: large traders have always been a regulatory focus, and their importance *increases* in an AI-driven landscape.

B is incorrect. Polarity flip: risks arise precisely when data fail to capture a full financial cycle.

D is incorrect. Role misassignment: NBFIs are the target of the disclosure requirement, because that is where transparency is falling.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.c and 100.d

Solution CI-1 d.9 **Answer: B**

Adapted from CI QB VI Q39 • SchweserNotes Book 5, LO 100.d

Step 1 – Data

Authorities could require the collection and dissemination of detailed information on OTC transactions by market participants.

Step 2 – The policy menu

Adjust liquidity frameworks; expand incentives for central clearing; enhance liquidity provision by market makers; introduce margin requirements for non-centrally cleared derivatives; develop backstop measures for liquidity and funding in extreme stress.

$\Rightarrow$ B pairs the data requirement with the backstops.

Interpretation

Every item on the OTC menu moves in one direction: more information, more clearing, more liquidity, more margin, and a backstop for the extreme case.

Why the other options are wrong

A is incorrect. Polarity flip on both halves: the reading would expand OTC data collection and *enhance* market makers' liquidity provision.

C is incorrect. Polarity flip: backstops are to be developed, and central clearing incentives are one tool among several, not the only one.

D is incorrect. Polarity flip: the option is to *introduce* margin requirements for non-centrally cleared derivatives, alongside more incentives for central clearing.

LO: CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** IMF (October 2024); SchweserNotes 2026 Book 5, Reading 100, LO 100.d

CI-2 • The Financial Stability Implications of Artificial Intelligence (FSB, November 2024) • Vardeez: VERY IMPORTANT

CI-2 a

P1 CONCEPT

Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A14 Artificial Intelligence and GenAI: AI came up in the August 2026 sitting, and Map #20 lists it in 6 of 16 sources. Shape A14·S1 maps to this LO. Workbook A14.1 (tagged CI-1 a; CI-2 a, c) is worked under CI-1 a, because its key rests on the IMF's two model groups; this block cross-references it rather than repeating it.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A14·S1** AI vs GenAI: what GenAI adds and its signature failure mode. The IMF's two model groups are CI-1 a (Workbook A14.1); here the shape is tested from the FSB's side: the Transformer, unstructured data, pre-trained models and hallucination (Questions CI-2 a.3 and CI-2 a.9).

What the questions look like: no calculations. Classification audits (which bucket a driver belongs to), the relative weight of the two sides, pairing each supply-side driver with its constraint, mapping initiatives to the three profitability channels, and the two-sided competition driver.

1. The reading and where this LO sits

The Financial Stability Board's November 2024 paper updates its 2017 report on AI in financial services. It runs in three steps, one per LO: what has driven adoption since 2017 (CI-2 a), where AI is now used (CI-2 b), and which vulnerabilities this creates for financial stability (CI-2 c). CI-1 is a different paper: the IMF's October 2024 study of AI in capital markets, with its own model grouping and market-participant outreach. An option that gives an IMF finding to the FSB, or the reverse, is a sibling swap.

AI tools have long been used in financial services, but adoption has accelerated in recent years. Applications now span operations (model risk management, capital optimisation), customer-focused activities (automated client interactions, credit assessments), portfolio management, trading, supervisory technology (SupTech) and regulatory technology (RegTech). GenAI in particular offers notable benefits in document processing, fraud detection, customer support, information retrieval, software development and market analysis. Alongside these advantages, AI introduces meaningful risks and potential vulnerabilities across the financial sector (CI-2 c).

2. What changed since 2017, and which side drove it

Since the 2017 report, major developments in **big data, computational power, deep learning and GenAI** have advanced the field significantly. Interest in AI has also grown considerably, and the launch of **ChatGPT in November 2022** was an important catalyst. The 2017 report examined both supply-side and demand-side factors expected to drive adoption. Both still matter today, but the evidence indicates that **supply-side forces have been the more influential driver**.

A quick way to keep the two lists apart: supply-side factors are what made AI more capable and cheaper to build and deploy (technology, data, business models); demand-side factors are why institutions want it (profit, rivals, rules). A driver is classified by what it is, not by who asks for it. Institutions buy cloud services, but cloud computing is a business model development on the supply side.

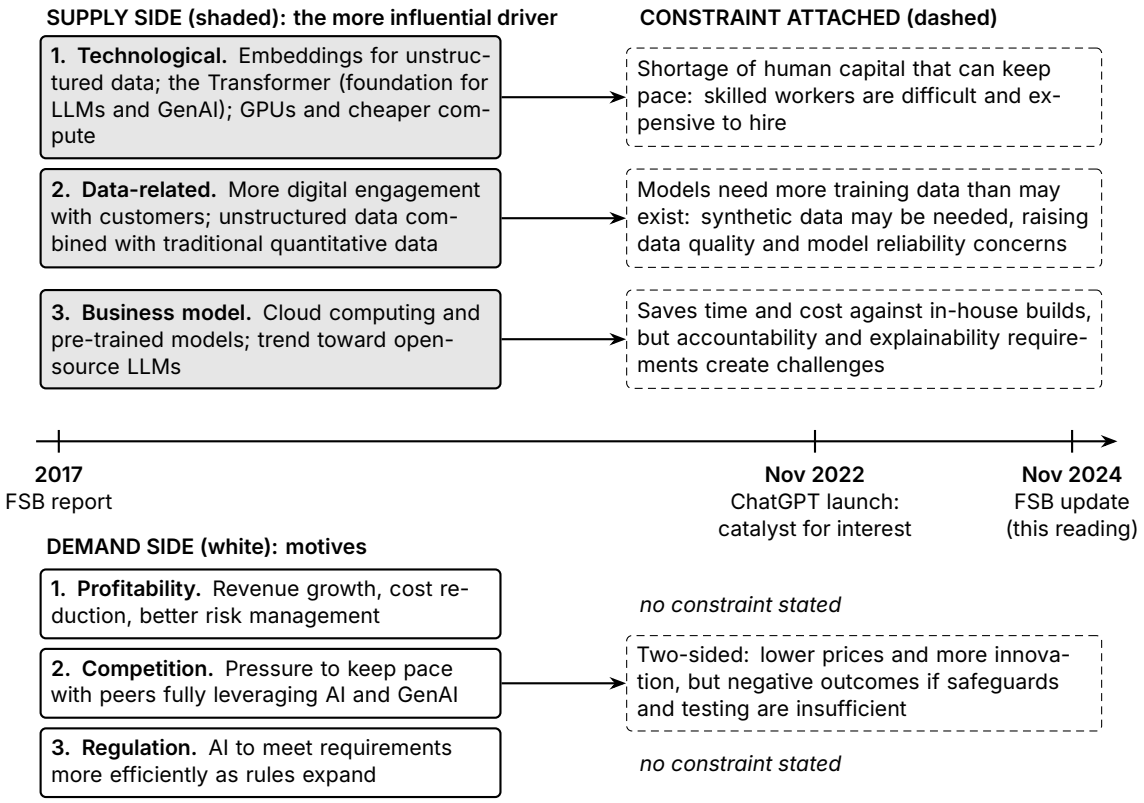


Figure 1. The split on the 2017 to 2024 timeline (years to scale). Shaded boxes are supply-side capabilities and each carries a constraint (dashed box). White boxes are demand-side motives. Of the three motives, only competition carries a stated downside, which is why the class notes' "demand side carries no constraint" rule needs the correction below.

3. Supply side: three capabilities, each with a constraint

1. **Technological developments.** Advances in software and hardware, together with the shift toward GenAI:
   - better processing of *unstructured* data through **embeddings**, numerical representations of data such as text, images or audio, enabled by advances in deep learning;
   - the **Transformer architecture**, which revolutionised natural language processing (NLP) and laid the foundation for large language models (LLMs) and GenAI. It uses an *attention mechanism* to focus on the most relevant parts of the input text and processes data in parallel rather than sequentially; *positional encoding* captures word order, and *feed-forward* components help predict

subsequent words from earlier ones (SchweserNotes Module Quiz 101.1);
- more **computational power** and wider use of graphics processing units (GPUs), so larger datasets and more complex models can be handled at lower cost.

Constraint: a shortage of human capital able to keep pace; workers with the necessary skills are difficult and expensive to hire.

2. **Data-related developments.** Digital engagement between institutions and their customers has increased and unstructured data has grown rapidly. AI can now combine traditional quantitative datasets with large volumes of unstructured data such as images and videos.
Constraint: modern AI and ML models often need more training data than is available, so a shortage of high-quality real-world data may emerge. **Synthetic data**, artificially generated to resemble real-world datasets, may be needed to close the gap, which raises questions about data quality and the reliability of the models built on it.

3. **Business model developments.** Wider use of **cloud computing** (flexible delivery of services) and **pre-trained models**. The high cost of training models and maintaining the infrastructure has pushed firms toward cloud providers that offer pre-trained AI capabilities. Pre-trained models save time and cost against building in-house. Pre-trained LLMs come as closed source or open source, and the industry trend has shifted toward **open source**, driven by its improving performance relative to closed-source models; open-source models are often free, easily accessible and customisable to a firm's needs.
Constraint: accountability and explainability requirements can create challenges for firms using models they did not build.

4. Demand side: three motives

1. **Profitability**, through three channels:
   - *revenue growth*: more tailored products and services, better customer onboarding, reduced churn (the rate at which customers stop using a product or service);
   - *cost reduction*: automating manual tasks, raising worker productivity, and reallocating resources toward revenue-generating activities that are hard to automate;
   - *improved risk management*: better forecasting of future cash flows, losses and delinquencies.

2. **Competition.** Pressure to keep up with competitors that are fully leveraging AI, and GenAI in particular. The outcome can be positive (lower prices, greater innovation) or negative if safeguards and testing are insufficient.

3. **Regulation.** Firms need to enhance compliance and meet regulatory requirements more efficiently. AI is increasingly seen as a tool that strengthens compliance, especially as rules on data protection, guidance on AI development and adoption, and international standards continue to expand. The driver is the firm's need to comply efficiently, not a mandate to adopt AI.

5. What GenAI adds: shape A14·S1 from the FSB's side

Shape A14·S1 asks what separates GenAI from established AI and ML. The IMF's answer, its two model groups, belongs to CI-1 a and is worked in Workbook A14.1. The FSB's answer can be assembled from the developments above and from its model risk section (CI-2 c):

- **What it is built on:** deep learning embeddings, which make unstructured data usable, and the Transformer, the foundation for LLMs and GenAI.

- **How firms get it:** mostly as pre-trained LLMs, either closed or open source, and often through cloud providers.
- **What it adds in use:** chatbots moved from rules-based systems to LLM-based ones, and operations use cases widened to content generation, information search and retrieval, code generation, legacy code modernisation and voice transcription (CI-2 b).
- **How it fails:** LLM outputs are typically unstructured text, so outcome analysis and error rates are hard to establish. The signature failure mode is the **hallucination**: a highly confident but incorrect response.

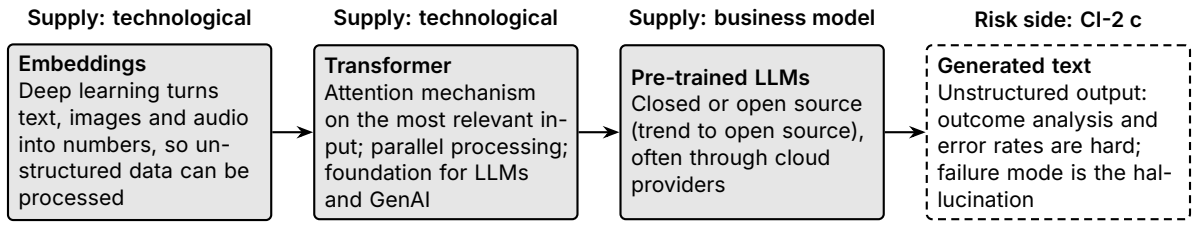


Figure 2. GenAI as the FSB describes it, read left to right. The three shaded boxes are supply-side developments from this LO; the dashed box is where the same technology turns into model risk (CI-2 c). For the IMF's split between established AI/ML and sophisticated AI, see CI-1 a and Workbook A14.1.

Correction to the source notes. The caption of the class notes' Figure CI-2.1 says that every supply-side driver carries a constraint and that "the demand-side drivers in the notes carry none", and offers this as the quickest way to tell the lists apart. The competition driver does carry a stated downside: competitive pressure can produce negative outcomes if safeguards and testing are insufficient (SchweserNotes Book 5, LO 101.a; the notes' own body text says "rushed implementation without proper safeguards"). The safer rule is that supply-side factors are capabilities and demand-side factors are motives; competition is a motive with two possible outcomes. Question CI-2 a.7 option C is the misreading the caption invites.

Results card: CI-2 a

Anchor dates	2017 FSB report; November 2022 ChatGPT launch (catalyst for interest); November 2024 FSB update
Since 2017	Big data, computational power, deep learning, GenAI
Relative weight	Both sides still matter; the evidence indicates the **supply side** has been more influential
Supply side (capabilities)	Technological: embeddings, Transformer, GPUs. *Constraint:* human capital shortage Data-related: digital engagement, unstructured data. *Constraint:* real data shortage, synthetic data Business model: cloud, pre-trained models, open-source trend. *Constraint:* accountability, explainability
Demand side (motives)	Profitability: revenue growth, cost reduction, risk management Competition: lower prices and innovation, or harm if safeguards and testing are insufficient Regulation: more efficient compliance as rules expand
GenAI (A14·S1)	Transformer-based LLMs; unstructured text output; failure mode: hallucination

Exam traps

- **Side swap.** Cloud computing and pre-trained models are a supply-side (business model) driver even though institutions buy them; competition and regulation are demand-side drivers even though they come from outside the firm.
- **Weight flattened or reversed.** "Equally influential" and "demand side dominates" are both wrong: both matter, supply has been more influential.
- **Date roles.** 2017 is the FSB report that examined both sides; November 2022 (ChatGPT) is a catalyst for interest, not the start of AI in finance.
- **Transformer parts swapped.** Attention focuses on the most relevant input and processing is parallel; positional encoding captures word order and feed-forward components help predict subsequent words (SchweserNotes Module Quiz 101.1 answer key).
- **Embeddings versus synthetic data.** Embeddings represent existing data as numbers; synthetic data is generated data that fills a shortage.
- **Constraint on the wrong driver.** Human capital shortage belongs to technology; synthetic data to data; accountability and explainability to business models.
- **Open versus closed source.** The trend is toward open source, on improving performance; customisability is an open-source trait.
- **Profitability channels.** Churn and onboarding are revenue growth; automation and reallocation are cost reduction; forecasting losses and delinquencies is risk management.
- **Competition made one-sided.** It can lower prices and spur innovation, or cause harm if safeguards and testing lag.
- **FSB for IMF.** The two model groups (established AI/ML against sophisticated AI) are the IMF's, in CI-1 a.

Practice questions: CI-2 a

Question CI-2 a.1 Adapted from CI QB VI Q41 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

The head of digital strategy at a Canadian bank is drafting a board paper that uses the FSB's framework to explain why AI adoption in financial services has accelerated since 2017. Which of the following classifications should the paper use?

A. Wider reliance on cloud providers and pre-trained models is a demand-side driver, because it reflects what institutions want to buy from vendors.

B. Pressure to keep pace with peers that are deploying GenAI is a supply-side driver, because it changes the AI tools that vendors bring to market.

C. Unstructured data growth is a supply-side driver because AI can combine it with quantitative data; efficient compliance is a demand-side driver.

D. The Transformer architecture and cheaper GPU computing are demand-side drivers, because they raise the expected profitability of AI projects.

Question CI-2 a.2 Adapted from CI QB VI Q42 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

A central bank economist in Australia is writing a retrospective on AI adoption in financial services between the FSB's 2017 report and its November 2024 update. Which of the following conclusions is most consistent with the FSB's assessment?

A. Demand-side forces have been the more influential driver, because profit pressure pushed firms to adopt AI before the technology was ready.

B. Both sides still matter, but supply-side forces have been the more influential driver, and the launch of ChatGPT was a catalyst for interest.

C. Supply-side forces drove adoption until the 2017 report, after which competition and regulation replaced them as the main drivers.

D. The two sides have been equally influential, and the FSB dates the start of AI use in financial services to the launch of ChatGPT in November 2022.

Question CI-2 a.3 Adapted from CI QB VI Q43 and Q44 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

A quantitative developer at a Hong Kong asset manager is briefing the investment committee on the technological developments behind the firm's new text-analytics platform. Which of the following statements is correct?

A. Embeddings are artificially generated datasets that resemble real-world data, and the Transformer processes text sequentially, one word at a time.

B. Embeddings are numerical representations of text, images or audio, and the Transformer uses an attention mechanism to focus on the most relevant input.

C. Embeddings are numerical representations of text, images or audio, and the Transformer's main contribution was cheaper computing through wider use of GPUs.

D. GPUs made larger models cheaper to train, but the Transformer's contribution was limited to structured data, so LLMs rest on earlier architectures.

Question CI-2 a.4 New • built from SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

A risk manager at a Swedish bank is preparing a training slide that pairs each supply-side driver of AI adoption in the FSB's framework with the constraint attached to it. Which of the following pairings is correct?

A. Technological developments: the shift toward open-source models has made accountability and explainability harder to achieve.

B. Business model developments: a shortage of skilled staff makes the people needed to build AI models difficult and expensive to hire.

C. Data-related developments: unstructured data still cannot be combined with traditional quantitative datasets in AI models.

D. Data-related developments: training needs may exceed the data available, so synthetic data may be needed, raising quality concerns.

Question CI-2 a.5 Adapted from CI QB VI Q46 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

The CTO of a mid-sized Brazilian bank is deciding whether to license a pre-trained large language model rather than build one in-house. Which of the following assessments is most consistent with the FSB's account of business model developments?

A. Licensing saves time and cost, though accountability and explainability requirements can create challenges; the trend is toward open-source models.

B. Licensing saves time and cost, and the trend is toward closed-source models, because only closed-source models can be customised to a firm's needs.

C. Licensing removes accountability and explainability concerns, because the vendor has already validated the pre-trained model before its release.

D. Licensing is becoming less common, as falling GPU costs lead most banks to train their own models in-house rather than rely on cloud providers.

Question CI-2 a.6 Adapted from CI QB VI Q47 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

The COO of a European insurer lists three AI initiatives in a business case: (1) tailored products that reduce customer churn; (2) automated claims handling, with staff moved to sales work that is hard to automate; (3) better forecasts of losses and delinquencies. Under the profitability driver in the FSB's framework, which of the following mappings is correct?

A. (1) cost reduction; (2) revenue growth; (3) improved risk management.

B. (1) revenue growth; (2) cost reduction; (3) improved risk management.

C. (1) revenue growth; (2) improved risk management; (3) cost reduction.

D. (1) improved risk management; (2) cost reduction; (3) revenue growth.

Question CI-2 a.7 Adapted from CI QB VI Q48 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a (correction box)

At a strategy offsite of a Singapore bank, a director argues that because rivals are rolling out GenAI, the bank should shorten its testing programme to keep pace. Which response by the CRO best reflects the FSB's view of the competition driver?

A. Competitive pressure can lower prices and spur innovation, and as a supply-side driver it leaves testing to the vendors that supply the AI tools.

B. Competitive pressure has been the most influential driver of adoption since 2017, so keeping pace with rivals should take priority over testing.

C. Competitive pressure can lower prices and spur innovation; the FSB ties its negative outcomes to falling behind rivals rather than to weak testing.

D. Competitive pressure can bring lower prices and more innovation, but it can also lead to negative outcomes if safeguards and testing fall short.

Question CI-2 a.8 Adapted from CI QB VI Q49 • SchweserNotes Book 5, LO 101.a; CI notes CI-2 a

The head of compliance at a Dubai-based bank is explaining to the audit committee why regulation appears among the drivers of AI adoption in the FSB's framework. Which of the following explanations is most consistent with the reading?

A. Regulators now require institutions to deploy AI in their compliance functions, so regulation acts as a mandate rather than as a motive.

B. Expanding rules raise the cost of experimenting with AI, so regulation acts mainly as a brake that offsets the supply-side drivers.

C. Firms use AI to meet requirements more efficiently as rules on data protection, guidance on AI and international standards expand.

D. Regulation is a supply-side driver, because it determines which AI models vendors are allowed to sell to regulated institutions.

Question CI-2 a.9 New • built from SchweserNotes Book 5, LO 101.a and 101.c; CI notes CI-2 a, c

A model risk officer at a Swiss private bank is drafting a training note on how the GenAI assistant the bank has just licensed differs from its long-standing ML credit models, drawing on the FSB's account of developments since 2017. Which of the following statements should she include?

A. GenAI builds on the Transformer architecture, the foundation for LLMs, and its outputs are mostly unstructured text, so error rates are harder to establish.

B. GenAI differs mainly in using only structured quantitative data, which is why it needs less training data than earlier ML models and no synthetic data at all.

C. GenAI's characteristic inaccuracy is data poisoning, in which the model produces a highly confident but incorrect answer without outside interference.

D. The FSB places LLMs with decision trees and clustering among established predictive tools, so the GenAI assistant needs no new validation approach.

Solutions • CI-2 a

Solution CI-2 a.1 — Answer: C

Adapted from CI QB VI Q41 • SchweserNotes Book 5, LO 101.a

Step 1 – Sort by what the driver is

Supply side: technological, data-related and business model developments. Demand side: profitability, competition and regulation.

⇒ Supply-side factors are capabilities; demand-side factors are motives.

Step 2 – Place the two items in C

The rapid growth of unstructured data, which AI can now combine with traditional quantitative datasets, is the data-related development (supply). Using AI to meet requirements more efficiently is the regulation driver (demand).

⇒ Both halves of C sit in the right bucket.

Interpretation

The classification rests on the nature of the driver, not on who pays for it or where it comes from. Every option offers a plausible "because" clause; in the wrong ones the reasoning sounds right and moves the driver across the line.

Why the other options are wrong

A is incorrect. Sibling swap: cloud computing and pre-trained models are business model developments on the supply side. That institutions buy them does not make them demand-side.

B is incorrect. Sibling swap: competition, the pressure to keep up with rivals fully leveraging AI, is a demand-side driver.

D is incorrect. Sibling swap: the Transformer and GPU computing are technological developments (supply side). Profitability is a separate demand-side driver.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024), The Financial Stability Implications of Artificial Intelligence; SchweserNotes 2026 Book 5, Reading 101, LO 101.a

Solution CI-2 a.2 — Answer: B

Adapted from CI QB VI Q42 • SchweserNotes Book 5, LO 101.a

Step 1 – The relative weight

The 2017 report examined both supply-side and demand-side factors. Both continue to matter, but the evidence indicates that supply-side forces have been the more influential driver.

Step 2 – The date that matters

Interest in AI has grown considerably since 2017, and the launch of ChatGPT in November 2022 was an important catalyst.

⇒ B states both facts in the reading's own terms.

Interpretation

The asymmetry is the tested point: technology, data and business models pushed adoption more than profit, competition and regulation pulled it. ChatGPT raised interest; it did not start AI use in finance, where AI tools have long been used.

Why the other options are wrong

A is incorrect. Polarity flip: it reverses the stated dominance; the supply side has been more influential.

C is incorrect. Scope error on time: 2017 is the date of the report that examined both sides. Since

then the supply side has been the more influential driver, not displaced.
D is incorrect. Two errors: "equally" flattens the stated asymmetry, and AI tools have long been used in financial services; ChatGPT was a catalyst for interest, not the start of use.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a

Solution CI-2 a.3 — Answer: B

Adapted from CI QB VI Q43 and Q44 • SchweserNotes Book 5, LO 101.a

Step 1 – Embeddings
Embeddings are numerical representations of data such as text, images or audio, enabled by advances in deep learning. They are what let models process unstructured data.

Step 2 – The Transformer
The Transformer revolutionised NLP and laid the foundation for LLMs and GenAI. It uses an attention mechanism to focus on the most relevant parts of the input text, and it processes data in parallel.
$\Rightarrow$ B states both correctly.

Interpretation
Of the three technological developments, embeddings and the Transformer are about what models can read; GPUs are about what it costs. Option C blurs that line, which is why it is tempting.

Why the other options are wrong
A is incorrect. Sibling swap and polarity flip: artificially generated look-alike data is synthetic data, a response to the data shortage. The Transformer processes data in parallel, not sequentially.
C is incorrect. Sibling swap inside the technological developments: cheaper computing through increased computational power and wider GPU use is the third development, separate from the Transformer, which revolutionised natural language processing and laid the foundation for LLMs and GenAI.
D is incorrect. Polarity flip on the second half: the Transformer revolutionised natural language processing and is the foundation for LLMs. The GPU half is true, which makes the option tempting.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a and Module Quiz 101.1

Solution CI-2 a.4 — Answer: D

New • built from SchweserNotes Book 5, LO 101.a

Step 1 – The three pairs
Technological developments: shortage of human capital that can keep pace. Data-related developments: models need more training data than may exist, so synthetic data may be needed, which raises data quality and model reliability considerations. Business model developments: accountability and explainability challenges.

Step 2 – Match
Only D pairs a driver with its own constraint.
$\Rightarrow$ Each supply-side capability arrived with a cost attached to that capability.

Interpretation
For a data-hungry bank, the practical message is that synthetic data closes a quantity gap and

opens a quality question. Validation has to cover the generated data as well as the model.

Why the other options are wrong

A is incorrect. Sibling swap: accountability and explainability are the business model constraint, tied to pre-trained models, not the technological one.

B is incorrect. Sibling swap: the human capital shortage is the constraint on technological developments.

C is incorrect. Polarity flip: combining traditional quantitative datasets with unstructured data such as images and videos is the data-related *advance*, not its constraint.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a

Solution CI-2 a.5 — Answer: A

Adapted from CI QB VI Q46 • SchweserNotes Book 5, LO 101.a

Step 1 – Why firms license

The high cost of training models and maintaining the infrastructure has pushed firms toward cloud providers that offer pre-trained AI capabilities. Pre-trained models save time and cost against building in-house.

Step 2 – The catch

Accountability and explainability requirements can create challenges for a firm using a model it did not build.

Step 3 – Which kind

Pre-trained LLMs come as closed source or open source; the industry trend has shifted toward open source, driven by improving performance relative to closed source. Open-source models are often free, accessible and customisable.

$\Rightarrow$ A has all three.

Interpretation

For the CTO, the decision is not build against risk-free buy. Licensing moves cost and time in the bank's favour, but the bank still has to explain and answer for the model.

Why the other options are wrong

B is incorrect. Polarity flip with a sibling swap: the trend is toward open source, and customisability is an open-source trait.

C is incorrect. Absolute claim: the reading says accountability and explainability requirements create challenges for pre-trained models; vendor testing does not remove them.

D is incorrect. Polarity flip: reliance on cloud providers and pre-trained models has increased. Cheaper computing is a technological driver of adoption, not a reason firms are leaving pre-trained models.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a

Solution CI-2 a.6 — Answer: B

Adapted from CI QB VI Q47 • SchweserNotes Book 5, LO 101.a

Step 1 – The three channels

Revenue growth: tailored products and services, better onboarding, reduced churn. Cost reduction: automating manual tasks, higher productivity, reallocating resources to revenue-generating work that is hard to automate. Improved risk management: better forecasts of cash flows, losses

and delinquencies.

Step 2 – Map the initiatives

(1) Lower churn through tailored products is revenue growth. (2) Automation with staff moved to hard-to-automate sales work is cost reduction. (3) Loss and delinquency forecasting is improved risk management.
⇒ B.

Interpretation

Two items invite a misfile. Churn sounds like a cost but sits under revenue growth, because it is about keeping customers. Moving staff to sales sounds like revenue but sits under cost reduction, because it is the second half of the automation story.

Why the other options are wrong

A is incorrect. Role misassignment: reduced churn is revenue growth, and automation with reallocation is cost reduction; the first two are swapped.
C is incorrect. Role misassignment: forecasting losses and delinquencies is risk management, not cost reduction, and automation is cost reduction.
D is incorrect. Role misassignment: tailored products that reduce churn are revenue growth, and forecasting is risk management; (1) and (3) are swapped.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a and Module Quiz 101.1

Solution CI-2 a.7 — Answer: D

Adapted from CI QB VI Q48 • SchweserNotes Book 5, LO 101.a

Step 1 – Place the driver

Competition is a demand-side driver: pressure to keep up with competitors that are fully leveraging AI, and GenAI in particular.

Step 2 – Its two outcomes

As competitive pressure increases, outcomes may be positive (lower prices, greater innovation) or negative if safeguards and testing are insufficient.
⇒ Cutting testing to keep pace is the route to the negative outcome.

Interpretation

The director's argument fails on the reading's own terms: competition is a reason to adopt, not a reason to test less. The same pressure reappears in CI-2 c, where competitive pressure and easy access to GenAI may speed deployment and introduce new risks.

Why the other options are wrong

A is incorrect. Sibling swap: competition is a demand-side driver, and the reading ties its outcome to safeguards and testing, not to who supplies the tools.
B is incorrect. Polarity flip on the weight: the evidence indicates that supply-side forces have been the more influential driver.
C is incorrect. Scope error: the negative outcomes the reading links to competition arise when safeguards and testing are insufficient, not from falling behind rivals. Dropping that downside is the misreading the class notes' Figure CI-2.1 caption invites (see the correction box).

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a and 101.c

Solution CI-2 a.8 — Answer: C

Adapted from CI QB VI Q49 • SchweserNotes Book 5, LO 101.a

Step 1 – What the driver is

Companies need to enhance compliance and meet regulatory requirements more efficiently, and AI is increasingly viewed as a tool that strengthens compliance capabilities.

Step 2 – Why it is growing

Rules on data protection, guidance on AI development and adoption, and international standards continue to expand.

$\Rightarrow$ More rules raise the value of efficient compliance, so they pull AI in: a demand-side motive.

Interpretation

Regulation here works like profitability: it gives firms a reason to want AI. The same idea seen from the firm's side is RegTech, an industry use case in CI-2 b.

Why the other options are wrong

A is incorrect. Scope error: the driver is the firm's need to comply efficiently; the reading describes no mandate to adopt AI.

B is incorrect. Polarity flip: expanding rules increase the demand for AI as a compliance tool; the reading lists regulation as a driver of adoption, not a brake.

D is incorrect. Sibling swap: regulation is one of the three demand-side factors.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a

Solution CI-2 a.9 — Answer: A

New • built from SchweserNotes Book 5, LO 101.a and 101.c

Step 1 – What GenAI is built on

The Transformer architecture revolutionised natural language processing and laid the foundation for LLMs and GenAI; embeddings from deep learning let models process unstructured data.

Step 2 – What changes for validation

LLM outputs are typically generated as unstructured text, so assessing their accuracy is hard and outcome analysis and error rates are often difficult to establish. The GenAI-specific inaccuracy is the hallucination, a highly confident but incorrect response.

$\Rightarrow$ A gives the foundation and the validation consequence correctly.

Interpretation

This is shape A14·S1 seen from the FSB's side: the assistant's text output needs different checks from a credit model's numerical forecasts. The IMF's version, its two model groups, is Workbook A14.1 under CI-1 a.

Why the other options are wrong

B is incorrect. Polarity flip: the advance is processing *unstructured* data, and modern models often need more training data than is available, hence synthetic data.

C is incorrect. Sibling swap on the failure mode: a highly confident but incorrect response is a hallucination. Data poisoning is a cyber attack that deliberately manipulates training data or model weights (CI-2 c).

D is incorrect. Role misassignment and sibling swap: grouping models into established AI/ML (decision trees, clustering, NLP) and sophisticated AI (including LLMs) is the IMF's scheme in CI-1 a, not the FSB's, and it places LLMs in the sophisticated group.

LO: CI-2 a: Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.a and 101.c

CI-2 • The Financial Stability Implications of Artificial Intelligence (FSB, November 2024) • Vardeez: VERY IMPORTANT

CI-2 b

P2 CONCEPT

Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall (reading section; no shape on this LO)

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter, but no reported shape lands on this LO. Workbook A14.3 (tagged CI-2 b, c; CI-1 a, c) is worked under CI-1 c, because its key is the financial-stability consequence of using one shared LLM for risk models; it uses this LO's volatility and liquidity risk use case as its setting. The nearest shape, A14·S3 (application of AI in market risk), is mapped to CI-1 a; its FSB half is in Section 3 below.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes: none on this LO.

What the questions look like: no calculations. Classify a use case into one of the four industry areas, tell RegTech (the firm runs it) from SupTech (the authority runs it), tell current supervisory uses from the emerging ones, and place a use case on the task and learning-type grid.

1. Where AI is used, and why adoption is hard to measure

CI-2 a explained why adoption accelerated after 2017. This LO asks where AI is now used, split by **who runs the tool**: the industry (banks, insurers, asset managers) or the authorities (regulators, supervisors and central banks). The FSB reads both as one trend: since 2017 the industry has seen broader adoption of AI tools, a more diversified set of use cases, and more reliance on both RegTech and SupTech.

The measurement problem. Firms have clearly increased their use of AI, but a key challenge is the lack of comprehensive data to assess how far adoption has gone. Competitive pressures may also contribute to a **gap between perceived and actual AI use**. The same gap limits the authorities (CI-2 c): representative and consistent data on how AI is being deployed are often lacking, and AI use is even less visible at institutions outside traditional regulatory frameworks.

2. Tasks and learning types

SchweserNotes Figure 101.2 pairs key financial-services tasks with example use cases and the type of learning or reasoning involved; Module Quiz 101.2 and its answer key name these four. The grid is a classification tool: given a use case, name the task, and the learning type follows.

Task	Learning or reasoning type
Recognition	Supervised classification of images, voice and text (facial recognition is the reading's example)
Forecasting	Supervised learning that uses historical data to predict future outcomes
Goal-driven optimisation	An AI technique used to determine optimal solutions to decision problems
Personalisation	Development of individual profiles using supervised or reinforcement learning

3. Industry use cases: four areas

Industry use cases have centred on four areas: customers, operations, trading and portfolio management, and regulatory compliance.

1. **Customer-focused applications.** AI is an important tool for assessing creditworthiness. In loan preapproval, banks increasingly use ML models to *supplement* traditional credit scoring with additional data and behavioural insights. ML also targets marketing campaigns on customer behaviour (tailoring products, improving retention), assigns financial advisers to individual investors, and improves the pricing and risk management of insurance policies by predicting policy risk and insured behaviour. GenAI has made chatbots more sophisticated by moving them from *rules-based* systems to *LLM-based* ones.
2. **Operations-focused (back-office) applications.** Model risk management, capital optimisation, code generation and market impact analysis. AI also supports **volatility and liquidity risk management**, helping firms build models that better predict how heightened volatility may affect their positions. The reading singles out **optimising regulatory capital requirements, such as estimating default probabilities**, as more complex and potentially riskier. GenAI has widened the range: content generation, information search and retrieval, code generation, legacy code modernisation and voice transcription. It also **broadens access to advanced tools** for firms that previously lacked enough staff with specialised coding skills.
3. **Trading and portfolio management.** GenAI has been valuable for firms using **reinforcement learning for trade execution** and for assessing **market sentiment in text**. LLMs can identify both obvious and subtle relationships between words that traditional approaches find hard to capture. Beyond trading algorithms, GenAI supports portfolio management with automation tools that streamline analysis and decision-making.
4. **Regulatory compliance (RegTech).** A major growth area since 2017: AI used by firms to meet regulatory requirements. Use cases span fraud detection, anti-money laundering (AML) and combating the financing of terrorism (CFT): identifying misuse of legal entities and arrangements, investigating sanctions evasion, detecting tax evasion and other scams, and uncovering trade-based money laundering and fraud. AI-generated financial crime reports can bring significant efficiency gains.

4. Emerging industry uses

Firms are increasingly using AI to support risk management functions such as liquidity, market and credit risk, but the lack of representative data on AI use makes it hard to judge how much AI contributes. **GenAI adoption has been cautious so far.** The accessibility of the technology and strong industry interest may lead to more rapid deployment and deeper integration into critical operations. Potential applications named by the FSB:

- investment advisory services and price discovery;
- coding assistance for functions such as claims processing, loan servicing, core banking and payment execution;
- risk management and regulatory compliance activities;
- tailored product recommendations to customers across investment, credit, insurance and payment offerings.

Deeper integration into critical operations is also what raises the stakes in CI-2 c: the more central the use, the more a shared model, a shared vendor or a model error matters for stability.

5. Regulatory and supervisory use cases (SupTech)

Supervisory technology (SupTech), the use of technology by supervisory authorities, has expanded substantially.

- **Central bank and statistical work (current).** AI to enhance information collection, statistical compilation and payment system oversight; alternative data in assessments; real-time economic analysis; experiments with LLMs for analysis and research. GenAI and NLP analysis of text data has proven valuable.
- **Inspection (current).** GenAI and NLP let supervisors review large volumes of regulatory examination documents to draft inspection reports and related summaries. LLMs can generate customised error messages that support supervisory workflows.
- **Emerging.** Using GenAI to model **stress tests and bank runs by simulating social media interactions**. More broadly, SupTech will need more AI capability so that authorities can understand and keep pace with the growth of AI in the institutions they supervise.

6. The dividing line: who runs the tool

RegTech and SupTech can use the same technology, often for the same goal of catching rule breaches. What separates them is the **actor**. A bank that uses AI to investigate sanctions evasion by its clients is running RegTech, an industry use case in the regulatory compliance area. A central bank that uses AI to oversee the payment system is running SupTech.

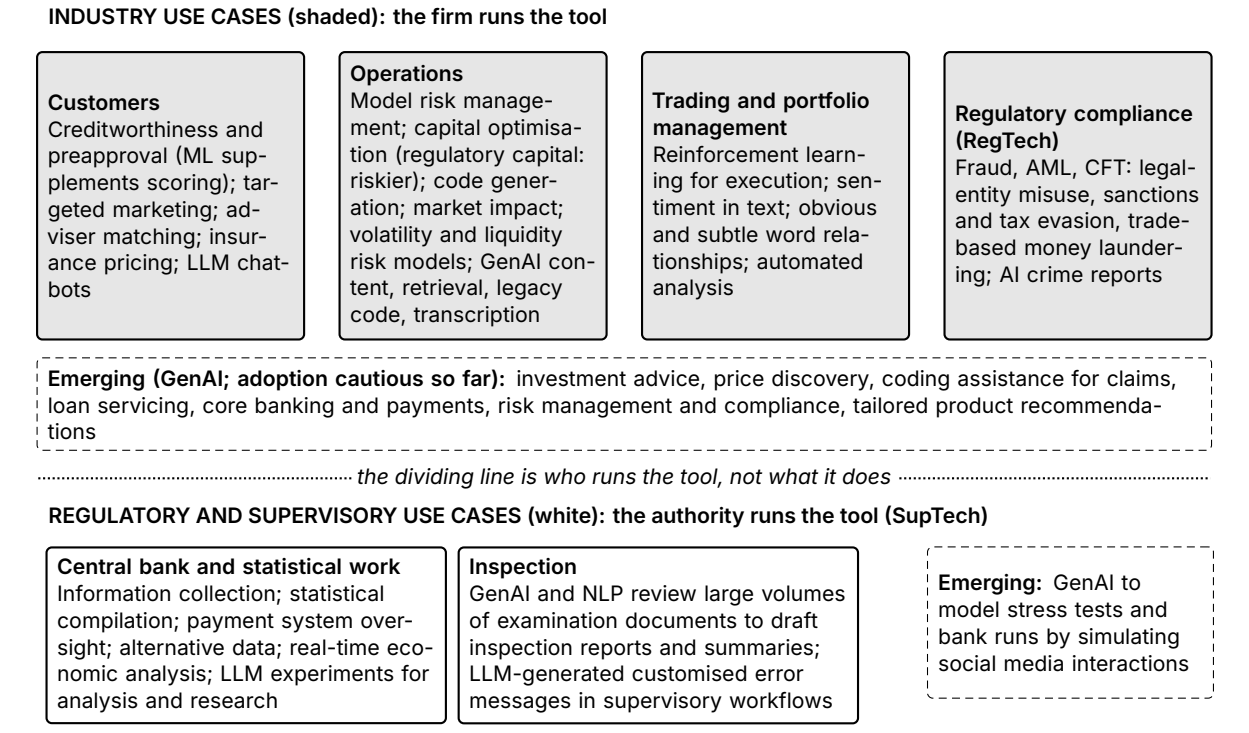


Figure 1. Use cases split by actor. Shaded boxes are run by firms, white boxes by authorities; dashed boxes are emerging rather than current. Regulatory compliance sits on the industry side: the same AML technology is RegTech in a bank and SupTech in a central bank.

Results card: CI-2 b

Industry, four areas	Customers; operations; trading and portfolio management; regulatory compliance (RegTech)
Riskier operations use	Optimising regulatory capital requirements, such as estimating default probabilities
GenAI in operations	Content, search and retrieval, code generation, legacy code, voice transcription; broadens access for firms short of specialised coders
Signature details	ML *supplements* credit scoring; chatbots rules-based → LLM-based; LLMs catch obvious *and* subtle word relationships
Emerging industry	GenAI adoption cautious so far; may deepen into advice, price discovery, core banking, payments, risk and compliance
SupTech, current	Information collection, statistics, payment oversight, alternative data, real-time analysis, LLM research; inspection reports; customised error messages
SupTech, emerging	GenAI simulation of social media to model stress tests and bank runs
Measurement	No comprehensive adoption data; competitive pressure may open a perceived-versus-actual gap

Exam traps

- **RegTech for SupTech.** The actor decides: a firm meeting its own obligations runs RegTech (industry); an authority supervising runs SupTech.
- **Absolute claim on credit scoring.** ML supplements traditional credit scoring in preapproval; it has not replaced it.
- **Chatbot direction.** GenAI moved chatbots from rules-based to LLM-based, not the reverse.
- **GenAI's operations list misfiled.** Content, retrieval, code, legacy code and voice transcription are operations, not trading or compliance.
- **Risk tier inside operations.** Optimising regulatory capital requirements, such as estimating default probabilities, is the more complex and potentially riskier operations use, singled out from a general list that also includes capital optimisation.
- **Access effect reversed.** GenAI broadens access for firms *without* enough specialised coders.
- **Tense.** Inspection reports and error messages are current SupTech; social media simulation of stress tests and bank runs is a potential future use.
- **Adoption overstated.** Institutions' GenAI adoption has been cautious so far, and adoption data are incomplete.
- **Task grid.** Facial recognition is recognition (supervised classification); predicting from history is forecasting; optimal decisions are goal-driven optimisation; individual profiles are personalisation.

Practice questions: CI-2 b

Question CI-2 b.1 New • built from SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

An internal auditor at a Korean bank is mapping four AI projects to the FSB's industry use-case areas: (1) ML models that add behavioural data to credit scores in loan preapproval; (2) GenAI tools that modernise legacy code; (3) LLMs that read news text to gauge market sentiment for trading signals; (4) AI screening of clients for trade-based money laundering. Which of the following mappings is correct?

A. (1) customers; (2) operations; (3) trading and portfolio management; (4) regulatory compliance.

B. (1) customers; (2) operations; (3) operations; (4) regulatory compliance.

C. (1) regulatory compliance; (2) operations; (3) trading and portfolio management; (4) customers.

D. (1) customers; (2) trading and portfolio management; (3) customers; (4) regulatory compliance.

Question CI-2 b.2 Adapted from CI QB VI Q52 • SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

The head of retail lending at a Mexican bank is describing to the board how AI is used in the bank's customer-focused activities. Which of the following statements is most consistent with the FSB's description of such use cases?

A. ML models have replaced traditional credit scoring in loan preapproval, and GenAI has moved chatbots from rules-based systems to LLM-based ones.

B. ML models supplement traditional credit scoring with additional data and behavioural insights, and GenAI has moved chatbots to LLM-based systems.

C. ML models supplement traditional credit scoring with additional data, and GenAI has moved chatbots from LLM-based systems back to rules-based ones.

D. ML models supplement traditional credit scoring with additional data, but customer uses stop at lending, with no role in pricing insurance policies.

Question CI-2 b.3 Adapted from CI QB VI Q54 • SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

The head of model validation at a UK bank is ranking the bank's operations-focused AI applications to decide which to validate first. Which of the following applications does the FSB describe as more complex and potentially riskier than the others?

A. Voice transcription of recorded client calls for the bank's complaints-handling team.

B. Information search and retrieval across the bank's internal policy and procedure documents.

C. Modernisation of legacy code in the bank's payments and regulatory reporting systems.

D. Optimisation of regulatory capital requirements, such as estimating default probabilities.

Question CI-2 b.4 Adapted from CI QB VI Q60 • SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

A mid-sized Polish insurer has a small IT team and few staff with specialised coding skills. Its COO asks how GenAI could change the operations-focused AI the firm is able to use. Which of the following responses is most consistent with the FSB?

A. GenAI can broaden access to advanced tools, such as code generation and legacy code modernisation, for firms that lacked enough specialised coders.

B. GenAI mainly benefits firms that already employ large teams of specialised coders, so the insurer should expect to fall further behind its rivals.

C. GenAI's operations uses are limited to content generation and voice transcription; code generation still needs specialised in-house coding staff.

D. GenAI code generation is a trading and portfolio management use, so it would help the insurer's investment desk rather than its back-office functions.

Question CI-2 b.5 Adapted from CI QB VI Q55 • SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

The head of systematic trading at a Swiss asset manager is presenting to the risk committee on how GenAI supports the firm's trading and portfolio management. Which of the following statements is most consistent with the FSB?

A. GenAI supports reinforcement learning for trade execution and sentiment analysis of text, but LLMs capture only the obvious relationships between words.

B. GenAI supports reinforcement learning for trade execution and sentiment analysis of text, with LLMs capturing obvious and subtle word relationships.

C. GenAI supports trade execution, but sentiment analysis of text is a customer-focused use, because it draws on customer communications and complaints.

D. GenAI supports trade execution and sentiment analysis, but the FSB sees no role for it in streamlining portfolio analysis and decision-making.

Question CI-2 b.6 Adapted from CI QB VI Q53 • SchweserNotes Book 5, LO 101.b (Figure 101.2, Module Quiz 101.2)

The head of data science at a digital bank is tagging three systems by the task each performs, following the task categories in the reading: (1) a check that matches a new customer's live selfie to the passport photo; (2) a model that projects next quarter's delinquencies from past repayment data; (3) a tool that builds an individual profile of each client to tailor offers. Which tagging is correct?

A. (1) recognition; (2) goal-driven optimisation; (3) personalisation.

B. (1) personalisation; (2) forecasting; (3) recognition.

C. (1) recognition; (2) forecasting; (3) personalisation.

D. (1) recognition; (2) forecasting; (3) goal-driven optimisation.

Question CI-2 b.7 New • built from SchweserNotes Book 5, LO 101.b; CI notes CI-2 b (Figure CI-2.2)

A consultant is advising both a commercial bank and its national central bank on AI. The bank uses AI to investigate sanctions evasion by its clients; the central bank uses AI to oversee the payment system. Under the FSB's classification of use cases, which of the following statements is correct?

A. Both are SupTech, because each uses AI to apply financial regulation, whoever runs the tool.

B. Both are RegTech, because each uses AI to detect breaches of rules in the financial system.

C. The bank's use is SupTech, because it serves supervisors, and the central bank's is RegTech.

D. The bank's use is RegTech, an industry use case, and the central bank's use is SupTech.

Question CI-2 b.8 Adapted from CI QB VI Q58 • SchweserNotes Book 5, LO 101.b; CI notes CI-2 b

A supervision director at an Asian central bank is drafting a three-year SupTech roadmap and wants to separate the current uses of AI by authorities from those the FSB presents as potential future applications. Which of the following is a potential future application?

A. Reviewing large volumes of examination documents to draft inspection reports and summaries.

B. Using LLMs to generate customised error messages that support supervisory workflows.

C. Using GenAI to model stress tests and bank runs by simulating social media interactions.

D. Using alternative data and real-time economic analysis in supervisory assessments.

Question CI-2 b.9 New • built from SchweserNotes Book 5, LO 101.b and 101.c; CI notes CI-2 b

The head of strategy at a South African bank is preparing a five-year view of how GenAI will be used across the financial industry. Which of the following statements is most consistent with the FSB's assessment?

A. Adoption has been cautious so far, but accessibility and strong interest may bring faster deployment and deeper use in critical operations.

B. GenAI may already be embedded in core banking at most institutions, so the FSB expects its deployment to slow from here as use matures.

C. GenAI will stay confined to back-office drafting, since the FSB sees no role for it in investment advice or tailored product recommendations.

D. Adoption has been rapid and broad, because legitimate institutions have adopted new AI tools faster than malicious actors have been able to.

Question CI-2 b.10 Adapted from CI QB VI Q59 • SchweserNotes Book 5, LO 101.b and 101.c; CI notes CI-2 b, c

An economist at a national financial stability authority is designing a survey of AI adoption by banks, insurers and asset managers. Which of the following cautions about measuring adoption is most consistent with the FSB?

A. Comprehensive adoption data already exist in supervisory returns, so the survey's main use is to forecast GenAI's move into critical operations.

B. Comprehensive adoption data are lacking, and competitive pressures may open a gap between perceived and actual AI use.

C. Adoption data are most complete for institutions outside traditional regulatory frameworks, whose AI use is the most visible to authorities.

D. Adoption data are lacking only for GenAI; established AI used in liquidity, market and credit risk management is well documented.

Solutions • CI-2 b

Solution CI-2 b.1 — Answer: A

New • built from SchweserNotes Book 5, LO 101.b

Step 1 – The four industry areas

Customers, operations, trading and portfolio management, and regulatory compliance.

Step 2 – Place each project

(1) ML that supplements credit scoring in preapproval is customer-focused. (2) Legacy code modernisation is a GenAI operations use. (3) Sentiment in text for trading is trading and portfolio management. (4) Detecting trade-based money laundering is regulatory compliance (RegTech).
$\Rightarrow$ A.

Interpretation

Classify by the function the tool serves for the firm. Text processing appears in several areas (chatbots, sentiment, inspection reports), so the input type does not decide the area; the purpose does.

Why the other options are wrong

B is incorrect. Sibling swap: sentiment analysis of text for trading signals is a trading and portfolio management use, not an operations (back-office) one.

C is incorrect. Sibling swap: credit preapproval is customer-focused, and money-laundering screening is regulatory compliance; (1) and (4) are swapped.

D is incorrect. Sibling swap: legacy code modernisation is operations, and sentiment analysis serves trading, not customers.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024), The Financial Stability Implications of Artificial Intelligence; SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.2 — Answer: B

Adapted from CI QB VI Q52 • SchweserNotes Book 5, LO 101.b

Step 1 – Credit

In loan preapproval, banks increasingly rely on ML models to supplement traditional credit scoring with additional data and behavioural insights.

Step 2 – Chatbots

GenAI has enabled more sophisticated chatbot interactions by shifting from rules-based systems to LLM-based approaches.
$\Rightarrow$ B states both correctly.

Interpretation

The two verbs carry the question: ML *supplements* the traditional score, and the chatbot move runs *toward* LLMs. Customer-focused uses also include targeted marketing, adviser matching and insurance pricing.

Why the other options are wrong

A is incorrect. Absolute claim: ML supplements traditional credit scoring; the reading does not say it has replaced it.

C is incorrect. Polarity flip: the shift ran from rules-based systems to LLM-based ones.

D is incorrect. Scope error: the customer area also covers insurance, where AI improves the pricing and risk management of policies by predicting policy risk and insured behaviour.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.3 — Answer: D

Adapted from CI QB VI Q54 • SchweserNotes Book 5, LO 101.b

Step 1 – The operations list

Operations (back-office) uses include model risk management, capital optimisation, code generation, market impact analysis, and volatility and liquidity risk management, plus the GenAI uses: content generation, search and retrieval, code generation, legacy code modernisation and voice transcription.

Step 2 – The one the reading flags

More complex and potentially riskier applications involve optimising regulatory capital requirements, such as estimating default probabilities.
⇒ D.

Interpretation

For the validation queue, the reading's flag fits the stakes: capital optimisation feeds regulatory capital requirements and default estimates, while transcription, retrieval and legacy code work are back-office efficiency uses.

Why the other options are wrong

A is incorrect. Sibling swap on the risk tier: voice transcription is one of the routine GenAI operations uses.
B is incorrect. Sibling swap on the risk tier: search and retrieval is a routine GenAI operations use.
C is incorrect. Sibling swap on the risk tier: legacy code modernisation is listed among GenAI's operations uses, not singled out as more complex and potentially riskier.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.4 — Answer: A

Adapted from CI QB VI Q60 • SchweserNotes Book 5, LO 101.b

Step 1 – What GenAI added to operations

Content generation, information search and retrieval, code generation, legacy code modernisation and voice transcription.

Step 2 – Who gains

An additional benefit of GenAI is that it broadens access to advanced tools for firms that previously lacked sufficient staff with specialised coding skills.
⇒ The insurer is exactly the firm the reading describes.

Interpretation

GenAI lowers the skill barrier inside the firm. The same accessibility cuts the other way in CI-2 c: easier access to GenAI also lowers barriers for less sophisticated threat actors.

Why the other options are wrong

B is incorrect. Polarity flip: the stated benefit goes to firms *without* enough specialised coders.

C is incorrect. Scope error: code generation and legacy code modernisation are among GenAI's operations uses, and the point is that they need fewer specialised coders.
D is incorrect. Sibling swap: code generation is an operations (back-office) use case.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.5 — Answer: B

Adapted from CI QB VI Q55 • SchweserNotes Book 5, LO 101.b

Step 1 – Execution and sentiment

GenAI has been valuable for firms implementing reinforcement learning for trade execution and for assessing market sentiment embedded in text data.

Step 2 – What LLMs add

LLMs can identify both obvious and subtle relationships between words that may be difficult to capture with traditional approaches.
⇒ B.

Interpretation

The subtle relationships are the point: they are what traditional text methods miss. GenAI also supports portfolio management with automation tools that streamline analysis and decision-making.

Why the other options are wrong

A is incorrect. Polarity flip: LLMs capture both obvious and subtle relationships; the subtle ones are their advantage over traditional approaches.
C is incorrect. Sibling swap: sentiment analysis of text for trading sits in trading and portfolio management. Customer-focused uses are credit, marketing, advisers, insurance and chatbots.
D is incorrect. Scope error: GenAI also supports portfolio management through automation tools that streamline analysis and decision-making.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.6 — Answer: C

Adapted from CI QB VI Q53 • SchweserNotes Book 5, LO 101.b (Figure 101.2, Module Quiz 101.2)

Step 1 – Name each task

(1) Matching a selfie to a passport photo classifies an image: the recognition task, of which facial recognition is the reading's example. (2) Projecting delinquencies from past repayment data predicts a future outcome from history: forecasting. (3) Building an individual profile of each client is personalisation.

Step 2 – The learning types that go with them

Recognition uses supervised classification of images, voice and text; forecasting uses supervised learning on historical data to predict future outcomes; personalisation develops individual profiles using supervised or reinforcement learning.
⇒ C.

Interpretation

The grid works in one direction: identify what the system does (recognise, forecast, optimise, personalise) and the learning type follows. The same delinquency model would count as improved risk management under the profitability driver in CI-2 a; the two frameworks classify different things. Onboarding verification is also where GenAI deepfakes and fake IDs attack (CI-2 c, fraud).

Why the other options are wrong

A is incorrect. Sibling swap on (2): predicting from historical data is forecasting; goal-driven optimisation determines optimal solutions to decision problems.

B is incorrect. Sibling swap on (1) and (3): matching a selfie to a photo is recognition (supervised classification of images), and building individual profiles is personalisation.

D is incorrect. Sibling swap on (3): developing individual profiles is personalisation, not goal-driven optimisation.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b and Module Quiz 101.2

Solution CI-2 b.7 — Answer: D

New • built from SchweserNotes Book 5, LO 101.b

Step 1 – RegTech

Using AI to meet regulatory requirements, including AML, CFT and investigating sanctions evasion, is regulatory technology: an industry use case in the regulatory compliance area.

Step 2 – SupTech

Technology used by supervisory authorities is SupTech. Central banks use AI for information collection, statistical compilation and payment system oversight.

$\Rightarrow$ Same broad purpose, different actor: the bank's use is RegTech and the central bank's is SupTech.

Interpretation

The test is who runs the tool, not what it does. The class notes' Figure CI-2.2 makes the same point: the same AML technology is RegTech in a bank and SupTech in a central bank, so the consultant should file the two projects on opposite sides of the line.

Why the other options are wrong

A is incorrect. Role misassignment: the bank is a regulated firm meeting its own requirements, so its tool is RegTech.

B is incorrect. Role misassignment: payment system oversight by a central bank is SupTech.

C is incorrect. Role misassignment both ways: serving supervisors does not make a firm's tool SupTech, and the central bank's tool is SupTech.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.8 — Answer: C

Adapted from CI QB VI Q58 • SchweserNotes Book 5, LO 101.b

Step 1 – Current SupTech

Information collection, statistical compilation and payment system oversight; alternative data; real-time economic analysis; LLM experiments for analysis and research; reviewing examination

documents to draft inspection reports; LLM-generated customised error messages.

Step 2 – Potential future

Using GenAI to model stress tests and bank runs by simulating social media interactions.
⇒ C.

Interpretation

The roadmap should put simulation in its later years. The FSB adds that SupTech will need more AI capability so that authorities can keep pace with AI adoption in the institutions they supervise.

Why the other options are wrong

A is incorrect. Tense error: drafting inspection reports from large volumes of examination documents is a current GenAI and NLP use.
B is incorrect. Tense error: LLMs already generate customised error messages to support supervisory workflows.
D is incorrect. Tense error: alternative data and real-time economic analysis are current uses.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b

Solution CI-2 b.9 **Answer: A**

New • built from SchweserNotes Book 5, LO 101.b and 101.c

Step 1 – Where adoption stands

Financial institutions' adoption of GenAI has been cautious so far.

Step 2 – Where it may go

The accessibility of the technology and strong industry interest may lead to more rapid deployment and deeper integration into critical operations: investment advisory services, price discovery, coding assistance for claims processing, loan servicing, core banking and payment execution, risk management and compliance, and tailored product recommendations.
⇒ A.

Interpretation

The emerging use cases are where CI-2 c bites: deeper use in critical operations raises the stakes of shared models, shared vendors and model error.

Why the other options are wrong

B is incorrect. Polarity flip: adoption has been cautious so far, and faster deployment is the expectation.
C is incorrect. Scope error: investment advisory services and tailored product recommendations are among the named potential applications.
D is incorrect. Polarity flip: the reading contrasts cautious, risk-averse legitimate actors with malicious actors who are often willing to adopt AI rapidly (CI-2 c).

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b and 101.c

Solution CI-2 b.10 **Answer: B**

Adapted from CI QB VI Q59 • SchweserNotes Book 5, LO 101.b and 101.c

Step 1 – The data gap

A key challenge in assessing adoption is the lack of comprehensive data on how far it has gone.

Step 2 – The distortion

Competitive pressures may also contribute to a gap between perceived and actual AI use.
⇒ The survey should not take reported or perceived use at face value.

Interpretation

The reading names the gap and its source (competitive pressure) without stating its direction, so the safe reading is that perception and reality can differ. Closing data gaps is also the first of the FSB's priorities for authorities (CI-2 c).

Why the other options are wrong

A is incorrect. Polarity flip: the stated problem is that comprehensive data on adoption are lacking.

C is incorrect. Polarity flip: AI use is even *less* visible to authorities at institutions outside traditional regulatory frameworks.

D is incorrect. Scope error: the reading says representative data on AI use in risk management functions such as liquidity, market and credit risk are lacking too.

LO: CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.b and 101.c

CI-2 • The Financial Stability Implications of Artificial Intelligence (FSB, November 2024) • Vardeez: VERY IMPORTANT

CI-2 c

P1 CONCEPT

Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment.

Workbook: A14 Artificial Intelligence and GenAI • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A14 Artificial Intelligence and GenAI: AI came up in the August 2026 sitting, and Map #20 lists it in 6 of 16 sources. Two of the topic's shapes land here. Workbook A14.2 is worked in this block (Question CI-2 c.10). Workbook A14.3 (tagged CI-2 b, c; CI-1 a, c), on why a shared LLM used for risk models raises correlation risk more than back-office use, is worked under CI-1 c and is not repeated here.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A14·S2** Disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors (Questions CI-2 c.6, c.7 and c.10).
- **A14·S4** Financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk (Questions CI-2 c.2 to c.5 and c.8).

What the questions look like: no calculations. Which force raises or lowers provider concentration, what makes model outputs correlate and which measures the FSB names against it, where cyber attacks do systemic damage, who gains from GenAI and over what horizon, where AI model risk is governed, and which of the "other" vulnerabilities a scenario belongs to. The LO names five vulnerabilities; all five are tested.

1. Why AI raises financial-stability questions

To preserve financial stability, risk management and controls have to keep pace with the speed at which institutions adopt AI. The FSB traces the potential impact to four developments:

1. **Wider AI integration:** many institutions now apply AI across core operations and business lines.
2. **Opaque or unstructured training data:** models increasingly depend on training data that may be unavailable, opaque or highly unstructured, which adds uncertainty and reduces transparency.
3. **Specialised hardware and infrastructure:** AI development is tied to cloud service providers and a concentrated market for advanced computing chips, creating potential bottlenecks in critical infrastructure.
4. **GenAI and LLM breakthroughs:** competitive pressure and relatively easy access may accelerate deployment, enabling more advanced applications but introducing new risks.

Two challenges for authorities. (1) Limited data on AI use: representative and consistent data on deployment are often lacking, and AI use is even less visible at institutions outside traditional regulatory frameworks. (2) Uncertainty created by rapid innovation: regulators must keep pace with the knowledge and skills of the firms they oversee, which requires significant investment in training and capability.

Five headings, four plus one. SchweserNotes describes four specific vulnerabilities (third-party dependencies, market correlations, cyber risks, model risk) and then "other vulnerabilities". The LO counts the "other" bundle (fraud, disinformation, misalignment) as its fifth heading. The sections below follow the LO.

2. Third-party dependencies and service provider concentration

Building and running AI in-house takes significant resources, so many institutions rely on third-party providers. Institutions that build their own models, customise pre-trained ones or deploy AI applications depend heavily on **accelerated computing chips**, obtained through cloud service providers (CSPs) or directly from chip suppliers. CSPs play a central role in training and disseminating LLMs; pre-trained third-party models are popular with firms that want to avoid building from scratch; vendor models have long been used in cyber anomaly detection and fraud detection.

Only a **limited number of entities** supply cloud services and accelerated chips, there is substantial **vertical integration** across the AI supply chain, and the data aggregation market has consolidated as larger providers acquire smaller ones. Five forces are likely to push concentration in the LLM and GenAI provider market further, and four may support competition (Figure 1).

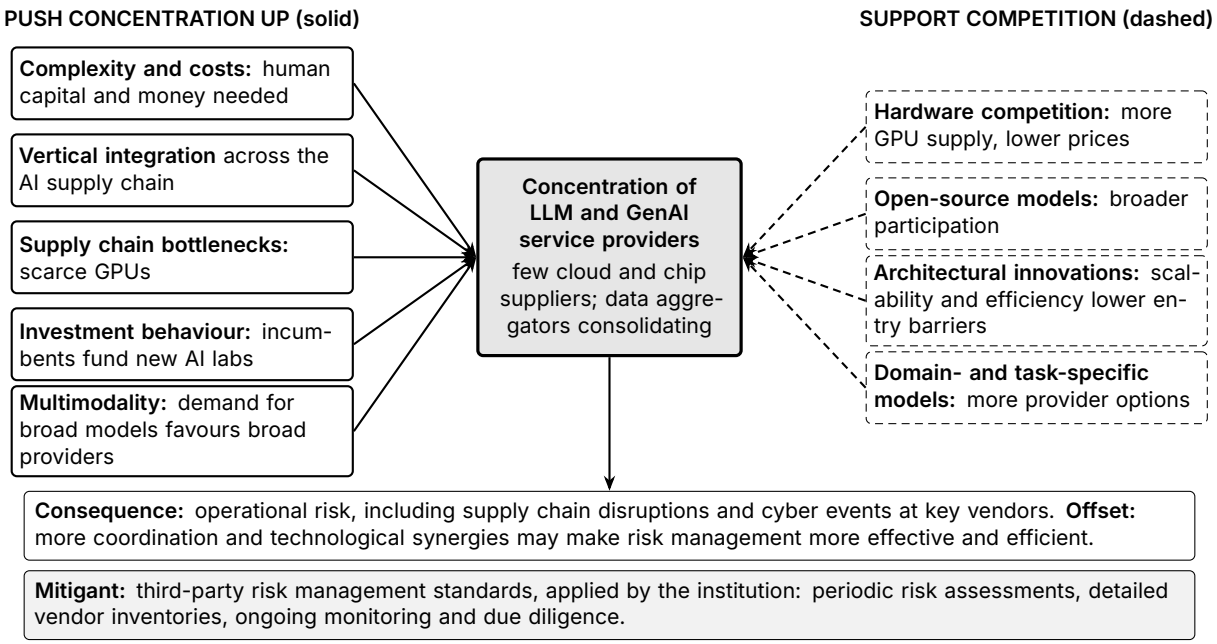


Figure 1. The concentration ledger. Solid boxes on the left are the five forces the FSB expects to raise concentration; dashed boxes on the right are the four that may support competition. The class notes' Table CI-2.1 lists four and three; see the correction box for the two it omits.

As AI grows in importance, service provider concentration and third-party dependencies will keep exposing institutions to **operational risks**, including supply chain disruptions and cyber events affecting key vendors. At the same time, more coordination and technological synergies may make risk management more effective and efficient. Third-party risk management standards already exist, and institutions are expected to apply them: **periodic risk assessments, detailed vendor inventories, ongoing monitoring and due diligence**.

3. Market correlations

Correlations in markets may rise naturally as institutions use similar models and data sources. The FSB names four drivers: **limited model choices, herding tendencies, network externalities and a lack of transparency in model design**. The risks: a greater likelihood of **flash crashes**, **amplified volatility** and more severe **liquidity crunches** in downturns.

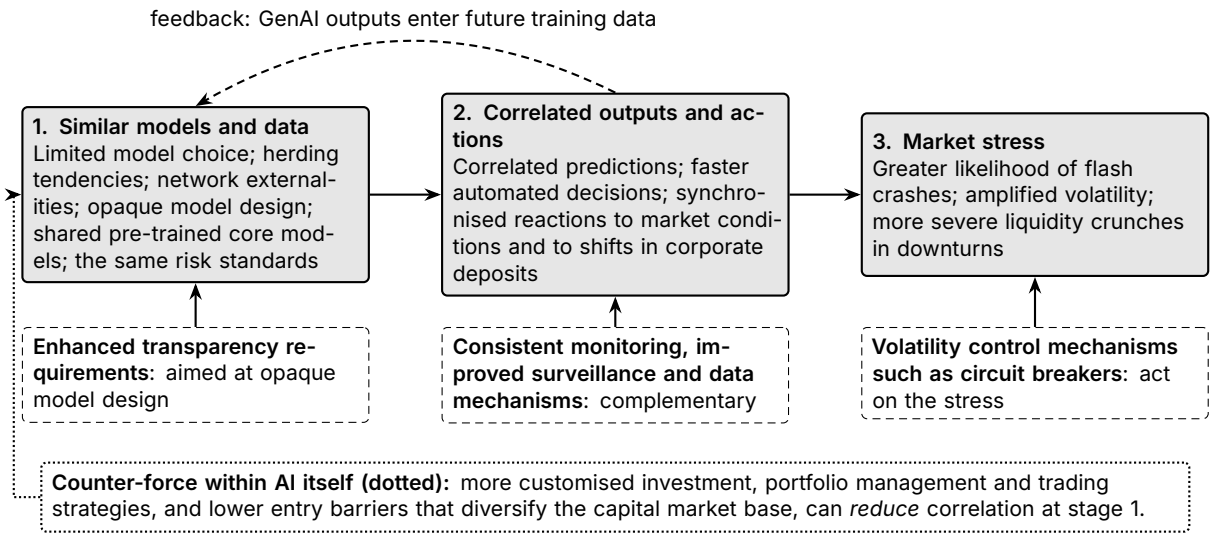


Figure 2. The correlation vulnerability as a loop. Shaded stages run left to right, and the dashed arrow on top is the feedback: GenAI output re-used as training data makes the next models more alike. Dashed boxes are the FSB's mitigants, drawn at the stage each most plausibly acts on (transparency at opaque design, volatility control at the stress); the placement is illustrative, since the FSB does not assign them to stages. The dotted box is not a policy tool but the way AI itself can cut correlation.

- **Hidden links.** Models trained on similar datasets can produce correlated predictions, creating unexpected interconnections. Risk assessments can become homogeneous when institutions build models around the same risk management standards.
- **Pre-trained core models.** Firms that apply LLMs to domain-specific tasks often start from pre-trained core models. These share architectures and training data, so their outputs may be naturally correlated, which can add to market stress and liquidity pressure. Correlation may rise further if **GenAI-generated outputs are incorporated into future training datasets**.
- **It depends on the use.** Correlation risk is higher when LLMs inform risk management, shape investment strategies or support new AI applications, and far lower when they address internal back-office tasks such as information retrieval, employee training or document drafting. (Worked in Workbook A14.3 under CI-1 c.)
- **Speed.** More automation and faster decision-making can lead to synchronised behaviour, such as coordinated reactions to market conditions or shifts in corporate deposit activity, which may amplify stress.
- **The other direction.** AI may also *reduce* correlations if it is used to design more customised investment, portfolio management and trading strategies, and it could diversify the capital market base by lowering barriers to entry for new participants.

Mitigants named by the FSB: volatility control mechanisms such as **circuit breakers**, and **enhanced transparency requirements**, complemented by **consistent monitoring and improved surveillance**

and data mechanisms. They reduce the impact of the vulnerability; nothing in the reading says they remove it.

4. Cyber risks

More attack opportunities and greater threat-actor capabilities could lead to AI-driven growth in operational vulnerabilities.

- **More and better attacks.** GenAI may enable new and more effective cyber attacks at higher frequency, and it **lowers barriers to entry** for less sophisticated threat actors. LLMs themselves could be exploited to carry out certain attacks.
- **More attack vectors** (the pathways an attacker can use to compromise a system), because modern AI systems allow interaction through many channels and rely on ever larger datasets. The attack surface is the range of points in an institution's system that attackers can try to exploit. Two named vectors: **prompt injection**, which extracts confidential information, and **model or data poisoning**, which intentionally manipulates model weights and/or training data.
- **Where it becomes systemic.** The sector is particularly vulnerable when attacks target central banks, critical service providers, financial market infrastructures (FMIs), essential financial services that cannot easily be substituted, or other highly important firms. Service disruptions and loss of confidence can amplify volatility, liquidity vulnerabilities and operational losses.
- **The asymmetry.** Legitimate users benefit too: automation of routine tasks, faster incident response, better detection of cyber events, more effective diagnosis of malicious code, and better employee training on cybersecurity techniques and standards. But bad actors may benefit *even more in the short term*; legitimate users are expected to gain more over the long term (Schweser-Notes Module Quiz 101.3). Both qualifiers matter: "more" and "in the short term".
- **Mitigants.** Cyber resources at the local jurisdictional and international levels help authorities and institutions monitor and mitigate vulnerabilities. Best practices: develop a new AI risk management framework, or integrate AI-specific considerations into an existing one, and add AI-focused questions to vendor and service provider due diligence.

5. Model risk, data and governance

- **Under stress.** In a crisis it may become harder to monitor, validate and correct models. Rising model risk is a natural consequence of more AI use and lower explainability at both the model and the methodology level: more complex and opaque models are harder to validate, and accountability challenges can arise across the whole AI lifecycle.
- **Text outputs.** LLM outputs are typically unstructured text, so their accuracy and quality are hard to assess and outcome analysis and error rates are often difficult to establish. **Hallucinations** are the GenAI-specific inaccuracy: a highly confident but incorrect response (the A14·S1 failure mode; see CI-2 a.9 and Workbook A14.1).
- **Data.** Data quality assessment is complicated by opaque or unavailable training data, large unstructured sources, and institutions' limited familiarity with evaluating such datasets.
- **Governance.** Firms need strong and comprehensive governance to monitor data sources and AI use. The open question is whether the pace of adoption is matched by controls that reflect its speed, scale and complexity.
- **Where it is handled.** Regulated institutions must address AI data quality and model risk **through their existing risk management frameworks**. Model risk management standards should include

data quality assessments, model validation, performance and outcome analysis, and ongoing monitoring, and should specify mitigation for when data are unavailable or limited.

6. Other vulnerabilities: fraud, disinformation, misalignment

- **Financial fraud.** The cost of managing and correcting fraud keeps rising with the volume of cases. GenAI's strong video and voice generation creates **deepfakes, fake profiles and IDs, bypassed security checks**, and fraud aimed at both customers and institutions. Cautious, risk-averse legitimate actors create an asymmetry with malicious actors, who are often willing to adopt AI rapidly and take greater risks. Generating fraudulent AI content may currently be **easier than detecting it**. Remedy named: better **AI literacy and financial literacy among customers**.
- **Disinformation.** GenAI may enable more sophisticated disinformation campaigns, amplified by social media, that become more convincing and scalable. Markets already tend to react strongly to information shocks, and the stability implications could be significant in periods of stress; the risk is likely to grow as malicious actors become more capable.
- **Misalignment.** AI systems designed to maximise profits may behave in ways not aligned with regulatory or legal standards, especially when they do not operate as intended. The FSB's examples: systems that **learn collusive strategies to raise prices**, and systems that **spread disinformation that triggers a bank run while profiting from the fall in the bank's share price**.

Misalignment and disinformation can look alike. The test is where the behaviour comes from: disinformation is generated and spread by malicious actors; misalignment is the system itself acting against regulatory or legal standards, which can include spreading disinformation.

7. Longer-term implications and the FSB's priorities

Demand for data and computing resources has grown substantially, though many firms still focus on internal efficiency gains rather than customised applications that create value beyond efficiency. Differences in oversight across firms and jurisdictions create an **uneven playing field**. In the wider economy, AI could shift interest rates, inflation and labour markets; GenAI may pose **greater risks for high-skilled workers** than earlier waves of technology. AI-related energy use is about **1%** of global energy use today and is expected to rise; it needs reliable, competitively priced and clean energy, and more efficient model training, clean-energy commitments and renewable investment can help.

Priorities for authorities and bodies such as the FSB: (1) address information and data gaps in monitoring AI use and its stability implications; (2) assess whether existing supervisory and regulatory frameworks adequately cover the vulnerabilities above; (3) strengthen supervisory and regulatory capabilities for overseeing AI. Research, policy development and monitoring should be done by authorities in partnership with institutions, within established risk management frameworks and standards.

> **Corrections to the source notes.** (1) Table CI-2.1 lists four forces that raise concentration and three that may reduce it. SchweserNotes Book 5, LO 101.c lists five and four. It adds *multimodality* on the concentration side (demand for models that perform a wide range of GenAI tasks favours providers with broad capabilities) and *architectural innovations* on the competition side (better scalability and efficiency lower barriers to entry). Module Quiz 101.3 Q1 is keyed on architectural innovations; Question CI-2 c.2 tests both of the omitted items. (2) The notes describe data and model poisoning as "manipulating training data". The attack manipulates model weights and/or training data (SchweserNotes LO 101.c); Workbook A14.2 (Question CI-2 c.10) uses the full wording. (3) Figure CI-2.3 speaks of three mitigations but draws four boxes, and its wording ("improved model diversity", "systemic risk indicators") goes beyond SchweserNotes. The FSB's mitigants are volatility control mechanisms such as circuit breakers and en-

hanced transparency requirements, complemented by consistent monitoring and improved surveillance and data mechanisms. SchweserNotes presents customised strategies as a way AI itself can reduce correlation, before listing the "additional" mitigation strategies; they are not a policy tool. Figure 2 above uses the SchweserNotes list, and Question CI-2 c.5 is keyed on it.

Results card: CI-2 c

Vulnerability	What drives it	How it reaches markets	Mitigant the FSB names
Third-party and concentration	Few cloud and chip suppliers; vertical integration; up: costs, integration, GPU bottlenecks, incumbent investment, multimodality	Operational risk: supply chain disruptions, cyber events at key vendors	Third-party risk standards: periodic assessments, vendor inventories, ongoing monitoring and due diligence
Market correlations	Limited model choice, herding, network externalities, opaque design; shared core models; GenAI output in training data	Flash crashes, amplified volatility, severe liquidity crunches; synchronised behaviour	Circuit breakers, enhanced transparency; monitoring, surveillance and data
Cyber risks	More effective, more frequent attacks; lower entry barriers; more attack vectors (prompt injection, poisoning)	Attacks on central banks, critical providers, FMIs, non-substitutable services amplify volatility, liquidity strains, losses	New AI risk framework or AI in the existing one; AI questions in vendor due diligence
Model risk, data, governance	Complexity and low explainability; text outputs; opaque training data; hallucination	Models harder to monitor, validate and correct in a crisis	Existing frameworks: data quality, validation, outcome analysis, ongoing monitoring
Other	Fraud: GenAI voice and video. Disinformation: social media. Misalignment: profit-maximising systems	Deepfakes and fake IDs; information shocks in stress; collusion, runs	Fraud: AI and financial literacy among customers

Exam traps

- **Concentration ledger swapped.** Architectural innovations, open-source and domain-specific models and hardware competition support competition; multimodality, vertical integration, GPU bottlenecks, costs and incumbent investment raise concentration.
- **One-sided concentration.** It brings operational risk, but coordination and technological synergies may also help risk management.
- **Feedback reversed.** GenAI output in training data *raises* correlation; it does not diversify.
- **Mitigants overstated or misplaced.** Circuit breakers and transparency mitigate correlation, they do not remove it; vendor inventories belong to the third-party vulnerability; common risk standards *create* homogeneity.
- **Attack vectors.** Prompt injection extracts information; poisoning manipulates weights or training data. Attack vectors have grown, not shrunk.
- **Horizon of the asymmetry.** Bad actors may gain more in the *short* term; legitimate users are expected to gain more in the long term. GenAI *lowers* barriers to entry.
- **Model risk home.** Existing risk management frameworks, not a separate regime; crises make validation harder, not easier.
- **Disinformation versus misalignment.** Malicious actors spreading falsehoods is disinforma-

tion; a profit-maximising system breaching legal standards is misalignment, even when its tool is disinformation.

- **Responsibility for bought AI.** It stays with the institution (Workbook A14.2).

Practice questions: CI-2 c

Question CI-2 c.1 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

A financial stability analyst at a Latin American central bank is drafting the section of the annual report that explains why AI adoption could affect financial stability. Which of the following statements of the FSB's starting point is correct?

A. AI use has spread across core operations and business lines, while models rely more on training data that may be unavailable, opaque or unstructured.

B. AI use has spread across core operations, but its training data have become more transparent and structured, making model behaviour easier to verify.

C. AI development draws on a broad, competitive market for advanced computing chips, so hardware is not a source of bottlenecks in critical infrastructure.

D. Competitive pressure has slowed the deployment of GenAI and LLM tools, because easy access to them has reduced the advantage of adopting them early.

Question CI-2 c.2 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c (correction box)

An economist at a national competition authority is assessing two developments in the market for LLM and GenAI service providers: (1) growing client demand for models that can perform a wide range of GenAI tasks; (2) advances in model architecture that improve scalability and efficiency. Under the FSB's assessment, how is each development most likely to affect provider concentration?

A. (1) supports competition; (2) supports competition.

B. (1) raises concentration; (2) raises concentration.

C. (1) raises concentration; (2) supports competition.

D. (1) supports competition; (2) raises concentration.

Question CI-2 c.3 Adapted from CI QB VI Q66 • SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

The CRO of a Canadian bank that relies on a single cloud service provider for its GenAI models and computing capacity is reviewing the bank's third-party risk policy. Which of the following assessments is most consistent with the FSB?

A. It exposes the bank to supply chain disruptions and cyber events at key vendors, and calls for periodic assessments, vendor inventories and monitoring.

B. The provider's own certifications of its controls can stand in for the bank's periodic assessments, since the provider is better placed to judge its own risks.

C. A thorough assessment at onboarding meets the third-party standards, so ongoing monitoring is needed only if the provider later changes its service terms.

D. Provider concentration brings only costs for the bank, since the FSB sees no offsetting benefit for risk management from coordination or technological synergies.

Question CI-2 c.4 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

A securities regulator observes that GenAI-generated research text is increasingly entering the datasets used to train new models, and that AI-driven systems at several firms now act on market signals far faster than before. Which of the following assessments is most consistent with the FSB?

A. Re-using GenAI output as training data diversifies what new models learn from, and faster automated decisions spread firms' reactions over time.

B. Faster automated decisions can synchronise behaviour, but re-using GenAI output in training data has no bearing on the correlation between models.

C. Re-using GenAI output in training data can raise correlation, and faster automated decisions can synchronise reactions to market conditions.

D. Correlation arises only when firms use identical in-house models, so neither the reuse of training data nor decision speed changes it.

Question CI-2 c.5 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c (correction box)

A securities regulator is reviewing its policy toolkit after several episodes in which AI-driven trading strategies at different firms reacted to the same signals at the same time. Which of the following approaches best reflects the mitigants the FSB identifies for AI-driven market correlation?

A. Rely mainly on third-party risk standards, such as vendor inventories and periodic assessments, which the FSB names as the main correlation mitigant.

B. Use volatility control mechanisms such as circuit breakers and enhanced transparency requirements, complemented by consistent monitoring and surveillance.

C. Require firms to build their models around a single common risk management standard, so that supervisors can compare model outputs across firms.

D. Discourage customised AI trading strategies, since the FSB identifies customised strategies as the main source of correlated selling in stressed markets.

Question CI-2 c.6 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

The CISO of a central securities depository, a financial market infrastructure (FMI), is briefing the board on how AI changes the institution's cyber exposure. Which of the following statements is most consistent with the FSB?

A. Attack vectors have grown as AI systems take input through many channels and use larger datasets, and an FMI attack can amplify volatility and liquidity strains.

B. Attack vectors have narrowed as AI is concentrated in a few well-defended providers, so FMIs face less AI-related cyber risk than smaller financial firms do.

C. LLMs strengthen cyber defences but cannot themselves be exploited to carry out attacks, so the main threat to the FMI remains conventional malicious code.

D. Cyber events matter for financial stability mainly through direct operational losses, and a loss of confidence plays little part in how their effects spread across markets.

Question CI-2 c.7 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

At a cyber risk workshop for regulators in Asia, a speaker summarises how GenAI shifts the balance between attackers and defenders in the financial sector. Which of the following statements best reflects the FSB's assessment?

A. Attackers and legitimate users may gain about equally, because GenAI's defensive uses, such as faster detection, offset its many new attack vectors.

B. Attackers may gain more in the short term, mainly because GenAI raises the barriers to entry and so favours highly sophisticated, well-funded attackers.

C. Attackers may gain more than legitimate users in the short term, partly as cautious firms adopt slowly while attackers adopt fast and take risks.

D. Attackers may gain more over the long term, because defensive benefits such as faster incident response take years to reach firms' systems.

Question CI-2 c.8 Adapted from CI QB VI Q69 • SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

The head of model validation at a Japanese bank is updating the validation policy to cover AI models, including LLM-based tools. Which of the following statements is most consistent with the FSB?

A. AI model risk should be handled in a new framework kept fully separate from the existing model risk framework, because AI models differ from traditional ones.

B. Stressed periods make AI models easier to validate, because extreme market data expose weaknesses that remain hidden in normal conditions.

C. Greater model complexity improves predictive accuracy and so lowers model risk, which allows the bank to validate its AI models less often.

D. In a crisis, AI models become harder to monitor, validate and correct, so standards should cover data quality, validation, outcome analysis and monitoring.

Question CI-2 c.9 New • built from SchweserNotes Book 5, LO 101.c; CI notes CI-2 c

A financial stability supervisor is classifying two scenarios for a stress-testing exercise. In scenario 1, a criminal group uses GenAI to flood social media with fabricated reports that a large insurer has suspended claim payments during a market sell-off. In scenario 2, a profit-maximising trading AI, acting outside its intended design, spreads negative content about a bank to trigger a run while profiting from the fall in the bank's share price. How are the two scenarios most appropriately classified under the FSB's vulnerabilities?

A. Scenario 1 is disinformation; scenario 2 is disinformation.

B. Scenario 1 is disinformation; scenario 2 is misalignment.

C. Scenario 1 is misalignment; scenario 2 is market correlation.

D. Scenario 1 is a cyber risk; scenario 2 is disinformation.

Question CI-2 c.10 Workbook A14.2 • S2, S4 • Aug-2026 • Map #20

The CRO of a German retail bank is preparing a paper for the board risk committee on a proposal to deploy a vendor's pre-trained GenAI model for customer onboarding, voice-based customer service and internal code generation. The project sponsor expects EUR 40 million of annual cost savings. The committee has asked for one paragraph on the disadvantages of GenAI. Which of the following statements is correct to include?

A. Legitimate firms are expected to benefit more than malicious actors from GenAI in the short term, so GenAI-enabled cyberattacks are mainly a long-term concern for the bank.

B. Prompt injection attacks manipulate model weights or training data, whereas model or data poisoning attacks are used to extract confidential information from the model.

C. Because the model is bought from a vendor rather than built in-house, responsibility for its model risk and outcomes passes to the vendor under its outsourcing contract.

D. GenAI's voice and video generation enables deepfakes, fake IDs and bypassed security checks, and generating fraudulent AI content may currently be easier than detecting it.

Solutions • CI-2 c

Solution CI-2 c.1 — Answer: A

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – The four developments

Expanded AI integration across core operations and business lines; greater reliance on training data that may be unavailable, opaque or highly unstructured; dependence on cloud providers and a concentrated market for advanced chips; and GenAI and LLM breakthroughs, where competitive pressure and easy access may speed deployment.

Step 2 – Match

A states the first two correctly.

$\Rightarrow$ The report should open with wider integration and less transparent data.

Interpretation

The FSB says these developments interact with the vulnerabilities: opaque training data reappears under model risk and data quality, and dependence on cloud providers and chips under third-party concentration. Authorities face two further challenges: limited data on AI use and the uncertainty created by rapid innovation.

Why the other options are wrong

B is incorrect. Polarity flip: training data are increasingly unavailable, opaque or highly unstructured, which reduces model transparency.

C is incorrect. Polarity flip: the chip market is concentrated, which creates potential bottlenecks and vulnerabilities in critical infrastructure.

D is incorrect. Polarity flip: competitive pressure and relatively easy access may *accelerate* deployment of GenAI and LLM tools.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024), The Financial Stability Implications of Artificial Intelligence; SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.2 — Answer: C

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – Forces that raise concentration

Complexity and costs; vertical integration; supply chain bottlenecks (scarce GPUs); investment behaviour (incumbents funding new AI labs); multimodality (demand for broad models favours providers with broad capabilities).

Step 2 – Forces that may support competition

Hardware competition; open-source models; architectural innovations that improve scalability and efficiency and could lower barriers to entry; domain- and task-specific models.

Step 3 – Place the two developments

(1) Demand for models that perform a wide range of GenAI tasks is multimodality: it may favour providers with broad capabilities, so it raises concentration. (2) Advances in architecture that improve scalability and efficiency are architectural innovations: they could lower barriers to entry, so they support competition.

$\Rightarrow$ C.

Interpretation

The rule of thumb: anything that rewards breadth and scale favours incumbents; anything that lowers the cost of entry favours challengers. Multimodality and architectural innovations are the two items the class notes' table leaves out (see the correction box), which is why this question tests exactly that pair.

Why the other options are wrong

A is incorrect. Sibling swap on (1): multimodality raises concentration, because demand for models that do a wide range of tasks favours providers with broad capabilities.

B is incorrect. Sibling swap on (2): architectural innovations that improve scalability and efficiency could lower barriers to entry, which supports competition.

D is incorrect. Sibling swap on both: the two developments sit on the opposite sides of the FSB's ledger.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c and Module Quiz 101.3

Solution CI-2 c.3 — Answer: A

Adapted from CI QB VI Q66 • SchweserNotes Book 5, LO 101.c

Step 1 – What the dependence exposes

Service provider concentration and third-party dependencies expose institutions to operational risks, including supply chain disruptions and cyber events affecting key vendors.

Step 2 – What the bank is expected to do

Institutions are expected to apply third-party risk management standards: periodic risk assessments, detailed vendor inventories, and ongoing monitoring and due diligence.

$\Rightarrow$ A.

Interpretation

The standards are the institution's to apply; the provider's size does not move the obligation. The FSB also notes an offset: more coordination and technological synergies may make risk management more effective and efficient.

Why the other options are wrong

B is incorrect. Role misassignment: the standards place periodic assessment and due diligence on the institution; nothing in the reading lets the provider's own certification replace them.

C is incorrect. Scope error: the standards call for periodic assessments and ongoing monitoring, not a single check at onboarding.

D is incorrect. Absolute claim: the FSB says increased coordination and technological synergies may enhance the effectiveness and efficiency of risk management.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.4 — Answer: C

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – The feedback

Correlations may increase if GenAI-generated outputs are incorporated into future training

datasets.

Step 2 – The speed channel

Greater automation and faster decision-making in financial markets can lead to more synchronised behaviour, such as coordinated reactions to market conditions or shifts in corporate deposit activity, which may amplify market stress.

⇒ C.

Interpretation

Both observations point the same way: correlation can build through what models learn from and through how fast they act on it. The third channel, pre-trained core models that share architectures and training data, is the subject of Workbook A14.3 under CI-1 c.

Why the other options are wrong

A is incorrect. Polarity flip on both halves: GenAI output re-used as training data can *increase* correlations, and faster automated decisions can synchronise behaviour rather than spread it.

B is incorrect. Scope error: it drops the feedback channel; correlations may also increase if GenAI-generated outputs are incorporated into future training datasets.

D is incorrect. Absolute claim: correlation arises from similar models and data sources, including shared pre-trained models, and from synchronised automated behaviour; it is not limited to identical in-house models.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.5 **Answer: B**

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – The named mitigants

Volatility control mechanisms such as circuit breakers, and enhanced transparency requirements.

Step 2 – The complements

Consistent monitoring and improved surveillance and data mechanisms.

⇒ B lists the FSB's toolkit.

Interpretation

One way to read the list (Figure 2): transparency at opaque model design, circuit breakers at the volatility itself, monitoring across the loop. The FSB presents them as reducing the impact; none removes the vulnerability. AI can also reduce correlation through more customised strategies, which is a reason not to discourage them.

Why the other options are wrong

A is incorrect. Sibling swap: vendor inventories and periodic assessments are the third-party risk mitigants, not the correlation ones.

C is incorrect. Polarity flip: homogeneity in risk assessments can arise when institutions build models around the same risk management standards, so a single standard would raise correlation.

D is incorrect. Polarity flip: AI may help *reduce* correlations if it is used to design more customised investment, portfolio management and trading strategies.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November

2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.6 Answer: A

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – Why the attack surface grows

Modern AI systems allow interaction through many channels and rely on increasingly large datasets, so the number of potential attack vectors, the pathways an attacker can use to compromise a system, has grown significantly.

Step 2 – Why an FMI matters

The sector is particularly vulnerable when attacks target central banks, critical service providers, FMIs, essential services that cannot easily be substituted, or other highly important firms. Service disruptions and declines in confidence can amplify volatility, liquidity vulnerabilities and operational losses.

⇒ A.

Interpretation

For the depository's board the message is two-sided: AI widens the ways in, and the institution's systemic role widens the damage. The FSB's best practices apply: an AI risk management framework (new, or AI added to the existing one) and AI questions in vendor due diligence.

Why the other options are wrong

B is incorrect. Polarity flip: attack vectors have grown, and FMIs are among the targets where the sector is most vulnerable.

C is incorrect. Absolute claim: LLMs themselves could be exploited to carry out certain types of attack.

D is incorrect. Scope error: declines in confidence, together with service disruptions, are what amplify volatility, liquidity vulnerabilities and operational losses.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.7 Answer: C

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – Both sides gain

Legitimate users benefit through automation of routine tasks, faster incident response, better detection of cyber events, more effective diagnosis of malicious code, and better employee training.

Step 2 – But not equally, and not forever

Bad actors may benefit even more in the short term, and malicious actors may be able to exploit AI tools more rapidly. Part of the reason is behavioural: cautious, risk-averse legitimate actors create an asymmetry with malicious actors who are often willing to adopt AI rapidly and take greater risks. Over the long term, legitimate users are expected to gain more (SchweserNotes Module Quiz 101.3 answer key).

⇒ C, with both qualifiers: "more" and "in the short term".

Interpretation

This is shape A14·S2's "short-term edge of bad actors". Every option is hedged, so the wording gives nothing away: what separates them is the direction of the edge, its horizon and its cause.

Why the other options are wrong

A is incorrect. Polarity flip on the asymmetry: legitimate users do benefit, but bad actors may benefit even more in the short term, so the gains are not equal.
B is incorrect. Polarity flip: GenAI *lowers* barriers to entry for less sophisticated threat actors.
D is incorrect. Polarity flip on the horizon: the bad actors' edge is a short-term one; the Module Quiz 101.3 answer key adds that legitimate users are expected to gain more over the long term.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c and Module Quiz 101.3

Solution CI-2 c.8

Answer: D

Adapted from CI QB VI Q69 • SchweserNotes Book 5, LO 101.c

Step 1 – Model risk under stress

In a crisis it may become more difficult to monitor, validate and correct models. More AI use and lower explainability raise model risk; more complex and opaque models are harder to validate.

Step 2 – Where it is handled

Regulated institutions must address AI data quality and model risk through their existing risk management frameworks. Standards should include data quality assessments, model validation, performance and outcome analysis, and ongoing monitoring, and should specify mitigation for limited data.
$\Rightarrow$ D.

Interpretation

For the LLM tools, outcome analysis is the hard part: their outputs are unstructured text, so error rates are difficult to establish, and hallucinations (highly confident but incorrect responses) have to be caught. The policy cannot assume that the tests used for numerical models carry over unchanged.

Why the other options are wrong

A is incorrect. Sibling swap: for cyber risk the FSB allows a new AI risk management framework, but for data and model risk it says regulated institutions must use their existing frameworks.
B is incorrect. Polarity flip: crises make models harder to monitor, validate and correct.
C is incorrect. Polarity flip: greater complexity and opacity raise model risk and make validation harder.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.9

Answer: B

New • built from SchweserNotes Book 5, LO 101.c

Step 1 – Scenario 1

Malicious actors use GenAI to generate and spread convincing, scalable false content, amplified by social media, during a period of stress. That is the FSB's disinformation vulnerability.

Step 2 – Scenario 2

A profit-maximising AI system behaves in a way not aligned with regulatory or legal standards,

outside its intended operation. The FSB's own example of misalignment is a system that spreads disinformation that triggers a bank run while profiting from the decline in the bank's stock price.
⇒ B.

Interpretation

The two scenarios use the same weapon, false content about a financial institution. The classification follows the source: a human malicious actor spreading it is disinformation; the AI system itself acting against legal standards is misalignment, even when its tool is disinformation. Option A is the trap for a candidate who classifies by the weapon rather than by the source.

Why the other options are wrong

A is incorrect. Sibling swap on scenario 2: the FSB's own example of misalignment is a profit-maximising system that spreads disinformation to trigger a bank run while profiting from the fall in the bank's share price. The source of the behaviour is the system, not a malicious actor.
C is incorrect. Sibling swap on both: scenario 1 has a human malicious actor, not a misaligned system, and scenario 2 is one system's behaviour, not many institutions using similar models.
D is incorrect. Sibling swap on both: scenario 1 attacks no system, so it is not a cyber event, and scenario 2's source is the AI system itself, which makes it misalignment.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • **Reference:** FSB (November 2024); SchweserNotes 2026 Book 5, Reading 101, LO 101.c

Solution CI-2 c.10 — Answer: D

Workbook A14.2 • S2, S4 • Aug-2026 • Map #20

Step 1 – Identify the vulnerability the proposal touches

Onboarding and voice service rely on identity checks. The FSB's "other vulnerabilities" include fraud: GenAI's strong video and voice generation creates deepfakes, fake profiles and IDs, bypassed security checks, and fraud targeting customers and institutions.
⇒ The bank's new voice channel is exactly the surface that GenAI-enabled fraud attacks.

Step 2 – The asymmetry

Generating fraudulent AI content may currently be easier than detecting it, and cautious legitimate actors adopt more slowly than risk-taking malicious actors. Mitigant named in the reading: AI literacy and financial literacy among customers.

Step 3 – Accountability for a bought model

Institutions remain fully responsible for AI they deploy, whether developed internally or obtained from external providers; AI questions belong in vendor due diligence.

Interpretation

D is correct. The EUR 40 million saving is irrelevant to the risk paragraph: the committee should weigh it against fraud and cyber controls on the onboarding and voice channels.

Why the other options are wrong

A is incorrect. Polarity flip on timing: bad actors may benefit more than legitimate users in the short term; legitimate users are expected to gain more over the long term.
B is incorrect. Sibling swap: the two attack vectors are reversed. Prompt injection extracts confidential information; model or data poisoning manipulates weights or training data.
C is incorrect. Role misassignment: the bank remains fully responsible for the AI systems it deploys regardless of whether they are developed internally or obtained from external providers.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities:

Third-party dependencies and service provider concentration; Market correlations; Cyber risks; Model risk, data, and governance; Other vulnerabilities, such as fraud, disinformation, and misalignment. • CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. • **Reference:** FSB (November 2024), Section on cyber risks and other vulnerabilities; IMF (October 2024), regulatory and supervisory considerations; SchweserNotes Book 5, Readings 101.3 and 100.3

CI-3 • The Global Drivers of Private Credit (BIS, February 2025) • Vardeez: VERY IMPORTANT

CI-3 a

P2 CONCEPT

Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall (reading section; no shape on this LO)

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter. The reading sits in Tier A section A28 (August 2026 recall, Map #49), but neither reported shape lands here: S1 is CI-3 b and S2 is CI-3 c.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Role in the reading. As the first LO of CI-3, this block carries the background on how private credit funds (PCFs) are built and who funds them. CI-3 b and CI-3 c recap it in a line and point back here.

Reported exam shapes: none on this LO.

What the questions look like: no calculations. Statement audits on the fund structure (closed-end, unlisted, institutional; BDCs as the listed, retail-facing exception), matching each lending strategy to its feature, the growth anchors ($0.2 trillion to $2.5 trillion of AUM; the United States at nearly 90% of originations), and the concentration trade-off, where the market has broadened while each fund stays narrow.

1. What a private credit fund is

Private credit funds are structured to provide loans *directly* to borrowers, such as small and medium-sized nonfinancial firms. From the fund's side, the loans are held as assets for their full term. Three structural features follow from that business model.

1. **Closed-end.** PCFs are typically closed-end vehicles: investor capital is committed for roughly **five to eight years**. Because the money cannot leave before the loans run off, the structure reduces *liquidity and maturity transformation risk*.
2. **Unlisted.** As investment products, PCFs are unlisted and therefore *relatively illiquid*. Keep the two features apart: the closed-end commitment is what protects the *fund*; being unlisted is what leaves the *investor* without an easy exit.
3. **Institutional investor base.** The underlying investors are predominantly institutional investors willing to commit capital for extended periods without needing regular liquidity. There is also a small but growing segment of retail investors.

The bank contrast. Banks rely heavily on short-term funding, which exposes them to significant asset-liability mismatches. PCFs avoid this today because their money is committed. If the retail investor base keeps expanding, PCFs *may begin to encounter* mismatch risks similar to banks'. Read the tense: the mismatch is a risk that could come with retail growth, not the current state.

Business development companies (BDCs)

In the United States, BDCs account for roughly **one-fifth** of the private credit market. They are built the other way round from a typical PCF: they are often **publicly listed** and structured to appeal to **retail investors**, offering more opportunities for earlier redemptions through **open-end features** rather than the long capital commitments of closed-end funds. The cost-of-capital comparison in CI-3 c is between US banks and BDCs.

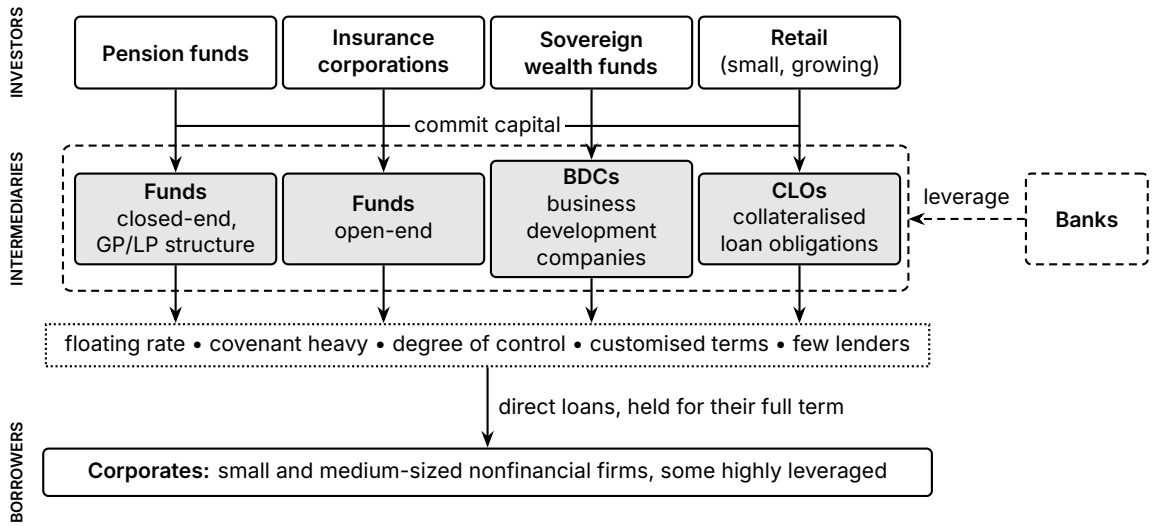


Figure 1. The intermediation chain, redrawn from the class notes' IMF figure (CI notes Figure CI-3.1). End investors pool committed capital into the intermediaries (solid line), which lend directly to middle-market corporates on the terms in the dotted band. Banks sit at the edge (dashed box): the dashed arrow into the dashed frame round the intermediaries shows them supplying leverage to the intermediaries rather than lending to these borrowers. GP = general partner, LP = limited partner.

2. Four lending strategies

PCFs frequently pursue a focused strategy of direct lending. The other three are common but less central to the typical business model.

Strategy	Defining feature
Direct lending (the focus)	Often floating-rate structures and extensive covenants
Mezzanine lending	Subordinated debt; may include equity rights
Asset-based lending	Requires substantial collateral
Distressed debt investing	Purchasing credit-risky debt at significant discounts

Keep two collateral facts apart. In CI-3 b, PCFs fill the gap left by banks for small firms that lack collateral by *reducing or removing* collateral requirements and relying on targeted covenants. Asset-based lending is a PCF strategy that *requires* substantial collateral. Both sit inside the PCF toolkit, so "PCFs never take collateral" is wrong.

3. Patterns in global growth

- **Size.** Over the past two decades, private credit assets under management (AUM) have grown from roughly $0.2 trillion to $2.5 trillion ("more than $2.5 trillion" in SchweserNotes' summary).
- **Geography.** The United States remains the dominant market, accounting for **nearly 90% of loan originations**. Europe, the United Kingdom and Canada make up most of the remaining activity.
- **Industries.** The sector has broadened from its earlier focus on manufacturing, technology and industrial firms to a wider range of industries, including cleantech (for example, environmental technologies) and life sciences.

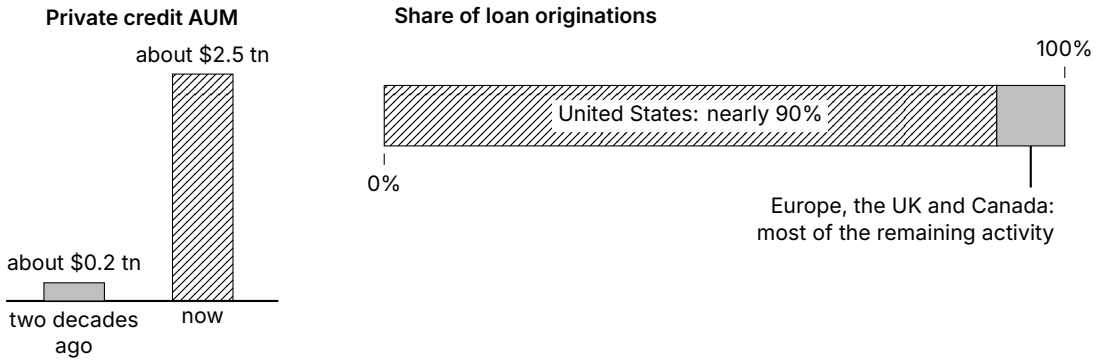


Figure 2. The growth anchors. Left: only the two endpoints are stated, so no path between them is drawn; the hatched bar is to scale against the grey one. Right: the hatched segment is drawn at 90%, the top of "nearly 90%"; the grey segment is the remainder, most of it Europe, the UK and Canada. The share is of loan *originations*, not of AUM.

Correction to the source notes. The class notes' growth bullet reads "The U.S. dominates (90%), while Europe, the UK, and Canada make up the rest" and does not say 90% of what. SchweserNotes (Book 5, LO 102.a) states it precisely: the United States accounts for *nearly 90% of loan originations*, and Europe, the United Kingdom and Canada make up *most of* the remaining activity, not all of it. The sources give no US share of AUM, so "90% of private credit AUM" is not a fact from this reading. Keep it apart from the other share in this LO: BDCs are about *one-fifth* of the *US* private credit market.

4. Concentration: the market broadened, the funds did not

Despite the growth and diversification of the market as a whole, individual PCFs tend to hold **highly concentrated loan portfolios**, often focused on a limited set of industries. PCFs are far less diversified than banks or other nonbank lenders in the syndicated loan market, as measured by the **Herfindahl-Hirschman Index (HHI)**.

- **Why managers concentrate.** Fund managers build specialised screening and monitoring capabilities within their chosen sectors. These let them assess risk better and tailor loan terms to riskier or more highly leveraged borrowers, who often have limited collateral and a higher likelihood of incurring net losses. Such borrowers are frequently rejected by banks but can still secure financing from PCFs.
- **What it costs.** The lack of diversification exposes PCFs to sector-specific downturns, which could pose significant risks to investors.
- **Who bears it well.** Institutional investors, who make up most of the investor base, are generally better positioned to manage the risk through broader portfolio diversification. Retail investors may

be less aware of the concentration risks inherent in PCFs.

- **The trade-off.** Greater diversification may enhance safety but can reduce the higher returns that concentrated strategies sometimes offer. As the asset class expands and managers seek more retail money, this trade-off is likely to become more important.

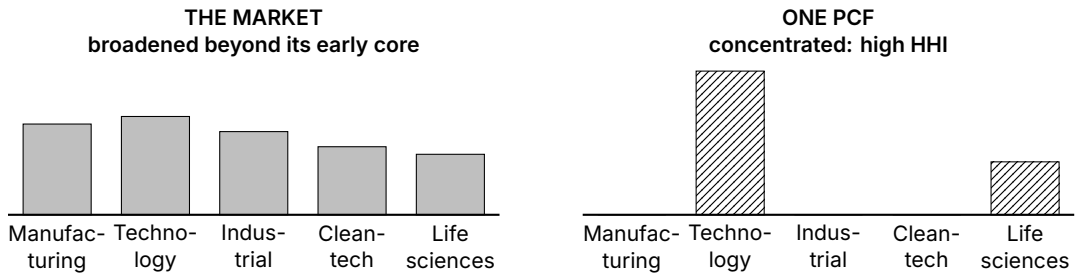


Figure 3. Two levels of diversification. Grey bars (left): lending across the market now spans the early core and the newer sectors. Hatched bars (right): a single fund lends to a few sectors only, which HHI measures show is far less diversified than a bank. Bar heights and the sectors chosen for the single fund are illustrative; the sources give no sector shares.

Results card: CI-3 a

	Typical PCF	BDC (US)	Bank
Structure	Closed-end; capital committed about 5 to 8 years	Open-end features; earlier redemptions	Heavy reliance on short-term funding
Listing	Unlisted, relatively illiquid	Often publicly listed	
Investors	Predominantly institutional; retail small but growing	Structured to appeal to retail	
Funding risk	Low liquidity and maturity transformation risk; bank-like mismatch *if* retail grows		Significant asset-liability mismatch
Portfolio	Highly concentrated (HHI)		More diversified than PCFs
Growth anchors	AUM about $0.2 trillion to $2.5 trillion in two decades; US nearly 90% of originations; BDCs about one-fifth of the US market		
Strategies	Direct lending (focus: floating rate, extensive covenants); mezzanine (subordinated, equity rights); asset-based (substantial collateral); distressed (deep discounts)		
Concentration	Buys specialised screening, tailored terms and access to borrowers banks reject; costs exposure to sector downturns		

Exam traps

- **Wrong-input twin on the shares.** One-fifth is the BDC share of the US market; nearly 90% is the US share of originations. Expect each number attached to the other quantity.
- **Sibling swap, BDC versus typical PCF.** Listed, open-end features and retail appeal describe BDCs; unlisted, closed-end and institutional describe the typical PCF.
- **Two kinds of liquidity.** The closed-end structure lowers the *fund's* liquidity and maturity transformation risk; being unlisted makes the *investment* illiquid for the investor.
- **Tense on the mismatch.** Bank-like asset-liability mismatch is what PCFs *may* face if retail grows, not their current state.
- **Aggregation slip.** The market broadened across industries; individual funds stayed concentrated.
- **Polarity on concentration.** It raises exposure to sector downturns. Diversification may add safety but can cut returns: the two do not rise together.
- **Strategy swaps.** Deep discounts mean distressed; subordinated with equity rights means mezzanine; substantial collateral means asset-based; floating rate with extensive covenants means direct lending.
- **90% of what.** Originations, not AUM.

Practice questions: CI-3 a

Question CI-3 a.1 New • built from SchweserNotes Book 5, LO 102.a; CI notes CI-3 a (Figure CI-3.1)

A supervisor at the Bank of England is mapping how UK banks are exposed to the private credit market. Based on how the private credit market is intermediated, which of the following descriptions is correct?

A. Banks typically originate most middle-market loans, and private credit funds then buy those loans from the banks to hold them to maturity.

B. Pension funds and insurers commit capital to funds that lend directly to mid-sized firms; banks supply leverage to the funds.

C. Retail investors are the predominant end investors, reaching mid-sized firms mainly through listed BDCs with open-end features.

D. Banks and private credit funds typically lend together in large syndicates, so banks hold shares of the same middle-market loans.

Question CI-3 a.2 New • built from SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

The investment committee of a Toronto-based pension plan is considering a commitment to a typical closed-end private credit fund. A trustee asks how liquid the fund and the plan's holding in it would be. Which of the following statements is correct?

A. Committed capital lowers the fund's liquidity and maturity transformation risk, but as it is unlisted the holding is relatively illiquid.

B. The fund's units are usually exchange-listed, so the plan could sell at any time, while the lock-up keeps the fund's own liquidity risk low.

C. The holding is liquid, because institutional investors in such funds need regular liquidity and the fund is built to provide it.

D. The fund already carries a bank-like asset-liability mismatch, because its loans run for their full term while investor capital is short-term.

Question CI-3 a.3 Adapted from CI QB VI Q72 • SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

A wealth adviser at a US broker-dealer is preparing a client note on business development companies (BDCs). Which of the following descriptions is consistent with the reading?

A. BDCs account for nearly 90% of the US private credit market, are often listed, and offer earlier redemptions through open-end features.

B. BDCs account for roughly one-fifth of the US private credit market, are often listed, and use long lock-ups to discourage retail redemptions.

C. BDCs account for roughly one-fifth of the US private credit market, are often listed, and offer earlier redemptions through open-end features.

D. BDCs account for roughly one-fifth of the US private credit market, are typically unlisted, and are sold mainly to institutions needing no liquidity.

Question CI-3 a.4 New • built from SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

A prudential supervisor at the Monetary Authority of Singapore learns that several private credit managers plan to widen distribution of their funds to retail investors. Which of the following consequences for the funds' funding side is most consistent with the reading?

A. Funds could lower their mismatch risk, because a broad retail base spreads redemption requests across many small investors.

B. Funds would face no new liquidity risk, because both their loans and their funding are structured on a floating-rate basis.

C. Funds' liquidity risk could stay low, because the five-to-eight-year closed-end commitment would apply to retail money as well.

D. Funds could begin to face asset-liability mismatch risks similar to banks', as retail money is less suited to long lock-ups.

Question CI-3 a.5 New • built from SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

A due diligence analyst at a Zurich family office is comparing the lending strategies of four private credit managers. Which of the following statements about PCF lending strategies is correct?

A. The focus strategy is asset-based lending, which requires substantial collateral; direct lending is usually fixed-rate with few covenants.

B. The focus strategy is direct lending, usually floating-rate with extensive covenants; mezzanine lending is subordinated and may carry equity rights.

C. The focus strategy is direct lending, usually floating-rate with extensive covenants; distressed debt investing makes new loans at par to stressed firms.

D. The focus strategy is mezzanine lending, which ranks ahead of senior loans; asset-based lending removes collateral in favour of covenants.

Question CI-3 a.6 Adapted from CI QB VI Q76 • SchweserNotes Book 5, LO 102.a; CI notes CI-3 a (correction box)

A strategist at a Hong Kong asset manager is drafting a market note on the global growth of private credit over the past two decades. Which of the following summaries is correct?

A. AUM rose from about $0.2 trillion to $2.5 trillion; the US accounts for about one-fifth of loan originations, with Europe the largest market.

B. AUM rose from about $0.2 trillion to $2.5 trillion; the US accounts for nearly 90% of originations, and lending has narrowed toward manufacturing and technology.

C. AUM rose from about $0.2 trillion to $2.5 trillion; the US accounts for nearly 90% of originations, with Europe, the UK and Canada most of the rest.

D. AUM rose from about $0.2 trillion to $2.5 trillion; the US accounts for nearly 90% of originations, with Asia most of the rest.

Question CI-3 a.7 Adapted from CI QB VI Q78 • SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

A risk analyst at a Dutch pension fund notes that private credit now reaches cleantech and life sciences as well as manufacturing and technology. She concludes that a holding in a single PCF is likely to be well diversified across industries. Which of the following responses is most appropriate?

A. The conclusion fails: HHI measures show individual PCFs are far less diversified than banks or other nonbank syndicated-loan lenders.

B. The conclusion holds: as the market broadened, HHI measures show that individual PCFs became about as diversified as banks.

C. The conclusion fails: individual PCFs are concentrated, yet HHI measures show them more diversified than banks, which lend to safer firms.

D. The conclusion holds: managers spread loans across sectors to offset the higher likelihood of net losses among their riskier borrowers.

Question CI-3 a.8 New • built from SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

The portfolio manager of a PCF that lends only to healthcare and software firms is asked by a prospective investor to justify the narrow focus. Which of the following rationales is most consistent with the reading?

A. A narrow focus lowers exposure to sector-specific downturns, because the manager can spot sector stress early and exit before losses appear.

B. A narrow focus lets the manager demand substantial collateral from each borrower, which is why it can lend to firms that banks reject.

C. A narrow focus is what keeps the fund's liquidity and maturity transformation risk low, because sector loans repay on predictable schedules.

D. A narrow focus builds specialised screening and monitoring, so the manager can tailor terms for riskier borrowers that banks often reject.

Question CI-3 a.9 Adapted from CI QB VI Q80 • SchweserNotes Book 5, LO 102.a; CI notes CI-3 a

A retail investor protection team at a European securities regulator is reviewing marketing material for a sector-focused PCF aimed at retail savers. Which of the following statements is most consistent with the reading's assessment of the concentration trade-off?

A. Retail investors are better placed than institutions to absorb the concentration, because each holds only a small stake in the fund.

B. Greater diversification within the fund would raise safety and expected returns together, so the trade-off may fade as retail grows.

C. Concentration matters little to investors, because loans held to maturity in a closed-end fund are insulated from sector downturns.

D. Institutions can offset fund concentration through broader portfolio diversification; retail investors may be less aware of it.

Solutions • CI-3 a

Solution CI-3 a.1 — Answer: B

New • built from SchweserNotes Book 5, LO 102.a; CI notes Figure CI-3.1

Step 1 – Who supplies the capital

End investors are pension funds, insurance corporations and sovereign wealth funds, with a small but growing retail segment. The investor base is predominantly institutional.

Step 2 – Who lends, and where banks sit

The intermediaries (closed-end GP/LP funds, open-end funds, BDCs and CLOs) lend directly to small and mid-sized corporates and hold the loans for their full term. In the class notes' intermediation chain (an IMF staff figure), banks supply leverage to the intermediaries rather than lending to these borrowers. The A28 revision notes on the IMF reading (IM-10) describe the same link: banks' exposure to private credit is secured lending and credit lines to the funds.
$\Rightarrow$ A bank's exposure to this market runs through its lending to the funds.

Interpretation

For the supervisor, the place to look is banks' financing of the intermediaries, not a book of middle-market loans that the funds bought from banks.

Why the other options are wrong

A is incorrect. Role misassignment: PCFs lend *directly* to borrowers and originate the loans themselves; nothing in the reading has them buying bank-originated loans.
C is incorrect. Polarity flip on the investor base: it is predominantly institutional, and retail is a small but growing segment. BDCs are about one-fifth of the US market, not the main channel.
D is incorrect. Sibling swap: a large multi-lender syndicate is the syndicated loan market. Private credit loans are direct loans with few lenders, often to borrowers that banks decline, and banks sit at the edge as providers of leverage to the funds.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; class notes Figure CI-3.1 (IMF staff); workbook A28 revision notes (IM-10 interconnectedness); SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.2 — Answer: A

New • built from SchweserNotes Book 5, LO 102.a

Step 1 – The fund's side

A typical PCF is closed-end: investor capital is committed for roughly five to eight years, so the fund is not funding long-held loans with money that can leave early. This reduces liquidity and maturity transformation risk.

Step 2 – The investor's side

PCFs are unlisted, so as investment products they are relatively illiquid.
$\Rightarrow$ The closed-end commitment protects the fund; being unlisted is what leaves the investor without an easy exit.

Interpretation

The trustee should expect to be locked in for the commitment period, with no listed market in which to sell the holding. That is the price of the structure that protects the fund.

Why the other options are wrong

B is incorrect. Sibling swap: listing is a feature of many BDCs. The typical PCF is unlisted,

which is why the holding is illiquid.
C is incorrect. Polarity flip on the investor base: PCF investors are predominantly institutions willing to commit capital for extended periods *without* needing regular liquidity, and the closed-end structure is built to lock that capital in.
D is incorrect. Tense error: a bank-like mismatch is what PCFs *may* face if retail participation grows. Today the capital is committed for five to eight years.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.3 — Answer: C

Adapted from CI QB VI Q72 • SchweserNotes Book 5, LO 102.a

Step 1 – Size
In the United States, BDCs account for roughly one-fifth of the private credit market.

Step 2 – Design
BDCs are often publicly listed and structured to appeal to retail investors, offering more opportunities for earlier redemptions through open-end features instead of the long commitments of closed-end funds.
⇒ BDCs are the listed, retail-facing exception to the typical PCF profile.

Interpretation
For the client note, the BDC's easier exit is exactly the feature that the reading links to rising retail participation and, potentially, bank-like mismatch risk.

Why the other options are wrong
A is incorrect. Wrong-input twin: the design half is right, but nearly 90% is the US share of global loan originations; BDCs are roughly one-fifth of the US market.
B is incorrect. Polarity flip: BDCs offer *earlier* redemptions to attract retail money; they do not lengthen lock-ups.
D is incorrect. Sibling swap: unlisted and institutional is the typical PCF profile, the one BDCs depart from.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.4 — Answer: D

New • built from SchweserNotes Book 5, LO 102.a

Step 1 – Why PCFs avoid mismatch today
The investor base is predominantly institutional, willing to commit capital for long periods without regular liquidity, and the funds are closed-end.

Step 2 – What retail growth changes
Banks rely heavily on short-term funding and carry significant asset-liability mismatches. If the retail base of PCFs continues to expand, PCFs may begin to encounter similar mismatch risks.
⇒ More retail money means more demand for exits against loans held for their full term.

Interpretation
The supervisor's concern is on the liability side: the funds' long-held loans stay the same, while the money behind them becomes less patient.

Why the other options are wrong

A is incorrect. Polarity flip: the reading links retail expansion to *more* mismatch risk, not less.
B is incorrect. Sibling swap: floating rates on both sides protect PCF margins from rate moves (CI-3 b). They say nothing about whether investors can withdraw money early.
C is incorrect. Scope error: the five-to-eight-year commitment describes institutional capital in closed-end funds; retail participation comes with earlier-redemption features, as BDCs show.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.5

Answer: B

New • built from SchweserNotes Book 5, LO 102.a

Step 1 – The focus strategy

PCFs frequently pursue a focused strategy of direct lending, which often involves floating-rate structures and extensive covenants.

Step 2 – The other three

Mezzanine: subordinated debt that may include equity rights. Asset-based: requires substantial collateral. Distressed: purchasing credit-risky debt at significant discounts.
⇒ Only B gets both halves right.

Interpretation

Each strategy has one defining feature. Match on it: discounts, subordination with equity rights, substantial collateral, or floating rate with covenants.

Why the other options are wrong

A is incorrect. Sibling swap and polarity flip: direct lending, not asset-based lending, is the focus, and it is floating-rate with extensive covenants.
C is incorrect. Sibling swap in the second half: distressed investing *buys* credit-risky debt at significant discounts; it is not new lending at par.
D is incorrect. Two errors: mezzanine debt is *subordinated*, and asset-based lending *requires* substantial collateral.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.6

Answer: C

Adapted from CI QB VI Q76 • SchweserNotes Book 5, LO 102.a

Step 1 – Size

AUM grew from roughly $0.2 trillion to $2.5 trillion over two decades. All four options share this half, so the question turns on the other half.

Step 2 – Geography

The United States accounts for nearly 90% of loan originations; Europe, the UK and Canada make up most of the remaining activity.

Step 3 – Industries

The sector has broadened from manufacturing, technology and industrial firms into cleantech and life sciences.
⇒ Only C gets the US share, the rest of the market and the industry pattern right.

Interpretation

The growth story has three anchors: size ($0.2 trillion to $2.5 trillion), geography (US nearly 90% of originations, Europe, the UK and Canada most of the rest) and breadth (more industries). A distractor usually keeps one and breaks another.

Why the other options are wrong

A is incorrect. Wrong-input twin: one-fifth is the BDC share of the US market, not the US share of originations.

B is incorrect. Polarity flip on industries: the sector has *broadened* beyond manufacturing and technology, not narrowed toward them.

D is incorrect. Sibling swap on the rest of the market: the share is right, but Europe, the UK and Canada make up most of the remaining activity; the reading names no Asian share.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.7 — Answer: A

Adapted from CI QB VI Q78 • SchweserNotes Book 5, LO 102.a

Step 1 – Separate the two levels

The market as a whole has broadened across industries. Individual PCFs still tend to hold highly concentrated portfolios focused on a limited set of industries.

Step 2 – What the HHI shows

Measured with the HHI, PCFs are far less diversified than banks or other nonbank lenders in the syndicated loan market.

$\Rightarrow$ Breadth across the market does not carry over to a single fund.

Interpretation

The analyst has made an aggregation error. A pension fund can diversify across several PCFs, but one fund on its own is a concentrated sector bet.

Why the other options are wrong

B is incorrect. Polarity flip: the HHI evidence shows PCFs to be *far less* diversified than banks.

C is incorrect. Polarity flip in the second half: the comparison with banks runs the other way, and banks are the lenders that turn down the riskier borrowers.

D is incorrect. Wrong rationale: managers concentrate in order to build sector expertise for riskier borrowers; they do not spread loans to offset that risk.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.8 — Answer: D

New • built from SchweserNotes Book 5, LO 102.a

Step 1 – The stated rationale

Concentration lets fund managers build specialised screening and monitoring capabilities in their chosen sectors.

Step 2 – What the capability buys

It allows better risk assessment and loan terms tailored to riskier or more highly leveraged borrowers, who often have limited collateral and a higher likelihood of net losses. Banks frequently

reject such borrowers.
⇒ Sector expertise stands in for the collateral these borrowers cannot offer.

Interpretation

The manager should also tell the investor about the cost: a two-sector book is exposed to downturns in those sectors.

Why the other options are wrong

A is incorrect. Polarity flip: concentration *increases* exposure to sector-specific downturns. That is its cost, not its benefit.
B is incorrect. Polarity flip on collateral: the borrowers PCFs win from banks often have *limited* collateral.
C is incorrect. Sibling swap: it is the closed-end structure with committed capital that keeps liquidity and maturity transformation risk low, not the sector focus.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

Solution CI-3 a.9 **Answer: D**

Adapted from CI QB VI Q80 • SchweserNotes Book 5, LO 102.a

Step 1 – Who can manage concentration risk

Institutional investors, who make up most of the base, are generally better positioned to manage it through broader portfolio diversification.

Step 2 – The retail concern

Retail investors may be less aware of the concentration risks inherent in PCFs. As managers seek more retail money, the trade-off between concentration and diversification becomes more important.
⇒ The same fund is a small tilt in an institutional portfolio and possibly a large, unrecognised bet for a retail saver.

Interpretation

For the regulator, this points to clear disclosure of sector concentration in material aimed at retail savers.

Why the other options are wrong

A is incorrect. Polarity flip: the reading places institutions, not retail investors, in the better position. The size of a stake does not diversify a sector bet.
B is incorrect. Denies the trade-off: diversification may enhance safety but can *reduce* the higher returns of concentrated strategies.
C is incorrect. Scope error: holding loans to maturity in a closed-end fund limits liquidity risk; it does not protect the loans from a downturn in their sector.

LO: CI-3 a: Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a

CI-3 • The Global Drivers of Private Credit (BIS, February 2025) • Vardeez: VERY IMPORTANT

CI-3 b

P1 CONCEPT

Discuss the potential drivers of private credit growth and interpret regression results about their impact.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: private credit came up in the August 2026 sitting, and Map #49 lists it twice in the same consolidated sheet. A28 shape S1 (growth drivers and the BIS regression) is this LO.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A28·S1** Growth drivers of private credit and the BIS regression results (Question CI-3 b.9: keyed on the CI-3 c cost-of-capital story; option A tests this LO).

What the questions look like: no calculations. Statement audits ("which of the following is correct"), scenario consequences ("which outcome is most consistent"), and reading the regression table: the sign of each coefficient together with the direction of the move, the ranking by economic effect, and one standard deviation rather than one unit.

1. Four drivers, one hinge

Private credit funds (PCFs) lend directly to borrowers such as small and mid-sized nonfinancial firms, and hold the loans to maturity (CI-3 a). The BIS names four forces behind the asset class's growth. Each works through the same hinge: it lowers banks' *ability or willingness* to lend to a group of borrowers, and PCFs fill the gap.

1. **Monetary policy (low policy rates).** Long periods of low rates compress bank margins. In SchweserNotes' words, the value of bank assets (for example floating-rate loans) declines more quickly than the value of their liabilities (for example fixed-rate customer deposits). One way to read it is as income: loan yields reset down with the policy rate while deposit costs move more slowly, so the margin narrows and banks lend less. PCFs escape the squeeze because both their assets and their liabilities are generally floating-rate. They may earn higher returns, which can attract more institutional investors.
2. **Banking-sector efficiency (the IMF FI index).** The IMF's financial institutions (FI) index measures *how effectively banks meet loan demand*. Banks are often reluctant to lend to small and medium-sized firms that cannot post enough collateral, which shows up as a lower FI index. PCFs offer an alternative by reducing or removing collateral requirements and relying instead on more targeted covenants.
3. **Regulatory stringency.** Higher capital requirements and stricter oversight make banks less willing to lend to firms they see as risky. Those firms turn to PCFs, which operate with greater flexibility.
4. **Corporate leverage.** Nonfinancial corporate leverage has risen alongside private credit, and PCFs are more willing than banks to lend to highly leveraged firms.

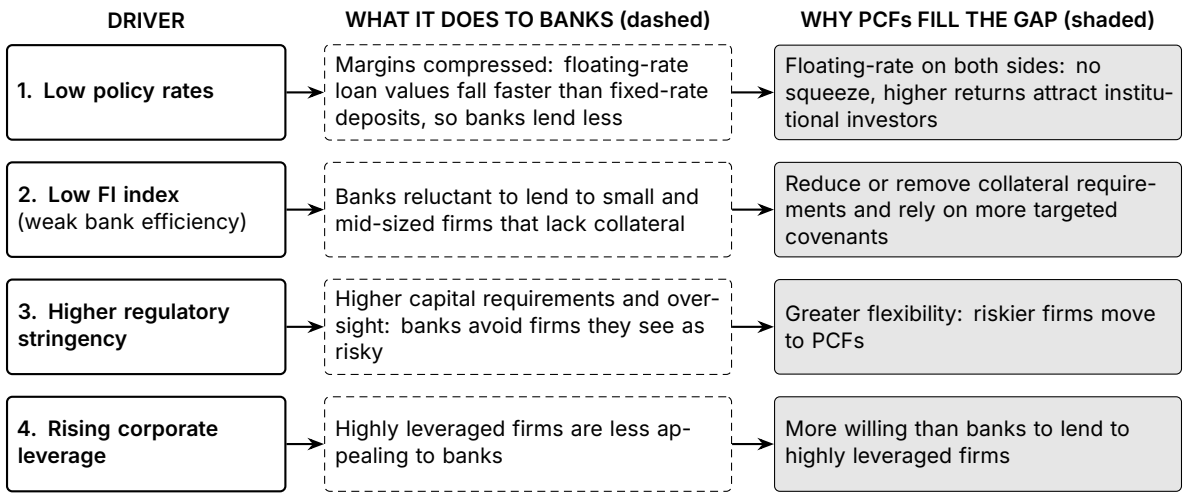


Figure 1. The four drivers read left to right. The middle column is the hinge the reading stresses: every driver reduces banks' ability or willingness to lend to some group of borrowers, and the right column is the PCF feature that lets the fund take that business.

2. The BIS regression

Design. 45 countries, 2010 to 2019. The dependent variable is the amount of *newly originated* private credit. The four explanatory variables are the four drivers: (1) the policy rate, (2) the IMF FI index, (3) bank regulatory stringency and (4) corporate leverage.

Variable	Coefficient	One s.d. move	Effect on new private credit
FI index	-1.679 to -1.906	**decline**	about $+33\%$: largest
Policy rate	-0.03	**decline**	about $+12\%$: second
Regulatory stringency	$+0.463$ to $+0.667$	**increase**	about $+2\%$ to $+4\%$: modest
Corporate leverage	$+0.001$	**increase**	under $+1\%$: weakest

The policy rate and the FI index are the two variables with the largest overall influence. Regulatory stringency and corporate leverage show much smaller economic effects.

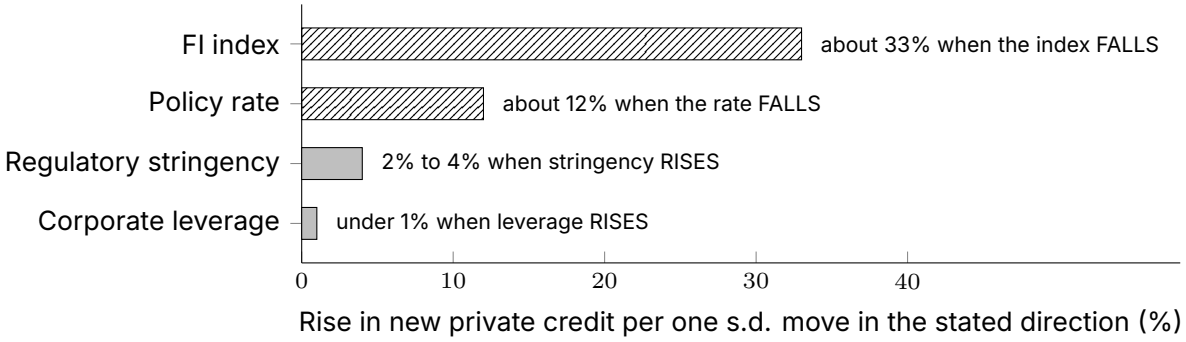


Figure 2. The four economic effects on one scale. Hatched bars follow a *fall* in the variable (negative coefficient); grey bars follow a *rise* (positive coefficient). The stringency bar is drawn at the top of its 2% to 4% range and the leverage bar at its 1% ceiling. The FI index is worth almost three times the policy rate; leverage barely registers.

3. Reading the results: four rules

1. **Read the sign and the direction together.** A negative coefficient means private credit rises when the variable *falls*. "33% increase" belongs to a *decline* in the FI index, and "12% increase" to a *decline* in the policy rate. Less effective banks and lower policy rates both go with more private credit.
2. **Rank by economic effect, not by coefficient size.** Each coefficient is in its own variable's units, so -1.8 on the FI index and -0.03 on the policy rate cannot be compared directly (their ratio is about 60). The one-standard-deviation effect puts all four on one scale: 33% against 12%, a ratio of 2.75. The policy rate has a tiny coefficient and the second-largest effect.
3. **One standard deviation, not one unit.** "About 12%" is stated for a one-standard-deviation decline in the policy rate. It need not equal the effect of a 1 percentage point cut (the sources give no standard deviation), and it is an estimated association across 45 countries over 2010 to 2019, not a guaranteed result in any single market.
4. **Co-movement is not the regression effect.** Corporate leverage rose alongside private credit and is "closely correlated" with it, yet in the regression, which includes the other three drivers, it has the weakest relationship: under 1% per standard deviation. A strong trend story can sit next to a small estimated effect.

Correction to the source notes. The class notes' regression table reads "Impact of 1 Std. Deviation Change: 33% Increase in PC activity" against the FI index coefficient of -1.68 to -1.91, and "12% Increase" against the policy rate coefficient of -0.03, with no direction for the move. Read the natural way, as a rise, it says that a *rise* in bank efficiency or in the policy rate increases private credit. Both coefficients are negative, so each increase follows a *decline* in the variable (SchweserNotes Book 5, LO 102.b). The stringency and leverage rows are increases and were right as printed, which is what makes the error easy to miss. Question CI-3 b.5 option B is this exact misreading.

Results card: CI-3 b

Study	BIS (February 2025): 45 countries, 2010 to 2019; dependent variable newly originated private credit
Ranking by economic effect	FI index decline $\approx 33\%$ $>$ policy rate decline $\approx 12\%$ $>$ stringency rise $\approx 2\%$ to 4% $>$ leverage rise $< 1\%$
Signs	FI index $-$, policy rate $-$, stringency $+$, leverage $+$
Hinge	Each driver cuts banks' ability or willingness to lend; PCFs fill the gap
PCF features that fill it	Floating-rate on both sides; collateral reduced or removed, targeted covenants; flexibility; appetite for leveraged borrowers

Exam traps

- **Sibling swap on the magnitudes.** 33% belongs to the FI index, 12% to the policy rate. Expect the other variable's number in a distractor.
- **Polarity flip on the direction.** "A one-standard-deviation *increase* in the FI index raises private credit by 33%" has the right number and the wrong direction.
- **Ranking by coefficient size.** -1.8 against -0.03 is a ratio of about 60; the economic effects are 33% against 12%.
- **One unit for one standard deviation.** The effects are per one-standard-deviation move, not per percentage point or index point.

- **Role misassignment on the margin squeeze.** Banks carry the fixed-rate deposits and get squeezed; PCFs are floating-rate on both sides and escape it.
- **Stringency read the wrong way.** Tighter *bank* regulation is associated with *more* private credit, modestly (2% to 4%), not less and not zero.
- **Trend mistaken for effect.** Leverage moved with private credit but has the weakest regression effect.
- **Overreach.** "In every market" or "will raise": the results are estimated associations across 45 countries over 2010 to 2019.

Practice questions: CI-3 b

Question CI-3 b.1

New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

The head of strategy at a Frankfurt-based private credit manager is explaining to prospective investors why the asset class grew during the long period of low policy rates after 2010. Based on the BIS analysis of the drivers of private credit, which of the following explanations is correct?

A. Low rates squeezed the margins of private credit funds, whose liabilities are mostly fixed-rate, pushing the funds toward riskier borrowers to restore returns.

B. Low rates widened bank margins as deposit costs fell faster than loan yields, so banks moved capital into large syndicated loans and left the middle market to funds.

C. Low rates squeezed bank margins as floating-rate loan values fell faster than fixed-rate deposits, while funds, floating-rate on both sides, avoided the squeeze.

D. Low rates reduced default rates on leveraged borrowers, so funds could relax covenants and undercut bank pricing on investment-grade corporate loans.

Question CI-3 b.2

Adapted from CI QB VI Q89 • SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

A World Bank consultant is reviewing an emerging economy whose score on the IMF financial institutions (FI) index is well below the regional average. Local banks routinely decline loan applications from small and medium-sized firms that cannot pledge enough collateral. According to the BIS analysis, which of the following best describes the role private credit funds play in such a market?

A. They fill the financing gap by reducing or removing collateral requirements and relying instead on more targeted covenants.

B. They fill the financing gap mainly because their lower cost of debt lets them price small-firm loans below what banks charge.

C. They play little role, because a low FI index signals weak loan demand from firms, which private credit funds cannot create.

D. They fill the financing gap mainly through asset-based lending, advancing against substantial collateral that banks will not accept.

Question CI-3 b.3 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

After a tightening of bank capital requirements and supervisory oversight, a central bank economist estimates that the country's index of bank regulatory stringency has risen by about one standard deviation. Based on the BIS regression results, which of the following outcomes is most consistent with this change?

A. Private credit origination falls by about 2% to 4%, because private credit funds become subject to the same tighter capital rules as banks.

B. Private credit origination rises by about 33%, because tighter regulation is the main reason banks withdraw from middle-market lending.

C. Private credit origination is broadly unchanged, because the regression finds no relationship between bank regulation and private credit.

D. Private credit origination rises by about 2% to 4%, because banks become less willing to lend to riskier firms.

Question CI-3 b.4 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

A research analyst at a pension fund wants to replicate the BIS study of the global drivers of private credit before citing its results in an allocation paper. Which of the following correctly describes the study's regression?

A. Private credit assets under management regressed on the policy rate, the FI index, bank regulatory stringency and corporate leverage, for the United States and Europe over 2000 to 2019.

B. Newly originated private credit regressed on the policy rate, the FI index, bank regulatory stringency and corporate leverage, across 45 countries over 2010 to 2019.

C. Newly originated private credit regressed on bank profitability, deposit growth, loan default rates and corporate leverage, across 45 countries over 2010 to 2019.

D. Private credit fund returns regressed on the policy rate, the FI index, bank regulatory stringency and corporate leverage, across 45 countries over 2010 to 2019.

Question CI-3 b.5 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b (correction box)

An analyst reads that the coefficient on the IMF financial institutions (FI) index in the BIS regression of newly originated private credit lies between -1.679 and -1.906. Which of the following is the correct interpretation?

A. A one-standard-deviation decline in the FI index is associated with about a 12% rise in private credit origination, the second-largest effect of the four variables.

B. A one-standard-deviation increase in the FI index is associated with about a 33% rise in private credit origination, the largest effect of the four variables.

C. Because the coefficient is negative, each one-standard-deviation change in the FI index lowers private credit origination by about 1.7% to 1.9%.

D. A one-standard-deviation decline in the FI index is associated with about a 33% rise in private credit origination, the largest effect of the four variables.

Question CI-3 b.6 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

A junior analyst compares the BIS coefficients on the policy rate (-0.03) and on the FI index (about -1.8) and writes in a memo that bank efficiency matters roughly 60 times as much as the policy rate for private credit growth. Which of the following is the most appropriate correction?

A. The memo overstates the gap: on one-standard-deviation effects, the FI index is about ten times the policy rate rather than sixty times.

B. The memo is right on the ranking but should use the low end of the FI index range, which gives a coefficient ratio of about 56 rather than 60.

C. Coefficients are in each variable's own units and cannot be compared; on standardised effects the FI index is under three times the policy rate.

D. The memo reverses the ranking: the policy rate's one-standard-deviation effect, about 33%, is larger than the FI index's effect of about 12%.

Question CI-3 b.7 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

A private credit fund's marketing deck states: "The BIS found that cutting the policy rate by 1 percentage point raises private credit origination by 12% in every market." A risk manager reviewing the deck flags the statement. Which of the following best describes what is wrong with it?

A. The 12% is stated for a one-standard-deviation decline, not a 1 percentage point cut, and is a cross-country association, not a rule for every market.

B. The 12% figure is right for a 1 percentage point cut, but it applies only to the United States, which accounts for nearly 90% of private credit loan originations.

C. The statement reverses the sign: the policy rate coefficient is positive, so a rate cut is associated with about a 12% fall in private credit origination.

D. The 12% figure belongs to the FI index; a one-standard-deviation move in the policy rate is associated with a change of less than 1%.

Question CI-3 b.8 New • built from SchweserNotes Book 5, LO 102.b; CI notes CI-3 b

A credit strategist observes that nonfinancial corporate leverage rose substantially in many countries alongside the expansion of private credit lending, and concludes that leverage must be one of the strongest drivers of private credit growth. Which of the following responses best reflects the BIS results?

A. The conclusion is supported: leverage carries the largest coefficient of the four variables, and a one-standard-deviation rise is associated with about a 12% increase.

B. The conclusion is not supported: in the regression, leverage has the weakest relationship, with a one-standard-deviation rise associated with less than a 1% increase.

C. The conclusion is not supported: the regression finds that higher corporate leverage reduces private credit, because funds avoid highly leveraged borrowers.

D. The conclusion is supported: funds are more willing than banks to lend to leveraged firms, so leverage and the FI index have similar economic effects.

Question CI-3 b.9 Workbook A28.3 • S1, S2 • Aug-2026 • Map #49

The treasurer of a US regional bank is preparing a board briefing on why business development companies (BDCs) have been winning middle-market loans that the bank used to make, drawing on the BIS study of private credit. Which of the following statements is correct to include?

A. In the BIS regression across 45 countries, the policy rate has the largest economic effect: a one-standard-deviation decline is linked to about a 33% rise in private credit origination.

B. The banks' funding advantage over BDCs narrowed by about 250 bp in 2010 to 2019 mainly because BDCs' cost of debt fell below the banks' cost of debt.

C. The roughly 250 bp erosion of the banks' funding advantage came about equally from banks' higher cost of equity and BDCs' higher leverage, while BDCs' cost of debt stayed marginally above banks'.

D. Because private credit funds are funded at floating rates on both sides of the balance sheet, a sustained rise in policy rates would further widen their cost-of-capital advantage over banks.

Solutions • CI-3 b

Solution CI-3 b.1 — Answer: C

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – Find the channel

The question is about the first driver, accommodative monetary policy. In the reading, prolonged low rates compress *bank* margins: the value of bank assets (floating-rate loans) declines more quickly than the value of their liabilities (fixed-rate customer deposits).
⇒ Banks earn less on lending and become less willing to extend credit, which opens a gap.

Step 2 – Why the funds escape

PCFs' assets and liabilities are generally both floating-rate, so falling rates move both sides together and there is no margin compression.
⇒ PCFs may earn higher returns, which can draw in institutional investors and fund more lending.

Interpretation

The channel is an asymmetry in balance-sheet structure: the same rate path squeezes the lender with sticky fixed-rate funding and leaves the matched floating-rate lender untouched. The same feature can work against BDCs when rates rise (CI-3 c).

Why the other options are wrong

A is incorrect. Role misassignment: the fixed-rate liabilities are the banks' deposits. PCFs are floating-rate on both sides, which is why they avoid the squeeze.
B is incorrect. Polarity flip: low rates compress bank margins rather than widen them; deposits are the leg that adjusts slowly.
D is incorrect. Invented mechanism: the reading's channel runs through bank margins, not default rates, and PCFs serve riskier borrowers with targeted covenants, not investment-grade borrowers with relaxed ones.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.b

Solution CI-3 b.2 — Answer: A

Adapted from CI QB VI Q89 • SchweserNotes Book 5, LO 102.b

Step 1 – What a low FI index says

The IMF FI index measures how effectively banks meet loan demand. A low score here comes from banks declining small and medium-sized firms that cannot post enough collateral.
⇒ The demand is there; the banks are not meeting it.

Step 2 – How PCFs meet it

PCFs reduce or remove collateral requirements and rely instead on more targeted covenants.

Interpretation

This is the largest of the four effects in the BIS regression (about 33% per one-standard-deviation decline in the index), so markets with inefficient banks are where private credit grows most.

Why the other options are wrong

B is incorrect. Polarity flip on pricing: the funds' cost of debt has been marginally *above* banks' (CI-3 c), and private credit rates are often higher than bank rates. Borrowers come for flexibility, not price.
C is incorrect. Misdefinition of the index: it scores how well banks *meet* loan demand, not how

much demand exists. A low score signals unmet demand, the opening for PCFs.
D is incorrect. Wrong strategy for the borrower: asset-based lending is a PCF strategy that *requires* substantial collateral, which is exactly what these firms lack.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.
• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.a, 102.b and 102.c

Solution CI-3 b.3 — Answer: D

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – Sign and size
Regulatory stringency carries a modest positive coefficient (0.463 to 0.667): a one-standard-deviation *increase* is associated with roughly a 2% to 4% *rise* in private credit origination.

Step 2 – Mechanism
Higher capital requirements and stricter oversight make banks less willing to lend to firms they see as risky; those firms turn to PCFs, which operate with greater flexibility.
⇒ Tighter *bank* rules push riskier borrowers to the funds.

Interpretation
The effect is positive but small next to the FI index and the policy rate. The variable measures the regulation of banks, and the result is consistent with the reading's explanation that riskier borrowers move from banks to funds.

Why the other options are wrong
A is incorrect. Polarity flip with a scope error: the variable is bank regulatory stringency and its coefficient is positive. Nothing in it puts the funds under bank capital rules.
B is incorrect. Wrong input twin: 33% is the effect of a decline in the FI index, the largest driver, not of stringency.
C is incorrect. Absolute claim: the effect is modest, not absent; the coefficient is positive and worth about 2% to 4% per standard deviation.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.
• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.b

Solution CI-3 b.4 — Answer: B

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – Dependent variable
The regression explains the amount of *newly originated* private credit: new lending, not the stock of assets under management and not fund returns.

Step 2 – Explanatory variables
The four drivers: the policy rate, the IMF FI index (banking-sector efficiency), bank regulatory stringency and corporate leverage.

Step 3 – Sample
45 countries, 2010 to 2019.
⇒ Only B has all three right.

Interpretation
Because the dependent variable is new origination, each coefficient describes how new private credit lending moves with a driver. Reading the results as effects on AUM or on returns answers a different question. The AUM figures in the reading (about $0.2 trillion to $2.5 trillion over two decades) describe the market's growth (CI-3 a), not the regression.

Why the other options are wrong

A is incorrect. Wrong dependent variable and wrong sample: AUM is a stock, the study covers 45 countries, and the period is 2010 to 2019.

C is incorrect. Wrong regressors: only corporate leverage is one of the four; the policy rate, the FI index and regulatory stringency are missing.

D is incorrect. Wrong dependent variable: the reading links higher fund returns to investor demand under the monetary policy driver, but the regression explains origination, not returns.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.

• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.a and 102.b

Solution CI-3 b.5 — Answer: D

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – Read the sign

A negative coefficient means the variable and private credit move in opposite directions: a lower FI index goes with more private credit.

Step 2 – Attach the economic effect to the right direction

A one-standard-deviation *decline* in the FI index is associated with about a 33% *rise* in newly originated private credit, the largest of the four effects.

$\Rightarrow$ Private credit grows most where banks are least effective at meeting loan demand.

Interpretation

The coefficient and the economic effect are different numbers. The coefficient (-1.679 to -1.906) is per unit of the index; the 33% is the standardised effect of a one-standard-deviation move. Read the row as a whole: negative coefficient, so the effect follows a fall.

Why the other options are wrong

A is incorrect. Sibling swap: right direction, wrong number. About 12% is the policy rate's effect; the FI index's is about 33%, the largest.

B is incorrect. Right number, wrong direction: the 33% follows a *decline* in the index. This is the misreading in the class notes' table (see the correction box).

C is incorrect. Reads the coefficient as a percentage effect: -1.7 to -1.9 is not a per-standard-deviation percentage, and a decline in the index raises private credit.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.

• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.b

Solution CI-3 b.6 — Answer: C

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – What the memo did

Ratio of coefficients: $1.679/0.03 \approx 56$ to $1.906/0.03 \approx 64$, hence "roughly 60 times".

Step 2 – Why that ratio means nothing

Each coefficient is the change in the dependent variable per unit of its own regressor, and the FI index and the policy rate are measured on different scales. Coefficients across differently scaled variables cannot be ranked directly.

Step 3 – Use the standardised effects

One-standard-deviation effects put both on one scale: FI index about 33%, policy rate about 12%, a ratio of $33/12 = 2.75$.

$\Rightarrow$ Bank efficiency matters most, but by under three times, not sixty.

Interpretation

The policy rate is the clearest case: the smallest-looking coefficient in the table carries the second-largest economic effect. Rank drivers by economic effect.

Why the other options are wrong

A is incorrect. Wrong input twin: about ten times is the FI index against regulatory stringency (33% against roughly 3%), not against the policy rate (12%).

B is incorrect. Keeps the memo's mistake: $1.679/0.03 \approx 56$ is still a ratio of coefficients in different units, so the choice of end of the range does not fix it.

D is incorrect. Sibling swap: it exchanges the two magnitudes; 33% belongs to the FI index and 12% to the policy rate.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.

• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.b

Solution CI-3 b.7

Answer: A

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – Check the unit of the move

The 12% is stated for a one-*standard-deviation* decline in the policy rate. The sources do not equate that to a 1 percentage point cut, so the deck's unit is not the study's.

Step 2 – Check the claim's reach

The BIS result is a cross-country regression over 45 countries from 2010 to 2019. It reports an estimated association ("is associated with"), not a guaranteed effect in every market.

$\Rightarrow$ The deck converts a standardised, sample-wide association into a per-point promise for every market.

Interpretation

The sign and the number in the deck are right; the unit and the reach are wrong, which is why the statement sounds like the reading.

Why the other options are wrong

B is incorrect. Keeps the unit error and misuses a CI-3 a fact: the US share of originations (nearly 90%) describes the market, not the scope of a regression that covers 45 countries.

C is incorrect. Polarity flip: the policy rate coefficient is negative (-0.03), so a decline goes with *more* private credit.

D is incorrect. Sibling swap: 12% is the policy rate's effect; under 1% belongs to corporate leverage.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact.

• **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.a and 102.b

Solution CI-3 b.8

Answer: B

New • built from SchweserNotes Book 5, LO 102.b

Step 1 – The narrative fact

Corporate leverage has risen alongside private credit and has been closely correlated with PCF lending; PCFs are more willing than banks to lend to highly leveraged firms.

Step 2 – The regression fact

In the regression, which includes the other three drivers, leverage has a very small positive coefficient (0.001) and the weakest relationship: a one-standard-deviation rise is associated with less than a 1% increase.

⇒ Co-movement over time is not the same as a large estimated effect.

Interpretation

The strategist reasons from a trend. Once the regression holds the other drivers fixed, leverage adds almost nothing; the FI index and the policy rate do the work.

Why the other options are wrong

A is incorrect. Wrong input twin: leverage has the *smallest* coefficient, and 12% is the policy rate's effect.

C is incorrect. Polarity flip: the leverage coefficient is positive, and PCFs are *more* willing than banks to lend to leveraged firms.

D is incorrect. True premise, wrong conclusion: PCFs' appetite for leveraged borrowers is in the reading, but leverage (under 1%) and the FI index (about 33%) are at opposite ends of the ranking.

LO: CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact. • **Reference:** BIS (February 2025); SchweserNotes 2026 Book 5, Reading 102, LO 102.b

Solution CI-3 b.9 — Answer: C

Workbook A28.3 • S1, S2 • Aug-2026 • Map #49

Step 1 – Rank the growth drivers by economic effect

FI index (bank efficiency): one s.d. decline, about $+33\%$ (largest). Policy rate: one s.d. decline, about $+12\%$ (second). Regulatory stringency: one s.d. increase, about $+2\%$ to 4%. Corporate leverage: under 1%.

⇒ Private credit grows most where banks are least effective at meeting loan demand.

Step 2 – Decompose the cost of capital

BDC cost of equity has stayed below banks' since the GFC; BDC cost of debt has been marginally above banks'. Bank CoE rose (market risk, stricter regulation); BDCs raised leverage, shifting the mix from expensive equity to cheaper debt, with only a modest rise in their CoE.

Step 3 – Attribute the 250 bp

The lower BDC CoE relative to banks and the increase in BDC leverage contributed roughly equally; the CoD played almost no role because the CoD gap stayed small while policy rates fell.

Interpretation

C is correct. For the board: the bank's lost advantage is an equity-cost and leverage story, not a deposit-cost story, and it may partly reverse if rates stay high.

Why the other options are wrong

A is incorrect. Sibling swap: the 33% effect belongs to the FI index; the policy rate is second, at about 12%.

B is incorrect. Polarity flip: BDCs' cost of debt has been marginally above banks', and the cost of debt played almost no role in narrowing the gap.

D is incorrect. Scope/condition error: floating-rate funding protects PCF margins when rates are low; when rates rise, BDC funding costs rise with policy rates faster than bank deposits, so BDCs could lose the cheap-debt benefit and the advantage may shift back.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes Book 5, Readings 102.1 and 102.2

CI-3 • The Global Drivers of Private Credit (BIS, February 2025) • Vardeez: VERY IMPORTANT

CI-3 c

P1 CONCEPT

Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them.

Workbook: A28 Private Credit, Private Markets and Their Drivers • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: private credit came up in the August 2026 sitting, and Map #49 lists it twice in the same consolidated sheet. A28 shape S2 (bank disintermediation) is this LO. **Vardeez:** "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **A28·S2** Bank disintermediation: private credit versus bank loans, cost of equity, cost of debt and leverage. Workbook A28.3 (tagged S1 and S2) is worked as **Question CI-3 b.9**; its key, the roughly equal split of the 250 bp erosion between the cost-of-equity gap and BDC leverage, is the core result of this LO. Do that question with this block. The questions below test the other elements and do not repeat its key: the equal split itself is never offered as an option here.

What the questions look like: no calculations. Statement audits on the three legs (which way each bank-BDC gap runs and who carries more leverage), the mechanism by which leverage lowered BDCs' cost of capital, the difference between the cost of debt's effect on a BDC's own cost of capital and its effect on the gap with banks, and what rising policy rates would do.

1. Private credit loans versus bank loans

Recap from CI-3 a: private credit funds (PCFs) lend directly to small and mid-sized firms and hold the loans for their full term; the core strategy is direct lending, typically floating-rate with extensive covenants. The reading's comparison with bank loans:

	Private credit loan	Bank loan
What it offers	More flexible terms, tailored covenants, faster execution; customised lending solutions	Stricter lending standards
Typical borrower	Often declined by banks: limited tangible collateral, high leverage and/or lower accounting transparency	Safer borrowers
Price	Interest rates often *higher* than on bank loans	Often lower
Collateral	May still be required; for collateral-poor small firms, reduced or removed in favour of targeted covenants (CI-3 b)	Banks reluctant to lend to small firms that lack sufficient collateral (CI-3 b)
Lender's funding	Equity plus debt whose cost moves in step with policy rates (BDCs)	Customer deposits, whose cost adjusts gradually; significantly higher leverage

Two further points from the sources. Repayment periods on private credit loans vary with the borrower's needs; they are not uniformly longer. And the loans in the intermediation chain (CI-3 a, Figure

1) are floating-rate and covenant heavy, with few lenders and a degree of control for the lender. The borrower's reason to choose private credit is **fit and speed, not price**.

2. The cost-of-capital framework

The cost of capital is the weighted average cost of capital (WACC): the cost of debt (CoD) and the cost of equity (CoE), weighted by their shares in the capital structure.

$$\text{WACC} = w_E\,\text{CoE} + w_D\,\text{CoD}, \qquad w_E + w_D = 1$$

Here, higher leverage means a higher debt share w_D. For both banks and BDCs debt is cheaper than equity, so shifting the mix toward debt pulls WACC down unless the cost of equity rises by enough to offset it. The reading's analysis compares **US banks** with **business development companies (BDCs)**, the often-listed, retail-facing part of the US private credit market (CI-3 a). The formula is the reading's verbal definition written out; the questions in this LO ask for no calculation.

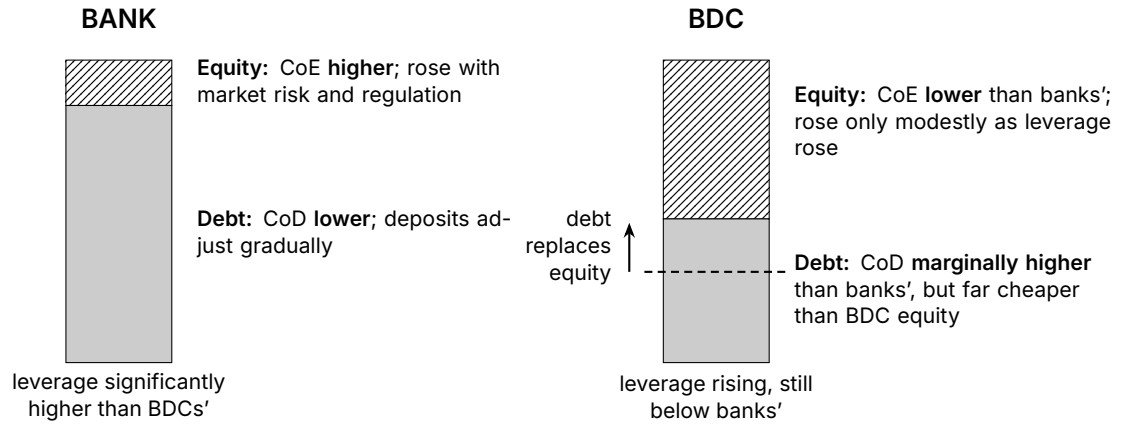


Figure 1. The capital stacks side by side, redrawn from the class notes' WACC sketches (Figure CI-3.3). Hatched: equity; grey: debt. The dashed line in the BDC bar marks the earlier, less levered mix; the arrow shows the debt share rising. The BDC wins on the expensive leg (cheaper equity) and loses on the cheap one (dearer debt), and by levering up it moves weight from the expensive leg to the cheap one. Proportions are illustrative; the sources give no capital ratios.

3. The three legs, bank against BDC

Leg	US banks	BDCs	Role in narrowing the gap
Cost of equity	Higher; rose significantly in 2010 to 2019 on heightened market risk and stricter regulatory requirements	Consistently lower than banks' since 2007 to 2009; rose only modestly as leverage rose	About half: the lower BDC CoE relative to banks
Cost of debt	Lower; customer deposits adjust gradually	Marginally higher for two decades; moves in step with policy rates; gap widens in tightening	Almost none: the CoD gap stayed small while rates fell
Leverage	Significantly higher: a high CoE pushed banks onto cheaper debt	Rising: equity replaced with relatively cheaper debt	About half: the increase in BDC leverage
WACC	Historically lower: the funding advantage came from a lower CoD	Historically higher; converged with banks' over the past decade	Banks' advantage narrowed by about 250 bp, 2010 to 2019

How leverage lowered BDCs' cost of capital. BDCs shifted their funding mix away from more expensive equity toward relatively cheaper debt. Their CoD is still higher than banks', but debt is still *substantially cheaper for BDCs than their own equity*. Substituting debt for equity therefore lowered their WACC. The extra leverage caused only a modest rise in BDCs' CoE, which was more than offset by the benefit of relying more on lower-cost debt. The comparison that matters is BDC debt against BDC equity, not BDC debt against bank debt.

What it did to competition. Lower funding costs let BDCs offer more competitive loan rates. This eroded banks' traditional funding advantage and supported the rising market share of private credit. "More competitive" need not mean below bank rates: SchweserNotes also notes that interest rates are often higher for private credit.

4. Attributing the 250 bp

Historically BDCs' cost of capital was higher than banks', but over the past decade the two have converged. Banks once had a substantial funding advantage because their CoD was lower; between 2010 and 2019 that advantage narrowed by roughly **250 basis points**. The reading splits the narrowing:

- the **lower CoE of BDCs relative to banks** (bank CoE rose with heightened market risk and stricter regulation), and
- the **increase in BDC leverage**,

contributed **roughly equally**. The **CoD played almost no role** in narrowing the gap, because the difference between bank and BDC CoD remained small while policy rates were falling.

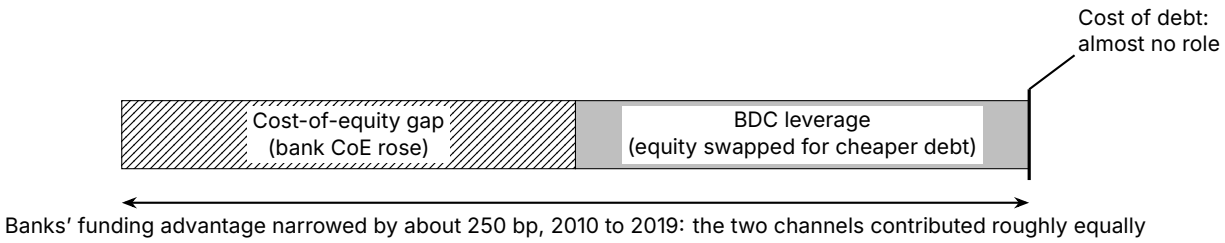


Figure 2. The attribution of the 250 bp. Hatched: the cost-of-equity channel; grey: the leverage channel; the heavy line at the right end is the cost-of-debt channel, drawn at zero width because it played almost no role. The two channels are drawn at half each to show "roughly equally"; the sources give no exact split.

Level versus gap. Cheap debt did lower BDCs' *own* cost of capital, which is how leverage worked. What played almost no role is the *difference* between bank and BDC CoD, which barely moved while rates fell. A statement that "cheap debt helped BDCs, so the cost of debt drove the narrowing" confuses the two.

Reading SchweserNotes' two statements together. The module quiz names the rise in banks' CoE as the primary reason the gap narrowed. The text says the lower BDC CoE relative to banks and the rise in BDC leverage contributed roughly equally. Read together: bank CoE rose significantly, while, in the quiz answer's words, "BDCs did not experience comparable increases in cost of equity", so the CoE gap in BDCs' favour widened *because* bank CoE rose. If a question offers the bank CoE rise but not BDC leverage, choose the CoE rise; if it offers both, the answer is "roughly equally".

5. The rate switch

The story depends on the direction of rates.

	Falling or low policy rates (2010 to 2019)	Rising policy rates (tightening)
CoD gap	Small and stable: BDC CoD only marginally above banks'	Tends to widen: BDC funding costs rise in step with policy rates; deposits adjust more gradually
Effect on BDCs	Cheap debt lowers BDC WACC; leverage pays	BDCs could gradually lose the benefit of cheap debt and face a higher WACC
Competitive dynamic	Banks' funding advantage erodes; private credit gains share	The dynamic may shift back toward banks

Do not import the CI-3 b channel into this table. In CI-3 b, *low* rates squeeze *bank* margins because floating-rate loan values fall faster than fixed-rate deposits, while PCFs, floating on both sides, escape. Here, *rising* rates raise *BDC* funding costs faster than bank deposit costs. Floating-rate funding helps the funds in one rate regime and hurts their cost of capital in the other.

Results card: CI-3 c

Private credit loan	Flexible terms, tailored covenants, faster execution; borrowers with limited tangible collateral, high leverage or low transparency; often priced above bank loans
Orderings (BDC vs bank)	CoE lower • CoD marginally higher • leverage lower (banks significantly higher)
Why bank CoE rose	Heightened market risk and stricter regulatory requirements
How leverage helped BDCs	Debt substantially cheaper than BDC equity; modest CoE rise more than offset
The 250 bp (2010 to 2019)	CoE gap and BDC leverage roughly equal; CoD almost no role
WACC	Historically BDC higher; converged over the past decade
Rate switch	Tightening widens the CoD gap; BDCs could lose cheap debt; dynamic may shift toward banks

Exam traps

- **Polarity flips on the two costs.** BDC CoE is *below* banks'; BDC CoD is *marginally above* banks'. Expect each reversed.
- **Level versus change in leverage.** BDCs *increased* leverage, but banks still carry *significantly higher* leverage. "BDCs are more levered than banks" is a role misassignment.
- **Wrong comparator.** Leverage lowered BDC WACC because BDC debt is cheaper than BDC *equity*, not because it is cheaper than bank debt.
- **Level versus gap.** Cheap debt lowered BDCs' own WACC; the CoD *gap* played almost no role in the narrowing.
- **Offset overstated.** Higher leverage raised BDC CoE only modestly; it did not cancel the gain.
- **Price.** Private credit loans are often *more* expensive than bank loans. Borrowers pay for flexi-

bility and speed.

- **Absolute claim on collateral.** Collateral may still be required; "no collateral" overreaches.
- **Convergence of what.** The costs of capital converged; the CoE gap persisted.
- **Rate direction.** Rising rates widen the CoD gap and threaten BDCs' advantage; the low-rate margin squeeze belongs to banks.

Practice questions: CI-3 c

Question CI-3 c.1 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

The CFO of a highly leveraged software company with few tangible assets is choosing between a bank term loan and a direct loan from a private credit fund. Based on the reading, which of the following statements best describes the trade-off she faces?

A. The private credit loan is likely to cost more, but it offers more flexible terms, tailored covenants and faster execution.

B. The private credit loan is likely to cost less, because BDCs' cost of debt has been consistently below banks' cost of debt.

C. The private credit loan will need no collateral at all, because private credit funds rely solely on covenants for protection.

D. The bank loan is likely to close faster, because deposit funding gives the bank a cheaper, slower-adjusting source of money.

Question CI-3 c.2 Adapted from CI QB VI Q92 • SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

A relationship manager at a UK commercial bank reviews why several mid-sized clients refinanced with private credit funds over the past year. Which of the following client profiles is most consistent with the borrowers that the reading says private credit serves?

A. A firm with abundant tangible collateral and audited accounts that is seeking the lowest available interest rate.

B. An investment-grade issuer seeking a large facility syndicated across many lenders at the tightest spread.

C. A firm with limited tangible collateral, high leverage and less transparent accounts that needs customised terms.

D. A highly leveraged firm seeking a fixed-rate loan with no covenants, so that it keeps full control of its operations.

Question CI-3 c.3 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c (Figure CI-3.3)

An equity analyst covering US banks and business development companies (BDCs) summarises the BIS comparison of their funding since the 2007 to 2009 financial crisis. Which of the following summaries is correct?

A. BDC cost of equity is lower than banks'; BDC cost of debt is marginally higher; BDCs carry more leverage than banks.

B. BDC cost of equity is higher than banks'; BDC cost of debt is marginally higher; banks carry more leverage than BDCs.

C. BDC cost of equity is lower than banks'; BDC cost of debt is lower; banks carry more leverage than BDCs.

D. BDC cost of equity is lower than banks'; BDC cost of debt is marginally higher; banks carry more leverage than BDCs.

Question CI-3 c.4 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

A strategy consultant tells the board of a US regional bank that the bank's historical cost-of-capital edge over BDCs "is intact, because our debt is still cheaper". Which of the following responses best reflects the BIS evidence?

A. The premise and the conclusion both hold: a lower cost of debt still gives banks the clearly lower overall cost of capital.

B. The debt edge remains, but the costs of capital have converged as banks' cost of equity rose with market risk and regulation.

C. The debt edge remains, but the costs of capital converged because BDCs' cost of equity fell sharply after the crisis.

D. The debt edge remains, but the advantage has gone because banks cut their leverage and replaced cheap debt with equity.

Question CI-3 c.5 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

A treasury analyst at a BDC reviews two decades of funding-cost data for BDCs and US banks. Which of the following statements is correct to include in her report?

A. BDC cost of debt has been marginally above banks', and the gap tends to widen when policy rates rise, as deposit costs adjust gradually.

B. BDC cost of debt has been marginally above banks', and the gap tends to narrow when policy rates rise, as deposits reprice faster than market debt.

C. BDC cost of debt has been marginally below banks', because banks pay a premium on deposits to offset their much higher leverage.

D. BDC cost of debt has been well above banks', so much so that debt has become more expensive than equity as a source of BDC funding.

Question CI-3 c.6 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

The CFO of a listed BDC is asked by analysts how the firm's move toward more debt financing affected its cost of capital. Which of the following explanations is correct?

A. More debt lowered the BDC's cost of equity, because lender covenants and monitoring reduced the risk borne by its shareholders.

B. More debt left the cost of capital unchanged, because the rise in the cost of equity fully offset the benefit of the cheaper debt.

C. More debt lowered it: BDC debt is much cheaper than BDC equity, and the modest rise in its cost of equity was more than offset.

D. More debt lowered it because BDC debt costs adjust as gradually as bank deposits, so they stayed low as rates moved.

Question CI-3 c.7 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

A junior analyst writes: "Cheap debt was central to the fall in BDCs' cost of capital, so the cost of debt must have been one of the main channels that narrowed BDCs' gap with banks." Which of the following corrections is most appropriate?

A. The reasoning is right: the cost of debt was the main channel, because BDC debt costs fell faster than bank deposit costs as rates declined.

B. Cheap debt did lower BDCs' own cost of capital, but the cost-of-debt gap with banks stayed small as rates fell, so it barely narrowed the gap.

C. The reasoning fails because cheap debt played no part in BDCs' cost of capital at all; the whole change came from banks' rising cost of equity.

D. The reasoning fails because the cost-of-debt gap widened in banks' favour over 2010 to 2019, which pushed the two costs of capital further apart.

Question CI-3 c.8 New • built from SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

The head of middle-market lending at a US bank asks how the fall in BDCs' funding costs has shown up in the competition for borrowers. Which of the following answers is most consistent with the reading?

A. Lower funding costs let BDCs offer more competitive loan rates, eroding banks' funding advantage and supporting private credit's share.

B. BDCs won share mainly by dropping covenants, since their higher cost of capital kept their loan rates far above what banks would charge.

C. Lower funding costs let BDCs undercut banks on investment-grade syndicated loans, the core of the banks' corporate lending book.

D. BDCs won share mainly through lower collateral demands, because their cost of capital has stayed well above banks' since the crisis.

Question CI-3 c.9 Adapted from CI QB VI Q100 • SchweserNotes Book 5, LO 102.c; CI notes CI-3 c

A strategist at a Canadian bank is assessing how a sustained rise in policy rates would affect the competitive position of US banks against BDCs. Which of the following assessments is most consistent with the reading?

A. BDCs could gain further ground, because higher rates would lift banks' cost of equity again, as regulation did in 2010 to 2019.

B. BDCs would be squeezed through the same margin-compression channel that hurt banks when policy rates were low.

C. Little would change, because the cost of debt played almost no role when the cost-of-capital gap narrowed in 2010 to 2019.

D. BDCs could gradually lose the benefit of cheap debt and face a higher cost of capital, shifting the dynamic toward banks.

Solutions • CI-3 c

Solution CI-3 c.1 — Answer: A

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – What private credit offers

Private credit loans typically offer more flexible terms, tailored covenants and faster execution, aimed at borrowers that banks often decline.

Step 2 – What it costs

Interest rates are often higher for private credit than for bank loans. Borrowers accept the higher price for flexibility and speed.

$\Rightarrow$ A highly leveraged firm with few tangible assets is the textbook private credit borrower, and it should expect to pay up.

Interpretation

The CFO is choosing between price and fit. The reading places the private credit advantage in terms and execution, not in cost.

Why the other options are wrong

B is incorrect. Polarity flip: BDC cost of debt has been *marginally above* banks', and private credit rates are often higher than bank rates.

C is incorrect. Absolute claim: collateral may still be required. PCFs reduce or remove collateral requirements for some borrowers; they do not drop them as a rule.

D is incorrect. True premise, wrong conclusion: bank deposit funding is cheaper and adjusts more gradually, but faster execution is a feature of private credit loans.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.2 — Answer: C

Adapted from CI QB VI Q92 • SchweserNotes Book 5, LO 102.c

Step 1 – The private credit borrower

Private credit targets borrowers often declined by banks: limited tangible collateral, high leverage and/or lower accounting transparency.

Step 2 – Why they fit PCFs

These traits make them less appealing to banks but better suited to PCFs that specialise in customised lending.

$\Rightarrow$ The client with thin collateral, high leverage and opaque accounts is the one the bank's standards screen out.

Interpretation

The relationship manager has not lost the bank's natural borrowers. The clients that moved are those the bank's stricter lending standards made hard to serve.

Why the other options are wrong

A is incorrect. Polarity flip: abundant collateral and audited accounts describe the bank-friendly borrower, and price is not the private credit draw.

B is incorrect. Sibling swap: a large investment-grade facility with many lenders is the syndicated market; private credit loans have few lenders.

D is incorrect. Polarity flip on the loan form: private credit direct lending is typically floating-

rate with extensive covenants, not fixed-rate and covenant-free.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.a and 102.c

Solution CI-3 c.3 — Answer: D

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – Cost of equity

Since the 2007 to 2009 financial crisis, BDC CoE has remained consistently lower than banks'.

Step 2 – Cost of debt

Over the past two decades, BDC CoD has been consistently, though only marginally, higher than banks'.

Step 3 – Leverage

Because banks face the higher CoE, they have relied increasingly on lower-cost debt, which has given them significantly higher leverage relative to BDCs.
$\Rightarrow$ BDC: cheaper equity, dearer debt, less leverage.

Interpretation

The leverage row is the one most often reversed. BDCs *increased* their leverage, which is part of the 250 bp story (CI-3 b.9), but they started from a lower level and banks remain the more levered party.

Why the other options are wrong

A is incorrect. Role misassignment on leverage: the first two cells are right, but banks, not BDCs, carry significantly higher leverage.
B is incorrect. Polarity flip on the cost of equity: BDC CoE has stayed consistently *below* banks' since the 2007 to 2009 crisis; the other two cells are right.
C is incorrect. Polarity flip on the cost of debt: BDC CoD is marginally *above* banks'.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.4 — Answer: B

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – Check the premise

Banks' CoD is still lower than BDCs'. Their historical funding advantage came from that lower CoD.

Step 2 – Check the conclusion

Historically BDCs' cost of capital was higher than banks', but over the past decade the two have converged. Between 2010 and 2019 banks' CoE rose significantly, driven by heightened market risk and stricter regulatory requirements.
$\Rightarrow$ A cheaper-debt edge on one leg no longer guarantees a lower overall cost of capital.

Interpretation

The consultant looks at one leg of the WACC. The bank lost ground on the other leg (equity) and on the mix (BDC leverage), so the board should not rely on the funding edge.

Why the other options are wrong

A is incorrect. Scope error: the advantage was real historically, but the two costs of capital have

since converged.

C is incorrect. Wrong mover on the cost of equity: BDC CoE has stayed below banks' since the crisis, but it was banks' CoE that rose; BDCs' CoE rose only modestly as they levered up. SchweserNotes' module quiz offers "decrease in BDC cost of equity" as a wrong answer to this question.

D is incorrect. Role misassignment: banks have *higher* leverage. It was BDCs that changed their mix, replacing equity with cheaper debt.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.5 — Answer: A

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – The ordering

Over two decades, BDC CoD has been consistently higher than banks', though only marginally.

Step 2 – The rate cycle

The gap tends to widen during monetary tightening. BDC funding costs rise in step with policy rates, while bank funding costs, particularly customer deposits, adjust more gradually.

⇒ Sticky deposits are the banks' cheap leg, and they matter most when rates rise.

Interpretation

This is the mechanism behind the rate switch: in a tightening cycle the CoD gap, which stayed small while rates fell, starts to work in banks' favour.

Why the other options are wrong

B is incorrect. Polarity flip on the cycle: deposits adjust *more gradually*, so the gap widens, not narrows, when rates rise.

C is incorrect. Polarity flip: BDC CoD is marginally *above* banks'. Deposits are the cheap source, not one that carries a premium.

D is incorrect. Overstates the gap: it is only marginal, and debt remains substantially cheaper than equity for BDCs. That is why BDC leverage lowered their WACC.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.6 — Answer: C

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – The right comparator

BDC debt costs more than bank debt, but it is still substantially cheaper than BDC *equity*. Shifting the mix from equity to debt moves weight onto the cheaper leg of the WACC.

Step 2 – The offset

The extra leverage caused only a modest rise in BDCs' CoE, and this was more than offset by relying more on lower-cost debt.

⇒ Net effect: a lower WACC, which supported more competitive loan rates.

Interpretation

BDC debt costs more than bank debt, which is the trap behind the question. That comparison is irrelevant to the BDC's own WACC; what matters is BDC debt against BDC equity.

Why the other options are wrong

A is incorrect. Polarity flip: higher leverage *raised* BDCs' CoE, if only modestly.
B is incorrect. Offset overstated: the CoE rise was more than offset, so the WACC fell.
D is incorrect. Polarity flip on the cost of debt: BDC funding costs move in step with policy rates, while it is bank deposits that adjust gradually. What made leverage pay is BDC debt against BDC equity.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.7

Answer: B

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – Separate the level from the gap

Cheap debt lowered BDCs' own cost of capital: that is how higher leverage worked. The narrowing of the bank-BDC gap is a different question.

Step 2 – What the cost of debt did to the gap

The CoD played almost no role in narrowing the gap, because the difference between bank and BDC CoD stayed small while policy rates were falling.
⇒ A channel can matter for one side's level and barely at all for the difference between the two sides.

Interpretation

The attribution itself (CoE gap and BDC leverage roughly equal) is Workbook A28.3, worked as Question CI-3 b.9. This question tests the distinction that the attribution rests on.

Why the other options are wrong

A is incorrect. True premise, wrong conclusion: BDC funding costs move in step with policy rates, so they can fall faster than deposit costs, but the CoD gap stayed small and the cost of debt was not the main channel.
C is incorrect. Overcorrection: cheap debt did help, through BDCs' own WACC. It did little to the *gap*.
D is incorrect. Condition error: the CoD gap tends to widen in *tightening*; while rates fell in 2010 to 2019 it stayed small and the two costs of capital converged.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.8

Answer: A

New • built from SchweserNotes Book 5, LO 102.c

Step 1 – From funding cost to loan price

The decline in BDCs' funding costs enabled them to offer more competitive loan rates.

Step 2 – From loan price to market share

That eroded banks' traditional funding advantage and supported the rising market share of private credit.
⇒ The cost-of-capital story is the bridge between the funding statistics and the share data.

Interpretation

"More competitive" need not mean cheaper than a bank: private credit rates are often higher, and

borrowers still come mainly for flexibility and speed.

Why the other options are wrong

B is incorrect. Polarity flip twice: private credit loans come with tailored, often extensive covenants, and BDCs' cost of capital has converged with banks' rather than staying far above it.

C is incorrect. Sibling swap: investment-grade syndicated lending is not the private credit market; PCFs serve borrowers banks often decline.

D is incorrect. Mixes a true feature with a false premise: collateral can be reduced for some borrowers, but BDCs' cost of capital has converged with banks', not stayed well above.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.c

Solution CI-3 c.9 Answer: D

Adapted from CI QB VI Q100 • SchweserNotes Book 5, LO 102.c

Step 1 – What rising rates do to the cost of debt

BDC funding costs rise in step with policy rates, while bank deposit costs adjust more gradually, so the CoD gap tends to widen in tightening.

Step 2 – What that does to the competitive dynamic

BDCs could gradually lose the benefit of cheap debt financing and face a higher overall cost of capital. The dynamic may shift back toward banks.

$\Rightarrow$ The BDC advantage was built in a falling-rate world and is rate-dependent.

Interpretation

For the strategist, the low-rate decade is not a guide to a tightening cycle: the leg that played no role then (the CoD gap) is the one that moves now.

Why the other options are wrong

A is incorrect. Condition error: banks' CoE rose in 2010 to 2019 on heightened market risk and stricter regulation, not because of rising rates. In a tightening cycle the moving leg is the cost of debt, and it moves against BDCs.

B is incorrect. Role misassignment: the margin compression from floating-rate loans against fixed-rate deposits was a bank problem in a low-rate world. The BDC risk in tightening runs through the cost of debt.

C is incorrect. Condition error: the CoD gap stayed small *while rates were falling*. Rising rates are the condition under which it widens.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. • **Reference:** BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes 2026 Book 5, Reading 102, LO 102.b and 102.c

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 a

P3 CONCEPT

Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D30 Geopolitical Risk and Financial Stability carries 2 questions (Vol 2 p.184). The Priority Map has no entry for CI-4, so the band follows the tier. D30·S1 (channels and policy responses) is this LO.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D30·S1** Channels from geopolitical risk to asset prices and financial stability, and policy responses (Question CI-4 a.8, workbook D30.1; its key also carries D30·S2, the bank chain that CI-4 f covers).

What the questions look like: no calculations. Classifying a scenario by channel (the trigger decides, not the outcome), statement audits on the scope of geopolitical risk, and matching each policy measure to the actor who takes it: country, financial institution or exposed entity.

Shared background for CI-4 sits here: the definition, the two channels, the policy set and a map of the reading's numbers (Section 4). CI-4 b to f recap in a line and point back.

1. What geopolitical risk is, and why it matters for stability

Definition. Geopolitical risk is the adverse effect on financial markets, trade relations and economic growth that arises from major conflicts: wars, terrorist actions, and trade disputes involving tariffs or sanctions. Trade disputes are inside the definition, which is why the US-China tariff measures are one of the reading's two case studies (CI-4 c).

The threat is enough. Even the threat of rising geopolitical tensions can depress asset prices and prompt a reallocation of capital flows, ultimately weakening financial stability. Nothing has to be destroyed or sanctioned first.

Why stability is at stake. Conflicts and other geopolitical shocks can cause disruption both at home and across borders through contagion. The vulnerabilities are more pronounced when economies are highly interconnected through trade-related revenue exposures and cross-border capital flows. Conflicts raise market, credit and liquidity risks, which leads to asset price declines, higher tail risk and tighter lending conditions in the affected economies (CI-4 f).

Modest on average, severe in the tail. Average stock market reactions to geopolitical tensions are often modest. Major conflicts or persistent shocks are different: they can trigger financial contagion, prolonged declines in economic activity and asset values, and higher risk premiums. The magnitudes are in CI-4 b and c.

2. Two channels to asset prices

1. **The economic channel** reflects disruptions to trade, capital flows, supply chains and production capacity. The reading's example is Russia's 2022 invasion of Ukraine: stock prices and sovereign bond values fell sharply, while commodity prices such as oil and grains rose on supply concerns. The same disruptions can raise sovereign debt burdens as governments boost military spending or lose fiscal revenue. Through this channel geopolitical risk depresses asset prices, raises sovereign risk premiums (CI-4 d) and weakens economic growth.
2. **The market sentiment channel** captures heightened uncertainty, increased risk aversion and reduced investor confidence. Even in the absence of direct conflict, the perception of instability can depress valuations and raise market volatility.

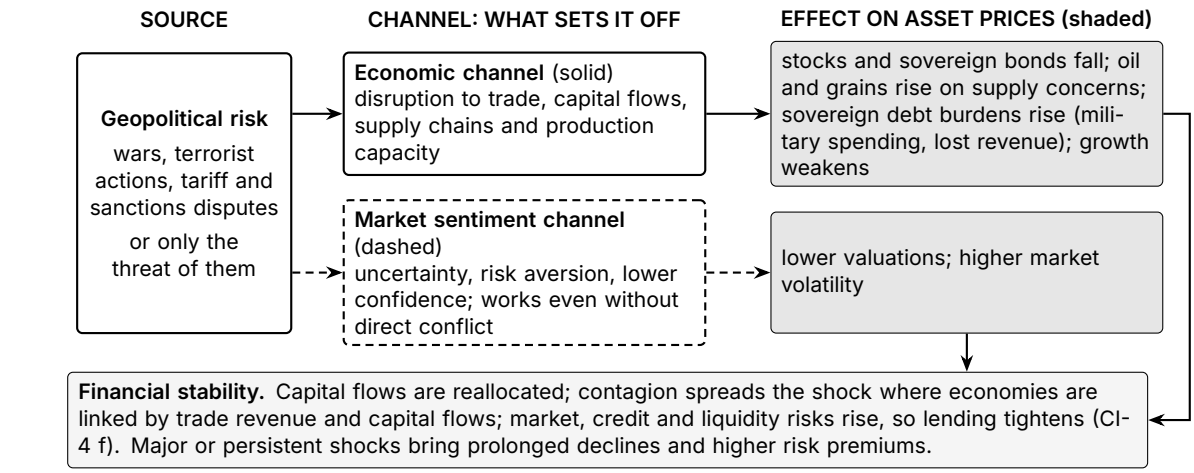


Figure 1. The two channels, after the class notes' Figure CI-4.1. Read the middle column: the channels differ in what sets them off, not in what they do to prices, since both can lower valuations. The economic channel (solid) runs through an actual disruption; the sentiment channel (dashed) can fire on the perception of instability alone. Both effect boxes feed the financial stability box.

How to tell them apart. Classify by the trigger. Cut trade routes, idle factories and lost fiscal revenue are the economic channel; uncertainty and risk aversion with no disruption are the sentiment channel. A fall in valuations on its own does not tell you which channel is at work. And the two channels are *economic* and *market sentiment*: fiscal and monetary channels belong to CI-5.

> **Correction to the source notes.** The caption to the class notes' Figure CI-4.1 says the sentiment channel "is the one that operates in markets far from the conflict". SchweserNotes says only that the sentiment channel works even in the absence of direct conflict; it does not make it the only route to distant markets. Cross-border effects are also transmitted through trade and financial linkages: countries geographically or economically close to the conflict, and firms with revenue exposure, subsidiaries or significant shareholders in the affected countries, see sharper declines (LO 103.c). SchweserNotes does not assign this transmission to either channel, so distance alone does not make a fall a sentiment effect.

3. Policy measures: who does what

The reading's policy advice is about resilience. Each measure belongs to an actor, and the questions test the allocation.

- **Entities exposed to geopolitical risk** should be able to assess, measure and manage those exposures.

- **Countries, particularly emerging market economies,** strengthen resilience by improving regulatory frameworks, preserving fiscal and monetary flexibility, and deepening financial markets.
- **Financial institutions** maintain adequate capital *and* liquidity buffers to absorb potential losses.

SchweserNotes' quiz answer adds what does *not* work on its own. Capital buffers alone are not sufficient, because institutions also need robust scenario analysis and stress testing. Fixed asset allocations limit responsiveness to rising risks. Reducing direct exposures does not address indirect channels of contagion or broader market spillovers. The same answer also has financial institutions supporting deeper and more liquid financial markets and stronger regulatory frameworks, so those two are not purely state measures; buffers, though, are always the institutions' own. CI-4 f adds the closing advice for banks and funds (avoid concentrated exposures, enhance risk management) and for policymakers (regulatory frameworks that account for geopolitical risk).

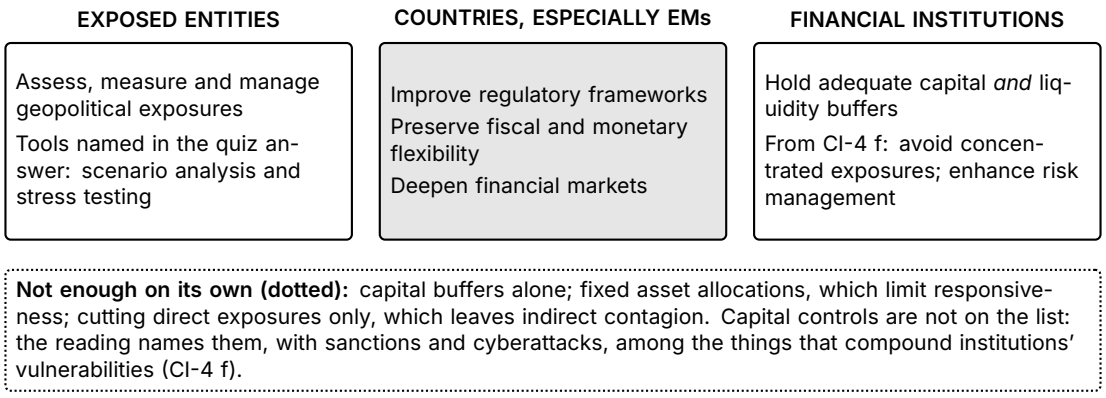


Figure 2. The policy set by actor. The shaded column is the country-level package in the LO text; buffers belong to the institutions, not to the state. The dotted box lists the partial answers that distractors offer as complete ones.

4. A map of the reading's numbers

CI-4 is the most number-heavy Current Issues reading, and several figures look alike but belong to different statistics. The numbers below are SchweserNotes'; the class notes thin them out. Each LO's results card repeats its own. The right-hand column names the look-alike that distractors use.

Statistic	Figure	LO	Keep apart from
Stock fall after major geopolitical events (event study)	about 3% immediately; up to 9% if severe	b	0.3% country-shock average
Aggregate stock fall after a country-specific shock	about 0.3% on average	c	3% event study; 1% domestic events
Large regional shock, at least 2 s.d. above normal	up to 7 times larger, more persistent	c	2.5 times proximity
Global shock hitting many countries at once	up to 20 times larger; up to six months	c	7 times regional
Domestic events within a month; EMs in military conflict	about 1%; about 5%	c	0.3% country-shock average; 1% bond fund returns
Countries close to the conflict	up to 2.5 times larger declines	c	2.5% subsidiary loss
Firms with subsidiaries in Russia or Ukraine, 2022	about 2.5% within a week	c	2.5 times proximity
China after US tariff announcements	about 4% average; nearly 8% in May 2019	c	8% fund loss, 8% claims

Statistic	Figure	LO	Keep apart from
US firms after tariff announcements	about 1.3% average; 1.6% to 1.8% in August 2019	c	China's 4%
Sovereign CDS within a month of major conflicts	EMs about 180 bp; AEs about 40 bp	d	100 and 120 bp
Below-median reserve adequacy; institutional quality	about 100 bp; about 120 bp	d	180 and 40 bp
Exposure to high-risk markets, 2024	8% bank claims; 10% bank liabilities; 13% equity fund assets	f	10% fund exposure
Funds' cut in direct Russia and Ukraine exposure	roughly 60% after 2022	f	6% one-week loss
Bond funds with 10% exposure to conflict countries	flows down 2.3%; returns down 1%	f	1% domestic events
Funds with 10% direct exposure to Russia and Ukraine	about 6% within a week; 8% over six months	f	60% exposure cut

Results card: CI-4 a

Definition	Adverse effects on markets, trade and growth from wars, terrorist actions, and tariff or sanctions disputes; the threat alone can depress prices and redirect capital flows
Economic channel	Disruption to trade, capital flows, supply chains, production capacity; Russia 2022: stocks and sovereign bonds down, oil and grains up; debt burdens up
Sentiment channel	Uncertainty, risk aversion, lower confidence; lower valuations and higher volatility even without direct conflict
Stability	Contagion stronger where economies are interconnected; modest average reactions, but major or persistent shocks bring contagion and higher risk premiums
Countries (esp. EMs)	Regulatory frameworks; fiscal and monetary flexibility; deeper financial markets
Institutions and entities	Capital and liquidity buffers; assess, measure, manage exposures; capital alone is not enough

Exam traps

- **Sibling swap with CI-5.** The channels are economic and market sentiment, not fiscal and monetary.
- **Classifying by outcome.** Both channels lower valuations. The trigger (actual disruption or perception) decides the channel.
- **Scope error on the definition.** Tariff and sanctions disputes and terrorist actions are inside it, not only wars.
- **Threat dismissed.** "Nothing happens until the conflict starts" is wrong: the threat alone can move prices and capital flows.
- **Role misassignment on policy.** Buffers belong to institutions, fiscal and monetary flexibility to the state. Market depth and regulatory frameworks are country measures in the LO text, though SchweserNotes' quiz answer also has institutions supporting them.

- **Absolute claim.** Capital buffers alone, fixed allocations, or cutting direct exposures only are partial answers.
- **Average for tail.** Average reactions are modest; that does not make major or persistent shocks harmless.

Practice questions: CI-4 a

Question CI-4 a.1 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

The head of country risk at a Toronto-based pension fund is drafting the fund's definition of geopolitical risk for its risk appetite statement, drawing on the IMF analysis. Which of the following statements is consistent with that analysis?

A. Geopolitical risk covers armed conflict and terrorist actions; tariff and sanctions disputes fall outside it and belong under trade policy risk.

B. Geopolitical risk covers wars, terrorism, and tariff or sanctions disputes, and its effects grow where economies are linked by trade revenue and capital flows.

C. Geopolitical risk covers wars, terrorism, and tariff or sanctions disputes, and its effects shrink where economies are linked, as links spread the shock thinly.

D. Geopolitical risk covers wars, terrorism, and tariff or sanctions disputes, and its effects stay confined to the countries directly involved.

Question CI-4 a.2 Adapted from CI QB Imp Q4 • SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

Tensions between two large economies escalate sharply, but no shots have been fired and no sanctions or tariffs have yet been imposed. A portfolio manager at a Hong Kong asset manager tells the investment committee that no repositioning is needed until a conflict or trade measure actually begins. Which response best reflects the IMF analysis?

A. The manager is right: asset prices and capital flows respond once trade, capital flows or production are actually disrupted, and not before.

B. The manager is partly right: threats can raise volatility, but capital flows are reallocated only once sanctions or tariffs are formally imposed.

C. The manager is wrong: even the threat of rising tensions can depress asset prices and redirect capital flows, weakening financial stability.

D. The manager is wrong: threats move commodities such as oil and gold, though equity and bond valuations wait for the conflict itself.

Question CI-4 a.3 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

A financial stability analyst at a central bank is classifying two recent episodes. In Market X, a conflict next door cut trade routes and idled factories, and asset prices fell. In Market Y, which has no direct involvement in any conflict, heightened uncertainty and risk aversion lowered valuations and raised volatility. Which classification is consistent with the IMF's two-channel framework?

A. Both episodes run through the economic channel, because in each market asset prices fell after the geopolitical event.

B. Market X runs through the market sentiment channel, and Market Y runs through the economic channel.

C. Market X runs through the economic channel; Market Y shows no geopolitical effect, because nothing there was disrupted.

D. Market X runs through the economic channel, and Market Y runs through the market sentiment channel.

Question CI-4 a.4 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

A commodities strategist at a Dubai trading house is writing a client note on how the IMF uses Russia's 2022 invasion of Ukraine to illustrate the economic channel. Which description of the market moves is consistent with the reading?

A. Stock prices and sovereign bond values fell sharply, while oil and grain prices rose on supply concerns.

B. Stock prices fell sharply, while the region's sovereign bonds rallied as investors sought safe-haven assets.

C. Stock prices and sovereign bond values fell sharply, while oil and grain prices fell as global demand weakened.

D. Oil and grain prices rose, which the reading treats as a sentiment effect because no supply was actually lost.

Question CI-4 a.5 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

At a macroprudential committee meeting, a board member argues that because average stock market reactions to geopolitical events are modest, geopolitical risk deserves no place in the financial stability review. Which response best reflects the IMF analysis?

A. The member is right: the reading finds that geopolitical shocks seldom move risk premiums or economic activity beyond the first few days.

B. The member is wrong: reactions are modest on average, but major or persistent shocks can bring contagion and higher risk premiums.

C. The member is wrong: the average reaction is large, a fall of about 9% after a typical event, so every shock threatens stability.

D. The member is right for advanced economies, because the reading confines geopolitical stability concerns to emerging markets.

Question CI-4 a.6 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

An IMF mission chief is drafting recommendations to strengthen an emerging market economy's resilience to geopolitical shocks. The economy has a large banking sector and shallow capital markets. Which allocation of measures is consistent with the reading?

A. The authorities hold capital and liquidity buffers on the banks' behalf; the banks deepen domestic markets and preserve the state's fiscal flexibility.

B. The authorities impose capital controls ahead of any conflict; the banks cut direct exposures to at-risk countries, removing the contagion risk.

C. The authorities improve regulation, keep fiscal and monetary flexibility and deepen markets; banks hold adequate capital and liquidity buffers.

D. The authorities use their fiscal and monetary room early, before any conflict; the banks hold adequate capital and liquidity buffers.

Question CI-4 a.7 New • built from SchweserNotes Book 5, LO 103.a; CI notes CI-4 a

The CRO of a Polish bank tells the board that, because the bank's capital ratio is well above the regulatory minimum, no further work on geopolitical risk is needed. Which response best reflects the IMF analysis?

A. The view is sound: capital buffers absorb all plausible geopolitical losses, so any further measures add cost without reducing risk.

B. The view is incomplete: the bank should also fix its asset allocation so that its holdings cannot drift as geopolitical risk rises.

C. The view is misplaced: buffers are a national policy tool, and the bank's part is limited to helping deepen domestic financial markets.

D. The view is incomplete: the bank also needs liquidity buffers, and it must assess, measure and manage its exposures.

Question CI-4 a.8 Workbook D30.1 • S1, S2 • self-gen

The CRO of a German bank is preparing a board briefing on the IMF's analysis of geopolitical risk and financial stability after a surge in military tensions in a region where the bank has EUR 3.1 billion of cross-border claims. Which of the following statements should the CRO include in the briefing?

A. Geopolitical risk moves asset prices through an economic channel and a market sentiment channel, and falling asset and collateral values combined with capital outflows can tighten bank lending.

B. Geopolitical risk moves asset prices through a fiscal channel and a monetary channel, so the main policy response the IMF recommends is that central banks cut rates as soon as conflicts begin.

C. Because US-China tariff announcements caused no meaningful fund losses, funds with 10% direct exposure to Russia and Ukraine also avoided material losses after the 2022 invasion.

D. Since the 2014 annexation of Crimea, banks have steadily increased their direct exposure to Russia and Ukraine, raising their vulnerability when the 2022 invasion began.

Solutions • CI-4 a

Solution CI-4 a.1 — Answer: B

New • built from SchweserNotes Book 5, LO 103.a

Step 1 – The scope

Geopolitical risk is the adverse effect on financial markets, trade relations and economic growth from major conflicts: wars, terrorist actions, and trade disputes involving tariffs or sanctions.
⇒ Tariff and sanctions disputes are inside the definition.

Step 2 – What interconnection does

The vulnerabilities become more pronounced when economies are highly interconnected through trade-related revenue exposures and cross-border capital flows, because shocks spread across borders through contagion.

Interpretation

For a risk appetite statement the two halves matter equally: what counts as geopolitical risk, and where it bites hardest. Linkages are the transmission route, not a diversifier.

Why the other options are wrong

A is incorrect. Scope error: trade disputes involving tariffs or sanctions are named in the definition, and the US-China tariff measures are one of the reading's two case studies.
C is incorrect. Polarity flip: interconnection through trade revenue and capital flows makes the vulnerabilities more pronounced, not smaller.
D is incorrect. Scope error: disruptions spread across borders through contagion; firms and countries linked to the conflict also suffer (CI-4 c).

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.a

Solution CI-4 a.2 — Answer: C

Adapted from CI QB Imp Q4 • SchweserNotes Book 5, LO 103.a

Step 1 – What the reading says about threats

Even the threat of rising geopolitical tensions can depress asset prices and prompt a reallocation of capital flows, ultimately weakening financial stability.

Step 2 – Which channel carries it

The market sentiment channel works through uncertainty, risk aversion and lower confidence, and it can depress valuations and raise volatility even in the absence of direct conflict.
⇒ Waiting for the conflict means repositioning after prices have already moved.

Interpretation

The manager treats geopolitical risk as if it ran only through the economic channel, which needs an actual disruption. The sentiment channel and the threat effect are why prices can move first.

Why the other options are wrong

A is incorrect. Channel confusion: the economic channel does need a disruption, but the threat alone can already move prices and capital flows.
B is incorrect. Scope or condition error: the threat itself can prompt a reallocation of capital flows; formal sanctions or tariffs are not required.
D is incorrect. Scope error: the threat depresses asset prices generally, and the perception of instability depresses valuations, not only commodity prices.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.a

Solution CI-4 a.3 Answer: D

New • built from SchweserNotes Book 5, LO 103.a

Step 1 – Market X

Cut trade routes and idled factories are disruptions to trade and production capacity: the economic channel.

Step 2 – Market Y

Uncertainty, risk aversion and lower confidence with no direct involvement in conflict: the market sentiment channel. Even without direct conflict, the perception of instability can depress valuations and raise volatility.

⇒ Same outcome, different triggers, different channels.

Interpretation

Classify by what set the move off. Lower asset prices appear in both markets, so the outcome cannot separate the channels.

Why the other options are wrong

A is incorrect. Classifies by outcome: both channels lower valuations, and Market Y had no disruption to trade or production.

B is incorrect. Sibling swap: the two channels are exchanged; disruption is economic and uncertainty is sentiment.

C is incorrect. Denies the sentiment channel: it operates even in the absence of direct conflict, which is exactly Market Y's case.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.a

Solution CI-4 a.4 Answer: A

New • built from SchweserNotes Book 5, LO 103.a

Step 1 – The economic channel

It reflects disruptions to trade, capital flows and production capacity.

Step 2 – The 2022 example

Russia's invasion led to sharp declines in stock prices and sovereign bond values, while commodity prices such as oil and grains rose because of supply concerns. The disruptions can also raise sovereign debt burdens as governments boost military spending or lose fiscal revenue.

⇒ Risk assets fall and the commodities whose supply is threatened rise.

Interpretation

Safe-haven sovereign bonds do appreciate in geopolitical shocks, but those are the bonds of countries such as the United States, the United Kingdom, Germany, Switzerland and Japan (CI-4 d), not of the region at the centre of the conflict.

Why the other options are wrong

B is incorrect. Sibling swap: sovereign bond values in the example fell; the safe-haven rally belongs to safe-haven countries' bonds.

C is incorrect. Polarity flip: oil and grain prices rose on supply concerns.

D is incorrect. Channel swap: the reading presents the commodity moves, driven by supply concerns, as the economic channel at work.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); Schweser-Notes 2026 Book 5, Reading 103, LO 103.a

Solution CI-4 a.5 Answer: B

New • built from SchweserNotes Book 5, LO 103.a and Exam Focus

Step 1 – The average

Average stock market reactions to geopolitical tensions are often modest.

Step 2 – The tail

Major conflicts or persistent shocks can trigger financial contagion, prolonged declines in economic activity and asset values, and higher risk premiums. Conflicts also raise market, credit and liquidity risks, which tightens lending (CI-4 f).

$\Rightarrow$ A modest average does not make the risk small for financial stability.

Interpretation

The member reasons from the mean of the distribution; stability analysis is about the tail. The same reading that reports modest averages reports far larger effects for large regional and global shocks (CI-4 c).

Why the other options are wrong

A is incorrect. Absolute claim: major or persistent shocks bring prolonged declines and higher risk premiums; global shocks can persist for up to six months (CI-4 c).

C is incorrect. Wrong input twin: up to 9% is the ceiling for *severe* events in the event study, and the immediate fall after major events is about 3% (CI-4 b). Neither is a typical reaction.

D is incorrect. Scope error: emerging markets are hit harder, but the reading does not exempt advanced economies; European markets fell more sharply than other advanced economies' markets after the 2022 invasion (CI-4 c).

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); Schweser-Notes 2026 Book 5, Reading 103, Exam Focus and LO 103.a

Solution CI-4 a.6 Answer: C

New • built from SchweserNotes Book 5, LO 103.a

Step 1 – Country-level measures

Countries, particularly emerging market economies, strengthen resilience by improving regulatory frameworks, preserving fiscal and monetary flexibility and deepening financial markets.

Step 2 – Institution-level measures

Financial institutions maintain adequate capital and liquidity buffers to absorb potential losses.

$\Rightarrow$ The state holds the policy levers; the banks hold the buffers.

Interpretation

Deepening financial markets is the measure that speaks most directly to this economy's shallow capital markets. Preserving flexibility means keeping fiscal and monetary room available, not spending it in advance.

Why the other options are wrong

A is incorrect. Role misassignment: buffers sit on financial institutions' balance sheets and fiscal flexibility is the state's; the authorities cannot hold buffers for the banks.

B is incorrect. Polarity flip and scope error: the reading lists capital controls among the things that compound vulnerabilities (CI-4 f), and cutting direct exposures does not remove indirect contagion.

D is incorrect. Polarity flip: the reading asks countries to preserve fiscal and monetary flexibility, not to spend it in advance of a conflict.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); Schweser-Notes 2026 Book 5, Reading 103, LO 103.a and 103.f

Solution CI-4 a.7 — Answer: D

New • built from SchweserNotes Book 5, LO 103.a

Step 1 – Buffers, plural

Financial institutions should maintain adequate capital *and* liquidity buffers.

Step 2 – Managing the exposure

Entities exposed to geopolitical risk should be able to assess, measure and manage those exposures. Capital buffers alone are not sufficient, because institutions also need robust scenario analysis and stress testing.

⇒ A strong capital ratio is necessary but not the whole answer.

Interpretation

Geopolitical shocks raise liquidity risk as well as credit and market risk (CI-4 f), so a capital-only view leaves the funding side and the measurement of the exposure uncovered.

Why the other options are wrong

A is incorrect. Absolute claim: capital buffers alone are not sufficient.

B is incorrect. Polarity flip: fixed asset allocations limit an institution's ability to respond to rising risks; they are not a protection.

C is incorrect. Role misassignment: capital and liquidity buffers are the institutions' own measure, not a national tool, and the bank's part is not limited to market depth.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • **Reference:** IMF (April 2025); Schweser-Notes 2026 Book 5, Reading 103, LO 103.a (Module Quiz 103.1)

Solution CI-4 a.8 — Answer: A

Workbook D30.1 • S1, S2 • self-gen

Step 1 – Name the channels

The economic channel works through disruptions to trade, capital flows and production capacity; the market sentiment channel through uncertainty, risk aversion and lower investor confidence.

Step 2 – Trace the effect to banks

Higher volatility triggers selloffs that erode asset and collateral values, leading to tighter lending; capital outflows amplify volatility in less liquid markets; weaker borrowers reduce cross-border lending and raise rollover risk on foreign debt.

⇒ For this bank, the regional claims are exposed through both credit quality and collateral values, and funding can come under rollover pressure.

Interpretation

A is correct. The board should hear that adequate capital and liquidity buffers, limits on concentrated exposures and stronger risk management are the IMF's prescriptions; cutting direct exposure alone does not address indirect contagion.

Why the other options are wrong

B is incorrect. Sibling swap (case cross-mapping): fiscal and monetary channels belong to CI-5 (Monetary and Fiscal Policy); CI-4's channels are economic and market sentiment, and the reading's policy advice is resilience (regulatory frameworks, fiscal and monetary flexibility, deeper markets, buffers), not rate cuts.

C is incorrect. Scope/condition error: extends the tariff result to a war. Funds with 10% direct exposure lost about 6% within a week and 8% over six months after the invasion; only the tariff announcements caused no meaningful fund losses.

D is incorrect. Polarity flip: banks reduced their direct exposure to Russia and Ukraine over the decade after 2014.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. • CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF, Global Financial Stability Report, April 2025, Ch.2, via SchweserNotes Book 5, Reading 103, Modules 103.1 and 103.2

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 b

P3 CONCEPT

Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D30 Geopolitical Risk and Financial Stability carries 2 questions (Vol 2 p.184), and the Priority Map has no CI-4 entry. D30·S4 pairs sovereign risk premiums (CI-4 d) with the importer and exporter split, which is this LO.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D30·S4** Sovereign risk premiums, and commodity importers vs exporters (Question CI-4 b.8, workbook D30.2: option D tests this LO; the key, B, is the CI-4 d fact, and option A tests CI-4 e).

What the questions look like: no calculations. Who gains and who loses across three cuts (countries, sectors, asset classes), reading across the row: the same price spike that helps an exporter hurts an importer. Numbers come in as recall anchors, with the other CI-4 magnitudes as distractors.

Background: the two channels and the definition are in CI-4 a; the map of the reading's numbers is CI-4 a, Section 4.

1. Uneven by design

Geopolitical risk events affect countries unevenly, largely according to their **economic structure and commodity exposure**. The same event moves sectors and asset classes in opposite directions, and the **nature of the event** sets how hard and how long. This LO asks you to read each cut across: who gains, who loses, and why.

2. Countries: involvement, importers and exporters

- **Directly involved or closely linked countries** tend to suffer the most severe consequences: physical damage in the directly affected countries, sanctions, and disruptions to trade and investment.
- **Commodity exporters, particularly energy producers,** often benefit from the higher prices that accompany global tensions. Oil exporters gained from surging energy prices after Russia's 2022 invasion of Ukraine.
- **Commodity importers** typically face currency depreciation, rising sovereign CDS spreads and greater financial pressure. After 2022, European importers faced currency depreciation and rising sovereign spreads, which increased their vulnerability. Exchange rates of commodity importers weaken, which amplifies the vulnerability further.



Figure 1. One shock, three country outcomes. The shaded and dashed boxes are the importer and exporter split the LO names: the price rise that improves an exporter's position is the one that weakens an importer's currency and widens its sovereign spread (CI-4 d).

3. Sectors

- **Defence and energy** often benefit from geopolitical shocks, as military spending increases and commodity prices rise. US defence firms have historically maintained stable cash flows during conflicts because of steady government contracting.
- **Banking, retail and utilities** typically decline, reflecting weaker consumer demand, tighter credit conditions and valuation pressures. Do not file utilities with the winners: the reading lists them with the decliners.

The sector story has a geographic condition, worked in CI-4 c: after the 2022 invasion, defence and energy firms *with revenue exposure to Russia and Ukraine* fell sharply while the same sectors rose elsewhere. Sector and exposure have to be read together.

4. Asset classes

- **Commodities** such as oil and gold typically rise, on supply concerns and safe-haven demand.
- **Equities** generally decline, particularly in emerging markets. Event-study evidence suggests stocks fall about 3% immediately after major geopolitical events, with severe events causing declines of up to 9%.
- **Safe-haven sovereign bonds** often appreciate, so their yields fall. A rise in price is a fall in yield.
- **Exchange rates**, especially of commodity-importing countries, tend to weaken.

Correction to the source notes. The class notes' Table CI-4.1 reads: "a standard event causes about a 3% drop, and a severe shock can trigger declines of up to 9%". SchweserNotes states the 3% for *major* geopolitical events, as the immediate fall in an event study. It is not the typical reaction: the reading's averages are far smaller, about 0.3% after a country-specific geopolitical risk shock and about 1% within a month after domestic events (CI-4 c). Read "standard" as "typical" and the two statistics collide. Question CI-4 b.4 option C uses exactly that confusion.

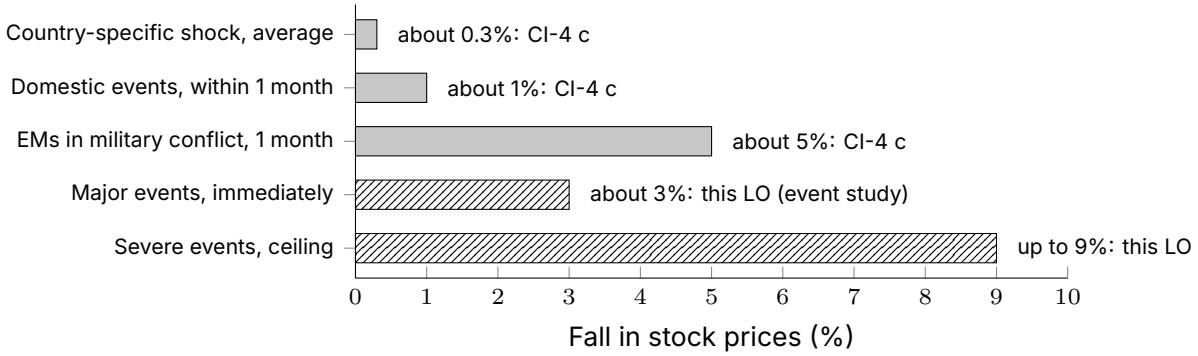


Figure 2. Five stock-market magnitudes from CI-4 on one scale. Only the hatched pair, from the event study of major events, belongs to this LO; the grey bars are CI-4 c statistics with different definitions and horizons. They are drawn together because distractors swap them, not because they measure the same thing.

5. Severity: the nature and length of the event

- **Wars and military interventions** produce more severe and longer-lasting economic consequences than sanctions or trade restrictions, because of loss of life, physical destruction and prolonged disruption.
- **Longer-lasting conflicts** lead to more persistent effects.
- **History since World War II.** Most major events produced modest and short-lived stock price declines. The exceptions were sharper and more sustained: the 1973 Arab oil embargo and Iraq's 1990 invasion of Kuwait, with declines that persisted for several months.

Results card: CI-4 b

Cut	Tends to gain	Tends to lose
Countries	Commodity exporters, especially energy producers (oil exporters after 2022)	Directly involved or closely linked countries most; commodity importers: currency depreciation, wider sovereign spreads (European importers)
Sectors	Defence (military spending; US firms' steady government contracting) and energy (commodity prices)	Banking, retail, utilities (weaker demand, tighter credit, valuation pressure)
Asset classes	Oil and gold; safe-haven sovereign bond prices (yields fall)	Equities, most in EMs: about 3% immediately after major events, up to 9% if severe; importer exchange rates
Severity	Wars and military interventions > sanctions or trade restrictions; longer conflicts, more persistent effects. Most post-1945 declines modest and brief; 1973 oil embargo and 1990 Kuwait invasion lasted several months.	

Exam traps

- **Role swap, importers and exporters.** Exporters gain from the price spike; importers suffer depreciation and wider spreads.
- **Utilities as a safe sector.** The reading lists utilities with banking and retail among the decliners.
- **Price-yield inversion.** Safe-haven bonds appreciate, so their yields *fall*.
- **EM versus AE equities.** Equity declines are most pronounced in emerging markets.
- **Wrong input twin on magnitudes.** 3% and 9% are the event study; 0.3%, 1%, 5%, 4% and 8% are other CI-4 statistics.
- **Severity reversed.** Wars outlast sanctions and trade restrictions, not the other way round.
- **Absolute history.** Most post-1945 events caused modest, short-lived declines; two named episodes lasted months.

Practice questions: CI-4 b

Question CI-4 b.1 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

A sovereign analyst at a Singapore bank is assessing three countries after a new war breaks out and pushes oil prices sharply higher. Country P borders the war zone and trades heavily with it; Country Q is a distant oil exporter and Country R a distant oil importer; neither has trade or financial links to the region. Which assessment is most consistent with the IMF analysis?

A. P suffers most; Q often benefits from higher oil prices; R tends to face currency depreciation and greater pressure.

B. P suffers most; Q and R are largely unaffected, because neither has any trade or financial links to the region.

C. Q suffers most, as sanctions and trade disruption fall mainly on oil producers; P and R are less affected by comparison.

D. All three are affected to a similar degree, because a global shock reprices risk in every market at once.

Question CI-4 b.2 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

An equity portfolio manager in Toronto is setting sector tilts ahead of a possible escalation of a regional war. Based on the IMF analysis of how sectors typically respond to geopolitical shocks, which set of tilts is most consistent?

A. Overweight utilities and defence as defensive holdings; underweight energy, which suffers as demand falls.

B. Overweight banking, which gains from wider lending margins; underweight both retail and defence.

C. Overweight retail on precautionary stockpiling by households; underweight energy and utilities.

D. Overweight defence and energy; underweight banking, retail and utilities.

Question CI-4 b.3 Adapted from CI QB Imp Q10 • SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

A multi-asset strategist at a Swiss private bank is writing a playbook for the first days after a major geopolitical shock. Which set of expected moves is consistent with the IMF analysis?

A. Oil and gold rise; equities fall, most in advanced markets; safe-haven bond yields rise; importer currencies weaken.

B. Oil and gold fall; equities fall, most in emerging markets; safe-haven bond yields fall; importer currencies strengthen.

C. Oil and gold rise; equities fall, most in emerging markets; safe-haven bond yields fall; importer currencies weaken.

D. Oil and gold rise; equities fall, most in emerging markets; safe-haven bond prices fall as investors sell duration; importer currencies weaken.

Question CI-4 b.4 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b (correction box)

A market risk manager at a Johannesburg bank is calibrating the equity leg of a geopolitical stress scenario to the event-study evidence on major geopolitical events cited by the IMF. Which calibration is consistent with that evidence?

A. An immediate fall in stock prices of about 1%, rising to about 5% for severe events.

B. An immediate fall in stock prices of about 3%, rising to as much as 9% for severe events.

C. An immediate fall in stock prices of about 0.3%, rising to as much as 7 times that for severe events.

D. An immediate fall in stock prices of about 4%, rising to nearly 8% for severe events.

Question CI-4 b.5 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

A scenario designer at a European insurer must rank three candidate scenarios by the severity and persistence of their economic effects: a short trade dispute with new tariffs, a broad sanctions package, and a prolonged military conflict. Which ranking and rationale are consistent with the IMF analysis?

A. The prolonged conflict ranks first, given loss of life, physical destruction and long disruption; sanctions and tariffs do less harm.

B. The sanctions package ranks first, because sanctions stay in place by design while conflicts end; the conflict and the tariffs rank below it.

C. All three rank about equally, because the reading finds that severity depends on market sentiment rather than on the type of event.

D. The trade dispute ranks first, because tariffs hit every trading partner at once, while a conflict's effects stay local and short-lived.

Question CI-4 b.6 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

A strategist at a sovereign wealth fund reviews how stock markets behaved after major geopolitical events since World War II. Which summary is consistent with the IMF analysis?

A. Most events produced sharp declines lasting several months; the 1973 Arab oil embargo and the 1990 invasion of Kuwait were rare short-lived exceptions.

B. Most events produced modest, short-lived declines, and no event in the period produced declines that persisted beyond a few weeks.

C. Most events produced sharp declines lasting several months, which is why the average fall after a country-specific shock is large.

D. Most events produced modest, short-lived declines; the 1973 Arab oil embargo and the 1990 invasion of Kuwait caused sharper falls lasting months.

Question CI-4 b.7 New • built from SchweserNotes Book 5, LO 103.b; CI notes CI-4 b

A credit analyst at a US asset manager is assessing a domestic defence contractor's debt ahead of a possible conflict. Which observation from the IMF analysis best supports the analyst's view that the contractor can keep servicing its debt during a conflict?

A. Defence firms gain from rising commodity prices, which lift their revenues in step with those of energy producers.

B. Defence stocks carry negative GPR betas, so their cash flows tend to rise when consumer demand weakens.

C. US defence firms have historically kept stable cash flows in conflicts thanks to steady government contracting.

D. Defence firms gain from safe-haven demand, which lowers their funding costs in the way it lowers safe-haven yields.

Question CI-4 b.8 Workbook D30.2 • S3, S4 • self-gen

A portfolio manager at an Australian multi-asset fund is reviewing how markets priced geopolitical risk in past conflicts before setting AUD 400 million of hedges for a potential new regional conflict. Which of the following conclusions from the IMF analysis is correct?

A. Energy and defense stocks typically have negative GPR betas, because consumer demand and private investment weaken when geopolitical tensions rise.

B. Within a month of major conflicts, sovereign CDS spreads widened about 180 bps in emerging markets versus about 40 bps in advanced economies, while safe-haven yields fell.

C. During the 2018 to 2019 tariff announcements, put premiums rose in the United States and China and spilled over strongly to most other global equity markets.

D. Commodity importers typically gain from geopolitical shocks through higher prices, while exporters face currency depreciation and wider sovereign spreads.

Solutions • CI-4 b

Solution CI-4 b.1 — Answer: A

New • built from SchweserNotes Book 5, LO 103.b

Step 1 – Country P

Countries directly involved in a conflict or closely linked to the affected region tend to suffer the most severe consequences: sanctions, physical damage where directly affected, and disrupted trade and investment.

Step 2 – Countries Q and R

Commodity exporters, particularly energy producers, often benefit from the higher prices that accompany global tensions. Commodity importers typically face currency depreciation and greater financial pressure; importers' exchange rates tend to weaken, which amplifies their vulnerability.
$\Rightarrow$ Distance does not shelter Q or R: the oil price reaches both, in opposite directions.

Interpretation

The analyst must read across: the impact depends on economic structure and commodity exposure, not only on proximity. Importers also face rising sovereign spreads (CI-4 d).

Why the other options are wrong

B is incorrect. Scope error: R and Q have no links to the region, but both are exposed to the oil price, which helps Q and hurts R.
C is incorrect. Role swap: energy exporters often benefit from the price rise; the countries directly involved or closely linked suffer most.
D is incorrect. Absolute claim: the reading's point is that effects are uneven across countries, by structure and commodity exposure.

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.2 — Answer: D

New • built from SchweserNotes Book 5, LO 103.b

Step 1 – The gainers

Defence and energy companies often benefit from geopolitical shocks, as military spending increases and commodity prices rise.

Step 2 – The decliners

Banking, retail and utilities typically decline, reflecting weaker consumer demand, tighter credit conditions and valuation pressures.
$\Rightarrow$ Tilt toward defence and energy, away from banking, retail and utilities.

Interpretation

A reasonable condition to add for a live portfolio: defence and energy firms with revenue exposure to the conflict zone fell after the 2022 invasion (CI-4 c), so the tilt is toward firms without revenue exposure to the conflict zone.

Why the other options are wrong

A is incorrect. Sibling swap: utilities are among the decliners, and energy is among the gainers.
B is incorrect. Polarity flip: banking declines on tighter credit conditions, and defence gains on higher military spending.
C is incorrect. Polarity flip: retail declines on weaker consumer demand, and energy gains on

higher commodity prices.

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.3 — Answer: C

Adapted from CI QB Imp Q10 • SchweserNotes Book 5, LO 103.b

Step 1 – Commodities and equities

Oil and gold typically rise, on supply concerns and safe-haven demand. Equities generally decline, particularly in emerging markets.

Step 2 – Bonds and currencies

Safe-haven sovereign bonds often appreciate, resulting in lower yields. Exchange rates, especially in commodity-importing countries, tend to weaken.

$\Rightarrow$ Four legs: commodities up, EM equities hit hardest, safe-haven yields down, importer currencies down.

Interpretation

Options A and D both get the bond leg wrong, one through the yield and one through the price: safe-haven bonds appreciate, so their prices rise and their yields fall.

Why the other options are wrong

A is incorrect. Two errors: equity declines are most pronounced in emerging markets, and safe-haven yields fall.

B is incorrect. Two polarity flips: oil and gold rise, and importer currencies weaken.

D is incorrect. Polarity flip: safe-haven bonds appreciate, so their prices rise and their yields fall.

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.4 — Answer: B

New • built from SchweserNotes Book 5, LO 103.b and 103.c

Step 1 – Identify the statistic

The stem asks for the event study of major geopolitical events: stocks fall about 3% immediately afterwards, and severe events cause declines of up to 9%.

Step 2 – Place the look-alikes

About 1% and about 5% are the declines within a month after domestic geopolitical risk events and in emerging markets in military conflict. About 0.3% and up to 7 times are the country-specific shock average and the large regional shock multiple. About 4% and nearly 8% are China's average fall after US tariff announcements and its fall after the May 2019 announcement. All are CI-4 c statistics.

$\Rightarrow$ Only the 3% and 9% pair comes from the event study.

Interpretation

Each distractor is a real pair from the same reading, attached to the wrong statistic. That is the shape to expect: know which statistic each number belongs to (CI-4 a, Section 4).

Why the other options are wrong

A is incorrect. Wrong input twin: 1% and 5% belong to domestic events and EMs in military

conflict, within a month (CI-4 c).
C is incorrect. Wrong input twin: 0.3% is the average after a country-specific shock and 7 times the large regional multiple (CI-4 c); the class notes' "standard event" wording invites this mix-up.
D is incorrect. Wrong input twin: 4% and nearly 8% are China's equity falls after US tariff announcements (CI-4 c).

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b and 103.c

Solution CI-4 b.5 — Answer: A

New • built from SchweserNotes Book 5, LO 103.b

Step 1 – Rank by the type of event

Extreme events such as wars and military interventions tend to produce more severe and longer-lasting economic consequences than sanctions or trade restrictions.

Step 2 – The reasons

Loss of life, physical destruction and prolonged disruption. Longer-lasting conflicts lead to more persistent effects.
⇒ The prolonged conflict ranks first on both severity and persistence.

Interpretation

The reading does not rank sanctions against tariffs, so a correct answer only needs to place both below the conflict.

Why the other options are wrong

B is incorrect. Polarity flip: wars and military interventions are more severe and longer-lasting than sanctions.
C is incorrect. Absolute claim: severity depends on the nature of the event, which is the point of the ranking.
D is incorrect. Polarity flip: trade restrictions rank below wars, and conflicts' effects are severe and long-lasting rather than local and brief.

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.6 — Answer: D

New • built from SchweserNotes Book 5, LO 103.b

Step 1 – The base rate

Most major geopolitical events since World War II resulted in modest and short-lived stock price declines.

Step 2 – The exceptions

The 1973 Arab oil embargo and Iraq's 1990 invasion of Kuwait caused sharper and more sustained market reactions, with declines that persisted for several months.
⇒ Modest is the rule; the two named episodes are the long-lasting exceptions.

Interpretation

The history matches the cross-section in CI-4 c: average reactions are modest, and the large, persistent moves come from a few major episodes.

Why the other options are wrong

A is incorrect. Polarity flip: it reverses the rule and the exceptions.

B is incorrect. Absolute claim: two named episodes produced declines that persisted for several months.

C is incorrect. Polarity flip with a wrong conclusion: most declines were modest, and the average fall after a country-specific shock is small, about 0.3% (CI-4 c).

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.7 — Answer: C

New • built from SchweserNotes Book 5, LO 103.b

Step 1 – Why defence gains

Defence companies often benefit from geopolitical shocks as military spending increases.

Step 2 – Why the cash flows are stable

US defence firms have historically maintained stable cash flows during conflicts because of steady government contracting.

⇒ The stability comes from the customer, not from markets.

Interpretation

Debt service rests on steady cash flows, and the reading's evidence on that is specific: US defence firms' cash flows held up in past conflicts because government is the customer. Sector gains alone, from whatever source, say nothing about stability.

Why the other options are wrong

A is incorrect. Not the reading's reason: it ties defence gains to military spending and the steadiness of defence cash flows to government contracting; revenue that moved with commodity prices would argue against stable cash flows, not for them.

B is incorrect. Polarity flip: defence and energy carry positive GPR betas, and weaker consumer demand is what hurts the negative-beta, consumer-oriented sectors (CI-4 e).

D is incorrect. Sibling swap: safe-haven demand lowers the yields of safe-haven sovereign bonds; it is not the reading's explanation for defence firms' cash flows.

LO: CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b

Solution CI-4 b.8 — Answer: B

Workbook D30.2 • S3, S4 • self-gen

Step 1 – Sovereign risk premiums (CI-4 d)

Sovereign CDS spreads rise sharply after military conflicts as investors worry about fiscal stability and debt sustainability: about 180 bps in EMs within one month vs about 40 bps in AEs. Safe-haven yields (US, UK, Germany, Switzerland, Japan) fall, more so the bigger the event.

⇒ EM sovereign exposure needs the larger hedge; safe-haven duration acts as a hedge.

Step 2 – Equity and option pricing (CI-4 e)

Energy and defense have positive GPR betas (they gain when tensions rise) and act as hedges, so they carry low or negative risk premiums; consumer sectors have negative GPR betas. Put premiums rise, tail-risk premiums the most; the tariff-week increases stayed in the US and China.

Step 3 – Importers and exporters (CI-4 b)
Exporters (especially energy producers) often benefit from higher prices; importers face currency depreciation and wider sovereign spreads.

Interpretation

B is correct. The manager should size sovereign hedges toward vulnerable, importing EMs, use safe-haven duration and tail protection, and treat energy and defense equity as partial hedges.

Why the other options are wrong

A is incorrect. Sibling swap: attributes the consumer-oriented sectors' negative GPR beta (weaker demand) to energy and defense, whose GPR betas are positive.
C is incorrect. Scope/condition error: the tariff-related rise in option premiums was confined to the United States and China and did not spill over meaningfully to other global markets.
D is incorrect. Role swap between importers and exporters: exporters gain from higher prices; importers suffer depreciation and wider spreads.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF, Global Financial Stability Report, April 2025, Ch.2, via SchweserNotes Book 5, Reading 103, Modules 103.1 and 103.2

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 c

P3 CONCEPT

Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D30 Geopolitical Risk and Financial Stability (2 questions, Vol 2 p.184), and the Priority Map has no CI-4 entry. D30 reports no shape for this LO, but its revision notes cover the stock market responses and both cases.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- None reported. Vardeez focus only: the general response ladder, the Russia-Ukraine and US-China cases, and the cross-border effects.

What the questions look like: no calculations. This LO holds most of the reading's magnitudes, so expect recall questions whose distractors are the other CI-4 numbers: the right figure attached to the wrong statistic, or two figures swapped between countries or horizons. Case questions test sector and exposure together.

Background: channels and definition in CI-4 a; the full map of numbers is CI-4 a, Section 4.

1. The general response: modest on average, much larger in the tail

Stock markets generally react only modestly to geopolitical events, but the severity and duration of the shock can amplify the response a great deal. The impact depends on both macroeconomic uncertainty and investors' risk attitudes.

- **Country-specific shock.** On average, aggregate stock prices fall by about 0.3% after a country-specific geopolitical risk shock.
- **Large regional shock.** Shocks at least two standard deviations above normal geopolitical risk levels, such as major conflicts, can produce losses up to 7 times larger, and with greater persistence.
- **Global shock.** Shocks that hit many countries at the same time can have an even stronger impact, as much as 20 times larger than typical returns, and their effects can persist for up to six months after the initial shock.
- **Volatility and tail risk.** The CBOE Volatility Index (VIX), a measure of expected market volatility, typically spikes after geopolitical shocks, signalling higher uncertainty and increased risk aversion. Tail risk, the likelihood of extreme market outcomes, also tends to rise.

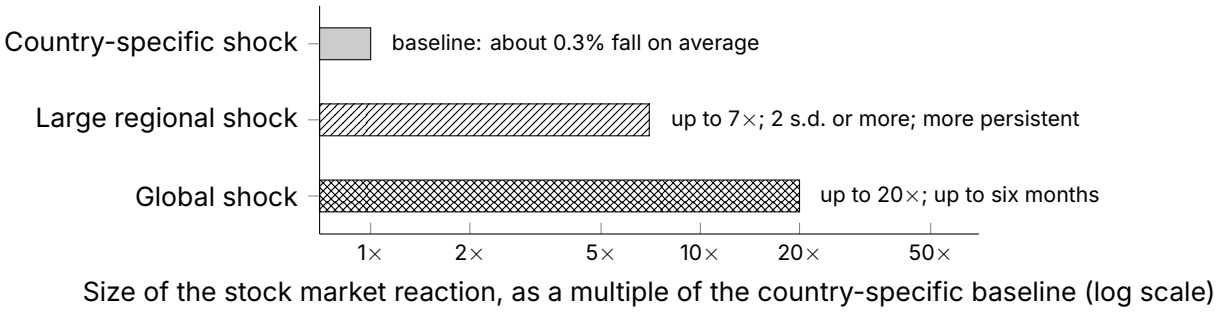


Figure 1. The shock ladder, after the class notes' Figure CI-4.2. The axis is logarithmic so the baseline stays visible next to the global case. Grey bar: country-specific baseline; single hatching: large regional shock; crosshatching: global shock. The multiples are ceilings ("up to"), not averages. SchweserNotes words the global figure as "20x larger than typical returns"; the class notes draw both multiples against the 0.3% baseline.

Corrections to the source notes. (1) The caption to the class notes' Figure CI-4.2 says "only the global case is described as persistent". SchweserNotes describes large regional shocks as producing losses up to 7 times larger "and with greater persistence"; global shocks then add effects that can last up to six months. Both large shocks are more persistent than the baseline. Question CI-4 c.2 option A is the class notes' reading. (2) The class notes' Table CI-4.2 gives the 2.5% one-week fall to "firms with direct exposure". SchweserNotes gives it to firms with *subsidiaries* in Russia or Ukraine; for defence and energy firms with revenue exposure it says only that they "experienced sharp declines".

2. Domestic events and military conflicts

International military conflicts tend to have a stronger impact on equity markets than other geopolitical events, particularly in emerging markets. On average, domestic geopolitical risk events lead to a decline of about 1% in equity markets within a month; the impact rises to roughly 5% in emerging market economies experiencing a military conflict.

Keep three small numbers apart. About 0.3% is the average fall in aggregate stock prices after a country-specific shock, the base of the 7 and 20 times multiples. About 1% is the fall within a month after domestic geopolitical risk events. About 3% is the event-study fall immediately after *major* events (CI-4 b). SchweserNotes reports them as separate statistics and does not reconcile them, so answer with the one whose definition the question uses.

3. Cross-border effects

Effects cross borders through **trade and financial linkages**.

- **Countries** geographically or economically close to conflict zones often face sharper declines, sometimes as much as 2.5 times larger.
- **Firms** with revenue exposure to affected countries, or with subsidiaries or significant shareholders located there, also see stock price declines, with the effects generally more pronounced in emerging markets. SchweserNotes' quiz answer adds that firms without such linkages are less likely to be affected, and that revenue diversification alone does not fully insulate a firm from spillovers.

4. Case study: Russia's 2022 invasion of Ukraine

A major global geopolitical risk event. Stock markets fell significantly, not only in Russia and Ukraine but also in countries with close economic and revenue ties to them.

- **Sector and exposure together.** Defence and energy firms with revenue exposure to Russia and Ukraine experienced sharp declines. At the same time, equity returns in the same sectors rose in other countries, because investors anticipated higher defence spending and rising energy prices.
- **Europe.** European markets fell more sharply than those of other advanced economies, because of their close economic links to the region and their reliance on energy imports.
- **Subsidiaries.** Firms with subsidiaries in Russia or Ukraine also had negative returns, with valuations declining by about 2.5% within a week of the invasion. Cross-border exposure amplifies the shock.
- **Since 2022.** Both the aggregate exposure of firms and countries to Russia and the number of firms with subsidiaries there have declined significantly. Exposure has increased for some firms, particularly in certain Asia-Pacific and European countries.



Figure 2. The 2022 sectoral split, after the class notes' Table CI-4.2. Shaded cell: falls; open cell: rises. Being in defence or energy was not enough to gain; being in those sectors *outside* the exposed economies was. Read sector and geography together.

5. Case study: US-China trade tensions, 2018 to 2024

A series of tariff and retaliatory tariff measures from 2018 to 2024, affecting firms in both countries and contributing to sizeable stock price declines.

- **China** took the larger hit: equity values fell by an average of about 4% after tariff announcements, including a drop of nearly 8% after the May 2019 US tariff announcement on Chinese products.
- **The United States** saw more moderate declines: about 1.3% on average between 2018 and 2024, and about 1.6% to 1.8% after China's retaliatory tariff announcement in August 2019.
- **Coverage.** The IMF report was published in April 2025, but the tariff analysis covers events from 2018 to 2024 only. No 2025 tariff events are in the study.

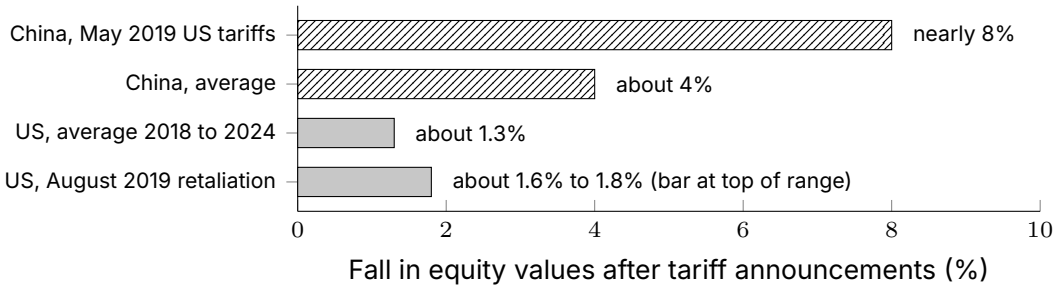


Figure 3. The tariff case in four numbers. Hatched bars: China; grey bars: United States. Within each country the single-event figure sits next to the average, which is the swap distractors use.

Results card: CI-4 c

Ladder	Country-specific about 0.3% $<$ large regional (at least 2 s.d.) up to 7×, more persistent $<$ global up to 20×, up to six months
Volatility	VIX spikes; tail risk rises
Domestic vs EM war	About 1% within a month; about 5% in EMs in military conflict
Cross-border	Trade and financial linkages; close countries up to 2.5×; firms with revenue, subsidiaries or major shareholders there; stronger in EMs
Russia 2022	Exposed defence and energy firms fell, same sectors rose elsewhere; Europe fell more than other AEs; subsidiaries about 2.5% in a week; exposure since down, up for some Asia-Pacific and European firms
US-China	China about 4% average, nearly 8% May 2019; US about 1.3% average, 1.6% to 1.8% August 2019; 2018 to 2024 only

Exam traps

- **Three multiples.** 2.5× proximity, 7× regional, 20× global. Expect each in the other's slot.
- **2.5% versus 2.5 times.** The percentage is the subsidiary firms' one-week loss; the multiple is the proximity effect.
- **Three small falls.** 0.3% country-specific average, 1% domestic within a month, 3% event study after major events.
- **Persistence.** Large regional and global shocks are both more persistent; global effects last up to six months.
- **Sector without geography.** Exposed defence and energy firms fell in 2022; the same sectors rose elsewhere.
- **Country swap and peak for average.** China about 4% average and nearly 8% in May 2019; US about 1.3% average.
- **Coverage.** The tariff study stops in 2024.

Practice questions: CI-4 c

Question CI-4 c.1

New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

An equity strategist at a Mumbai asset manager is building a table of how large the fall in aggregate stock prices after a geopolitical shock can be, by the scope of the shock. Which set of figures is consistent with the IMF analysis?

A. About 0.3% on average for a country-specific shock; up to 7 times larger for a large regional shock; up to 20 times for a global shock.

B. About 1% on average for a country-specific shock; up to 7 times larger for a large regional shock; up to 20 times for a global shock.

C. About 0.3% on average for a country-specific shock; up to 20 times larger for a large regional shock; up to 7 times for a global shock.

D. About 0.3% on average for a country-specific shock; up to 2.5 times larger for a large regional shock; up to 7 times for a global shock.

Question CI-4 c.2

New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c (correction box)

A stress-testing lead at a Canadian bank is deciding how long the equity shock should last in three scenarios: a country-specific shock, a large regional shock at least two standard deviations above normal geopolitical risk, and a global shock that hits many countries at once. Which design is most consistent with the IMF analysis?

A. Make only the global shock persistent; the large regional shock is bigger than the country-specific one but just as short-lived.

B. Make the country-specific shock the most persistent, because country-level effects compound while regional and global shocks diversify.

C. Make both the regional and the global shocks more persistent than the country shock, with global effects lasting up to six months.

D. Make all three shocks fade within one month, the horizon over which domestic events lower equity markets by about 1%.

Question CI-4 c.3

New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

A volatility trader at a London hedge fund expects a sharp rise in geopolitical risk. Which pattern does the IMF analysis lead the trader to expect?

A. The VIX falls as investors move into safe-haven assets, while the likelihood of extreme market outcomes rises.

B. The VIX spikes as uncertainty and risk aversion rise, and tail risk, the chance of extreme market outcomes, rises too.

C. The VIX spikes as uncertainty rises, but tail risk falls because extreme outcomes are already priced into equity markets.

D. The VIX is broadly unchanged, because average stock market reactions to geopolitical events are modest.

Question CI-4 c.4 Adapted from CI QB Imp Q13 • SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

A country risk analyst at a Frankfurt asset manager compares equity market responses to two kinds of event: domestic geopolitical risk events in general, and military conflicts in emerging market economies. Which comparison is consistent with the IMF analysis?

A. About a 5% decline within a month for domestic events, falling to about 1% for emerging markets in military conflict.

B. About a 1% decline within a month for domestic events, with military conflicts hitting advanced economies harder than emerging ones.

C. About a 1% decline within a month for domestic events, rising to as much as 9% for emerging markets in military conflict.

D. About a 1% decline within a month for domestic events, rising to about 5% for emerging markets in military conflict.

Question CI-4 c.5 New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

A sovereign strategist at a Brazilian bank compares two countries that are not party to a new conflict. Country K is geographically and economically close to the conflict zone; Country L is distant and has few links to it. Which comparison of their equity market declines is consistent with the IMF analysis?

A. Country K's declines can be up to 7 times larger than Country L's, because K sits inside the region of the shock.

B. Country K's declines can be as much as 2.5 times larger than Country L's, through trade and financial linkages.

C. Country K's declines can be up to 20 times larger than Country L's, because K is linked through both trade and finance.

D. The two countries' declines are similar, because global markets price conflict risk uniformly across countries.

Question CI-4 c.6 New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

A portfolio manager at a Singapore fund holds shares in firms based in countries that are not party to a new military conflict. Which firms does the IMF analysis suggest are most at risk of stock price declines?

A. Firms with revenue exposure, subsidiaries or significant shareholders in the affected countries, especially in emerging markets.

B. Firms headquartered in advanced economies, since cross-border effects are generally stronger in advanced than in emerging markets.

C. Firms in the defence and energy sectors in every country, since those two sectors fall wherever a new conflict begins.

D. No firms outside the conflict countries, since diversified revenue streams fully insulate them from geopolitical spillovers.

Question CI-4 c.7 New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

An equity analyst at a Paris asset manager compares two energy companies listed outside Russia and Ukraine. Firm P earns a large share of its revenue in Russia and Ukraine; Firm Q, based in another country, has no such exposure. Based on the IMF analysis of the 2022 invasion, which outcome is most consistent?

A. Both firms' shares rose, because investors expected energy prices to rise.

B. Both firms' shares fell, because risk aversion hit the whole energy sector.

C. Firm P's shares fell sharply, while Firm Q's rose on expected higher energy prices.

D. Firm P's shares rose on its pricing power in the region, while Firm Q's shares fell.

Question CI-4 c.8 Adapted from CI QB Imp Q15 • SchweserNotes Book 5, LO 103.c; CI notes CI-4 c (correction box)

A risk analyst at a Zurich bank is reviewing equity market reactions to Russia's 2022 invasion of Ukraine. Which statement is consistent with the IMF analysis?

A. European markets fell more than other advanced economies', and firms with subsidiaries in Russia or Ukraine lost about 2.5% of value within a week.

B. European markets fell more than other advanced economies', and firms with subsidiaries in Russia or Ukraine fell up to 2.5 times as much as other firms.

C. European markets fell less than other advanced economies', helped by hedged energy imports, and subsidiary firms lost about 2.5% within a week.

D. European markets fell more than other advanced economies', and firms with subsidiaries in Russia or Ukraine lost about 6% within a week and 8% in six months.

Question CI-4 c.9 Adapted from CI QB Imp Q16 • SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

An analyst at a Hong Kong broker summarises the equity market effects of the US-China tariff measures of 2018 to 2024 studied by the IMF. Which summary is correct?

A. US equities fell about 4% on average after tariff announcements and Chinese equities about 1.3%, so the US bore most of the cost.

B. Chinese equities fell nearly 8% on average after tariff announcements, and US firms fell about 1.6% to 1.8% on average over the period.

C. Chinese equities fell about 4% on average, and the effects intensified with the 2025 tariff announcements included in the study.

D. Chinese equities fell about 4% on average, nearly 8% after the May 2019 US announcement; US firms fell about 1.3% on average.

Question CI-4 c.10 New • built from SchweserNotes Book 5, LO 103.c; CI notes CI-4 c

A compliance officer at a Tokyo trading company is reviewing how corporate exposure to Russia has changed since the 2022 invasion, as described in the IMF analysis. Which statement is correct?

A. Aggregate exposure to Russia has risen, led by firms in Asia-Pacific and Europe, while the number of firms with Russian subsidiaries is broadly unchanged.

B. Aggregate exposure and the number of firms with subsidiaries in Russia have fallen sharply, though exposure rose for some Asia-Pacific and European firms.

C. Aggregate exposure and the number of firms with Russian subsidiaries have fallen to near zero in every region, Asia-Pacific and Europe included.

D. Aggregate exposure has fallen, but only for European firms; the exposure of firms in Asia-Pacific countries has risen across the board.

Solutions • CI-4 c

Solution CI-4 c.1 — Answer: A

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The baseline

On average, aggregate stock prices fall by about 0.3% after a country-specific geopolitical risk shock.

Step 2 – The multiples

Large regional shocks, at least two standard deviations above normal geopolitical risk levels, can produce losses up to 7 times larger. Global shocks hitting many countries at once can be as much as 20 times larger.

$\Rightarrow$ The wider the shock's reach, the larger the multiple: 0.3%, then up to 7 times, then up to 20 times.

Interpretation

The ladder is the reading's answer to "modest on average, severe in the tail". The multiples are ceilings, which is why the figure in CI-4 c Section 1 uses a log scale.

Why the other options are wrong

B is incorrect. Wrong input twin: about 1% is the fall in equity markets within a month after domestic geopolitical risk events, a separate statistic; the country-specific average is about 0.3%.

C is incorrect. Sibling swap: the regional and global multiples are exchanged; global shocks carry the larger one.

D is incorrect. Wrong input twin: 2.5 times is the proximity effect for countries close to a conflict, and 7 times belongs to the regional shock.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.2 — Answer: C

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – Large regional shocks

Shocks at least two standard deviations above normal geopolitical risk levels can produce losses up to 7 times larger, "and with greater persistence".

Step 2 – Global shocks

Global events that affect many countries can have effects that persist for up to six months after the initial shock.

$\Rightarrow$ Both large shocks should last longer than the country-specific one, and the global one up to six months.

Interpretation

The severity and the duration of a shock both amplify the market response. A stress scenario that scales the size but not the length of the large shocks understates them.

Why the other options are wrong

A is incorrect. The class notes' misreading (see the correction box): SchweserNotes gives greater persistence to large regional shocks too.

B is incorrect. Polarity flip: the country-specific shock is the small, brief baseline; the larger

shocks are the persistent ones.
D is incorrect. Wrong input twin on the horizon: "within a month" belongs to the 1% fall after domestic events, while global effects can last up to six months.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.3 — Answer: B

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The VIX
The VIX, an important measure of expected market volatility, typically spikes following geopolitical shocks, signalling higher uncertainty and increased risk aversion.

Step 2 – Tail risk
During such periods tail risk, the likelihood of extreme market outcomes, also tends to rise.
⇒ Both move up together.

Interpretation
The same pattern shows up in option prices: premiums for tail-risk protection rise the most in major shocks (CI-4 e).

Why the other options are wrong
A is incorrect. Polarity flip: safe-haven flows do not lower the VIX; it spikes.
C is incorrect. Polarity flip: tail risk rises in these periods.
D is incorrect. Scope error: the modest *average* stock reaction does not stop the VIX from spiking after shocks.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.4 — Answer: D

Adapted from CI QB Imp Q13 • SchweserNotes Book 5, LO 103.c

Step 1 – Domestic events
On average, domestic geopolitical risk events lead to a decline of about 1% in equity markets within a month.

Step 2 – Military conflict in EMs
International military conflicts have a stronger impact than other events, particularly in emerging markets: the decline rises to roughly 5% in emerging market economies experiencing a military conflict.
⇒ About 1% against about 5%, both within a month.

Interpretation
The two numbers share a horizon, which makes them easy to swap and to confuse with the 0.3% country-specific average and the 3% event study.

Why the other options are wrong
A is incorrect. Sibling swap: the two figures are exchanged.
B is incorrect. Polarity flip: military conflicts hit emerging markets hardest.
C is incorrect. Wrong input twin: up to 9% is the ceiling for severe events in the event study (CI-4 b).

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.5 — Answer: B

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The route

Cross-border effects are transmitted through trade and financial linkages.

Step 2 – The size

Countries geographically or economically close to conflict zones often face sharper declines, sometimes as much as 2.5 times larger.

⇒ Proximity, measured by geography or economic ties, raises the hit by up to 2.5 times.

Interpretation

The reading has three multiples: 2.5 times for proximity, 7 times for large regional shocks and 20 times for global shocks. Only the first compares countries hit by the same event.

Why the other options are wrong

A is incorrect. Wrong input twin: 7 times is the large regional shock multiple, which compares shock sizes, not near and distant countries.

C is incorrect. Wrong input twin: trade and financial links are the right route, but 20 times is the global shock multiple; proximity raises declines by up to 2.5 times.

D is incorrect. Absolute claim: cross-border effects are uneven, travelling through trade and financial linkages.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.6 — Answer: A

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The firm-level linkages

Firms with revenue exposure to affected countries, or with subsidiaries or significant shareholders there, see stock price declines.

Step 2 – Where it is stronger

The effects are generally more pronounced in emerging markets.

⇒ The conflict reaches a firm through its own revenue and ownership ties, wherever it is listed.

Interpretation

SchweserNotes' quiz answer adds the converse: firms without such linkages are less likely to be affected, and revenue diversification alone does not fully insulate a firm.

Why the other options are wrong

B is incorrect. Polarity flip: effects are generally more pronounced in emerging markets.

C is incorrect. Scope error: defence and energy firms fall when they are exposed to the conflict, but the same sectors rose elsewhere in 2022.

D is incorrect. Absolute claim: revenue diversification alone does not fully insulate firms from geopolitical spillovers.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. •

Reference: IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c (Module Quiz 103.1)

Solution CI-4 c.7 — Answer: C

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The exposed firm

Defence and energy firms with revenue exposure to Russia and Ukraine experienced sharp declines after the invasion.

Step 2 – The unexposed firm

Equity returns in the same sectors rose in other countries, because investors anticipated higher defence spending and rising energy prices.

⇒ Same sector, opposite outcomes, sorted by exposure.

Interpretation

Two forces with opposite signs act on an exposed energy firm: the sector tailwind and the exposure penalty. For firms earning revenue in the conflict zone, the exposure penalty dominated.

Why the other options are wrong

A is incorrect. Scope error: keeps the sector effect and drops the exposure condition.

B is incorrect. Scope error from the other side: the sector effect was positive for firms without exposure.

D is incorrect. Polarity flip: the exposed firm fell and the unexposed firm rose.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.8 — Answer: A

Adapted from CI QB Imp Q15 • SchweserNotes Book 5, LO 103.c

Step 1 – Europe

European markets fell more sharply than those of other advanced economies after the invasion, because of close economic links to the region and reliance on energy imports.

Step 2 – Subsidiaries

Firms with subsidiaries in Russia or Ukraine had negative returns, with valuations declining by about 2.5% within a week.

⇒ A percentage over a week, not a multiple.

Interpretation

Two figures in this reading read "2.5": the subsidiary firms' 2.5% one-week loss and the 2.5 times proximity effect for countries. Note the unit before you answer.

Why the other options are wrong

B is incorrect. Wrong input twin: 2.5 times is the proximity multiple for countries close to a conflict, not the subsidiary firms' loss.

C is incorrect. Polarity flip: European markets fell *more*, because of energy import reliance and close links.

D is incorrect. Wrong input twin: about 6% in a week and 8% over six months are the losses of investment funds with 10% direct exposure (CI-4 f).

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c and 103.f

Solution CI-4 c.9 — Answer: D

Adapted from CI QB Imp Q16 • SchweserNotes Book 5, LO 103.c

Step 1 – China

Chinese equity values fell by an average of about 4% after tariff announcements, including nearly 8% after the May 2019 US tariff announcement on Chinese products.

Step 2 – United States

US firms had more moderate declines, about 1.3% on average between 2018 and 2024, and about 1.6% to 1.8% after China's retaliatory announcement in August 2019.

$\Rightarrow$ China bore the larger cost; each country has an average and a single-event figure.

Interpretation

The study covers tariff events from 2018 to 2024 only, although the report was published in April 2025.

Why the other options are wrong

A is incorrect. Sibling swap: the two countries' figures are exchanged; China fell more.

B is incorrect. Single event used as the average: nearly 8% is the May 2019 drop and 1.6% to 1.8% the August 2019 US move; the averages are about 4% and 1.3%.

C is incorrect. Scope error: no 2025 tariff events are included in the study.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c

Solution CI-4 c.10 — Answer: B

New • built from SchweserNotes Book 5, LO 103.c

Step 1 – The aggregate trend

Since 2022, both the aggregate exposure of firms and countries to Russia and the number of firms with subsidiaries there have declined significantly.

Step 2 – The exception

Exposure has increased for some firms, particularly in certain Asia-Pacific and European countries.

$\Rightarrow$ Down overall, with a named exception.

Interpretation

Banks had already cut their direct exposure after 2014 and funds cut theirs by roughly 60% after 2022 (CI-4 f). The firm-level picture here is the same direction with exceptions.

Why the other options are wrong

A is incorrect. Polarity flip: aggregate exposure and the number of subsidiary firms fell.

C is incorrect. Absolute claim: exposure rose for some firms in Asia-Pacific and European countries.

D is incorrect. Scope error: the decline is aggregate, and the increase is limited to some firms in certain Asia-Pacific *and* European countries.

LO: CI-4 c: Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.c and 103.f

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 d

P3 CONCEPT

Evaluate the effects of geopolitical shocks on sovereign risk premiums.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D30 Geopolitical Risk and Financial Stability (2 questions, Vol 2 p.184), and the Priority Map has no CI-4 entry. D30·S4 (sovereign risk premiums, and commodity importers vs exporters) is this LO.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D30·S4** Sovereign risk premiums, and commodity importers vs exporters. Workbook D30.2 is worked under CI-4 b as Question CI-4 b.8: its key, option B (CDS spreads about 180 bp wider in EMs against about 40 bp in AEs within a month, with safe-haven yields falling), is this LO's headline fact. No question here repeats that key; the 180 and 40 bp figures return as distractors.

What the questions look like: no calculations. Which sovereigns are repriced most and why (fundamentals, trading partners, commodity position), how CDS spreads and yields move, and the conditional CDS figures kept apart from the EM and AE averages. The class notes give the direction of every effect but no magnitudes; the numbers here are SchweserNotes'.

1. What raises a sovereign's risk premium

Geopolitical risks, particularly those involving emerging markets or military conflicts, can increase financial instability and lead to higher sovereign risk premiums. Three routes run through the reading:

- **Direct exposure and the budget.** Through the economic channel (CI-4 a), conflicts can raise sovereign debt burdens as governments boost military spending or lose fiscal revenue.
- **Trading partners.** Conflicts that affect a country's key trading partners can also raise its sovereign risk premium, by weakening expected economic activity and heightening external vulnerabilities. The country need not be involved itself.
- **Commodity position.** Sovereign risk premiums tend to rise after major shocks particularly in emerging markets and commodity-importing countries. Importers face currency depreciation and rising spreads, while exporters often benefit from higher prices (CI-4 b).

2. Measuring the premium: sovereign CDS spreads

Sovereign risk premiums are commonly assessed with sovereign credit default swap (CDS) spreads. The spreads tend to rise sharply after military conflicts, because investors become more concerned about fiscal stability and debt sustainability.

- **By economy type, within one month of major conflicts:** emerging market spreads widened by roughly 180 bp; advanced economy spreads by about 40 bp.

- **By vulnerability, after major geopolitical shocks:** countries with international reserve adequacy ratios below the median saw increases of about 100 bp; countries with institutional quality below the median, about 120 bp.

Sovereign risk is higher where debt-to-GDP ratios are elevated, fiscal and reserve buffers are limited and institutional quality is weaker. Overall, a sovereign's risk premium reflects both its **direct exposure** to geopolitical shocks and its **underlying fiscal strength**; rising fiscal vulnerabilities heighten the sensitivity of the premium to geopolitical events.

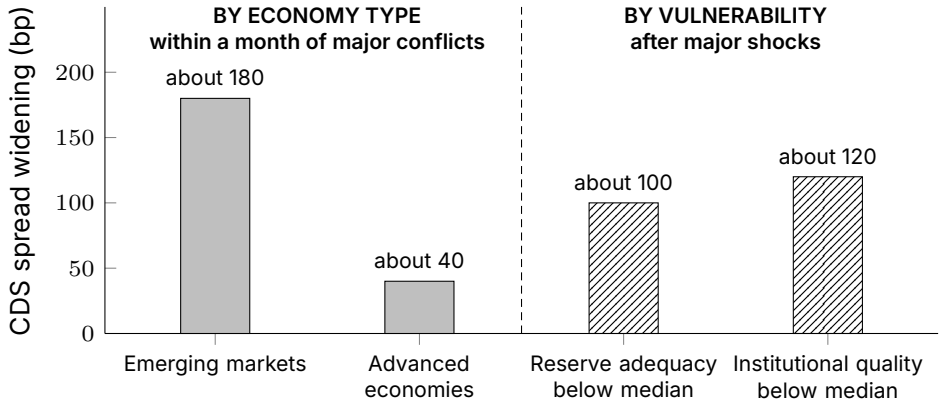


Figure 1. Four CDS figures from two different cuts of the evidence, separated by the dashed line. Grey bars: the average by economy type (the 180 against 40 bp is the D30.2 key). Hatched bars: the conditional figures for weak fundamentals; weak institutions (about 120 bp) cost more than weak reserves (about 100 bp). Do not move a number from one side of the line to the other.

3. The yield split: two directions at once

- **Vulnerable and conflict-affected emerging markets.** Consistent with wider sovereign risk premiums, long-term sovereign yields tend to rise.
- **Safe havens**, such as the United States, the United Kingdom, Germany, Switzerland and Japan. Yields generally fall as investors seek safety during global shocks. The safe-haven response grows with the size and severity of the risk event.

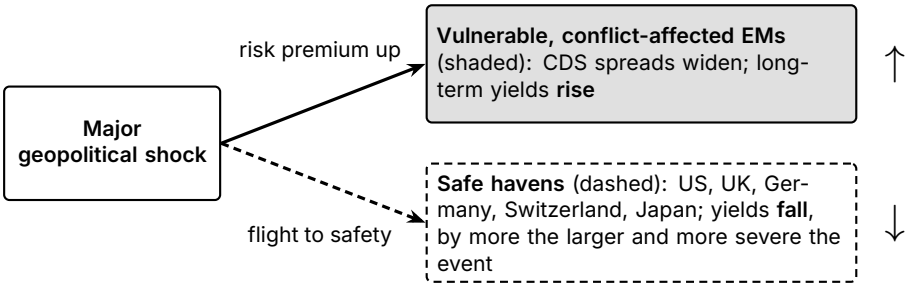


Figure 2. The same shock moves sovereign yields in opposite directions. "Advanced economy" and "safe haven" are not the same group: advanced economies' CDS spreads still widen by about 40 bp on average, while the named safe havens see their yields fall.

Results card: CI-4 d

Measure	Sovereign CDS spreads; widen sharply after military conflicts on fiscal stability and debt-sustainability concerns
EM vs AE, one month	About 180 bp vs about 40 bp
Conditional	Reserve adequacy below median about 100 bp; institutional quality below median about 120 bp
Amplifiers	High debt-to-GDP; limited fiscal and reserve buffers; weak institutions; commodity importer
Two determinants	Direct exposure *and* underlying fiscal strength; rising fiscal vulnerability raises sensitivity
Trading partners	Their conflicts raise your premium: weaker expected activity, higher external vulnerability
Yields	Vulnerable EMs up; safe havens (US, UK, Germany, Switzerland, Japan) down, more for bigger events

Exam traps

- **Wrong input twin.** 180 and 40 bp are the EM and AE averages; 100 and 120 bp the weak-reserves and weak-institutions figures.
- **Sibling swap on the conditional pair.** Weak institutions (about 120 bp) cost more than weak reserves (about 100 bp).
- **Polarity flip on CDS.** Spreads widen after military conflicts; they do not narrow.
- **Yields one way.** Vulnerable EM yields rise while safe-haven yields fall.
- **Scope error on involvement.** A conflict among key trading partners can raise a premium without direct involvement.
- **One determinant only.** Premiums reflect exposure *and* fiscal strength.
- **Exporter for importer.** Commodity importers, not exporters, see premiums rise most when prices jump.
- **Size-blind safe havens.** The safe-haven response grows with the size and severity of the event.

Practice questions: CI-4 d

Question CI-4 d.1 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A sovereign credit analyst at a Paris asset manager wants a market measure of how geopolitical shocks change sovereign risk premiums. Which measure does the IMF analysis use, and how does it typically respond to a military conflict?

A. The GPR betas of each country's equity index, which turn negative as investors price higher sovereign risk.

B. The VIX, which spikes after military conflicts as investors price higher sovereign default risk.

C. Sovereign CDS spreads, which widen sharply after military conflicts on fiscal and debt-sustainability worries.

D. Sovereign CDS spreads, which narrow after military conflicts as investors expect governments to support growth.

Question CI-4 d.2 Adapted from CI QB Imp Q21 • SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A sovereign debt portfolio manager in New York compares two sovereigns hit by the same major geopolitical shock. Sovereign X has an international reserve adequacy ratio below the median; Sovereign Y has institutional quality below the median. Based on the historical evidence cited by the IMF, which expectation is most consistent?

A. X's CDS spread widens by about 100 bp and Y's by about 120 bp, so weak institutions carry the larger increase.

B. X's CDS spread widens by about 120 bp and Y's by about 100 bp, so weak reserves carry the larger increase.

C. Both CDS spreads widen by about 180 bp, the reading's average for emerging markets within a month of a major conflict.

D. Both CDS spreads widen by about 40 bp, because reserve adequacy and institutional quality are not priced in CDS markets.

Question CI-4 d.3 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A rates strategist at a Tokyo life insurer holds long-term government bonds of a conflict-affected emerging market and of Switzerland. A major military conflict breaks out. Which outcome is most consistent with the IMF analysis?

A. Both yields rise, as global risk premiums increase across all sovereign issuers.

B. Both yields fall, as investors expect central banks everywhere to ease policy.

C. The emerging market's yield falls and Switzerland's rises, as investors rotate to higher yields.

D. The emerging market's yield rises and Switzerland's falls, as investors seek safety.

Question CI-4 d.4 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A sovereign analyst at a Malaysian bank covers Country M, which is not involved in a new conflict but whose key trading partners are. Which assessment of Country M's sovereign risk premium is most consistent with the IMF analysis?

A. It is unaffected, because sovereign risk premiums respond only to a country's own direct involvement in a conflict.

B. It can rise, because conflicts affecting key trading partners weaken expected activity and heighten external vulnerabilities.

C. It is likely to fall, because capital leaving the partners' markets flows into Country M as a nearby safe haven.

D. It can rise, but only if Country M's banks hold the partners' sovereign bonds, since the channel runs through bank balance sheets.

Question CI-4 d.5 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A risk manager at a global bond fund must decide which sovereign exposure to hedge first ahead of a possible major geopolitical shock. Based on the IMF analysis, which sovereign is MOST likely to see the largest rise in its CDS spread?

A. An advanced economy with deep financial markets and moderate trade ties to the region at risk.

B. A safe-haven issuer such as Switzerland, whose bonds attract inflows during global shocks.

C. An emerging market with high debt-to-GDP, thin fiscal and reserve buffers and weak institutions.

D. An energy-exporting emerging market with large reserves and strong institutions, as energy prices rise.

Question CI-4 d.6 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

Two sovereigns have the same direct exposure to a regional conflict; one has strong fiscal buffers and the other rising fiscal vulnerabilities. A ratings analyst at a Madrid asset manager argues that their CDS spreads should therefore respond identically. Which correction best reflects the IMF analysis?

A. Premiums reflect exposure and fiscal strength, so the sovereign with rising fiscal vulnerabilities should be more sensitive.

B. Premiums reflect fiscal strength alone, so direct exposure to the conflict is irrelevant once fiscal positions are known.

C. Premiums reflect direct exposure alone, but the sovereign with the larger defence sector should see the larger widening.

D. Premiums reflect both factors, but the sovereign with stronger fiscal buffers should see more widening, having more to lose.

Question CI-4 d.7 New • built from SchweserNotes Book 5, LO 103.d; CI notes CI-4 d

A treasury strategist at a Danish pension fund compares two global geopolitical shocks, one moderate and one severe. Based on the IMF analysis, how should yields on safe-haven government bonds such as US Treasuries and German Bunds respond?

A. They fall in the moderate shock but rise in the severe one, as investors sell every asset to raise cash.

B. They fall by the same amount in both, since the safe-haven response does not depend on the event's size or severity.

C. They rise in both, as higher global risk premiums push up the yields on all sovereign debt alike.

D. They fall in both, and by more in the severe shock, as the safe-haven response grows with severity.

Question CI-4 d.8 New • built from SchweserNotes Book 5, LO 103.b and 103.d; CI notes CI-4 d

A sovereign strategist at a Johannesburg bank compares two emerging market sovereigns with similar debt, reserves and institutional quality. One is a large energy importer and the other a large energy exporter. A global geopolitical shock pushes energy prices sharply higher. Which outcome is most consistent with the IMF analysis?

A. The exporter's spread widens more, as its currency weakens while the importer tends to gain from prices.

B. The importer's spread widens more, as its currency weakens while the exporter tends to gain from prices.

C. Both spreads widen by the same amount, because the two share an emerging market's fundamentals.

D. Both spreads narrow, because higher energy prices lift global nominal growth and fiscal revenues.

Solutions • CI-4 d

Solution CI-4 d.1 — Answer: C

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – The measure

Sovereign risk premiums are commonly assessed using sovereign CDS spreads.

Step 2 – The response

The spreads tend to rise sharply after military conflicts, because investors become more concerned about fiscal stability and debt sustainability.
⇒ A wider CDS spread is a higher sovereign risk premium.

Interpretation

Each market in this reading has its own gauge: CDS spreads for sovereigns, GPR betas for equities, put and deep out-of-the-money put premiums for options, and the VIX for expected equity volatility. Matching gauge to market is a common distractor pattern.

Why the other options are wrong

A is incorrect. Sibling swap: GPR betas measure the sensitivity of equity returns to geopolitical shocks (CI-4 e), not sovereign risk premiums.
B is incorrect. Sibling swap: the VIX measures expected equity market volatility (CI-4 c).
D is incorrect. Polarity flip: spreads widen sharply after military conflicts.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.2 — Answer: A

Adapted from CI QB Imp Q21 • SchweserNotes Book 5, LO 103.d

Step 1 – Weak reserves

Countries with international reserve adequacy ratios below the median have experienced sovereign CDS spread increases of about 100 bp following major geopolitical shocks.

Step 2 – Weak institutions

Where institutional quality is below the median, the increase has been larger, roughly 120 bp.
⇒ Y widens more than X.

Interpretation

Both are conditional figures for a weak fundamental. They sit on a different cut of the evidence from the averages by economy type (about 180 bp for EMs and 40 bp for AEs within a month of major conflicts), so the stem gives no reason to reach for those.

Why the other options are wrong

B is incorrect. Sibling swap: the two conditional figures are exchanged.
C is incorrect. Wrong input twin: 180 bp is the EM average by economy type, not the conditional figure for either weakness.
D is incorrect. Wrong input twin and polarity flip: 40 bp is the AE average, and the reading shows that both weaknesses are priced, with larger increases.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.3 — Answer: D

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – The emerging market bond

Consistent with wider sovereign risk premiums, long-term sovereign yields tend to rise in conflict-affected and more vulnerable emerging markets.

Step 2 – The Swiss bond

Yields in safe-haven countries such as the United States, the United Kingdom, Germany, Switzerland and Japan generally fall as investors seek safety during global shocks.
⇒ The two holdings move in opposite directions: the Swiss bond gains in price while the EM bond loses.

Interpretation

For the insurer, the Swiss holding is a partial hedge against the EM holding in a conflict, which is the reading's yield split.

Why the other options are wrong

A is incorrect. Absolute claim: it denies the safe-haven flow that pushes Swiss yields down.
B is incorrect. Invented mechanism: the reading does not describe global easing, and the EM yield rises.
C is incorrect. Polarity flip: both directions are reversed.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.4 — Answer: B

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – The trading-partner route

Conflicts that affect a country's key trading partners can raise its sovereign risk premium.

Step 2 – Why

They weaken the country's expected economic activity and heighten its external vulnerabilities.
⇒ Involvement is not required; trade ties are enough.

Interpretation

This is the sovereign version of the cross-border effect in CI-4 c, where countries economically close to a conflict see sharper equity declines.

Why the other options are wrong

A is incorrect. Scope error: the reading names key trading partners' conflicts as a separate route to a higher premium.
C is incorrect. Polarity flip: the named safe havens are large advanced issuers; a country linked to the conflict through trade sees its premium rise.
D is incorrect. Scope error: the route runs through expected activity and external vulnerability, not only through bank holdings.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.5 — Answer: C

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – Who is repriced most

Emerging market spreads widen far more than advanced economy spreads after major conflicts.

Step 2 – The amplifiers

Sovereign risk tends to be higher in countries with elevated debt-to-GDP ratios, limited fiscal and reserve buffers and weaker institutional quality.

$\Rightarrow$ An EM that combines all three weaknesses is the first hedge.

Interpretation

SchweserNotes' own quiz on this LO keys the same profile: emerging markets with low fiscal buffers. Fiscal strength is the shock absorber the market prices.

Why the other options are wrong

A is incorrect. Sibling swap: advanced economies' spreads rise much less, about 40 bp against about 180 bp for EMs.

B is incorrect. Polarity flip: safe-haven yields fall in global shocks.

D is incorrect. Polarity flip: energy exporters often benefit from higher prices, and strong reserves and institutions reduce the widening.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d (Module Quiz 103.2)

Solution CI-4 d.6 — Answer: A

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – Two determinants

Sovereign risk premiums reflect both a country's direct exposure to geopolitical shocks and its underlying fiscal strength.

Step 2 – Sensitivity

Rising fiscal vulnerabilities can undermine financial stability and heighten the sensitivity of sovereign risk premiums to geopolitical events.

$\Rightarrow$ Equal exposure, unequal fiscal positions: unequal responses.

Interpretation

The analyst holds one determinant fixed and forgets the other. The conditional CDS figures (about 100 bp for weak reserves, 120 bp for weak institutions) are the evidence that fundamentals change the response.

Why the other options are wrong

B is incorrect. Absolute claim: direct exposure is one of the two determinants.

C is incorrect. Absolute claim with an invented mechanism: fiscal strength is the second determinant, and a defence sector is not a CDS amplifier in the reading.

D is incorrect. Polarity flip: weaker fiscal positions, not stronger ones, raise sensitivity.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.7 — Answer: D

New • built from SchweserNotes Book 5, LO 103.d

Step 1 – Direction

Yields in safe-haven countries generally fall as investors seek safety during global shocks.

Step 2 – Size

The magnitude of the safe-haven response increases with the size and severity of the risk event.
⇒ A larger fall in the severe shock.

Interpretation

The safe-haven response is graded, not a switch. That is why hedging with safe-haven duration pays most in the scenarios that hurt most.

Why the other options are wrong

A is incorrect. Polarity flip in the severe case: the safe-haven response grows with severity; it does not reverse.
B is incorrect. Scope error: the response depends on the size and severity of the event.
C is incorrect. Polarity flip: safe-haven yields fall; rising yields belong to vulnerable EMs.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.d

Solution CI-4 d.8 — Answer: B

New • built from SchweserNotes Book 5, LO 103.b and 103.d

Step 1 – Hold fundamentals fixed

The stem equalises debt, reserves and institutions, so the commodity position decides.

Step 2 – Importer against exporter

Sovereign risk premiums tend to rise after major shocks particularly in commodity-importing countries, which face currency depreciation and rising sovereign spreads. Commodity exporters often benefit from the higher prices.
⇒ The importer's premium rises more.

Interpretation

European importers after 2022 are the reading's example: currency depreciation and rising sovereign spreads. The same price move improves an exporter's position.

Why the other options are wrong

A is incorrect. Role swap: the price rise tends to help energy exporters; it is importers whose currencies weaken and whose spreads widen.
C is incorrect. Scope error: equal fundamentals do not equalise commodity exposure, which the reading treats as a separate driver.
D is incorrect. Polarity flip: sovereign premiums tend to rise after major shocks, especially for importers.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. • CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.b and 103.d

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 e

P3 CONCEPT

Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D30 Geopolitical Risk and Financial Stability (2 questions, Vol 2 p.184), and the Priority Map has no CI-4 entry. D30·S3 (pricing in equity and option markets) is this LO.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D30·S3** Pricing in equity and option markets: GPR betas, downside and tail-risk premiums. Workbook D30.2 is worked under CI-4 b as Question CI-4 b.8: its options A (energy and defence GPR betas) and C (tariff-week put premiums) test this LO. Questions CI-4 e.3 and e.8 key the facts that those two options reverse; they stay because the sector signs and the tariff-week result are core to this LO, and their distractors differ from D30.2's.

What the questions look like: no calculations. The sign rule for risk premiums, the sign of each sector's GPR beta and what it implies for the premium, the 2022 shift in the energy GPR beta premium, which option buys which protection, and where each case's option-market effect did and did not spread.

Background: channels in CI-4 a; the VIX and tail risk in CI-4 c.

1. The pricing rule

Investors price geopolitical risk into both equity and option markets through risk premiums. Under modern asset-pricing theory:

- **Assets that lose value when geopolitical risk rises** carry *positive* risk premiums: investors demand compensation for bearing those losses.
- **Assets expected to gain value when geopolitical risk is elevated** carry *negative* risk premiums: investors accept lower, or even negative, premiums in exchange for their protective value.

The sign of the premium follows from how the asset behaves in the bad state, not from how risky it looks.

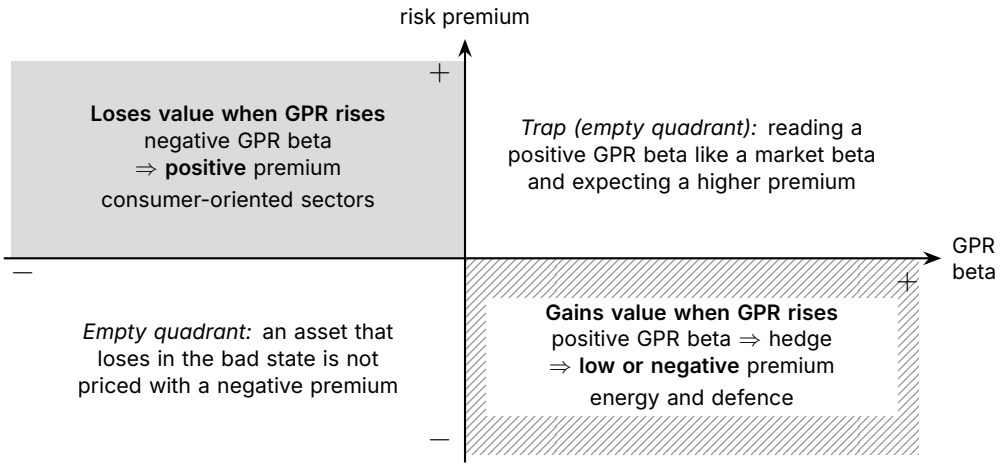


Figure 1. The sign rule on one chart, after the class notes' Figure CI-4.3. Grey quadrant: assets that lose when geopolitical risk rises, priced with a positive premium. Hatched quadrant: hedges, priced with a low or negative premium. A positive GPR beta lowers the premium; it does not raise it.

2. Equity markets: GPR betas

Geopolitical risk (GPR) betas measure the sensitivity of equity returns to geopolitical shocks. They differ by sector, reflecting underlying economic sensitivities:

- **Energy and defence** stocks typically show *positive* GPR betas: these firms often benefit when tensions rise.
- **Consumer-oriented sectors** show *negative* GPR betas: consumer demand tends to weaken when uncertainty is high.

Put the two sections together and a positive GPR beta marks a stock that gains when geopolitical risk rises, so it works as a hedge and carries a low or negative risk premium. Do not carry over the market-beta intuition, where a higher beta earns a higher premium.

The 2022 shift. GPR beta premiums can shift substantially in response to major geopolitical events, particularly wars. After Russia's 2022 invasion of Ukraine, the GPR beta premium between the energy sector and other sectors moved from negative to positive. The shift indicates a change in investor preferences toward stocks that provide protection, or act as hedges, when geopolitical risk is elevated.

> **Correction to the source notes.** The class notes' Figure CI-4.3 says that after the 2022 invasion "the energy sector's premium shifted from negative to positive as it became a critical hedge". Read as energy's own risk premium, that sits against the sign rule in the same figure: a hedge carries a low or negative premium. SchweserNotes names the object more precisely, the *GPR beta premium between the energy sector and other sectors*, does not define it further, and reads the move as a shift in investor preferences toward stocks that act as hedges. Learn the fact and that interpretation; do not derive the sign of energy's own premium from it. Question CI-4 e.5 option C has investors demanding compensation for holding a hedge.

3. Option markets: downside and tail-risk premiums

Geopolitical risk shows up in option prices, particularly in equity put options that protect against stock price declines.

- **Downside risk protection** is bought with put options; **tail-risk protection** with deep out-of-the-money put options.

- Both premiums typically **rise sharply** during major geopolitical shocks, and **tail-risk protection premiums rise the most**, reflecting heightened concern about extreme losses.

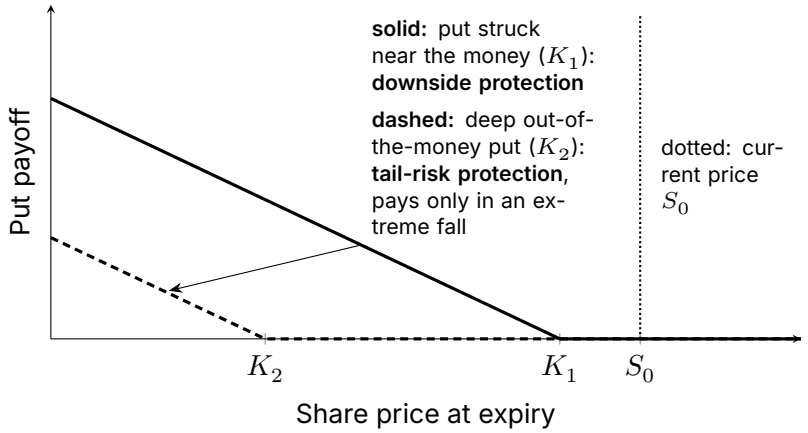


Figure 2. Which put buys which protection (standard payoff diagram, no data). The solid put pays on any meaningful fall below K_1; the dashed put pays only below the far strike K_2, so it insures the tail. In major geopolitical shocks both premiums rise sharply, and the tail-risk premium rises the most.

The two cases.

- **Russia 2022.** Put option premiums rose sharply after the invasion, not only for Russian firms but also for European firms with revenue or subsidiary exposure to Russia and Ukraine.
- **US-China tariffs, 2018 to 2019.** The tariff announcements raised option premiums in both countries, but the increase did not spill over meaningfully to other global markets. SchweserNotes shows the changes around tariff announcement weeks in two figures (downside and tail protection premiums) and reports no values in the text, so none are given here.

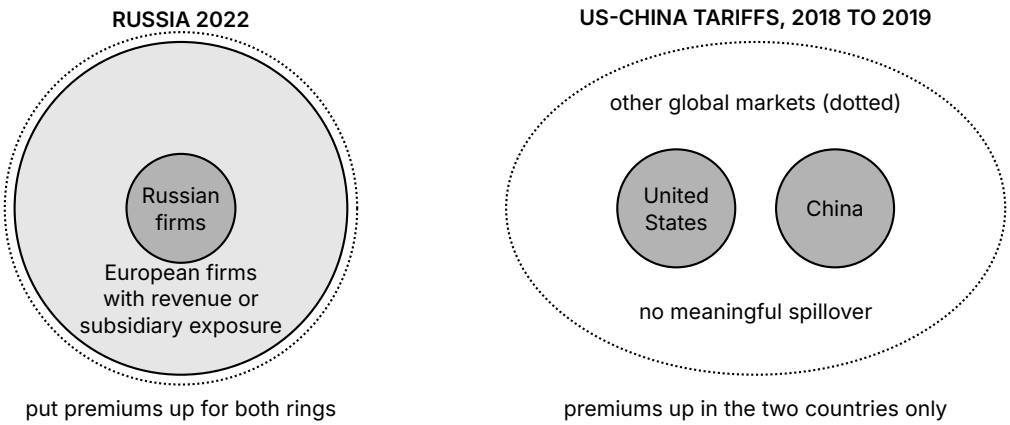


Figure 3. Where option premiums rose. Shaded areas saw higher put premiums; the dotted outer boundary marks markets outside the episode. The war spread along exposure ties into Europe; the bilateral tariff fight stayed bilateral in option markets.

Results card: CI-4 e

Sign rule	Loses value when GPR rises: positive premium; gains: lower or negative premium (protective value)
GPR beta	Sensitivity of equity returns to geopolitical shocks; energy and defence positive, consumer-oriented negative
Implication	Positive GPR beta = hedge = low or negative premium (not a market-beta premium)
2022 shift	Energy vs other sectors GPR beta premium: negative to positive; preference for hedging stocks
Options	Puts: downside protection; deep OTM puts: tail-risk protection; both up sharply in major shocks, tail the most
Cases	Russia 2022: Russian firms and exposed European firms; US-China 2018 to 2019: US and China, no meaningful spillover

Exam traps

- **Polarity flip on the sign rule.** Assets that lose in the bad state earn positive premiums; hedges accept negative ones.
- **Sibling swap on GPR betas.** Energy and defence are positive; consumer-oriented sectors are negative.
- **Market-beta intuition.** A positive GPR beta lowers the premium, because the stock hedges geopolitical risk.
- **The 2022 object.** The energy-versus-others GPR beta premium turned positive; it signals a preference for hedges.
- **Instrument swap.** Puts and deep out-of-the-money puts, not calls, buy downside and tail protection.
- **Ranking flip.** Tail-risk premiums rose the most, more than ordinary downside premiums.
- **Case swap on spillovers.** The war spread to exposed European firms; the tariff effect stayed bilateral.

Practice questions: CI-4 e

Question CI-4 e.1 New • built from SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

A new analyst on the geopolitical risk desk of a Chicago asset manager is asked which measures the IMF uses to show how geopolitical risk is priced in equity markets and in option markets. Which pairing is correct?

A. Equity markets: the VIX index of expected volatility; option markets: sovereign CDS spreads.

B. Equity markets: put option premiums; option markets: the GPR betas of each sector.

C. Equity markets: GPR betas; option markets: call premiums on energy and defence stocks.

D. Equity markets: GPR betas; option markets: downside and tail-risk protection premiums.

Question CI-4 e.2 New • built from SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

A quantitative strategist at a Zurich asset manager classifies two assets by how they behave when geopolitical risk rises: Asset X tends to lose value, and Asset Y tends to gain. Under the asset-pricing logic applied in the IMF analysis, which statement about their risk premiums is correct?

A. X carries a negative premium and Y a positive one, since investors reward the asset that gains in a shock.

B. X carries a positive premium; Y a lower or even negative one, which investors accept for its protective value.

C. Both carry positive premiums, since geopolitical risk cannot be diversified away and every exposure must be paid for.

D. Neither carries a premium, since geopolitical events are too unpredictable to be priced in equity or option markets.

Question CI-4 e.3 Adapted from CI QB Imp Q24 • SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

A quant at a Seoul hedge fund estimates geopolitical risk (GPR) betas by sector. Which pattern and explanation are consistent with the IMF analysis?

A. Energy and defence show positive GPR betas, gaining when tensions rise; consumer-oriented sectors show negative ones, as demand weakens.

B. Defence shows positive GPR betas but energy shows negative ones, as weaker consumer demand cuts fuel sales whenever geopolitical tensions rise.

C. All sectors show positive GPR betas, since a geopolitical shock raises the systematic risk of every listed equity alike.

D. GPR betas measure the volatility of a geopolitical risk index rather than of returns, so they are the same for every sector.

Question CI-4 e.4 New • built from SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

An analyst at a Johannesburg asset manager notes that energy and defence stocks carry positive GPR betas, and concludes that, like stocks with high market betas, they must offer investors a higher risk premium. Which response best reflects the IMF analysis?

A. The conclusion is right: a positive GPR beta means more exposure to geopolitical risk, which investors must be paid extra to bear.

B. The conclusion is right for defence but not for energy, since energy's GPR beta turned negative after the 2022 invasion.

C. The conclusion is wrong: a positive GPR beta means the stock gains as risk rises, so it is a hedge with a low or negative premium.

D. The conclusion is wrong: GPR betas are not priced, because geopolitical risk can be diversified away across sectors and countries.

Question CI-4 e.5 Adapted from CI QB Imp Q25 • SchweserNotes Book 5, LO 103.e; CI notes CI-4 e (correction box)

After Russia's 2022 invasion of Ukraine, the GPR beta premium between the energy sector and other sectors moved from negative to positive. A strategist at a Norwegian fund is asked what the shift most likely indicates. Which answer is consistent with the IMF analysis?

A. Energy stocks lost their value as a geopolitical hedge, so investors moved toward consumer-oriented sectors instead.

B. Investor preferences shifted toward stocks that provide protection, or act as hedges, in times of elevated risk.

C. Investors began to demand extra compensation for holding hedging stocks, which made energy shares less attractive to own.

D. Energy cash flows became more sensitive to consumer demand, the main driver of the negative-beta sectors.

Question CI-4 e.6 Adapted from CI QB Imp Q26 • SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

A derivatives strategist at a Frankfurt bank reviews how equity option prices behaved during major geopolitical shocks. Which statement is consistent with the IMF analysis?

A. Premiums for downside protection rose sharply, while premiums for tail-risk protection stayed stable, as extreme outcomes were already priced.

B. Premiums for downside and tail-risk protection, bought through calls on safe-haven assets, both rose, downside protection the most.

C. Premiums for downside and tail-risk protection both fell, as investors sold equities outright rather than buying protection.

D. Premiums for downside protection (puts) and tail-risk protection (deep out-of-the-money puts) both rose sharply, tail the most.

Question CI-4 e.7 Adapted from CI QB Imp Q27 • SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

An equity derivatives trader in Paris sees put premiums on a French industrial group jump in the days after Russia's 2022 invasion of Ukraine. The group earns revenue in Russia and has a subsidiary in Ukraine. Which interpretation is consistent with the IMF analysis?

A. The move fits the evidence: put premiums rose not only for Russian firms but also for European firms with revenue or subsidiary exposure.

B. The move is an outlier: put premiums rose only for firms domiciled in Russia and Ukraine, not for exposed firms listed elsewhere in Europe.

C. The move fits the evidence only for defence and energy firms; in other sectors, exposure to Russia did not affect option prices at all.

D. The move is an outlier: put premiums for exposed European firms fell, as investors sold the shares rather than buying protection.

Question CI-4 e.8 New • built from SchweserNotes Book 5, LO 103.e; CI notes CI-4 e

A portfolio manager at a Brazilian pension fund is considering buying index put protection on Brazilian equities after a new round of US-China tariff announcements. Which evidence from the IMF analysis of the 2018 to 2019 tariff announcements is most relevant to the decision?

A. Option premiums rose most in third countries linked to both economies by supply chains, so Brazilian protection repriced sharply.

B. Option premiums did not rise even in the United States or China, because tariff measures are not military conflicts.

C. Option premiums rose in the United States and China, but the increase did not spill over meaningfully to other markets.

D. Option premiums rose in the United States and China, with tail-risk premiums rising by less than downside premiums.

Solutions • CI-4 e

Solution CI-4 e.1 — Answer: D

New • built from SchweserNotes Book 5, LO 103.e

Step 1 – Equity markets

GPR betas measure the sensitivity of equity returns to geopolitical shocks; their sector pattern and the shifts in GPR beta premiums show how equity investors price the risk.

Step 2 – Option markets

Geopolitical risk is reflected in the premiums for downside risk protection (put options) and for tail-risk protection (deep out-of-the-money put options).
⇒ GPR betas for equities; protection premiums for options.

Interpretation

Each market has its own gauge in this reading: GPR betas for equities, put premiums for options, CDS spreads for sovereigns (CI-4 d), and the VIX for expected equity volatility (CI-4 c).

Why the other options are wrong

A is incorrect. Sibling swap: the VIX is the volatility gauge in CI-4 c, and CDS spreads are the sovereign risk premium measure in CI-4 d, not an option price.
B is incorrect. Sibling swap: the two measures are exchanged between the markets.
C is incorrect. Instrument swap: protection against declines is bought with puts, not calls.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.2 — Answer: B

New • built from SchweserNotes Book 5, LO 103.e

Step 1 – Asset X

Risk premiums are positive for assets that lose value when geopolitical risk increases, because investors demand compensation for bearing those losses.

Step 2 – Asset Y

Risk premiums are negative for assets expected to gain value when geopolitical risk is elevated, because investors accept lower, or even negative, premiums in exchange for their protective value.
⇒ Insurance is paid for: the asset that pays off in the bad state earns less.

Interpretation

The rule is the base of everything else in this LO: GPR betas tell you which way a stock moves in the bad state, and the rule turns that into the sign of its premium.

Why the other options are wrong

A is incorrect. Polarity flip: the signs are reversed.
C is incorrect. Absolute claim: it denies the negative-premium case for assets with protective value.
D is incorrect. Polarity flip: the reading shows geopolitical risk priced in both equity and option markets.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.3 — Answer: A

Adapted from CI QB Imp Q24 • SchweserNotes Book 5, LO 103.e

Step 1 – What a GPR beta is

The sensitivity of a stock's returns to geopolitical shocks.

Step 2 – The sector pattern

Energy and defence stocks typically show positive GPR betas, since these firms often benefit when tensions rise. Consumer-oriented sectors show negative GPR betas, since consumer demand tends to weaken when uncertainty is high.

$\Rightarrow$ The sign follows each sector's economic sensitivity to the shock.

Interpretation

The cross-section is the point of the measure: if every sector had the same GPR beta, it would not separate hedges from exposed stocks.

Why the other options are wrong

B is incorrect. Sibling swap: energy and defence both show positive GPR betas; weaker consumer demand is the reason the consumer-oriented sectors' GPR betas are negative.

C is incorrect. Absolute claim: GPR betas differ across sectors, with negative values for consumer-oriented sectors.

D is incorrect. Misdefinition: GPR betas measure the sensitivity of equity returns to geopolitical shocks, so they vary by sector.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.4 — Answer: C

New • built from SchweserNotes Book 5, LO 103.e

Step 1 – What the positive GPR beta says

Energy and defence stocks tend to gain when geopolitical risk rises.

Step 2 – Apply the sign rule

Assets expected to gain value when geopolitical risk is elevated carry lower, or even negative, risk premiums, because investors value their protection.

$\Rightarrow$ Positive GPR beta, hedge, low or negative premium.

Interpretation

The analyst imports the market-beta intuition, where more beta means a higher expected return. A GPR beta runs the other way: a positive value means the stock pays off in the bad state, which investors pay for.

Why the other options are wrong

A is incorrect. Polarity flip: a positive GPR beta means the stock gains in the bad state, so investors accept a lower premium.

B is incorrect. Sibling swap and polarity flip: the 2022 shift is in the energy-versus-other-sectors GPR beta premium, which moved from negative to *positive*; the reading never says energy's GPR beta turned negative.

D is incorrect. Polarity flip: the reading prices geopolitical risk through risk premiums in both equity and option markets.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e; Workbook D30 revision notes

Solution CI-4 e.5 Answer: B

Adapted from CI QB Imp Q25 • SchweserNotes Book 5, LO 103.e

Step 1 – The fact

The GPR beta premium between the energy sector and other sectors moved from negative to positive after the 2022 invasion.

Step 2 – The reading's interpretation

The shift indicates a change in investor preferences toward stocks that provide protection, or act as hedges, when geopolitical risk is elevated.
⇒ Investors came to value energy stocks as geopolitical hedges.

Interpretation

GPR beta premiums can shift substantially after major events, particularly wars. The reading gives the interpretation directly; it does not say what happened to energy's own required return.

Why the other options are wrong

A is incorrect. Polarity flip: the shift reflects a stronger preference for energy's hedging role, not a loss of it.
C is incorrect. Sign-rule error (see the correction box): investors accept, not demand, a lower premium on a hedge, and the reading reads the shift as a move toward hedges, not away from them.
D is incorrect. Sibling swap: sensitivity to consumer demand describes the negative-beta, consumer-oriented sectors.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.6 Answer: D

Adapted from CI QB Imp Q26 • SchweserNotes Book 5, LO 103.e

Step 1 – The instruments

Downside risk protection is bought with put options; tail-risk protection with deep out-of-the-money put options.

Step 2 – The response

Both premiums typically rise sharply during major geopolitical shocks, and tail-risk protection premiums rise the most, reflecting concern about extreme losses.
⇒ The deeper the protection, the larger the repricing.

Interpretation

This matches the rise in tail risk after geopolitical shocks in CI-4 c: when extreme outcomes become more likely, the insurance against them reprices most.

Why the other options are wrong

A is incorrect. Ranking flip: tail-risk premiums rose the most, not stayed stable.
B is incorrect. Instrument swap and ranking flip: protection is bought with puts on equities, and tail protection rose the most.
C is incorrect. Polarity flip: protection premiums rose sharply.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.7 Answer: A

Adapted from CI QB Imp Q27 • SchweserNotes Book 5, LO 103.e

Step 1 – The Russia case in option markets

Put option premiums increased sharply after Russia's invasion, not only for Russian firms but also for European firms with revenue or subsidiary exposure to Russia and Ukraine.

Step 2 – Apply it

The French group has both kinds of exposure.
⇒ Its put premiums rising is what the evidence predicts.

Interpretation

The option market followed the same exposure ties as the equity market in CI-4 c, where firms with subsidiaries in Russia or Ukraine lost value within a week.

Why the other options are wrong

B is incorrect. Scope error: the increase reached European firms with revenue or subsidiary exposure, not only firms in Russia and Ukraine.
C is incorrect. Scope error: the reading ties the option-market effect to revenue or subsidiary exposure, not to the defence and energy sectors.
D is incorrect. Polarity flip: put premiums rose for exposed European firms.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

Solution CI-4 e.8 Answer: C

New • built from SchweserNotes Book 5, LO 103.e

Step 1 – Where premiums rose

The 2018 to 2019 tariff announcements led to higher option premiums in both the United States and China.

Step 2 – Where they did not

The increase did not spill over meaningfully to other global markets.
⇒ On that evidence, a bilateral tariff round is not a strong reason to expect Brazilian protection to reprice.

Interpretation

The contrast with the war is the lesson: Russia 2022 spread along exposure ties into Europe, while the tariff episode stayed with the two countries. The evidence is historical, for 2018 to 2019, and the reading draws no forecast from it.

Why the other options are wrong

A is incorrect. Polarity flip: there was no meaningful spillover beyond the two countries.
B is incorrect. Polarity flip: option premiums did rise in both countries.
D is incorrect. Ranking flip: tail-risk protection premiums rose the most in the reading's evidence.

LO: CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.e

CI-4 • Global Financial Stability Report, Chapter 2: Geopolitical Risk (IMF, April 2025) • Vardeez: IMPORTANT

CI-4 f

P3 CONCEPT

Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows.

Workbook: D30 Geopolitical Risk and Financial Stability • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D30 Geopolitical Risk and Financial Stability (2 questions, Vol 2 p.184), and the Priority Map has no CI-4 entry. D30·S2 (effects on banks and nonbank financial institutions) is this LO.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D30·S2** Effects on banks and nonbank financial institutions: valuations, lending, fund flows. Workbook D30.1 is worked under CI-4 a as Question CI-4 a.8: the second half of its key, option A (falling asset and collateral values with capital outflows tighten bank lending), and its options C (fund losses after 2022) and D (bank exposure since 2014) test this LO. Questions CI-4 f.3 and f.6 key the bank-exposure and fund-loss facts that options D and C reverse; they stay because those facts are core to this LO, and their distractors differ from D30.1's.

What the questions look like: no calculations. The risk chain from volatility to collateral to lending and fund flows, the 2024 exposure shares and the post-2014 and post-2022 retreats from Russia and Ukraine, the fund evidence with its numbers kept apart, and the resilience prescriptions by actor.

Background: channels and policy set in CI-4 a; the map of numbers is CI-4 a, Section 4.

1. Why both banks and nonbanks are exposed

Both bank and nonbank financial institutions face elevated **market, credit and liquidity risks** during geopolitical conflicts, because market volatility and macroeconomic uncertainty rise.

- Increased volatility can trigger **selloffs** that erode asset and collateral values, which leads to tighter lending conditions.
- **Capital outflows** may further amplify volatility, particularly in less liquid markets.
- The vulnerabilities can be compounded by **sanctions, capital controls** and the growing frequency of **cyberattacks**.

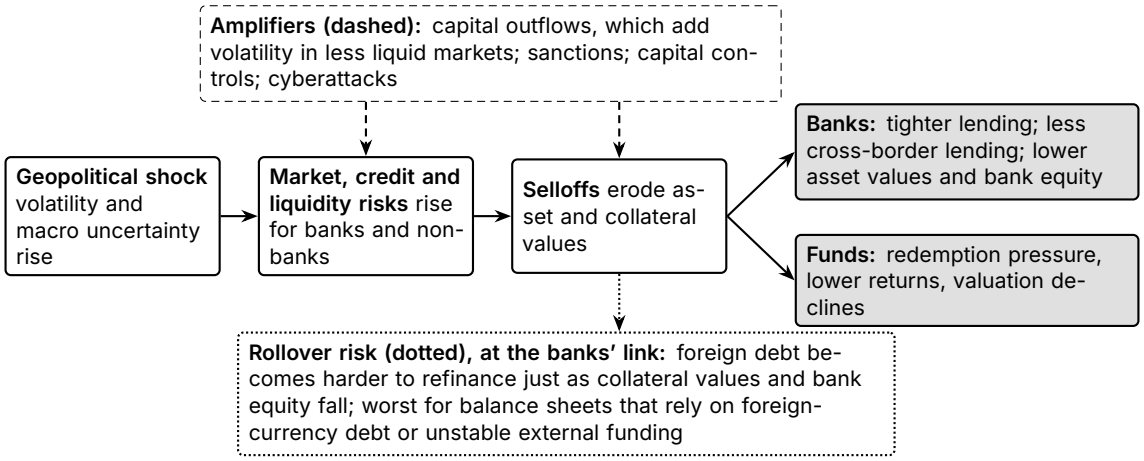


Figure 1. The transmission chain, after the class notes' Figure CI-4.4. It runs from prices to collateral to credit and flows, and the last link is where banks and funds part company (shaded): banks feel it as tighter lending, funds as redemptions. Dashed: what amplifies the chain; dotted: where rollover risk sits.

2. How large the exposure is

Markets with elevated geopolitical risk. As of 2024, financial institutions' exposure to these markets had increased markedly. They accounted for about 8% of total cross-border bank claims, 10% of total cross-border bank liabilities and 13% of equity fund assets.

Russia and Ukraine. The direct exposures moved the other way. Banks have reduced their direct exposure to Russia and Ukraine over the past decade, following Russia's 2014 annexation of Crimea. Investment funds cut their direct exposure by roughly 60% after the 2022 invasion, and also lowered their indirect exposure through asset holdings linked to the two countries.

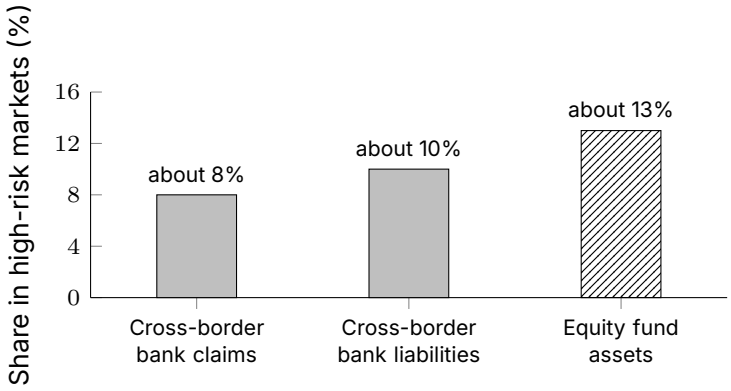


Figure 2. Exposure to markets with elevated geopolitical risk, 2024. Grey bars: banks; hatched bar: equity funds. The three shares rise in the order claims, liabilities, equity fund assets (8, 10, 13), and the exposure had increased markedly, even as direct exposure to Russia and Ukraine was cut.

3. Equity valuations and lending

Bank lending and financial stability suffer from heightened geopolitical shocks and weakening borrower creditworthiness. These conditions can lead to:

- reduced **cross-border lending**;
- declines in **asset valuations and bank equity**;

- greater **rollover risk on foreign debt**. SchweserNotes' key concepts add higher funding risk, particularly for balance sheets that rely on foreign-currency debt or unstable external funding.

4. Investment fund flows

Investment funds with significant exposure to countries in military conflicts are particularly vulnerable to **redemption pressure and valuation declines**, which raises the risk of spillovers and cross-market contagion. SchweserNotes' quiz answer adds that such funds are unlikely to see inflows during heightened uncertainty and generally do not receive direct liquidity support from central banks.

- **Bond funds with 10% exposure to conflict-affected countries**: a 2.3% drop in asset flows and a 1% decline in returns. Equity funds saw smaller effects.
- **Funds with 10% direct exposure to Russia and Ukraine**: about 6% lost in cumulative returns within one week of the 2022 invasion, and 8% over the next six months.
- **US-China tariff announcements, 2018 to 2024**: no meaningful declines in fund returns.

Correction to the source notes. The class notes say funds face "stronger impacts observed during major conflicts like the Russia-Ukraine war compared to more localised events like US-China trade tensions", which reads as a large effect against a smaller one. SchweserNotes is sharper: the 2018 to 2024 tariff announcements "did not lead to meaningful declines in fund returns". Treat the contrast in fund returns as present against absent; SchweserNotes gives no tariff-period figure for fund flows. The class notes also drop the third exposure share (10% of cross-border bank liabilities), which Question CI-4 f.2 uses.

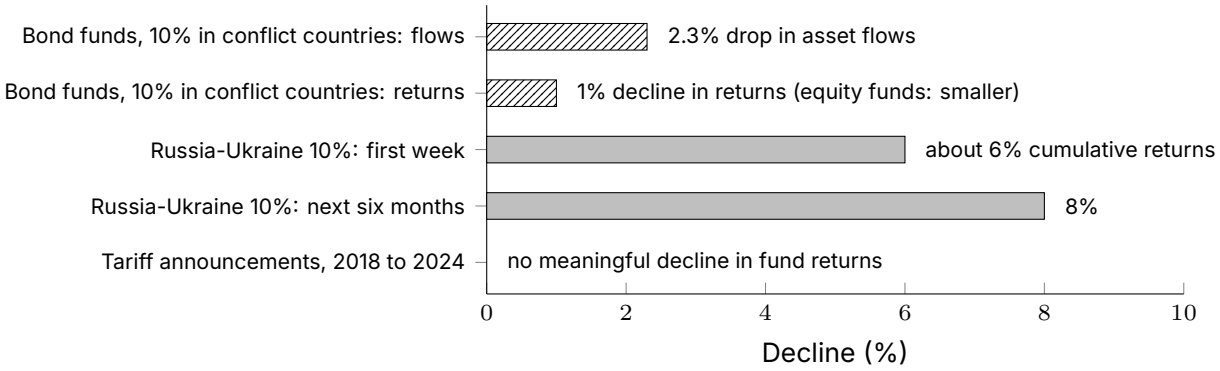


Figure 3. The fund evidence. Hatched bars: the bond fund estimate for 10% exposure to conflict-affected countries; grey bars: funds with 10% direct exposure to Russia and Ukraine after the 2022 invasion; no bar: the tariff episode. The two 10% conditions are different estimates, and the 1% here is not the 1% equity fall after domestic events in CI-4 c.

5. Resilience: what the reading asks for

The broader financial system can be destabilised if institutions lack sufficient capital and liquidity buffers, which increases the risk of contagion across markets. To withstand large but plausible losses from geopolitical events:

- **Banks and investment funds** should maintain adequate buffers, avoid concentrated exposures and enhance their risk management practices.
- **Policymakers** should ensure that regulatory frameworks account for geopolitical risks and help safeguard financial stability during periods of heightened global stress.

The country-level measures (regulatory frameworks, fiscal and monetary flexibility, deeper markets) are in CI-4 a.

Results card: CI-4 f

Risks	Market, credit and liquidity risks rise for banks *and* nonbanks; selloffs cut asset and collateral values; lending tightens
Amplifiers	Capital outflows (less liquid markets); sanctions; capital controls; cyber-attacks
Exposure, 2024	High-risk markets: 8% cross-border bank claims, 10% bank liabilities, 13% equity fund assets; increased markedly
Russia and Ukraine	Banks cut direct exposure since 2014; funds cut direct exposure by roughly 60% after 2022, indirect too
Lending	Weaker borrowers: less cross-border lending, lower asset values and bank equity, higher rollover risk on foreign debt
Funds	Bond funds, 10% exposure: flows -2.3%, returns -1% (equity funds less); 10% Russia-Ukraine: about -6% in a week, -8% over six months; tariffs: no meaningful loss
Remedies	Buffers, no concentrated exposures, better risk management; regulation that accounts for geopolitical risk

Exam traps

- **Nonbanks insulated.** Both banks and nonbanks face higher market, credit and liquidity risks.
- **Outflows calm markets.** Capital outflows amplify volatility, especially in less liquid markets.
- **Order of the exposure triple.** 8% claims, 10% liabilities, 13% equity fund assets, and rising, not falling.
- **60% versus 6%.** Funds cut direct Russia and Ukraine exposure by about 60%; 6% is the one-week return loss.
- **Flows versus returns.** Bond funds: 2.3% drop in flows, 1% in returns; equity funds less.
- **Recovery assumed.** Russia-exposed funds lost about 6% within a week and 8% over the next six months; the loss did not reverse.
- **Wrong case.** The tariff announcements caused no meaningful fund losses; do not carry that to a war.
- **Inflows and central bank support.** Exposed funds face redemptions and generally get no direct central bank support.

Practice questions: CI-4 f

Question CI-4 f.1 New • built from SchweserNotes Book 5, LO 103.f; CI notes CI-4 f

A supervisor at an Asian central bank is briefing the financial stability committee on how a geopolitical conflict affects banks and nonbank financial institutions. Which statement is most consistent with the IMF analysis?

A. Banks face higher market, credit and liquidity risks, but nonbanks are insulated because they do not rely on deposit funding.

B. Both face higher market, credit and liquidity risks; outflows amplify volatility in thin markets, and sanctions, capital controls and cyberattacks compound the strain.

C. Both face higher market, credit and liquidity risks, but capital outflows dampen volatility by reducing the open positions that investors hold in less liquid local markets.

D. Both face higher market risk only; credit and liquidity risks stay contained because sanctions and capital controls stop money from leaving the affected local markets.

Question CI-4 f.2 New • built from SchweserNotes Book 5, LO 103.f; CI notes CI-4 f (correction box)

A financial stability analyst at a European supervisor is summarising financial institutions' exposure, as of 2024, to markets with elevated geopolitical risk, as reported by the IMF. Which summary is correct?

A. About 13% of cross-border bank claims, 10% of cross-border bank liabilities and 8% of equity fund assets, after a marked increase.

B. About 8% of cross-border bank claims, 10% of cross-border bank liabilities and 13% of equity fund assets, after a marked fall as institutions left Russia.

C. About 10% of cross-border bank claims, 8% of cross-border bank liabilities and 13% of equity fund assets, after a marked increase.

D. About 8% of cross-border bank claims, 10% of cross-border bank liabilities and 13% of equity fund assets, after a marked increase.

Question CI-4 f.3 Adapted from CI QB Imp Q30 • SchweserNotes Book 5, LO 103.f; CI notes CI-4 f

A risk officer at a Luxembourg fund manager is reviewing how institutions' direct exposure to Russia and Ukraine has changed. Which statement is consistent with the IMF analysis?

A. Banks have cut direct exposure since the 2014 annexation of Crimea; funds cut theirs by roughly 60% after 2022 and lowered indirect exposure too.

B. Banks have cut direct exposure since the 2014 annexation of Crimea; funds cut theirs by roughly 6% after 2022 and left their indirect exposure unchanged.

C. Banks began cutting direct exposure only after the 2022 invasion; funds cut theirs by roughly 60% and lowered indirect exposure too.

D. Banks have cut direct exposure since the 2014 annexation of Crimea; funds cut theirs by roughly 60% after 2022 but raised indirect exposure through linked assets.

Question CI-4 f.4 New • built from SchweserNotes Book 5, LO 103.f; CI notes CI-4 f

A bank supervisor in an emerging market where several banks rely on short-term foreign-currency wholesale funding asks how a major geopolitical shock is likely to affect them. Which assessment is most consistent with the IMF analysis?

A. Rollover risk falls, because foreign lenders seeking higher spreads expand their cross-border lending to the affected market.

B. Rollover risk rises mainly for banks funded by stable domestic deposits, while banks using foreign wholesale funding are little affected.

C. Rollover risk rises, as weaker borrower credit cuts cross-border lending and lowers asset values and bank equity as debt must roll.

D. The effect is confined to market risk on the banks' trading books, so lending, funding and bank equity are not materially affected.

Question CI-4 f.5 New • built from SchweserNotes Book 5, LO 103.f; CI notes CI-4 f

A fund risk manager in Dublin uses the IMF's estimate for bond funds with 10% of their portfolios in conflict-affected countries to gauge how such a fund might be hit. Which estimate is consistent with that evidence?

A. A drop in asset flows of about 2.3% and a fall in returns of about 1%, with smaller effects for equity funds.

B. A drop in asset flows of about 1% and a fall in returns of about 2.3%, with smaller effects for equity funds.

C. A drop in asset flows of about 2.3% and a fall in returns of about 1%, with larger effects for equity funds.

D. A drop in asset flows of about 13% and a fall in returns of about 8%, with smaller effects for equity funds.

Question CI-4 f.6 New • built from SchweserNotes Book 5, LO 103.f; CI notes CI-4 f

The investment committee of a Canadian pension fund reviews how funds with 10% direct exposure to Russia and Ukraine performed after the 2022 invasion. Which summary is consistent with the IMF analysis?

A. They lost about 6% in cumulative returns within a week, and the loss was recovered over the following six months.

B. They lost about 8% in cumulative returns within a week, narrowing to about 6% over the following six months.

C. They lost about 2.3% in cumulative returns within a week, in line with the drop in bond fund asset flows.

D. They lost about 6% in cumulative returns within a week and about 8% over the following six months.

Question CI-4 f.7 Adapted from CI QB Imp Q32

A liquidity risk manager at an investment fund with large holdings in countries hit by a major military conflict is preparing for the weeks after the event. According to the IMF analysis, which outcome is MOST likely?

A. Inflows from investors seeking discounted valuations in the affected markets.

B. Improved liquidity from direct central bank support provided to the fund.

C. Redemption pressures and valuation declines that may force asset sales.

D. Little change in net asset value, since diversification offsets the exposure.

Question CI-4 f.8 New • built from SchweserNotes Book 5, LO 103.a and 103.f; CI notes CI-4 f

A supervisor at a European securities regulator is reviewing a bond fund that holds a concentrated position in a region where geopolitical risk is rising. Which recommendation best reflects the IMF analysis?

A. Sell the direct holdings in the region, since indirect exposure through assets linked to the region carries no material risk.

B. Hold adequate buffers, avoid concentrated exposures and strengthen risk management, to absorb large but plausible losses.

C. Add to the region's sovereign bonds, whose rising yields would cushion the fund if the conflict were to escalate further.

D. Take no action at fund level, since the reading assigns buffers and exposure limits to policymakers rather than to the funds.

Solutions • CI-4 f

Solution CI-4 f.1 — Answer: B

New • built from SchweserNotes Book 5, LO 103.f

Step 1 – Who is exposed

Both bank and nonbank financial institutions face elevated market, credit and liquidity risks during geopolitical conflicts, because volatility and macroeconomic uncertainty rise.

Step 2 – What amplifies it

Capital outflows may further amplify volatility, particularly in less liquid markets; sanctions, capital controls and cyberattacks compound the vulnerabilities.

$\Rightarrow$ Three risk types, both kinds of institution, and four amplifiers.

Interpretation

For the committee the amplifiers are the point: the same shock does more damage in thin markets and where sanctions, capital controls or cyber incidents are in play.

Why the other options are wrong

A is incorrect. Scope error: nonbanks face the same elevated risks; investment funds face redemptions and valuation declines.

C is incorrect. Polarity flip: capital outflows amplify volatility.

D is incorrect. Scope error and polarity flip: credit and liquidity risks rise too, and sanctions and capital controls compound the vulnerabilities rather than contain them.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025), Global Financial Stability Report, Chapter 2: Geopolitical Risk; SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.2 — Answer: D

New • built from SchweserNotes Book 5, LO 103.f

Step 1 – The three shares

As of 2024, markets with elevated geopolitical risk accounted for about 8% of total cross-border bank claims, 10% of total cross-border bank liabilities and 13% of equity fund assets.

Step 2 – The direction

The exposure had increased markedly.

$\Rightarrow$ 8, 10, 13, and rising.

Interpretation

Two trends run side by side: exposure to high-risk markets in general rose, while direct exposure to Russia and Ukraine fell. A candidate who knows only the second will pick option B.

Why the other options are wrong

A is incorrect. Sequence reversed: 13% is the equity fund share and 8% the bank claims share.

B is incorrect. Polarity flip: the shares are right, but the exposure increased markedly; the retreat from Russia is a separate trend.

C is incorrect. Sibling swap: the bank claims and bank liabilities shares are exchanged.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.3 — Answer: A

Adapted from CI QB Imp Q30 • SchweserNotes Book 5, LO 103.f

Step 1 – Banks

Banks have reduced their direct exposure to Russia and Ukraine over the past decade, following Russia's 2014 annexation of Crimea.

Step 2 – Funds

Investment funds cut their direct exposure by roughly 60% after the 2022 invasion and lowered their indirect exposure through asset holdings linked to the two countries.
⇒ Banks started a decade earlier, after 2014; funds cut deeper and on both fronts.

Interpretation

Cutting both direct and indirect exposure matters because, as the reading's policy advice notes, cutting direct exposure alone leaves indirect contagion (CI-4 a).

Why the other options are wrong

B is incorrect. Wrong input twin: about 6% is the one-week return loss of funds with 10% direct exposure, not the exposure cut, and funds also lowered indirect exposure.
C is incorrect. Timing error: banks have reduced their direct exposure over the past decade, following the 2014 annexation of Crimea.
D is incorrect. Polarity flip: funds lowered their indirect exposure as well.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.4 — Answer: C

New • built from SchweserNotes Book 5, LO 103.f

Step 1 – The lending effect

Heightened geopolitical shocks and weakening borrower creditworthiness can lead to reduced cross-border lending, declines in asset valuations and bank equity, and greater rollover risk on foreign debt.

Step 2 – Who is most exposed

The key concepts add higher rollover and funding risks, particularly for balance sheets that rely on foreign-currency debt or unstable external funding.
⇒ These banks are the textbook case.

Interpretation

The pressure comes from both sides at once: the foreign debt has to be refinanced just as collateral values and bank equity are falling.

Why the other options are wrong

A is incorrect. Polarity flip: cross-border lending is reduced, not expanded.
B is incorrect. Role misassignment: the risk is concentrated in foreign-currency and unstable external funding, not stable domestic deposits.
D is incorrect. Scope error: the reading names effects on lending, funding and bank equity, not only on trading books.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.5 — Answer: A

New • built from SchweserNotes Book 5, LO 103.f

Step 1 – Flows and returns

Bond funds with 10% exposure to conflict-affected countries experienced a 2.3% drop in asset flows and a 1% decline in returns.

Step 2 – Equity funds

Equity funds saw smaller effects.

$\Rightarrow$ Flows fell more than returns, and bond funds were hit harder than equity funds.

Interpretation

The estimate is the general one for conflict-affected countries; the larger 6% and 8% losses belong to funds with 10% direct exposure to Russia and Ukraine after 2022.

Why the other options are wrong

B is incorrect. Sibling swap: the flow and return figures are exchanged.
C is incorrect. Polarity flip: equity funds saw smaller effects than bond funds.
D is incorrect. Wrong input twin: 13% is the equity fund asset share in high-risk markets and 8% the bank claims share.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.6 — Answer: D

New • built from SchweserNotes Book 5, LO 103.f

Step 1 – The first week

Funds with 10% direct exposure to Russia and Ukraine lost about 6% in cumulative returns within one week of the invasion.

Step 2 – The next six months

They lost 8% over the next six months.

$\Rightarrow$ The losses did not reverse.

Interpretation

The contrast the reading draws is with the tariff episode, which caused no meaningful fund losses. A war in the funds' own markets does lasting damage; a bilateral tariff fight did not.

Why the other options are wrong

A is incorrect. Polarity flip on the path: the losses were not recovered over six months.
B is incorrect. Sibling swap: the two horizons' figures are exchanged.
C is incorrect. Wrong input twin: 2.3% is the drop in bond fund asset flows for 10% exposure to conflict-affected countries.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f

Solution CI-4 f.7 — Answer: C

Adapted from CI QB Imp Q32

Step 1 – What exposed funds face

Investment funds with significant exposure to countries involved in military conflicts are partic-

ularly vulnerable to redemption pressures and valuation declines.

Step 2 – Why that spreads

Redemptions into falling markets can force asset sales, raising the risk of spillovers and cross-market contagion.

⇒ Outflows and lower NAVs, not support or inflows.

Interpretation

SchweserNotes' quiz on this LO keys the same answer and adds that such funds are unlikely to see inflows in heightened uncertainty and generally do not receive direct liquidity support from central banks.

Why the other options are wrong

A is incorrect. Polarity flip: inflows are unlikely during periods of heightened uncertainty.

B is incorrect. Role misassignment: investment funds generally do not receive direct liquidity support from central banks.

D is incorrect. Absolute claim: exposed funds suffer valuation declines; diversification does not offset the exposure.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.f (Module Quiz 103.2)

Solution CI-4 f.8 **Answer: B**

New • built from SchweserNotes Book 5, LO 103.a and 103.f

Step 1 – The prescription for funds

To strengthen resilience against large but plausible losses from geopolitical events, banks and investment funds should maintain adequate buffers, avoid concentrated exposures and enhance their risk management practices.

Step 2 – The supervisor's own part

Policymakers should ensure that regulatory frameworks account for geopolitical risks.

⇒ The concentration itself is the problem the supervisor should address.

Interpretation

Without sufficient buffers, losses at one institution can spread: the reading warns that the broader system can be destabilised and contagion across markets can rise.

Why the other options are wrong

A is incorrect. Scope error: cutting direct holdings alone does not address indirect contagion, and funds after 2022 cut indirect exposure too.

C is incorrect. Polarity flip: rising yields mean falling bond prices, so more of the region's sovereign bonds adds to the loss.

D is incorrect. Role misassignment: the reading asks banks and investment funds themselves to hold buffers, avoid concentrated exposures and enhance risk management; policymakers make sure regulatory frameworks account for geopolitical risk.

LO: CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. • **Reference:** IMF (April 2025); SchweserNotes 2026 Book 5, Reading 103, LO 103.a and 103.f

CI-5 • Monetary and Fiscal Policy: Safeguarding Stability and Trust (BIS, June 2023) • Vardeez: IMPORTANT

CI-5 a

P3 CONCEPT

Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances.

Workbook: D16 Monetary and Fiscal Policy • Tier D • Map #67

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D16 Monetary and Fiscal Policy (Map #67, 2/16: "Reported: fiscal and monetary policy"). Shape D16·S1 is filed under this LO.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Role in the reading. As the first LO of CI-5, this block carries the shared background: the two policy functions, their channels and the region of stability. CI-5 b and CI-5 c recap it in a line and point back here.

Reported exam shapes

- **D16·S1** Fiscal and monetary policy: how the two interact, including the tension that high debt creates (fiscal dominance). The notes below introduce it (Section 3); its questions sit in CI-5 c: the sources and safeguard of fiscal dominance are Question CI-5 c.5, and Workbook D16.1 (tagged S1 and S2) is worked as **Question CI-5 c.10**. Do both with this block as well.

What the questions look like: no calculations. Classifying measures as fiscal or monetary, the direct-versus-indirect contrast, reading a transmission chain with the right actor and direction, the two-way overlap between the policies, and the properties of the region of stability (joint, country-specific, moving, hard to measure, with a margin of safety), including a case where only one policy moves.

1. Two functions of the state

Fiscal and monetary policy are the two core economic functions of the state. Both are critical to **trust in policymakers** and to **economic stability**, the two words in the reading's title.

- **Fiscal policy** is the government's authority to raise money through taxes or borrowing and then spend it in pursuit of its objectives.
- **Monetary policy** controls the money supply and policy interest rates.

Either can push the economy toward expansion or contraction, depending on the actions taken in its channels.

2. The channels compared

	Monetary policy	Fiscal policy
Managed by	Central bank	Government
Route	**Indirect**: through the price and quantity of money	**Direct**: through the government's own spending and taxing
Tools	Open market transactions; policy interest rates	Government spending; tax policy; production of some services; transfer payments to individuals and businesses (subsidies)
First impacts	Spill over into bond yields, borrowing costs, asset prices and exchange rates	Aggregate demand, employment, income distribution

The route row carries the whole contrast. Monetary policy reaches the economy *indirectly*, through the price of money (policy rates) and its quantity (open market operations), so its first effects are market prices: **financial markets**. Fiscal policy reaches it *directly*, through what the government spends, taxes and transfers, so its first effects are real-economy quantities: **economic activity**. Each still reaches the other side in the end; the monetary chains in Figure 1 run through to consumer spending and demand.

Classify by **who acts and through what**, not by what money does afterwards. Transfer payments put money in households' hands, but they are a fiscal tool. Open market operations trade government bonds, but they are a monetary tool.

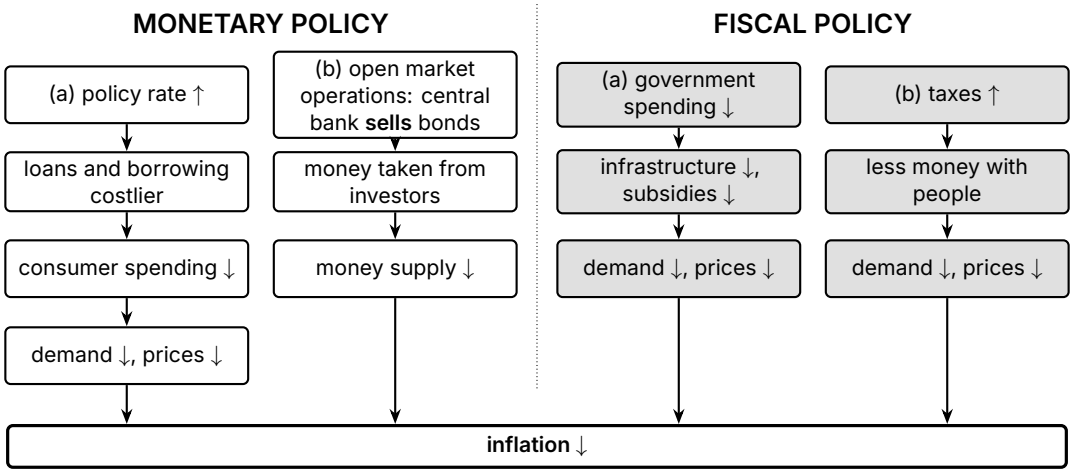


Figure 1. Four routes to the same goal, redrawn from the class notes' hand-written working (Figure CI-5.1), with the fiscal spending chain run through demand, as the reading frames fiscal policy (see the box below). White boxes: monetary; shaded boxes: fiscal. Read the second step: the two monetary routes act on the price or the quantity of money; the two fiscal routes act on the government's own spending or on households' after-tax income. In the open market chain, the central bank *sells* bonds to cut inflation, draining money from the public.

Reading the class notes' sketch. The class notes' hand-drawn fiscal spending chain runs "government spending ↓ → infrastructure ↓, subsidies ↓ → *money supply* ↓ → inflation ↓". Read "money supply" there loosely, as less money spent in the economy, in the same sense as the tax chain's "less money with people". In the reading's own terms the money supply is managed by *monetary* policy (open market transactions and policy rates), while fiscal policy acts directly through spending, taxes, services and transfers, with impacts on aggregate demand, employment and income distribution (SchweserNotes Book 5, LO 104.a; CI notes Table CI-5.1). Figure 1 therefore runs the spending chain through demand and prices, as the tax

chain already does. This is a presentational alignment, not a factual error in the notes. Expect "transfers are a monetary channel because they add to the money supply" as a distractor.

3. How the two overlap

The two policies overlap naturally, in both directions.

1. **Monetary to fiscal (cost).** If the central bank raises interest rates, government borrowing costs rise (on new debt and on refinancing), the economy slows and tax revenues decrease.
2. **Monetary to fiscal (revenue).** If monetary policy operations produce positive revenues, those revenues flow through to the government and support fiscal spending.
3. **Fiscal to monetary.** Too much fiscal stimulus can produce inflation, which requires a monetary policy response.

The class notes call this a policy feedback loop: fiscal expansion triggers monetary tightening, and monetary tightening strains fiscal budgets. The overlap can foster stability, but under the wrong circumstances it can undermine it.

When debt is high, the overlap becomes a tension (D16·S1). High public debt can work against the monetary tightening needed to fight inflation: the tightening raises the cost of government borrowing and constrains growth. In the extreme, monetary policy is constrained by fiscally imposed constraints. The reading calls this **fiscal dominance**. Its two sources (political spending pressure and the risk that rate rises spark a sovereign debt crisis) and the safeguard (protected central bank independence) are covered in CI-5 c. The direction never changes: in fiscal dominance it is *monetary* policy that is constrained.

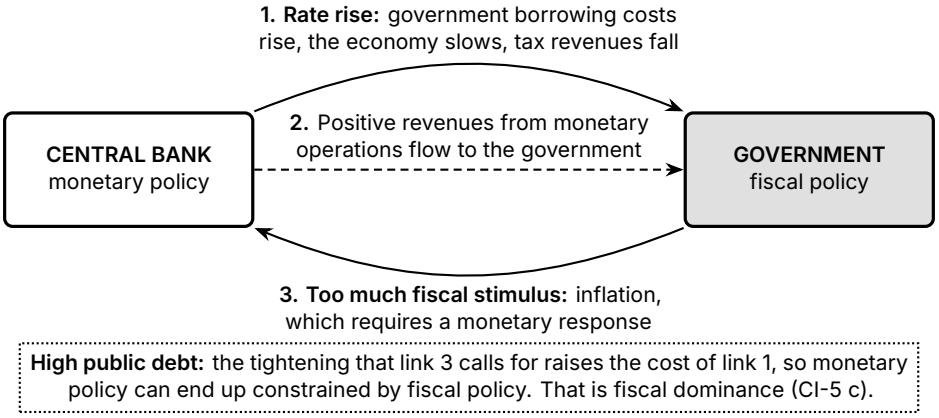


Figure 2. The overlap in three links. Solid arrows: links 1 and 3, which close the feedback loop; dashed arrow: link 2, the revenue channel. The dotted box is how high debt turns the loop into a constraint on the central bank.

4. The region of stability

Definition. The region of stability is the safe zone, defined over the *joint* stances of fiscal and monetary policy, within which stability is maintained: the economy is growing and healthy, with stable inflation and stable growth. The class notes call it a balanced zone where both policies support stable, healthy growth.

Because the region is defined over the *pair* of stances, it is an area, not two separate thresholds. A fiscal stance that is safe alongside tight monetary policy may sit outside the region alongside loose monetary policy (Figure 3).

Why it is hard to manage.

- The boundaries **vary by country**.
- The region is difficult to quantify precisely in a **single metric**.
- Its **shape, size and locus** are constantly changing. Some changes are **gradual** (demographic shifts, financial innovations); others are **rapid** (a commodity price spike from a geopolitical event, a sudden-onset pandemic).
- **Regulation and foreign exchange interventions** can alter the region.

The margin of safety. Because the boundaries blur easily, policy should keep a margin of safety: fiscal policy should not be pushed to the outer limit of stability, but should stop short of the expected limit to allow for unexpected outcomes. The reading's example: prolonged monetary expansion (ultra-low interest rates) can encourage governments to build excessive debt, and too much debt can lead to inflation that hinders the economy. Moderation matters, because testing the boundaries can trigger painful outcomes. What happens when they are breached, and how that has differed between advanced and emerging economies, is CI-5 b.

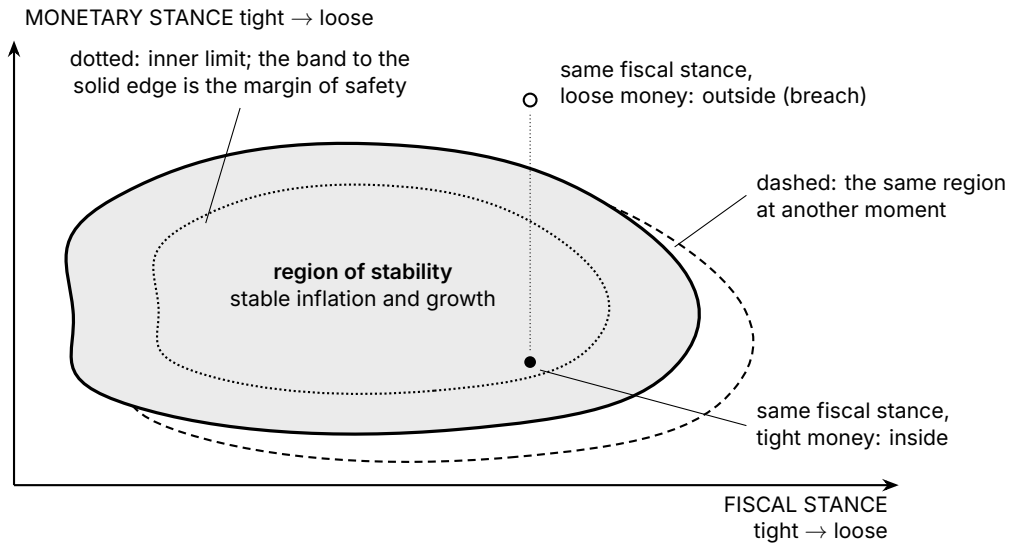


Figure 3. The region of stability over the joint stance, redrawn from the class notes (Figure CI-5.2). Solid outline (shaded): the region now; dotted outline: the inner limit that policy keeps to, so the band between it and the solid edge is the margin of safety; dashed outline: the same region at another moment, since demographics, shocks, regulation and FX intervention move the boundary. The filled and open dots share a fiscal stance: inside the region with tight money, outside it with loose money. Shape and position are illustrative, not measured.

Results card: CI-5 a

Fiscal policy	Government; *direct*: spending, tax policy, production of some services, transfers; first impacts on demand, employment, income distribution
Monetary policy	Central bank; *indirect*: open market transactions and policy rates; spills into bond yields, borrowing costs, asset prices, exchange rates
Overlap	Rate rise raises government borrowing costs and cuts tax revenue; central bank revenues flow to the government; fiscal excess forces a monetary response
High-debt tension	Tightening becomes costly for the budget; monetary policy constrained by fiscal policy is fiscal dominance (CI-5 c)
Region of stability	Joint fiscal and monetary stances that keep inflation and growth stable
Its properties	Country-specific; no single metric; shape, size and locus change gradually (demographics, innovation) or rapidly (commodity spike, pandemic); altered by regulation and FX intervention
Margin of safety	Stop short of the expected limit; ultra-low rates can encourage excessive debt

Exam traps

- **Role misassignment on tools.** Transfers, taxes, spending and services are fiscal even though they move money to or from households. Open market operations are monetary even though they trade government bonds.
- **Direct and indirect reversed.** Fiscal is the *direct* channel; monetary works *indirectly* through rates and money.
- **Direction in the open market chain.** To cut inflation the central bank *sells* bonds, draining money from the public.
- **Polarity on the overlap.** A rate rise *raises* government borrowing costs and *lowers* tax revenue; fiscal excess *raises* inflation.
- **Joint, not separate.** The region is defined over the pair of stances, not as a debt ceiling plus a rate ceiling.
- **Absolute claims on the region.** Not the same in all countries, not one metric, not fixed in size or shape.
- **Gradual versus rapid swap.** Demographics and financial innovation are gradual; a commodity spike or a pandemic is rapid.
- **Margin of safety.** The reading applies it to fiscal policy in particular; "run policy at the edge" is the wrong answer.
- **Fiscal dominance direction.** Monetary policy is the constrained party.

Practice questions: CI-5 a

Question CI-5 a.1

New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a

An analyst in a central bank's research department classifies four measures announced in the same week: (i) higher transfer payments to households; (ii) central bank sales of government bonds in open market operations; (iii) a rise in the policy rate; and (iv) a cut in public infrastructure spending. Which of the following classifications is correct?

A. (i) and (iv) are fiscal measures; (ii) and (iii) are monetary measures.

B. (ii) and (iv) are fiscal measures; (i) and (iii) are monetary measures.

C. (iv) is a fiscal measure; (i), (ii) and (iii) are monetary measures.

D. (i), (ii) and (iv) are fiscal measures; (iii) is a monetary measure.

Question CI-5 a.2

New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a (Table CI-5.1)

An economist at a finance ministry is contrasting how the two policies reach the economy and financial markets. Which of the following statements is correct?

A. Monetary policy acts directly through government spending and taxes, while fiscal policy works indirectly through bond yields and exchange rates.

B. Fiscal policy acts directly via spending, taxes and transfers; monetary policy acts indirectly via money and rates, spilling into market prices.

C. Both policies act only through interest rates, and they differ mainly in whether the government or the central bank sets the rate.

D. Fiscal policy acts first on bond yields, asset prices and exchange rates, while monetary policy acts first on employment and income distribution.

Question CI-5 a.3

New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a (Figure CI-5.1)

A trainer at a central bank's staff college asks new economists to identify a correctly described route from a policy action to lower inflation, with the right actor and the right direction. Which of the following chains is correct?

A. The central bank buys bonds from the public, which drains money from investors, lowers the money supply and so reduces inflation.

B. The central bank raises taxes, which leaves households with less money to spend after tax, so demand and prices fall.

C. The central bank raises the policy rate, which makes loans costlier, reduces consumer spending and lowers demand and prices.

D. The central bank cuts its spending on infrastructure and subsidies, which lowers demand and prices and so reduces inflation.

Question CI-5 a.4 New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a

A newly hired sovereign analyst at a London asset manager asks how monetary and fiscal policy overlap in practice. Which of the following illustrations is most consistent with the reading?

A. A policy rate rise lowers government borrowing costs, because higher rates attract more buyers to new government bond issues.

B. Revenues from monetary policy operations stay on the central bank's balance sheet, so they cannot support fiscal spending.

C. Fiscal stimulus lowers inflation by raising output, which in turn allows the central bank to ease monetary policy.

D. A policy rate rise raises government borrowing costs and, as the economy slows, lowers the government's tax revenues.

Question CI-5 a.5 New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a (Figure CI-5.2)

A sovereign strategist at a Swiss private bank is explaining the BIS concept of the "region of stability" to clients. Which of the following definitions is correct?

A. A zone defined over the joint fiscal and monetary stance, within which the economy keeps stable inflation and stable growth.

B. The range of policy rates that a central bank can set without missing its inflation target, judged separately from fiscal policy.

C. A pair of separate ceilings, one on public debt and one on the policy rate, each of which can be checked against its own limit.

D. A fixed zone, common to all countries, whose boundary can be measured precisely with a single published metric.

Question CI-5 a.6 New • built from SchweserNotes Book 5, LO 104.a and 104.b; CI notes CI-5 a (Figure CI-5.2)

A sovereign analyst at a Milan asset manager observes that a government has kept the same fiscal stance for several years while its central bank moved from tight policy to prolonged ultra-low rates. Which of the following assessments is most consistent with the reading?

A. The fiscal stance cannot have left the region of stability, because only a change in fiscal policy can cause a fiscal breach.

B. Prolonged ultra-low rates can widen the region of stability, because cheaper borrowing gives the government more room for deficits.

C. Because the region is defined over the joint stance, an unchanged fiscal stance can fall outside it once money is very loose.

D. The fiscal stance stays inside the region while inflation is at target, because the region is defined by inflation alone.

Question CI-5 a.7 New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a

A strategist at an emerging-market bond fund notes that a country's region of stability appears to have shifted between two policy meetings. Which of the following statements about such shifts is consistent with the reading?

A. Shifts occur only gradually, through demographic change or financial innovation, so a sudden move points to measurement error.

B. Shifts can be gradual, as with demographic change, or rapid, as with a pandemic; regulation or FX intervention can also alter it.

C. Shifts can be gradual, as with a commodity price spike, or rapid, as with demographic change, but policy actions cannot alter it.

D. Shifts in size and shape are rare, because each country's region is anchored by the mandate it gives its central bank.

Question CI-5 a.8 New • built from SchweserNotes Book 5, LO 104.a; CI notes CI-5 a

A finance ministry adviser proposes running fiscal policy right up to the estimated outer limit of the region of stability, to give growth the most support. Which of the following responses best reflects the reading?

A. The plan is sound as long as monetary policy stays tight, because a tight monetary stance fixes the region's fiscal limit in place.

B. The plan is sound for advanced economies, whose regions of stability are large enough that no margin of safety is required.

C. The plan is sound, because the reading applies the margin of safety to monetary policy alone, not to fiscal policy.

D. The plan should stop short of the expected limit, since blurred, shifting boundaries call for a margin against surprises.

Solutions • CI-5 a

Solution CI-5 a.1 — Answer: A

New • built from SchweserNotes Book 5, LO 104.a

Step 1 – Fiscal tools

Fiscal policy acts through government spending, tax policy, the production of some services and transfer payments to individuals and businesses. So (i) transfers and (iv) infrastructure spending are fiscal.

Step 2 – Monetary tools

Monetary policy manages the money supply through open market transactions and policy interest rates. So (ii) open market sales and (iii) the policy rate are monetary.

$\Rightarrow$ Classify by who acts and through what, not by what happens to money afterwards.

Interpretation

The two traps are mirror images: transfers "feel" monetary because they put money in households' hands, and open market operations "feel" fiscal because they involve government bonds.

Why the other options are wrong

B is incorrect. Double sibling swap: it moves transfers to monetary and open market operations to fiscal.

C is incorrect. Role misassignment: transfers are a fiscal channel even though they raise households' money balances.

D is incorrect. Role misassignment: open market operations are a monetary tool even though the securities traded are government bonds.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2, Monetary and Fiscal Policy: Safeguarding Stability and Trust; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.2 — Answer: B

New • built from SchweserNotes Book 5, LO 104.a

Step 1 – Fiscal: direct

Fiscal policy has a direct impact on the economy through government spending, tax policy, the production of some services and transfer payments.

Step 2 – Monetary: indirect

Monetary policy manages the money supply through open market transactions and policy rates, which spill over into bond yields, borrowing costs, asset prices and exchange rates.

$\Rightarrow$ Fiscal acts on economic activity first; monetary acts on financial markets first.

Interpretation

This is the "compare and contrast" half of the LO in one sentence. Once the route (direct or indirect) is right, the first impacts follow.

Why the other options are wrong

A is incorrect. Sibling swap: it reverses the two routes.

C is incorrect. Scope error: fiscal policy works through spending, taxes, services and transfers, not only through interest rates.

D is incorrect. Sibling swap on impacts: market prices are where monetary policy shows up first; employment and income distribution are fiscal impacts.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Table CI-5.1; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.3 — Answer: C

New • built from SchweserNotes Book 5, LO 104.a; CI notes Figure CI-5.1

Step 1 – The rate chain

A higher policy rate makes loans and borrowing costlier, consumer spending falls, demand and prices fall, and inflation comes down.

Step 2 – Check the actor and the direction of the other three

To cut inflation the central bank *sells* bonds, draining money. Taxes and spending on infrastructure and subsidies are government (fiscal) tools, not central bank tools.

$\Rightarrow$ Only C has the right actor and every arrow pointing the right way.

Interpretation

Each distractor breaks one thing: the direction of the open market operation (A), or who acts (B and D). Under exam pressure, check the first box of each chain, actor and verb, before reading on.

Why the other options are wrong

A is incorrect. Polarity flip: buying bonds *adds* money to the public. The contractionary open market operation is a sale.

B is incorrect. Role misassignment: this is the class notes' tax chain, but tax policy is a fiscal tool of the government, not of the central bank.

D is incorrect. Role misassignment: the chain is right, but government spending on infrastructure and subsidies is a fiscal tool; the central bank acts through open market transactions and policy rates.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Figure CI-5.1; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.4 — Answer: D

New • built from SchweserNotes Book 5, LO 104.a

Step 1 – Monetary to fiscal

If the central bank raises interest rates, government borrowing costs rise, the economy slows and tax revenues decrease.

Step 2 – The other links

Positive revenues from monetary policy operations flow through to the government and support fiscal spending. Too much fiscal stimulus can produce inflation that requires a monetary response.

$\Rightarrow$ The overlap runs both ways, and a rate rise hits the budget twice: higher interest costs and lower revenue.

Interpretation

This is why the two policies cannot be set in isolation, and why the region of stability is defined over their *joint* stance.

Why the other options are wrong

A is incorrect. Polarity flip: higher policy rates *raise* government borrowing costs.

B is incorrect. Polarity flip: positive revenues from monetary operations *do* flow to the government.

C is incorrect. Polarity flip: excessive fiscal stimulus *produces* inflation and calls for tightening, not easing.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.5 — Answer: A

New • built from SchweserNotes Book 5, LO 104.a; CI notes Figure CI-5.2

Step 1 – What it is defined over

The region is defined in terms of the *joint* stances of fiscal and monetary policy, as the LO wording says.

Step 2 – What holds inside it

Within the region, stability is maintained: the economy is growing and healthy, with stable inflation and stable growth.

⇒ It is an area over the pair of stances, not a threshold for either policy alone.

Interpretation

A definition built on one policy alone misses the point. Question CI-5 a.6 applies the joint definition to a case in which only monetary policy moves.

Why the other options are wrong

B is incorrect. Scope error: it defines a monetary-only range and ignores the fiscal stance.

C is incorrect. Scope error: two separate ceilings, each checked on its own, are not a region defined over the joint stance.

D is incorrect. Absolute claim: the boundaries vary by country, are hard to quantify in a single metric, and constantly change.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Figure CI-5.2; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.6 — Answer: C

New • built from SchweserNotes Book 5, LO 104.a and 104.b; CI notes Figure CI-5.2

Step 1 – What the region is defined over

The region of stability is defined in terms of the *joint* stances of fiscal and monetary policy. A fiscal stance that is safe alongside tight monetary policy may sit outside the region alongside loose monetary policy (class notes Figure CI-5.2).

Step 2 – What prolonged ultra-low rates do

The reading's margin-of-safety example: prolonged monetary expansion (ultra-low interest rates) can encourage the government to build excessive debt, and too much debt can lead to inflation. CI-5 b adds that the region is narrowed when low policy rates have encouraged high risk-taking and debt build-up.

⇒ The fiscal stance has not moved, but the joint stance has.

Interpretation

For the analyst, an unchanged budget is not evidence of safety: the monetary half of the pair has moved, and with it the pressure to build debt. This is the "joint policy stances" part of the LO wording applied to a case.

Why the other options are wrong

A is incorrect. Scope error: the region is defined over the pair of stances, so a move in monetary policy alone can take the joint stance outside it.

B is incorrect. Polarity flip: ultra-low rates can encourage excessive debt, and the region is *narrowed* when low policy rates have encouraged risk-taking and debt build-up.

D is incorrect. Scope error: the region is defined over the joint stance and calls for stable inflation *and* stable growth; inflation at target on its own does not place the stance inside it.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Figure CI-5.2; SchweserNotes 2026 Book 5, Reading 104, LO 104.a and 104.b

Solution CI-5 a.7

Answer: B

New • built from SchweserNotes Book 5, LO 104.a

Step 1 – Two speeds

The region's shape, size and locus are constantly changing. Some changes are gradual (demographic shifts, financial innovations); others are rapid (a commodity price spike from a geopolitical event, a sudden-onset pandemic).

Step 2 – Policy can move it

Regulation and foreign exchange interventions can alter the region.

⇒ A sudden shift is consistent with the reading; it need not be a measurement problem.

Interpretation

For the strategist, a shifting region is a reason to keep a margin of safety, not a reason to distrust the concept.

Why the other options are wrong

A is incorrect. Absolute claim: rapid changes are explicitly recognised.

C is incorrect. Sibling swap on the examples and a false absolute: commodity spikes are rapid, demographics gradual, and regulation and FX intervention can alter the region.

D is incorrect. Polarity flip: shape and size are constantly changing; the reading ties the region to no mandate.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

Solution CI-5 a.8

Answer: D

New • built from SchweserNotes Book 5, LO 104.a

Step 1 – Why the edge is unsafe

The boundaries of the region can easily become blurred, and they keep moving.

Step 2 – The rule

There should always be a margin of safety: fiscal policy should not be pushed to the outer limit of stability but should stop short of what is expected to be the limit, to allow for unexpected outcomes.
⇒ Testing the boundaries can trigger painful outcomes; moderation is the stated principle.

Interpretation

The reading's cautionary example runs through the other policy: prolonged ultra-low rates can encourage governments to build excessive debt, which can lead to inflation.

Why the other options are wrong

A is incorrect. Invented mechanism: no policy stance fixes the region in place; its boundaries change with shocks, demographics, regulation and FX intervention.
B is incorrect. Scope error: advanced economies have a larger region than emerging ones (CI-5 b), but the margin of safety applies to them too.
C is incorrect. Role misassignment: the reading states the margin of safety for *fiscal* policy in particular: it should not be pushed to the outer limit of stability.

LO: CI-5 a: Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.a

CI-5 • Monetary and Fiscal Policy: Safeguarding Stability and Trust (BIS, June 2023) • Vardeez: IMPORTANT

CI-5 b

P3 CONCEPT

Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies.

Workbook: D16 Monetary and Fiscal Policy • Tier D • Map #67

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D16 Monetary and Fiscal Policy (Map #67, 2/16). On the map this LO carries Vardeez focus only: no workbook shape and no workbook question.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes: none on this LO.

What the questions look like: no calculations. Matching each historical phase to its symptom and its end state, the asymmetric rate policy of the low-inflation era, the two reasons policy kept drifting (loss of traction, self-reinforcement), and the four implications of foreign exchange dependency for emerging market economies (EMEs), including how they have responded. The SchweserNotes summary gives the most detail, and these questions draw on it.

1. The starting point: boundaries already stretched

Recap from CI-5 a: the region of stability is the set of joint fiscal and monetary stances that keep inflation and growth stable; its boundaries blur and move, so policy should keep a margin of safety.

Before the COVID-19 pandemic, both policies were already pushing the boundaries:

- **Monetary policy** had kept interest rates low and let central bank balance sheets balloon in the years since the 2007 to 2009 global financial crisis (GFC).
- **Fiscal policy** had produced excessive public debt and persistently high annual deficits.

The combination left a **small margin for error** and made the economy more prone to **shocks**, **high inflation** and **financial fragility**. These are the consequences of operating near or beyond the boundary. The reading adds that there may have been overconfidence about how far the boundaries could be stretched.

2. Advanced economies: two phases, then the pandemic

Phase 1: the 1960s to the mid-1980s. Rising inflation was the signal that policy was testing the boundaries. Policymakers (politicians and central bankers) believed they could **fine-tune** the economy through fiscal and monetary channels, aiming at stable unemployment and inflation. The result was **very high inflation**, fought with **very high policy rates**. The period brought a **surge in public debt** and a shift in mindset toward prioritising **price stability**.

Phase 2: the mid-1980s to the pandemic. Financial systems moved from government-led to market-led (**financial liberalisation**), and **globalisation** accelerated. Cheaper labour in EMEs helped drive inflation down. Boundary testing therefore shifted from excessive inflation to **risk-taking and financial imbalances**, such as outsized expansions in credit and asset prices. **Financial recessions replaced inflation-induced recessions**, and they had a **longer-lasting impact on growth**, as the economy carried a heavier burden of public debt.

The Phase 2 asymmetry. Monetary policy cut rates in contractions but did not tighten in the recoveries, because the economy was expanding without higher inflation (partly a result of globalisation). The low-rate regime enabled public debt to increase massively, as fiscal policy was used to stimulate growth further while low rates on their own were not lifting inflation to target.

After Phase 2: the pandemic. Significant fiscal stimulus was used to offset the expected economic effects of lockdowns. The economy rebounded rapidly, and inflation rose to elevated levels. The surge in inflation came on top of **historically high public and private debt** and **elevated asset prices**, all of which increased **financial vulnerability**.

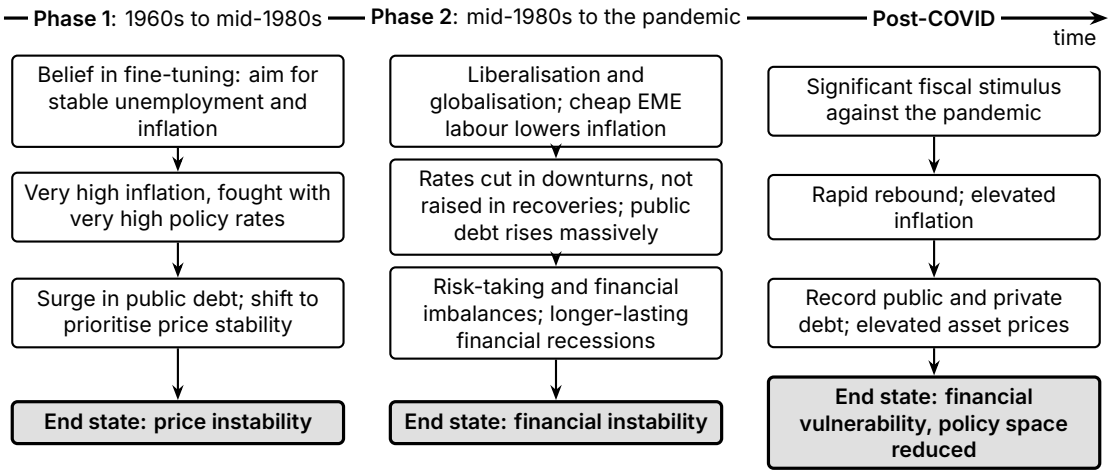


Figure 1. Three episodes, three ways of leaving the region, redrawn from the class notes (Figure CI-5.3) with the Phase 1 dates corrected (see the box below). The shaded end-state row is the one to memorise: Phase 1 ends in price instability, Phase 2 in financial instability, and the post-COVID period in financial vulnerability with much of the policy space already used. Phase 2 is the counter-intuitive one: low inflation and low rates were the conditions under which the imbalances built.

Correction to the source notes. The class notes' three-phase sketch (Figure CI-5.3) labels Phase 1 "1980 to mid-1980s", a span of about five years. SchweserNotes (Book 5, LO 104.b) and the D16 workbook notes both date Phase 1 from the **1960s** to the mid-1980s. The two-decade span matters: Phase 1 is the whole era of fine-tuning and high inflation, not a brief episode before liberalisation. Figure 1 uses the corrected dates.

3. Why policy kept drifting toward the boundary

The reading gives two reasons for the path that monetary and fiscal stimulus took.

1. **Loss of policy traction as the boundaries were approached.** Policy rate changes have less impact on aggregate demand when rates are low, so larger adjustments (for example 50 or 75 basis point moves) are needed to have an effect. Fiscal stimulus also becomes less powerful when public debt is high, because the sustainability of the stimulus and default risk enter the conversation. In SchweserNotes' summary: with high sovereign debt, fiscal stimulus has less effect and more extreme monetary action is needed. The region itself narrows when low rates have encouraged high risk-taking and debt build-up.
2. **Self-reinforcing interaction between the two policies.** Lower policy rates reduce fiscal borrowing constraints and encourage higher public debt. After the pandemic, central bank holdings of government debt soared in advanced economies. Higher public debt makes monetary normalisation (of interest rates and of central bank balance sheets) more difficult, and lower rates are sought to keep debt service costs down.

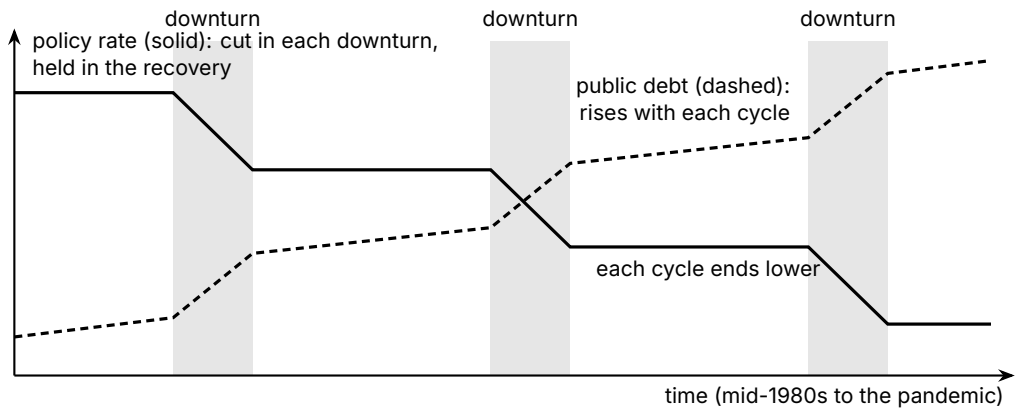


Figure 2. The Phase 2 ratchet, stylised. Shaded bands are downturns. The solid line is the policy rate: cut in each downturn and not raised in the recovery, because inflation stayed low. The dashed line is public debt, which the low-rate regime let rise cycle after cycle. The shape follows the reading's description; there are no data on the axes.

4. Emerging market economies

What EMEs share with AEs. Financial liberalisation increased the incidence of financial cycles and banking crises in EMEs too: the **Southern Cone** crisis in Latin America in the early 1980s, the **Tequila** crisis in Mexico in 1994 and the **Asian** crisis in the late 1990s. Globalisation broadly held inflation down, but political and institutional factors produced different experiences across EMEs: **Asia** enjoyed relatively low inflation, while **Latin America** experienced hyperinflation in the 1980s and 1990s, often paired with sovereign debt defaults.

Why EMEs are more exposed. EMEs are more sensitive to global financial conditions because their financial markets are less developed and they rely more on foreign exchange funding. The degree of reliance, and so the impact, varies across EMEs. Foreign exchange dependency has four broad implications:

1. **A smaller region.** The region of stability is much smaller for EMEs than for AEs, so the margin for error is smaller. Stepping outside the boundaries produces a rapid and potentially dramatic response, and the speed and intensity of a breach can force a rapid reversal of policy.
2. **The symptoms of a breach.** Breaches often lead to capital outflows and currency depreciation; exchange rate dependency can cause currency crises. Over time, EMEs have shifted toward more flexible exchange rate goals and inflation-targeting frameworks.
3. **Spillovers from AE policy.** Because of their exchange rate dependency, monetary policy in AEs, especially the United States, can have a big impact on EMEs. A US policy rate change can move exchange rates unfavourably for EMEs and raise concerns about domestic growth. Flexible exchange rate goals can help mitigate this.
4. **Exposure to global financial conditions.** As financial crises become more global, EMEs focus more than ever on flexible exchange rates and inflation targeting.

How EMEs have responded. Flexible exchange rates and inflation targeting are building shock resilience and letting EMEs run countercyclical policies when times get tough. During the pandemic, for example, Latin America tightened monetary policy **earlier and more forcefully** than its AE counterparts. These improvements relax policy constraints, which **widens the region of stability**, though it may also introduce other risks.

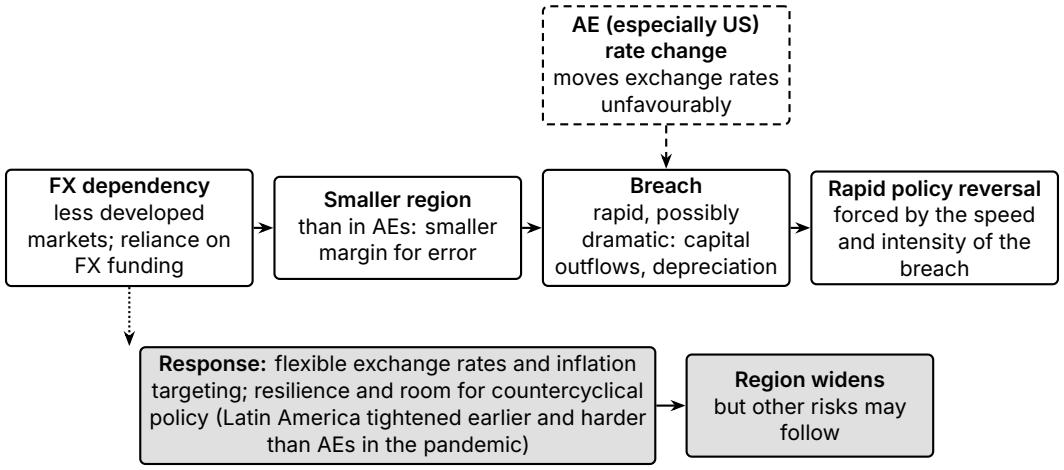


Figure 3. Why breaches play out faster in EMEs. Top row: the chain from foreign exchange dependency to a forced policy reversal; dashed box and arrow: the spillover from AE policy that can trigger it. Shaded row: the EME response and its payoff. The arrows follow the reading's four implications; the reading does not rank them.

Results card: CI-5 b

	Advanced economies	Emerging market economies
Size of region	Larger	Much smaller: smaller margin for error
How a breach shows	Phase 1 inflation; Phase 2 financial imbalances and longer-lasting financial recessions; post-COVID financial vulnerability	Rapid, possibly dramatic: capital outflows, currency depreciation, possible currency crises; forced policy reversal
Crisis record	Inflation-induced recessions, then financial recessions	Southern Cone (early 1980s), Tequila (1994), Asian (late 1990s); Latin American hyperinflation with defaults, Asia low inflation
External exposure	Source of spillovers (especially US policy)	Sensitive to global conditions and AE policy via FX dependency
Response	Shift to price stability after Phase 1	Flexible exchange rates and inflation targeting; earlier, firmer tightening in the pandemic (Latin America)
Why policy drifted	Loss of traction (low rates need 50 or 75 bp moves; high debt weakens fiscal stimulus); self-reinforcement (low rates encourage debt; debt makes normalisation harder)	

Exam traps

- **Phase swap.** Phase 1 is fine-tuning, high inflation and high rates; Phase 2 is liberalisation, globalisation, low inflation and financial imbalances.
- **Dates.** Phase 1 runs from the 1960s, not 1980, to the mid-1980s.
- **Polarity on the asymmetry.** In Phase 2 rates were cut in downturns and *not* raised in recoveries;

that is what let debt rise.

- **Recession type and length.** Financial recessions *replaced* inflation-induced ones and lasted *longer*.
- **Polarity on traction.** Low rates mean *less* effect per move, so *larger* moves; high debt makes fiscal stimulus *weaker*.
- **Size of the EME region.** Smaller, not larger; breaches bring depreciation and outflows, not appreciation.
- **Region swap.** Hyperinflation with defaults was Latin America; Asia had relatively low inflation.
- **Crisis dates.** Southern Cone early 1980s, Tequila 1994, Asian late 1990s.
- **Response.** Flexible exchange rates and inflation targeting, not pegs; Latin America tightened *earlier and harder* than AEs.

Practice questions: CI-5 b

Question CI-5 b.1 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

An economist at a global asset manager is briefing the investment committee on the policy mix in advanced economies just before the COVID-19 pandemic. Which of the following statements is consistent with the reading?

A. Low rates and large central bank balance sheets met high public debt and persistent deficits, leaving little margin for error.

B. Monetary policy had tightened steadily after the GFC while fiscal policy consolidated, rebuilding a wide margin of safety for both.

C. Low rates had kept monetary policy near the boundary, but fiscal consolidation had pulled public debt back to safe levels.

D. Both arms were near the boundary, which made the economy more prone to shocks but not to inflation or financial fragility.

Question CI-5 b.2 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b (correction box)

An economic historian at a pension fund is describing how advanced-economy policy tested the region of stability from the 1960s to the mid-1980s. Which of the following descriptions of that phase is correct?

A. Financial liberalisation and globalisation lowered inflation, so boundary testing showed up as credit booms and financial recessions.

B. Belief in fine-tuning led to very high inflation and policy rates, a surge in public debt and a shift to price stability.

C. Policymakers held rates low to support growth, so public debt fell while inflation stayed close to target throughout the phase.

D. Belief in fine-tuning led to very high inflation, which policymakers accommodated by holding policy rates low throughout.

Question CI-5 b.3 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

A macro strategist is explaining why public debt in advanced economies rose so much between the mid-1980s and the pandemic. Which of the following explanations is most consistent with the reading?

A. Rates were raised promptly in each recovery, so governments borrowed heavily to offset the resulting drag on growth.

B. High inflation driven by globalisation forced repeated rounds of fiscal stimulus to protect households' real incomes.

C. Rates were cut in downturns but not raised in recoveries as inflation stayed low, and low rates let public debt rise massively.

D. Rates were cut in downturns but not raised in recoveries, which kept inflation above target and so eroded the real value of debt.

Question CI-5 b.4 Adapted from CI QB Imp Q38 • SchweserNotes Book 5, LO 104.b; CI notes CI-5 b (Figure CI-5.3)

A risk manager at a Canadian insurer is comparing the recessions that followed boundary breaches in the two pre-pandemic phases in advanced economies. Which of the following comparisons is correct?

A. Phase 1 breaches ended in financial recessions with lasting growth effects; Phase 2 breaches ended in brief inflation-induced recessions.

B. Phase 2 breaches ended in inflation-induced recessions that lasted longer than those of Phase 1, because public debt was higher.

C. In Phase 2, financial recessions replaced inflation-induced ones but were shorter, because low rates allowed quicker recoveries.

D. In Phase 2, financial recessions replaced inflation-induced ones and had a longer-lasting impact on growth, under heavier debt.

Question CI-5 b.5 Adapted from CI QB Imp Q42 • SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

An equity strategist at a Paris-based asset manager reviews how the pandemic period played out in advanced economies. Which of the following statements is consistent with the reading?

A. Large fiscal stimulus brought a rapid rebound and elevated inflation, on top of high public and private debt and asset prices.

B. Large fiscal stimulus was offset by lockdowns, so inflation stayed low and debt ratios fell back toward pre-pandemic levels.

C. Fiscal support was modest, so the rebound was slow and inflation stayed below target, although asset prices still rose.

D. Large fiscal stimulus brought a rapid rebound, and low pre-pandemic public debt left ample policy space to absorb the inflation.

Question CI-5 b.6 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

A rates strategist is assessing how the effectiveness of monetary and fiscal policy changes as the boundaries of the region of stability are approached. Which of the following statements is consistent with the reading?

A. When rates are low, rate changes affect demand less, so larger moves are needed; high debt makes fiscal stimulus more powerful.

B. When rates are low, rate changes affect demand more, so smaller moves suffice; high debt makes fiscal stimulus less powerful.

C. When debt is high, monetary policy needs smaller moves, because each rate rise has a larger effect on the interest bill.

D. When rates are low, rate changes affect demand less, so 50 or 75 bp moves are needed; high debt weakens fiscal stimulus.

Question CI-5 b.7 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

A sovereign analyst at a German bank describes how high public debt feeds back into monetary policy in advanced economies, as part of the self-reinforcing interaction between the two policies. Which of the following descriptions is correct?

A. Lower rates tighten fiscal borrowing constraints, which forces governments to cut debt and so eases monetary normalisation.

B. Higher debt makes normalising rates and balance sheets harder, so lower rates are then sought to contain debt service costs.

C. Higher debt makes monetary normalisation easier, because central banks can shrink their balance sheets by selling the bonds.

D. Central bank holdings of government debt fell sharply after the pandemic, which broke the link between the two policies.

Question CI-5 b.8 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

An emerging-market sovereign analyst is reviewing how financial liberalisation played out in emerging market economies. Which of the following statements is consistent with the reading?

A. Liberalisation lowered the incidence of banking crises by deepening markets, and inflation stayed uniformly low across EMEs.

B. Liberalisation raised the incidence of crises; Asia had hyperinflation with sovereign defaults, while Latin America kept inflation low.

C. Liberalisation raised the incidence of crises such as the Southern Cone, Tequila and Asian crises; Latin America had hyperinflation.

D. Liberalisation raised the incidence of crises, including the Tequila crisis of the late 1990s and the Asian crisis of 1994.

Question CI-5 b.9 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

A portfolio manager at an emerging-market debt fund asks why breaches of the region of stability play out differently in EMEs than in advanced economies. Which of the following statements is correct?

A. EMEs' region of stability is larger, because faster growth gives them more margin for error before a breach occurs.

B. A breach in an EME tends to bring currency appreciation and capital inflows, as investors chase the higher yields on offer.

C. AE monetary policy has little effect on EMEs, because their reliance on foreign exchange funding shields them from US rate moves.

D. EMEs' region is smaller, so breaches bring rapid, possibly dramatic responses (outflows, depreciation) that can force a reversal.

Question CI-5 b.10 New • built from SchweserNotes Book 5, LO 104.b; CI notes CI-5 b

A central bank adviser in Latin America summarises how emerging market economies have responded to their exposure to global financial conditions. Which of the following summaries is consistent with the reading?

A. EMEs moved to flexible exchange rates and inflation targeting; Latin America tightened earlier and harder than AEs in COVID-19.

B. EMEs moved to fixed exchange rate pegs to block spillovers from US policy, and Latin America tightened later than AEs.

C. EMEs moved to flexible exchange rates and inflation targeting, which removed all risk by fixing the size of the region for good.

D. EMEs moved to flexible exchange rates, but Latin America tightened later and less forcefully than AEs during the pandemic.

Solutions • CI-5 b

Solution CI-5 b.1 — Answer: A

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – Both arms

Before the pandemic, monetary policy had kept rates low and central bank balance sheets had ballooned since the GFC; fiscal policy had built excessive public debt and persistently high deficits.

Step 2 – The consequence

The combination left a small margin for error and made the economy more prone to shocks, high inflation and financial fragility.

$\Rightarrow$ Both arms were close to the boundary at the same time.

Interpretation

This is the starting point for every later consequence in the LO: when the pandemic shock arrived, there was little room left inside the region.

Why the other options are wrong

B is incorrect. Polarity flip on both arms: rates stayed low and balance sheets grew; debt and deficits rose.

C is incorrect. Polarity flip on the fiscal arm: public debt was excessive and deficits persistent.

D is incorrect. Scope error: the reading lists shocks, high inflation *and* financial fragility.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2, Monetary and Fiscal Policy: Safeguarding Stability and Trust; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.2 — Answer: B

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – The belief

In the 1960s to the mid-1980s, policymakers believed they could fine-tune the economy through fiscal and monetary channels, aiming for stable unemployment and inflation.

Step 2 – The breach and its legacy

The result was very high inflation, fought with very high policy rates. Public debt surged, and the mindset shifted to prioritising price stability.

$\Rightarrow$ Phase 1 left the region through prices: its end state is price instability.

Interpretation

Note the dates: Phase 1 starts in the 1960s. The class notes' sketch labels it "1980 to mid-1980s" (see the correction box).

Why the other options are wrong

A is incorrect. Phase swap: liberalisation, globalisation, low inflation and financial recessions describe Phase 2.

C is incorrect. Polarity flip: inflation was very high and fought with very high rates, and public debt surged.

D is incorrect. Polarity flip on the response: high inflation was fought with very high policy rates, not accommodated.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June

2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.3 — Answer: C

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – The asymmetry

In Phase 2, monetary policy lowered rates in contractions and did not tighten in recoveries, because the economy expanded without higher inflation (partly a result of globalisation).

Step 2 – The debt build-up

The low interest rate regime enabled public debt to increase massively, as fiscal policy was used to stimulate growth further while low rates alone were not lifting inflation to target.
⇒ Each cycle left rates lower and debt higher (Figure 2).

Interpretation

This is the self-reinforcing interaction at work: lower rates relaxed fiscal constraints, and more debt then made it harder to raise rates.

Why the other options are wrong

A is incorrect. Polarity flip: rates were *not* raised in the recoveries.
B is incorrect. Polarity flip on inflation: globalisation helped push inflation *down* in Phase 2.
D is incorrect. Polarity flip on the outcome: inflation stayed *low*, below what the stimulus was aiming for, so it did not erode the debt.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.4 — Answer: D

Adapted from CI QB Imp Q38 • SchweserNotes Book 5, LO 104.b; CI notes Figure CI-5.3

Step 1 – What changed

With inflation low in Phase 2, boundary testing shifted from excessive inflation to risk-taking and financial imbalances. Inflation-induced recessions were replaced by financial recessions.

Step 2 – How they differed

Financial recessions had a longer-lasting impact on economic growth, as the economy suffered under the burden of higher public debt.
⇒ Phase 2 breaches showed up in finance rather than in prices, and cost more growth.

Interpretation

The end states differ: Phase 1 price instability, Phase 2 financial instability. A question that pairs the right recession type with the wrong phase or the wrong length is testing exactly this row.

Why the other options are wrong

A is incorrect. Phase swap: Phase 1 breaches showed up in inflation; financial recessions belong to Phase 2.
B is incorrect. Sibling swap: Phase 2's recessions were financial, not inflation-induced.
C is incorrect. Polarity flip on length: financial recessions had a *longer*-lasting impact on growth.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Figure CI-5.3; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.5 Answer: A

Adapted from CI QB Imp Q42 • SchweserNotes Book 5, LO 104.b

Step 1 – The policy response

Significant fiscal stimulus was used to offset the expected economic effects of the pandemic and lockdowns.

Step 2 – The outcome

The economy rebounded rapidly and inflation rose to elevated levels. The surge came on top of historically high public and private debt and elevated asset prices, all of which increased financial vulnerability.

$\Rightarrow$ The post-COVID end state is financial vulnerability with much of the policy space already used.

Interpretation

The pandemic arrived when both arms were already near the boundary (Question CI-5 b.1). The stimulus that worked on output also pushed the economy toward the inflation and fragility the reading warned of.

Why the other options are wrong

B is incorrect. Polarity flip: inflation became elevated after the rebound, and debt was at historic highs.

C is incorrect. Polarity flip on the stimulus and its result: support was significant and the rebound rapid.

D is incorrect. Polarity flip on the starting point: public debt was already excessive before the pandemic.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.6 Answer: D

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – Monetary traction

Policy rate changes have less impact on aggregate demand when rates are low, so larger adjustments, such as 50 or 75 basis point moves, are needed.

Step 2 – Fiscal traction

Fiscal stimulus becomes less powerful when public debt is high, because sustainability and default risk enter the conversation.

$\Rightarrow$ Near the boundaries, both levers need more force for the same effect.

Interpretation

SchweserNotes' quiz states the same point from the debt side: when sovereign debt is high, fiscal stimulus has less effect and more extreme monetary action is needed.

Why the other options are wrong

A is incorrect. Polarity flip on the fiscal half: the monetary half is right, but high debt makes fiscal stimulus *less* powerful, as sustainability and default risk enter the conversation.

B is incorrect. Polarity flip on the monetary half: the fiscal half is right, but low rates give rate changes *less* impact on demand, so larger moves are needed.

C is incorrect. Polarity flip: the reading says *more* extreme monetary action is needed, not less. The interest-bill effect of rate rises is the tension of CI-5 c, not a reason for smaller moves.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these conse-

quences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.7 — Answer: B

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – From monetary to fiscal

Lower policy rates reduce fiscal borrowing constraints and encourage higher public debt loads.

Step 2 – From fiscal back to monetary

Higher public debt makes monetary normalisation (of rates and of central bank balance sheets) more challenging, and lower rates are sought to minimise debt service costs.

$\Rightarrow$ The question asks about Step 2, the return leg: debt holds rates down, which closes the loop that Step 1 opens.

Interpretation

After the pandemic, central bank holdings of government debt soared in AEs, which ties the balance sheet side of normalisation to the debt as well. Question CI-5 b.3 tests the first leg, low rates letting public debt rise.

Why the other options are wrong

A is incorrect. Polarity flip on the first leg: lower rates *ease* fiscal constraints and encourage more debt, which makes normalisation harder, not easier.

C is incorrect. Polarity flip: higher debt makes normalisation *harder*.

D is incorrect. Polarity flip: central bank holdings of government debt *soared* after the pandemic.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.8 — Answer: C

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – Crises

As in AEs, financial liberalisation increased the incidence of financial cycles and banking crises: the Southern Cone crisis (early 1980s), the Tequila crisis in Mexico (1994) and the Asian crisis (late 1990s).

Step 2 – Inflation

Political and institutional factors produced different experiences: Asia had relatively low inflation, while Latin America experienced hyperinflation in the 1980s and 1990s, often with sovereign defaults.

$\Rightarrow$ EMEs shared the crisis pattern but split on inflation.

Interpretation

Two anchors do most of the work: the three crises with their dates, and the Asia versus Latin America inflation contrast.

Why the other options are wrong

A is incorrect. Polarity flip and absolute claim: liberalisation *raised* the incidence of crises, and inflation outcomes differed across EMEs.

B is incorrect. Region swap: hyperinflation with defaults was Latin America's experience; Asia had relatively low inflation.

D is incorrect. Wrong-input twin on the dates: the Tequila crisis was 1994 and the Asian crisis

the late 1990s.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.9 — Answer: D

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – The size of the region

Because of foreign exchange dependency, the region of stability is much smaller for EMEs than for AEs, so the margin for error is smaller.

Step 2 – What a breach looks like

Stepping outside produces a rapid and potentially dramatic response, often capital outflows and currency depreciation, and the speed and intensity of the breach can force a rapid policy reversal.
$\Rightarrow$ The same policy mistake costs an EME more, and sooner.

Interpretation

The AE policy spillover is the third implication: a US rate change can move exchange rates unfavourably and push an EME toward its boundary.

Why the other options are wrong

A is incorrect. Polarity flip: the EME region is *smaller*; growth is not the stated determinant.
B is incorrect. Polarity flip: breaches bring capital *outflows* and currency *depreciation*.
C is incorrect. Polarity flip: foreign exchange dependency is the reason AE, and especially US, policy has a *big* impact on EMEs.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

Solution CI-5 b.10 — Answer: A

New • built from SchweserNotes Book 5, LO 104.b

Step 1 – The frameworks

Over time, EMEs have shifted to more flexible exchange rate goals and inflation-targeting frameworks, which build shock resilience and allow countercyclical policy when times get tough.

Step 2 – The pandemic evidence

Latin America tightened monetary policy earlier and more forcefully during the pandemic than its AE counterparts.
$\Rightarrow$ The response relaxes policy constraints and widens the region of stability, though it may introduce other risks.

Interpretation

This is the "evolved over time" half of the LO for EMEs: from crisis-prone, FX-dependent economies toward frameworks that give them more room.

Why the other options are wrong

B is incorrect. Polarity flip on the framework and the timing: the shift is toward *flexible* exchange rates, and Latin America tightened *earlier*.
C is incorrect. Absolute claim: the region widens, but the reading warns that other risks may follow, and the region keeps moving.
D is incorrect. Polarity flip on the evidence: Latin America tightened earlier and more forcefully

than AEs.

LO: CI-5 b: Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.b

CI-5 • Monetary and Fiscal Policy: Safeguarding Stability and Trust (BIS, June 2023) • Vardeez: IMPORTANT

CI-5 c

P3 CONCEPT

Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy.

Workbook: D16 Monetary and Fiscal Policy • Tier D • Map #67

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D16 Monetary and Fiscal Policy (Map #67, 2/16). The map entry names this LO directly: "LO 104.c as introduced but unresolved: the interest-growth differential and the risks of high public debt."
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D16·S2** The interest-growth differential and the risks of high public debt (Questions CI-5 c.9 and CI-5 c.10).
- **D16·S1** How the two policies interact, including the tension that high debt creates (fiscal dominance). Introduced in the CI-5 a notes (Section 3); tested here through fiscal dominance's sources and safeguard (Question CI-5 c.5) and Workbook D16.1, which is placed here as Question CI-5 c.10.

What the questions look like: no calculations. The three challenges, the spending and inflation facts with their recall numbers (4% to 5% of GDP, 25% to 50%, 6% of GDP), the definition and sign logic of the interest-growth differential, the countercyclical loop and fiscal dominance, the bank-sovereign loop, and the policy implications.

1. Three challenges

Recap from CI-5 a: fiscal policy acts directly, monetary policy indirectly, and the two overlap in both directions. That overlap creates a natural tension in balancing price stability against financial stability risks. The reading names three primary challenges, which organise this LO:

1. **Strained public finances**, from the intersection of record-high public debt, pressure to maintain spending, higher interest rates and slowing growth prospects (Sections 2 and 3).
2. **High public debt working against monetary tightening** when inflation must be fought (Section 4).
3. **Financial stability risks** from the level of public debt (Section 5).

2. Strained public finances

Government spending, debt, interest rates and inflation are connected.

- **Ratings.** Since the GFC, high public debt has brought declining sovereign credit ratings, more dramatically in advanced economies (AEs), though the decline is present in EMEs too.
- **Inflation flatters the balances.** Nominal prices and wages respond fairly quickly to inflation, and so do nominal tax revenues. Fiscal balances therefore look better than the underlying reality.

- **Spending is hard to cut.** Rising costs push spending up; inflation creates a need for higher public wages and transfer payments; age-related expenditure is expected to rise by **4% to 5% of GDP over the next 20 years**; the global push for sustainability is costly; and rising geopolitical tension calls for more defence spending.

If inflation stays stubbornly high, higher policy rates must be maintained. Two factors **amplify the effect of policy rates on public borrowing**:

1. **Central bank balance sheets.** After the GFC, central bank purchases in major AEs rose to **25% to 50% of outstanding government debt**. This monetisation *shortened the maturity* of public debt, which reduced borrowing costs and *lowered remittances* from central banks to governments. SchweserNotes lists it among the factors that amplify the effect of policy rates on public borrowing without spelling out why; the natural reading is a duration point, that shorter-maturity debt reprices sooner when the policy rate changes.
2. **Growth.** High debt raises the cost of debt service in annual budgets, estimated to grow to **6% of GDP**. The problem would be amplified if medium-term growth disappoints, for structural reasons such as slower labour force growth or slowdowns in EMEs.

The reading adds a historical regularity: higher inflation has led to higher public debt.

3. The interest-growth differential (D16·S2)

The **interest-growth differential** is the difference between interest rates and economic (GDP) growth. SchweserNotes defines it with policy rates.

- **Rates below growth**, as before the pandemic: public finances can thrive.
- **Rates above growth**: the economy can slow, which creates a natural challenge for public spending.

The monetary tightening needed to fight persistent inflation works against public finances **on both legs**: it raises the cost of government borrowing and constrains economic growth.

> **Addition to the source notes.** The class notes' CI-5 c section never mentions the interest-growth differential, although it is the reported shape for this LO (Map #67, D16·S2). It is a gap in the notes rather than an error in them, and it is added here from SchweserNotes (Book 5, LO 104.c). On the definition: SchweserNotes writes "the difference between policy rates and economic (i.e., GDP) growth"; the D16 workbook notes that the BIS original frames it as the interest rate paid on public debt minus growth (marked there as outside-source). The sign logic, rates below growth helping public finances and rates above growth straining them, is the same under either wording, and it is what the questions test.

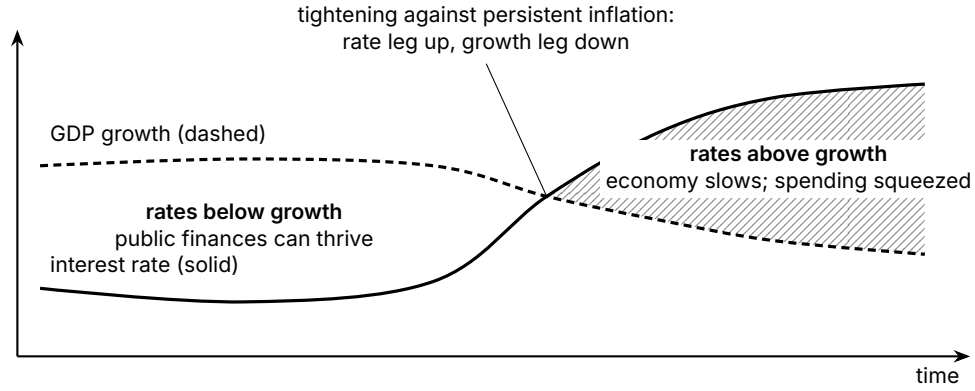


Figure 1. The interest-growth differential, stylised. Solid line: the interest rate; dashed line: GDP growth. Left of the crossing, rates sit below growth (the pre-pandemic case). Right of it, the hatched gap is the differential working against public finances. Tightening moves both legs the wrong way at once. There are no data on the axes.

4. Debt against monetary tightening

Countercyclical effects. In the reading's sense, monetary and fiscal policy are sometimes countercyclical: they work in *opposite* directions. Fiscal deficits remain large in many countries to support aggregate demand, which adds to inflation. The normal reaction to higher inflation is higher policy rates, which can weaken the economy, against the aim of fiscal policy. The tension becomes more acute the longer inflation remains high.

EMEs. The inflationary effect of fiscal stimulus is more pronounced in EMEs, because of their greater reliance on exchange rates. When the currency depreciates, domestic prices rise, which feeds ongoing higher inflation and slower growth.

Fiscal dominance. High debt can also stop monetary policy from taking the action it wants. **Fiscal dominance** exists when monetary policy is constrained by fiscally imposed constraints. It can arise from:

1. **Political spending pressure.** Pressure to maintain or increase spending on fiscal priorities can place a floor under fiscal flexibility unless priorities are adjusted.
2. **Sovereign debt implications.** Under certain circumstances, interest rate increases could spark a sovereign debt crisis.

The safeguard: strong institutional safeguards protecting the independence of monetary policy, so that fiscal policymakers cannot hijack monetary policy to pursue fiscal priorities.

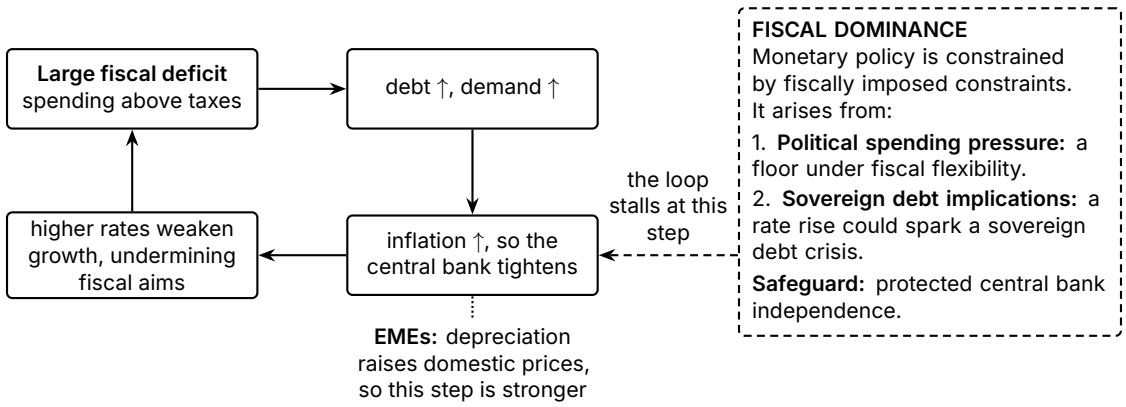


Figure 2. The countercyclical loop (left) and fiscal dominance (right), redrawn from the class notes (Figure CI-5.4). The notes' deficit box also lists "money supply ↑"; it is left out here as a simplification, because the reading runs this loop through demand and inflation (see the CI-5 a note on the money supply). Each arm's remedy is the next arm's problem, and the circuit closes. Fiscal dominance (dashed box and arrow) is the state in which the loop stalls at the tightening step, because the central bank can no longer take the action the loop calls for.

5. Financial stability risks

Sovereign debt vulnerability passes through to the financial system for two reasons.

1. **Direct exposure.** Sovereign debt helps keep the financial system working: it is a store of value, it serves as collateral, it is widely seen as the most liquid asset, and it helps price risk for other financial assets. Sovereign credit events (downgrades, defaults) can spill over into the pricing of risk across the economy.
2. **The government as ultimate backstop.** The government backstops the economy and the solvency of the financial system; the central bank often acts as **lender of last resort** and **market maker of last resort**. Banks likely have *higher* credit ratings with this backstop than without it.

The bank-sovereign feedback loop. Banks rely on the government for liquidity and on sovereign bonds as collateral. A credit downgrade at one can spark trouble at the other and reinforce a downward spiral. The loop is **most pronounced in EMEs**.

Sharp yield moves. The sheer size of high sovereign debt makes sharp bond yield moves more likely. When monetary policy tightens, long-term yields generally rise more in countries with high debt burdens. Ultra-low rates before the pandemic encouraged governments to issue long-term debt to lock in low rates, and long-term fixed-income assets are the most volatile when policy rates change. Sharp yield adjustments can also come from liquidity events, when fearful investors would rather hold cash than sovereign debt.

Repricing after loose policy. The financial sector is much more vulnerable after periods of loose fiscal and monetary policy. If investors judge that the boundaries of the region of stability are close to being breached, sovereign bonds could be repriced rapidly and aggressively, and the repricing could spread across the financial system.

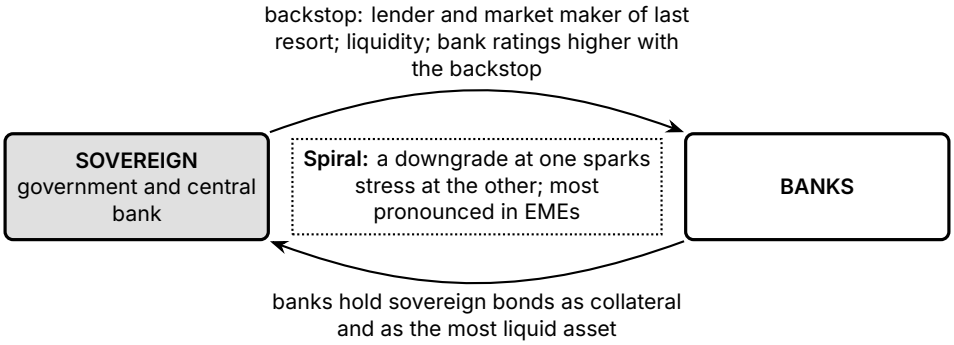


Figure 3. The bank-sovereign feedback loop. Each side props up the other (upper and lower arrows), which is why a downgrade at either end can reinforce a downward spiral (dotted box). The backstop raises bank ratings in good times and is the channel of contagion in bad times.

6. Policy implications

- **Stay inside the region.** Monetary and fiscal policy play complementary roles in financial stability, so both should aim to stay within the region of stability.
- **Monetary policy** should stay focused on price stability, because curing high inflation can be very costly. It should tolerate inflation below target while watching for signs of overheating.
- **Fiscal policy** should be as conservative as possible, to keep headroom for crises and avoid putting monetary policymakers in difficult positions. Relying on inflation to reduce sovereign debt burdens is **counterproductive**, because high inflation would severely weaken the economy.
- **Central bank balance sheets** should be kept as small as possible, for three reasons: (1) a smaller footprint on the economy and no crowding out of private lenders; (2) less budget reliance on transfers from the central bank; (3) more room to expand the balance sheet temporarily in a future crisis.
- **Quality of spending.** High-quality, productivity- and growth-enhancing spending (infrastructure, education) should come first; transfers and entitlement programmes have a place but should not be prioritised over growth.
- **Oversight.** Central bank independence guards against politicised monetary policy; guardrails such as fiscal rules or oversight panels keep spending focused on productivity and growth; regulation should be thoughtful but not overly restrictive.

Results card: CI-5 c

Three challenges	Strained public finances; debt against monetary tightening; financial stability risks
Recall numbers	Age-related spending $+4\%$ to 5% of GDP over 20 years • central bank purchases 25% to 50% of government debt (major AEs, post-GFC) • debt service to 6% of GDP
Amplifiers of policy rates	Central bank purchases shortened maturity and lowered remittances; disappointing growth (slower labour force growth, EME slowdowns)
Interest-growth differential	Interest rates minus GDP growth: below growth, finances thrive; above growth, strain. Tightening hits both legs
Countercyclical	Policies pull in opposite directions; worse the longer inflation lasts; stronger in EMEs via depreciation
Fiscal dominance	Monetary policy constrained by fiscal constraints: spending pressure; rate rises could spark a debt crisis. Safeguard: independence
Financial stability	Sovereign debt as collateral, liquid asset and benchmark; backstop raises bank ratings; bank-sovereign loop worst in EMEs; long yields rise more in high-debt countries
Policy	Price stability first; conservative fiscal policy; no inflating away; small central bank balance sheets; growth spending over transfers; independence and guardrails

Exam traps

- **Wrong-input twins on the numbers.** 4% to 5% of GDP is the rise in age-related spending; 6% of GDP is debt service; 25% to 50% is the central bank share of government debt.
- **Polarity on central bank purchases.** They *shortened* maturity and *lowered* remittances; limiting the balance sheet does the opposite.
- **Polarity and definition of the differential.** Rates below growth help; rates above growth strain. It compares interest rates with GDP growth, not real with nominal rates and not long with short yields.
- **Role misassignment on fiscal dominance.** Monetary policy is the constrained party; independence is the safeguard.
- **Two senses of "countercyclical".** In CI-5 c it means the two policies pulling in opposite directions; in CI-5 b, EMEs' "countercyclical policies" lean against the cycle.
- **Sibling swap, AE versus EME.** Ratings fell more in AEs; the bank-sovereign loop and the inflation effect of stimulus are stronger in EMEs.
- **Backstop polarity.** The backstop raises bank ratings, not lowers them.
- **Absolute claim.** "Inflating away the debt" is counterproductive, not a safe option.

Practice questions: CI-5 c

Question CI-5 c.1 Adapted from CI QB Imp Q43 • SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

An economist at a Gulf sovereign wealth fund lists the three primary challenges that the BIS links to high public debt. Which of the following lists is correct?

A. Currency crises, banking crises and sovereign defaults, as seen in the Southern Cone, Tequila and Asian episodes.

B. Strained public finances; high debt working against monetary tightening; and financial stability risks from the debt.

C. Ageing populations, the cost of the push for sustainability, and higher defence spending as geopolitical tensions rise.

D. Loss of central bank independence, a return to fixed exchange rates and the abandonment of inflation targeting.

Question CI-5 c.2 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A sovereign ratings analyst notes that several governments' fiscal balances improved during the recent inflation surge. A colleague concludes that public spending can now be cut easily. Which of the following responses is most consistent with the reading?

A. The improvement is genuine and lasting, because inflation lowers the real cost of public wages and transfer payments.

B. Inflation flattered the balances, but cuts will be easy, because age-related spending is projected to fall by about 6% of GDP.

C. The improvement reflects rising sovereign ratings, which have improved most in advanced economies since the GFC.

D. Inflation flattered the balances as nominal revenues rose quickly, while ageing, sustainability and defence push spending up.

Question CI-5 c.3 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A fixed-income strategist at a Tokyo life insurer is assessing how the cost of servicing public debt could evolve if policy rates stay high. Which of the following statements is consistent with the reading?

A. Debt service is projected to rise to about 4% to 5% of GDP, and faster medium-term growth would amplify the burden.

B. Debt service is projected to rise to about 6% of GDP, but slower labour force growth would ease the burden by cutting borrowing.

C. Debt service is projected to rise to about 6% of GDP, and weaker medium-term growth from structural factors would amplify it.

D. Debt service is projected to rise to about 6% of GDP, but inflation will offset it for good by lifting nominal tax revenues.

Question CI-5 c.4 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A macro economist at a Johannesburg-based bank is analysing an emerging market economy that is running a large fiscal deficit while its currency depreciates. Which of the following outcomes is most consistent with the reading?

A. Depreciation lifts domestic prices and adds to inflation, so the central bank tightens, weakening growth against the stimulus.

B. Depreciation lowers domestic prices by making exports cheaper, so the central bank can ease alongside the fiscal stimulus.

C. Fiscal stimulus feeds inflation less in EMEs than in AEs, because their flexible exchange rates absorb most of the extra demand.

D. The two policies reinforce each other: the stimulus lifts inflation, and higher rates then lift growth by attracting capital.

Question CI-5 c.5 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c (Figure CI-5.4)

A governance adviser to an emerging-market central bank is asked what creates fiscal dominance and what safeguard the reading recommends. Which of the following answers is correct?

A. Central bank pressure that forces spending cuts creates it; the safeguard is a fiscal rule limiting the central bank's room to raise rates.

B. Policies pulling in opposite directions create it; the safeguard is for the central bank to tolerate inflation above target until debt falls.

C. Spending pressure and sovereign debt risk create it; the safeguard is a mandate to set rates low enough to stabilise debt service.

D. Spending pressure and the risk that rate rises spark a debt crisis create it; the safeguard is protected central bank independence.

Question CI-5 c.6 Adapted from CI QB Imp Q46 • SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A bank supervisor is drafting a note on how sovereign debt vulnerability passes through to the financial system. Which of the following statements is LEAST accurate?

A. Sovereign debt serves as collateral and as the most liquid asset, and it helps price risk for other financial assets.

B. The central bank acts as the lender of last resort and as the market maker of last resort for the financial system.

C. The government backstop lowers banks' credit ratings, because it ties each bank's standing to the sovereign's credit.

D. As banks rely on the state for liquidity and on sovereign bonds as collateral, a downgrade at one can spiral into the other.

Question CI-5 c.7 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A rates strategist at a European pension fund is assessing how high public debt affects sovereign bond markets when monetary policy tightens. Which of the following assessments is consistent with the reading?

A. Long-term yields generally rise less in high-debt countries, because large and liquid markets absorb selling more easily.

B. Long-term yields generally rise more in high-debt countries, and the long-term debt issued at ultra-low rates is the most volatile.

C. Sharp yield moves come only from policy rate changes, because in a panic investors always prefer sovereign debt to cash.

D. Long-term yields generally rise more in high-debt countries, but loose policy beforehand leaves the financial sector less vulnerable.

Question CI-5 c.8 New • built from SchweserNotes Book 5, LO 104.c; CI notes CI-5 c

A finance ministry adviser recommends that the central bank keep its large balance sheet and that the government "let inflation run a little to shrink the debt". Which of the following responses best reflects the reading's policy implications?

A. Letting inflation run is acceptable for a short period, provided the central bank keeps a large balance sheet to support bond prices.

B. Central banks should keep large balance sheets, because the remittances they send to the budget reduce the need for fiscal restraint.

C. Fiscal policy should favour transfer payments over infrastructure and education, because transfers support demand fastest.

D. Relying on inflation is counterproductive, and smaller central bank balance sheets avoid crowding out and keep room for crises.

Question CI-5 c.9 Adapted from CI QB Imp Q44 • SchweserNotes Book 5, LO 104.c (interest-growth differential)

A sovereign credit analyst at a Singapore bank is asked by the risk committee what the interest-growth differential measures and what it signalled for public finances before the pandemic. Which of the following answers is correct?

A. It compares interest rates with GDP growth; with rates below growth, as before the pandemic, public finances can thrive.

B. It compares real and nominal policy rates; a low differential before the pandemic meant inflation was eroding public debt.

C. It compares interest rates with GDP growth; with rates below growth, as before the pandemic, public finances can come under strain.

D. It is the spread between long and short sovereign yields, so it measures the slope of the curve rather than public finances.

Question CI-5 c.10 Workbook D16.1 • S1, S2 • Map #67

A sovereign risk analyst at a Brazilian asset manager is preparing a briefing for the investment committee on the BIS analysis of high public debt. Policy rates in several advanced economies have recently risen above nominal GDP growth after a long period below it, while government debt remains near record highs. Which of the following statements should the analyst include in the briefing?

A. With policy rates now above GDP growth, the interest-growth differential works against public finances, and high debt that makes tightening harder to sustain raises the risk of fiscal dominance.

B. Central bank purchases of 25% to 50% of government debt after the GFC lengthened the maturity of public debt, which raised central bank remittances to governments.

C. Fiscal dominance describes fiscal policy being constrained by the central bank, because high policy rates force governments to cut spending to protect price stability.

D. The feedback loop between banks and their sovereign is most pronounced in advanced economies, because their larger debt stocks make sharp yield moves more likely.

Solutions • CI-5 c

Solution CI-5 c.1 — Answer: B

Adapted from CI QB Imp Q43 • SchweserNotes Book 5, LO 104.c

Step 1 – The budget

Public finances are strained by the intersection of record-high public debt, pressure to maintain spending, higher interest rates and slowing growth prospects.

Step 2 – The central bank and the system

High public debt can work against monetary tightening when inflation must be fought, and the level of public debt introduces financial stability risks.

⇒ The debt strains the budget, fights the central bank and endangers the financial system.

Interpretation

Everything else in the LO hangs on these three: spending pressure and amplifiers under the first, countercyclical effects and fiscal dominance under the second, the bank-sovereign loop under the third.

Why the other options are wrong

A is incorrect. Sibling swap: these are the EME crisis episodes of CI-5 b, not the challenges of high public debt.

C is incorrect. Sibling swap: these are spending pressures inside the first challenge, promoted to the list itself.

D is incorrect. Sibling swap: central bank independence, flexible exchange rates and inflation targeting are safeguards and frameworks; the reading does not list their loss as the challenges.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy.

• **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2, Monetary and Fiscal Policy: Safeguarding Stability and Trust; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.2 — Answer: D

New • built from SchweserNotes Book 5, LO 104.c

Step 1 – Why the balances look better

Nominal prices and wages respond fairly quickly to inflation, and so do nominal tax revenues. Inflation has made fiscal balances look better than reality.

Step 2 – Why cuts are hard

Rising costs push spending up; inflation calls for higher public wages and transfers; age-related spending is expected to rise by 4% to 5% of GDP over 20 years; the push for sustainability is costly; geopolitical tension needs more defence spending.

⇒ The improvement is a nominal effect, and the spending pressures run the other way.

Interpretation

For the ratings analyst, the right read is that headline balances overstate fiscal space while the structural pressures build.

Why the other options are wrong

A is incorrect. Polarity flip: inflation creates a need for *higher* public wages and transfer payments.

B is incorrect. Wrong-input twin and polarity flip: age-related spending is expected to *rise* by 4% to 5% of GDP; 6% of GDP is the projected cost of debt service.

C is incorrect. Polarity flip: since the GFC, high debt has brought *declining* sovereign ratings, most dramatically in AEs.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.3 — Answer: C

New • built from SchweserNotes Book 5, LO 104.c

Step 1 – The projection

Historically high debt burdens raise the cost of debt service in annual budgets, which is estimated to grow to 6% of GDP.

Step 2 – The amplifier

The problems from high public debt would be amplified if medium-term growth disappoints. Structural factors such as slower labour force growth and slowdowns in EMEs could lower growth forecasts.

$\Rightarrow$ Growth is the denominator that can turn a heavy debt bill into a crisis.

Interpretation

The other amplifier is the central bank balance sheet: post-GFC purchases of 25% to 50% of government debt shortened its maturity and lowered remittances; read as a duration point, shorter-maturity debt reprices sooner when rates rise. Question CI-5 c.10 option B tests its reversed form.

Why the other options are wrong

A is incorrect. Wrong-input twin and polarity flip: 4% to 5% of GDP is the projected rise in age-related spending, and it is *disappointing* growth that amplifies the problem.

B is incorrect. Polarity flip: slower labour force growth is one of the structural factors that *lower* growth and *amplify* the burden.

D is incorrect. Absolute claim on inflation: the inflation boost to nominal revenues makes fiscal balances look better than reality; it does not offset the debt bill for good, and relying on inflation to lessen debt burdens is counterproductive.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.4 — Answer: A

New • built from SchweserNotes Book 5, LO 104.c

Step 1 – The EME twist

The impact of fiscal stimulus on inflation is more pronounced in EMEs because of their higher reliance on exchange rates. Depreciation raises domestic prices, feeding ongoing higher inflation and slower growth.

Step 2 – The countercyclical response

The normal reaction to higher inflation is higher policy rates, which can weaken the economy, against the aim of fiscal policy.

$\Rightarrow$ The two policies pull in opposite directions, and the EME exchange rate makes the pull stronger.

Interpretation

The longer inflation stays high, the more acute this becomes. Note the reading's use of "countercyclical" here: policies working in opposite directions, not leaning against the cycle.

Why the other options are wrong

B is incorrect. Polarity flip: depreciation *raises* domestic prices.

C is incorrect. Sibling swap: the inflation effect of fiscal stimulus is *more* pronounced in EMEs, through exchange rates.

D is incorrect. Polarity flip: higher rates weaken growth and work *against* the fiscal stimulus.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.5 — Answer: D

New • built from SchweserNotes Book 5, LO 104.c; CI notes Figure CI-5.4

Step 1 – Definition

Fiscal dominance exists when monetary policy is constrained by fiscally imposed constraints.

Step 2 – Sources

Political spending pressure can place a floor under fiscal flexibility; and under certain circumstances, interest rate increases could spark a sovereign debt crisis.

Step 3 – Safeguard

Strong institutional safeguards protecting the independence of monetary policy, so fiscal policymakers cannot hijack it for fiscal priorities.

$\Rightarrow$ The cure protects the constrained party, the central bank.

Interpretation

The CI-5 a notes (Section 3) show how high debt turns the overlap into this constraint; this question tests where it comes from and what stops it, and Question CI-5 c.10 links it to the interest-growth differential.

Why the other options are wrong

A is incorrect. Role misassignment: it is monetary policy that is constrained, and a rule tying the central bank's hands is the problem, not the safeguard.

B is incorrect. Sibling swap and counterproductive remedy: opposite directions describes the countercyclical tension, and relying on inflation to reduce debt is counterproductive.

C is incorrect. Right sources, wrong safeguard: setting rates to stabilise debt service would *be* fiscal dominance.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; class notes Figure CI-5.4; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.6 — Answer: C

Adapted from CI QB Imp Q46 • SchweserNotes Book 5, LO 104.c

Step 1 – Check each channel

Sovereign debt is a store of value, collateral, widely seen as the most liquid asset, and a benchmark for pricing risk (A is accurate). The central bank acts as lender of last resort and market maker

of last resort (B is accurate). Banks rely on the state for liquidity and on sovereign bonds as collateral, so a downgrade at one can spiral into the other (D is accurate).

Step 2 – The backstop and ratings

Banks likely have *higher* credit ratings with the government backstop than without it.
⇒ C reverses the effect, so it is the least accurate statement.

Interpretation

The backstop is why the loop is dangerous: it lifts bank ratings while the sovereign is strong and transmits stress when it weakens, most of all in EMEs.

Why the other options are wrong

A is incorrect. The statement is accurate, so it is not the answer: these are the direct-exposure functions of sovereign debt.
B is incorrect. The statement is accurate, so it is not the answer: these are the central bank's backstop roles.
D is incorrect. The statement is accurate, so it is not the answer: this is the bank-sovereign feedback loop.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.7 — Answer: B

New • built from SchweserNotes Book 5, LO 104.c

Step 1 – Yields under tightening

High sovereign debt makes sharp yield moves more likely because of its sheer size. When monetary policy tightens, long-term yields generally rise more in countries with high debt burdens.

Step 2 – Why the long end

Ultra-low pre-pandemic rates encouraged governments to issue long-term debt to lock in low rates, and long-term fixed-income assets are the most volatile when policy rates change.
⇒ High debt plus long maturities means larger price moves when rates turn.

Interpretation

Holders of that long-dated paper, banks included, carry the price risk. If investors sense the region's boundary is near, the repricing can be rapid and spread through the system.

Why the other options are wrong

A is incorrect. Polarity flip: long-term yields rise *more* in high-debt countries.
C is incorrect. Absolute claim: sharp moves can also come from liquidity events, when fearful investors prefer cash to sovereign debt.
D is incorrect. Polarity flip in the second half: the financial sector is *more* vulnerable after periods of loose fiscal and monetary policy.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.8 — Answer: D

New • built from SchweserNotes Book 5, LO 104.c

Step 1 – Inflation as a debt tool

Relying on inflation to lessen sovereign debt burdens is counterproductive, because high inflation would severely weaken the economy. Monetary policy should stay focused on price stability.

Step 2 – Balance sheets

Central banks should keep their balance sheets as small as possible: to limit their footprint and avoid crowding out private lenders, to reduce budget reliance on central bank transfers, and to keep room to expand temporarily in a future crisis.

$\Rightarrow$ Both halves of the adviser's proposal run against the reading.

Interpretation

The reading also asks fiscal policy to be as conservative as possible, so that monetary policymakers are not put in a difficult position. The adviser's plan does the reverse on both arms.

Why the other options are wrong

A is incorrect. Counterproductive remedy: inflating away debt is rejected, and a large balance sheet is what the reading advises against.

B is incorrect. Polarity flip: reducing budget reliance on central bank transfers is one of the reasons to keep the balance sheet *small*.

C is incorrect. Polarity flip: high-quality, growth-enhancing spending such as infrastructure and education should come before transfers.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c

Solution CI-5 c.9 — Answer: A

Adapted from CI QB Imp Q44 • SchweserNotes Book 5, LO 104.c (interest-growth differential); workbook D16 notes

Step 1 – The definition

SchweserNotes defines the interest-growth differential as the difference between policy rates and economic (GDP) growth. The D16 workbook notes add that the BIS original uses the interest rate paid on public debt (marked there as outside-source). Either way, it compares interest rates with growth.

Step 2 – The pre-pandemic sign

When rates are below the growth rate, as they were before the pandemic, public finances can thrive. When rates rise above growth, the economy can slow, which creates a natural challenge for public spending.

$\Rightarrow$ Rates below growth are the favourable side; rates above growth are the strain.

Interpretation

Shape D16·S2. Workbook D16.1 (Question CI-5 c.10) takes the next step: rates have now risen above growth while debt stays near record highs, and it links that shift to the risk of fiscal dominance.

Why the other options are wrong

B is incorrect. Sibling swap: the differential compares interest rates with GDP growth, not real with nominal rates, and the reading treats relying on inflation to lessen debt burdens as counterproductive.

C is incorrect. Polarity flip: with rates below growth public finances can thrive; it is rates *above*

growth that strain them.
D is incorrect. Sibling swap: the slope of the yield curve is a different concept. The differential compares interest rates with GDP growth, and its sign is read for public finances.

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS (June 2023), Annual Economic Report 2023, Section 2; SchweserNotes 2026 Book 5, Reading 104, LO 104.c; workbook D16 revision notes

Solution CI-5 c.10 — Answer: A

Workbook D16.1 • S1, S2 • Map #67

Step 1 – Read the interest-growth differential

The differential is the difference between interest rates and GDP growth. Below growth (pre-pandemic) public finances can thrive; above growth the economy can slow and public spending comes under pressure.
⇒ The stem's move from rates below growth to rates above growth is exactly the shift that strains public finances.

Step 2 – Link high debt to the policy tension

High debt works against monetary tightening: the tightening needed to fight persistent inflation raises the cost of government borrowing and slows growth. Fiscal dominance exists when monetary policy is constrained by fiscally imposed constraints (spending pressure; the risk that rate rises spark a sovereign debt crisis).

Interpretation

A is correct. The briefing should flag that the committee's AE sovereign exposures now face both a worse differential and a higher risk that central banks are pressed to go easy; the BIS remedy is fiscal restraint and protected central bank independence.

Why the other options are wrong

B is incorrect. Polarity flip: central bank monetization shortened the maturity of public debt and lowered remittances from central banks to governments.
C is incorrect. Role misassignment: fiscal dominance is monetary policy constrained by fiscal policy; the stated mechanism reverses the actors.
D is incorrect. Sibling swap: the costly feedback loop between banks and governments is most pronounced in EMEs, not AEs. (High debt does raise the risk of sharp yield moves, but that is a separate point.)

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. • **Reference:** BIS, Annual Economic Report 2023, Section 2, via SchweserNotes Book 5, Reading 104, Module 104.2

CI-6 • Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets (IMF, September 2022) • Vardeez: IMPORTANT

CI-6 a

P3 CONCEPT

Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 Crypto, Tokenisation and Digital Resilience (Map #68, 1/16, notes only). The class notes tag CI-6 DEFER, with a historical question frequency of zero, and the map explains why that carries little weight: "Because Current Issues rotates every year, the absence of recall evidence carries almost no information." The reading is built in full. There are no workbook questions for CI-6, so every question in its four LOs is new or adapted from the CI question bank.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so every LO here gets 8 to 10 questions.

Reported exam shapes

- **D29·S1** CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda. The map gives no further split. This LO carries the categories, plus the background (DLT, unbacked crypto assets) that CI-6 b to d build on.

What the questions look like: no calculations. Statement audits on DLT and blockchain, the money test for unbacked crypto assets, and classification scenarios: given who issues a token, what right it carries and what (if anything) keeps its value stable, name the category and the reason.

1. What a crypto asset is

Crypto assets are a form of *digital asset* with two features: they use **cryptography** to secure information, and they rely on technologies such as **distributed ledger technology** (**DLT**) to record transactions.

- **DLT** broadly refers to a sequenced and standardised record of an activity that is cryptographically secured on a network that is typically decentralised. It is called *distributed* because the record is made available (distributed) *simultaneously to all network participants*.
- **Blockchain** is the best-known form of DLT. It is a chain of blocks: each block contains the records of multiple transactions and is linked to the previous block. Records are generally **immutable**, meaning they cannot be altered.

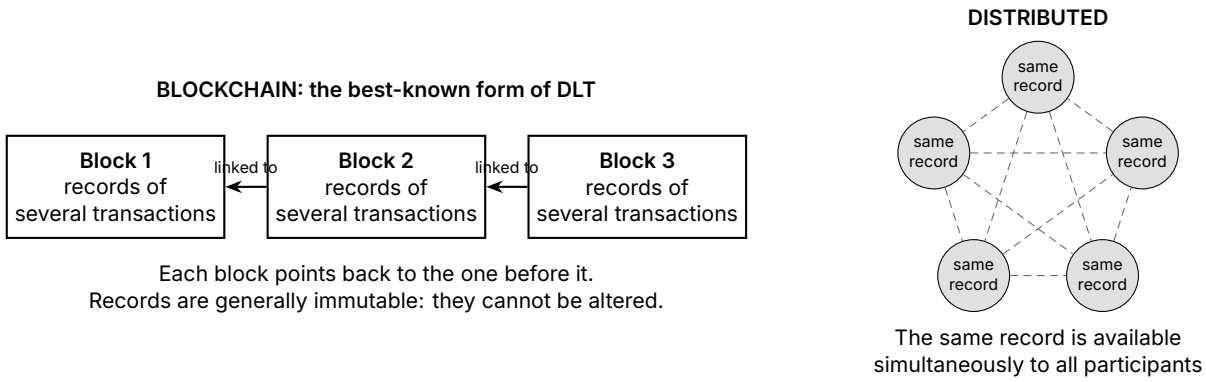


Figure 1. Left: blockchain, a chain of blocks each holding several transaction records and linked to the previous block (solid arrows). Right: what "distributed" means. Every participant on the typically decentralised network holds the same cryptographically secured record at the same time (dashed links). DLT is the general technology; blockchain is one form of it.

2. Backed and unbacked crypto assets

Crypto assets can be *backed* by another asset or *unbacked*. Unbacked crypto assets, such as Bitcoin and Ether, do not rely on the backing of another asset: their value is determined through **supply and demand**. They typically show **high volatility** and **low intrinsic value**, and they are often used for **speculative** activities.

The money test. Unbacked crypto assets are not considered money, because they are generally not used as a *medium of exchange*, a *store of value* or a *unit of account*. Be precise about where they fall short:

- **Not transferability.** They are transferable, and the class notes' category table describes them as designed to be used as a means of exchange. Transferability could meet the medium-of-exchange definition, so "they cannot be transferred" is the wrong reason.
- **The value.** Set by supply and demand, highly volatile, with low intrinsic value, and mainly used for speculation rather than for payments. That is why they do not fully meet all three definitions of money.
- **Not the technology.** The volatility comes from supply and demand, not from DLT. Immutability concerns the security of the records, not the asset's function as money.

> **Correction to the source notes.** The class notes say that unbacked assets like Bitcoin "have no intrinsic value". SchweserNotes (Book 5, LO 105.a) says they "have low intrinsic value", and the workbook's D29 revision notes agree. Use *low*. An option that rests on "no intrinsic value" states more than the reading does.

3. The six categories regulators use

There is **no single internationally accepted classification** of crypto assets. Regulators broadly use **six categories**, grouped on the basis of taxonomy and risks. The table follows the IMF note's Figure 1 as the class notes set it out (Table CI-6.1), with SchweserNotes' one-line definitions folded in.

Category	Defining characteristics	Examples
Unbacked crypto assets	Usually decentralised, sometimes centrally issued. Not backed by another asset; designed to be used as a means of exchange; transferable. Limited rights for the holder and no single issuer to enforce rights against. Mainly used for speculation, not payments.	Bitcoin, Ether
Nonfungible tokens (NFTs)	Usually centrally issued. A right to ownership of a specific product; collectible and non-substitutable.	
Security tokens	Centrally issued. A blockchain-based ownership interest in an asset, with rights similar to traditional securities, such as profit-sharing or ownership. Meets the definition of a security in each jurisdiction, so the definition varies by jurisdiction; within the regulatory perimeter.	
Utility tokens	Centrally issued. A blockchain-based right to a product or service; transferable and can be used as a means of exchange; can be accepted across multiple ecosystems but is often used within a single or closed network.	Gaming tokens, tokenised store cards
Stablecoins	Designed to be value stable. The stability mechanism can be backing or collateralisation with a commodity, fiat currency, multiple currencies or crypto assets, or algorithms.	Tether, USDC, Binance USD
Central bank digital currencies (CBDCs)	Centrally issued by a state or central bank as a digital form of money. Designed to be value stable; the stability mechanism is usually sovereign fiat currency.	

Two discriminators do most of the work (class notes):

1. **Who issues it.** Every category is centrally issued except unbacked crypto assets, which are usually decentralised. That is exactly why an unbacked crypto asset has no single issuer to enforce rights against.
2. **What keeps the value stable.** Only stablecoins and CBDCs are described as designed to be value stable. A stablecoin gets there through backing, collateralisation or an algorithm; a CBDC, issued by a state or central bank, usually through sovereign fiat currency. Same design goal, different issuer and mechanism.

Backing can itself be in crypto assets. A token designed to be value stable and collateralised with other crypto assets is a stablecoin: it relies on backing, which an unbacked crypto asset by definition does not. SchweserNotes' one-line definition (stablecoins are "backed by other currencies or assets") and the class notes' table agree on this case.

The rest of the reading focuses on four of the six: **unbacked crypto assets, security tokens, utility tokens and stablecoins**.

Results card: CI-6 a

Crypto asset	Digital asset that uses cryptography to secure information and relies on DLT to record transactions
DLT	Sequenced, standardised, cryptographically secured record on a typically decentralised network; "distributed" = available simultaneously to all participants
Blockchain	Best-known form of DLT: linked blocks of transaction records; records generally immutable
Unbacked crypto assets	Value from supply and demand; high volatility, low intrinsic value; speculative. Transferable, but generally not used as medium of exchange, store of value or unit of account, so not money
Classification	No single international classification; six broad categories based on taxonomy and risks
The six	Unbacked crypto assets, NFTs, security tokens, utility tokens, stablecoins, CBDCs
Two discriminators	Who issues it (all centrally issued except unbacked, usually decentralised); what keeps the value stable (stablecoin: backing, collateral or algorithm; CBDC: sovereign fiat)
Reading's focus	Unbacked crypto assets, security tokens, utility tokens and stablecoins

Exam traps

- **Hierarchy inverted.** Blockchain is a form of DLT, not the reverse.
- **"Distributed" misread.** It means the same record is available simultaneously to all participants, not that each holds a fragment or that the record reaches them in turn.
- **Immutability stretched.** It secures the record; it does not stabilise the price and it is not why unbacked crypto assets fail as money.
- **Wrong reason for "not money".** Unbacked crypto assets are transferable. They fall short because their volatile, supply-and-demand value means they are generally not used as medium of exchange, store of value or unit of account.
- **"No" for "low".** Intrinsic value is low, not absent.
- **Stablecoin and CBDC swapped.** Both are designed to be value stable. The CBDC is issued by a state or central bank and usually relies on sovereign fiat; the stablecoin relies on backing, collateral or an algorithm.
- **Crypto collateral, so unbacked?** No. A value-stable token backed by other crypto assets is a stablecoin; unbacked crypto assets rely on no backing at all.
- **One global definition.** There is no single international classification, and whether a token is a security depends on each jurisdiction's definition.
- **Classification confused with capital treatment.** The BCBS lower-risk and higher-risk groups (CI-6 c) are a prudential treatment of banks' exposures, not the regulators' taxonomy.

Practice questions: CI-6 a

Question CI-6 a.1 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

The chief technology officer of a Hong Kong custodian bank is briefing the board on the technology that records crypto asset transactions. Which of the following statements would the CTO be correct to make?

A. Blockchain is the broad family of ledger technologies, and distributed ledger technology is one specific form of blockchain used to record crypto assets.

B. A ledger is called distributed because each participant stores a different fragment of the record, so that no single participant holds the full history.

C. A distributed ledger is a cryptographically secured record available simultaneously to all participants, and blockchain records are generally immutable.

D. Blockchain records can be amended by the participants that validate them, which allows errors in earlier blocks to be corrected without breaking the chain.

Question CI-6 a.2 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A central bank economist in Seoul is drafting a speech arguing that unbacked crypto assets such as Bitcoin should not be treated as money. Which of the following arguments is most consistent with the IMF note?

A. Their value, set by supply and demand, is highly volatile, so they are generally not used as a medium of exchange, store of value or unit of account.

B. They cannot be transferred from one user to another without a regulated intermediary, so they are unable to serve as a medium of exchange for payments.

C. Their transaction records are immutable, so a payment made in error can never be reversed, and that feature on its own rules them out as money.

D. Their value is backed by a pool of other crypto assets whose prices are unstable, so they fail the store-of-value test that money must meet.

Question CI-6 a.3 Adapted from CI QB Imp Q51 • SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A policy analyst at an international standard-setting body is drafting a background note on how crypto assets are classified across jurisdictions. Which of the following statements should the note contain?

A. Regulators have adopted the BCBS split between lower-risk and higher-risk crypto assets as the single international classification of crypto assets.

B. There is no single internationally accepted classification, but regulators broadly use six categories based on taxonomy and risks.

C. A single international classification applies to stablecoins and CBDCs, while the other four categories are left to the national regulators.

D. The six broad categories sort crypto assets by the type of ledger they run on, permissioned or permissionless, and by their market size.

Question CI-6 a.4 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

After a sharp fall in the price of Ether, a retail investor complains to a financial ombudsman in Australia and asks whether she has a claim against the token's issuer. Based on the IMF note's description of unbacked crypto assets, which of the following responses is most accurate?

A. She can enforce profit-sharing rights against the issuing foundation, in the same way that the holder of a security token can against its issuer.

B. She has no claim, because although unbacked crypto assets are centrally issued, their white papers grant holders no enforceable rights.

C. She can redeem her tokens at their original value from the reserve of assets that backs them and is meant to keep their price stable.

D. She has no claim against an issuer, because holders have limited rights and issuance is usually decentralised, with no single issuer.

Question CI-6 a.5 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A compliance officer at a Singapore bank is classifying a token issued by a video game company. The company issues the token centrally, the token gives holders the right to buy in-game services, and players can transfer it to each other within the company's own network. Under the categories broadly used by regulators, the token is best classified as:

A. A security token, because it is centrally issued and transferable, which places it inside the regulatory perimeter in each jurisdiction.

B. A utility token, because it gives a right to a product or service and is used mainly within a single or closed network.

C. An unbacked crypto asset, because no other asset backs it and its holders have only limited rights against the company.

D. A nonfungible token, because it is centrally issued and can be used only within the issuer's own ecosystem of products.

Question CI-6 a.6 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A property developer plans to issue blockchain-based tokens that entitle holders to a share of the rental profits from an office building, and to sell them to investors in several countries. Its legal adviser is asked how regulators are likely to classify the tokens. Which of the following responses is most accurate?

A. They are likely to be security tokens, but whether they meet the definition of a security depends on each jurisdiction's own definition.

B. They are likely to be utility tokens, because a share of rental income is a right to a service that the issuer provides to holders.

C. They are security tokens under a single internationally agreed definition, so the same classification will apply in every country of sale.

D. They are likely to be stablecoins, because the rental income from the building backs the tokens and is what keeps their value stable.

Question CI-6 a.7 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A digital art platform issues tokens, each conferring ownership of one specific artwork. Each token is unique, cannot be substituted for another, and carries no profit-sharing rights; collectors buy and hold them. A regulator reviewing the platform asks which category the tokens fall into. Which of the following answers is correct?

A. Utility tokens, because each confers a right to a product, and such tokens can be used as a means of exchange across several ecosystems.

B. Stablecoins, because each token is backed by a specific asset, and backing by an asset is one of the mechanisms that stabilise value.

C. Unbacked crypto assets, because the price of each token is set only by supply and demand among collectors, with limited holder rights.

D. Nonfungible tokens, because each confers a right to ownership of a specific product and is collectible and non-substitutable.

Question CI-6 a.8 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A payments strategist at a French bank compares a fiat-backed stablecoin such as USDC with a central bank digital currency (CBDC). Which of the following statements correctly distinguishes the two?

A. Only the CBDC is designed to be value stable; the stablecoin's value is set by supply and demand, as for an unbacked crypto asset.

B. Both are designed to be value stable; the CBDC holds its value through an algorithm, while the stablecoin is backed by the state that issues it.

C. Both are designed to be value stable; the CBDC is issued by a state or central bank and usually relies on sovereign fiat currency.

D. Both are designed to be value stable; the stablecoin is issued by the central bank, while the CBDC is issued by commercial banks under licence.

Question CI-6 a.9 New • built from SchweserNotes Book 5, LO 105.a; CI notes CI-6 a

A risk analyst at an asset manager reviews a privately issued token that is designed to keep a stable value against the US dollar and is collateralised with a pool of other crypto assets. A colleague argues that, because its collateral is made up of crypto assets, the token must itself be an unbacked crypto asset. Which of the following responses best reflects the categories in the IMF note?

A. The colleague is right: a token's category follows its collateral, so a token backed by crypto assets is itself unbacked.

B. The token is a CBDC, because its value is pegged to a sovereign fiat currency, which is the usual CBDC stability mechanism.

C. The token is a stablecoin: it is designed to be value stable, and crypto asset collateral is one of the listed mechanisms.

D. The token fits none of the six categories, since only fiat currency or a commodity can back a token designed to be stable.

Solutions • CI-6 a

Solution CI-6 a.1 — Answer: C

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Define DLT and "distributed"

DLT is a sequenced and standardised record of an activity, cryptographically secured on a network that is typically decentralised. It is distributed because the record is made available simultaneously to all network participants.

Step 2 – Place blockchain

Blockchain is the best-known form of DLT: a chain of blocks, each holding several transaction records and linked to the previous block. Records are generally immutable.

$\Rightarrow$ C states both parts correctly.

Interpretation

The family-and-member relation and the meaning of "distributed" are the two definitional points a board briefing is most likely to garble. Immutability is what makes the history trustworthy.

Why the other options are wrong

A is incorrect. Sibling swap of the hierarchy: blockchain is a form of DLT, not the family that contains it.

B is incorrect. Misreads "distributed": each participant receives the same record simultaneously; the record is not split into fragments.

D is incorrect. Polarity flip: blockchain records are generally immutable and cannot be altered.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022), Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets; SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.2 — Answer: A

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Recall the reason the note gives

Unbacked crypto assets are not considered money because they are generally not used as a medium of exchange, store of value or unit of account.

Step 2 – Trace it to the value

Their value is determined through supply and demand; they show high volatility and low intrinsic value and are often used for speculation.

$\Rightarrow$ The failure lies in the value and in how the assets are used, not in transferability or technology.

Interpretation

The argument that works in the speech links the asset's unstable, supply-and-demand value to the three functions of money. They are designed to be used as a means of exchange; in practice they are mainly held for speculation, not used for payments.

Why the other options are wrong

B is incorrect. Polarity flip: unbacked crypto assets are transferable, and transferability could meet the medium-of-exchange definition.

C is incorrect. Sibling swap: immutability concerns the security and trustworthiness of the records, not the asset's function as money.

D is incorrect. Category error: unbacked crypto assets do not rely on the backing of another asset; their value comes from supply and demand.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.3 — Answer: B

Adapted from CI QB Imp Q51 • SchweserNotes Book 5, LO 105.a

Step 1 – State of agreement

There is no single internationally accepted classification of crypto assets.

Step 2 – What regulators use instead

Six broad categories based on taxonomy and risks: unbacked crypto assets, NFTs, security tokens, utility tokens, stablecoins and CBDCs.

⇒ Regulators share a broad map but have not signed one common classification.

Interpretation

The lack of a single classification is also why security tokens are defined by each jurisdiction's own definition of a security.

Why the other options are wrong

A is incorrect. Role misassignment: the BCBS lower-risk and higher-risk groups are a proposed prudential treatment of banks' crypto exposures (CI-6 c), not an international taxonomy of crypto assets.

C is incorrect. Scope error: no category has a single international classification; the note says none exists.

D is incorrect. Wrong basis: the categories rest on taxonomy and risks. Permissioned and permissionless ledgers are a CI-6 b distinction about technology risk.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.4 — Answer: D

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Classify Ether

Ether is one of the class notes' examples of an unbacked crypto asset.

Step 2 – Recall the holder's position

Unbacked crypto assets are usually decentralised, carry limited rights for the token holder, and have no single issuer to enforce rights against.

⇒ The ombudsman has no issuer to direct the claim to.

Interpretation

Decentralised issuance is the feature that separates unbacked crypto assets from the other five categories, all centrally issued. It brings access and anonymity, and it leaves the holder without a counterparty.

Why the other options are wrong

A is incorrect. Sibling swap: profit-sharing rights enforceable against an issuer describe security tokens, not unbacked crypto assets.

B is incorrect. Right conclusion, wrong reason: unbacked crypto assets are usually decentralised and only sometimes centrally issued. The claim fails because there is no single issuer to enforce rights against, not because of what a central issuer's white paper says.

C is incorrect. Sibling swap: redemption from a reserve that keeps the price stable describes a backed stablecoin; Ether's value comes from supply and demand.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.5 — Answer: B

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Read the features

Centrally issued; gives a right to a product or service (in-game services); transferable; used inside the issuer's own network.

Step 2 – Match the category

Utility tokens are centrally issued and give a right to a product or service; they are transferable and often used within a single or closed network. Gaming tokens are the class notes' own example.

⇒ Utility token.

Interpretation

Central issuance and transferability are shared by several categories, so they cannot decide the case. The right the holder gets does.

Why the other options are wrong

A is incorrect. Sibling swap: a security token carries rights similar to traditional securities, such as profit-sharing or ownership, and must meet a jurisdiction's definition of a security. Central issuance and transferability do not make a token a security.

C is incorrect. Wrong category: a right to a product or service is the utility-token hallmark, and gaming tokens are the class notes' own example. Unbacked crypto assets carry only limited rights and are usually decentralised, with no single issuer.

D is incorrect. Sibling swap: an NFT gives ownership of a specific, collectible, non-substitutable product. Use within a closed network is a utility-token feature.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.6 — Answer: A

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Match the rights

A share of rental profits is a profit-sharing right similar to those of traditional securities, so the tokens look like security tokens.

Step 2 – Apply the jurisdiction test

A security token meets the definition of a security in each respective jurisdiction, so the definition varies by jurisdiction. There is no single international classification.

⇒ The adviser must check each country of sale.

Interpretation

A cross-border token sale is where the variation bites: the same token can sit inside the regulatory perimeter in one country and be treated differently in another.

Why the other options are wrong

B is incorrect. Sibling swap: utility tokens give a right to a product or service. Profit-sharing is the security-token right.

C is incorrect. Absolute claim: the definition varies by jurisdiction, and there is no single internationally accepted classification.

D is incorrect. Category error: a stablecoin is designed to be value stable through backing, collateral or an algorithm. Rental income paid to holders is a profit share, not a stability mechanism.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.7 — Answer: D

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Read the features

Each token confers ownership of one specific artwork; the tokens are collected and held.

Step 2 – Match the category

NFTs are usually centrally issued and give a right to ownership of a specific product; they are collectible and non-substitutable.

⇒ Nonfungible tokens.

Interpretation

"Specific" and "non-substitutable" separate an NFT from a utility token, whose holder has a right to a product or service of a kind, not to one unique item. Nor is it a security token: those carry rights similar to traditional securities, such as profit-sharing, and must meet a jurisdiction's definition of a security, whereas these tokens convey one collectible item and no profit share.

Why the other options are wrong

A is incorrect. Sibling swap: a right to a product or service and use as a means of exchange describe utility tokens; these tokens each convey one unique artwork.

B is incorrect. Category error: stablecoins are designed to be value stable. Nothing here stabilises the token's value.

C is incorrect. Wrong category: unbacked crypto assets are usually decentralised with no single issuer and limited rights. These tokens are issued by the platform and convey ownership of a specific product.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.8 — Answer: C

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – Shared design goal

Both stablecoins and CBDCs are designed to be value stable.

Step 2 – Issuer and mechanism

A CBDC is centrally issued by a state or central bank as a digital form of money; its stability mechanism is usually sovereign fiat currency. A stablecoin's stability comes from backing or collateralisation (a commodity, fiat currency, multiple currencies or crypto assets) or from algorithms.

⇒ Same goal, different issuer and mechanism.

Interpretation

USDC is one of the class notes' stablecoin examples. Being backed by fiat does not make a stablecoin a CBDC: the issuer does.

Why the other options are wrong

A is incorrect. Polarity flip: stablecoins are designed to be value stable; supply-and-demand pricing describes unbacked crypto assets.

B is incorrect. Mechanism swap: the CBDC usually relies on sovereign fiat currency; algorithms are a stablecoin mechanism, and a stablecoin is not issued by the state.
D is incorrect. Role misassignment: the CBDC is the instrument issued by the state or central bank.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

Solution CI-6 a.9 — Answer: C

New • built from SchweserNotes Book 5, LO 105.a

Step 1 – What defines a stablecoin

A stablecoin is designed to be value stable. In the class notes' table, the stability mechanism can be backing or collateralisation with a commodity, fiat currency, multiple currencies or crypto assets. SchweserNotes: stablecoins are "backed by other currencies or assets".

Step 2 – Apply it

The token is designed to hold a stable value against the US dollar, and its collateral is a pool of crypto assets.

Step 3 – Why it is not unbacked

Unbacked crypto assets do not rely on the backing of another asset, and their value is set by supply and demand. This token relies on backing.
$\Rightarrow$ It is a stablecoin.

Interpretation

The category turns on the token's design and on what keeps its value stable, not on the category of the assets that back it. Both sources agree on this case.

Why the other options are wrong

A is incorrect. Category error: the token's category follows its own design and backing, not the category of its collateral. Unbacked crypto assets rely on no backing at all.
B is incorrect. Role misassignment: a CBDC is issued by a state or central bank. Tracking a fiat currency does not make a private token a CBDC.
D is incorrect. Scope error: the class notes list crypto assets, alongside a commodity, fiat currency and multiple currencies, among the assets that can back or collateralise a stablecoin.

LO: CI-6 a: Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.a

CI-6 • Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets (IMF, September 2022) • Vardeez: IMPORTANT

CI-6 b

P3 CONCEPT

Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 (Map #68, 1/16, notes only), as set out under CI-6 a. There are no workbook questions for CI-6, so every question here is new or adapted from the CI question bank.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S1** CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda. The map gives no further split. This LO carries "ecosystem entities and their regulation".

What the questions look like: no calculations. Scenario questions that match a risk to the entity that generates it and a regulatory response to the right entity. Role misassignment is the natural distractor: suitability tests belong to issuers, the IOSCO framework to trading platforms, segregation and wind-down rules to wallet providers, and capital deductions and risk weights to regulated financial institutions.

1. The ecosystem in five entity groups

Background from CI-6 a: crypto assets use cryptography and DLT; unbacked crypto assets have no backing, a supply-and-demand value and no single issuer. The key components of the ecosystem are its **entities**, the **risks** their activities create and the **regulation** that responds. SchweserNotes groups the entities into five:

1. **Issuers**
2. **Crypto asset service providers**, chiefly crypto asset exchanges
3. **Wallet providers**
4. **Validators, miners and the underlying technology**
5. **Regulated financial institutions**

A sixth theme cuts across them: in crypto markets one provider often performs functions that traditional markets spread across many specialised entities (Section 7).

2. Issuers

Risks from decentralised issuance. Unbacked crypto issuance is often decentralised. That moves it away from central control and provides *anonymity, liquidity and broad access*. The same features create risks that are difficult for authorities to address and monitor:

- disclosure about the issuers of unbacked crypto assets is limited, and even the intended purpose of the assets or the technology used can be hard to determine;

- failure risk can arise at any node, and networks are susceptible to operational risk and cyberattacks.

Risks from issuance and distribution. Crypto assets can be issued through *pre-mining* (a batch sale of tokens or coins ahead of public distribution, usually to a small group), *continuous mining*, or a hybrid of the two. Distribution channels include pre-token sales, *initial coin offerings* (ICOs), mining, *airdrops* (the issuer sends tokens or coins to certain entities for marketing) and forks. **Pre-mining and ICOs expose participants to price manipulation and pump-and-dump schemes.**

Risks from marketing. White papers set out the developers, objectives and technical data. They are often incomplete or inaccurate and may deliberately overstate benefits and understate risks.

Regulatory response for issuers (improve white-paper information, the notification process and marketing disclosures, using suitability tests):

(a) **White papers.** Clear, accurate, easy-to-understand language, with the necessary details: why the asset was issued, the technology used and the potential risks.

(b) **Notification process.** Regulators may want to be notified when a white paper is available, to understand the asset's parameters and monitor risks. They may choose to require notification only of major developments (for example, building a framework) or when the issuance is small or restricted (for example, NFTs).

(c) **Suitability.** Issuers can be required to conduct suitability tests on whether the crypto asset is appropriate for *retail users*.

(d) **Marketing.** Fair, accurate, clear and simple, and containing the information in the white paper. Some jurisdictions require the regulator to approve crypto marketing materials; others prohibit the marketing of riskier assets such as unregulated crypto assets.

(e) **Consumer education.** Together with financial and technology literacy programmes.

3. Crypto asset exchanges

Exchanges facilitate the buying and selling of crypto assets and often provide services beyond those of a securities exchange. The largest are typically **centralised intermediaries**: they foster trust in a centralised counterparty but hinder innovation in the decentralised space. Unlike securities exchanges, crypto exchanges often (1) allow **direct access to retail users**, (2) **trade from their own inventory** to provide liquidity and (3) offer **custody services** (Figure 1).

Mitigants and their limits. Exchanges buy *cyber insurance*, keep tokens in *cold wallets* (offline storage) and set up *compensation funds*. These work with varying effectiveness, for three reasons: the level and terms of cyber coverage vary; users find it hard to convert crypto assets to fiat currency; and assets held in cold wallets still need online connectivity to be accessed.

Market integrity. The crypto market is relatively new, usually illiquid and has less sophisticated intermediaries, so trading schemes can manipulate it. Exchanges' very large holdings give them control when dealing with networks that use a *proof-of-stake* consensus mechanism. Enforcement against manipulation is hard because transactions are usually anonymous and run on decentralised platforms across multiple jurisdictions.

Regulatory gaps. Frameworks for exchanges vary globally, because local markets are small and regulatory priorities differ, and many jurisdictions have not yet developed exchange regulation. Regulation often focuses only on *lower-risk security tokens*, so an exchange can circumvent oversight by listing only unbacked crypto assets.

Correction to the source notes. (1) SchweserNotes writes that regulation "often focuses only on lower-risk security tokens rather than unbacked security tokens", so exchanges can circumvent oversight "by only listing unbacked security tokens". "Unbacked security tokens" does not fit the CI-6 a taxonomy: security tokens are centrally issued, meet a jurisdiction's definition of a security and sit inside the regulatory perimeter. The class notes give the consistent reading, used here: rules that cover only lower-risk tokens let an exchange avoid oversight by listing unbacked crypto assets. (2) SchweserNotes' key concepts list "write-down arrangements" among the wallet rules. Its own body text, the class notes and the workbook say *wind-down* arrangements, which is correct.

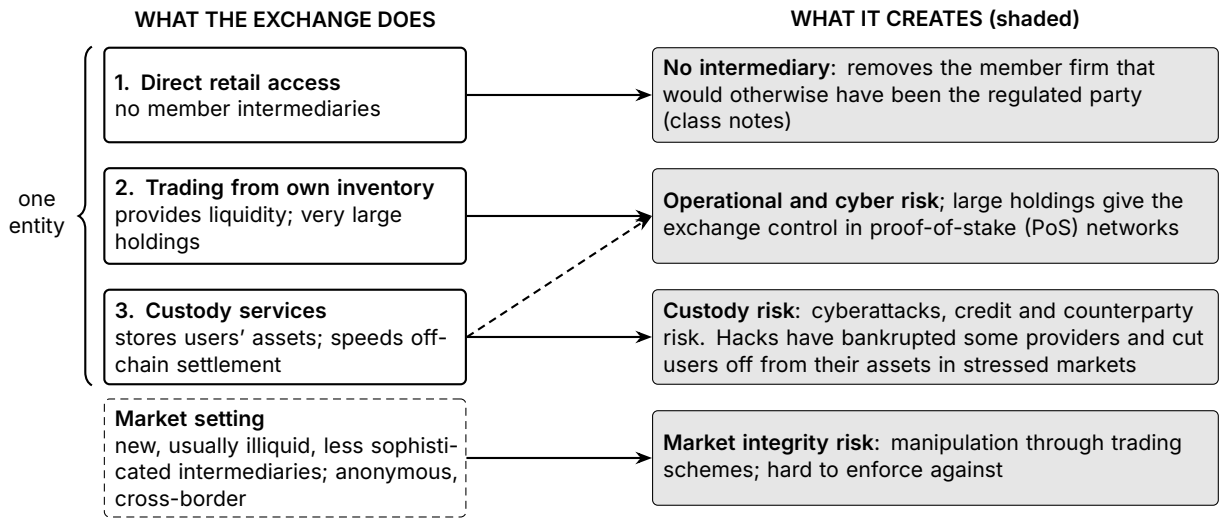


Figure 1. Exchange functions (solid boxes) and what each creates (shaded). Trading from own inventory and combining exchange and custody services expose exchanges to operational and cyber risk (solid and dashed arrows into the second row). The dashed box is the market setting, not a function. The brace marks the reading's core concern: one entity performs all three functions.

The IOSCO framework for crypto asset platforms. The International Organization of Securities Commissions (IOSCO) has outlined a framework built on nine requirements:

1. Prudent **governance** of platform operators: rules for control functions, operations and capital
2. **Regulated access** to platforms: protecting users, ensuring orderly trading
3. Robust **controls and processes** against key risks, including cyber hacks
4. Strengthened **market integrity**: less manipulation and fewer conflicts of interest
5. Improved **transparency**: trading efficiency, integrity and information
6. **AML/CFT** rules: compliance with domestic and international rules
7. Clear **product offering criteria**, helping authorities decide which assets should be regulated
8. Mitigating **conflicts of interest** through strong governance frameworks
9. Better regulation of **off-chain transactions**

What the framework does *not* do (SchweserNotes, Module Quiz 105.1): it does not require suitability tests for exchange participants (suitability tests belong to the issuer framework and concern retail end users), and it does not impose cold wallets, although it highlights their advantages. Even within a global framework, some variation across jurisdictions would be needed to reflect differences in prudential approaches and in the nature and size of local exchanges.

4. Wallet providers

Digital wallets store the **private keys** needed to access a user's crypto assets. Losing the wallet means losing the private key, and therefore the assets. Wallets are classified along two independent dimensions (Figure 2).

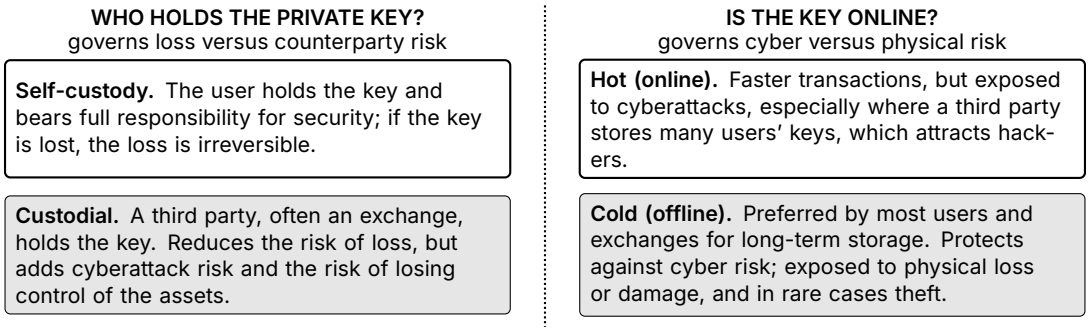


Figure 2. The two wallet dimensions, kept apart by the dotted rule. Left: who holds the key (self-custody, white; custodial, shaded). Right: whether the key is online (hot, white; cold, shaded). The class notes classify wallets on each dimension separately: a custodial wallet can be hot or cold.

Regulation of wallet providers. Wallet security protects the safeguarding of funds and the security of the system. Regulation should follow best practice in cryptography and focus on *key protection and lifecycle management controls*. Frameworks cover eight factors:

1. **Segregation and safekeeping.** User funds and crypto assets are kept separate from the provider's own. Reusing or lending client assets should require user consent.
2. **Operational and cyber resilience.** Robust cybersecurity frameworks for custodied assets, effective incident management and prompt reporting of operational or cyber incidents.
3. **Record keeping.** Providers rely on DLT to improve processes; regulation should make those processes secure, resilient, timely and accurate.
4. **AML/CFT compliance.**
5. **Reporting and prudential regulation.** Authorities may impose reporting and prudential rules covering risk management, client asset protection, and minimum capital and liquidity.
6. **Safeguards.** Cyber insurance for a designated proportion of holdings.
7. **Wind-down arrangements.** Together with regulation of third-party asset providers, they protect customers from fraud and reduce contagion to the financial sector. Private keys stored by third-party custodians can complicate legal issues in bankruptcy, because client and custodian funds become commingled.
8. **Conflicts of interest.** Stringent governance, especially when the wallet provider is part of a group.

5. Validators, miners and technology

Using DLT in financial services helps **reduce** operational and cyber risks: transactions are immutable and leave a clear audit trail. Different forms of DLT carry different risks:

- **Private or permissioned blockchains.** Their closed nature reduces interoperability and can lead to fragmentation, concentration and financial stability risks.
- **Public or permissionless blockchains.** Better protected against single points of failure, but exposed to other failure risks related to user behaviour.

- **Cross-network interoperability** is a problem: most crypto assets can operate only on their own blockchain network and may lack fungibility even when offered on separate networks.

Regulation typically aims to be **technology neutral**. That aim may create friction with other policy objectives and may hinder future innovation.

6. Regulated financial institutions

Financial institutions' direct and indirect exposures to crypto assets can be significant and can expose the financial system to more risk.

- **Direct exposure** carries the high volatility, and potentially significant losses, of unbacked crypto assets, although the number of institutions exposed and the size of their exposures are currently limited. Regulators take a *conservative approach*: capital deductions and higher risk weights. The BCBS proposal splits banks' holdings into lower-risk and higher-risk groups (CI-6 c).
- **Indirect exposure** arises through derivatives, loans to crypto users, or other services such as cyber insurance for wallet providers. Institutions should monitor these exposures carefully.
- **Broker-dealers, asset managers and insurers** are regulated financial institutions too. Their exposure has grown as they extend services and products linked to unbacked crypto assets; oversight should focus on enhanced monitoring.

7. Provider functions and market infrastructure

In traditional markets, a large number of specialised entities each perform one infrastructure function: record keeping, clearing, settlement, custody. In crypto markets the same provider often performs these tasks, globally. That raises operational risk and concentrates it, with the potential for broad market disruption.

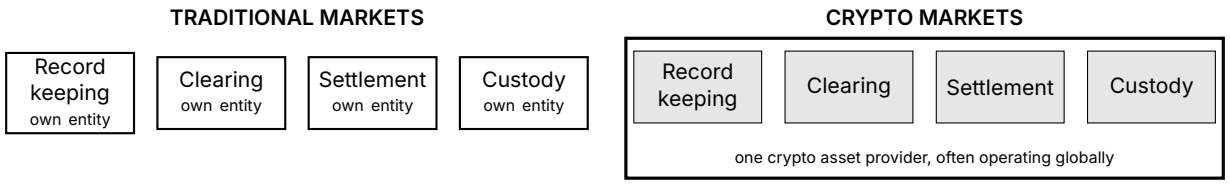


Figure 3. Traditional markets give each function to its own specialised entity (white boxes); in crypto markets one provider often performs all four (shaded boxes inside the heavy frame), which concentrates operational risk and can lead to broad market disruption.

Systemic designation. Regulation should first determine whether a crypto asset provider is "systemic" and should be designated a **financial market infrastructure (FMI)**, because such entities could cause significant disruption. For stablecoins, designation could rest on the *number of users, transaction volume, risk profile and level of interconnectedness* with the wider financial system. Designation alone does not fully mitigate the risks: users can replace one crypto exchange with another, or move transactions "on-chain", and so circumvent regulation. Close regulatory supervision and oversight of crypto service providers is therefore paramount.

Results card: CI-6 b

Entity	Main risks	Regulatory response
Issuers	Decentralised issuance: limited disclosure, unclear purpose and technology, failure at any node, cyberattacks. Pre-mining and ICOs: manipulation, pump-and-dump. Misleading white papers	White papers, notification, suitability tests for retail users, fair marketing, consumer education
Exchanges	Direct retail access, own-inventory trading and custody in one entity: operational, cyber, credit and counterparty risk; manipulation; PoS control; gaps where rules cover only security tokens	IOSCO: governance, regulated access, controls, market integrity, transparency, AML/CFT, product offering criteria, conflicts of interest, off-chain transactions
Wallet providers	Custodial: cyber, loss of control. Hot: cyberattacks. Cold: physical loss	Segregation and consent, cyber resilience, record keeping, AML/CFT, prudential rules, cyber insurance, wind-down, conflicts
Validators, technology	Private: interoperability, fragmentation, concentration. Public: user behaviour. Cross-network fungibility	Technology neutral as an aim, with possible friction and innovation costs
Regulated FIs	Direct: volatile losses. Indirect: derivatives, loans to crypto users, services such as cyber insurance	Direct: conservative (capital deductions, higher risk weights). Indirect: careful monitoring
Infrastructure	One provider performs every function: concentrated operational risk	Systemic test and FMI designation, plus close supervision

Exam traps

- **Role misassignment across frameworks.** Suitability tests on retail appropriateness: issuers. Governance, access, integrity and off-chain rules: IOSCO, for platforms. Segregation, consent and wind-down: wallet providers. Capital deductions and higher risk weights: regulated financial institutions (BCBS).
- **Centralised, not decentralised.** The largest exchanges are centralised intermediaries; decentralisation describes unbacked crypto *issuance*.
- **Custody misread.** Custody speeds off-chain settlement and *adds* cyber, credit and counterparty risk for users.
- **Cold and hot swapped.** Cold is offline: cyber-protected, physically exposed, preferred for long-term storage, and still needs online access to use. Hot is online, fast and hackable.
- **Concentration as defence.** A third party storing many users' keys online is a target, not a safer vault.
- **"Reduces" read as "removes".** Custodial wallets reduce the risk of loss; exchange mitigants work with varying effectiveness; FMI designation does not fully mitigate risk.
- **Mandates that do not exist.** IOSCO highlights cold wallets but does not require them, and it contains no participant suitability test.
- **DLT direction.** DLT reduces operational and cyber risk. Private chains fragment and concentrate; public chains resist single points of failure.

- **Direct and indirect swapped.** Holdings are direct; derivatives, loans to crypto users and cyber insurance for wallet providers are indirect.
- **Which tokens the rules cover.** Rules often cover only lower-risk security tokens, so exchanges escape by listing unbacked crypto assets, not the reverse.

Practice questions: CI-6 b

Question CI-6 b.1 Adapted from CI QB Imp Q54 • SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A market surveillance officer at a securities regulator in Brazil is reviewing a new token. Before public distribution, the developers sold a large batch of the tokens to a small group of early buyers; the token was then offered to the public through an initial coin offering (ICO). Which risk does the IMF note most directly associate with this pattern of issuance and distribution?

A. Price manipulation and pump-and-dump schemes, to which pre-mining and initial coin offerings both expose participants.

B. Failure risk at a single node, which arises because pre-mined tokens are held on a server run by the developers.

C. Manipulation risk at the ICO stage only, because a pre-mined batch sale to a small group never reaches the public market.

D. Marketing risk only, since pre-mining is a form of airdrop in which tokens are sent to selected entities for promotion.

Question CI-6 b.2 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A securities regulator in South Africa is concerned that retail investors are buying newly issued unbacked crypto assets whose risks they do not understand. Which of the following responses is most consistent with the regulatory measures for issuers in the IMF note?

A. Require trading platforms, under the IOSCO framework, to run suitability tests on each participant before allowing them to trade.

B. Require issuers to test whether the asset is suitable for retail users, and to publish white papers that are clear and accurate.

C. Require banks that hold the asset to apply a 1,250% risk weight, so that the higher capital cost deters retail investors from buying.

D. Require the regulator to approve every white paper before issuance, in place of rules on marketing materials and consumer education.

Question CI-6 b.3 Adapted from CI QB Imp Q58 • SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A market integrity officer at a European securities regulator is explaining to new staff how the largest crypto asset exchanges differ from traditional securities exchanges. Which of the following statements is correct?

A. They route retail orders through licensed member brokers, as securities exchanges do, but also hold those brokers' crypto assets in custody.

B. They are mostly decentralised protocols, which removes counterparty risk because no central party ever holds the users' crypto assets.

C. They give retail users direct access, trade from their own inventory and offer custody, exposing users to cyber, credit and counterparty risk.

D. They offer custody so that trades settle on-chain, which removes the cyber and counterparty risks that faster off-chain settlement would create.

Question CI-6 b.4 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A securities commission in Kenya plans to adopt the IOSCO regulatory framework for crypto asset trading platforms. Which of the following requirements belongs to that framework?

A. A 1,250% risk weight on each platform's own crypto asset inventory, applied to the maximum of its long and short positions.

B. Horizon scanning for emerging technologies through innovation offices, as the basis for setting the scope of crypto regulation.

C. A rule that platforms hold all client tokens in cold wallets, given the protection that cold storage offers against cyber risk.

D. Clear product offering criteria, the mitigation of conflicts of interest, and better regulation of off-chain transactions.

Question CI-6 b.5 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b (correction box)

A researcher at a financial policy institute is studying why crypto asset exchanges escape effective oversight in many countries. Which of the following explanations is most consistent with the IMF note?

A. Many countries lack exchange rules, and rules often cover only lower-risk security tokens, so exchanges can list only unbacked crypto assets.

B. IOSCO has issued binding global rules for exchanges, so gaps arise only where national authorities have failed to transpose them into domestic law.

C. Existing rules often cover unbacked crypto assets but leave out security tokens, so exchanges can escape oversight by listing only security tokens.

D. Enforcement fails mainly because trades sit on immutable ledgers, which stops regulators from reversing manipulative trades once they are detected.

Question CI-6 b.6 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A family office in Dubai plans to hold Bitcoin for several years and must choose between holding the private keys itself and using an exchange as custodian, and between hot and cold storage. Which of the following statements best describes the trade-offs?

A. Holding the keys itself lowers the risk of loss compared with a custodian, and hot storage suits a multi-year holding as it allows faster transactions.

B. A custodian lowers the risk of losing the keys but adds cyber risk and loss of control; cold storage resists cyber risk but can be lost or damaged.

C. A custodian is better placed using hot storage, because an exchange that keeps many users' keys together in one online wallet can defend them better.

D. Cold storage removes both cyber and physical risk, so the choice between holding the keys and using a custodian no longer changes the risk.

Question CI-6 b.7 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A wallet provider that is part of a larger crypto group proposes to lend its clients' crypto assets to a trading affiliate to earn a yield. Which of the following responses from its regulator best reflects the regulatory framework for wallet providers?

A. The lending can go ahead without client consent, provided the provider holds cyber insurance for a designated proportion of its holdings.

B. The lending can go ahead without client consent, provided the provider meets the minimum capital and liquidity requirements of the authority.

C. Client assets must be kept apart from the provider's own, lending needs client consent, and group conflicts call for stringent governance.

D. Client assets may be pooled with the provider's own, since one commingled pool makes a wind-down faster and simpler if the provider fails.

Question CI-6 b.8 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A consortium of Canadian banks is considering building a settlement system on a private, permissioned blockchain rather than a public, permissionless one. A supervisor is asked to summarise the risk trade-off. Which of the following summaries is most consistent with the IMF note?

A. Closed private chains cut interoperability and can fragment and concentrate risk; public chains resist single points of failure but face user-behaviour risk.

B. A private chain is better protected against single points of failure, while a public chain's closed nature tends to fragment the market and to concentrate risk.

C. Either choice avoids cross-network interoperability problems, because crypto assets are generally fungible across separate blockchain networks.

D. Either form of DLT raises operational and cyber risk relative to existing systems, because the transactions recorded on it cannot later be altered.

Question CI-6 b.9 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A supervisor at a European regulator reviews a financial group whose bank lends to crypto asset users and whose insurer sells cyber insurance to a wallet provider. The group holds no crypto assets itself. Which of the following assessments is most consistent with the IMF note?

A. The group has direct exposures, because the loan book and the insurance policy both tie its losses to the value of crypto assets.

B. The exposures need no attention, because the number of exposed institutions and the size of their crypto exposures are currently limited.

C. These are indirect exposures to be monitored carefully; the capital deductions and higher risk weights apply to direct holdings.

D. The loans attract the 1,250% risk weight, while the cyber insurance is outside scope because insurers are not regulated institutions.

Question CI-6 b.10 New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

A central bank in Asia is deciding whether to designate a large crypto asset provider, which performs record keeping, clearing, settlement and custody itself, as a financial market infrastructure (FMI). Which of the following assessments best reflects the IMF note?

A. Designation fully mitigates the provider's risks, because once it is designated its users can no longer move their transactions elsewhere.

B. Designation is possible only if the provider is a bank, since FMI status applies to regulated financial institutions, not to crypto providers.

C. Designation is usually unnecessary, because performing several functions within one entity tends to spread operational risk across them.

D. Designation can be warranted if the provider is systemic, but it does not fully mitigate risk, as users can switch exchange or go on-chain.

Solutions • CI-6 b

Solution CI-6 b.1 — Answer: A

Adapted from CI QB Imp Q54 • SchweserNotes Book 5, LO 105.b

Step 1 – Name the methods

A batch sale to a small group ahead of public distribution is *pre-mining*. The later public offer is an *initial coin offering*.

Step 2 – Recall the risk

Pre-mining and ICOs expose participants to price manipulation and pump-and-dump schemes.
⇒ Both steps of this token's issuance are on the note's list of manipulation risks.

Interpretation

Issuance risk is one of three issuer risks in the note, alongside decentralised issuance (limited disclosure, failure at any node) and misleading white papers. The regulatory answer is better white papers, notification, suitability tests, fair marketing and consumer education.

Why the other options are wrong

B is incorrect. Sibling swap: failure risk at any node is a feature of decentralised networks, not a consequence of pre-mining, and nothing in the note ties pre-mining to a single server.
C is incorrect. Scope error: the note says pre-mining *and* ICOs both expose participants to price manipulation and pump-and-dump schemes; the early batch sale to a small group is itself a source of the risk.
D is incorrect. Sibling swap: an airdrop is the issuer sending tokens to certain entities for marketing. Pre-mining is a batch sale.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022), Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets; SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.2 — Answer: B

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Identify the entity

The concern is the *issuance* of new unbacked crypto assets to retail buyers, so the relevant toolkit is the issuer framework.

Step 2 – Pick the tool

Regulation can require issuers to conduct suitability tests on whether the crypto asset is appropriate for retail users, and white papers must be clear, accurate and easy to understand, including the risks.
⇒ Suitability tests sit with the issuer and concern the asset's fit for retail users.

Interpretation

The other issuer tools (notification, fair marketing consistent with the white paper, consumer education) complement these two. None of the other options is an issuer measure.

Why the other options are wrong

A is incorrect. Role misassignment: the IOSCO framework governs platforms and does not contemplate suitability tests for exchange participants.
C is incorrect. Role misassignment: the proposed 1,250% risk weight is the BCBS treatment of banks' own higher-risk crypto holdings, a prudential rule for banks, not a retail-protection tool.

D is incorrect. Scope error: the note describes a notification process for white papers, not approval of every white paper, and adds marketing rules and consumer education rather than replacing them.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.3 — Answer: C

Adapted from CI QB Imp Q58 • SchweserNotes Book 5, LO 105.b

Step 1 – The three differences

Unlike securities exchanges, crypto exchanges often allow direct access to retail users, trade from their own inventory to provide liquidity, and offer custody services.

Step 2 – The risk from custody

Custody can expedite off-chain settlement but exposes users to cyberattacks and to credit and counterparty risk, which can result in losses.

⇒ One entity is marketplace, dealer and custodian at once.

Interpretation

The largest exchanges are centralised intermediaries: that builds trust in a centralised counterparty, and it is also why the exchange's own failure or hack falls on its users.

Why the other options are wrong

A is incorrect. Polarity flip: crypto exchanges often give retail users direct access, with no member brokers in between.

B is incorrect. Polarity flip: the largest exchanges are typically centralised intermediaries, and their custody services create counterparty risk.

D is incorrect. Polarity flip on custody: custody speeds off-chain settlement and adds cyber, credit and counterparty risk for users; it removes none of them.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.4 — Answer: D

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Recall the IOSCO list

Governance of platform operators; regulated access; robust controls; market integrity; transparency; AML/CFT; clear product offering criteria; mitigating conflicts of interest; better regulation of off-chain transactions.

Step 2 – Match the option

D lists three of the nine.

⇒ IOSCO's framework is about how a platform is run and what it lists.

Interpretation

Three standard setters appear in this reading, each with its own product: IOSCO for trading platforms, the BCBS for banks' crypto exposures, and the IMF with the World Bank for the Bali Fintech Agenda. Matching the requirement to the body is the test.

Why the other options are wrong

A is incorrect. Role misassignment: the 1,250% weight on the maximum of long and short

positions is the BCBS proposal for banks' higher-risk crypto holdings.
B is incorrect. Role misassignment: horizon scanning through innovation offices is a monitoring tool under the Bali Fintech Agenda (CI-6 d).
C is incorrect. Mandate that does not exist: the framework highlights the advantages of cold wallets but does not require them.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.5 — Answer: A

New • built from SchweserNotes Book 5, LO 105.b; CI notes CI-6 b

Step 1 – Why rules differ

Frameworks for exchanges vary globally because local markets are small and regulatory priorities differ, and many jurisdictions have not yet developed exchange regulation.

Step 2 – Where the gap opens

Regulation often focuses only on lower-risk security tokens, so an exchange can circumvent oversight by listing only unbacked crypto assets.
⇒ The riskiest assets fall outside rules written for the safest ones.

Interpretation

SchweserNotes' wording here ("unbacked security tokens") is garbled; the class notes and the taxonomy in CI-6 a make clear the gap is unbacked crypto assets (see the correction box).

Why the other options are wrong

B is incorrect. Role misassignment and absolute claim: IOSCO has outlined a framework; it is not binding global law, and many jurisdictions have no exchange rules at all.
C is incorrect. Polarity flip: the rules tend to cover lower-risk security tokens, and the escape route is listing unbacked crypto assets.
D is incorrect. Sibling swap: enforcement against manipulation is hard because transactions are usually anonymous and run on decentralised platforms across jurisdictions. Immutability is a strength of DLT records, not the stated obstacle.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.6 — Answer: B

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Who holds the key

A custodial wallet, often held by an exchange, reduces the risk of losing the key but introduces cyberattack risk and the risk of losing control of the assets.

Step 2 – Online or offline

Cold wallets are stored offline and are preferred by most users and exchanges for long-term storage: they protect against cyber risk but are exposed to physical loss or damage, and in rare cases theft.
⇒ For a multi-year holding, cold storage is the natural choice; the custody decision is a separate trade-off.

Interpretation

The two dimensions are independent. Self-custody versus custodial decides loss versus counterparty risk; hot versus cold decides cyber versus physical risk.

Why the other options are wrong

A is incorrect. Polarity flip twice: it is the custodial wallet that reduces the risk of loss (self-custody leaves the full loss risk with the holder), and hot (online) storage suits fast transactions, not long-term holding.

C is incorrect. Polarity flip: third parties storing many users' keys in hot wallets makes them attractive to hackers.

D is incorrect. Absolute claim: cold storage protects against cyber risk but remains exposed to physical loss or damage, and the custody choice still matters.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.7 — Answer: C

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Segregation and consent

Wallet providers should segregate user funds and crypto assets from their own, and reusing or lending client assets should require user consent.

Step 2 – Group conflicts

Stringent governance procedures help reduce conflicts of interest, especially when the wallet provider is part of a group.

$\Rightarrow$ Lending to an affiliate needs consent and strong governance.

Interpretation

Cyber insurance, prudential requirements and wind-down arrangements are also on the list, but each addresses a different risk. None replaces client consent.

Why the other options are wrong

A is incorrect. Condition swap: cyber insurance for a designated proportion of holdings is a separate safeguard against cyber loss; it does not remove the consent requirement.

B is incorrect. Sibling swap: minimum capital and liquidity are prudential requirements that authorities may impose; they do not replace consent either.

D is incorrect. Polarity flip: commingling of client and custodian funds is what complicates legal issues in bankruptcy. Segregation is the rule.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.8 — Answer: A

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Private or permissioned

The closed nature of private blockchains reduces interoperability and can lead to fragmentation, concentration and financial stability risks.

Step 2 – Public or permissionless

Public blockchains are better protected against single points of failure but are exposed to other failure risks related to user behaviour.

⇒ Closed systems fragment and concentrate; open systems trade single-point risk for user-behaviour risk.

Interpretation

For a bank consortium, the private chain's control comes at the cost of interoperability with other networks, a problem the note already flags: most crypto assets operate only on their own network.

Why the other options are wrong

B is incorrect. Sibling swap: it reverses the two profiles. Protection against single points of failure belongs to public chains; the closed nature belongs to private ones.

C is incorrect. Polarity flip: cross-network interoperability is a problem, and crypto assets may lack fungibility even when offered on separate networks.

D is incorrect. Polarity flip: DLT in financial services helps *reduce* operational and cyber risks, through immutable transactions and a clear audit trail.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.9 — Answer: C

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Classify the exposures

Indirect exposure arises through derivatives, loans to crypto users, or other services such as cyber insurance for wallet providers. Both of the group's links are on that list.

Step 2 – Match the response

Regulated financial institutions should carefully monitor indirect exposures. The conservative approach (capital deductions and higher risk weights) is the response to direct exposures, for which the BCBS proposal sets the risk weights (CI-6 c).

⇒ Monitor the loans and the insurance book; there is no crypto holding to deduct.

Interpretation

Insurers, broker-dealers and asset managers are regulated financial institutions too, and their crypto-related business has grown, so enhanced monitoring applies across the group.

Why the other options are wrong

A is incorrect. Sibling swap: direct exposure means holding crypto assets; loans to crypto users and services such as cyber insurance for wallet providers are the note's examples of indirect exposure.

B is incorrect. True premise, wrong conclusion: the number of institutions and the size of exposures are limited for now, but indirect exposures should still be carefully monitored.

D is incorrect. Role misassignment: the 1,250% weight is the BCBS proposal for banks' holdings of higher-risk crypto assets (CI-6 c), not for loans to crypto users, and insurers are regulated financial institutions.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

Solution CI-6 b.10 Answer: D

New • built from SchweserNotes Book 5, LO 105.b

Step 1 – Why the question arises

In traditional markets, specialised entities each perform one function; in crypto markets one provider often performs them all, globally, which raises and concentrates operational risk.

Step 2 – The designation test

Regulation should first determine whether the provider is systemic and should be designated an FMI.

Step 3 – Its limit

Designation alone does not fully mitigate the risks, because users can replace one exchange with another or move transactions on-chain.

$\Rightarrow$ Close supervision and oversight remain paramount.

Interpretation

For stablecoins, the note lists the factors that could drive designation: number of users, transaction volume, risk profile and interconnectedness with the financial system.

Why the other options are wrong

A is incorrect. Absolute claim: designation does not fully mitigate risk, precisely because users can move elsewhere.

B is incorrect. Scope error: the note applies the systemic test and FMI designation to crypto asset providers themselves.

C is incorrect. Polarity flip: performing several functions in one provider increases and concentrates operational risk rather than spreading it.

LO: CI-6 b: Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.b

CI-6 • Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets (IMF, September 2022) • Vardeez: IMPORTANT

CI-6 c

P3 CONCEPT

Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 (Map #68, 1/16, notes only), as set out under CI-6 a. There are no workbook questions for CI-6, so every question here is new.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S1** CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda. The map gives no further split. This LO carries "national approaches" and "BCBS treatment".

What the questions look like: no calculations. Jurisdiction-matching (which country's approach is which), why national rules are bypassed, why a blanket ban is judged ineffective, and how the BCBS proposal treats a given bank holding. The one number is the 1,250% risk weight; know which group it applies to and how long and short positions enter.

1. National approaches

Background: unbacked crypto assets (CI-6 a) and the conservative treatment of banks' direct exposures (CI-6 b, Section 6).

Regulation of unbacked crypto assets and crypto activities is often done in a **national setting**, which **may be sufficient and necessary** if global regulation does not exist. National approaches differ. Some regulators give *high-level guidance* only; others write *bespoke* rules for specific activities, such as the transfer and safekeeping of assets, and AML/CFT. SchweserNotes gives six examples.

Jurisdiction	What the regulation focuses on	Key-concept grouping
European Union	Primarily stablecoins, onshoring, and licensing of issuing entities.	Products: stablecoins and tokens
United Kingdom	Initial guidance regulated *security tokens*, but not utility and exchange tokens. Proposed future guidance is broader: stablecoins, market infrastructure and cryptocurrencies.	Products: stablecoins and tokens
Japan	Bespoke rules for crypto asset providers: asset segregation, operational risk, cyber risk, audit and capital requirements. Stablecoin regulation introduced since 2020.	Operational aspects
Albania	Licensing, monitoring and supervision of crypto asset service providers. Crypto assets legalised for trading purposes in 2020.	Operational aspects
Switzerland	More progressive crypto regulation than most jurisdictions: guidelines for initial token offerings, clearer token trading rules, asset segregation, and ledger-based securities on a blockchain.	Crypto friendly
Nigeria	Central bank guidelines: crypto assets are not legal tender; regulated financial institutions are prohibited from dealing in, or facilitating payments in, crypto assets.	Restrictive

Two dates are easy to cross: **2020** belongs to both Albania (crypto legalised for trading) and Japan (stablecoin regulation introduced since then); **2021** belongs to China's trading ban (Section 2).

2. Why national rules leak

Crypto activities often involve global markets, which makes effective regulation and supervision hard. Without harmonised regulation there are three bypass routes:

- trading crypto assets on **unregulated exchanges** can bypass sanctions and capital flow restrictions;
- crypto service providers can **shift their services to less regulated jurisdictions**;
- users and providers can use **virtual private networks (VPNs)** to obscure their locations.

Bypassing is often successful even where stringent rules exist, when monitoring and enforcement are ineffective because resources are limited. Both **Egypt and Nigeria** have strict restrictions, yet crypto assets and activities are widespread. Crypto trading has been banned in **China since 2021**, yet evidence suggests significant mining activity.

3. What international bodies recommend

Several international financial bodies, including the **IMF and the BCBS**, have called for **stronger international cooperation** among regulatory authorities. Their position has three parts:

1. **Regulate what the asset does and how it is built.** Regulation should primarily focus on the *economic functions* of crypto assets and their *technical design*.
2. **No blanket ban.** A blanket ban on crypto activities is likely ineffective and would hinder innovation. SchweserNotes' Module Quiz 105.2 puts it as: a general ban would likely be ineffective in *both* the short and the long run, because monitoring and enforcement are difficult.
3. **Targeted restrictions.** Restricting only certain aspects of crypto assets is considered more effective in the short term. Examples: restricting crypto asset derivatives (Japan and the UK), or restricting the use of crypto assets as payment for goods and services.

The short-term effectiveness belongs to *targeted restrictions*, not to a general ban. That pairing is the classic distractor.

4. The BCBS proposal for banks' crypto exposures

The BCBS aims to help standardise banks' holdings of crypto assets, given the high-risk nature of these investments. Holdings are split into two groups:

- **Lower-risk assets**: *tokenised assets*, with risk weights similar to those on traditional assets, and *stablecoins*, which may benefit from lower risk weighting. Overall, risk weights similar to or lower than those on traditional assets.
- **Higher-risk assets**: *cryptocurrencies and unbacked crypto assets*. More conservative requirements, with a proposed **1,250% risk weight applied to the maximum of long and short positions**.

The class notes and the workbook label the two groups Group 1 and Group 2; the workbook flags those names as coming from the BCBS standard rather than the IMF note. The class notes add one interpretation: at an 8% capital ratio, a 1,250% risk weight is a full capital deduction ($12.5 \times 8\% = 100\%$), the mechanism behind the "conservative approach" to direct exposures in CI-6 b.

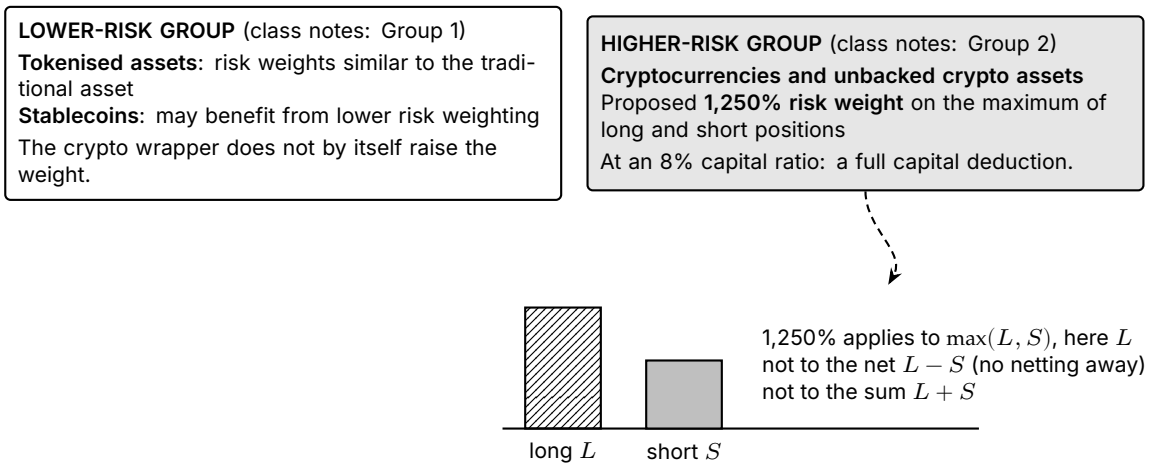


Figure 1. The BCBS proposal as the IMF note reports it. Left (white): the lower-risk group. Right (shaded): the higher-risk group and its 1,250% weight. Below: how long and short positions in a higher-risk asset enter, with the long position hatched and the short position grey; the weight attaches to the larger of the two.

Correction to the source notes. (1) The class notes' Figure CI-6.2 says the 1,250% weight is "applied to both long and short positions". That is imprecise rather than wrong: SchweserNotes ("for maximum long and short positions") and the workbook's D29 notes ("applied to the maximum of long and short positions") say it applies to the larger of the two, not to their sum. The class notes' qualifier survives: a short position does not net the charge away. (2) SchweserNotes and the class notes label the second group "higher-risk (traditional) assets". Do not read "traditional" as putting tokenised traditional assets there: the same sources place tokenised assets in the lower-risk group and list cryptocurrencies and unbacked crypto assets as the higher-risk group.

Direct and indirect exposures, and the conservative approach to direct holdings, are covered under CI-6 b, Section 6.

Results card: CI-6 c

National setting	May be sufficient and necessary absent global rules; high-level guidance or bespoke activity rules (transfer, safekeeping, AML/CFT)
Six jurisdictions	EU: stablecoins, onshoring, licensing issuers. UK: security tokens first, broader proposed. Japan: bespoke provider rules (segregation, operational, cyber, audit, capital); stablecoins since 2020. Albania: CASP licensing and supervision; legal for trading 2020. Switzerland: progressive. Nigeria: not legal tender; regulated FIs barred
Bypass routes	Unregulated exchanges (sanctions, capital flow restrictions), relocation, VPNs; weak enforcement. Egypt, Nigeria; China (ban since 2021, mining)
IMF and BCBS	Stronger international cooperation; regulate economic function and technical design; blanket ban likely ineffective and hinders innovation; targeted restrictions (derivatives in Japan and UK, crypto as payment) more effective short term
BCBS lower-risk	Tokenised assets: similar to traditional; stablecoins: possibly lower
BCBS higher-risk	Cryptocurrencies and unbacked crypto assets: proposed 1,250% on the maximum of long and short positions

Exam traps

- **Jurisdiction swaps.** EU: stablecoins, onshoring, licensing of issuing entities. Japan: provider operations and capital. Albania: licensing and supervision of service providers. UK: security tokens first. Switzerland: progressive. Nigeria: restrictive.
- **UK polarity.** Initial UK guidance covered security tokens, *not* utility and exchange tokens.
- **The dates.** 2020: Albania legalised trading; Japan's stablecoin rules. 2021: China's trading ban.
- **Strict rules, no activity?** Egypt and Nigeria are strict, yet activity is widespread; China still shows significant mining.
- **Ban horizons.** A general ban is ineffective in both the short and long run. Short-term effectiveness belongs to targeted restrictions.
- **Tokenised wrapper.** A tokenised traditional asset is lower-risk with a similar risk weight, not 1,250%, and not necessarily lower (that possibility is for stablecoins).
- **Long and short.** The 1,250% weight attaches to the maximum of long and short positions: not the net, not the sum.
- **Attribution.** The bank capital treatment is the BCBS's. The IMF and the BCBS together call for cooperation and against blanket bans; IOSCO's framework is for platforms (CI-6 b).

Practice questions: CI-6 c

Question CI-6 c.1 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

A policy adviser at a finance ministry in Southeast Asia is reviewing how other countries regulate unbacked crypto assets before drafting national rules. No harmonised global regulation yet exists. Which of the following statements is most consistent with the IMF note?

A. National rules may be sufficient and necessary absent global rules, from high-level guidance to bespoke rules on safekeeping and AML/CFT.

B. National rules are likely to be ineffective while crypto markets are global, so drafting should wait for a binding global framework.

C. National approaches have converged on one model of bespoke activity rules, so the adviser can copy the framework of any major jurisdiction.

D. National rules are needed only for security tokens, because unbacked crypto assets already fall under BCBS rules in every jurisdiction.

Question CI-6 c.2 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

The head of compliance at a crypto asset exchange headquartered in Singapore is preparing to apply for a licence in Japan. Which of the following descriptions of the Japanese approach, as set out in the IMF note, should she expect?

A. Rules focused mainly on stablecoins, onshoring and the licensing of issuing entities, with less attention to the operations of providers.

B. Bespoke rules for crypto providers on asset segregation, operational and cyber risk, audit and capital, with stablecoin rules since 2020.

C. A prohibition on regulated financial institutions dealing in crypto assets or facilitating payments in them, as they are not legal tender.

D. A more progressive regime than most, with guidelines for initial token offerings, clearer token trading rules and ledger-based securities.

Question CI-6 c.3 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

A start-up in London issues a utility token that gives holders access to its software platform. Its founder asks a consultant how the UK approach described in the IMF note has treated tokens of this kind. Which of the following answers is correct?

A. The UK's initial guidance covered utility and exchange tokens but not security tokens, which were left to existing securities law.

B. The UK legalised crypto assets for trading in 2020 and has since focused on licensing and supervising crypto asset service providers.

C. The UK runs a more progressive regime than most, with guidelines for initial token offerings and rules for ledger-based securities.

D. The UK's initial guidance focused on security tokens and did not cover utility and exchange tokens; proposed guidance is broader.

Question CI-6 c.4 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

An IMF mission to a country with strict legal restrictions on crypto assets finds that crypto trading there is nonetheless widespread. Which of the following explanations is most consistent with the IMF note?

A. Strict restrictions eventually end crypto activity, as China's experience since its 2021 trading ban shows, so the finding reflects a temporary transition.

B. Widespread activity under strict rules shows the rules are still too lenient; a complete ban on all crypto activity is the effective next step.

C. Monitoring and enforcement often fail for lack of resources, and users trade on unregulated exchanges and hide their location with VPNs.

D. Strict rules tend to attract crypto service providers seeking regulatory certainty, so activity rises as providers relocate into the country.

Question CI-6 c.5 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

A finance ministry adviser proposes a complete ban on all crypto asset activities to protect the domestic financial system. Which of the following recommendations best reflects the position of international bodies such as the IMF and the BCBS?

A. Adopt the ban as a short-term measure while international rules develop, since a general ban may be effective in the short run.

B. Prefer targeted limits, such as on crypto derivatives or on crypto as payment for goods and services, which may work short term.

C. Adopt the ban, since a blanket ban is what these bodies recommend once a jurisdiction faces systemic risk from crypto assets.

D. Avoid any restriction, since these bodies hold that restricting single crypto activities hinders innovation as much as a full ban.

Question CI-6 c.6 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

A working group of supervisors from several countries is drafting principles for regulating unbacked crypto assets. Which of the following principles is most consistent with the approach called for by international bodies such as the IMF and the BCBS?

A. Regulate each crypto asset according to the label its issuer gives it, so that firms can predict their treatment in every jurisdiction.

B. Leave crypto regulation wholly to national authorities, since cross-border cooperation slows the response to fast-changing technology.

C. Base the rules mainly on the economic functions and technical design of crypto assets, with stronger cooperation between authorities.

D. Base the rules on the size of each national crypto market, since jurisdictions with small markets face no risks that need cooperation.

Question CI-6 c.7 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c

The treasurer of a German bank plans to buy a tokenised version of a government bond recorded on a blockchain. Under the BCBS proposal described in the IMF note, which of the following treatments should the treasurer expect?

A. The 1,250% risk weight, because any asset recorded on a blockchain is treated as a higher-risk crypto asset under the proposal.

B. A risk weight below the bond's own, because tokenised assets are lower-risk and the proposal rewards the use of DLT for them.

C. A full deduction from capital, because the conservative approach for direct crypto exposures applies to every tokenised asset.

D. A risk weight similar to that of the traditional asset, because tokenised assets fall within the proposal's lower-risk group.

Question CI-6 c.8 New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c (correction box)

A bank holds a long position in an unbacked crypto asset and a smaller short position in the same asset. Its head of regulatory capital asks how the BCBS proposal would treat the two positions. Which of the following answers is correct?

A. A 1,250% risk weight applies to the larger of the long and short positions, so the short position does not net the charge away.

B. A 1,250% risk weight applies to the net of the long and short positions, so the short position reduces the charge one for one.

C. A 1,250% risk weight applies to the sum of the long and short positions, so the short position adds to the bank's capital charge.

D. A risk weight similar to that on traditional assets applies, because the long and short positions together form a hedged exposure.

Solutions • CI-6 c

Solution CI-6 c.1 — Answer: A

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – Is national regulation enough?

Regulating unbacked crypto assets is often done in a national setting, which may be sufficient and necessary if global regulation does not exist.

Step 2 – How national approaches differ

Some regulators give high-level guidance only; others write bespoke rules for specific activities, including transfer and safekeeping of assets and AML/CFT.

$\Rightarrow$ The adviser can act nationally and must choose where on that range to sit.

Interpretation

National rules are not the end of the story: global markets make them easy to bypass (Section 2), which is why international bodies call for stronger cooperation. That is a reason to cooperate, not a reason to wait.

Why the other options are wrong

B is incorrect. Polarity flip: the note says national regulation may be sufficient and necessary in the absence of global regulation. Bypass risk is a reason to cooperate across borders, not to wait.

C is incorrect. Absolute claim: approaches differ, from high-level guidance to bespoke rules, and the six examples focus on different things.

D is incorrect. Role misassignment and scope error: the BCBS proposal covers banks' crypto exposures, not the regulation of unbacked crypto assets in general.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022), Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets; SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.2 — Answer: B

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – Recall Japan

Japanese regulation is bespoke for crypto asset providers, focusing on asset segregation, operational risk, cyber risk, audit and capital requirements. Stablecoin regulation has been introduced since 2020.

$\Rightarrow$ An exchange entering Japan should expect provider-level rules on how it runs.

Interpretation

The key concepts group Japan with Albania under "operational aspects", against the EU and UK focus on stablecoins and tokens.

Why the other options are wrong

A is incorrect. Sibling swap: stablecoins, onshoring and licensing of issuing entities describe the EU.

C is incorrect. Sibling swap: crypto assets not being legal tender, and the ban on regulated institutions dealing in them, describe Nigeria.

D is incorrect. Sibling swap: a more progressive regime, with guidelines for initial token offerings and ledger-based securities on a blockchain, describes Switzerland.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including

the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.3 — Answer: D

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – Recall the UK

Initial UK guidance focused on regulating security tokens, but not utility and exchange tokens.

Step 2 – The direction of travel

Proposed future guidance is broader, also including stablecoins, market infrastructure and cryptocurrencies.

⇒ A utility token sat outside the initial guidance.

Interpretation

The UK started from the token type that already looks like a security and has been widening the perimeter since.

Why the other options are wrong

A is incorrect. Polarity flip: it reverses which tokens the initial guidance covered.

B is incorrect. Sibling swap: legalising crypto for trading in 2020 and licensing service providers describe Albania.

C is incorrect. Sibling swap: the progressive regime with initial token offering guidelines and ledger-based securities is Switzerland's.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.4 — Answer: C

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – Why strict rules leak

Without harmonised regulation, trading on unregulated exchanges can bypass sanctions and capital flow restrictions, providers can shift to less regulated jurisdictions, and VPNs can obscure locations.

Step 2 – Why enforcement fails

Bypassing is often successful even where stringent rules exist, because monitoring and enforcement are ineffective with limited resources.

⇒ The mission's finding fits the note's evidence: Egypt and Nigeria have strict restrictions, yet activity is widespread.

Interpretation

The lesson the note draws is cooperation across borders, not tougher national rules on their own.

Why the other options are wrong

A is incorrect. Polarity flip: China has banned crypto trading since 2021, yet evidence suggests significant mining activity. The ban did not end activity.

B is incorrect. Polarity flip: international bodies judge a blanket ban likely ineffective and harmful to innovation.

D is incorrect. Polarity flip: providers shift their services *to less regulated* jurisdictions.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022);

SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.5 Answer: B

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – The verdict on a ban

International bodies including the IMF and the BCBS judge a blanket ban on crypto activities likely ineffective, and say it would hinder innovation. SchweserNotes: ineffective in both the short and the long run, because monitoring and enforcement are difficult.

Step 2 – The alternative

Restricting only certain aspects is considered more effective in the short term: for example, restricting crypto asset derivatives (Japan and the UK) or restricting crypto assets as payment for goods and services.

$\Rightarrow$ Targeted restrictions, not a general ban.

Interpretation

Read the horizons carefully: "effective in the short term" is said of targeted restrictions only.

Why the other options are wrong

A is incorrect. Sibling swap: short-run effectiveness belongs to targeted restrictions; a general ban is ineffective in both horizons.

C is incorrect. Polarity flip: these bodies argue against a blanket ban.

D is incorrect. Absolute claim: they favour targeted restrictions; it is the blanket ban that would hinder innovation.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.6 Answer: C

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – The regulatory lens

Regulation should primarily focus on the economic functions of crypto assets and their technical design.

Step 2 – The institutional ask

The IMF, the BCBS and other international bodies call for stronger international cooperation among regulatory authorities.

$\Rightarrow$ What the asset does and how it is built, regulated jointly.

Interpretation

A function-and-design lens is what allows a regulator to restrict particular activities (derivatives, payments) without banning crypto assets outright.

Why the other options are wrong

A is incorrect. Wrong basis: the recommended focus is the asset's economic function and technical design, not the issuer's label.

B is incorrect. Polarity flip: these bodies call for stronger international cooperation.

D is incorrect. Wrong basis and absolute claim: the recommended focus is economic function and technical design, and the call for stronger cooperation is not limited to large markets.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022);

SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.7 — Answer: D

New • built from SchweserNotes Book 5, LO 105.c

Step 1 – Place the asset

A tokenised government bond is a tokenised asset, which the BCBS proposal puts in the lower-risk group.

Step 2 – Its weight

Tokenised assets would carry risk weights similar to those on traditional assets.
⇒ The bond is weighted much as the bond itself would be.

Interpretation

The higher-risk group, with the proposed 1,250% weight, is for cryptocurrencies and unbacked crypto assets. Being recorded on a blockchain does not move an asset there.

Why the other options are wrong

A is incorrect. Scope error: 1,250% is proposed for higher-risk assets (cryptocurrencies and unbacked crypto assets), not for every asset on a blockchain.
B is incorrect. Sibling swap: it is stablecoins that may benefit from lower risk weighting; tokenised assets get weights similar to traditional assets.
C is incorrect. Scope error: the capital deductions of the conservative approach target direct exposures to unbacked crypto assets, not tokenised traditional assets.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.c

Solution CI-6 c.8 — Answer: A

New • built from SchweserNotes Book 5, LO 105.c; CI notes CI-6 c (correction box)

Step 1 – Place the asset

Unbacked crypto assets fall in the BCBS higher-risk group, with a proposed 1,250% risk weight.

Step 2 – How the positions enter

The weight applies to the maximum of the long and short positions: here, the long position, because it is the larger.
⇒ The short position neither nets the charge down nor adds to it.

Interpretation

At an 8% capital ratio, 1,250% is a full capital deduction for the larger position (class notes). The rule stops a bank offsetting the charge by holding the other side.

Why the other options are wrong

B is incorrect. Polarity flip: netting is what the maximum rule rules out; the short position does not reduce the charge.
C is incorrect. Scope error: the weight applies to the larger of the two positions, not to their sum (see the correction box).
D is incorrect. Sibling swap: similar weights are for tokenised assets in the lower-risk group; holding both sides does not move an unbacked crypto asset out of the higher-risk group.

LO: CI-6 c: Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.c

CI-6 • Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets (IMF, September 2022) • Vardeez: IMPORTANT

CI-6 d

P3 CONCEPT

Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 (Map #68, 1/16, notes only), as set out under CI-6 a. There are no workbook questions for CI-6, so every question here is new or adapted from the CI question bank.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S1** CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda. The map gives no further split. This LO carries "Bali Fintech Agenda".

What the questions look like: no calculations. Who developed the BFA and what it is (against the BCBS and IOSCO products), then one scenario per consideration: which consideration a regulator's action falls under, and what the BFA recommends for a given market (small, centralised, niche, speculative, cross-border).

1. What the BFA is

Background: the risks by entity (CI-6 b) and the national approaches and BCBS treatment (CI-6 c).

The **Bali Fintech Agenda (BFA)** is a global regulatory and supervisory framework *for regulators*, jointly developed by the **IMF and the World Bank**. It sets out the standards for addressing fintech risks related to crypto assets and technologies. It encourages regulators to ensure **legal certainty** about crypto assets in order to **build investor confidence**, which matters because in many jurisdictions investors do not have sufficient rights. In the class notes' words, it is a framework for regulators to organise themselves, not a set of rules imposed on issuers.

Keep the attributions apart. The BFA is the IMF and the World Bank's. The proposed treatment of banks' crypto exposures, with the 1,250% weight for the higher-risk group, is the BCBS's (CI-6 c). The framework for crypto asset platforms is IOSCO's (CI-6 b). The calls for stronger international cooperation and against a blanket ban come from international bodies including the IMF and the BCBS (CI-6 c).

2. The five considerations

The BFA proposes that domestic regulatory frameworks for crypto assets be built around five considerations. Neither source separates "steps" from these five considerations; with the legal-certainty goal in Section 1, they are all the sources give for this LO.

(a) Monitoring

Regulators should monitor new developments in the crypto space, including *investor knowledge and behaviour*. Tools include **web searches** and **horizon scanning** for emerging technologies using **innovation offices**.

(b) Prioritising resources

Regulators have limited resources, so the **cost and feasibility** of their approach matter. In countries with **small or undeveloped crypto markets**, it may be sufficient to devote **limited resources initially**, while monitoring new developments and assessing whether more resources are needed. Schweser-Notes' key concepts put the other side: most resources go where crypto presence is significant, to contain systemic risk.

(c) Scope

Alongside prioritisation, the scope of regulation must be set. **In general, all entities with key functions relating to crypto assets should be regulated.** Within that general rule, the intensity of regulation follows systemic risk (Figure 1).

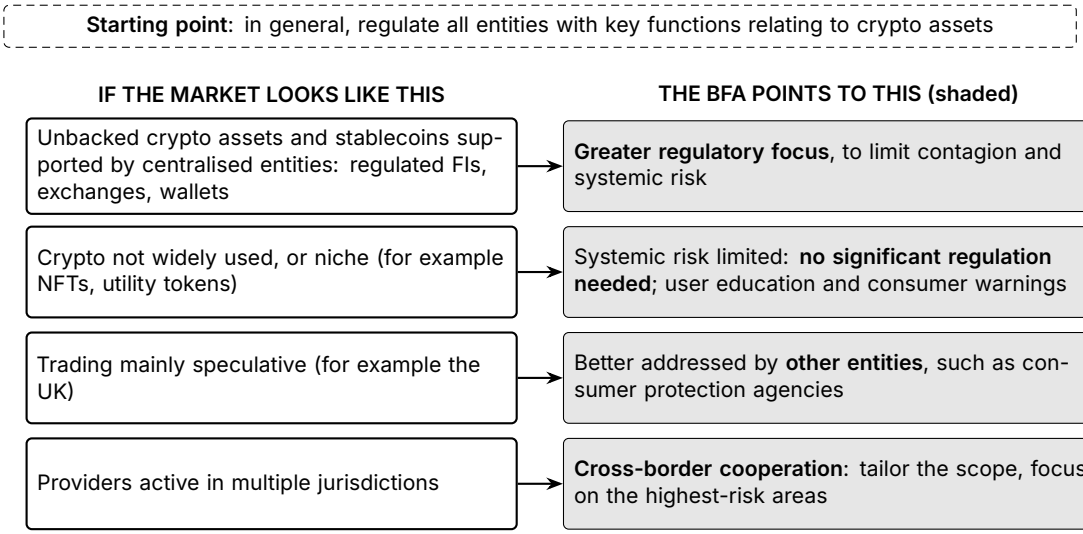


Figure 1. The scope consideration. The dashed box is the general rule; each row is a market situation (white) and the BFA's response (shaded). Intensity follows systemic risk: heaviest where centralised entities carry the exposure, lightest where crypto is niche.

(d) Domestic collaboration

Regulating crypto assets often involves **several domestic authorities**, each addressing specific risks. Coordination between them is important and should be tailored to specific risks, including **determining responsibility for AML/CFT mandates and legal issues**.

(e) Continuous risk assessment

Regulators often aim for **technological neutrality**: neither favouring nor discriminating against particular technologies. But crypto technology changes quickly and materially, so **regulation may need to shift** to keep addressing the greatest risks.

- DLT is the example: it let users trade fractional assets on a decentralised platform with significantly greater liquidity than in traditional asset markets.
- DLT networks often have **different objectives**: some focus on speed and efficiency, others on transparency. Regulation should be tailored and customised to their unique risks.
- Because providers often operate internationally, **domestic and international cooperation** is needed to reduce inconsistencies in laws and regulations and to sanction and prosecute crypto cases effectively. International bodies such as the IMF can play a significant role in assisting this cooperation.

Results card: CI-6 d

Consideration	The question it answers	What the BFA says
(a) Monitoring	What is happening?	New developments, investor knowledge and behaviour; web searches, horizon scanning, innovation offices
(b) Prioritising resources	How much to spend?	Cost and feasibility matter; small or undeveloped market: limited resources first, keep monitoring, reassess
(c) Scope	Who and what to regulate?	All key-function entities in general; greater focus where centralised entities support unbacked crypto and stablecoins; niche: education and warnings; speculative (UK): consumer protection agencies; cross-border cooperation to target the highest risks
(d) Domestic collaboration	Who does what at home?	Coordination tailored to risks; who holds AML/CFT mandates and legal issues
(e) Continuous risk assessment	Are the rules still right?	Technology neutral as an aim, but rules shift and are tailored as technology and risks change; domestic and international cooperation; IMF can assist
Overall	Who and why	IMF and World Bank; a framework for regulators; legal certainty to build investor confidence

Exam traps

- **Attribution.** The BFA is the IMF and the World Bank's. The BCBS sets the bank capital treatment; IOSCO, the platform framework.
- **What it is.** A framework for regulators, not binding rules on issuers or platforms.
- **Monitoring versus continuous risk assessment.** Web searches, horizon scanning and innovation offices are monitoring tools. Continuous risk assessment is about changing the rules as risks change.
- **Small market.** Limited resources at first, with continued monitoring; not a full regime at once, and not zero.
- **Scope polarity.** Heaviest focus where centralised entities support unbacked crypto and stablecoins; lightest where crypto is niche.
- **Speculative markets.** Where trading is mainly speculative (the UK example), regulation can be better addressed by other entities such as consumer protection agencies.
- **Neutrality is not rigidity.** The aim is not to favour a technology; the rules still shift and are tailored to each network's risks.
- **The IMF's role.** It can assist cross-border cooperation; it does not take over national supervision.

Practice questions: CI-6 d

Question CI-6 d.1 Adapted from CI QB Imp Q68 • SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A regulator at a central bank in Latin America is introducing the Bali Fintech Agenda (BFA) to her team. Which of the following descriptions is correct?

A. A framework developed by the BCBS that sets the prudential treatment of banks' crypto exposures, including a 1,250% risk weight.

B. A framework developed by IOSCO for crypto asset trading platforms, covering governance, market integrity and off-chain transactions.

C. A framework for regulators, developed jointly by the IMF and World Bank, that sets out standards for addressing fintech risks.

D. A framework agreed by the IMF and the BCBS that calls for a blanket ban on unbacked crypto assets until global rules are in place.

Question CI-6 d.2 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A finance ministry official in a country where crypto investors have few enforceable rights asks what the BFA encourages regulators to do to build investor confidence in crypto assets. Which of the following answers is most consistent with the BFA?

A. Require issuers to test whether each asset suits retail users, since the BFA is a set of rules imposed on issuers.

B. Restrict crypto derivatives and crypto payments, since targeted restrictions are the BFA's main tool for confidence.

C. Ensure legal certainty about crypto assets, since in many jurisdictions investors do not have sufficient rights.

D. Leave investor rights to the courts, since the BFA deals with supervisory resources, not the legal status of assets.

Question CI-6 d.3 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A securities regulator in Indonesia sets up an innovation office to scan the horizon for emerging crypto technologies, and uses web searches to track investor knowledge and behaviour. Under the BFA, this work falls most directly under which consideration?

A. Monitoring, which follows new developments in the crypto space, including investor knowledge and behaviour.

B. Continuous risk assessment, which uses horizon scanning to rewrite the rules each time a new technology appears.

C. Scope, which uses horizon scanning to decide which entities with key crypto functions should be regulated.

D. Domestic collaboration, which uses innovation offices to share AML/CFT duties among the domestic authorities.

Question CI-6 d.4 Adapted from CI QB Imp Q69 • SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

The head of the financial regulator in a small island state, whose crypto market is small and undeveloped, asks how the BFA suggests she allocate her limited supervisory staff to crypto assets. Which of the following approaches is most consistent with the BFA?

A. Build a full supervisory regime at once, because the BFA treats the cost and feasibility of an approach as secondary to full coverage.

B. Devote limited resources at first, while monitoring new developments and assessing whether more resources are needed later.

C. Stop monitoring crypto assets until the market becomes systemic, because a small market does not justify any supervisory resources.

D. Copy a large market's rulebook unchanged, since the BFA is one uniform set of rules that every jurisdiction must apply.

Question CI-6 d.5 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

In a large economy, regulated banks, major exchanges and custodial wallet providers support trading in unbacked crypto assets and stablecoins. A review under the BFA asks where the regulator should concentrate. Which of the following recommendations best reflects the BFA's scope consideration?

A. Give greater regulatory focus to these centralised entities, to limit the potential for contagion and systemic risk.

B. Leave the activity to the exchanges' own risk controls, since centralised entities already manage crypto risk for their users.

C. Wait for cross-border agreement on scope, since a national regulator cannot act on entities that also operate abroad.

D. Regulate only the issuers of the crypto assets, since exchanges and wallets merely facilitate trading and hold private keys.

Question CI-6 d.6 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

In another jurisdiction, crypto assets are not widely used, and the activity that exists is mainly in NFTs and utility tokens. Which of the following approaches does the BFA's scope consideration suggest?

A. Prudential capital and liquidity rules for all NFT and utility token issuers, since early rules prevent later contagion to banks.

B. Little or no action, since limited systemic risk means significant regulation is not needed and warnings add nothing.

C. A ban on NFTs and utility tokens, since niche markets are where a ban on crypto activity is most effective in the short term.

D. Light regulation, since systemic risk is limited, with the focus on user education and on consumer warnings.

Question CI-6 d.7 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A policy adviser in a country where crypto trading is mainly speculative retail activity, with little involvement from regulated financial institutions, asks which body may be best placed to address the main crypto risks. Based on the BFA's scope consideration, which of the following answers is most consistent?

A. The banking supervisor, applying capital deductions to the speculative crypto positions that retail traders hold.

B. Other entities such as consumer protection agencies, as in the UK, where trading is mainly speculative.

C. The IMF, which takes over supervision wherever crypto trading is speculative and crosses national borders.

D. The securities regulator alone, applying the IOSCO platform framework in place of any domestic coordination.

Question CI-6 d.8 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

In one country, the central bank, the securities regulator and the financial intelligence unit each claim part of the crypto perimeter, and none has clear ownership of AML/CFT supervision of crypto service providers. Which BFA consideration addresses this most directly, and how?

A. Monitoring: horizon scanning by each authority's innovation office, so that each tracks AML/CFT risks on its own.

B. Continuous risk assessment: reassign AML/CFT leadership whenever new technology shifts where the greatest risks lie.

C. Scope: regulate all entities with key crypto functions, so each authority keeps AML/CFT duties for the firms it licenses.

D. Domestic collaboration: coordination tailored to specific risks, including deciding who holds AML/CFT mandates.

Question CI-6 d.9 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A regulator's crypto rulebook was written to be technology neutral. Since then, a new DLT network built for speed and efficiency has gained wide use alongside an older network built for transparency. Under the BFA's continuous risk assessment consideration, which of the following responses is most appropriate?

A. Keep the rulebook unchanged, because technology neutrality means that regulation should not respond to new technologies.

B. Apply one uniform rule set to both networks, because DLT networks share the same objectives and therefore the same risks.

C. Shift and tailor the rules to each network's unique risks, while still aiming not to favour either technology.

D. Favour the faster network in the rules, because DLT lets users trade fractional assets with greater liquidity than before.

Question CI-6 d.10 New • built from SchweserNotes Book 5, LO 105.d; CI notes CI-6 d

A crypto service provider operates in many jurisdictions whose laws on its activities differ, which has made misconduct hard to sanction and prosecute. Which of the following statements best reflects the BFA's approach?

A. Domestic and international cooperation is needed to reduce inconsistent laws and to sanction and prosecute; the IMF can assist.

B. Each jurisdiction should act alone, since cross-border cooperation would widen the scope of regulation beyond the highest-risk areas.

C. The BCBS should prosecute the provider directly, since it sets the international standard for the treatment of crypto exposures.

D. Inconsistent laws fall outside the BFA, which covers domestic frameworks only and leaves all cross-border issues to IOSCO.

Solutions • CI-6 d

Solution CI-6 d.1 — Answer: C

Adapted from CI QB Imp Q68 • SchweserNotes Book 5, LO 105.d

Step 1 – Who developed it

The BFA was jointly developed by the IMF and the World Bank.

Step 2 – What it is

A global regulatory and supervisory framework for regulators that sets out standards for addressing fintech risks related to crypto assets and technologies.
⇒ C has both the authors and the content.

Interpretation

The reading names three standard setters with three different products. Attribution is the quickest way to eliminate options here.

Why the other options are wrong

A is incorrect. Role misassignment: the prudential treatment of banks' crypto exposures, with the 1,250% weight, is the BCBS proposal (CI-6 c).
B is incorrect. Role misassignment: the platform framework covering governance, market integrity and off-chain transactions is IOSCO's (CI-6 b).
D is incorrect. Sibling swap and polarity flip: the IMF and the BCBS are among the bodies that call for stronger cooperation, and they judge a blanket ban likely ineffective.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022), Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets; SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.2 — Answer: C

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – The BFA's starting point

The BFA encourages regulators to ensure there is legal certainty relating to crypto assets, in order to build investor confidence.

Step 2 – Why it matters

In many jurisdictions, investors do not have sufficient rights.
⇒ Legal certainty is the BFA's route to investor confidence.

Interpretation

The five considerations then shape how the domestic framework is built: monitoring, prioritising resources, scope, domestic collaboration and continuous risk assessment.

Why the other options are wrong

A is incorrect. Role misassignment: suitability tests are part of the issuer framework (CI-6 b), and the BFA is a framework for regulators, not a set of rules imposed on issuers.
B is incorrect. Sibling swap: targeted restrictions on derivatives and payments are the short-term alternative to a blanket ban described in CI-6 c, not the BFA's answer to weak investor rights.
D is incorrect. Polarity flip: legal certainty about crypto assets is exactly what the BFA encourages regulators to ensure.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.3 — Answer: A

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – Match the tools

Under the BFA, regulators should monitor new developments in the crypto space, including investor knowledge and behaviour. Monitoring tools could include web searches and horizon scanning for emerging technologies using innovation offices.
⇒ Every element of the regulator's work is on the monitoring list.

Interpretation

Monitoring also runs through prioritising resources: a regulator with a small market devotes limited resources at first while it keeps monitoring new developments, to judge whether more are needed.

Why the other options are wrong

B is incorrect. Sibling swap: continuous risk assessment is about shifting and tailoring regulation as technology and risks change; the tools named in the stem are monitoring tools.
C is incorrect. Sibling swap: scope decides which entities and activities to regulate; horizon scanning is not its tool.
D is incorrect. Sibling swap: domestic collaboration coordinates the authorities, including AML/CFT responsibilities; innovation offices are a monitoring tool.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.4 — Answer: B

Adapted from CI QB Imp Q69 • SchweserNotes Book 5, LO 105.d

Step 1 – The consideration

Prioritising resources: regulators have limited resources, so the cost and feasibility of their approach are important.

Step 2 – Apply it to a small market

In countries with small or undeveloped crypto markets, it may be sufficient to devote limited resources initially, while monitoring new developments and assessing whether more resources are needed.
⇒ Start small, keep watching, scale up if needed.

Interpretation

Prioritising and monitoring work together: the small-market regulator spends little but does not stop looking.

Why the other options are wrong

A is incorrect. Polarity flip: cost and feasibility are central to the BFA's prioritising consideration.
C is incorrect. Polarity flip: the BFA keeps monitoring running in small markets, to judge whether more resources are needed.
D is incorrect. Sibling swap on what the BFA is: a framework for regulators to organise themselves around considerations such as cost, feasibility and scope, not one uniform rulebook.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.5 — Answer: A

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – The general rule

In general, all entities with key functions relating to crypto assets should be regulated.

Step 2 – Where the focus goes

Greater focus should be given where unbacked crypto assets and stablecoins are supported by centralised entities such as regulated financial institutions, exchanges and wallets, to limit potential contagion and systemic risk.

⇒ The stem describes exactly that market.

Interpretation

Centralised entities are the channel through which crypto losses could spread to the financial system, so intensity follows them.

Why the other options are wrong

B is incorrect. Polarity flip: where centralised entities support unbacked crypto assets and stablecoins, the BFA calls for greater regulatory focus, because they are the channel for contagion and systemic risk.

C is incorrect. Scope error: in general all entities with key crypto functions should be regulated. Cross-border cooperation helps tailor the scope and target the highest-risk areas; it does not replace domestic regulation.

D is incorrect. Scope error: exchanges and wallets are among the centralised entities the BFA singles out for greater focus, not intermediaries to leave aside.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.6 — Answer: D

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – Read the market

Crypto assets are not widely used, and activity is in niche forms (NFTs, utility tokens).

Step 2 – The BFA's response

In such jurisdictions systemic risk is limited, so significant regulation is not necessary; regulation should focus on user education and consumer warnings.

⇒ Light regulation, aimed at users.

Interpretation

This is the low end of the scope scale in Figure 1. It sits well with prioritising resources, under which a small or undeveloped market may need only limited resources at first.

Why the other options are wrong

A is incorrect. Polarity flip: significant (prudential) regulation is not necessary where systemic risk is limited.

B is incorrect. Half right: significant regulation is not necessary where systemic risk is limited, but the BFA still directs regulation to user education and consumer warnings.

C is incorrect. Sibling swap: targeted restrictions, not bans, are what the reading calls more effective in the short term, and the BFA's answer for niche markets is education and warnings.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.7 Answer: B

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – Read the market

Trading is mainly speculative retail activity, with little involvement from regulated financial institutions.

Step 2 – The BFA's response

In some markets, like the UK, crypto trading is mainly speculative, and regulation can be better addressed by other entities such as consumer protection agencies.
⇒ The risk is to consumers, so a consumer body is the natural lead.

Interpretation

Scope is not only about how much to regulate but also about who is best placed to do it.

Why the other options are wrong

A is incorrect. Role misassignment: capital deductions apply to regulated financial institutions' own direct exposures, not to retail traders.
C is incorrect. Role misassignment: the IMF can assist international cooperation; it does not take over supervision.
D is incorrect. Scope error: the IOSCO framework is for trading platforms, and the BFA asks for coordination among domestic authorities, not one authority acting in place of it.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.8 Answer: D

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – Name the problem

Several domestic authorities each address part of the crypto risk, and responsibility for AML/CFT is unclear.

Step 2 – The consideration

Domestic collaboration: coordination between the authorities is important and should be tailored to specific risks, including determining responsibility for AML/CFT mandates and legal issues.
⇒ Settle who owns AML/CFT.

Interpretation

Domestic collaboration is the at-home counterpart of the cross-border cooperation stressed under scope and continuous risk assessment.

Why the other options are wrong

A is incorrect. Sibling swap: horizon scanning by innovation offices is monitoring, and separate tracking by each authority is the opposite of coordination.
B is incorrect. Sibling swap: continuous risk assessment shifts and tailors the rules as technology and risks change; it does not allocate mandates among authorities.
C is incorrect. Sibling swap: scope decides which entities and activities are regulated. Leaving each authority with AML/CFT for its own licensees is the gap the stem describes, which domestic collaboration resolves by settling who holds the mandate.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.9 — Answer: C

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – The aim

Regulators often aim for technological neutrality, neither favouring nor discriminating against particular technologies.

Step 2 – The limit of the aim

Crypto technology changes quickly and materially, so regulation may need to shift to address the greatest risks. DLT networks often have different objectives (speed and efficiency, or transparency), and regulation should be tailored to their unique risks.
⇒ Neutral about which technology, not about where the risk sits.

Interpretation

Continuous risk assessment keeps neutrality as the aim while letting the rules move with the risks.

Why the other options are wrong

A is incorrect. Scope error: neutrality is about not favouring a technology, not about freezing the rules; regulation may need to shift.
B is incorrect. Polarity flip: DLT networks often have different objectives, so their risks differ and rules should be tailored.
D is incorrect. Polarity flip: the greater liquidity DLT enabled shows how fast technology changes; it is no reason to favour one network.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

Solution CI-6 d.10 — Answer: A

New • built from SchweserNotes Book 5, LO 105.d

Step 1 – The problem

Providers often operate internationally, and differing national laws make misconduct hard to pursue.

Step 2 – The BFA's answer

Domestic and international cooperation is needed to reduce inconsistencies in laws and regulations and to sanction and prosecute crypto cases effectively; the IMF can play a significant role in assisting.
⇒ Cooperate across borders, with the IMF's help.

Interpretation

Under scope, the same cooperation lets authorities focus on the highest-risk areas.

Why the other options are wrong

B is incorrect. Polarity flip: cross-border cooperation lets authorities focus on the highest-risk areas; it does not widen the scope beyond them.
C is incorrect. Role misassignment: the BCBS proposes the treatment of banks' crypto exposures; it does not prosecute crypto providers.
D is incorrect. Scope error: the BFA expressly calls for international cooperation, and IOSCO's framework is for platforms.

LO: CI-6 d: Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets. • **Reference:** IMF Fintech Notes (September 2022); SchweserNotes 2026 Book 5, Reading 105, LO 105.d

CI-7 • Tokenization and Financial Market Inefficiencies (IMF, January 2025) • Vardeez: VERY IMPORTANT

CI-7 a

P2 CONCEPT

Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading Very Important. The recall workbook covers it only in D29 Crypto, Tokenisation and Digital Resilience (Tier D, Map #68, notes only): revision notes and traps, no questions. So every question in CI-7 is new or adapted from the CI question bank. Map #68 explains why the thin recall record should not lower the priority: "Because Current Issues rotates every year, the absence of recall evidence carries almost no information. Treat the provider ratings as the only signal available and cover all eight readings."
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." The class notes tag CI-7 DEFER with an SRC of 0 (no historical question frequency) and still build all five objectives in full. This book does the same: 8 to 10 questions per LO.

Reported exam shapes

- **D29·S2** CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures. This LO carries the first part of the shape: what a token is, the three routes onto a ledger, and the three features (every question here).

What the questions look like: no calculations. Classification (native, non-native or migrated), matching a concern to the feature it belongs to (trust, sharedness or programmability), and "which design meets these requirements" items in the style of the SchweserNotes module quiz.

0. The reading in one paragraph

The IMF Fintech Note *Tokenization and Financial Market Inefficiencies* (January 2025) is SchweserNotes Book 5, Reading 106, and CI-7 on the 2026 list. It asks one question: when assets move onto digital ledgers, which market inefficiencies get smaller and which get bigger? The five LOs follow the argument in order.

LO	Question it answers	Core content
CI-7 a	What is a token, and what must the ledger provide?	Native, non-native and migrated tokens; trust, sharedness, programmability
CI-7 b	Which frictions does tokenization reduce, phase by phase?	Four frictions across issuance, trading, servicing and redemption; what survives
CI-7 c	Which externalities can it make worse or better?	Shock transmission, infrastructure investment, network and knowledge effects
CI-7 d	What changes for retail investors and brokers?	Internalities from direct trading; broker market power; private-ledger market power
CI-7 e	How are ledgers actually built, and how does trust fail?	Single, common and compatible ledgers; single-operator, permissionless and permissioned trust

1. Tokens and tokenization

- **Token.** An asset, or a representation of an asset, created on a digital ledger. Put simply, a token works like a digital certificate of ownership that can be transferred between parties through a secure digital system.
- **Tokenization.** The process of creating tokens. It is not a single technology: it is a collection of tools and approaches designed to make financial assets easier to transfer, automate and manage securely.
- **Features vary by degree.** The key features of a tokenized system (trust, sharedness and programmability, Section 3) can vary by degree from one ledger to another, and tokens may move across ledgers to find a more suitable combination of them.

2. Three routes onto a ledger

The class notes set the reading's "What is tokenization?" figure as a two-column table: the state the ledger must be in (shared, programmable, trusted) and the three ways a token can be created on it.

1. **Native tokens: new issuance.** Created directly on a ledger, they exist only within that system. Examples: cryptocurrencies such as Bitcoin and Ether, and digital art.
2. **Non-native tokens: a new virtual representation of an existing asset.** The asset exists outside the ledger and the token is its digital representation. Virtually any asset can be tokenized this way: financial instruments such as equities, bonds and commodities, and also real estate and physical artwork.
3. **Migration: destruction and reissuance.** A token can represent an asset migrated from another ledger. The original is *destroyed* on the source ledger and *reissued as a native token* on the new ledger.

The class notes add that migration is the route most easily forgotten and the one with the most legal weight, since the original is destroyed rather than mirrored.

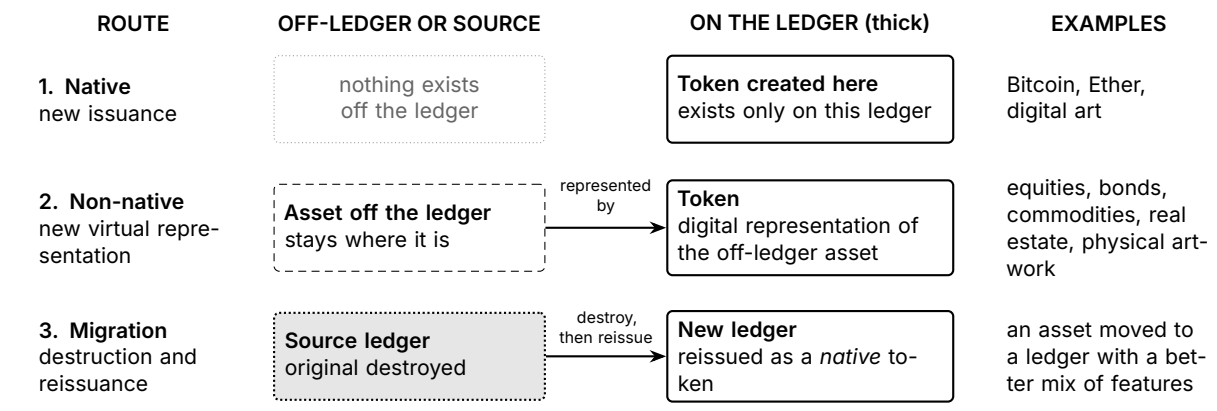


Figure 1. The three routes onto a ledger. Thick boxes are the token on the ledger; the dashed box is an asset that keeps living off the ledger; the dotted shaded box is an original that is destroyed. Only the non-native route leaves an asset outside the ledger. Migration leaves no copy behind, and the reissued token is native on its new ledger.

Route	Where the asset lives	Any original?	Examples
Native	On the ledger only	None exists	Bitcoin, Ether, digital art
Non-native	Outside the ledger; the token represents it	Stays off the ledger	Equities, bonds, commodities, real estate, physical artwork
Migration	On the new ledger, as a native token	Destroyed on the source ledger	An asset moved from another ledger

Two pairs are easy to cross. *Digital art* created on a ledger is native, while a *physical* painting is non-native. And migration is not representation: a non-native token mirrors an asset that keeps existing, whereas migration destroys the original. The escrow-and-certificate arrangement, in which the original survives locked in escrow, is a different thing again: the common ledger model of CI-7 e.

3. Three features the ledger must provide

For tokenization to function effectively, the ledger must let users own and transfer assets, users must trust it to record ownership and transactions accurately, and it must be programmable so that it can automate instructions such as paying interest or enforcing trading rules. That gives three features.

1. **Trust.** Users can be confident that their assets cannot be moved without their consent, and that transactions will be executed reliably and accurately. In the key-concept wording: ownership and transactions are recorded accurately and cannot be altered without the owner's consent.
2. **Sharedness.** The ledger is accessible to all relevant parties, asset ownership can be transferred between them, and ownership records are updated directly on the ledger.
3. **Programmability.** The ledger can incorporate *smart contracts*: immutable code that automates instructions. Smart contracts can create assets, transfer assets only when certain conditions are met, and ensure compliance with rules and regulations. They make it possible to bundle complex transactions together and ensure they operate exactly as intended. The reading's examples: coupon-paying transferable bonds, and restricting transactions to qualified investors.

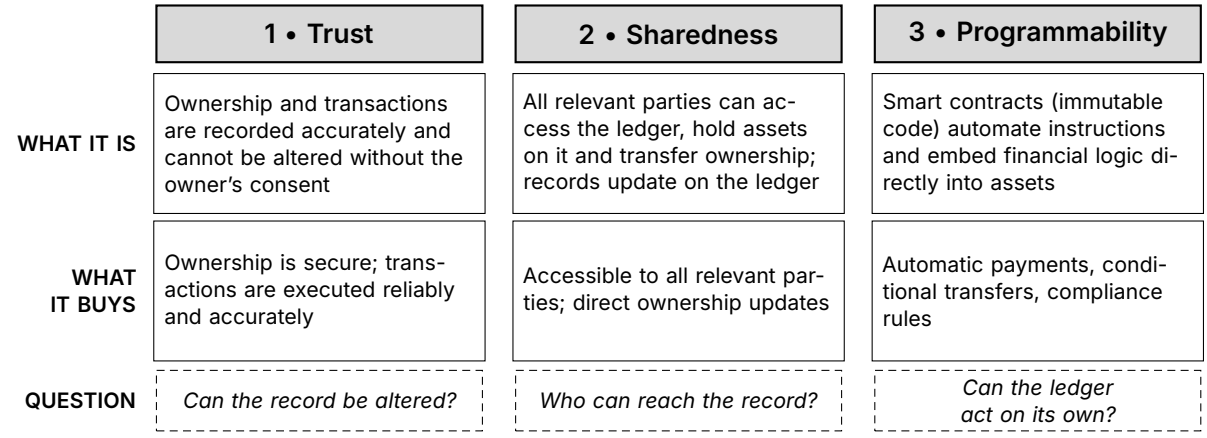


Figure 2. The three features (the class notes' Figure CI-7.1, with a row added). Shaded boxes name the feature, plain boxes say what it is and what it buys, and dashed boxes give the one question each feature answers. The three are not interchangeable: trust is about whether the record can be altered, sharedness about who can reach it, programmability about whether the ledger can act by itself.

4. Reading the features: four rules

1. **All three, each by degree.** Tokenization functions effectively only when the ledger provides all three features. "Vary by degree" describes how much of each a ledger offers; it is not permission to drop one. The class notes put it plainly: a ledger missing any one of the three cannot support tokenization.
2. **Match the concern to the feature.** Consent and accurate records point to trust; access and direct transfer between parties point to sharedness; anything the ledger does by itself (paying, gating, enforcing) points to programmability.
3. **Smart contracts are immutable.** The code is fixed in advance and executes as written, and smart contracts make it possible to bundle complex transactions and ensure they operate exactly as intended. Code that an operator rewrites on request is not what the reading calls a smart contract.
4. **"All relevant parties", not "everyone".** Sharedness asks that the relevant parties can reach the ledger, transfer between themselves and see ownership update on it. It does not ask that the ledger be open to anyone: who may join and validate is a question of trust design, compared in CI-7 e.

Correction to the source notes. The class notes define tokens as "digital representations of ownership recorded on a ledger" and tokenization as "a set of tools built on distributed ledgers". The first echoes the reading's own "digital certificate of ownership" analogy but is narrower than its definition; the second is narrower than the reading. A token is an asset *or* a representation of an asset: a native token such as Bitcoin is the asset itself and represents nothing off the ledger. And the ledger is a *digital* ledger, not necessarily a distributed one: in CI-7 e, trust can rest on a single-operator system (one entity manages the records and confirms transactions) as well as on permissionless or permissioned DLT (SchweserNotes Book 5, LO 106.a and 106.e). Question CI-7 a.7 turns on this point.

Results card: CI-7 a

Token	An asset, or a representation of an asset, created on a digital ledger; a transferable digital certificate of ownership
Tokenization	The process of creating tokens; a collection of tools and approaches, not one technology
Native	Created on the ledger and exists only there: Bitcoin, Ether, digital art
Non-native	Represents an asset outside the ledger: equities, bonds, commodities, real estate, physical artwork
Migration	Original destroyed on the source ledger, reissued as a *native* token on the new ledger
Trust	Assets cannot be moved without consent; reliable, accurate execution
Sharedness	All relevant parties can access the ledger and transfer ownership; records update on the ledger
Programmability	Smart contracts (immutable code): create assets, conditional transfers, compliance; coupon-paying bonds, qualified-investor limits
Rule	All three are needed; each varies by degree; tokens can move ledgers to find a better mix

Exam traps

- **Sibling swap on token type.** Digital art made on the ledger is native; a physical painting is non-native. Bitcoin represents nothing off the ledger.
- **Migration read as mirroring.** The original is destroyed, not kept, and the reissued token is native on its new ledger.
- **Migration read as escrow.** Locking the original in escrow and trading a certificate is the common ledger model (CI-7 e).
- **Feature swap.** Consent and accuracy is trust; access and transfer is sharedness; automation is programmability.
- **"Vary by degree" read as "optional".** Strong sharedness and programmability do not make up for missing trust.
- **Editable smart contracts.** Smart contracts are immutable code.
- **Scope error.** Not only financial instruments: virtually any asset, including real estate and physical artwork.
- **Distributed assumed.** Tokens sit on a digital ledger; a single-operator system can provide the trust.
- **Shared read as public.** Sharedness means all *relevant* parties.

Practice questions: CI-7 a

Question CI-7 a.1 Adapted from CI QB VI Q107 • SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

The chief operating officer of a Luxembourg fund administrator tells the board that tokenization is "one new technology that we either adopt or do not". Which statement about tokenization is most consistent with the IMF note?

A. Tokenization is one technology, but its features are fixed by the protocol, so the only real choice for the board is which ledger vendor to adopt.

B. Tokenization is a set of tools, and ledgers offer trust, sharedness and programmability by degree, so tokens can move ledgers for a better mix.

C. Tokenization is a set of tools that applies only to assets created on a ledger, so the fund units it already administers could not be tokenized.

D. Tokenization is one technology defined by decentralised validation, so adopting it means giving up any role in keeping the fund's records.

Question CI-7 a.2 New • built from SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

A specialist at a London auction house is preparing three tokens: (1) a digital artwork created directly on a ledger, with no existence outside it; (2) a token giving title to an oil painting stored in the auction house's vault; (3) a token representing a share in a commercial office building. Which classification is correct?

A. (1) native; (2) native, because the title to the painting is issued on the ledger; (3) non-native.

B. (1) non-native, because it represents a work of art; (2) non-native; (3) non-native.

C. (1) native; (2) non-native; (3) cannot be tokenized, because only financial instruments and art can be.

D. (1) native; (2) non-native; (3) non-native, because the building exists outside the ledger.

Question CI-7 a.3 Adapted from CI QB VI Q102 • SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

A digital-securities manager at a European bank plans to move a bond from the bank's legacy ledger to a new ledger with more programmable features. The legal team asks what migration involves under the reading's definitions. Which answer is correct?

A. The bond is destroyed on the legacy ledger and reissued as a native token on the new ledger, so no copy of it remains behind.

B. The bond stays live on the legacy ledger, and a non-native token representing it is issued on the new ledger, so that both can trade.

C. The bond cannot move, because a token's mix of trust, sharedness and programmability is fixed by the ledger that first issued it.

D. The bond is destroyed on the legacy ledger and reissued on the new ledger as a non-native token, because it was first issued elsewhere.

Question CI-7 a.4 New • built from SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

The head of digital assets at a Toronto pension fund lists three concerns about a proposed ledger: (1) the fund's usual counterparties are not on the ledger, so ownership cannot pass to them on it; (2) the operator could move the fund's tokens without the fund's consent; (3) coupon payments would still have to be calculated and sent by hand. Which features of a tokenized ledger do the three concerns relate to?

A. (1) trust; (2) sharedness; (3) programmability.

B. (1) sharedness; (2) programmability; (3) trust.

C. (1) sharedness; (2) trust; (3) programmability.

D. (1) programmability; (2) trust; (3) sharedness.

Question CI-7 a.5 Adapted from CI QB VI Q104 • SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

A structurer at a Hong Kong bank proposes a tokenized note whose smart contract the ledger operator would rewrite by hand whenever the parties want to change its terms. A colleague objects that the proposal misreads how the IMF note describes smart contracts. Which description supports the colleague?

A. Smart contracts are instructions that users send to a separate database operator, who executes them once the parties have agreed the terms.

B. Smart contracts are immutable code that automates instructions, so the note's transfers and payments operate exactly as intended.

C. Smart contracts are digitised legal agreements attached to the token record, which give holders a claim but execute nothing by themselves.

D. Smart contracts are the ledger's trust feature, which ensures that the operator cannot move a holder's assets without the holder's consent.

Question CI-7 a.6 New • built from SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

A consultant reviewing a ledger for a Mumbai exchange finds that all relevant parties can access it, ownership records update directly on it, and smart contracts automate payments. However, the ledger's rules let the operator reassign holdings without the owner's consent. Which conclusion is most consistent with the reading?

A. It can support tokenization effectively, because the features vary by degree and two strong features can make up for a weak third.

B. It can support tokenization effectively, because trust comes from contract law rather than from the ledger, so only two features matter.

C. It cannot support tokenization effectively, because it lacks sharedness: holdings that an operator can reassign are not truly shared.

D. It cannot support tokenization effectively, because it lacks trust: assets must not be movable without the owner's consent.

Question CI-7 a.7 New • built from SchweserNotes Book 5, LO 106.a and 106.e; CI notes CI-7 a (correction box)

A fintech supervisor at a central bank is reviewing a firm's claim that its centrally operated platform "cannot host tokens, because tokens can exist only on a distributed ledger". Which response is most consistent with the reading?

A. The claim is right for native tokens only: non-native tokens can sit on a central platform, because their assets exist off the ledger.

B. The claim is right: trust in a tokenized system can come only from consensus among many validators, which a single operator cannot supply.

C. The claim is wrong: tokens sit on a digital ledger, and trust can come from one operator's credibility and oversight, not only from DLT.

D. The claim is wrong, but only because a central platform is a common ledger model, which holds the original assets in escrow for users.

Question CI-7 a.8 Adapted from CI QB VI Q108 • SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

A board member at a Seoul brokerage argues that a ledger restricted to approved participants can never provide sharedness, because sharedness means that anyone in the world can read and use the ledger. Which response best reflects the reading's definition?

A. The board member is wrong: sharedness requires access for all relevant parties, transfers between them and records updated on the ledger.

B. The board member is right: sharedness means no party can move assets without the owner's consent, which a restricted ledger cannot ensure.

C. The board member is right: sharedness requires a ledger open to anyone, so a restricted ledger can be trusted and programmable but not shared.

D. The board member is wrong: sharedness means that the code and the database are integrated, which any ledger running smart contracts provides.

Question CI-7 a.9 Adapted from CI QB VI Q106 • SchweserNotes Book 5, LO 106.a; CI notes CI-7 a

The head of capital markets at a development bank wants a new bond that (1) exists only on a ledger, (2) can be transferred only with the holder's authorisation, and (3) pays coupons and restricts transfers to qualified investors automatically. Which design best meets all three requirements?

A. A non-native token on a shared ledger, backed by a conventional bond held with a custodian, which pays the coupons to holders.

B. A native token on a ledger whose smart contracts pay coupons and screen buyers, and whose holdings cannot move without consent.

C. A native token on a shared ledger, with the qualified-investor limit enforced through the ledger's trust feature rather than code.

D. A bond migrated from a conventional register but kept live there, and mirrored on the ledger so that both records can be updated.

Solutions • CI-7 a

Solution CI-7 a.1 — Answer: B

Adapted from CI QB VI Q107 • SchweserNotes Book 5, LO 106.a

Step 1 – What tokenization is

The reading is explicit: tokenization is not a single technology but a collection of tools and approaches designed to make financial assets easier to transfer, automate and manage securely.

Step 2 – How the features behave

Trust, sharedness and programmability can vary by degree across ledgers, and tokens may move across ledgers to find a more suitable combination of them.

⇒ The board is choosing among designs with different mixes of features, not making a yes-or-no technology decision.

Interpretation

"A set of tools whose features vary by degree" is the frame for the whole reading: CI-7 e compares the specific tools (three sharedness models and three trust models) that produce different mixes.

Why the other options are wrong

A is incorrect. Absolute claim twice over: tokenization is not one technology, and its features are not fixed; they vary by degree and tokens can move between ledgers.

C is incorrect. Scope error: assets created on a ledger are only the native route. Existing assets can be tokenized as non-native tokens, and virtually any asset qualifies.

D is incorrect. Two errors: it is not one technology, and nothing in the reading requires decentralised validation. A single-operator system, in which one entity manages the records, is one of the trust models (CI-7 e).

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025), Tokenization and Financial Market Inefficiencies; SchweserNotes 2026 Book 5, Reading 106, LO 106.a

Solution CI-7 a.2 — Answer: D

New • built from SchweserNotes Book 5, LO 106.a

Step 1 – Ask where each asset lives

(1) The digital artwork is created on the ledger and has no existence outside it: native. The reading names digital art as a native example.

Step 2 – Off-ledger assets

(2) The oil painting sits in a vault; the token represents it: non-native. (3) The office building exists outside the ledger; the token represents a share in it: non-native.

⇒ Native means born on the ledger and living only there; non-native means a ledger representation of something that lives elsewhere.

Interpretation

The art pair is the trap the reading sets up by naming both "digital art" (native) and "physical artwork" (non-native). Real estate is one of the reading's own non-native examples: virtually any asset can be tokenized as a non-native token.

Why the other options are wrong

A is incorrect. Sibling swap on (2): issuing the token on the ledger does not make it native. The painting exists outside the ledger, so the token is a representation.

B is incorrect. Sibling swap on (1): the digital artwork does not represent anything off the ledger;

it exists only on it, which is the definition of native.
C is incorrect. Scope error on (3): virtually any asset can be tokenized as a non-native token, and real estate is named explicitly.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a

Solution CI-7 a.3 — Answer: A

Adapted from CI QB VI Q102 • SchweserNotes Book 5, LO 106.a

Step 1 – The migration route
Tokens can represent assets migrated from another ledger: the original asset is destroyed on the source ledger and reissued as a native token on the new ledger.

Step 2 – Why move at all
Tokens may move across ledgers to find a more suitable combination of trust, sharedness and programmability, here a ledger with more programmable features.
⇒ After migration the bond lives only on the new ledger, as a native token.

Interpretation
The class notes flag migration as the route with the most legal weight, because the original is destroyed rather than mirrored. That is the contrast with a non-native token, where the underlying asset keeps existing off the ledger.

Why the other options are wrong
B is incorrect. Sibling swap: a live original with a ledger representation is the non-native route, not migration. Migration destroys the original.
C is incorrect. Absolute claim: the reading says tokens may move across ledgers precisely to find a better mix of features.
D is incorrect. Right first half, wrong label: the reissued token is *native* on the new ledger. "First issued elsewhere" does not make it a representation, because no original survives to be represented.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a

Solution CI-7 a.4 — Answer: C

New • built from SchweserNotes Book 5, LO 106.a

Step 1 – Concern (1)
Sharedness: the ledger is accessible to all relevant parties and ownership can be transferred between them. Counterparties who are not on the ledger cannot receive ownership on it.

Step 2 – Concern (2)
Trust: users can be confident that their assets cannot be moved without their consent.

Step 3 – Concern (3)
Programmability: smart contracts automate instructions such as paying interest.
⇒ Access and transfer, consent, automation: sharedness, trust, programmability.

Interpretation
Each feature answers a different question (Figure 2): who can reach the record, can it be altered, can the ledger act by itself. Matching a concern to its question is the quickest way through a feature-mapping item.

Why the other options are wrong

A is incorrect. Sibling swap on (1) and (2): access for counterparties is sharedness, and protection against moves without consent is trust.
B is incorrect. Sibling swap on (2) and (3): consent is trust, and automated coupons are programmability.
D is incorrect. Sibling swap on (1) and (3): reaching counterparties is sharedness, and automating coupons is programmability.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a

Solution CI-7 a.5 — Answer: B

Adapted from CI QB VI Q104 • SchweserNotes Book 5, LO 106.a

Step 1 – The reading's definition

Programmability lets the ledger incorporate smart contracts: *immutable* code that automates instructions. They can create assets, transfer assets only when conditions are met and ensure compliance with rules.

Step 2 – What smart contracts make possible

This capability makes it possible to bundle complex transactions together and ensure they operate exactly as intended.
⇒ A contract the operator rewrites on request is not what the reading calls a smart contract.

Interpretation

The reading's own examples (coupon-paying transferable bonds, transfers restricted to qualified investors) work because the rules are written in advance and execute automatically.

Why the other options are wrong

A is incorrect. Sibling swap: users sending instructions to a separate database operator is the traditional set-up that the ledger's internal programmability replaces (CI-7 e).
C is incorrect. Form without function: a document attached to a record automates nothing, whereas a smart contract executes instructions.
D is incorrect. Feature swap: protection against moves without consent is trust. Smart contracts belong to programmability.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a and 106.e

Solution CI-7 a.6 — Answer: D

New • built from SchweserNotes Book 5, LO 106.a

Step 1 – Check the features present

Access for all relevant parties and direct record updates: sharedness. Smart contracts automating payments: programmability.

Step 2 – Check the feature missing

Trust requires that assets cannot be moved without the owner's consent. An operator who can reassign holdings breaks exactly that.

Step 3 – Apply the rule

Tokenization functions effectively only when the ledger provides all three features.
⇒ Two out of three is not enough.

Interpretation

"The features vary by degree" means a ledger can offer more or less of each; it does not mean one

can be absent. The class notes: a ledger missing any one of the three cannot support tokenization.

Why the other options are wrong

A is incorrect. Misreads "vary by degree" as permission to trade one feature off against another; all three are essential.

B is incorrect. Scope error: trust is one of the three features of the ledger itself, covering accurate records and consent, so it cannot be dropped from the ledger.

C is incorrect. Feature swap: the defect is the operator's power to move assets without consent, which is trust. Access and direct record updates (sharedness) are present.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a

Solution CI-7 a.7 — Answer: C

New • built from SchweserNotes Book 5, LO 106.a and 106.e

Step 1 – Where tokens live

Tokens are assets or representations of assets created on a *digital* ledger. Nothing in the definition requires the ledger to be distributed.

Step 2 – Where trust can come from

Trust in digital ledgers can be established through single-operator systems, permissionless DLT or permissioned DLT. In a single-operator system one entity manages records and executes and confirms transactions, and trust rests on its credibility, supervision and oversight.

$\Rightarrow$ A centrally run platform can host tokens if it supplies trust, sharedness and programmability.

Interpretation

The class notes' phrase "a set of tools built on distributed ledgers" is narrower than the reading (see the correction box). The design question is how well each trust model works and how it fails (CI-7 e), not whether a central ledger is allowed at all.

Why the other options are wrong

A is incorrect. Invented split: the native and non-native distinction is about where the asset lives, not about which kind of ledger may host the token.

B is incorrect. Absolute claim: consensus is how permissionless and permissioned DLT build trust; a single-operator system builds it through the operator's credibility and oversight.

D is incorrect. Axis confusion: the common ledger model (escrow and certificates) is a way of achieving sharedness. It does not describe a centrally operated platform as such.

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a and 106.e

Solution CI-7 a.8 — Answer: A

Adapted from CI QB VI Q108 • SchweserNotes Book 5, LO 106.a

Step 1 – The definition

Sharedness: the ledger is accessible to all *relevant* parties, asset ownership can be transferred between them, and ownership records are updated directly on the ledger.

Step 2 – Apply it

A ledger limited to approved participants provides sharedness if the relevant parties are on it and can transfer ownership on it. Who may join or validate is a separate design choice about trust (CI-7 e).

$\Rightarrow$ "All relevant parties" is not "anyone in the world".

Interpretation

The board member has merged two questions: who can reach the record (sharedness) and who is allowed to join and validate (the trust model, for example permissioned DLT).

Why the other options are wrong

B is incorrect. Feature swap: consent protection is the definition of trust, not of sharedness.

C is incorrect. Scope error: the definition asks for access by all relevant parties, not openness to anyone.

D is incorrect. Feature swap: integration of code and database is how the reading describes programmability (CI-7 e).

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a and 106.e

Solution CI-7 a.9 **Answer: B**

Adapted from CI QB VI Q106 • SchweserNotes Book 5, LO 106.a (Module Quiz 106.1 shape)

Step 1 – Requirement (1) fixes the token type

"Exists only on a ledger" is the definition of a native token. A non-native token represents an asset that exists outside the ledger.

Step 2 – Requirements (2) and (3) fix the features

Transfer only with the holder's authorisation is trust. Automatic coupons and a qualified-investor restriction are programmability: the reading's own smart-contract examples. The ledger must also be shared so that ownership can pass between holders.

$\Rightarrow$ A native token on a trusted, shared and programmable ledger, with smart contracts doing the paying and the screening.

Interpretation

Read requirement lists one at a time and map each to a token type or a feature. The SchweserNotes module quiz uses the same shape.

Why the other options are wrong

A is incorrect. Fails (1) and (3): the conventional bond exists off the ledger, so the token is non-native, and coupons paid by a custodian are not automated on the ledger.

C is incorrect. Feature swap: restricting transfers to qualified investors is a smart-contract use, which is programmability, not trust.

D is incorrect. Misdefines migration: migration destroys the original, and a bond kept live off the ledger with a mirror on it is a non-native representation, which fails (1).

LO: CI-7 a: Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a

CI-7 • Tokenization and Financial Market Inefficiencies (IMF, January 2025) • Vardeez: VERY IMPORTANT

CI-7 b

P2 CONCEPT

Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading Very Important; the workbook covers it in D29 Crypto, Tokenisation and Digital Resilience (Tier D, Map #68, notes only). There are no workbook questions for this reading, so every question here is new or adapted from the CI question bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S2** CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures. This LO is the "lifecycle frictions" part: the four frictions, the intermediaries that reduce them, the three further inefficiencies, and what tokenization does at each phase (every question here).

What the questions look like: no calculations. Classifying a situation as a friction or as an externality, internality or market power; "which benefit is most consistent" at a given phase; and the one thing tokenization cannot remove, residual counterparty risk, in the SchweserNotes "least effective" shape.

1. Four frictions

Recap from CI-7 a: tokens sit on a ledger that must be trusted, shared and programmable. This LO asks what those features do to the costs of owning and trading assets.

Market *frictions* are a subset of market inefficiencies. They arise at the stages of an asset's lifecycle: issuance, trading, servicing and redemption. The reading names four.

Friction	What it is	The reading's example
Information asymmetries	One party has more information than others	A company owner during an IPO
Search frictions	Difficulty finding suitable trading partners	OTC markets, where information and liquidity are limited
Transaction costs	Fees, trading costs and opportunity costs such as the time spent completing a transaction	Intermediaries' fees
Counterparty risk	The risk that one party fails to meet its obligations	Derivatives: exposure at initiation (paying a fee or premium) and through the life of the contract (the buyer holds a future right to transact)

2. Where the frictions sit in the lifecycle

The class notes redraw the IMF lifecycle figure as a wheel (Figure 1). The load is not evenly spread: issuance and exchange (trading) carry three frictions each, while servicing and redemption carry only transaction costs. Transaction costs are the one friction present in all four phases.

SchweserNotes does not map frictions to phases: it says they can arise at any stage of the lifecycle. The mapping here is the IMF figure's, as redrawn in the class notes, and SchweserNotes' phase-by-phase discussion is consistent with it (servicing and redemption: transaction costs; trading: counterparty risk, transaction costs and, in OTC markets, search frictions).

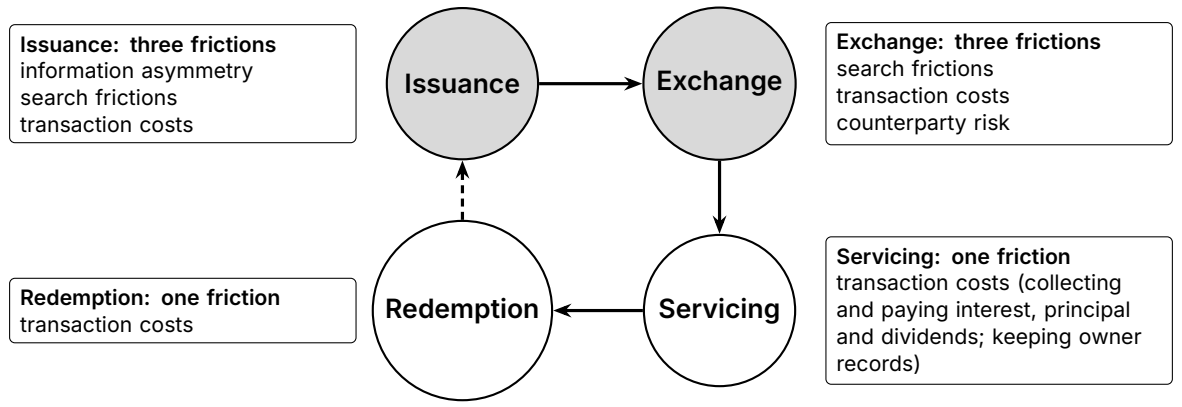


Figure 1. Frictions along the lifecycle of an asset (the class notes' Figure CI-7.2, from the IMF note). Shaded circles carry three frictions, plain circles carry one. The dashed arrow closes the wheel. Information asymmetry is placed at issuance and counterparty risk at exchange; transaction costs appear everywhere.

3. Intermediaries reduce frictions, for a fee

Specialised intermediaries reduce these frictions, but each charges fees that add to overall transaction costs.

- **Investment banks** reduce information asymmetries by conducting extensive due diligence on issuers and by underwriting new stocks or bonds.
- **Central securities depositories (CSDs) and central clearinghouses (CCHs)** manage settlement and clearing. They reduce counterparty risk by ensuring both sides of a trade have sufficient funds or assets to meet their obligations.
- **Central counterparties (CCPs)**, a subcategory of CCHs, add a further layer. A CCP interposes itself as the counterparty to all trading parties in a derivatives contract. On a default it covers nonpayment or nondelivery using the *defaulting party's* margin and collateral first and, if needed, the CCP's guarantee fund.
- **At issuance**, registrars (which keep the formal register of security holders), banks and trust companies create records and manage transactions. They lower costs through economies of scale, given large volumes, but they charge for their services.

4. Three further inefficiencies

Transacting in financial markets also gives rise to externalities, internalities and market power. The reading defines them here and applies them in CI-7 c (externalities) and CI-7 d (internalities and market power).

Concept	Definition	The reading's examples	Who bears it
Externality	One participant's actions affect others; positive or negative. Even positive externalities can lead to socially inefficient outcomes	*Positive:* network effects, liquidity improves as more participants join, yet participation may stay below the socially optimal level. *Negative:* moral hazard, intermediaries take excessive risk expecting guarantees or bailouts, keeping the profits and sharing the losses	Others
Internality	Participants take actions that are not in their own best interest	A retail investor trading without fully understanding the risks	The participant
Market power	The ability to influence prices or terms	An institution charging above-market fees because barriers to switching keep users from moving	The users who pay

5. How tokenization mitigates frictions, phase by phase

Tokenization can reduce certain frictions more effectively and at lower cost than traditional markets, in (1) issuance, (2) trading and (3) servicing and redemption. The working features are *sharedness* and *programmability*.

Phase	Traditional friction and intermediary	What a shared, programmable ledger does
Issuance	Registrars, banks and trust companies keep records and manage transactions, for fees; information on smaller issuances is often limited	Links assets directly to owners' accounts: no separate recordkeeping and no stand-alone custodial services. Smart contracts automate complex processes such as auctions for new shares. Key data can be embedded in the token: BNP Paribas's 2022 tokenized bond carried its term sheet and ESG data, increasing transparency. Most valuable for *smaller* issuances
Trading	Counterparty risk, nonsimultaneous settlement and transaction costs; CSDs and CCHs reduce counterparty risk but charge fees; search frictions in OTC markets	A self-executing smart contract completes a trade only if both the buyer's cash and the seller's securities are locked in: simultaneous or even instantaneous settlement. Smart contracts also reduce counterparty risk in multistep trades such as cross-currency trades. Direct access to a single shared ledger makes counterparties easier to find, lowers costs such as bid-ask spreads and reduces the need for broker-dealers
Servicing and redemption	Costs of collecting and transmitting interest, principal and dividends, and of keeping owner records up to date	Smart contracts automate payments and link assets directly to owners. The ledger can specify when payments are due and how much to pay per token, and can even embed dividend distribution decisions

Instantaneous settlement means a trade settles as soon as the parties agree. Traditional markets cannot do this because CCHs need time to coordinate and update multiple ledgers. On a shared ledger, instantaneous settlement reduces costs and transaction times and can improve liquidity and capital allocation.

6. The other direction: what tokenization cannot remove

The LO asks how the features can both *mitigate* and *increase* risks. The class notes point out that, within this LO, only one item runs the other way.

- **Residual counterparty risk.** Even with shared and programmable ledgers, tokenization cannot fully eliminate counterparty risk when a party fails to meet its obligations *between contract initiation and execution*.
- **The fixes.** A tokenized system could incorporate a CCP, or require that collateral be locked up for the duration of the contract.
- **The cost of the fix.** Locking up assets as collateral creates opportunity costs and may discourage investors from using tokenized trades. Closing one friction (counterparty risk) reopens another (transaction costs, in the form of opportunity cost).

The fuller set of risks that tokenization can increase sits in the next two LOs: faster shock transmission, infrastructure and fragmentation externalities (CI-7 c), and amplified internalities and new market power (CI-7 d).

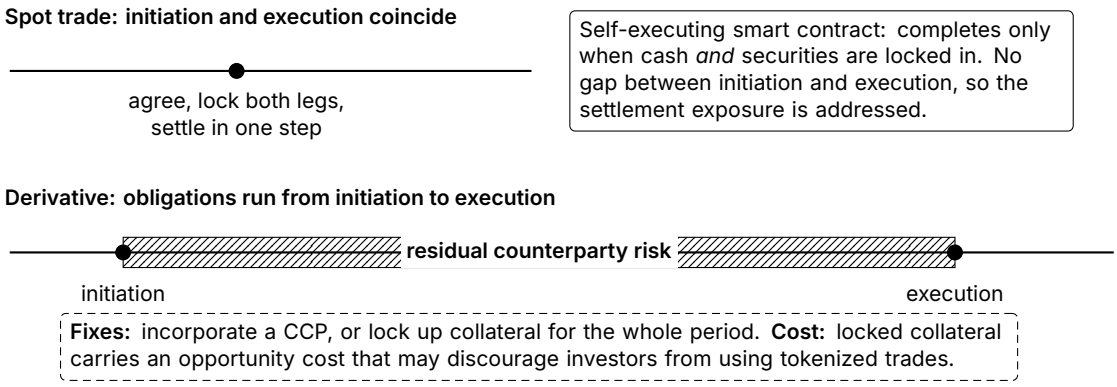


Figure 2. Why counterparty risk survives tokenization. Top line: when initiation and execution are one step, the smart contract addresses the settlement exposure. Bottom line: the hatched band is the period between initiation and execution, which no single settlement step can cover; the dashed box gives the fixes and their cost.

Correction to the source notes. The caption to the class notes' lifecycle figure says counterparty risk "appears at exactly one point, exchange, which is why it is the friction tokenization cannot fully remove." The placement follows the IMF figure; the reasoning does not. At the point of exchange a self-executing smart contract can complete the trade only when both legs are locked in, which addresses the settlement exposure. What survives is the risk that a party fails *between contract initiation and execution*, as with a derivative, where exposure arises both at initiation and through the life of the contract (SchweserNotes Book 5, LO 106.b). The residual risk is a matter of time, not of position on the wheel. Questions CI-7 b.8 and b.10 turn on this point.

Results card: CI-7 b

Four frictions	Information asymmetries, search frictions, transaction costs, counterparty risk
Lifecycle	Issuance and exchange: three frictions each; servicing and redemption: transaction costs only (IMF figure, class notes)
Intermediaries	Investment banks (due diligence, underwriting); CSDs and CCHs (settlement, clearing); CCPs (interpose; defaulter's margin and collateral, then guarantee fund)
Three inefficiencies	Externality (affects others; network effects, moral hazard); internality (not in own interest); market power (prices or terms)
Issuance fix	Direct owner links, no separate records or custody; automated auctions; embedded data (BNP Paribas, 2022), most useful for smaller issues
Trading fix	Both legs locked: simultaneous or instantaneous settlement; cross-currency steps; OTC search frictions, spreads and broker-dealers down
Servicing fix	Payment timing and amount per token coded; dividend decisions can be embedded
What survives	Counterparty risk between initiation and execution: CCP or locked collateral, at an opportunity cost

Exam traps

- **Friction swap.** Not finding a partner is search; knowing less is information asymmetry; a partner failing to perform is counterparty risk.
- **Phase swap.** Information asymmetry sits at issuance; servicing and redemption carry transaction costs only.
- **Role misassignment.** Investment banks fight information asymmetry; CSDs, CCHs and CCPs fight counterparty risk; registrars keep the holder register.
- **Waterfall reversed.** A CCP uses the defaulter's margin and collateral first, its guarantee fund only if needed.
- **Externality versus internality.** Harm to others is an externality (moral hazard); harm to oneself is an internality.
- **Size reversed.** Embedded token data matters most for *smaller* issuances.
- **Overreach on counterparty risk.** Tokenization reduces it at settlement but cannot remove it between initiation and execution.
- **Costless collateral.** Locked-up collateral has an opportunity cost that may deter use.

Practice questions: CI-7 b

Question CI-7 b.1 New • built from SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

A lecturer at a Frankfurt business school gives students three situations: (1) the founder of a firm going public knows far more about its prospects than the investors buying its shares; (2) a dealer cannot find a willing buyer for a thinly traded OTC bond; (3) an option buyer holds a right to transact at a future date and the option writer may not perform. Which classification of the three frictions is correct?

A. (1) search friction; (2) transaction cost; (3) counterparty risk.

B. (1) information asymmetry; (2) counterparty risk; (3) search friction.

C. (1) information asymmetry; (2) search friction; (3) counterparty risk.

D. (1) information asymmetry; (2) search friction; (3) transaction cost.

Question CI-7 b.2 New • built from SchweserNotes Book 5, LO 106.b; CI notes CI-7 b (Figure CI-7.2)

A consultant at a Johannesburg exchange maps market frictions to the phases of an asset's lifecycle, following the IMF note's lifecycle figure. Which statement is consistent with that mapping?

A. Servicing and redemption carry only transaction costs; counterparty risk sits at exchange, information asymmetry at issuance.

B. Counterparty risk is the main friction at servicing, because coupon and dividend payments can fail once the asset is held.

C. Information asymmetry sits at exchange, where buyers know less than sellers, while issuance carries transaction costs only.

D. Search frictions arise only at exchange, because finding a counterparty matters only once an asset has started trading on a market.

Question CI-7 b.3 Adapted from CI QB VI Q112 • SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

A clearing member defaults on a large derivatives position cleared through a central counterparty (CCP). A risk manager at another clearing member asks how the IMF note describes the CCP's role. Which description is correct?

A. The CCP, acting as a central securities depository, records the default and leaves the loss with the defaulter's original counterparty.

B. The CCP covers the default first from its own guarantee fund, and only then draws on the defaulting member's margin and collateral.

C. The CCP limits information asymmetry by doing due diligence on members, but it does not stand between the parties to a trade.

D. The CCP stands between both sides and covers the default from the defaulter's margin and collateral, then its guarantee fund.

Question CI-7 b.4 New • built from SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

A policy adviser at a finance ministry is classifying three cases: (1) a large bank takes excessive risk because it expects a government bailout, keeping the profits and sharing the losses; (2) a retail investor buys a complex product whose risks she does not understand; (3) a custodian charges above-market fees because clients find switching to a rival too costly. Which classification is correct?

A. (1) internality; (2) negative externality; (3) market power.

B. (1) negative externality; (2) internality; (3) market power.

C. (1) market power; (2) internality; (3) negative externality.

D. (1) negative externality; (2) market power; (3) internality.

Question CI-7 b.5 Adapted from CI QB VI Q114 • SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

The treasurer of a small Nordic utility is considering its first bond issue in tokenized form, similar to the bond BNP Paribas issued in 2022. Which benefit of tokenized issuance is most consistent with the reading?

A. The token removes the need for due diligence and underwriting, because a shared ledger eliminates information asymmetry.

B. Tokenized issuance raises registrars' economies of scale, because registrars and trust companies can handle larger volumes.

C. Embedding the term sheet and ESG data in the token adds transparency, most useful where issuances are small.

D. Embedding key data in the token mainly helps the largest issuers, whose information is hardest for investors to obtain.

Question CI-7 b.6 Adapted from CI QB VI Q115 • SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

The head of post-trade at a Singapore exchange explains to the board why a tokenized market could settle trades as soon as the parties agree, which the exchange's current market cannot do. Which explanation is most consistent with the reading?

A. A CCP on the ledger guarantees every trade from its fund at once; clearinghouses must first collect margin from both sides of every trade.

B. A smart contract extends intraday credit to whichever party is slower; clearinghouses must wait for both parties to post funds in full.

C. A shared ledger removes search frictions; in the current market most settlement delay comes from finding a counterparty after the trade.

D. A smart contract executes only once both cash and securities are locked in; clearinghouses need time to update multiple ledgers.

Question CI-7 b.7 Adapted from CI QB VI Q117 • SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

A dealer at a London bank trades an OTC product in which information and liquidity are limited. The platform operator proposes moving trading onto a single shared, programmable ledger to which all participants have direct access. Which outcome is most consistent with the reading?

A. Counterparties become easier to find and costs such as bid-ask spreads fall, so the need for broker-dealers declines.

B. Search frictions are unchanged, because OTC prices are already centralised, but counterparty risk is fully eliminated.

C. Reliance on broker-dealers rises, because participants need dealers to interpret the information now held on the ledger.

D. Bid-ask spreads widen, because the dealers that provided liquidity are disintermediated and lose the incentive to quote.

Question CI-7 b.8 New • built from SchweserNotes Book 5, LO 106.b (Module Quiz 106.1 shape); CI notes CI-7 b (correction box)

A derivatives risk manager at a Chicago trading firm is assessing how far smart contracts on a programmable ledger can reduce counterparty risk. In which of the following trades would programmability be LEAST effective at mitigating counterparty risk?

A. A spot bond trade in which the contract executes only once both the cash and the bonds have been locked in.

B. A forward contract with delivery long after initiation, in which neither party locks up any collateral.

C. A cross-currency trade completed in several linked steps, each of them automated by a smart contract.

D. An option contract in which the writer's collateral stays locked in the smart contract for its full life.

Question CI-7 b.9 Adapted from CI QB VI Q120 • SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

The head of corporate actions at a custodian bank asks what a programmable ledger would change in the servicing and redemption phase of a tokenized security. Which statement is most consistent with the reading?

A. It automates interest and principal, but dividends still need an off-ledger decision, so dividend distribution cannot be embedded.

B. Its main benefit at this stage is lower search frictions, because holders can find buyers more easily before the asset is redeemed.

C. It can specify when payments are due and how much to pay per token, and can embed dividend decisions, cutting servicing costs.

D. It adds a second, on-ledger record of owners beside the registrar's, which raises recordkeeping costs but makes payments accurate.

Question CI-7 b.10 New • built from SchweserNotes Book 5, LO 106.b; CI notes CI-7 b

A tokenized derivatives venue in Paris decides that collateral for every contract will be locked up on the ledger for the full life of the contract. The venue's head of risk is asked to describe the trade-off. Which description is most consistent with the reading?

A. Counterparty risk is mitigated, but locked collateral creates opportunity costs that may discourage investors from trading.

B. Counterparty risk and transaction costs both fall to zero, because the smart contract now covers the whole period before execution.

C. Counterparty risk rises, because concentrating all collateral on one ledger creates a single point of failure for the venue.

D. Counterparty risk is mitigated, and participation rises, because investors prefer contracts whose collateral cannot be withdrawn early.

Solutions • CI-7 b

Solution CI-7 b.1 — Answer: C

New • built from SchweserNotes Book 5, LO 106.b

Step 1 – Situation (1)

One party (the founder) has more information than others: information asymmetry. The reading's own example is a company owner during an IPO.

Step 2 – Situation (2)

Difficulty finding a suitable trading partner: search friction.

Step 3 – Situation (3)

The risk that one party fails to meet its obligations: counterparty risk. In derivatives it arises at initiation and through the life of the contract, when the buyer holds a future right to transact.

⇒ Information asymmetry, search friction, counterparty risk.

Interpretation

Ask what goes wrong: knowing less, not finding a partner, or the partner not performing. Transaction costs are the fees, trading costs and time that every phase carries.

Why the other options are wrong

A is incorrect. Sibling swap on (1) and (2): the founder's informational edge is asymmetry, and the missing buyer is a search problem.

B is incorrect. Sibling swap on (2) and (3): the missing buyer is a search friction, and the writer's possible failure is counterparty risk.

D is incorrect. Sibling swap on (3): non-performance by the writer is counterparty risk, not a fee or trading cost.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025), Tokenization and Financial Market Inefficiencies; SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.2 — Answer: A

New • built from SchweserNotes Book 5, LO 106.b; CI notes Figure CI-7.2

Step 1 – Read the figure

Issuance: information asymmetry, search frictions, transaction costs. Exchange: search frictions, transaction costs, counterparty risk. Servicing: transaction costs. Redemption: transaction costs.

Step 2 – Check the description of servicing

The reading describes servicing and redemption frictions as transaction costs: collecting and transmitting interest, principal and dividend payments, and keeping owner records up to date.

⇒ Three, three, one, one.

Interpretation

The uneven load explains where the reading's tokenization benefits are richest: issuance and trading get several fixes each, servicing and redemption get one, payment automation.

Why the other options are wrong

B is incorrect. Phase swap: counterparty risk is placed at exchange; the servicing friction is transaction costs.

C is incorrect. Phase swap: information asymmetry sits at issuance (the IPO example), and issuance carries three frictions, not one.

D is incorrect. Scope error: the figure places search frictions at issuance as well as at exchange, and SchweserNotes says frictions can arise at any stage.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025), lifecycle figure as redrawn in the class notes (Figure CI-7.2); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.3 — Answer: D

Adapted from CI QB VI Q112 • SchweserNotes Book 5, LO 106.b

Step 1 – What a CCP is

A CCP is a subcategory of central clearinghouses. It interposes itself as the counterparty to all trading parties in a derivatives contract.

Step 2 – How it absorbs a default

It covers nonpayment or nondelivery using the defaulting party's margin and collateral and, if needed, the CCP's guarantee fund.

$\Rightarrow$ The defaulter's own resources go first; the guarantee fund is the backstop.

Interpretation

This is the traditional answer to counterparty risk, and it carries fees. CI-7 b.8 and b.10 show why tokenized markets may still need a CCP or locked collateral.

Why the other options are wrong

A is incorrect. Role misassignment: CSDs manage settlement; the CCP's defining act is interposing itself between the parties, so the loss does not stay with the original counterparty.

B is incorrect. Waterfall reversed: the defaulter's margin and collateral are used first, the guarantee fund only if needed.

C is incorrect. Role misassignment: due diligence against information asymmetry is the investment bank's role, and standing between the parties is exactly what a CCP does.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.4 — Answer: B

New • built from SchweserNotes Book 5, LO 106.b

Step 1 – Case (1)

Excessive risk-taking in expectation of a bailout, with profits kept and losses shared, is moral hazard: the reading's example of a *negative externality*, because the losses land on others.

Step 2 – Case (2)

Acting against one's own best interest, such as trading without understanding the risks, is an *internality*.

Step 3 – Case (3)

Above-market fees sustained by switching barriers is *market power*.

$\Rightarrow$ Others harmed, self harmed, users overcharged.

Interpretation

Who bears the cost separates the three: an externality falls on others, an internality on the actor, and market power on the customers who pay the price set.

Why the other options are wrong

A is incorrect. Sibling swap on (1) and (2): moral hazard harms others (externality); the investor's mistake harms herself (internality).
C is incorrect. Sibling swap on (1) and (3): bailout-driven risk-taking is moral hazard, a negative externality, and switching-barrier fees are market power.
D is incorrect. Sibling swap on (2) and (3): the investor's mistake is an internality, and the custodian's fees are market power.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.5 — Answer: C

Adapted from CI QB VI Q114 • SchweserNotes Book 5, LO 106.b

Step 1 – The issuance benefits

A shared, programmable ledger links assets directly to owners' accounts, removing separate recordkeeping and stand-alone custody; smart contracts automate processes such as share auctions; and key data can be embedded in the token.

Step 2 – Who gains most from embedded data

Embedding data makes asset and company information easier to access, which is particularly valuable for *smaller* issuances, where such information is often limited. BNP Paribas's 2022 tokenized bond embedded its term sheet and ESG data.
$\Rightarrow$ A small utility is the kind of issuer the reading has in mind.

Interpretation

Embedded data attacks information asymmetry at issuance by making information easier to reach. The reading does not say it replaces the due diligence and underwriting of investment banks.

Why the other options are wrong

A is incorrect. Overreach: the reading says embedded data increases transparency, not that it eliminates information asymmetry or the need for due diligence and underwriting.
B is incorrect. Polarity flip: economies of scale are the advantage of the *traditional* intermediaries; the tokenized ledger removes the need for separate recordkeeping.
D is incorrect. Size reversed: embedded data is most valuable for smaller issuances, where information is often limited.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.6 — Answer: D

Adapted from CI QB VI Q115 • SchweserNotes Book 5, LO 106.b

Step 1 – The tokenized mechanism

A self-executing smart contract completes a trade only if both the buyer's cash and the seller's securities are locked in, so the two legs move together: simultaneous or even instantaneous settlement.

Step 2 – Why the current market cannot match it

Instantaneous settlement is not possible in traditional markets because CCHs need time to coordinate and update multiple ledgers.

⇒ One shared ledger with conditional execution replaces several ledgers that must be reconciled.

Interpretation

The benefit is lower costs and shorter transaction times, and it can improve liquidity and capital allocation. It does not cover obligations that run past the settlement step (CI-7 b.8).

Why the other options are wrong

A is incorrect. Role misassignment: speed comes from conditional execution on a shared ledger, not from a CCP's guarantee fund.

B is incorrect. Invented mechanism: the smart contract locks both legs; it does not lend to the slower party.

C is incorrect. Phase swap: search frictions concern finding a counterparty before a trade is agreed, not settling it afterwards.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.7 — Answer: A

Adapted from CI QB VI Q117 • SchweserNotes Book 5, LO 106.b

Step 1 – The OTC friction

Search frictions bite hardest in OTC markets, where information and liquidity are limited.

Step 2 – What a single shared ledger does

Direct access to a single shared ledger makes it easier to find trading counterparties and helps lower costs such as bid-ask spreads, which reduces the need for broker-dealers.

⇒ Sharedness turns a search problem into a visibility problem that the ledger solves.

Interpretation

CI-7 c develops the same move as a network externality: as liquidity grows, spreads narrow and more users join. Note what the option does not claim: counterparty risk is not fully removed.

Why the other options are wrong

B is incorrect. False premise and overreach: OTC information is limited, not centralised, and counterparty risk is never fully eliminated (between initiation and execution it survives).

C is incorrect. Polarity flip: the need for broker-dealers falls.

D is incorrect. Polarity flip: spreads narrow; the reading links the shared ledger to lower bid-ask spreads.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.8 — Answer: B

New • built from SchweserNotes Book 5, LO 106.b (Module Quiz 106.1 shape)

Step 1 – Where programmability works

A smart contract reduces counterparty risk at the settlement step by completing the trade only when both legs are locked in (A), and it reduces counterparty risk in multistep trades such as cross-currency trades (C).

Step 2 – Where it cannot reach

Tokenization cannot fully eliminate counterparty risk when a party fails between contract initiation and execution. The fixes are a CCP or collateral locked up for the duration of the contract

(D has that).

Step 3 – Find the uncovered case

The forward contract has a long gap between initiation and delivery and no locked collateral.
⇒ Programmability alone cannot manage that exposure.

Interpretation

The residual risk is about time: obligations that run past the settlement step (see the correction box). The forward contract is the only option with such a gap and no safeguard.

Why the other options are wrong

A is incorrect. Effective case: locking cash and bonds before execution is the reading's own example of a self-executing smart contract.
C is incorrect. Effective case: the reading says smart contracts can reduce counterparty risk in multistep transactions such as cross-currency trades.
D is incorrect. Effective case: collateral locked for the life of the contract is one of the two fixes the reading names for the residual risk.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b and Module Quiz 106.1

Solution CI-7 b.9 **Answer: C**

Adapted from CI QB VI Q120 • SchweserNotes Book 5, LO 106.b

Step 1 – The servicing friction

Servicing and redemption carry transaction costs: collecting and transmitting interest, principal and dividend payments, and keeping owner records up to date.

Step 2 – The tokenized fix

A shared, programmable ledger automates these payments through smart contracts, linking assets directly to owners. It can specify when payments are due and how much to pay per token, and can even embed dividend distribution decisions.
⇒ Lower servicing costs, with the paying logic in the code.

Interpretation

Servicing is the phase with the simplest story: one friction (transaction costs) and one fix (payment automation).

Why the other options are wrong

A is incorrect. Scope error: the reading says dividend distribution decisions can be embedded.
B is incorrect. Phase swap: lower search frictions are a trading-phase benefit, not a servicing one.
D is incorrect. Polarity flip: linking assets directly to owners removes the need for separate recordkeeping, so recordkeeping costs fall.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

Solution CI-7 b.10 **Answer: A**

New • built from SchweserNotes Book 5, LO 106.b

Step 1 – What the rule fixes

Requiring collateral to be locked up for the duration of the contract is one of the two ways the

reading names for mitigating counterparty risk between initiation and execution (the other is a CCP).

Step 2 – What it costs

Locking up assets as collateral creates opportunity costs and may discourage investors from using tokenized trades.

$\Rightarrow$ One friction falls (counterparty risk) and another rises (transaction costs, as opportunity cost).

Interpretation

This is the clearest case in this LO of a tokenized design both mitigating and increasing frictions at once, which is what the LO wording asks you to explain.

Why the other options are wrong

B is incorrect. Absolute claim: locked collateral carries an opportunity cost, so transaction costs do not fall to zero.

C is incorrect. Sibling swap: a single point of failure is the infrastructure externality of a dominant ledger (CI-7 c), not the effect of collateral lock-up on counterparty risk.

D is incorrect. Polarity flip: the reading says locked collateral creates opportunity costs that may discourage investors from using tokenized trades, not that it draws them in.

LO: CI-7 b: Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b

CI-7 • Tokenization and Financial Market Inefficiencies (IMF, January 2025) • Vardeez: VERY IMPORTANT

CI-7 c

P2 CONCEPT

Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading Very Important; the workbook covers it in D29 Crypto, Tokenisation and Digital Resilience (Tier D, Map #68, notes only). There are no workbook questions for this reading, so every question here is new or adapted from the bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S2** CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures. This LO is the "externalities" part: shock transmission, infrastructure investment, network and knowledge effects (every question here). One of the D29 likely distractors comes from this LO: fragmented, non-interoperable ledgers do not mainly amplify the transmission of shocks (Question CI-7 c.7).

What the questions look like: no calculations. "Which chain of effects is most consistent" (cheaper debt to leverage, deposits to wholesale funding), naming the externality behind a scenario, and the two-sided items: network effects that still fall short of the optimum, and open code that spreads knowledge but weakens the incentive to create it.

1. The double edge

Recap from CI-7 b: an externality arises when one participant's actions affect others. It can be positive or negative, and even a positive externality can lead to a socially inefficient outcome. Tokenization and tokenized ledgers can significantly improve interconnectedness, trading liquidity and transaction speeds, but they can also introduce new risks. The reading works through four externalities.

Externality	Direction	What tokenization does
Shock transmission	Negative	Can amplify shocks through speed, cheaper debt and leverage, deposit migration, linked assets and rehypothecation, and automated trading
Infrastructure investment	Negative if operators underinvest	Raises the stakes of operational and cyber failures as users and linked transactions grow; single point of failure; concentration risk
Network	Positive	More users bring more liquidity and innovation; fractionalisation widens access; non-interoperable ledgers weaken the effect
Knowledge	Positive, with an incentive cost	Open code lets others build cheaply; easy copying weakens the incentive to fund R&D

2. Shock transmission externalities

The traditional amplifiers. In traditional markets, high leverage, low funding liquidity and strong interconnectedness make it easier for shocks to spread, increasing risks to overall stability. Highly leveraged institutions are more likely to take on greater risk, shifting their funding mix toward debt relative to equity. They keep the upside from equity returns while shifting potential losses from failure onto other institutions, or onto the public through bailouts. When institutions are closely interconnected, a shock to one can trigger a domino effect.

What tokenization adds. Tokenization can increase the impact of shocks, especially as transaction speeds and leverage increase. The reading gives five channels.

1. **Speed.** Faster transactions spread shocks more quickly and make it more difficult for policymakers to contain the costs associated with externalities.
2. **Cheaper debt, more leverage.** Tokenization can lower the issuance cost of debt compared with conventional bonds. Borrowing becomes more attractive, which can encourage higher leverage and raises the risk of instability if institutions take on excessive debt.
3. **Deposit migration.** Tokenization makes it easier for retail investors to place funds in financial assets outside bank deposits. Banks may see retail deposits decline and rely more on wholesale funding, which can dry up quickly in stress, raising the risk of defaults and spillovers.
4. **Linked assets and rehypothecation.** The interconnectedness of shared, programmable ledgers gives investors easier access to complex and linked assets, including assets tied to other assets or pegged to different currencies. Complex products may involve *rehypothecation*, where collateral is reused across multiple transactions. When many investors can move funds quickly and cheaply across assets and markets, a sudden drop in one asset can trigger rapid selloffs in several markets, making it harder for regulators to respond.
5. **Automated trading.** Smart contracts running rules-based algorithms can execute trades instantly or nearly instantly and bundle complex transactions, which may lead to rapid market movements. A chain of self-executing smart contracts that triggers large volumes of trades can contribute to a *flash crash*, a sharp price drop over a short period.

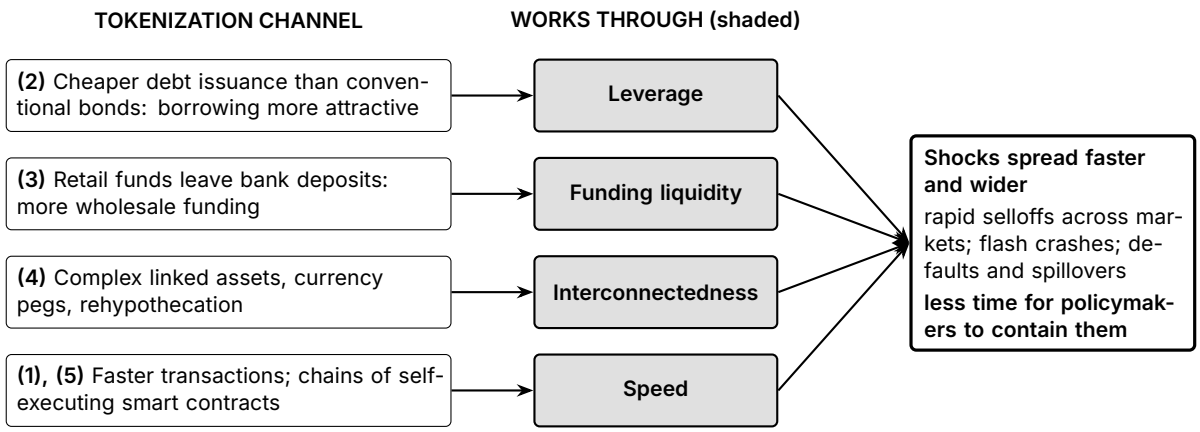


Figure 1. The five shock-transmission channels read left to right. Each tokenization channel (plain boxes) works through one of the traditional amplifiers or through speed (shaded boxes); all of them end in the same outcome (heavy box). Channel numbers follow the list in the text.

Correction to the source notes. The class notes list the leverage channel as "Lower issuance cost of tokens: encourages higher leverage". The reading is more specific: tokenization can lower the issuance cost of *debt* compared with conventional bonds, which makes borrowing more attractive and can encourage higher leverage (SchweserNotes Book 5, LO 106.c and Module Quiz 106.2). The link to leverage runs through cheaper borrowing; the class notes' wording drops the word that makes the mechanism work. Question CI-7 c.1 option D shows why the word matters: cheaper equity issuance would push leverage the other way.

3. Market infrastructure investment externalities

- As more users rely on digital ledgers, interconnectedness rises and transactions speed up. This also raises the cost of infrastructure investment, because operators must spend more to mitigate the risks of operational failures and cyberattacks.
- **The externality.** If private operators underinvest in operational security, the resulting risks can affect *all* participants. A single point of failure in a dominant ledger could disrupt the entire market.
- **Why it grows.** The potential impact of failures grows with the number of users and interconnected transactions, so tokenization makes infrastructure investment more important, not less. Risks include human and software errors, fraud and system failures.
- **Concentration risk.** A widely used digital ledger can become a target for cyberattacks, with potential for broader disruption across the financial system.

The operator chooses how much to spend on security, while a failure would land on every participant: that gap between who decides and who bears the cost is what makes underinvestment an externality.

4. Network externalities

Definition. Network externalities (network effects) are the value an agent gains from joining a ledger. A larger number of users increases market liquidity and strengthens incentives for innovation, which makes the ledger more attractive to new participants. The sharedness feature is closely connected to liquidity.

The positive cycle. Moving an OTC market onto a digital ledger can improve liquidity by standardising contracts, centralising price information and reducing search frictions. As liquidity grows, transaction costs fall and bid-ask spreads narrow, which encourages more users to join and reinforces the cycle. Positive does not mean optimal: as CI-7 b notes, with network effects participation may still remain below the socially optimal level.

Fractionalisation. Tokenized ledgers support asset fractionalisation, which lets retail investors buy fractional shares and diversify more easily. Some intermediaries already offer fractional shares of certain equities; tokenization enables fractionalisation across a wider range of assets, including equities and bonds. When the issuer itself issues the tokens, intermediary risks can be reduced or even eliminated, improving access particularly for smaller investors.

Interoperability is the condition. The strength of the effect depends on whether ledgers are interoperable. Multiple non-interoperable ledgers split users into segmented pools, so liquidity can decline, and fragmentation raises transaction costs and search frictions, weakening the network benefits. The risk is greater when private entities set up ledgers without coordination or without regulatory requirements for interoperability, which lowers sharedness and market efficiency. Fragmentation is a network-externality problem: it reduces liquidity but does not amplify shock transmission (SchweserNotes Module Quiz 106.2).

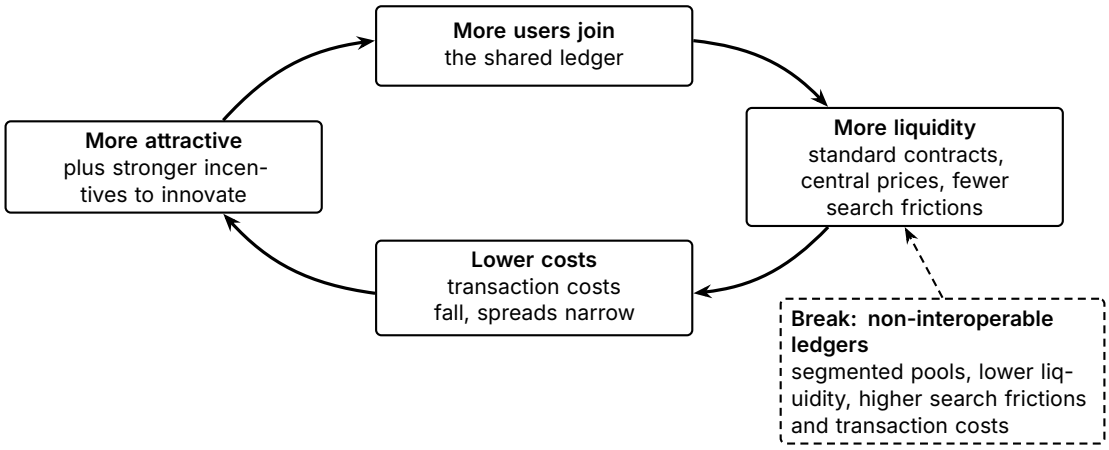


Figure 2. The network-externality cycle (solid arrows) and what breaks it (dashed box and dashed arrow). Each turn of the cycle makes the ledger more valuable to the next user; fragmentation across non-interoperable ledgers attacks the liquidity step, and the whole cycle weakens with it.

5. Knowledge externalities

Positive knowledge externalities arise when innovations or new knowledge benefit market participants beyond the original creator, and they grow as the number of platform users increases. Tokenization supports open code development, so innovators can build on existing smart contracts and applications at low or no cost, creating wider benefits for users.

Side	Effect (class notes Table CI-7.2, with SchweserNotes detail)
Positive	Innovators build on existing smart contracts at low or no cost, accelerating progress for the whole community; the effect grows with the number of users
Negative	Because code can be quickly copied and improved by others, open code lowers the incentive for significant R&D; incentives may hold up only if the shared user base is large enough. Licensing and patenting can protect innovations but limit the benefits of open-source development

The class notes make the key point: one property, code on a shared ledger that is visible and copyable, produces both rows. Faster diffusion and a weaker incentive to fund the original research are one mechanism seen from the copier's side and from the inventor's side.

Results card: CI-7 c

Shock transmission	Speed; cheaper debt, so leverage; deposits to wholesale funding; linked assets and rehypothecation; automated trading
Infrastructure	Underinvestment hits all participants; single point of failure; concentration risk
Network	More users, more liquidity; fractional shares on more assets; may stay below the social optimum
Fragmentation	Segmented pools, higher search frictions and costs; not a shock amplifier
Knowledge	Cheap building on open code, weaker R&D incentive; patents protect but limit open source

Exam traps

- **Leverage channel misread.** It is cheaper *debt* issuance that makes borrowing attractive and raises leverage.
- **Deposit direction flipped.** Retail deposits *fall*; banks lean on wholesale funding, which is the funding that dries up in stress.
- **Speed as pure benefit.** Faster execution helps settlement (CI-7 b) but also spreads shocks and shortens the time policymakers have.
- **Infrastructure risk diluted.** The impact of failures *grows* with users and linked transactions.
- **Fragmentation as shock amplifier.** Fragmentation weakens network effects and raises search frictions; it is not a shock-transmission channel.
- **Positive means optimal.** Network effects are positive, yet participation can stay below the social optimum.
- **Fractional shares are new.** Some intermediaries already offer them for certain equities; tokenization widens the range.
- **Immutable means uncopyable.** Open code can be copied and improved; that is what weakens R&D incentives.

Practice questions: CI-7 c

Question CI-7 c.1 New • built from SchweserNotes Book 5, LO 106.c; CI notes CI-7 c (correction box)

A financial stability analyst at the European Central Bank is assessing how widespread tokenized bond issuance could affect systemic risk. Which chain of effects is most consistent with the reading?

A. Debt becomes cheaper to issue, firms replace bank loans with bonds, and bank leverage falls, which reduces the risk of contagion.

B. Debt becomes cheaper to issue than conventional bonds, borrowing grows more attractive, and higher leverage can raise instability.

C. Debt becomes costlier to issue because of security spending, borrowing falls, and lower leverage slows the spread of shocks.

D. Equity becomes cheaper to issue relative to debt, firms shift toward equity funding, and lower leverage dampens shock transmission.

Question CI-7 c.2 Adapted from CI QB VI Q122 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

The treasurer of a Spanish retail bank asks how the spread of tokenization could change the bank's funding. Which outcome is most consistent with the reading?

A. Retail deposits rise as tokenized deposits become more attractive, so the bank can rely less on wholesale funding.

B. Retail deposits fall, but the bank becomes more stable, because wholesale funding is less prone than deposits to run in stress.

C. Retail deposits are unaffected, because tokenization changes how securities settle rather than where households keep funds.

D. Retail deposits fall as funds move into financial assets, so the bank leans on wholesale funding that can dry up in stress.

Question CI-7 c.3 Adapted from CI QB VI Q123 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

A supervisor at a securities regulator finds that investors on a shared, programmable ledger hold complex products linked to other assets and to assets pegged to different currencies, with the same collateral reused across several transactions. Which concern does the reading attach to this setting?

A. Rehypothecation and quick, cheap movement of funds across linked assets mean a fall in one asset can set off rapid selloffs in several markets.

B. Fractionalisation of the linked products concentrates retail portfolios in single assets, so that losses remain with the investors who bought them.

C. Reusing collateral across transactions is escrow certification, so each product is backed by an original asset locked on a separate ledger.

D. Linked products slow the transmission of shocks, because losses on one asset are offset by gains on the assets to which it is pegged.

Question CI-7 c.4 Adapted from CI QB VI Q124 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

A market surveillance officer at an Australian regulator reviews an episode in which prices on a tokenized market fell sharply within a few minutes. Which explanation and implication are most consistent with the reading?

A. An operational failure at the dominant ledger's single point of failure halted trading, so the fall reflects underinvestment.

B. Smart contracts pause automatically in volatile markets, so the fall must reflect human order-entry errors rather than automated trading.

C. A chain of self-executing smart contracts set off large volumes of trades, and faster execution makes such shocks harder to contain.

D. Fragmentation across non-interoperable ledgers caused the fall, and merging the ledgers would let liquidity flow between the pools.

Question CI-7 c.5 Adapted from CI QB VI Q125 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

A board member of the private company that operates a dominant tokenized ledger argues that spending on operational security is "our own business decision, with no consequences for anyone else". Which response best reflects the reading?

A. The board member is right: the impact of a failure falls as more users join, because a larger ledger spreads losses more thinly.

B. The board member is wrong: underinvestment creates risks for every participant, and one point of failure could disrupt the market.

C. The board member is right: automation removes human and software errors, so the only remaining risk is fraud borne by the operator.

D. The board member is wrong: security spending is a knowledge externality, because rival ledgers can copy the operator's code for free.

Question CI-7 c.6 Adapted from CI QB VI Q119 • SchweserNotes Book 5, LO 106.b and 106.c; CI notes CI-7 b, c

A junior economist at a central bank writes that because network effects on a tokenized ledger are a positive externality, participation will reach the socially optimal level without any policy action. Which correction is most accurate?

A. Even positive externalities can be inefficient: liquidity improves as users join, yet participation may stay below the social optimum.

B. Network effects on tokenized ledgers are negative, because each new user adds congestion, so participation tends to exceed the optimum.

C. Network effects are an internality, because each user's decision to join affects only that user's own liquidity, so no optimum applies.

D. The economist is right, because each new user raises liquidity for all users, which gives every potential user a private reason to join.

Question CI-7 c.7 Adapted from CI QB VI Q129 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

Several banks in one region each launch their own tokenized bond ledger, without coordination and with no regulatory requirement for interoperability. A regional regulator's economist is asked for the most likely consequence. Which outcome is most consistent with the reading?

A. Shock transmission is amplified, because fragmented ledgers tie institutions together more tightly and speed up the spread of losses.

B. Programmability falls on each ledger, because smart contracts can run only where one ledger serves the whole of the market.

C. Liquidity improves, because competition among the ledgers deepens each pool and narrows bid-ask spreads on every platform.

D. Liquidity splits into segmented pools, search frictions and transaction costs rise, and network externalities weaken.

Question CI-7 c.8 Adapted from CI QB VI Q128 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

The head of a retail wealth platform tells clients that tokenization will, for the first time, allow small investors to buy fractional shares. Which statement is the most accurate correction?

A. On tokenized ledgers, fractional ownership is still limited to equities, so the gain for small investors is lower fees, not wider access.

B. Fractionalisation raises intermediary risk for small investors, because tokens issued by the issuer add a layer between investor and asset.

C. Some intermediaries already offer fractional shares of certain equities; tokenization extends this to more assets, including bonds.

D. The claim is right, but fractionalisation leaves small investors less diversified, which is the internality that the reading warns about.

Question CI-7 c.9 Adapted from CI QB VI Q130 • SchweserNotes Book 5, LO 106.c; CI notes CI-7 c

A venture investor is considering a start-up that builds smart-contract applications on an open-code tokenized ledger, and asks how the reading views knowledge externalities. Which statement is most consistent with the reading?

A. Knowledge externalities shrink as the ledger's user base grows, because more users mean more competitors copying each application.

B. Open code lets others build on existing smart contracts at low or no cost, but easy copying weakens the incentive to fund R&D.

C. Smart contracts are immutable and so cannot be copied, which means open code strengthens the incentive to invest in R&D.

D. Licensing and patenting widen the benefits of open-source development, because protected code can be shared more freely.

Solutions • CI-7 c

Solution CI-7 c.1 — Answer: B

New • built from SchweserNotes Book 5, LO 106.c

Step 1 – The cost effect

Tokenization can lower the issuance cost of debt compared with conventional bonds.

Step 2 – The behavioural effect

Cheaper debt makes borrowing more attractive and can encourage higher leverage.

Step 3 – The stability effect

In a highly interconnected system, higher leverage raises the risk of instability if institutions take on excessive debt.

$\Rightarrow$ Cheaper debt, more borrowing, more leverage, more fragility.

Interpretation

This is the channel the SchweserNotes module quiz names first for a highly interconnected market. Leverage was already a traditional amplifier: highly leveraged institutions keep the upside and shift losses onto others. Tokenization feeds it by making debt cheaper.

Why the other options are wrong

A is incorrect. Invented chain with a polarity flip: the reading says cheaper debt encourages *higher* leverage, not lower.

C is incorrect. Polarity flip: tokenization lowers the issuance cost of debt. Security spending is the infrastructure externality, a separate channel.

D is incorrect. Misreads the channel (see the correction box): the reading's mechanism is cheaper *debt* issuance, which raises leverage.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025), Tokenization and Financial Market Inefficiencies; SchweserNotes 2026 Book 5, Reading 106, LO 106.c and Module Quiz 106.2

Solution CI-7 c.2 — Answer: D

Adapted from CI QB VI Q122 • SchweserNotes Book 5, LO 106.c

Step 1 – Where retail money goes

Tokenization makes it easier for retail investors to place funds in financial assets outside traditional bank deposits, so banks may see retail deposits decline.

Step 2 – What replaces it

Banks become more reliant on wholesale funding.

Step 3 – Why that matters

Wholesale funding can dry up quickly in market stress, raising the risk of defaults and spillovers across the system.

$\Rightarrow$ Deposits out, wholesale in, fragility up.

Interpretation

This channel works through funding liquidity, one of the three traditional amplifiers (Figure 1). It is a bank funding effect, distinct from the broker effects of CI-7 d.

Why the other options are wrong

A is incorrect. Polarity flip: the reading has deposits declining; "tokenized deposits" attracting funds is not its story.

B is incorrect. Polarity flip on stability: wholesale funding is the funding that can dry up quickly in stress.
C is incorrect. Scope error: the reading explicitly links tokenization to retail funds moving out of bank deposits.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c

Solution CI-7 c.3 — Answer: A

Adapted from CI QB VI Q123 • SchweserNotes Book 5, LO 106.c

Step 1 – Name the practice

Collateral reused across multiple transactions is rehypothecation, a feature of the complex products that shared, programmable ledgers make easier to reach.

Step 2 – Name the risk

As more investors move funds quickly and cheaply across linked assets and markets, shocks spread more rapidly: a sudden drop in one asset could trigger rapid selloffs in several markets.
⇒ The same interconnectedness that improves access speeds up contagion, and makes it harder for regulators to respond.

Interpretation

This channel works through interconnectedness. Note the combination the reading stresses: linked assets, reused collateral, and fast, cheap movement of funds.

Why the other options are wrong

B is incorrect. Sibling swap: fractionalisation helps retail investors diversify (a network-externality point); it does not describe linked products or reused collateral.
C is incorrect. Sibling swap: escrow certificates belong to the common ledger model (CI-7 e); reusing collateral is rehypothecation.
D is incorrect. Polarity flip: linked assets let shocks spread more rapidly; the reading describes no offsetting effect.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c

Solution CI-7 c.4 — Answer: C

Adapted from CI QB VI Q124 • SchweserNotes Book 5, LO 106.c

Step 1 – The mechanism

Programmable ledgers allow automated trading at much faster speeds. Smart contracts using rules-based algorithms can execute trades instantly or nearly instantly and bundle complex transactions. A chain of self-executing smart contracts that triggers large volumes of trades can contribute to a flash crash.

Step 2 – The implication

Faster transaction speeds make it more difficult for policymakers to contain the costs of such externalities.
⇒ Automation explains the speed of the fall, and speed explains why containment is harder.

Interpretation

Programmability is a benefit in CI-7 b and a risk here: same feature, opposite sign.

Why the other options are wrong

A is incorrect. Sibling swap: a single point of failure is the infrastructure externality; the reading's account of a sharp, fast fall is a chain of self-executing smart contracts.

B is incorrect. Invented property: nothing in the reading says smart contracts pause in volatility; they execute as coded, and chained execution is the reading's mechanism.

D is incorrect. Sibling swap: fragmentation is a network-externality problem that reduces liquidity; it is not the reading's account of rapid price falls.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c

Solution CI-7 c.5 — Answer: B

Adapted from CI QB VI Q125 • SchweserNotes Book 5, LO 106.c

Step 1 – Who bears the risk

If private operators underinvest in operational security, the resulting risks can affect all participants. A single point of failure in a dominant ledger could disrupt the entire market.

Step 2 – Why it grows with scale

The potential impact of failures grows with the number of users and interconnected transactions, so operators must spend more to mitigate operational failures and cyberattacks.

⇒ The operator decides the spending, but everyone bears the failure: an externality.

Interpretation

Concentration adds to it: a widely used ledger becomes a target for cyberattacks, with potential disruption across the financial system.

Why the other options are wrong

A is incorrect. Polarity flip: the impact of failures grows, not shrinks, as users and linked transactions increase.

C is incorrect. Scope error: the reading lists human and software errors, fraud and system failures, and the risks fall on all participants.

D is incorrect. Sibling swap: knowledge externalities concern innovation spread through open code, not the spillover from security underinvestment.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c

Solution CI-7 c.6 — Answer: A

Adapted from CI QB VI Q119 • SchweserNotes Book 5, LO 106.b and 106.c

Step 1 – The definition

Externalities can be positive or negative, but even positive externalities can lead to socially inefficient outcomes.

Step 2 – The network case

Network effects are the reading's example: liquidity improves as more participants join, yet the

number of participants may still remain below the socially optimal level.
⇒ Positive is not the same as optimal.

Interpretation

The benefit a new user brings to others does not enter that user's own decision, which is why participation can fall short.

Why the other options are wrong

B is incorrect. Polarity flip: network effects are the reading's example of a *positive* externality.
C is incorrect. Sibling swap: an internality is a participant acting against its own interest; network effects spill over to others.
D is incorrect. Keeps the economist's error: a private reason to join exists, but the reading says participation may still fall short of the social optimum.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.b and 106.c

Solution CI-7 c.7 — Answer: D

Adapted from CI QB VI Q129 • SchweserNotes Book 5, LO 106.c (Module Quiz 106.2 shape)

Step 1 – The condition

The strength of network externalities depends on the interoperability of ledgers. Multiple non-interoperable ledgers leave users with access only to segmented pools of assets.

Step 2 – The consequences

Liquidity can decline, and fragmentation raises transaction costs and search frictions, weakening the benefits of network effects. The risk is greater when private entities set up ledgers without coordination or without regulatory requirements for interoperability.
⇒ Fragmentation lowers sharedness and market efficiency.

Interpretation

Fragmentation attacks the liquidity step of the network cycle (Figure 2). SchweserNotes adds that fragmentation does not amplify shock transmission and is unrelated to programmability.

Why the other options are wrong

A is incorrect. Sibling swap: shock amplification comes from speed, leverage, deposit migration, linked assets and automated trading. Fragmentation reduces liquidity; it is not a transmission channel.
B is incorrect. Sibling swap: programmability depends on ledger design, not on interoperability.
C is incorrect. Polarity flip: segmentation reduces liquidity and raises costs.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c and Module Quiz 106.2

Solution CI-7 c.8 — Answer: C

Adapted from CI QB VI Q128 • SchweserNotes Book 5, LO 106.c

Step 1 – What already exists

Some intermediaries already offer fractional shares of certain equities, so fractional ownership is not new.

Step 2 – What tokenization adds

Tokenization enables fractionalisation across a wider range of assets, including equities and bonds. When the issuer issues the tokens, intermediary risks can be reduced or even eliminated, improving access particularly for smaller investors.

⇒ The novelty is breadth, not existence.

Interpretation

Fractionalisation sits under network externalities because wider participation feeds liquidity. The retail downside (internalities) is CI-7 d.

Why the other options are wrong

A is incorrect. Scope error: tokenization widens fractionalisation beyond equities, to bonds among others.

B is incorrect. Polarity flip: tokens issued directly by the issuer can reduce or eliminate intermediary risks.

D is incorrect. Two errors: fractional shares are not new, and fractionalisation helps investors diversify. Insufficient diversification is the reading's consequence of bypassing advisers (CI-7 d), not of fractionalisation.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c and 106.d

Solution CI-7 c.9 **Answer: B**

Adapted from CI QB VI Q130 • SchweserNotes Book 5, LO 106.c

Step 1 – The positive side

Because tokenization supports open code development, innovators can build on existing smart contracts and applications at low or no cost, creating wider benefits for users. These knowledge externalities grow as the number of platform users increases.

Step 2 – The negative side

Open code access lowers the incentive for significant R&D, because code can be quickly copied and improved by others. Incentives may hold up only if the shared user base is large enough.

⇒ One property, copyable code, produces both effects.

Interpretation

The start-up gains from building cheaply on others' code, and others gain from its code. Patents protect, at the cost of open-source benefits.

Why the other options are wrong

A is incorrect. Polarity flip: knowledge externalities grow as the user base grows.

C is incorrect. Misuses immutability: deployed smart contracts are immutable, but open code can still be copied and improved, which is the source of the weaker incentive.

D is incorrect. Polarity flip: licensing and patenting protect innovations but *limit* open-source benefits.

LO: CI-7 c: Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c

CI-7 • Tokenization and Financial Market Inefficiencies (IMF, January 2025) • Vardeez: VERY IMPORTANT

CI-7 d

P2 CONCEPT

Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading Very Important; the workbook covers it in D29 Crypto, Tokenisation and Digital Resilience (Tier D, Map #68, notes only). There are no workbook questions for this reading, so every question here is new or adapted from the bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S2** CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures. This LO is the "retail and broker effects" part: internalities from direct trading, broker market power, and the new market power of privately owned ledgers (every question here).

What the questions look like: no calculations. Naming the inefficiency behind a scenario (internality or market power, and not an externality), "which assessment is most consistent" for a broker or a regulator, and the two-turn argument: ledgers erode brokers' market power, then a dominant private ledger can rebuild it.

1. Two groups, two inefficiencies

Distributed ledger technology (DLT), including shared ledgers that support tokenized assets, changes how assets are traded, transferred and serviced. The reading follows its effect on two groups, each through one of the inefficiencies defined in CI-7 b.

- **Retail investors: internalities.** Participants taking actions that are not in their own best interest.
- **Brokerage firms and ledger owners: market power.** The ability to influence prices or terms. A market power inefficiency occurs when a dominant entity can charge above-market prices or fees because competition is limited.

2. Retail investors

The internality. Tokenized ledgers that let retail investors trade directly can *amplify* internalities, particularly when investors make decisions that are not in their own best interest.

- **How it arises.** Investors trade securities whose risks they do not fully understand, especially when they bypass financial advisers to avoid fees.
- **What it leads to.** Insufficient diversification, and exposure to large and sudden losses from risky or complex assets and derivatives.
- **When it is worst.** Especially in the absence of strong consumer financial protections.

The benefit side, from CI-7 c. Fractionalisation lets retail investors buy fractional shares across a wider range of assets and diversify more easily, and tokens issued directly by the issuer can reduce or even eliminate intermediary risks, improving market access particularly for smaller investors.

For retail investors	What improves	What can get worse
Access	Direct trading; tokens issued by the issuer cut intermediary risk	Direct trading in complex assets and derivatives whose risks are not understood
Cost	Advisory fees can be avoided; broker fees come under pressure	The adviser who screened the risk goes with the fee
Diversification	Fractional shares across more asset classes make it easier	Investors who bypass advisers may end up insufficiently diversified
Protection		Large, sudden losses, worst where consumer financial protections are weak

The class notes sum it up: the benefit is cheaper direct access, and the cost is that the adviser who was screening the risk has been removed along with the fee. Note that the harm falls on the investor who makes the decision, which is what makes it an internality rather than an externality.

3. Brokerage firms: market power

Why traditional brokers can keep fees high. Brokers charge fees to cover costs and generate profit, and retail investors are often discouraged from switching because the process is time-consuming and cumbersome. Transferring assets *in kind*, meaning moving the securities themselves rather than selling them and transferring cash, may require going through a clearinghouse in the United States, which adds delays and complexity. Brokers can therefore keep charging higher fees without losing clients.

What a digital ledger changes.

1. **Instant switching.** Changing brokers and transferring assets can be instantaneous, because transactions and ownership records are immediately verifiable.
2. **No clearinghouse.** The same features remove the need for a clearinghouse, which reduces costs.
3. **Competition and scale.** Further cost reductions may come if brokers lower their fees under greater competitive pressure, or if increased market participation lets them achieve economies of scale.
4. **Direct trading.** If retail investors can buy assets directly from issuers, brokers face additional pressure to lower fees. However, in most countries trading is still required to be conducted through licensed brokers.

4. Privately owned ledgers: market power in a new place

Privately owned ledgers can create market power inefficiencies of their own.

- **Monopoly rents.** Private owners may charge high fees, and the fees can rise as the ledger becomes more widely shared and dominant.
- **Privileged data.** Owners can monetise privileged data by offering credit, selling the data, or charging higher fees to the most active traders.
- **Free, then fees.** A private owner might provide free access when the ledger is new, to attract users, and start charging once the ledger is popular and holds a strong market position.

5. Reading the argument: three rules

1. **Name the inefficiency by who bears it.** A retail investor hurting herself through a trade she does not understand is an internality. A dominant entity charging above-market fees is market power. Neither is an externality.
2. **Two turns, not one.** Ledgers remove the switching barrier that protected brokers; a dominant privately owned ledger can rebuild the barrier at the ledger (Figure 1).
3. **Pressure is not elimination.** Direct trading and instant switching push broker fees down, but most countries still require trading through licensed brokers, so brokers do not simply disappear.

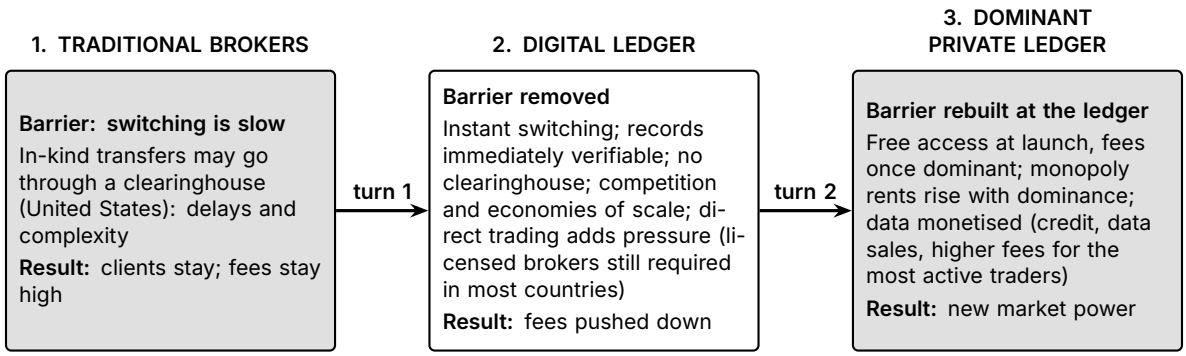


Figure 1. "The argument turns around twice" (class notes). Shaded boxes are the two states in which market power is high; the plain box is the state in which it is eroded. A statement that stops at the middle box, "DLT reduces market power", is only half of what the reading says.

Results card: CI-7 d

Retail internalities	Direct trading; risks not understood; advisers bypassed to avoid fees; under-diversification; large, sudden losses; worst without strong consumer protection
Retail benefits (CI-7 c)	Fractional shares across more assets; issuer-issued tokens cut intermediary risk
Broker market power	Switching slow and cumbersome; in-kind transfer via a clearinghouse in the United States
Ledger effect on brokers	Instant switching, records verifiable, no clearinghouse; competition; economies of scale; direct trading pressure
Limit	Most countries still require trading through licensed brokers
Private ledgers	Monopoly rents rising with dominance; data monetised (credit, data sales, higher fees for active traders); free then fees

Exam traps

- **Internality called an externality.** The retail loss falls on the investor who made the decision.
- **Verification mistaken for protection.** An immediately verifiable record confirms the trade, not the wisdom of it.
- **One-sided diversification.** Fractional shares make diversifying easier; bypassing advisers can leave investors under-diversified. Both are in the reading.

- **In-kind transfers read as instant.** Traditionally they may need a clearinghouse; instant transfer is the ledger outcome.
- **Brokers abolished.** Licensed brokers are still required in most countries.
- **Cause confused with conduct.** Network effects build a ledger's dominance; charging monopoly rents afterwards is market power.
- **Rents falling with scale.** Fees on a dominant private ledger can *rise* as it becomes more widely shared.
- **Half the argument.** "DLT reduces market power" stops after the first turn.

Practice questions: CI-7 d

Question CI-7 d.1 Adapted from CI QB VI Q137 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A consumer protection official at a European securities regulator is reviewing a tokenized platform that lets retail investors trade complex assets and derivatives directly, without an adviser. Which assessment is most consistent with the reading?

A. The main risk is a negative externality, because retail losses on the platform spill over to the other investors who trade the same assets.

B. The risk is small, because the ledger verifies each trade at once and so stops investors from entering trades whose risks they do not understand.

C. Direct access can amplify internalities, and the risk of large, sudden losses is greatest where consumer financial protections are weak.

D. The main risk is market power, because retail investors who bypass advisers are charged higher fees by the platform for complex products.

Question CI-7 d.2 New • built from SchweserNotes Book 5, LO 106.c and 106.d; CI notes CI-7 d

A research analyst at an Indian wealth manager is asked whether tokenized ledgers will leave retail investors better or worse diversified. Which answer best reflects the reading?

A. Either: fractional shares ease diversifying across more assets, but investors who bypass advisers to save fees may end up under-diversified.

B. Better diversified, because fractional shares across more assets let small investors spread funds widely even without an adviser.

C. Worse diversified, because direct trading pushes investors into complex products, and fractional shares stay limited to equities.

D. Neither: diversification depends only on which assets exist, and tokenization changes how assets settle rather than which ones investors can reach.

Question CI-7 d.3 Adapted from CI QB VI Q132 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A retail investor in the United States wants to move her portfolio to a cheaper broker without selling the securities. A financial journalist uses her case to explain why brokers can keep fees high. Which explanation is most consistent with the reading?

A. Brokers keep fees high because regulators set minimum commissions, so moving to a cheaper broker changes little for the investor.

B. In-kind transfers are not allowed between brokers, so she must sell the securities and move the cash, and that is what deters switching.

C. In-kind transfers settle instantly between brokers, so fees reflect the value of the broker's service rather than a barrier to switching.

D. Moving the securities themselves may require going through a clearinghouse, which adds delays and complexity and discourages switching.

Question CI-7 d.4 Adapted from CI QB VI Q133 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

The chief executive of a Toronto online broker asks how the spread of distributed ledgers is likely to affect the fees the firm can charge. Which assessment is most consistent with the reading?

A. Fees can rise, because verifying ownership on a ledger costs more than clearing through a clearinghouse, and clients bear that cost.

B. Fees face pressure: switching can be instant, no clearinghouse is needed, competition grows, and scale economies may follow.

C. Fees are unaffected, because switching brokers on a ledger still requires every transfer to be reconciled through a clearinghouse.

D. Fees rise as participation grows, because brokers lose economies of scale once clients can switch between them instantly.

Question CI-7 d.5 Adapted from CI QB VI Q136 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A fintech founder pitches a platform on which retail investors buy tokenized assets directly from issuers, and tells prospective investors that "brokers will soon be unnecessary everywhere". Which response best reflects the reading?

A. Direct trading would push brokers to lower fees, but in most countries trading must still be conducted through licensed brokers.

B. Direct trading would have no effect on broker fees, because issuers selling directly would simply charge the same fees as brokers.

C. Direct trading would end the internality problem, because retail investors who buy from issuers receive the issuer's data in the token.

D. Direct trading is barred in most countries, because issuers are not permitted to create tokens that represent their own securities.

Question CI-7 d.6 Adapted from CI QB VI Q139 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A privately owned ledger offered free access at launch and now dominates trading in tokenized bonds. An analyst at a competition authority is asked what the reading implies for the fees users will pay from here. Which assessment is most consistent with the reading?

A. Fees stay low, because each new participant adds liquidity that lowers the owner's costs, a positive network externality.

B. Fees fall as the ledger grows, because the owner gains economies of scale and competitive pressure pushes it to pass the savings on.

C. Fees can rise as the ledger grows more widely shared and dominant: the owner extracts monopoly rents, a market power inefficiency.

D. Fees rise only to fund security spending, because underinvestment on a dominant ledger would expose every participant.

Question CI-7 d.7 Adapted from CI QB VI Q135 • SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A competition economist is reviewing how the owner of a dominant privately owned ledger uses the trading data it collects from users. Which set of practices does the reading identify as a source of market power inefficiency?

A. Publishing central price information to all users, standardising contracts and narrowing bid-ask spreads across the ledger.

B. Offering credit based on the data, selling the data, and charging higher fees to the most active traders on the ledger.

C. Paying rebates to the most active traders, sharing data with regulators, and lowering fees as the ledger keeps growing.

D. Embedding term sheets and ESG data in tokens, so that investors can see issuer information directly on the ledger.

Question CI-7 d.8 New • built from SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

A brokerage strategist writes: "Distributed ledgers remove the switching barriers that let brokers charge high fees, so market power inefficiencies in retail investing will disappear." Which response best reflects the reading?

A. The statement is correct, because once ownership records are immediately verifiable no entity can charge above-market fees.

B. The statement is wrong, because instant switching covers only cash, so in-kind transfers still need a clearinghouse on a ledger.

C. The statement is correct, because ledger owners compete for users and so pass on the savings from removing the clearinghouse.

D. The statement is half right: ledgers can erode brokers' market power, but a dominant private ledger can create new market power.

Solutions • CI-7 d

Solution CI-7 d.1

Answer: C

Adapted from CI QB VI Q137 • SchweserNotes Book 5, LO 106.d

Step 1 – Name the inefficiency

Tokenized ledgers that let retail investors trade directly can amplify *internalities*: decisions that are not in the investor's own best interest, such as trading securities whose risks they do not fully understand, especially after bypassing advisers to avoid fees.

Step 2 – The consequences and the condition

Insufficient diversification and exposure to large, sudden losses from risky or complex assets and derivatives, especially in the absence of strong consumer financial protections.
$\Rightarrow$ The harm falls on the investor, and weak protections make it worse.

Interpretation

For the official, the strength of consumer financial protections is the condition that decides how large losses get.

Why the other options are wrong

A is incorrect. Sibling swap: the loss falls on the investor who made the decision, which is an internality, not a spillover to others.
B is incorrect. Confuses verification with protection: an immediately verifiable record confirms the trade, not that the investor understood its risks.
D is incorrect. Sibling swap: market power is about a dominant entity charging above-market fees; the reading's retail concern here is the investor's own decisions.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025), Tokenization and Financial Market Inefficiencies; SchweserNotes 2026 Book 5, Reading 106, LO 106.d

Solution CI-7 d.2

Answer: A

New • built from SchweserNotes Book 5, LO 106.c and 106.d

Step 1 – The diversification benefit

Tokenized ledgers support fractionalisation, which lets retail investors buy fractional shares and diversify more easily, across a wider range of assets than intermediaries offer today (CI-7 c).

Step 2 – The diversification risk

Direct trading can amplify internalities: investors who bypass advisers to avoid fees may trade risks they do not understand, which can result in insufficient diversification (CI-7 d).
$\Rightarrow$ The reading supports both directions; which one dominates depends on how investors use the access.

Interpretation

The class notes' summary fits: cheaper direct access is the benefit, and losing the adviser who screened the risk is the cost.

Why the other options are wrong

B is incorrect. One-sided: fractionalisation helps, but the reading warns that investors who bypass advisers may end up insufficiently diversified.
C is incorrect. One-sided and a scope error: bypassing advisers can lead to complex, poorly understood trades, but tokenization extends fractional shares beyond equities, to bonds among others.

D is incorrect. Scope error: the reading links tokenization to which assets retail investors can reach, through fractionalisation and direct trading.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.c and 106.d

Solution CI-7 d.3 — Answer: D

Adapted from CI QB VI Q132 • SchweserNotes Book 5, LO 106.d

Step 1 – What an in-kind transfer is

Moving the securities themselves rather than selling them and moving cash.

Step 2 – Why it is a barrier

In the United States an in-kind transfer may require going through a clearinghouse, which adds delays and complexity. Switching is time-consuming and cumbersome, so investors are discouraged from moving.

$\Rightarrow$ Brokers can keep charging higher fees without losing clients.

Interpretation

This is the barrier the ledger removes in the first turn of the argument (Figure 1).

Why the other options are wrong

A is incorrect. Invented mechanism: the reading's explanation is switching friction, not regulated minimum commissions.

B is incorrect. Scope error: in-kind transfers are possible; the problem is that they may need a clearinghouse and are slow and cumbersome.

C is incorrect. Polarity flip: in-kind transfers are slow in traditional markets; instant transfer is what a digital ledger offers.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d

Solution CI-7 d.4 — Answer: B

Adapted from CI QB VI Q133 • SchweserNotes Book 5, LO 106.d

Step 1 – Switching becomes easy

On a digital ledger, changing brokers and transferring assets can be instantaneous, because transactions and ownership records are immediately verifiable. These features also remove the need for a clearinghouse, which reduces costs.

Step 2 – Further sources of lower fees

Brokers may lower fees under greater competitive pressure, and increased market participation may let them achieve economies of scale.

$\Rightarrow$ The broker's fee-setting power weakens from several directions at once.

Interpretation

Direct trading would add more pressure, limited by the licensed-broker requirement (CI-7 d.5).

Why the other options are wrong

A is incorrect. Polarity flip: the ledger removes the clearinghouse and reduces costs.

C is incorrect. Polarity flip: the ledger removes the clearinghouse step.

D is incorrect. Polarity flip: wider participation may give brokers economies of scale.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d

Solution CI-7 d.5 — Answer: A

Adapted from CI QB VI Q136 • SchweserNotes Book 5, LO 106.d

Step 1 – The economic effect

Direct trading by retail investors, buying assets directly from issuers, would put additional pressure on brokers to lower their fees.

Step 2 – The legal limit

In most countries, trading is still required to be conducted through licensed brokers.
⇒ Pressure on fees, yes; brokers unnecessary everywhere, no.

Interpretation

The founder's claim fails on regulation, not on economics: the reading accepts the fee pressure but notes the licensed-broker requirement.

Why the other options are wrong

B is incorrect. Polarity flip: the reading says direct trading would add pressure on broker fees.
C is incorrect. Polarity flip: direct access can amplify internalities; embedded data improves transparency (CI-7 b) but does not end self-harming decisions.
D is incorrect. Invented prohibition: the limit is that trading must go through licensed brokers, not a ban on issuers tokenizing their securities.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d

Solution CI-7 d.6 — Answer: C

Adapted from CI QB VI Q139 • SchweserNotes Book 5, LO 106.d (Module Quiz 106.2 shape)

Step 1 – The pattern

A private owner may offer free access while the ledger is new to attract users, then begin charging once the ledger is popular and holds a strong market position.

Step 2 – The direction of fees

Private owners may charge high fees, known as monopoly rents, with fees rising as the ledger becomes more widely shared and dominant: a market power inefficiency.
⇒ Dominance first, then rents that grow with it.

Interpretation

Network effects help build the dominance; the fee-setting that follows is market power. Scale economies and competition are the reading's reasons for lower *broker* fees.

Why the other options are wrong

A is incorrect. Cause confused with conduct: network effects build the dominance; they do not keep the owner's fees low.
B is incorrect. Sibling swap: scale economies and competition explain lower broker fees; a dominant ledger's fees can rise.
D is incorrect. Invented justification: the reading's infrastructure externality concerns *underinvestment* in security; it does not tie the owner's fees to security spending.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d and Module Quiz 106.2

Solution CI-7 d.7 — Answer: B

Adapted from CI QB VI Q135 • SchweserNotes Book 5, LO 106.d

Step 1 – The source of the power

The owner of a privately owned ledger sees privileged data on its users' trading.

Step 2 – The practices

The reading lists three ways to monetise it: offering credit, selling the data, and charging higher fees to the most active traders.

⇒ Data held by a dominant owner is a second source of market power, alongside fees.

Interpretation

Data monetisation sits beside monopoly rents: both are market power of privately owned ledgers.

Why the other options are wrong

A is incorrect. Sibling swap: central price information, standard contracts and narrower spreads are network-externality benefits (CI-7 c).

C is incorrect. Polarity flip: the reading has the most active traders charged *more*, and fees on a dominant ledger can rise, not fall.

D is incorrect. Sibling swap: embedding term sheets and ESG data is an issuance benefit (CI-7 b), not a use of users' trading data.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d

Solution CI-7 d.8 — Answer: D

New • built from SchweserNotes Book 5, LO 106.d; CI notes CI-7 d

Step 1 – The first turn

On a digital ledger switching brokers can be instantaneous and needs no clearinghouse, which reduces costs and puts pressure on broker fees. The strategist has this part right.

Step 2 – The second turn

Privately owned ledgers can introduce new market power inefficiencies: high fees on dominant platforms and monetised privileged data.

⇒ Market power can move from the broker to the ledger owner rather than disappear.

Interpretation

The class notes' argument that turns around twice (Figure 1): stopping at "DLT reduces market power" is half the answer.

Why the other options are wrong

A is incorrect. Absolute claim: verifiable records weaken the brokers' switching barrier, but a dominant ledger owner can still charge above-market fees.

B is incorrect. Scope error: on a digital ledger, assets move between brokers instantly, with no clearinghouse.

C is incorrect. Scope error: privately owned ledgers can create new market power, so savings need not be passed on.

LO: CI-7 d: Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.d

CI-7 • Tokenization and Financial Market Inefficiencies (IMF, January 2025) • Vardeez: VERY IMPORTANT

CI-7 e

P2 CONCEPT

Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading Very Important; the workbook covers it in D29 Crypto, Tokenisation and Digital Resilience (Tier D, Map #68, notes only). There are no workbook questions for this reading, so every question here is new or adapted from the bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up."

Reported exam shapes

- **D29·S2** CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures. This LO is the "ledger models and trust failures" part (every question here). One of the D29 likely distractors comes from it: the attack on more than 50% of computing power belongs to permissionless proof-of-work ledgers, not to permissioned DLT (Questions CI-7 e.5 and e.6).

What the questions look like: no calculations. Matching a design to its model, and the model to the way its trust fails, in the SchweserNotes "most vulnerable to collusion" shape; comparing a single-operator system with a permissioned DLT; and keeping the two families of names apart (single, common and compatible *ledgers* for sharedness; single-operator, permissionless and permissioned *systems* for trust).

1. Three features, three toolkits

Recap from CI-7 a: tokenization works only on a ledger that is trusted, shared and programmable. This LO compares the methods available for each feature.

- **Sharedness** comes from one of three *tokenization models*: single ledger, common ledger or compatible ledgers. Each model also differs in how it establishes trust and programmability.
- **Programmability** comes from integrating code and database within the ledger.
- **Trust** comes from one of three designs: a single-operator system, permissionless DLT or permissioned DLT.

The two families of three are easy to cross. "Single ledger" is a sharedness model (everyone on one system); "single-operator system" is a trust model (one entity keeps the records). "Compatible ledgers" describes interoperability; it does not by itself determine how trust is established.

2. Sharedness: single, common and compatible ledgers

1. **Single ledger model.** All parties, or their intermediaries, are placed on one system, and all assets are created directly on that system. The class notes: high efficiency, but a centralised structure.
2. **Common ledger model.** The common ledger carries representations of non-native assets that exist on other ledgers. The original assets are placed in escrow and represented through *escrow certificates*. The class notes: the original is locked, and the tokenized version trades elsewhere.

3. **Compatible ledger model.** Different ledgers use standardised technology that lets them run the same programs and coordinate transfers, which makes them interoperable. The class notes: interoperability across systems without centralisation.

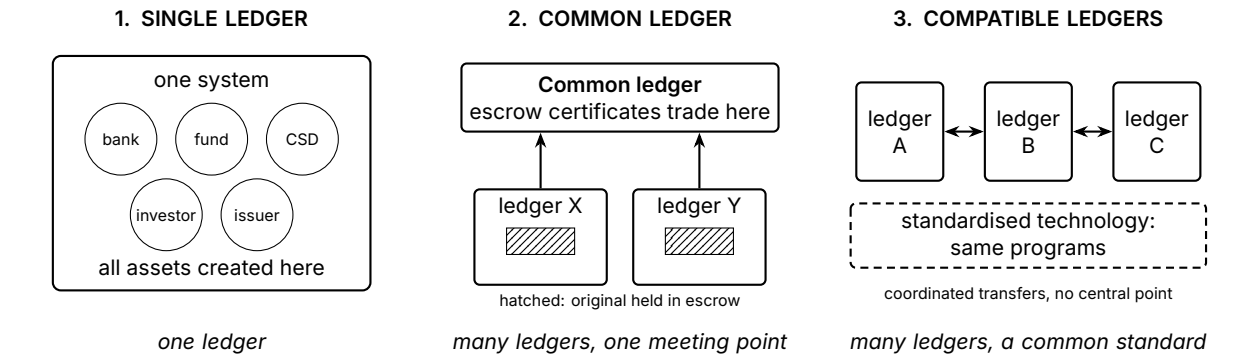


Figure 1. The three sharedness models (the class notes' Figure CI-7.3). Left: one system holds every party and every asset. Middle: originals stay on their own ledgers, held in escrow (hatched), and escrow certificates trade on the common ledger. Right: separate ledgers linked by a shared standard (dashed band) and coordinated transfers (double arrows), with no central point.

Model	Where the assets are	How sharedness is bought (class notes)
Single ledger	Created directly on the one system	With centralisation: high efficiency, centralised structure
Common ledger	Originals on their own ledgers, in escrow; certificates on the common ledger	By locking the original and trading a certificate; this adds a second asset, the certificate, and so a second thing that can fail
Compatible ledgers	On separate, interoperable ledgers	With a standard and no central point: same programs, coordinated transfers

Keep the common ledger apart from migration (CI-7 a): migration destroys the original, while the common ledger keeps it alive in escrow and trades a certificate.

3. Programmability: code and database in one place

Programmability in digital ledgers means that the code and the database are integrated within the system; in traditional set-ups, external users must send instructions to a separate database operator. Internal programmability ensures that transactions are coherent and efficient, and makes it easier to bundle or link smart contracts.



Figure 2. Where programmability lives. Left (dashed boxes): instructions pass through a separate operator. Right (thick boxes in one frame): code and records sit in the same system.

4. Trust: three designs

1. **Single-operator systems** are centralised: one entity manages the records and executes and confirms transactions. Trust rests on the operator's credibility, supervision and oversight.
2. **Permissionless DLT** relies on a decentralised network of nodes to verify and validate transactions, with no single party able to alter the ledger. Anyone can join, transact and take part in validation, and validators are typically rewarded.
3. **Permissioned DLT** combines elements of both. Only authorised participants, approved by a central entity or consortium, can access or validate transactions.

All three are designed to establish trust, and each offers different trade-offs in openness, security and control. The level of trust depends on the design and on the mechanisms in place to prevent abuse, operational failures and other technical problems.

5. How trust fails

Each design fails in its own way. Two terms first. *Front running* is positioning one's own trades ahead of others in the validation queue; a *sandwich attack* is a form of front running that places transactions immediately before and after a victim's transaction to generate a profit.

Why the single operator ranks worse than permissioned DLT. The risks of trust failure in a fully centralised single-operator system are similar to those in a permissioned DLT, but higher: trust does not rely on consensus among participants and depends entirely on the operator's credibility and the strength of legal protections.

Model	Trust rests on	How trust fails	What limits it
Single-operator system	One entity's credibility, supervision and oversight	The whole system depends on one entity: operational failure or misuse, with no consensus to spread trust. Risks similar to permissioned DLT but *even higher*	Only the operator's credibility and the strength of legal protections
Permissionless DLT	Consensus among an open network; validators rewarded	Attack on the consensus mechanism: in proof of work, control of more than 50% of the computing power used to validate. Manipulation of transaction fees to jump the validation queue: front running, including sandwich attacks. External information more exposed to cyberattacks without a central authority. Regulatory compliance harder to enforce	Many independent validators make collusion difficult
Permissioned DLT	Controlled access; participants certified by a central entity or consortium	As a hybrid, it can show the failures of either design. Authorised participants colluding to modify the ledger or its programming for their own benefit	Reputation (to earn future profits); regulatory oversight and penalties. The structure can improve security and simplify regulation

The class notes give the reading rule: read the "rests on" column first, because each failure is the failure of exactly that thing. The single operator rests on one entity, so the failure is that entity; permissionless DLT rests on consensus, so the failure is capturing the consensus; permissioned DLT rests on a closed group, so the failure is the group colluding.

Precision note on the source notes. The class notes' trust table lists the permissionless failure as a "51% attack (majority control of network)". The reading's condition is narrower: in a proof-of-work system, an attacker who gains control of more than 50% of the *computing power used to validate transactions* can compromise the consensus mechanism (SchweserNotes Book 5, LO 106.e). It is not a majority of tokens held, of nodes, or of consortium seats; the last describes the permissioned design, whose failure is collusion. Questions CI-7 e.5 and e.6 turn on this.

Results card: CI-7 e

Single ledger	All parties or their intermediaries on one system; all assets created on it; efficient but centralised
Common ledger	Non-native assets from other ledgers held in escrow, traded as escrow certificates
Compatible ledgers	Standardised technology: same programs, coordinated transfers, interoperable; no central point
Programmability	Code and database integrated, versus instructions sent to a separate operator; coherent, efficient, easy to bundle
Single-operator	One entity; trust from credibility, supervision, oversight; fails with that entity; risk higher than permissioned
Permissionless DLT	Anyone joins and validates, rewarded; fails by $> 50\%$ PoW computing power, fee manipulation (front running, sandwich), cyber exposure, weak compliance
Permissioned DLT	Authorised participants only; fails by collusion; checked by reputation and regulatory penalties

Exam traps

- **Axis confusion.** Single, common and compatible are sharedness models; single-operator, permissionless and permissioned are trust models.
- **Common ledger read as migration.** Escrow keeps the original alive; migration destroys it.
- **Compatible read as common.** Compatible ledgers share a standard; nothing is escrowed.
- **Programmability reversed.** Integrated code and database is the ledger; a separate operator is the traditional set-up.
- **Collusion misplaced.** Collusion is the permissioned failure; permissionless has many independent validators, and a single operator has no one to collude with.
- **Wrong 50%.** More than 50% of the *computing power* in proof of work, not of tokens or of consortium seats.
- **Single operator as the safe choice.** Its risks are similar to permissioned DLT but even higher.
- **Mitigants misassigned.** Reputation and regulatory penalties restrain permissioned participants; certification by a consortium is not a permissionless feature.

Practice questions: CI-7 e

Question CI-7 e.1 New • built from SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

A market infrastructure consultant is classifying three proposals for sharing tokenized assets: (1) all banks and their clients join one system on which every asset is created directly; (2) bonds held on several existing ledgers are placed in escrow and trade on a central ledger as escrow certificates; (3) existing ledgers adopt a standardised technology so that they run the same programs and coordinate transfers. Which classification is correct?

A. (1) single ledger; (2) common ledger; (3) compatible ledgers.

B. (1) single ledger; (2) compatible ledgers; (3) common ledger.

C. (1) common ledger; (2) single ledger; (3) compatible ledgers.

D. (1) single-operator system; (2) common ledger; (3) permissionless DLT.

Question CI-7 e.2 Adapted from CI QB VI Q142 • SchweserNotes Book 5, LO 106.a and 106.e; CI notes CI-7 e

A risk officer at an asset manager is asked to approve trading a tokenized bond on a common ledger. Which description of what the asset manager would hold is most accurate?

A. A native token reissued on the common ledger after the bond is destroyed on its home ledger, so nothing else needs to be relied on.

B. A copy of the bond that trades freely on both ledgers at once, so the holder can sell on whichever ledger offers the better price.

C. An escrow certificate for the bond, which stays in escrow on its home ledger, so the holder relies on that escrow too.

D. A bond on its home ledger that runs the same programs as the common ledger, so transfers are coordinated with no representation.

Question CI-7 e.3 Adapted from CI QB VI Q144 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

A systems architect at a Tokyo securities firm is contrasting programmability on a digital ledger with the firm's current set-up. Which contrast is most consistent with the reading?

A. On the ledger, users send instructions to a separate database operator; in the current set-up, code and data sit in one system.

B. On the ledger, code and database are kept apart for security, which makes smart contracts harder to bundle than in the current set-up.

C. On the ledger, programmability means all relevant parties can reach the same records; the current set-up keeps records apart.

D. On the ledger, code and database are integrated, so transactions are coherent and efficient and smart contracts are easy to bundle.

Question CI-7 e.4 Adapted from CI QB VI Q145 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

A governance consultant at a Swiss bank compares three ways of establishing trust on a ledger: (1) one entity manages the records and executes and confirms transactions; (2) anyone can join, transact and validate, and validators are rewarded; (3) only participants approved by a central entity or consortium can access or validate transactions. Which matching is correct?

A. (1) single ledger model; (2) permissionless DLT; (3) permissioned DLT.

B. (1) single-operator system; (2) permissionless DLT; (3) permissioned DLT.

C. (1) single-operator system; (2) permissioned DLT; (3) permissionless DLT.

D. (1) permissioned DLT; (2) permissionless DLT; (3) common ledger model.

Question CI-7 e.5 Adapted from CI QB VI Q148 • SchweserNotes Book 5, LO 106.e (Module Quiz 106.2 shape)

A supervisor at an Asian securities regulator is reviewing a ledger whose validators are a small group of banks certified by a consortium. Which trust failure is this design most exposed to, and what limits it?

A. Authorised participants colluding to alter the ledger or its code, limited by reputation and by regulatory oversight and penalties.

B. An attacker gaining more than 50% of the network's computing power, limited by the rewards paid to the validators of the network.

C. Users manipulating transaction fees to front-run others, limited by the consortium certifying each transaction in the queue.

D. Failure of the single entity that runs the ledger, limited only by its credibility and by the strength of users' legal protections.

Question CI-7 e.6 Adapted from CI QB VI Q146 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e (precision note)

A cybersecurity analyst at a crypto-asset exchange explains how the consensus mechanism of a permissionless proof-of-work ledger could be compromised. Which condition does the reading identify?

A. An attacker acquires more than 50% of the tokens outstanding on the ledger, which gives it a majority of the votes on each block.

B. An attacker gains a majority of the seats on the consortium that certifies the ledger's validators and approves its participants.

C. An attacker obtains the credentials of the single entity that confirms transactions, so that it can alter records undetected.

D. An attacker gains control of more than 50% of the computing power that is used to validate transactions on the ledger.

Question CI-7 e.7 Adapted from CI QB VI Q147 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

A trader on a permissionless ledger notices that her large orders are repeatedly preceded and followed by trades from the same address, which profits at her expense. Which explanation is most consistent with the reading?

A. Rehypothecation: the collateral backing her orders is reused by the other address across several linked transactions.

B. A sandwich attack: by manipulating transaction fees, the attacker places trades just before and after hers in the queue.

C. Collusion among authorised validators, who change the ledger's programming to reorder transactions for their own gain.

D. A consensus attack: the other address controls more than 50% of computing power and rewrites her trades after validation.

Question CI-7 e.8 Adapted from CI QB VI Q149 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

The board of a Dubai bank must choose between running a tokenized platform as a single-operator system or as a permissioned DLT with several certified participants. A director argues that the single-operator design carries less risk of trust failure, because one accountable entity is easier to supervise. Which response best reflects the reading?

A. The director is right: a single operator has no one to collude with, so its risk of trust failure is lower than a permissioned DLT's.

B. The director is right in effect: both designs restrict participation, so the risks are identical and supervision decides the choice.

C. The director is wrong: risks are similar but higher for one operator, whose trust rests only on its credibility and legal protections.

D. The director is wrong: permissioned DLT has no trust risk, because reputation and regulatory penalties remove the incentive to collude.

Question CI-7 e.9 Adapted from CI QB VI Q150 • SchweserNotes Book 5, LO 106.e; CI notes CI-7 e

A compliance director at a global bank is weighing whether to settle tokenized trades on a permissionless DLT. Beyond attacks on the consensus mechanism and on transaction ordering, which further weaknesses does the reading attribute to this design?

A. Collusion among the authorised validators, and reliance on reputational incentives that weaken once validators expect no future profits.

B. Dependence on one operator's credibility, and a legal framework that gives users weaker protection than the operator's own rules.

C. External information more exposed to cyberattacks with no central authority, and regulatory compliance harder to enforce.

D. Lower sharedness than a single ledger, because anyone can join, which splits the ledger's liquidity into segmented validator pools.

Solutions • CI-7 e

Solution CI-7 e.1 — Answer: A

New • built from SchweserNotes Book 5, LO 106.e

Step 1 – Proposal (1)
All parties or their intermediaries on one system, with all assets created directly on it: the single ledger model.

Step 2 – Proposal (2)
Representations of non-native assets that exist on other ledgers, with the originals in escrow and represented through escrow certificates: the common ledger model.

Step 3 – Proposal (3)
Standardised technology that lets different ledgers run the same programs and coordinate transfers: the compatible ledger model.
⇒ One system; one meeting point for many ledgers; one standard for many ledgers.

Interpretation
The three are the reading's ways of achieving *sharedness*. The class notes read them as a spectrum: sharedness bought with centralisation, with escrow and certificates, or with a standard and no central point.

Why the other options are wrong
B is incorrect. Sibling swap on (2) and (3): escrow and certificates define the common ledger; a shared standard defines compatible ledgers.
C is incorrect. Sibling swap on (1) and (2): one system with all assets created on it is the single ledger; escrowed originals define the common ledger.
D is incorrect. Axis confusion: single-operator systems and permissionless DLT are trust models, not ways of achieving sharedness.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025), Tokenization and Financial Market Inefficiencies; SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.2 — Answer: C

Adapted from CI QB VI Q142 • SchweserNotes Book 5, LO 106.a and 106.e

Step 1 – The common ledger mechanism
Common ledger models allow representations of non-native assets that exist on other ledgers. The original asset is placed in escrow and represented through an escrow certificate.

Step 2 – What the holder relies on
The bond stays on its home ledger, locked in escrow; the asset manager holds the certificate that represents it.
⇒ The arrangement adds a second asset, the certificate, and so a second thing that can fail.

Interpretation
The class notes make the point: the common ledger is the model that introduces the escrow certificate, and with it a dependence on the escrow behind it. This is the contrast with migration, which leaves no original behind.

Why the other options are wrong
A is incorrect. Sibling swap: destroying the original and reissuing a native token is migration

(CI-7 a), not the common ledger model.
B is incorrect. Scope error: the original is locked in escrow precisely so that it is not traded alongside its representation.
D is incorrect. Sibling swap: running the same programs and coordinating transfers describes compatible ledgers, which need no escrow certificate.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.a and 106.e

Solution CI-7 e.3 — Answer: D

Adapted from CI QB VI Q144 • SchweserNotes Book 5, LO 106.e

Step 1 – The ledger
Programmability in digital ledgers means the code and the database are integrated within the system.

Step 2 – The traditional set-up
External users must send instructions to a separate database operator.

Step 3 – What integration buys
Internal programmability ensures transactions are coherent and efficient, and makes it easier to bundle or link smart contracts.
⇒ No external system has to process the transaction.

Interpretation
Smart contracts can act on the records directly because code and records live in one place (Figure 2).

Why the other options are wrong
A is incorrect. Mirror reversal: the separate database operator is the traditional set-up, and integration is the ledger.
B is incorrect. Polarity flip: integration makes bundling easier, not harder.
C is incorrect. Feature swap: access to the same records by all relevant parties is sharedness, not programmability.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.4 — Answer: B

Adapted from CI QB VI Q145 • SchweserNotes Book 5, LO 106.e

Step 1 – Design (1)
One entity manages records and executes and confirms transactions: a single-operator system. Trust rests on the operator's credibility, supervision and oversight.

Step 2 – Design (2)
A decentralised network that anyone can join, transact on and help validate, with validators rewarded: permissionless DLT.

Step 3 – Design (3)
Only participants approved by a central entity or consortium can access or validate: permissioned DLT, the hybrid of the other two.
⇒ Centralised, open, and closed group.

Interpretation

Each design rests trust on something different, and each fails when that thing fails.

Why the other options are wrong

A is incorrect. Axis confusion on (1): the single ledger model is a way of achieving sharedness; one entity running the records is the single-operator trust model.

C is incorrect. Sibling swap on (2) and (3): open validation with rewards is permissionless; consortium approval is permissioned.

D is incorrect. Two errors: (1) is the single-operator system, and (3) is permissioned DLT; the common ledger model is a sharedness model.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.5 — Answer: A

Adapted from CI QB VI Q148 • SchweserNotes Book 5, LO 106.e

Step 1 – Identify the design

Validators limited to a small group certified by a consortium: permissioned DLT.

Step 2 – Its failure

A permissioned DLT can fail if authorised participants collude to modify the ledger or its programming for their own benefit.

Step 3 – What limits it

Participants generally want to protect their reputations to earn future profits, and regulatory oversight and penalties for noncompliance discourage abuse.

$\Rightarrow$ Collusion, checked by reputation and regulation.

Interpretation

The structure can also improve security and simplify regulation. Because a permissioned DLT is a hybrid, it can show failures of either design, but collusion is the one the reading attaches to it specifically.

Why the other options are wrong

B is incorrect. Sibling swap: a 50% computing-power attack is the consensus attack on a permissionless proof-of-work ledger.

C is incorrect. Sibling swap with a misassigned mitigant: fee manipulation is a permissionless weakness, and consortium certification is a permissioned feature.

D is incorrect. Sibling swap: dependence on a single entity's credibility and legal protections describes the single-operator system.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e and Module Quiz 106.2

Solution CI-7 e.6 — Answer: D

Adapted from CI QB VI Q146 • SchweserNotes Book 5, LO 106.e

Step 1 – Why permissionless is exposed

A permissionless DLT lets anyone participate and validate transactions, so it is vulnerable to attacks that compromise the consensus mechanism.

Step 2 – The condition

In a proof-of-work system, the attack can succeed if the attacker gains control of more than 50% of the computing power used to validate transactions.
⇒ Computing power, not token holdings, is the resource that matters in proof of work.

Interpretation

The class notes' "51% attack" label is looser; see the precision note.

Why the other options are wrong

A is incorrect. Wrong input twin: the same 50% threshold applied to the wrong resource; the reading's condition is computing power used to validate.
B is incorrect. Model confusion: a consortium that certifies validators belongs to permissioned DLT; a permissionless ledger has none.
C is incorrect. Model confusion: one entity confirming transactions is the single-operator system.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.7 — Answer: B

Adapted from CI QB VI Q147 • SchweserNotes Book 5, LO 106.e

Step 1 – The mechanism

On a permissionless ledger, trust can be undermined if users manipulate transaction fees offered by others, positioning their own trades ahead of others in the validation queue. This enables front running.

Step 2 – The pattern

A sandwich attack is a form of front running in which the attacker places transactions before and after a victim's transaction to generate a profit.
⇒ Trades just before and just after hers, from the same address, at her expense: a sandwich attack.

Interpretation

This failure needs no control of the consensus. It exploits the ordering of the validation queue, which is why the reading lists it separately from the 50% attack.

Why the other options are wrong

A is incorrect. Sibling swap: rehypothecation is the reuse of collateral across transactions (CI-7 c), not an ordering attack.
C is incorrect. Model confusion: collusion among authorised validators is the permissioned failure; the trader is on a permissionless ledger.
D is incorrect. Sibling swap: a 50% computing-power attack compromises consensus itself; the pattern described is queue positioning through fees.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.8 — Answer: C

Adapted from CI QB VI Q149 • SchweserNotes Book 5, LO 106.e

Step 1 – The comparison in the reading

The risks of trust failure in a fully centralised single-operator system are similar to those in a

permissioned DLT.

Step 2 – The ranking

They are even higher, because trust in a single-operator system does not rely on consensus among participants and depends entirely on the operator's credibility and the strength of legal protections.

⇒ Similar in kind, worse in degree.

Interpretation

The director's "one accountable entity" is the reason for the higher risk: everything rides on it, whereas a permissioned DLT needs several participants to collude.

Why the other options are wrong

A is incorrect. True premise, wrong conclusion: a single operator cannot collude, but it fails through its own failure or misuse, and the reading ranks that risk higher.

B is incorrect. Absolute claim: "identical" contradicts the reading, which ranks single-operator risk higher.

D is incorrect. Absolute claim: reputation and regulatory oversight help limit collusion in a permissioned DLT; they do not remove the risk of trust failure.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

Solution CI-7 e.9 **Answer: C**

Adapted from CI QB VI Q150 • SchweserNotes Book 5, LO 106.e

Step 1 – Set aside what the stem excludes

Attacks on consensus (more than 50% of proof-of-work computing power) and on ordering (fee manipulation, front running, sandwich attacks) are already named.

Step 2 – The remaining permissionless weaknesses

The absence of a central authority leaves external information more vulnerable to cyberattacks, and the decentralised structure makes regulatory compliance more challenging to enforce.

⇒ Open and headless: no guardian for outside information, no single point for a regulator to act on.

Interpretation

The openness that removes the need to trust one operator also removes the party a regulator would hold to account.

Why the other options are wrong

A is incorrect. Model confusion: collusion among authorised validators, and the reputational check on it, belong to permissioned DLT.

B is incorrect. Model confusion: dependence on one operator's credibility and legal protections is the single-operator system.

D is incorrect. Axis confusion: open participation is a trust-design feature; segmented liquidity pools come from non-interoperable ledgers (CI-7 c), not from permissionless validation.

LO: CI-7 e: Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models. • **Reference:** IMF Fintech Note (January 2025); SchweserNotes 2026 Book 5, Reading 106, LO 106.e

CI-8 • Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector (Banco de España, April 2023) • Vardeez: IMPORTANT

CI-8 a

P3 CONCEPT

Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 Crypto, Tokenisation and Digital Resilience (Map #68, 1/16, notes only). The workbook gives revision notes and likely distractors for CI-8 but no questions, so every question in this LO is new or adapted from the CI question bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." Current Issues is covered in full, so P3 LOs here are not thinned and get 8 to 10 questions.

Reported exam shapes

- **D29·S3** CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools and their challenges. This LO carries the first two parts: cyber versus ICT risk (Questions CI-8 a.1 to a.3) and cloud concentration (Questions CI-8 a.4 to a.9). Contagion is CI-8 b; the tools are CI-8 c.

What the questions look like: no calculations. Classifying an incident by its cause, statement audits on what cloud adoption did to each risk, and the reading's recall anchors: about 80% and roughly 20% for cloud outages, and all major banks against less than half of asset managers and insurers for multiclouds.

1. The reading in brief

Jose Ramon Martinez (Banco de España, April 2023) asks which policy tools can protect financial stability now that financial institutions are moving toward an almost completely digital system. The goal is *digital resilience*, which covers both technological and financial aspects. The argument runs in three steps, one per LO.

1. **CI-8 a (this LO).** What cyber risk and ICT risk are, why financial institutions are exposed to them, and how the move to cloud computing has changed the mix.
2. **CI-8 b.** How a technology event turns into financial risk (illiquidity, excessive leverage, loss of customer trust) and then into systemic risk through contagion.
3. **CI-8 c.** The policy measures and micro-oriented tools already in use, the three macroprudential tools (circuit breakers, cooperative arrangements, structural measures) and the challenge to the adoption of each.

Shared background for the reading sits here; CI-8 b and CI-8 c recap it in a line.

2. Cyber risk versus ICT risk: told apart by cause

- **Cyber risk** is the loss of confidentiality, integrity and availability (the **CIA triad**) of data and data systems that results from a *cyberattack*.
- **ICT risk** is disruption to information and communication systems that may have similar effects to cyber risk but, strictly speaking, does *not* stem from a cyberattack.

The test is the cause, not the consequence. A service outage looks the same to a customer whether an attacker or a failed upgrade caused it; only the first is cyber risk. Two further points follow from the definitions. Cyber risk spans the whole triad, so it is not confined to confidentiality, and an ICT event can hit the same data and systems with similar effects.

	Cyber risk	ICT risk
Cause	A cyberattack	Not a cyberattack, strictly speaking
Effect	Loss of confidentiality, integrity and availability of data and data systems	Disruption to information and communication systems, with effects that may be similar
Examples	Breaches of client information, unauthorised access, theft of funds, denial-of-service (DoS) attacks	System breakdowns, communication failures, service outages
Why it matters	Some institutions rank it as more significant than market risk	Essential for uninterrupted transaction services
Challenge	No reliable loss estimates for lack of data, which has hindered cyber insurance	Dependence on technology; vulnerability to system failures

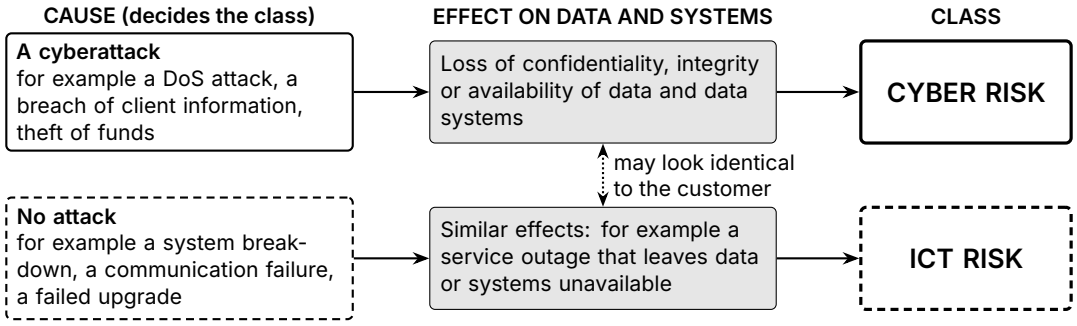


Figure 1. Two incidents, one effect, two classes. The top row (solid cause box) is cyber risk; the bottom row (dashed cause box) is ICT risk. The shaded middle column can be identical in both rows, which is why the class is read from the left-hand column only.

3. Why financial institutions are exposed

1. **Digitalisation meets essential services.** The overwhelming level of digitalisation at financial institutions, combined with the essential nature of their services, leaves them vulnerable. The sector manages an immense amount of funds and is expected to provide uninterrupted transaction services and to maintain confidentiality. Both point to a greater risk of cyberattacks and of ICT breakdowns.
2. **Cyber risk ranked above market risk.** Serious cybersecurity incidents have increased in general, and on that basis some financial institutions have ranked cyber risk as *more* significant than market risk. Major incidents worldwide include confidentiality breaches of client information, theft of funds and DoS attacks.
3. **No reliable loss estimates.** Reliable estimates of losses from cyber risk are generally not possible because the data are lacking. As a result, the development of cyber insurance has been significantly hindered.

Read points 2 and 3 together. The ranking rests on the rise in serious incidents; it cannot rest on reliable loss estimates, because the reading says reliable estimates are generally not possible.

Correction to the source notes. (1) Table CI-8.1 in the class notes says some financial institutions rank cyber risk "as significant as market risk". SchweserNotes Book 5, LO 107.a says some have ranked it as *more* significant than market risk, and the D29 workbook notes agree ("above market risk"). Use "more significant"; Question CI-8 a.3 option B reverses the ranking. (2) The class notes' cloud list says "widespread adoption increases systemic cyberattack risk". Do not read it as "cloud adoption raises cyber risk". SchweserNotes, LO 107.a: adoption has brought a corresponding *decrease* in cyber risks and an *increase* in ICT risks, and has integrated the two to an extent that there *could* be systemic implications. The systemic concern is the integration of both risks, stated as a possibility; the notes' line drops ICT risk and hardens "could be" into "increases". Questions CI-8 a.4 and a.9 test the two halves, and Question CI-8 a.3 option D is the misreading.

4. Cloud computing: less cyber risk, more ICT risk

What it is. Cloud computing is the provision of computer resources, such as storage, servers and software, through the internet ("the cloud"). It runs on a shared system with multiple users. Its benefits are lower costs and greater operating flexibility, and cloud service providers are expected to have extremely robust controls in place.

What adoption did to the two risks.

- Financial institutions have strongly embraced cloud computing. The result is a corresponding *decrease* in cyber risks and an *increase* in ICT risks.
- **Partial transfer.** Cloud computing moves some, but not all, of an organisation's cyber risks to cloud service providers, because the organisation's providers and customers might not always have adequate controls.
- **Integration.** Adoption has been so significant that cyber and ICT risks have become integrated to an extent that there could be systemic implications (CI-8 b).

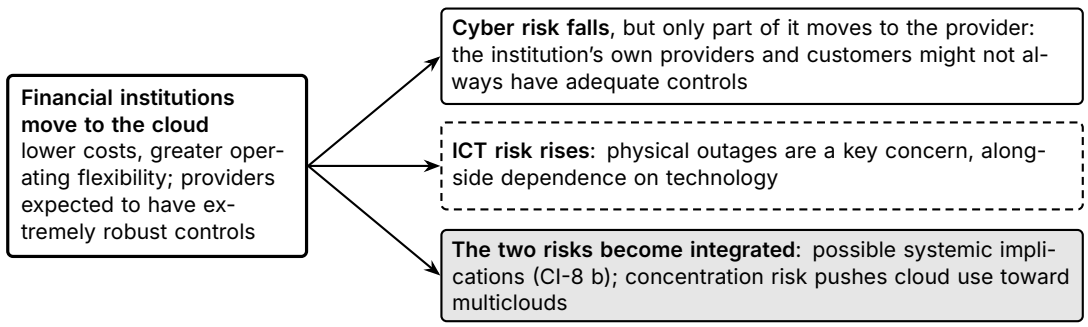


Figure 2. What the move to the cloud does. Solid box: the cyber leg falls but is only partly transferred. Dashed box: the ICT leg rises. Shaded box: the two legs merge, which is the bridge to systemic risk in CI-8 b. "Both risks fall" and "cyber up, ICT down" are the two classic distractors.

Concentration and multiclouds.

- Cloud computing has needed to evolve, for example to **multiclouds**, to deal with concentration risks. Spreading computing assets over multiple clouds, rather than one, limits the damage a cyberattack can do.
- **All major banks** have adopted multiclouds, compared with **less than half** of asset management companies and insurance companies.
- The evolution brings unintended additional risks, such as **weaker stewardship** over the distributed computing processes.

Outages.

- ICT risk in the form of physical outages has become a key concern. A cloud services audit firm reports that about **80%** of cloud customers suffered at least one outage within the prior three years. Roughly **20% of those outages** were significant enough to noticeably disrupt operations.
- Climate change and rising energy costs may be causing *longer and more frequent* outages.
- In short, cloud use is high despite its numerous and significant drawbacks.

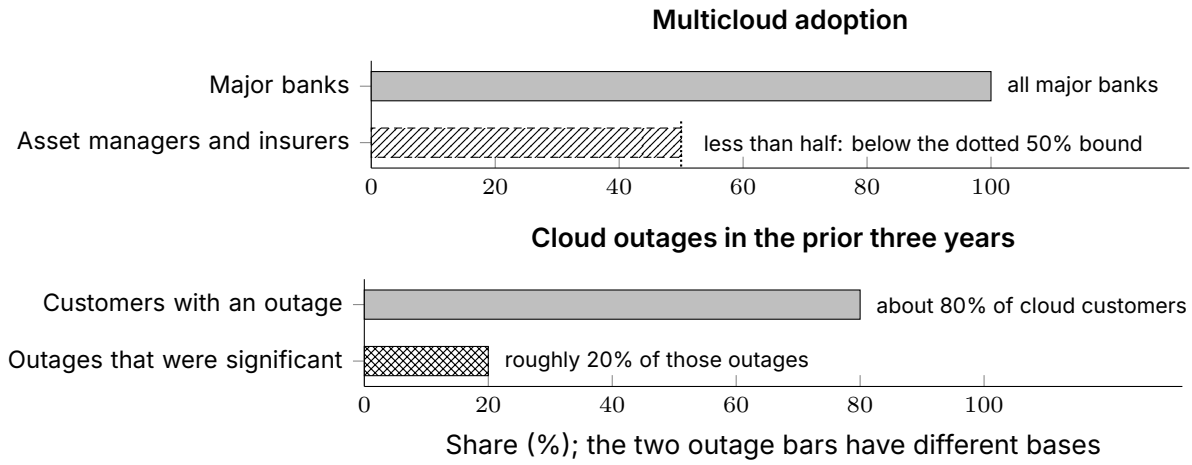


Figure 3. The reading's recall anchors. Top: grey bar for all major banks. For asset managers and insurers the source gives only "less than half", so the hatched bar has no right edge and stops at the dotted 50% upper bound: the true share lies somewhere below that line, not on it. Bottom: the grey bar is a share of cloud customers; the cross-hatched bar is a share of *those outages*, not of customers, so the two bars cannot be swapped or combined.

Results card: CI-8 a

Cyber risk	Loss of confidentiality, integrity and availability of data and data systems from a cyberattack
ICT risk	Disruption to information and communication systems with similar effects, not from a cyberattack
The test	Cause, not consequence; cyber risk covers the whole CIA triad
Why exposed	Overwhelming digitalisation plus essential services: immense funds, uninterrupted transactions, confidentiality
Ranking	Some institutions rank cyber risk as *more* significant than market risk, after a rise in serious incidents
Loss data	No reliable estimates for lack of data; cyber insurance significantly hindered
Cloud	Cyber risk down, ICT risk up; only some cyber risk moves to providers; the two integrate, with possible systemic implications
Multiclouds	Answer to concentration risk: limit attack damage; all major banks against less than half of asset managers and insurers; cost: weaker stewardship
Outages	About 80% of cloud customers had at least one in three years; roughly 20% of those outages noticeably disrupted operations; may become longer and more frequent

Exam traps

- **Classifying by effect.** A service outage is ICT risk unless an attack caused it; the CIA effect alone does not decide.
- **Shrinking cyber risk to confidentiality.** Cyber risk is loss of all three legs of the CIA triad, not breaches of client data only.
- **Polarity flip on cloud.** Cyber risk down, ICT risk up. "Both fell" and "cyber up, ICT down" are wrong.
- **Absolute claim on transfer.** Some, not all, cyber risk moves to the provider; the weak links are the institution's own providers and customers.
- **Integration reversed.** Cloud adoption has merged cyber and ICT risk; it has not made them easier to analyse apart.
- **Sector swap on multiclouds.** All major banks; less than half of asset managers and insurers.
- **Wrong-input twin on outages.** 80% is customers with at least one outage; 20% is the share of *those outages* that were significant.
- **"As significant as" market risk.** The reading says *more* significant, and only for some institutions.
- **Free diversification.** Multiclouds limit attack damage but bring weaker stewardship over distributed processes.
- **Outage trend reversed.** Climate change and rising energy costs may mean longer and more frequent outages, not fewer.

Practice questions: CI-8 a

Question CI-8 a.1

New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

The operational risk officer of a Lisbon-based retail bank is classifying two incidents from the same quarter. In the first, a denial-of-service attack made online banking unavailable for an afternoon. In the second, a failed software upgrade at the bank's cloud provider made online banking unavailable for an afternoon. Based on the distinction drawn in the Banco de España paper, which classification is correct?

A. Both are cyber risk events, because each caused a loss of availability of data systems, and cyber risk is defined by loss of the CIA triad.

B. The first is a cyber risk event and the second an ICT risk event, because only the first stems from a cyberattack.

C. Both are ICT risk events, because each disrupted information and communication systems, whatever caused the disruption.

D. Neither is a cyber risk event unless client information was also breached, because cyber risk concerns the confidentiality of data.

Question CI-8 a.2 New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

A non-executive director on the risk committee of a Nordic bank asks why the Banco de España paper treats financial institutions as particularly vulnerable to cyber and ICT risks. Which answer best reflects the reading?

A. Their systems are less digitalised than those of most other sectors, so older platforms fail more often under modern attack techniques.

B. Reliable loss data show their cyber losses exceeding their market risk losses, which is why some of them now rank cyber risk first.

C. Their cloud service providers are expected to have weaker controls than the institutions themselves, which widens their exposure.

D. Digitalisation meets essential services: they manage immense funds and must transact without interruption and keep data confidential.

Question CI-8 a.3 Adapted from CI QB Imp Q75 • SchweserNotes Book 5, LO 107.a; CI notes CI-8 a (correction box)

The CRO of a mid-sized Canadian bank wants to transfer more of the bank's cyber risk through insurance and finds that the market for such cover is thin. Which explanation is most consistent with the Banco de España paper?

A. Moving to the cloud has transferred all of the bank's cyber risk to its providers, so little cyber risk is left for an insurer to cover.

B. Loss data are plentiful, but banks rank cyber risk below market risk and so see little need to buy cover against it.

C. Reliable estimates of cyber losses are generally not possible for lack of data, and this has significantly hindered cyber insurance.

D. Cloud adoption has raised banks' cyber risk faster than insurers can assess it, so cover has not kept pace with demand.

Question CI-8 a.4 New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a (correction box)

After migrating its core systems to public cloud providers, a Frankfurt bank's CRO is preparing a note for the board on how the move has changed the bank's risk profile. Based on the Banco de España paper, which statement should the note contain?

A. Cyber risk is likely to have fallen and ICT risk to have risen, so the bank has traded part of one exposure for more of the other.

B. Both cyber risk and ICT risk are likely to have fallen, because cloud providers are expected to have extremely robust controls in place.

C. Cyber risk is likely to have risen and ICT risk to have fallen, because the bank's data now sit on a system shared with many users.

D. Cyber and ICT risk can now be assessed apart, because the providers carry the cyber risk while the bank keeps the ICT risk.

Question CI-8 a.5 Adapted from CI QB Imp Q77 • SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

A vendor risk manager at a Singapore asset manager argues that, because the firm's data now sit with a cloud provider that is expected to have extremely robust controls, the firm's cyber risk has passed entirely to the provider. Which response best reflects the Banco de España paper?

A. The argument holds for cyber risk, and what the firm keeps is ICT risk, because outages at the provider would still disrupt its operations.

B. Only some of the cyber risk moves to the provider, because the firm's own providers and customers may not always have adequate controls.

C. None of the cyber risk moves to the provider, because the firm keeps legal ownership of its data and stays the target of any attack.

D. Only some of the cyber risk moves, because the provider's controls are expected to be weaker than the firm's own controls.

Question CI-8 a.6 New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

A consultant benchmarking cloud strategies for a group of European insurers cites the Banco de España paper's figures on multicloud adoption across the financial sector. Which statement is accurate?

A. About 80% of major banks have adopted multiclouds, compared with about 20% of asset management and insurance companies.

B. Less than half of major banks have adopted multiclouds, compared with all major asset management and insurance companies.

C. All major banks have adopted multiclouds, compared with less than half of asset management and insurance companies.

D. About half of major banks, asset managers and insurers have adopted multiclouds, with little difference between sectors.

Question CI-8 a.7 Adapted from CI QB Imp Q76 • SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

The chief information officer of a Zurich wealth manager proposes moving from a single cloud provider to a multicloud set-up. The risk committee asks what, according to the Banco de España paper, the move would buy and what it would cost. Which answer is most accurate?

A. It removes concentration risk entirely and adds no new risk, because no single provider would hold all of the firm's computing assets.

B. It raises the damage a single cyberattack can do, because an attack on one cloud can spread across the firm's connected clouds.

C. It limits the damage one cyberattack can do and also tightens stewardship, because each provider oversees a smaller share of the processes.

D. It limits the damage one cyberattack can do by spreading assets across clouds, at the cost of weaker stewardship over the processes.

Question CI-8 a.8 New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

An operational resilience lead at a Toronto bank is preparing a board paper on the bank's dependence on cloud providers and wants to quote the outage figures in the Banco de España paper. Which statement is accurate?

A. About 80% of cloud customers had an outage in the prior three years, and roughly 20% of those outages noticeably disrupted operations.

B. About 20% of cloud customers had at least one outage in the prior three years, and roughly 80% of those outages disrupted operations.

C. About 80% of cloud customers had an outage significant enough to noticeably disrupt their operations at some point in the prior three years.

D. About 80% of cloud customers had at least one outage in the prior three years, and climate change is expected to make outages shorter.

Question CI-8 a.9 New • built from SchweserNotes Book 5, LO 107.a; CI notes CI-8 a

A supervisor at a national central bank finds that most banks in her jurisdiction run critical services on the same small group of cloud providers. Which concern raised in the Banco de España paper does this finding most directly illustrate?

A. Cloud adoption has separated cyber risk from ICT risk, so the supervisor can now assess each bank's cyber risk on its own.

B. Concentration lowers systemic risk, because each of the providers is expected to have extremely robust controls in place.

C. Heavy adoption has merged cyber and ICT risks, with possible systemic implications; multiclouds evolved to deal with concentration risk.

D. Concentration matters mainly for confidentiality, because a system shared with multiple users exposes each bank's client data to the others.

Solutions • CI-8 a

Solution CI-8 a.1 — Answer: B

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – Apply the definitions

Cyber risk is the loss of confidentiality, integrity and availability of data and data systems that results from a cyberattack. ICT risk is disruption to information and communication systems with similar effects that does not stem from a cyberattack.

Step 2 – Read the cause, not the effect

Both incidents had the same effect: online banking was unavailable. The DoS attack is a cyberattack, so it is cyber risk. The failed upgrade is not an attack, so it is ICT risk.
⇒ Same consequence, different cause, different class.

Interpretation

The reading's distinction is about the origin of the event. A customer locked out of online banking cannot tell the two apart, which is exactly why the class must be read from the cause.

Why the other options are wrong

A is incorrect. Classifies by effect: loss of availability is common to both, but only an attack makes it cyber risk. An ICT event may have similar effects by definition.
C is incorrect. Scope error: ICT risk is the disruption that does *not* stem from a cyberattack. The DoS attack is the textbook cyber case; DoS attacks are among the reading's examples of cyber incidents.
D is incorrect. Scope error: cyber risk covers all three legs of the CIA triad, availability included, not confidentiality alone.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023), Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector; SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.2 — Answer: D

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – The reading's two reasons

The overwhelming level of digitalisation at financial institutions, combined with the essential nature of their services, places them in a vulnerable position.

Step 2 – What "essential" means here

The sector manages an immense amount of funds and is expected to provide uninterrupted transaction services and to maintain confidentiality.
⇒ Both raise the risk of cyberattacks and of ICT breakdowns.

Interpretation

The vulnerability comes from what the sector does and how digital it is, not from weak providers or from measured losses.

Why the other options are wrong

A is incorrect. Polarity flip: the reading stresses the *overwhelming* digitalisation of financial institutions, not a lack of it.
B is incorrect. Invented evidence: reliable estimates of cyber losses are generally not possible for lack of data. The ranking by some institutions follows the general increase in serious incidents.
C is incorrect. Polarity flip: cloud service providers are expected to have *extremely robust* controls

in place.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.3 — Answer: C

Adapted from CI QB Imp Q75 • SchweserNotes Book 5, LO 107.a

Step 1 – The data gap

Reliable estimates of losses arising from cyber risks are generally not possible to compute because of a lack of data.

Step 2 – The consequence

As a result, the development of cyber insurance has been significantly hindered.
⇒ Without loss data there is little basis for pricing cover, and the market stays thin.

Interpretation

The CRO's experience is the reading's point: without reliable estimates of cyber losses, the development of cyber insurance has been significantly hindered.

Why the other options are wrong

A is incorrect. Absolute claim: cloud computing moves some, but not all, of an organisation's cyber risks to providers.
B is incorrect. Two polarity flips: loss data are lacking, not plentiful, and some institutions rank cyber risk *above* market risk (see the correction box).
D is incorrect. Polarity flip: cloud adoption has brought a corresponding *decrease* in cyber risks (and an increase in ICT risks). What has hindered cyber insurance is the lack of loss data (see the correction box, item 2).

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.4 — Answer: A

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – The cyber leg

Financial institutions have strongly embraced cloud computing, and this has brought a corresponding *decrease* in cyber risks. Cloud service providers are expected to have extremely robust controls in place.

Step 2 – The ICT leg

The same move has *increased* ICT risks: physical outages have become a key concern.
⇒ The bank has exchanged part of one exposure for more of the other.

Interpretation

The note should not report a simple risk reduction. The move also merges the two risks (CI-8 a.9), which is the route to systemic implications in CI-8 b.

Why the other options are wrong

B is incorrect. True premise, wrong conclusion: providers' robust controls go with lower cyber risk, but ICT risk has risen. This is the workbook's corrupted statement.
C is incorrect. Polarity flip on both legs: cyber risk has fallen and ICT risk has risen. The shared system is a feature of the cloud, not the reason cyber risk rose.

D is incorrect. Two errors: the risks have become integrated rather than separable, and only some of the cyber risk moves to the providers.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.5 — Answer: B

Adapted from CI QB Imp Q77 • SchweserNotes Book 5, LO 107.a

Step 1 – Test the claim of full transfer

The reading's main issue with cloud computing is that it moves some, but not all, of an organisation's cyber risks to the cloud service providers.

Step 2 – Why the transfer is partial

The organisation's own providers and customers might not always have adequate controls. Those weak links stay with the firm, however strong the cloud provider is.
⇒ The provider's controls do not cover the rest of the firm's chain.

Interpretation

Strong controls at the provider are an expectation, not a transfer of every exposure. The vendor risk manager has confused the provider's robustness with the firm's whole exposure.

Why the other options are wrong

A is incorrect. True fact, wrong conclusion: ICT risk has indeed risen with cloud use, but the claim that cyber risk has passed entirely to the provider is still wrong.
C is incorrect. Polarity flip: some cyber risk *is* moved to the provider; the transfer is partial, not nil.
D is incorrect. Right conclusion, wrong reason: the transfer is partial, but cloud service providers are expected to have *extremely robust* controls. The weak links are the firm's own providers and customers.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.6 — Answer: C

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – Recall the anchor

All major banks have adopted multiclouds, compared with less than half of asset management companies and insurance companies.

Step 2 – Check the direction

Banks lead; asset managers and insurers lag.
⇒ For the insurers in the stem, the benchmark shows them behind the major banks.

Interpretation

Multiclouds are the cloud's answer to concentration risk. The gap between banks and the other two sectors is the reading's only cross-sector figure in CI-8, so expect it to be swapped or replaced by another number from the reading.

Why the other options are wrong

A is incorrect. Wrong-input twin: 80% and 20% are the outage figures (customers with an outage, and the share of those outages that were significant).

B is incorrect. Polarity flip: the ranking is reversed; it is all major banks against less than half of the other two sectors.
D is incorrect. Wrong number and wrong conclusion: "all" against "less than half" is a clear difference between sectors.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.7 — Answer: D

Adapted from CI QB Imp Q76 • SchweserNotes Book 5, LO 107.a

Step 1 – What multiclouds buy

Cloud computing has needed to evolve, for example to multiclouds, to deal with concentration risks. Spreading computing assets over multiple clouds rather than one limits the amount of damage a cyberattack could do.

Step 2 – What they cost

The evolution comes with unintended additional risks, such as weaker stewardship over the distributed computing processes.
⇒ Less damage from any one attack, less control over the whole.

Interpretation

Diversifying the infrastructure buys resilience and costs oversight. The risk committee should expect both in the proposal.

Why the other options are wrong

A is incorrect. Absolute claim: multiclouds address concentration risk but bring unintended additional risks.
B is incorrect. Polarity flip on damage: spreading assets over several clouds *limits* the damage a cyberattack can do.
C is incorrect. Right benefit, cost reversed: multiclouds do limit attack damage, but the evolution brings *weaker* stewardship over the distributed computing processes, not tighter.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.8 — Answer: A

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – Customers with an outage

About 80% of cloud customers suffered at least one outage within the prior three years.

Step 2 – Significant outages

Roughly 20% of *those outages* were significant enough to noticeably disrupt operations.
⇒ The 20% is a share of outages, not of customers.

Interpretation

Physical outages are the form of ICT risk the reading calls a key concern, and climate change and rising energy costs may be making them longer and more frequent. The board paper should keep the two figures on their own bases.

Why the other options are wrong

B is incorrect. Sibling swap: the two percentages are exchanged.

C is incorrect. Scope error: 80% had at least one outage of any kind; only roughly 20% of the outages were significant enough to noticeably disrupt operations.
D is incorrect. Polarity flip: climate change and rising energy costs may be causing *longer* and *more frequent* outages.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a

Solution CI-8 a.9 Answer: C

New • built from SchweserNotes Book 5, LO 107.a

Step 1 – Integration

The significant adoption of cloud computing by financial institutions has integrated cyber and ICT risks to an extent that there could be systemic implications.

Step 2 – Concentration

Cloud computing has needed to evolve, for example to multiclouds, to deal with concentration risks.
⇒ Many banks on a few providers is the concentration problem the reading describes, multi-clouds are the evolution meant to deal with it, and the merged risks are what could make it systemic.

Interpretation

This is the bridge to CI-8 b, where a technology event at a shared provider reaches many institutions at once, and to CI-8 c, where structural measures target systemically important technological institutions.

Why the other options are wrong

A is incorrect. Polarity flip: adoption has *integrated* the two risks, not separated them.
B is incorrect. True premise, wrong conclusion: providers are expected to have robust controls, but the reading treats concentration as a risk to be dealt with, not a source of lower systemic risk.
D is incorrect. Scope error: the shared, multi-user system is the definition of the cloud; the reading's concern with concentration is systemic, not a confidentiality leak between users.

LO: CI-8 a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a and LO 107.c

CI-8 • Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector (Banco de España, April 2023) • Vardeez: IMPORTANT

CI-8 b

P3 CONCEPT

Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 Crypto, Tokenisation and Digital Resilience (Map #68, 1/16, notes only). The workbook has no CI-8 questions, so every question here is new or adapted from the CI question bank.
Vardeez: "Please solve book-wise Current Issues questions as anything can show up." P3 LOs in Current Issues are not thinned: this one gets 9 questions.

Reported exam shapes

- **D29·S3** CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools and their challenges. This LO is the "contagion to financial risk" part: every question (CI-8 b.1 to b.9) is on it.

What the questions look like: no calculations. The direction of the chain (technology risk to financial risk), the counter-intuitive CIA severity ordering, why contagion cannot be extrapolated, what agent-based models can and cannot yet do, and scenario readings of the blocked-withdrawal example and the liquidity buffer caveat.

Source note. The original class notes carried this material unlabelled inside 107.a; the consolidated notes reattribute it to CI-8 b (corrections record). This LO also uses SchweserNotes detail the five points omit: qualitative CIA effects, the infancy of agent-based models and their use for stress tests, the "not necessarily 100% reliability" standard and the liquidity buffer caveat.

1. The question the reading asks: technology risk to financial risk

CI-8 a showed that cloud adoption has integrated cyber and ICT risk to an extent that there could be systemic implications. This LO follows that through to financial stability.

- **The direction.** The most important consideration is how *technology* risks can lead to the realisation of *financial* risks. The arrow runs from technology to finance.
- **The financial risks.** Cyber vulnerabilities that are triggered can cause negative financial impacts such as **illiquidity**, **excessive leverage** and **loss of customer trust**. SchweserNotes' module quiz answer puts it the same way: shocks to the CIA triad trigger vulnerabilities focused on a few main areas such as liquidity, leverage and trust, not profitability.
- **The join is the CIA triad.** The effects of such systemic risks on confidentiality, integrity and availability are wide-ranging, and they are not equal. Severity seems *lower* for data confidentiality and *higher* for data integrity and for the availability of data and systems for operational use.
- **Quantity and quality.** Quantifying systemic risk is important, but the reading also asks for the qualitative effects, judged by their impact on the CIA triad.

The class notes call this ordering counter-intuitive, and it is the examinable detail. What weighs most is records that cannot be relied on and systems that cannot be used for operations, not information that has been exposed.

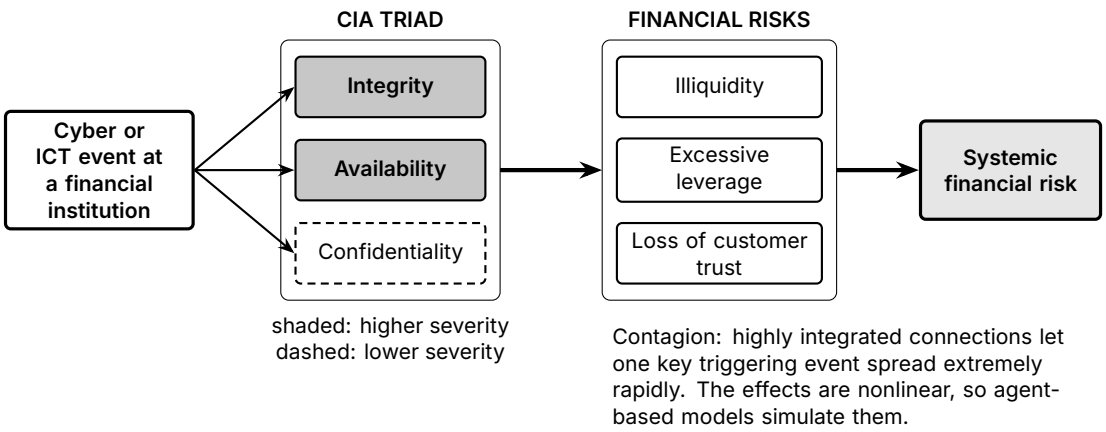


Figure 1. The join between technology and finance runs through the CIA triad (redrawn from class notes Figure CI-8.1). Shaded boxes are the legs with higher severity; the dashed box is the leg with lower severity. The triad and the financial risks are linked by one thick arrow, not one-to-one, because neither source pairs a specific leg with a specific financial risk.

2. Contagion: fast, nonlinear and hard to predict

- **Integration opens the door.** The modern design of cyber and ICT connections is highly integrated, which opens the potential for significant contagion effects. Financial contagion can occur extremely rapidly through the industry with the onset of just *one* key triggering event.
- **Nonlinearity.** Because contagion effects are nonlinear, it is challenging to predict the probability of key events with reasonable accuracy.
- **The proposed tool: agent-based models (ABMs).** ABMs are computer simulations of the actions and interactions among agents. Applied to cyber risk, they quantify the impact of a cyber incident in terms of its CIA effects and how those effects reach the liquidity and asset values of the institutions involved.
- **Their maturity.** ABMs are still in their infancy, but their basic principles could be applied to developing cyber and ICT stress tests.

3. Why the surface story misleads

ABMs are also useful for *tracing the source* of systemic risk, because what appears to happen in real life between technology and finance is not always accurate. The reading's example:

- A cyber incident triggers a run on a bank's deposits, but no withdrawals follow, because the bank's systems are disrupted and cannot process transactions.
- On the surface, technology has suppressed the contagion. That is clearly misleading: the bank's customers have alternatives, such as withdrawing funds they hold on deposit with other banks.

Measured withdrawals at the disrupted bank understate the run. The pressure has not gone away; it has moved to institutions whose systems still work.

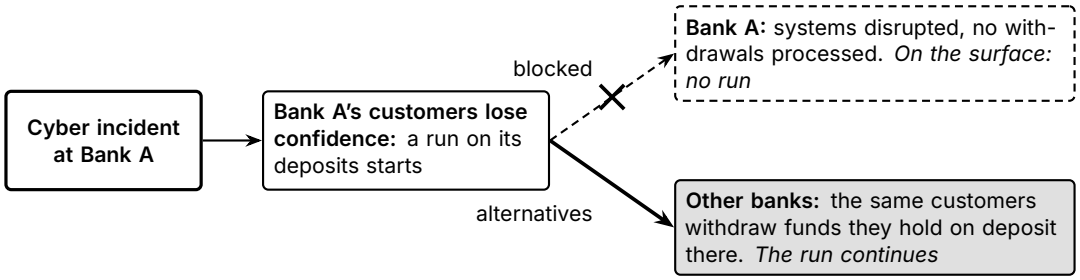


Figure 2. The blocked-withdrawal example. The dashed arrow, crossed, is the withdrawal channel at the disrupted bank; the thick arrow is the channel the surface reading misses. Agent-based models are the reading's way of tracing where the pressure actually goes.

4. Interdependence and the reliability standard

- Virtually all financial institutions are now digitalised, and they have become integrated and more reliant on each other through **cloud computing, networks and shared software**.
- So each participating institution must be reliable enough to **recover quickly** from a cyber incident, to ensure the continuity of key functions in the financial system.
- Given the importance of ICT functions, each institution must be "tested" to ensure it is **sufficiently reliable**, which does *not* necessarily mean 100% reliability.

The standard is quick recovery that keeps key functions running, not perfection. It applies to every participant, because in an interdependent system a failure anywhere can be the one key triggering event of section 2.

5. The end point: loss of confidence

- **The ultimate result.** Cyber and ICT risks, combined with the vulnerabilities of financial institutions, end in an overwhelming loss of customer confidence. The same combination has often been seen in the past, in runs on bank deposits that stemmed from customer panic or from financial fundamentals.
- **Liquidity buffers.** In theory, building up liquidity buffers helps to mitigate the effects of bank runs caused by cyber shocks. But history shows a high level of liquidity in the financial system for a reasonable amount of time, so such buffers have not had to be used.
- **Measuring the impact.** A more accurate assessment of the impact of a loss of confidence requires agent-based models.

The buffer point cuts both ways for an exam. Buffers are a sound idea in principle, but there is no history of drawing on them against cyber-driven runs to calibrate from, and for assessing the impact of a loss of confidence the reading points to agent-based models.

Results card: CI-8 b

Direction	Technology risk leads to the realisation of financial risk, not the reverse
Financial impacts	Illiquidity, excessive leverage, loss of customer trust
CIA severity	Lower for confidentiality; higher for integrity and for availability of data and systems for operational use; assess qualitatively as well as quantitatively
Contagion	Highly integrated cyber and ICT connections; one key triggering event; extremely rapid; nonlinear, so probabilities are hard to predict
Agent-based models	Simulate agents' actions and interactions; map CIA effects onto liquidity and asset values; trace the source of systemic risk; in their infancy; principles usable for cyber and ICT stress tests
Surface trap	Blocked withdrawals at a disrupted bank do not mean suppressed contagion: customers withdraw deposits held at other banks
Reliability	Interdependence through cloud, networks and shared software; each institution tested to recover quickly; not necessarily 100%
End point	Overwhelming loss of customer confidence and runs; liquidity buffers help in theory but high system liquidity has meant they have not had to be used

Exam traps

- **Reversed arrow.** The key question is how technology risk leads to financial risk, not how financial stress leads to technology failure.
- **Profitability as the channel.** The main areas of vulnerability are liquidity, leverage and trust, not profitability.
- **Severity flip.** Confidentiality is the *least* severe leg; integrity and availability are the more severe.
- **Linear contagion.** Contagion is nonlinear, which is why probabilities are hard to predict and the reading turns to agent-based models.
- **Maturity overstated.** Agent-based models are in their infancy; stress tests can borrow their basic principles.
- **Quantification only.** The reading wants the qualitative CIA effects as well as numbers.
- **Blocked run read as no run.** Customers of the disrupted bank withdraw funds they hold at other banks.
- **100% reliability.** The test is sufficient reliability and quick recovery, not necessarily 100%.
- **Buffers overstated.** Buffers help in theory, but high system liquidity has meant they have not had to be used; "drawn on repeatedly" is wrong.
- **Several triggers needed.** One key triggering event is enough for rapid contagion.
- **Sibling swap into CI-8 c.** A disrupted bank that cannot process withdrawals is not a circuit breaker.

Practice questions: CI-8 b

Question CI-8 b.1 New • built from SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

The head of a central bank's financial stability department is setting the scope of a new workstream on cyber risk. Based on the Banco de España paper, which framing should guide the workstream?

A. How financial risks such as illiquidity and excessive leverage at institutions lead to the realisation of technology risks.

B. How cyber incidents erode the profitability of the institutions hit, which the paper treats as the main channel to finance.

C. How technology risks lead to financial risks: illiquidity, excessive leverage and loss of customer trust.

D. How technology risks lead to financial risks, with the most severe effects arising from breaches of data confidentiality.

Question CI-8 b.2 New • built from SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

A supervisor at a European national competent authority is ranking three scenarios for a cyber exercise at a large bank: (i) theft of client records, (ii) corruption of the payment ledger so that balances cannot be trusted, and (iii) a prolonged outage of core systems. Using the severity the Banco de España paper associates with each leg of the CIA triad, which ranking is most consistent with the paper?

A. Scenarios (ii) and (iii) ahead of (i), because severity tends to be higher for integrity and availability than for confidentiality.

B. Scenario (i) ahead of (ii) and (iii), because breaches of client confidentiality do the most damage to customers' trust in the bank.

C. Scenarios (i) and (ii) ahead of (iii), because an outage leaves the data intact and is ICT risk rather than cyber risk.

D. All three scenarios equally, because the paper treats the effects of systemic risk on the CIA triad as impossible to rank.

Question CI-8 b.3 New • built from SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

A research economist at a central bank proposes to estimate the probability that a cyber incident at one bank spreads to others by extrapolating linearly from past incidents. Which response best reflects the Banco de España paper?

A. The approach is sound, because the paper finds that cyber contagion grows in proportion to the number of institutions affected.

B. The approach is sound in principle, but it should draw on the extensive cyber loss data that financial institutions already report.

C. Contagion effects are nonlinear, so key events are hard to predict; the paper proposes relying on liquidity buffers instead.

D. Contagion effects are nonlinear, so key events are hard to predict accurately; the paper proposes agent-based models instead.

Question CI-8 b.4 Adapted from CI QB Imp Q82 • SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

A supervisor plans to develop a cyber and ICT stress test and asks a team to assess agent-based models as its basis. Which assessment is most accurate?

A. They are mature, validated tools already embedded in supervisory stress tests, so the supervisor can adopt them as they stand.

B. They are still in their infancy, but their basic principles could be applied to developing cyber and ICT stress tests.

C. They capture only the technology layer, so they cannot show how CIA effects reach institutions' liquidity and asset values.

D. Once an incident's effects are quantified with them, no qualitative assessment of the effects on the CIA triad is needed.

Question CI-8 b.5 Adapted from CI QB Imp Q79 • SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

A cyber incident at a Spanish regional bank triggers a run on its deposits, but the bank's systems are too disrupted to process any withdrawals. A board member concludes that the outage protected the banking system from contagion. Which assessment best reflects the Banco de España paper?

A. The conclusion is right: the disruption worked as a circuit breaker, pausing normal rules while the bank planned its recovery.

B. The conclusion is right: with no withdrawals processed, the incident did not turn into a funding problem for the banking system.

C. The conclusion is misleading: customers can still withdraw funds they hold on deposit at other banks, so contagion can continue.

D. The conclusion is misleading only because the blocked withdrawals will all be processed at the bank once its systems are restored.

Question CI-8 b.6 Adapted from CI QB Imp Q80 • SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

A supervisor is drafting operational expectations for a banking system in which institutions rely on the same cloud providers, networks and software. She asks what standard of reliability the Banco de España paper implies for each institution. Which answer is most consistent with the paper?

A. Each institution should be tested to show it can recover quickly enough from a cyber incident, which need not mean 100% reliability.

B. Each institution should show 100% reliability, because a single key triggering event in an integrated system is enough to start contagion.

C. Only the largest cloud providers need to be tested, because individual institutions' reliability does not affect the continuity of the system.

D. Each institution should be tested against cyberattacks only, because ICT disruptions that do not stem from attacks do not threaten continuity.

Question CI-8 b.7 New • built from SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

The treasurer of a bank proposes to size a liquidity buffer against cyber-driven deposit runs by studying how often such buffers have been drawn on in past episodes. Which comment is most consistent with the Banco de España paper?

A. The approach is sound, because buffers against cyber-driven runs have often been drawn on, which gives a long history to calibrate from.

B. The approach is flawed, because the paper finds that liquidity buffers do not help to mitigate cyber-driven runs, even in theory.

C. The approach is flawed, because the paper replaces liquidity buffers with circuit breakers as the tool against cyber-driven runs.

D. The approach is flawed: buffers help in theory, but ample system liquidity has meant they have not been used; use agent-based models.

Question CI-8 b.8 Adapted from CI QB Imp Q81 • SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

Several banks in one country run their core systems on the same software from a single vendor. A financial stability analyst is asked how a cyber incident at one of the banks could spread. Which assessment is most consistent with the Banco de España paper?

A. Contagion is unlikely, because highly integrated connections give the system redundant pathways that slow the spread of a shock.

B. Highly integrated connections open the way to significant contagion, which can spread extremely rapidly from one key trigger.

C. Contagion needs several simultaneous triggering events, so a single incident at one bank is likely to stay within that bank.

D. Contagion is possible, but its probability can be predicted accurately from the number of banks that share the vendor's software.

Question CI-8 b.9 New • built from SchweserNotes Book 5, LO 107.b; CI notes CI-8 b

In a speech, a deputy governor describes where the chain from cyber and ICT risks to systemic financial risk ends. Which statement is most consistent with the Banco de España paper?

A. It ends in higher operational losses at the institutions hit, a channel the paper treats as separate from the deposit runs of past crises.

B. It ends in a system-wide loss of data confidentiality, because client data are the asset most severely affected by cyber incidents.

C. It ends in an overwhelming loss of customer confidence, which in the past has often shown up as runs driven by panic or fundamentals.

D. It ends in concentration at a few cloud providers, which then become systemically important technological institutions.

Solutions • CI-8 b

Solution CI-8 b.1 — Answer: C

New • built from SchweserNotes Book 5, LO 107.b

Step 1 – The direction

The reading names the most important consideration: how technology risks can lead to the realisation of financial risks.

Step 2 – The financial risks

Triggered cyber vulnerabilities can cause illiquidity, excessive leverage and loss of customer trust.
⇒ The workstream should start from the technology event and trace it to these three.

Interpretation

This is the frame for the whole LO. The CIA triad is the join between the two sides (Figure 1), and contagion and loss of confidence are how the financial effects become systemic.

Why the other options are wrong

A is incorrect. Polarity flip: it reverses the direction. The reading's arrow runs from technology risk to financial risk.
B is incorrect. Scope error: the main areas of financial vulnerability are liquidity, leverage and trust, not the profitability of the institutions hit (SchweserNotes, Module Quiz 107.1, answer to question 2).
D is incorrect. Right direction, severity flipped: confidentiality is the leg with the *lower* severity; integrity and availability are higher.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023), Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector; SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.2 — Answer: A

New • built from SchweserNotes Book 5, LO 107.b

Step 1 – Map each scenario to a leg of the triad

(i) Theft of client records: confidentiality. (ii) A corrupted ledger whose balances cannot be trusted: integrity. (iii) A prolonged outage of core systems: availability.

Step 2 – Apply the severity ordering

Severity seems lower for data confidentiality and higher for data integrity and for the availability of data and systems for operational use.
⇒ Scenarios (ii) and (iii) rank ahead of scenario (i).

Interpretation

The ordering is counter-intuitive and is the examinable detail. A bank whose balances cannot be relied on, or whose systems cannot be used, has a harder problem than one whose data have been exposed.

Why the other options are wrong

B is incorrect. Polarity flip: confidentiality is the *least* severe leg, not the most severe.
C is incorrect. Severity flip with a classification error: availability is among the more severe legs. Whether an outage is cyber or ICT risk depends on its cause (CI-8 a), and either way it is an availability effect.
D is incorrect. Absolute claim: the effects are wide-ranging, but the reading does rank them,

lower for confidentiality and higher for integrity and availability.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.3 — Answer: D

New • built from SchweserNotes Book 5, LO 107.b

Step 1 – Why extrapolation fails

Contagion effects are nonlinear, so it is challenging to predict the probability of key events with reasonable accuracy.

Step 2 – The reading's response

Use agent-based models: computer simulations of the actions and interactions among agents, which quantify a cyber incident's CIA effects and how they reach institutions' liquidity and asset values.

⇒ Simulation replaces extrapolation.

Interpretation

A linear rule assumes that twice the trigger gives twice the spread. With nonlinear contagion, one key triggering event can spread extremely rapidly, so the past is a poor straight-line guide.

Why the other options are wrong

A is incorrect. Polarity flip: contagion effects are nonlinear, not proportional.

B is incorrect. Wrong input: reliable estimates of cyber losses are generally not possible for lack of data (CI-8 a), so there is no extensive loss record to extrapolate from.

C is incorrect. True premise, wrong tool: contagion is nonlinear, but the reading's answer to the prediction problem is agent-based models. Liquidity buffers help to mitigate cyber-driven runs in theory; they do not estimate the probability of contagion.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a and LO 107.b

Solution CI-8 b.4 — Answer: B

Adapted from CI QB Imp Q82 • SchweserNotes Book 5, LO 107.b

Step 1 – Maturity

Agent-based models are still in their infancy.

Step 2 – Usefulness for the stress test

Their basic principles could be applied to developing cyber and ICT stress tests.

⇒ A promising foundation, not an off-the-shelf tool.

Interpretation

The models are designed to do exactly what a cyber stress test needs: take the CIA effects of an incident and follow them into liquidity and asset values. What is missing is maturity, not scope.

Why the other options are wrong

A is incorrect. Polarity flip on maturity: the models are in their infancy, not mature and embedded.

C is incorrect. Scope error: mapping CIA effects onto institutions' liquidity and asset values is the models' stated purpose.

D is incorrect. Overreach: the reading says quantification is important but that the qualitative

effects, in terms of their impact on the CIA triad, must also be looked at.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.5 — Answer: C

Adapted from CI QB Imp Q79 • SchweserNotes Book 5, LO 107.b

Step 1 – What the surface shows

A cyber incident triggers a run, but no withdrawals happen because the bank's systems cannot process transactions. It appears that technology has suppressed the contagion.

Step 2 – Why that is misleading

The bank's customers have other alternatives, such as withdrawing funds they have on deposit with other banks.

⇒ The run is not stopped; it shows up elsewhere.

Interpretation

This is the reading's reason for agent-based models: what appears to happen between technology and finance is not always accurate, and a simulation of agents' behaviour traces where the pressure goes.

Why the other options are wrong

A is incorrect. Sibling swap into CI-8 c: a circuit breaker is a deliberate, temporary pause on normal rules and regulations, not a system failure.

B is incorrect. Polarity flip: triggered cyber vulnerabilities can cause illiquidity and loss of trust, and the run can continue at other banks.

D is incorrect. Right verdict, wrong reason: the reading's point is that customers can act through other banks now, not that the blocked withdrawals are only delayed.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b and LO 107.c

Solution CI-8 b.6 — Answer: A

Adapted from CI QB Imp Q80 • SchweserNotes Book 5, LO 107.b

Step 1 – Why every institution matters

Virtually all financial institutions are digitalised and have become integrated and more reliant on each other through cloud computing, networks and shared software.

Step 2 – The standard

Each participating institution must be reliable enough to recover quickly from a cyber incident, to keep key functions of the financial system running. Each must be tested to ensure it is sufficiently reliable, which does not necessarily mean 100% reliability.

⇒ Quick recovery, tested, short of perfection.

Interpretation

The standard is about continuity of the system's key functions. It does not ask any institution to be immune; it asks each to come back fast.

Why the other options are wrong

B is incorrect. Absolute claim: sufficient reliability does not necessarily mean 100%. The premise about one key triggering event is true, but it does not imply perfection.

C is incorrect. Scope error: the reading requires each participating institution to be tested, because all are interdependent.
D is incorrect. Scope error: each institution must be tested *given the importance of ICT functions* in the financial system, and ICT disruptions such as physical outages are a key concern (CI-8 a). Testing cannot stop at cyberattacks.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.7 — Answer: D

New • built from SchweserNotes Book 5, LO 107.b

Step 1 – Buffers in theory
Building up liquidity buffers helps to mitigate the effects of bank runs caused by cyber shocks, in theory.

Step 2 – Buffers in history
There has been a high level of liquidity in the financial system for a reasonable amount of time, so such buffers have not had to be used.

Step 3 – What to use instead
A more accurate assessment of the impact of a loss of confidence requires agent-based models.
⇒ There is no record of drawdowns to calibrate from; simulation fills the gap.

Interpretation
The treasurer's method needs a history that does not exist. The idea of a buffer is sound; the calibration has to come from modelling the loss of confidence.

Why the other options are wrong
A is incorrect. Polarity flip: the buffers have *not* had to be used, because system liquidity has been high.
B is incorrect. Polarity flip on the theory: the reading says buffers do help mitigate cyber-driven runs in theory.
C is incorrect. Sibling swap into CI-8 c: the reading says buffers do help in theory, and circuit breakers *supplement* existing deposit controls; nothing in the reading has them replace buffers.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.8 — Answer: B

Adapted from CI QB Imp Q81 • SchweserNotes Book 5, LO 107.b

Step 1 – Integration
The modern design of cyber and ICT connections is highly integrated, which opens the potential for significant contagion effects. Shared software is one of the links that make institutions reliant on each other.

Step 2 – Speed and trigger
Financial contagion can occur extremely rapidly through the industry with the onset of just one key triggering event.
⇒ One incident at one bank can be enough.

Interpretation

The vendor in the stem is a shared node. The reading's point is that integration raises the potential for contagion, not that it adds redundancy.

Why the other options are wrong

A is incorrect. Polarity flip: high integration opens the potential for contagion rather than slowing it.

C is incorrect. Scope error: one key triggering event is enough.

D is incorrect. Polarity flip on predictability: contagion effects are nonlinear, so the probability of key events is hard to predict with reasonable accuracy.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

Solution CI-8 b.9 — Answer: C

New • built from SchweserNotes Book 5, LO 107.b

Step 1 – The end point

The ultimate result of cyber and ICT risks, combined with the vulnerabilities of financial institutions, is an overwhelming loss of customer confidence.

Step 2 – The precedent

The combination has often been seen in the past, in runs on bank deposits that stemmed from customer panic or financial fundamentals.

$\Rightarrow$ A cyber shock ends in the familiar way, as a run.

Interpretation

The first macroprudential tool in CI-8 c, the circuit breaker, is aimed at exactly this run. The class notes put it the same way: cyber risks can diminish customer confidence, potentially leading to bank runs and systemic instability.

Why the other options are wrong

A is incorrect. Scope error: the reading ties the end point to loss of confidence and runs of the familiar kind, not to a separate channel of operational losses.

B is incorrect. Severity flip: confidentiality is the leg with the lower severity, so it is not the end point of the chain.

D is incorrect. Sibling swap into CI-8 c: systemically important technological institutions are the target of structural measures, not the end point of the chain from cyber risk to systemic risk.

LO: CI-8 b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.b

CI-8 • Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector (Banco de España, April 2023) • Vardeez: IMPORTANT

CI-8 c

P3 CONCEPT

Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one.

Workbook: D29 Crypto, Tokenisation and Digital Resilience • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D29 Crypto, Tokenisation and Digital Resilience (Map #68, 1/16, notes only). The workbook has no CI-8 questions, so every question here is new or adapted from the CI question bank.

Vardeez: "Please solve book-wise Current Issues questions as anything can show up." P3 LOs in Current Issues are not thinned: this one, the richest of the reading, gets 10 questions.

Reported exam shapes

- **D29·S3** CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools and their challenges. This LO is the "macroprudential tools and their challenges" part: policy measures (Questions CI-8 c.1 to c.4) and the three tools with their challenges (Questions CI-8 c.5 to c.10).

What the questions look like: no calculations. Classifying a tool (Sheltered Harbor is the classic case), matching each tool to *its own* challenge, attributing each policy measure to the right body or instrument (ITU, ESRB, DORA, the United States), and direction traps: temporary not long-term, supplement not substitute, tick-the-box to judgment and not the reverse.

1. Digital resilience: two layers of policy

Recap from CI-8 a and b: cloud adoption has merged cyber and ICT risk, and a technology event can reach liquidity, leverage and customer trust and then spread through integrated connections. Digital resilience therefore covers both technological and financial aspects, and the reading builds it from two kinds of policy:

1. **Micro-oriented tools:** general operational guidance.
2. **Framework policies:** the fundamental rules for network and system operations. Because of the significant threats to digital resilience in recent years, framework policies have seen major revisions to strengthen defences against new cyber and ICT risks.

The reading then takes the *policy measures and micro-oriented tools* already in use (section 2) before the three *macroprudential tools* (section 3). Section 2 also shows why the second set is needed: principles-based rules work better for the risk of a given entity than for systemic risk.

2. Policy measures and micro-oriented tools

- **ITU: certification and standardisation.** The International Telecommunications Union (ITU) covers a wide range of legal, technical and other measures to promote cyber resilience. Because of the complexity involved, certification and standardisation of solutions are needed as a *starting point* for building cybersecurity defences.
- **Consolidation of bodies.** With many entities involved in cyber resilience, a well-coordinated effort is needed to maximise efficiency and minimise redundancy. The result has been the consolidation

of numerous cybersecurity entities at national level.

- **Forward-looking, principles-based rules.** Recent operational resilience rules are more forward-looking and principles-based than past rules, which they need to be to evolve with the growing complexity of cyber resilience. *The caveat:* a principles-based approach is more useful for the risk of a given entity than for systemic risk, which involves multiple entities.
- **EU incident reporting and the ESRB.** Policies for incident reporting and information sharing have long been in place in the European Union. Going a step further, the European Systemic Risk Board (ESRB) has suggested forming a European-wide structure for coordinating incident reporting.
- **DORA.** The EU's Digital Operational Resilience Act has provisions to respond to ICT third-party risks associated with cloud computing. It also covers penetration stress testing (cyberattack simulation), a useful exercise that keeps entities conscious of security issues on an ongoing basis.

Measure	Who or where	What it does	Limit stated
Certification and standardisation	ITU	Starting point for cybersecurity defences, given the complexity of the measures involved	None stated
Consolidation of bodies	National level	Coordinates many entities: more efficiency, less redundancy	None stated
Principles-based rules	Recent operational resilience rules	Forward-looking; evolve with growing complexity	Suit a given entity's risk better than systemic risk
Incident reporting	EU; ESRB proposal	EU reporting and information sharing long in place; ESRB suggests a European-wide coordinating structure	None stated
DORA	EU	ICT third-party risk from cloud computing; penetration stress testing	None stated

3. Macroprudential tools and the challenge to each

Circuit breakers

- **What.** A circuit breaker is a *temporary* pause on normal rules and regulations. Circuit breakers have been used *more* in the banking industry in recent years.
- **Why.** After a major cyber incident in the banking industry, a circuit breaker minimises the negative impact of customers' loss of confidence and the resulting run on deposits (the end point of CI-8 b). It *supplements* the controls already in place for bank deposits; it does not replace them.
- **Analogy.** A temporary stoppage to analyse the fallout from a major attack and to decide the next steps for recovery.
- **Challenge.** The bank must prove that the cyberattack is serious enough to be considered a **force majeure** event, which would give it some recovery time. The counter-argument is that proper cyber controls and other preventative measures could have avoided the damage, so the attack is not force majeure.
- **Case for.** Its ability to control and minimise the negative effects of a potential bank run after a significant cyberattack.

Cooperative arrangements

- **Slow start.** Cooperation may seem an obvious solution for small and medium-sized organisations, but it has taken time in the financial sector, where most organisations are *large*. With information and knowledge-sharing resources now developing, there is evidence of growing cooperation to build more robust defences against cyberattacks.
- **United States.** There has been progress in unifying the various cybersecurity standards and in moving *from* tick-the-box compliance *to* a judgment-based system.
- **Challenge.** Bringing standardisation and thorough reporting to an **international** level, although cooperation on incident reporting has been rising in Europe and internationally.
- **ICT buffers.** A good example of cooperation against systemic risk: they allow a much quicker recovery for a bank that has suffered a major cyberattack. ICT buffers have been used before in the United States, in its **tri-party repo** system.
- **Sheltered Harbor, LLC.** A collectively owned US entity. If a co-owner bank suffers a serious cyberattack, its customers can transact on their accounts at another co-owner bank. With consistent data formats and operating procedures for storing and restoring data, the risk to a co-owner bank from a cyberattack is substantially reduced.

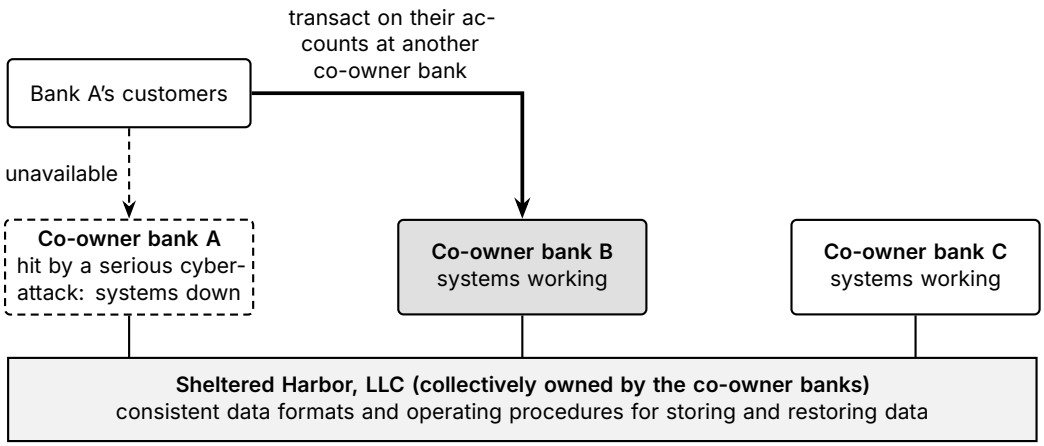


Figure 1. An ICT buffer, the reading's example of a cooperative arrangement. The dashed bank is the one under attack; the shaded bank serves its customers; the bar underneath is the shared standard that makes the switch possible. It is cooperation among banks, not a pause on rules (circuit breaker) and not a mandate on providers (structural measure).

Structural measures

- **SITIs.** The reach of macroprudential regulation has inevitably spread to **systemically important technological institutions** (**SITIs**), given the numerous externalities they create and the financial sector's significant dependence on them.
- **Alternatives to macroprudential regulation.** More thorough evaluations of critical providers; strongly encouraging multicloud services; using private clouds.
- **Forced diversification.** Regulators could push firms to use multiclouds or multiple types of software. Penalties for concentrated software use could act as negative motivators.
- **Challenge.** Penalties introduce the issue of **enforceability**.

One link back to CI-8 a: multiclouds are not free. Encouraging them answers concentration risk but

brings the unintended cost the reading names there, weaker stewardship over the distributed computing processes.

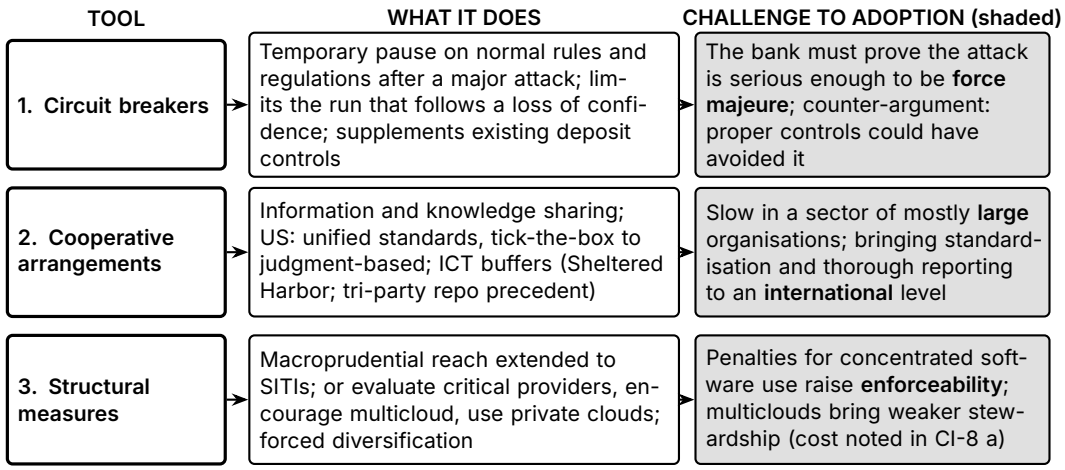


Figure 2. The three macroprudential tools with the challenge to each (redrawn from class notes Figure CI-8.2, with the challenges completed from SchweserNotes). The shaded right-hand column is the one the LO asks for; each challenge belongs to its own row, and swapping rows is the commonest distractor.

Completing the source notes. The class notes (Figure CI-8.2) attach a challenge only to the circuit breaker, and state it as defining how severe an attack must be before the breaker fires. That is incomplete rather than wrong. The LO asks for the challenge to *each* tool, and SchweserNotes Book 5, LO 107.c gives one for each. It frames the circuit breaker's as a burden of proof: the bank must show the attack is serious enough to be force majeure, against the argument that proper controls could have prevented it (Question CI-8 c.6). For cooperation it gives slow progress in a sector of mostly large organisations and the need for international standardisation and reporting; for structural measures, the enforceability of penalties. This LO also adds what the class notes leave out: the ESRB proposal, the principles-based caveat, the tri-party repo precedent and the name Sheltered Harbor, LLC. Only the last is listed in the notes' corrections record, as considered and not made.

Results card: CI-8 c

Tool or measure	What it does	Challenge to adoption
Two layers	Micro-oriented tools: general operational guidance. Framework policies: fundamental rules for network and system operations, recently revised	
Policy measures	ITU: certification and standardisation as a starting point; national consolidation of bodies; forward-looking, principles-based rules; ESRB: European-wide incident reporting structure; DORA: cloud third-party risk, penetration testing	Principles-based rules suit a given entity better than systemic risk
Circuit breakers	Temporary pause on normal rules; limits the run after a loss of confidence; supplements deposit controls; used more in recent years	Prove force majeure, against "proper controls would have avoided it"
Cooperative arrangements	Information sharing; US: unified standards, tick-the-box to judgment-based; ICT buffers: Sheltered Harbor, tri-party repo precedent	Slow in a sector of large organisations; international standardisation and reporting
Structural measures	SITIs within macroprudential reach; provider evaluations, multicloud, private clouds; forced diversification	Enforceability of penalties; multiclouds bring weaker stewardship (cost noted in CI-8 a)

Exam traps

- **Sheltered Harbor misfiled.** It is a cooperative arrangement (an ICT buffer), not a circuit breaker and not a structural measure.
- **Circuit breaker polarity.** Temporary, not long-term; supplements deposit controls, does not substitute for them; used *more in recent years*, not for decades.
- **Burden swapped.** The bank, not the supervisor, must prove the attack is serious enough to be force majeure.
- **Challenge swap.** Force majeure belongs to circuit breakers, international standardisation and reporting to cooperation, enforceability to penalties under structural measures.
- **Principles-based reversed.** More useful for a given entity than for systemic risk.
- **US direction reversed.** From tick-the-box to judgment-based, not the other way.
- **Role misassignment.** ESRB: European-wide incident reporting structure. DORA: cloud third-party risk and penetration testing. ITU: certification and standardisation. Sheltered Harbor and tri-party repo: United States.
- **Size flipped.** Cooperation was slow because most financial organisations are *large*.
- **SITIs outside the perimeter.** Macroprudential reach has *inevitably spread* to them.
- **Layers swapped.** Micro-oriented tools give operational guidance; framework policies set the fundamental rules.
- **Sequence error.** Certification and standardisation are the *starting point*, not a final step.

Practice questions: CI-8 c

Question CI-8 c.1 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A policy adviser at a finance ministry is drafting a national digital resilience strategy around the Banco de España paper's two-part approach. Which description of the two parts is correct?

A. Framework policies give operational guidance to individual entities, while micro-oriented tools set the fundamental rules for system operations.

B. Micro-oriented tools give operational guidance to entities, while framework policies set fundamental rules that have changed little in recent years.

C. Both parts deal only with network and system operations, because digital resilience covers the technological but not the financial side.

D. Micro-oriented tools give general operational guidance, while framework policies set the fundamental rules and have been revised substantially.

Question CI-8 c.2 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

An official at a national cybersecurity agency is reviewing how the country's many cyber resilience bodies and standards should be organised. Based on the Banco de España paper, which statement is correct?

A. Overlapping bodies should be kept, because the redundancy they create is a useful defence when one of the bodies is itself attacked.

B. Certification and standardisation of solutions are a starting point, and coordination has led to consolidating cybersecurity bodies.

C. Certification and standardisation should come only after penetration stress testing has shown where each institution's defences fail.

D. The ITU is the EU body that runs penetration stress testing of cloud providers, so national bodies should defer to its results.

Question CI-8 c.3 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A macroprudential supervisor argues that recent forward-looking, principles-based operational resilience rules are enough on their own to contain systemic cyber risk. Which response best reflects the Banco de España paper?

A. The premise is wrong: recent operational resilience rules are more prescriptive and backward-looking than the rules they replaced.

B. The supervisor is right, because a principles-based approach is at its most useful when a risk spans many entities at once.

C. The supervisor is wrong: principles-based rules suit one entity's risk better than systemic risk, which spans multiple entities.

D. The supervisor is right, because the rules replaced tick-the-box compliance with judgment, which removes the need for other tools.

Question CI-8 c.4 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

The head of compliance at a Dutch bank is mapping EU initiatives on digital resilience for the board. Which statement is correct?

A. The ESRB has proposed EU-wide coordination of incident reporting; DORA covers ICT third-party risks from cloud computing.

B. DORA has proposed EU-wide coordination of incident reporting; the ESRB runs penetration stress tests of EU banks.

C. The EU has only recently begun incident reporting and information sharing, under the ESRB's proposal for a European-wide structure.

D. DORA covers penetration stress testing but leaves ICT third-party risks from cloud computing to Sheltered Harbor-type arrangements.

Question CI-8 c.5 Adapted from CI QB Imp Q84 • SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A deposit insurer's policy team is evaluating circuit breakers as a response to a major cyber incident at a bank. Which statement about circuit breakers is most consistent with the Banco de España paper?

A. They are a long-term suspension of normal rules and regulations, giving the bank as much time as it needs to rebuild its systems.

B. They are a temporary pause on normal rules that supplements deposit controls and targets the run a loss of confidence brings.

C. They are a valid substitute for existing deposit controls, because a pause in normal rules makes those controls unnecessary.

D. They have been used extensively in banking for many decades, so the paper treats them as a long-proven tool for cyber incidents.

Question CI-8 c.6 Adapted from CI QB Imp Q87 • SchweserNotes Book 5, LO 107.c; CI notes CI-8 c (completion box)

The general counsel of a bank hit by a severe cyberattack is preparing the bank's request to its supervisor to activate a circuit breaker. Which challenge to the adoption of this tool does the Banco de España paper identify?

A. The supervisor must prove the attack is serious enough to be force majeure, because pausing normal rules is the supervisor's decision.

B. It works only once standardisation and thorough incident reporting have been brought to an international level, which has not yet happened.

C. The pause weakens stewardship over the bank's operations, because normal rules and oversight are suspended while it lasts.

D. The bank must prove the attack is serious enough to be force majeure, against the argument that proper controls could have avoided it.

Question CI-8 c.7 Adapted from CI QB Imp Q83 • SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A US bank's board is asked to join Sheltered Harbor, LLC. Under the arrangement, if a co-owner bank suffers a serious cyberattack, its customers can transact on their accounts at another co-owner bank, supported by consistent data formats and procedures for storing and restoring data. How does the Banco de España paper classify this tool?

A. As a cooperative arrangement: an ICT buffer that allows a much quicker recovery for a co-owner bank hit by a major cyberattack.

B. As a circuit breaker: a temporary pause on normal rules that limits the deposit run after the attacked bank loses customer confidence.

C. As a structural measure: forced diversification that requires each bank to spread its customers' accounts across several providers.

D. As a micro-oriented tool: general operational guidance on data storage and restoration that each co-owner bank applies on its own.

Question CI-8 c.8 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A senior official at an international standard-setting body is reviewing progress on cooperation against cyber threats in the financial sector. Which statement is most consistent with the Banco de España paper?

A. Cooperation came quickly in finance, because most financial organisations are small or medium-sized and gain most from sharing resources.

B. The United States has moved from judgment-based cybersecurity standards to tick-the-box compliance, which eases international comparison.

C. The US has moved toward unified, judgment-based standards; the remaining challenge is international standardisation and reporting.

D. Cooperation on incident reporting has declined in Europe and internationally as authorities consolidate around national bodies.

Question CI-8 c.9 New • built from SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A regulator proposes penalties for banks that rely on a single software vendor for critical services, to push them toward using multiple types of software. Which assessment is most consistent with the Banco de España paper?

A. This is a cooperative arrangement, and its main challenge is proving that a vendor failure is serious enough to be a force majeure event.

B. This is forced diversification, a structural measure; penalties act as negative motivators but raise the issue of enforceability.

C. This is a circuit breaker for software concentration, and the paper finds penalties of this kind straightforward to enforce.

D. This is a structural measure, but the paper rules out penalties and limits regulators to evaluating critical providers more thoroughly.

Question CI-8 c.10 Adapted from CI QB Imp Q86 • SchweserNotes Book 5, LO 107.c; CI notes CI-8 c

A financial stability committee is discussing the banking sector's heavy dependence on a handful of large cloud and technology providers. Which recommendation best reflects the Banco de España paper's discussion of structural measures?

A. Leave the providers outside macroprudential regulation, because their externalities fall on technology markets rather than on the financial sector.

B. Encourage banks to consolidate on the single most robust provider, so that supervisors can concentrate their oversight in one place.

C. Treat them as SITIs within macroprudential reach or, as alternatives, evaluate critical providers closely and encourage multiclouds.

D. Bring them within macroprudential reach as SITIs, but discourage multiclouds, which the paper finds add to concentration risk.

Solutions • CI-8 c

Solution CI-8 c.1

Answer: D

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – Name the two layers

Micro-oriented tools deal with general operational guidance. Framework policies deal with the fundamental rules for network and system operations.

Step 2 – What has happened to them

Because of the significant threats to digital resilience in recent years, framework policies have seen major revisions to strengthen defences against new cyber and ICT risks.
⇒ Guidance at one level, rules at the other, and the rules have been revised.

Interpretation

Digital resilience covers both technological and financial aspects, which is why the reading goes on from these two layers to macroprudential tools aimed at the financial system.

Why the other options are wrong

A is incorrect. Sibling swap: the two definitions are exchanged.
B is incorrect. Polarity flip on the second half: framework policies have seen major revisions, not little change.
C is incorrect. Scope error: digital resilience covers both technological and financial aspects.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023), Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector; SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.2

Answer: B

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – The starting point

The ITU covers a wide range of legal, technical and other measures to promote cyber resilience. Because of the complexity involved, certification and standardisation of solutions are needed as a starting point for building cybersecurity defences.

Step 2 – Organising the bodies

With so many entities involved, a well-coordinated effort is needed to maximise efficiency and minimise redundancy. The result has been the consolidation of numerous cybersecurity entities at national level.
⇒ Standardise first; consolidate the bodies.

Interpretation

Both points are about coherence: common standards and fewer, better-coordinated bodies. Neither is yet a macroprudential tool; they are the groundwork.

Why the other options are wrong

A is incorrect. Polarity flip: the reading seeks to *minimise* redundancy, and the trend is consolidation.
C is incorrect. Sequence error: certification and standardisation are the *starting point*, not a step that follows testing.
D is incorrect. Role misassignment: penetration stress testing is a DORA provision; the ITU is cited for its wide range of legal, technical and other measures.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.3 — Answer: C

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – Grant the premise

Recent operational resilience rules are indeed more forward-looking and principles-based than past rules, which they need to be to evolve with the growing complexity of cyber resilience.

Step 2 – The caveat

A principles-based approach is more useful when working with the risk of a given entity than with systemic risk, which involves multiple entities.

⇒ Entity-level principles do not by themselves contain a system-wide risk.

Interpretation

This is the bridge from the policy measures to the macroprudential tools: the systemic part of cyber risk needs instruments that act across institutions.

Why the other options are wrong

A is incorrect. Polarity flip: recent rules are more forward-looking and principles-based, not more prescriptive and backward-looking.

B is incorrect. Polarity flip on the caveat: principles-based rules are more useful for a given entity than for risk spanning many entities.

D is incorrect. Sibling swap with overreach: the move from tick-the-box compliance to a judgment-based system is the reading's point about cooperation in the United States, and nothing in the reading says it removes the need for other tools.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.4 — Answer: A

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – The ESRB

Policies for incident reporting and information sharing have long been in place in the EU. Going a step further, the European Systemic Risk Board has suggested forming a European-wide structure for coordinating incident reporting.

Step 2 – DORA

The Digital Operational Resilience Act has provisions to respond to ICT third-party risks associated with cloud computing, and provisions on penetration stress testing.

⇒ ESRB: coordinated reporting. DORA: cloud third-party risk and penetration testing.

Interpretation

Attribution questions turn on who did what. Keep the four names apart: ITU (certification and standardisation), ESRB (incident reporting structure), DORA (cloud third-party risk, penetration testing), Sheltered Harbor (US ICT buffer).

Why the other options are wrong

B is incorrect. Role misassignment: the two bodies' roles are swapped. The ESRB made the incident reporting proposal; penetration testing sits in DORA.

C is incorrect. Polarity flip on timing: EU incident reporting and information sharing policies have long been in place; the ESRB proposal goes a step further.
D is incorrect. Role misassignment and jurisdiction error: DORA covers ICT third-party risks from cloud computing, and Sheltered Harbor is a US arrangement among banks.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.5 — Answer: B

Adapted from CI QB Imp Q84 • SchweserNotes Book 5, LO 107.c

Step 1 – What it is

A circuit breaker is a temporary pause on normal rules and regulations, analogous to a temporary stoppage to analyse the fallout from a major attack and decide the next steps for recovery.

Step 2 – What it is for

After a major cyber incident, it minimises the negative impact of customers' loss of confidence and the resulting bank run on deposits. It supplements the controls already in place for deposits.
⇒ Temporary, targeted at the run, added to existing controls.

Interpretation

The circuit breaker is the macroprudential answer to the end point of CI-8 b: the loss of confidence that turns into a run.

Why the other options are wrong

A is incorrect. Polarity flip on duration: the pause is temporary.
C is incorrect. Sibling swap: circuit breakers supplement deposit controls; they are not a substitute.
D is incorrect. Timeline error: circuit breakers have been used *more* in the banking industry in recent years, not extensively for decades.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.6 — Answer: D

Adapted from CI QB Imp Q87 • SchweserNotes Book 5, LO 107.c

Step 1 – The burden on the bank

The challenge is for the bank to prove that the attack is serious enough to be considered a force majeure event, which would give it some recovery time.

Step 2 – The counter-argument

It could be argued that proper cyber controls and other preventative measures could have avoided the negative effects, so the attack is not force majeure.

Step 3 – The case for

Circuit breakers can control and minimise the negative effects of a potential bank run after a significant attack.
⇒ The difficulty is the trigger, not the usefulness of the tool.

Interpretation

The burden of proof sits with the bank, and the obstacle is the preventability argument. The other options either move that burden or borrow a challenge or cost that the reading attaches to

something else; learn the shaded column of Figure 2 row by row.

Why the other options are wrong

A is incorrect. Role misassignment: the reading puts the burden on the bank, which must prove the attack is serious enough to be force majeure; the supervisor does not have to prove it.
B is incorrect. Sibling swap: bringing standardisation and thorough reporting to an international level is the challenge for cooperative arrangements.
C is incorrect. Sibling swap: weaker stewardship is the unintended cost of multiclouds (CI-8 a), not of a circuit breaker, whose stated challenge is the force majeure test.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a and LO 107.c

Solution CI-8 c.7 — Answer: A

Adapted from CI QB Imp Q83 • SchweserNotes Book 5, LO 107.c

Step 1 – Identify the mechanism

Customers of an attacked co-owner bank transact on their accounts at another co-owner bank, with consistent data formats and operating procedures for storing and restoring data.

Step 2 – Classify it

This is an ICT buffer, which the reading gives as a good example of cooperation against systemic risk. It allows a much quicker recovery for a bank that has suffered a major cyberattack. (ICT buffers have also been used in the past in the US tri-party repo system.)
$\Rightarrow$ A cooperative arrangement.

Interpretation

The value of the arrangement is shared: each co-owner bank's risk from a cyberattack is substantially reduced, because the others stand ready to serve its customers on a common data standard.

Why the other options are wrong

B is incorrect. Sibling swap: a circuit breaker pauses normal rules; Sheltered Harbor keeps customers transacting through another bank.
C is incorrect. Sibling swap: forced diversification pushes firms toward multiclouds or multiple types of software. Nothing in Sheltered Harbor spreads accounts across providers.
D is incorrect. Layer swap: micro-oriented tools are general operational guidance. The common data formats here serve an arrangement that works across co-owner banks against systemic risk; no bank applies it on its own.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.8 — Answer: C

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – Progress

In the United States there has been progress in unifying the various cybersecurity standards and in moving from tick-the-box compliance to a judgment-based system.

Step 2 – The challenge

The challenge remains to bring standardisation and thorough reporting to an international level, although cooperation on incident reporting has been rising in Europe and internationally. Option

C states both halves: the national progress and the international challenge.
⇒ National progress; the international step is the hard part.

Interpretation

Cooperation was also slow to start: it may seem obvious for small and medium-sized organisations, but it took time in a sector where most organisations are large.

Why the other options are wrong

A is incorrect. Polarity flip on size: cooperation took time precisely because most financial organisations are large.

B is incorrect. Polarity flip on direction: the move is from tick-the-box compliance to a judgment-based system.

D is incorrect. Polarity flip on the trend: cooperation on incident reporting has been rising in Europe and internationally.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.9 — Answer: B

New • built from SchweserNotes Book 5, LO 107.c

Step 1 – Classify the proposal

Pushing firms toward multiclouds or multiple types of software is what the reading calls forced diversification, one of the structural measures.

Step 2 – The lever and its challenge

Penalties for concentrated software use could be used as negative motivators; however, penalties introduce the issue of enforceability.
⇒ Right tool for concentration, hard to enforce.

Interpretation

The regulator's other options under structural measures are softer: more thorough evaluations of critical providers, strongly encouraging multicloud services and using private clouds.

Why the other options are wrong

A is incorrect. Sibling swap twice: the proposal is a structural measure, not a cooperative arrangement, and force majeure is the circuit breaker's challenge.

C is incorrect. Sibling swap with a polarity flip: it is not a circuit breaker, and penalties raise the issue of enforceability rather than being straightforward.

D is incorrect. Absolute claim: the reading says penalties *could* be used, with an enforceability problem; it does not rule them out.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.c

Solution CI-8 c.10 — Answer: C

Adapted from CI QB Imp Q86 • SchweserNotes Book 5, LO 107.c

Step 1 – Why the providers come into scope

The reach of macroprudential regulation has inevitably spread to systemically important technological institutions (SITIs), given the numerous externalities they create and the financial sector's significant dependence on them.

Step 2 – The alternatives

Alternatives to macroprudential regulation include more thorough evaluations of critical providers, strongly encouraging multicloud services and using private clouds.

⇒ Bring the providers in, or manage the dependence through evaluation and diversification.

Interpretation

Encouraging multiclouds answers concentration risk, but it carries the cost named in CI-8 a: weaker stewardship over distributed computing processes. The committee should weigh both.

Why the other options are wrong

A is incorrect. Polarity flip: macroprudential reach has *spread* to SITIs, precisely because of their externalities and the financial sector's dependence on them.

B is incorrect. Polarity flip: concentrating on one provider is the concentration risk the reading wants reduced through multiclouds.

D is incorrect. Half right, half reversed: macroprudential reach has spread to SITIs, but multiclouds are the evolution meant to deal with concentration risk, and strongly encouraging them is one of the alternatives.

LO: CI-8 c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one. • **Reference:** Martinez, Banco de España (April 2023); SchweserNotes 2026 Book 5, Reading 107, LO 107.a and LO 107.c